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第一篇 概 述 篇

本系统是我公司最新开发的普及型铣床、钻床、镗床数控系统。控制电路采用最新工业级高速 ARM 处理器、大规模现场可编程门列矩阵 FPGA 技术，多层 PCB 印制电路板，整机采用高集成芯片和表贴元件，结构更为紧凑合理，更好的保证了系统的可靠性和稳定性。

实时控制高速度(最高快移速度可达 240 米/分、最高进给插补速度可达 30 米/分)、高精度；采用 800x600 点阵 TFT LCD 亮度自适应液晶显示技术，LED 背光亮度均匀使用寿命长，且克服了以往显示亮度受环境温度变化的缺点。全屏幕中文菜单显示，操作极为简单方便。

本系统是以铣床为代表的三轴、四轴、五轴、六轴、七轴、八轴坐标联动，总线闭环控制普及型全数字式数控系统，功能强大，指令丰富，编程代码符合 ISO 国际代码标准。直接控制交流伺服系统，如果选配本公司生产的双通道交流伺服驱动器性价比更优。

本使用手册详细介绍了本数控系统的编程和使用方法。

重要提示：

1、所有第 A、B、C、Xs、Ys 轴的功能在选配第四、五轴、六轴、七轴、八轴系统时才有效。

2、对本数控系统操作不当可能引起意外事故，在使用前务必仔细阅读本使用手册。

3、所有绝对值功能必须是选配的系统是绝对值系统才有效，所有总线功能必须是选配的系统是总线系统才有效。

4、系统面板上的“启动”按键作为调试时使用（杂类参数 P9-D14 位设置“有效”或“无效”，在正常使用时必须设置为“0”），在系统装配时必须外加“启动”按钮，否则由于“启动”按键寿命问题可能引起意外事故!!! 所以，本系统禁止长时间使用系统面板上的“启动”按键，否则由此造成的安全后果与我公司无关!!!

Chapter I. Introduction

This system represents our company's latest development in a user-friendly CNC control solution for milling, drilling, and broaching machines. Utilizing cutting-edge industrial-grade ARM processors and large-scale field-programmable gate array (FPGA) technology, it features multi-layer PCB boards with integrated high-density chips and surface-mount components. The compact design significantly enhances system reliability and stability while maintaining optimal operational efficiency.

Real-time control with high-speed operation (maximum linear speed: 240 m/min, maximum feed interpolation speed: 30 m/min) and high precision. Featuring an 800x600 dot-matrix TFT LCD display with adaptive brightness technology, the LED backlight ensures uniform illumination and extended lifespan, effectively addressing previous limitations of ambient temperature-dependent brightness in displays. The full-screen Chinese menu interface provides exceptionally intuitive operation.

This system is a fully digital CNC solution featuring three-axis, four-axis, five-axis, six-axis, seven-axis, and eight-axis coordinate linkage with milling machine as the core component. It employs closed-loop bus control technology and delivers robust functionality with extensive command capabilities, programming standards compliant with ISO international specifications. The system directly controls AC servo motors, and when paired with our company's dual-channel AC servo driver, it achieves superior cost-effectiveness through optimized performance.

This manual introduces in detail the programming and use of this CNC system.

Important note:

1、 All functions of the A, B, C, Xs and Ys axes are valid only when the fourth, fifth, sixth, seventh and eighth axis systems are selected.

2. Improper operation of this CNC system may cause accidents, so please read this manual carefully before use.

3. All absolute value functions must be optional. The system is valid only when it is an absolute value system. All bus functions must be optional. The system is valid only when it is a bus system.

4. The "Start" button on the system panel is used during debugging (Miscellaneous Parameters P9-D14: Set the bit to "Enabled" or "Disabled". During normal operation, it must be set to "0"). When assembling the system, an additional "Start" button must be added. Otherwise, due to the lifespan limitations of the "Start" button, accidents may occur!!! Therefore, this

The system prohibits the long-term use of the "start" button on the system panel, otherwise the security consequences caused by this are not related to our company!!!

第二篇 编程篇

数控机床是按照事先编好的数控程序自动地对工件进行加工的高效自动化设备。编制是根据被加工工件的图纸和加工工艺的要求，用所使用数控系统的数控语言来描述加工轨迹，及其辅助动作的过程。理想的数控程序不仅应该保证能加工出符合图纸要求的合格工件，还应使数控机床的功能得到合理的应用和充分的发挥，所以编好数控程序非常重要，本篇将详细介绍数控程序的各种指令及用法，请仔细阅读。

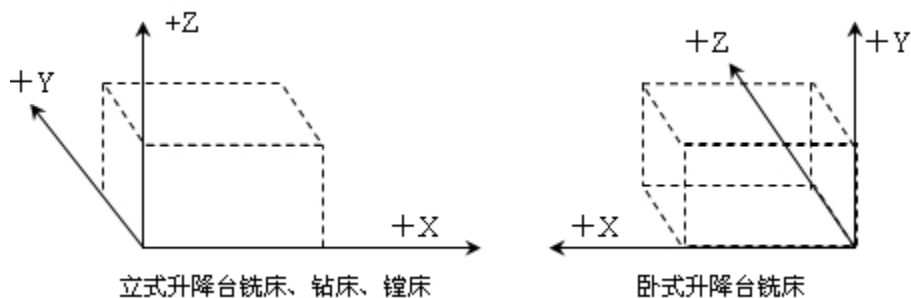
2.1 基本概念

程序段：（程序行）是由指令段和数据段共同组成的一个完整命令行。

程序：是为了完成工件的加工，按加工的逻辑结构组合的全部程序段的集合。

机床坐标系：是以机床零点为原点建立的坐标系。铣床坐标轴和其方向按“ISO841”的规定。标准中规定的基本确定方法，即：以右手法则确定的笛卡尔直角坐标系作为编程标准坐标系，对于刀具旋转的机床（如铣床）等，Z轴是平行于主轴的坐标轴，X轴一般是水平的，与工件卡装面平行，Y轴由右手法则确定，各轴的运动使工件的尺寸增加的方向为各轴的正方向。

当未设定工件坐标系时，将机床坐标系作为工件坐标系。



机床坐标轴及方向示意图

工件坐标系：工件加工时使用的坐标系称作工件坐标系，由CNC 预先设置。工件坐标系可以用移动它的原点来改变。

设置工件坐标系有二种方法：

1 、G50/G52指令

程序中在G52指令之后指定一个值来设定工件坐标系。

2 、G54/G54.1-G54.48/G55/G56/G57/G58/G59指令

使用参数设置中的坐标系可以设置6个工件坐标系。

用绝对值指令时必须用上述方法建立工件坐标系

Part II: Programming

A CNC machine is an efficient automated device that processes workpieces automatically according to pre-programmed numerical control (NC) instructions. Programming involves describing the machining path and auxiliary actions using the NC language of the system, based on the workpiece's blueprint and manufacturing requirements. An ideal NC program must not only ensure production of qualified parts meeting design specifications but also optimize the machine's capabilities. This makes proper programming crucial. This article provides a detailed explanation of various NC instructions and their applications. Please read carefully.

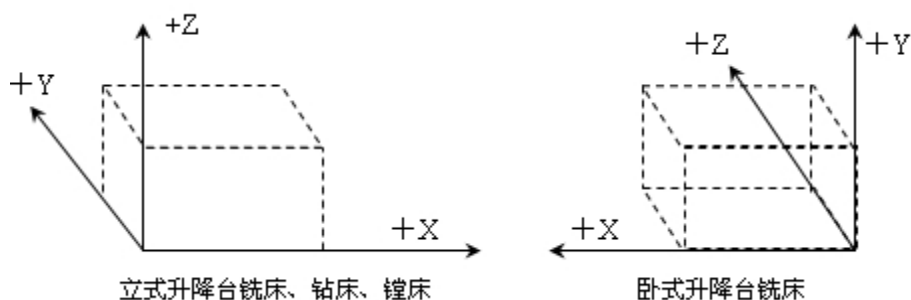
2.1 Conceptual Framework

Program section: (program line) is a complete command line composed of instruction section and data section.

Program: is the collection of all program segments combined according to the logical structure of processing in order to complete the machining of workpiece.

Machine tool coordinate system: A coordinate system established with the machine tool's zero point as the origin. The milling machine's coordinate axes and their orientations follow the ISO841 standard. The basic determination method specified in the standard is: using the Cartesian rectangular coordinate system defined by the right-hand rule as the programming reference system. For machine tools with rotating tools (such as milling machines), the Z-axis is parallel to the spindle, the X-axis is generally horizontal and parallel to the workpiece clamping surface, while the Y-axis is determined by the right-hand rule. The direction of motion that increases the workpiece's dimensions is defined as the positive direction of each axis.

When the workpiece coordinate system is not set, the machine tool coordinate system is taken as the workpiece coordinate system.



Schematic diagram of machine tool coordinate axis and direction

Workpiece coordinate system: The coordinate system used in the machining of a workpiece is called the workpiece coordinate system, which is preset by the CNC. The workpiece coordinate system can be changed by moving its origin.

There are two ways to set the workpiece coordinate system:

1 、 G50/G52 instructions

In the program, a value is specified after the G52 instruction to set the workpiece coordinate system.

2 、 G54/G54.1-G54.48/G55/G56/G57/G58/G59 Directive

Use the coordinate system in the parameter Settings to set up 6 workpiece coordinate systems.

When using the absolute value instruction, the workpiece coordinate system must be established by the above method

局部坐标系: 当在工件坐标系中编制程序时, 为容易编程可以设定工件坐标系的子坐标系, 该子坐标系称为局部坐标系。

绝对坐标编程 是以建立的绝对坐标系为基准确定的坐标数据编程方式, 绝对坐标是指刀具相对于加工坐标系原点的坐标。用 G90 指令, X、Y、Z、A、B、C、Xs、Ys 坐标字表示。

例如: G00 G90 X200 Y200 Z300

相对坐标编程 也叫增量坐标编程, 是指运行终点相对于运行起点的距离和方向。相对坐标是指刀具运动位置的坐标值是相对于前位置, 而不是相对于固定的加工坐标系原点。用 G91 指令, X、Y、Z、A、B、C、Xs、Ys 坐标字表示。

例如: G00 G91 X200 Y200 Z300

最小编程单位: 最小编程单位为 0.001mm。

模态指令: 是指在程序中具有保留功能属性的指令, 它不仅在本程序行起作用, 而且在以后的程序行中也起作用。直到其它同组的模态指令替代为止。

在同样操作类型中可能具有多个模态指令, 如 M03 (主轴顺转)、M04 (主轴逆转)、M05 (主轴停) 它们同样用来控制主轴, 都是模态, 在指令系统中把同类型的模态指令划分为一个模态组。任何时候只能是其中之一。而且必然有其中之一, 因此存在一个初始状态问题, 初始选用的模态指令, 叫做模态初态。如上述模态组中, 模态初态为 M05。

中止模态 (破坏模态): 是指具有把模态指令变为模态初态或模态可能被破坏的指令。比如 M20 (程序结束指令), 表示运行结束, 回到开始状态, 把全部模态指令变为模态初态。

鑫科瑞数控

非模态指令: 是指不具备保留功能, 仅在本程序段内起作用。

2.2 程序格式的一般表达式

G02, T02, H02, D02, M02, S04, F04, X-043, Y-043, Z-043, A-043, B-043, C-043, Xs-043, Ys-043, I-043, J-043, K-043, L04, P4, R043。

注 1: 地址符后有“-”号表示该数据可以用“-”符号 (正号不写)。

注 2: 数字前为 0, 表示该数据可以只写有效数据位。

注 3: 数字表示的是位数, 当为两位时, 高位表示整数位数的最多位数, 低位表示小数点后的最多位数。

2.3 程序指令代码及其功能说明

本节将详细介绍本系统中所有指令代码的功能及使用方法。

2.3.1 指令及数据说明

功 能	地址符	意 义	数据范围
-----	-----	-----	------

Local coordinate system: When programming in the workpiece coordinate system, a sub-coordinate system of the workpiece coordinate system can be set to facilitate programming. This sub-coordinate system is called the local coordinate system.

Absolute coordinate programming: This is a programming method that determines coordinate data based on the established absolute coordinate system. Absolute coordinates refer to the coordinates of the tool relative to the origin of the machining coordinate system. The G90 command uses X, Y, Z, A, B, C, Xs, and Ys coordinates to represent this.

For example: G00 G90 X200 Y200 Z300

Relative coordinate programming, also known as incremental coordinate programming, refers to the distance and direction between the end point and the starting point during machining. This method determines tool movement positions relative to their previous locations rather than a fixed reference coordinate system origin. The G91 command specifies these coordinates using X, Y, Z, A, B, C, Xs, and Ys parameters.

For example: G00 G91 X200 Y200 Z300

Minimum program unit: minimum program unit is 0.001mm.

Modal instructions: These are instructions in a program that have reserved functional attributes. They not only work in the current line of the program, but also in later lines of the program until they are replaced by other modal instructions in the same group.

Within the same operational type, multiple mode commands may exist, such as M03 (spindle forward rotation), M04 (spindle reverse rotation), and M05 (spindle stop). These commands all control the spindle and are classified as modes within the instruction system. The same type of mode commands are grouped into a mode group, with only one applicable at any given time. Since there must be one selected mode, an initial state issue arises—this initial mode command is called the mode initial state. For example, in the aforementioned mode group, the mode initial state is M05.

Termination mode (destructive mode): refers to the instruction that changes the mode instruction to the mode initial state or the mode may be destroyed. For example, M20 (program termination instruction), which indicates the end of operation and returns to the starting state, changing all mode instructions to the mode initial state.

Non-modular instruction: refers to the instruction that does not have the retention function and only takes effect in this program segment.

2.2 A General Expression for Program Formatting

G02, T02, H02, D02, M02, S04, F04, X-043, Y-043, Z-043, A-043, B-043, C-043, Xs-043, Ys-043, I-043, J-043, K-043, L04, P4, R043.

Note 1: A "-" after the address indicates that the data can be written with the "-" symbol (the plus sign is not written).

Note 2: A number with a zero in front indicates that the data can only write valid data bits.

Note 3: The number indicates the number of bits. When it is two bits, the high bit indicates the maximum number of bits in the integer number and the low bit indicates the maximum number of bits after the decimal point.

2.3 Program Instruction Code and Its Function Description

This section will detail the functions and usage of all instruction codes in this system.

2.3.1 Instructions and Data Descriptions

function	address character	meaning	data area
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文件名		加工工件的文件名	0-9、A-Z 组成
程序行号	N	程序段的顺序号	0000-99999 行距可设
准备功能	G	指定动作状态 (G01、G02 等)	00-99
辅助功能	M	控制机床辅助动作	00-99
刀具功能	TDH	指定刀具号	换刀 00-99
主轴功能	S	指定第一主轴转速	0-99999
	SS	指定第二主轴转速	0-99999
切削速度	F	指定每分钟的切削速度	0.01-15000mm/min
坐标字	XYZABCX sYs	分别为 XYZABCXsYs 轴坐标移动值	±99999.999mm
圆心坐标	IJK	X、Y、Z 轴圆心坐标增量值	±99999.999mm
圆弧半径	R	圆弧半径值	0.001-99999.999mm
延时时间	P、X	指定 G04 的延时时间	0.001-65s
程序入口	P	子程序调用或转移入口行	0000-9999
重复次数	L	循环次数或子程序调用次数 L 也可多头螺纹头数	1-99999 1-99
程序行跳步功能	/	如果程序行首有“/”表示此行不执行	杂类参数 P9 的 D12=1 时可以屏蔽跳步功能

2.3.2 程序指令表

表 1 G 指令代码及功能

G 代码	功能
G00	快速定位
G01	直线插补
G31	检测跳转不报警
G311	检测跳转要报警
G02	顺圆插补/螺旋线插补 CW: 螺旋移动的螺旋线插补可指令 2 个圆弧插补轴同步移动的其他轴。指令方法只是简单地加上一个不是圆弧插补轴的移动轴。
G03	逆圆插补/螺旋线插补 CCW
G01(G00) X I	自动倒斜角
G01(G00) YI	自动倒斜角
G01(G00) ZI	自动倒斜角

document name		The file name of the workpiece being processed	0-9, A-Z composition
Line number of the program	N	The sequence number of the program segment	000-99999 line spacing can be set
g function	G	Specify the action status (G01, G02, etc.)	00-99
additional function	M	Control machine tool auxiliary actions	00-99
tool function	TDH	Specify the tool number	Change the blade 00-99
Spindle function	S	Specify the spindle speed	0-99999
	SS	Specify the second spindle speed	0-99999
cutting speed	F	Specify the cutting speed per minute	0.01-15000mm/min
Coordinate letter	XYZABCXsYs	Are the axis coordinate movement values of XYZABC-XsYs respectively	±99999.999mm
Center of circle coordinates	IJK	X, Y, Z axis center of circle coordinate increment value	±99999.999mm
Arc radius	R	Arc radius value	0.001-99999.999mm
Delay time	P、X	Specify the delay time for G04	0.001-65s
program entry	P	A subroutine call or transfer entry line	0000-9999
number of replication	L	The number of cycles or subroutine calls L can also be the number of threads in a multi-head thread	1-99999 1-99
Procedure: Walk function	/	If the line starts with a /, it is not executed	When D12=1 in the miscellaneous category parameter P9, the skip function can be shielded

2.3.2 Program Instruction Table

Table 1 G Instruction Code and Function

G code	function
G00	quick positioning
G01	linear interpolation
G31	No alarm is raised when the test jump is detected
G311	The test jump should alarm
G02	Circular interpolation/spiral interpolation CW: The spiral movement of the spiral interpolation can be instructed to move two arc interpolation axes synchronously with other axes. The instruction method is simply to add a moving axis that is not an arc interpolation axis.
G03	Counterclockwise circular interpolation/spiral line interpolation
G01(G00) X I	Automatic reverse angle
G01(G00) YI	Automatic reverse angle
G01(G00) ZI	Automatic reverse angle

G01(G00)X R	自动倒圆角	
G01(G00)Y R	自动倒圆角	
G01(G00)Z R	自动倒圆角	
G04	延时	
G15	极坐标指令取消	
G16	极坐标指令: 极坐标（半径和角度），角度的正向是所选平面的第1轴正向的逆时针转向，而负向是顺时针转向。	
G17	选择 XpYp 平面	Xp: X 轴或其平行轴 Yp: Y 轴或其平行轴 Zp: Z 轴或其平行轴
G18	选择 ZpXp 平面	
G19	选择 YpZp 平面	
G20	英寸（英制）输入	
G21	毫米（公制）输入	
G28/G281-G288	返回机床零点	
M881	加工中心主轴准停，输出 M61，检测 M22	
M884-M887	X 轴、Y 轴自动分中指令	
M880	自动对刀指令	
G30/G301-G308	回坐标零点	
G26/G261-G268	返回程序起点	
G25	记忆当前点 XYZABCXsYs 坐标	
G61/G611-G618	返回 G25 记忆点的坐标	
G40	刀具半径补偿取消	
G41	刀具半径左补偿	
G42	刀具半径右补偿	
G43	正向刀具长度补偿	
G44	负向刀具长度补偿	
G49	刀具长度补偿取消	
G45	刀具位置偏置加	
G46	刀具位置偏置减	
G47	刀具位置偏置加 2 倍	
G48	刀具位置偏置减 2 倍	
G37	比例缩放取消	
G36	比例缩放有效	
G12	可编程镜像取消	

G01(G00)X R	Automatically round corners
G01(G00)Y R	Automatically round corners
G01(G00)Z R	Automatically round corners
G04	delayed
G15	The polar coordinate command is cancelled
G16	Polar coordinate instructions: Polar coordinates (radius and Angle), where the positive direction of the Angle is a counterclockwise turn from the positive direction of the first axis of the selected plane, and the negative direction is a clockwise turn.
G17	Select the XpYp plane
G18	Select the ZpXp plane
G19	Select the YpZp plane
G20	Inch (imperial) input
G21	Millimeters (metric) input
G28/G281-G288	Return to machine zero point
M881	The spindle of the machining center is stopped, output M61 and detect M22
M884-M887	X-axis and Y-axis automatic centering instructions
M880	Automatic alignment command
G30/G301 - G308	Return to the origin
G26/G261-G268	Return to the starting point of the procedure
G25	Remember the coordinates of the current point XYZABCXsYs
G61/G611-G618	Return the coordinates of the G25 memory point
G40	Tool radius compensation is cancelled
G41	Left tool radius compensation
G42	Right tool radius compensation
G43	Forward tool length compensation
G44	Negative tool length compensation
G49	Blade length compensation is cancelled
G45	Add tool position offset
G46	Tool position offset reduced
G47	The tool position offset is added by 2 times
G48	The tool position offset is reduced by 2 times
G37	Proportional scaling is cancelled
G36	Proportional scaling works
G12	Programmable mirroring is cancelled

G11	可编程镜像有效：实现坐标轴的对称加工。	
G50/G52	坐标系设定：在工件坐标系中编制程序时，为容易编程，可以设定工件坐标系的子坐标系	
G53	选择机床坐标系	
G54.1-G54.48	选择工件坐标系 1	说明：这六个工件坐标系是用直接存储在计算机中，需用时可选择其中一个
G55	选择工件坐标系 2	
G56	选择工件坐标系 3	
G57	选择工件坐标系 4	
G58	选择工件坐标系 5	
G59	选择工件坐标系 6	
G60	准确停止定位	
G64	连续路径加工	
G68	坐标旋转	
G69	坐标旋转取消	
G65	非模态宏指令	
G66	模态宏指令	
G67	取消模态宏指令	
G73	深孔钻循环：间歇进给，回退快速移动	
G80	固定循环取消/外部操作功能取消	
G81	钻孔循环：切削进给， 镗循环或外部操作功能，回退快速移动	
G82	排屑钻孔循环或反镗循环：切削进给，在孔底停刀，回退快速移动	
G83	小孔深孔钻循环：间歇进给，回退快速移动	
G85	镗孔循环：切削进给，回退切削进给	
G86	镗孔循环：切削进给，在孔底主轴停止，回退快速移动	
G89	镗孔循环：在孔底停刀，回退切削进给	
G74	左旋啄式深孔攻丝循环	
G84	右旋啄式深孔攻丝循环	
G76	精镗循环	
G90	绝对值编程	
G91	增量值编程	
G98	固定循环返回到初始点：用于最后钻孔	
G99	固定循环返回到 R 点： 用于第一次钻孔	
G22	程序循环指令	
G800	取消程序循环指令	

G11	Programmable mirror effect: achieve symmetrical machining of coordinate axes.	
G50/G52	Coordinate system setting: When programming in the workpiece coordinate system, you can set the sub-coordinate system of the workpiece coordinate system for easy programming	
G53	Select the machine tool coordinate system	
G54.1-G54.48	Select the workpiece coordinate system 1	Note: The coordinate systems for these six workpieces are stored directly in the computer and one of them can be selected when needed
G55	Select the workpiece coordinate system 2	
G56	Select the workpiece coordinate system 3	
G57	Select the workpiece coordinate system 4	
G58	Select the workpiece coordinate system 5	
G59	Select the workpiece coordinate system 6	
G60	Stop positioning accurately	
G64	Continuous path processing	
G68	Coordinate rotation	
G69	Coordinate rotation cancelled	
G65	Non-modal macro instructions	
G66	Modulator macro instructions	
G67	Remove the modal macro command	
G73	Deep hole drill cycle: intermittent feed, rapid retraction	
G80	Fixed cycle cancellation/external operation function cancellation	
G81	Drill cycle: Cutting feed, boring cycle or external operation function, quick retreat	
G82	Drill hole or reverse bore cycle: Cutting feed, stop cutting at the bottom of the hole, and move back quickly	
G83	Small hole deep hole drill cycle: intermittent feed, quick move back	
G85	Boring cycle: cutting feed, retraction cutting feed	
G86	Bore hole cycle: Cutting feed, spindle stops at the bottom of the hole, and rapid retraction	
G89	Bore hole cycle: stop the tool at the bottom of the hole and back off the cutting feed	
G74	Left rotary deep hole tapping cycle	
G84	Right rotation twist deep hole tapping cycle	
G76	Borehole circulation	
G90	Absolute value programming	
G91	Incremental value programming	
G98	Fixed cycle back to the starting point: for the final borehole	
G99	Fixed cycle back to point R: for the first borehole	
G22	Program loop instructions	
G800	Cancel the program loop instruction	

表 2 M 代码及功能

主 轴	M03 M04 M05 M203 M204 M205	主轴正转 主轴反转 主轴停止刹车 M7053/M7054 Pxxxx ; 主轴自 动正转/反转一段时间后自动 停止, 时间由参数 P 指定, 单 位为毫秒; 定时精度可达到 毫秒级。 例如: M7053 P2000; 表主轴正 转 2 秒钟后停止。	√	√ √ √	M203 L3/L4/L5/L6 M204 L3/L4/L5/L6 M205 L3/L4/L5/L6 第二到第六主轴正、 反转、停止
冷 却	M08 M09	冷却开 冷却关	√	√ √	
卡 盘	M10 M11	松紧刀夹紧 松紧刀松开	√	√ √	
自定 义	M79 M78	自定义输出 7 开, 输出 M79 自定义输出 7 关, 输出 M73	√	√ √	双信号输出
润 滑	M32 M33	机床润滑开 机床润滑关	√	√ √	控制 M32 信号
吹 气	M59 M58	吹气开 吹气关	√	√ √	控制 M59 信号
自定 义 输 出	M61 M60	自定义 1 开 自定义 1 关		√ √	控制 M61 信号
	M63 M62	自定义 2 开 自定义 2 关		√ √	控制 M63 信号
	M65 M64	自定义 3 开 自定义 3 关		√ √	控制 M65 信号
	M67 M66	自定义 4 开 自定义 4 关		√ √	控制 M67 信号
	M69 M68	自定义 5 开 自定义 5 关		√ √	控制 M69 信号
	M71 M70	自定义 6 开 自定义 6 关		√ √	控制 M71 信号
	M75 M74	自定义 8 开 自定义 8 关		√ √	控制 M75 信号
	M38xx M48xx	输出 xx 口有效 输出 xx 口无效			
自 定	M12 M13	检测 M12 有效 检测 M12 无效		√ √	条件成立转移, 如 M12 P120 即: 当

Table 2 M Code and Function

principal axis	M03 M04 M05 M203 M204 M205	Main shaft forward rotation main shaft reverse main shaft stop brake M7053/M7054 Pxxxx; The spindle automatically stops after a period of automatic forward/reverse rotation, and the time is specified by parameter P, with the unit of millisecond; The timing accuracy can reach millisecond level. For example: M7053 P2000; the table spindle stops after 2 seconds of forward rotation.	√	√ √ √	M203 L3/L4/L5/L6 M204 L3/L4/L5/L6 M205 L3/L4/L5/L6 The second to sixth main shafts are positive, reversed and stopped
cold step back	M08 M09	Cool down Cooling guard	√	√ √	
block tray	M10 M11	Set the jaws of the screwdriver Loosen the wrench	√	√ √	
user-defined	M79 M78	Open custom output 7 and output M79 Custom output 7, output M73	√	√ √	Dual signal output
lubricating	M32 M33	Open the machine tool lubrication Machine tool lubrication	√	√ √	Control the M32 signal
scavenging	M59 M58	Turn on the air blowers Blow off the lid	√	√ √	Control the M59 signal
user-defined output	M61 M60	Customize 1 open Customization 1 concerns		√ √	Control the M61 signal
	M63 M62	Customize 2 open Customization 2		√ √	Control the M63 signal
	M65 M64	Custom 3 open Customization 3		√ √	Control the M65 signal
	M67 M66	Customize 4 open Customization 4		√ √	Control the M67 signal
	M69 M68	Customize 5 open Customize 5		√ √	Control the M69 signal
	M71 M70	Customize 6 open Customization 6		√ √	Control the M71 signal
	M75 M74	Customize 8 open Customize 8		√ √	Control the M75 signal
	M38xx M48xx	Output xx is valid Output xx port is invalid			
customize	M12 M13	The M12 is valid for testing The M12 is invalid		√ √	Conditions for transfer, such as M12 P120, are: when

义 输入	M14	检测 M14 有效		√	M12 信号有效时, 程序跳转到 120 行执行; 如: M1810 检测 X10 口有效
	M15	检测 M14 无效		√	
	M16	检测 M16 有效		√	
	M17	检测 M16 无效		√	
	M18	检测 M18 有效		√	
	M19	检测 M18 无效		√	
	M28	检测 M28 有效		√	
	M29	检测 M28 无效		√	
主 轴 换 档	M22	检测 M22 有效		√	M41/M44 输出 S01/S02 同时调整 主轴模拟电压
	M23	检测 M22 无效		√	
	M18xx	检测 xx 口有效		√	
	M28xx	检测 xx 口无效		√	
	M41	第一档			
	M42	第二档			
	M43	第三档			
	M44	第四档			
子 程 序	M97	程序跳转			L=1-99 P 为转移程序行号
	M98	子程序调用			
	M99	子程序调用返回			
程 序 控 制	M87	工件数加 1			杂类参数 P10=0 设 为工件数不自动加 1 时, 指令 M87 工件 数加 1
	M188	M188 进给保持手动干预 坐标后不返回起点			
	M189	M189 进给保持回起点			
	M00	程序暂停			
	M01	程序准停, 输入 X46 有效暂 停			
	M02	程序结束			
	M30	M05、M09 程序结束			
	M20	程序结束, 根据参数设置运 行次数自动重复运行程序, 用于考机			
	S	设定第一主轴速度		√	
	SS	设定第二主轴转速		√	
主 轴 速 度	SC/SD	第三/四主轴转速		√	S=0-99999 SS=0-99999
	SG/SH	第五/六四主轴转速		√	

Right to entry	M14 M15	The M14 is valid for testing The M14 is invalid		√ √	When the M12 signal is valid, the program jumps to 120 and line is executed; For example, M1810 detects X10, and the port is valid
	M16 M17	The M16 was tested and found to be effective The M16 is invalid		√ √	
	M18 M19	The M18 is valid for testing The M18 is invalid		√ √	
	M28 M29	The M28 was tested and found to be effective The M28 is invalid		√ √	
	M22 M23	The M22 was tested and found to be effective The M22 is invalid		√ √	
	M18xx M28xx	Test xx is valid Test xx port is invalid		√ √	
Change the main axis	M41 M42 M43 M44	first gear second gear third gear Episode IV			M41/M44 outputs S01/S02 and adjusts the spindle simulation voltage at the same time
subprogram	M97 M98 M99	Program jump subprogram call Subroutine call returned			L=1-99 P is the transfer program line number
programmed control	M87 M188 M189 M00 M01 M02 M30 M200	Add 1 to the number of jobs M188 feed hold handwheel intervention coordinates without returning to the starting point M189 feed is maintained at the starting point program halt The program is ready to stop, and the input X46 is valid for stopping end of program M05, M09 program end At the end of the program, the program will be automatically repeated according to the number of times set by parameters for computer testing			When the miscellaneous parameter P10=0 is set to not automatically add 1 to the number of workpieces, the instruction M87 adds 1 to the number of workpieces
Axis speed	S S SC/SD SG/SH	Set the first spindle speed Set the second spindle speed and third/fourth spindle speed Fifth/sixth main shaft speed		√ √	S=0-99999 SS=0-99999

读绝对式电机位置	M499 M500 M501 M502 M503 M504 M505 M506 M507 M508	M499: 绝对伺服旋转轴归算的数据溢出进行处理, 用在 M97/M02/M30/M20/GOTO 之前; M500: 读取全部进给轴绝对电机位置并重置当前机床坐标。 M501-M508: 读取 XYZABCXsYs 轴绝对电机位置并重置当前机床坐标。			
清工件坐标	M312 M313 M314 M315 M316 M317 M318 M319 M320	M312: 清零 C 轴工件坐标 M313: 清零 Xs 轴工件坐标 M314: 清零 Ys 轴工件坐标 M315: 清零 A 轴工件坐标 M316: 清零 B 轴工件坐标 M317: 清零 X 轴工件坐标 M318: 清零 Y 轴工件坐标 M319: 清零 Z 轴工件坐标 M320: 清零所有轴工件坐标			
清机床坐标	M412 M413 M414 M415 M416 M417 M418 M419 M420	M412: 清零 C 轴机床坐标 M413: 清零 Xs 轴机床坐标 M414: 清零 Ys 轴机床坐标 M415: 清零 A 轴机床坐标 M416: 清零 B 轴机床坐标 M417: 清零 X 轴机床坐标 M418: 清零 Y 轴机床坐标 M419: 清零 Z 轴机床坐标 M420: 清零所有轴机床坐标			
总线驱动指令	M133 M135	M133: 驱动器以指定速度转动 (必须是 30 的倍数); 驱动参数 P30=0 恒转速, P30=1 往复摆动, 见轴类参数。 M135: 使能指定驱动器			例子: M133 X300; M135 Z0; 关闭 Z 轴驱动 M135 X1; 使能 X 轴驱动
倍率指令	M100 M101	倍率无效, 固定为 100%; 倍率有效;			

M1xxx: 等待辅助继电器有效, 例如: M1076 表示等待 M76 有效;

M2xxx: 等待辅助继电器无效, 例如: M1078 表示等待 M78 无效;

M3xxx: 置辅助继电器有效, 例如: M3330 表示置 M330 有效;

Read the absolute motor position	M499 M500 M501 M502 M503 M504 M505 M506 M507 M508	M499: Process the data overflow of absolute servo rotation axis, which is used before M97/M02/M30/M20/GOTO; M500: Read all feed axis absolute motor positions and reset the current machine coordinate. M501-M508: Read the absolute motor position of the XYZABCXsYs axis and reset the current machine tool coordinates.			
Set the work coordinates	M312 M313 M314 M315 M316 M317 M318 M319 M320	M312: Zero the C axis workpiece coordinates M313: Zero the Xs axis workpiece coordinates M314: Zero the Ys axis workpiece coordinates M315: Zero the A axis workpiece coordinates M316: Zero the B axis workpiece coordinates M317: Zero the X axis workpiece coordinates M318: Zero the Y axis workpiece coordinates M319: Zero the Z axis workpiece coordinates M320: Zero the coordinates of all shaft workpieces			
Clear the machine seat coordinates	M412 M413 M414 M415 M416 M417 M418 M419 M420	M412: Zero the C axis machine tool coordinates M413: Zero the Xs axis machine tool coordinates M414: Zero the Ys axis machine tool coordinates M415: Zero the A axis machine tool coordinates M416: Zero the B axis machine tool coordinates M417: Zero the X axis machine tool coordinates M418: Zero the Y axis machine tool coordinates M419: Zero the Z axis machine tool coordinates M420: Zero all axis machine tool coordinates			
Bus drive instruct	M133 M135	M133: The driver rotates at a specified speed (must be a multiple of 30); drive parameter P30=0 for constant speed, P30=1 for reciprocating swing, see shaft parameters. M135: Enable the specified driver			example : M133 X300; M135 Z0; Turn off Z-axis drive M135 X1; Enable X-axis drive
multiplying power instruct	M100 M101	The multiplier is invalid and fixed at 100%; The multiplier is effective;			

M1xxx: Wait for the auxiliary relay to be valid, for example: M1076 indicates that M76 is valid;
M2xxx: The waiting auxiliary relay is invalid, for example: M1078 indicates that the waiting M78 is invalid;
M3xxx: The auxiliary relay is set to be effective. For example, M3330 indicates that M330 is set to be effective;

M4xxx: 置辅助继电器无效, 例如: M4331 表示置 M331 无效;

M38xx Pxxxx: Yxx 输出点置为有效, 例如: M3809 表示将输出点 Y9 置为有效;

P 为正值时表示输出信号延时再向后执行;

P 为负值时表示短信号, 如: P3821 P-2000, 输出 Y21, 2 秒后关闭;

注意:使用此指令时 PLC 对应的输出点程序行要删除;

M48xx: Yxx 输出点置为无效, 例如: M4807 表示将输出点 Y7 置为无效;

M18xx: 直接等待输入点 Xxx 有效后再向后执行, 例如: M1809 表示 X09 有效后再向后执行;

M28xx: 直接等待输入点 Xxx 无效后再向后执行, 例如: M2807 表示 X07 无效后再向后执行;

M18xx Pxx: 直接根据 Xxx 判断: 无效时顺序执行; 有效时跳转到 Pxx;

例如: M1809 P234 表示输入点 X09 无效时顺序执行; 有效时跳转到 P234;

M28xx Pxx: 直接根据 Xxx 判断: 有效时顺序执行; 无效时跳转到 Pxx;

例如: M2807 P456 表示输入点 X07 有效时顺序执行; 无效时跳转到 P456;

2.4 编程指令及使用方法

2.4.1 编程约定

NC/NCe 数控
鑫科瑞数控

(1) 多指令共段: 在同一个程序行内可以有多个指令同时存在, 以减少编程行数, 但同组指令不能共段。

(2) 在同一条程序段内指令与参数项可以任意摆放。

例如: M03 G01 X20 Y-30 可以写成 G01 Y-30 X20 M03

(3) 在同一条程序段内指令不允许重复出现。

(4) 在同一条程序段内不能有与指令无关的参数和操作数存在。

(5) 指令代码的前“0”可以不写, 例如: G01 M03 可写成 G1 M3。

(6) 任意点起动、任意行起动、换刀指令后的命令行必须用绝对编程。

(7) 非模态指令只在被指定的程序段中有效, 例如: G04。

(8) 模态指令在同组指令出现前一直有效。

例如: N0000 G01 X300 F100 ; G01 指令

N0001 X260 ; G01 指令

N0002 G00 Y200 ; G00 指令, G01 无效

2.4.2 G 功能指令

(1) 选择坐标系指令 (G53/G54/G55/G56/G57/G58/G59)

M4xxx: The auxiliary relay is invalid, for example: M4331 indicates that M331 is invalid;

M38xx Pxxxx: Yxx output point is set to valid, for example: M3809 indicates that the output point Y9 is set to valid;

When P is positive, the output signal is delayed and executed again;

If P is negative, it indicates a short signal, such as P3821 P-2000, and the output Y21 is closed after 2 seconds;

Note: When using this instruction, the corresponding output point program line of the PLC should be deleted; M48xx: Yxx output point is invalid, for example: M4807 indicates that the output point Y7 is invalid;

M18xx: Directly wait for the input point Xxx to be valid before executing backward, for example: M1809 means that X09 is valid before executing backward;

M28xx: Directly wait for the invalid input point Xxx and then execute backward, for example: M2807 means X07 is invalid and then execute backward;

M18xx Pxx: Directly judged according to Xxx: execute in sequence when invalid; jump to Pxx when valid;

For example, M1809 P234 indicates that the input point X09 is invalid and is executed in sequence; when valid, it jumps to P234;

M28xx Pxx: directly determines according to Xxx: when valid, it is executed in sequence; when invalid, it jumps to Pxx;

For example, M2807 P456 means that the input point X07 is valid and executed in sequence; if invalid, it jumps to P456;

2.4 Programming Instructions and Usage

2.4.1 Programming Agreement

(1) Multiple instructions in a common section: Multiple instructions can exist in the same program line at the same time to reduce the number of programming lines, but instructions in the same group cannot be shared.

(2) Instructions and parameters can be placed arbitrarily within the same program segment.

For example, M03 G01 X20 Y-30 can be written as G01 Y-30 X20 M03

(3) Instructions are not allowed to repeat within the same program segment.

(4) There can be no instruction-independent parameters and operands in the same program segment.

(5) The "0" in the instruction code can be omitted. For example, G01 M03 can be written as G1 M3.

(6) Absolute programming must be used in the command line for any point start, any line start, and tool change instructions.

(7) Non-modal instructions are only valid in the specified program segment, such as G04.

(8) Modulator instructions remain valid until the same group of instructions appears.

For example: N0000 G01 X300 F100; G01 instruction

N0001 X260 ; G01 instruction

N0002 G00 Y200 ; G00 instruction, G01 invalid

2.4.2 G Function Command

(1) Select the coordinate system command (G53/G54/G55/G56/G57/G58/G59)

用来选择工件坐标系或机床坐标系。

指令格式: G53 (G54/G55/G56/G57/G58/G59) (设定后为模态)

G53 机床坐标系

G54 工件坐标系 1

G54.1/G54.48 工件坐标系 54.1/54.48

G55 工件坐标系 2

G56 工件坐标系 3

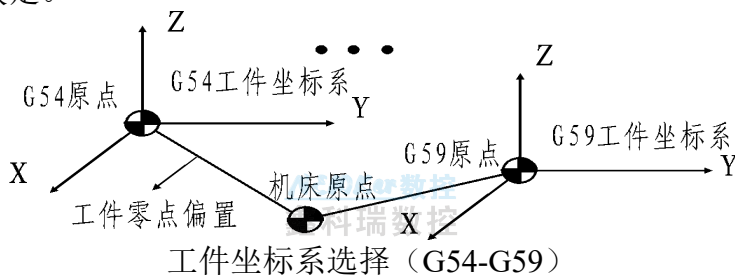
G57 工件坐标系 4

G58 工件坐标系 5

G59 工件坐标系 6

G53 机床坐标系是指由机床第一参考点（即机床零点）确定的坐标系。若修改机床坐标系中坐标值后全部的工件坐标将随之偏置，通常情况建议不要修改 G53 坐标值，G53 为默认坐标系。

G54/G55/G56/G57/G58/G59 工件坐标系在机床坐标系中的偏移量由坐标系参数中的设置决定。



例如: G00 G54 X50 Y60 Z70

以 G00 快速移动到工件坐标系 G54 下的 X50、Y60、Z70 坐标点。

(2) 坐标系指令 (G50/G52)

当在工件坐标系中编制程序时，为容易编程可以设定工件坐标系的子坐标系称为局部坐标系。G50是工件坐标系绝对坐标设定，G52是局部坐标系相对坐标设置。

指令格式: G50/G52 X- Y- Z- ; 设定坐标系。(设定后为模态)
G52 X0 Y0 Z0; 取消局部坐标系。

用指令 G50/G52 可以在机床坐标系 (G53) 或工件坐标系 (G54~G59) 中设定局部坐标系。局部坐标的原点设定在机床坐标系或工件坐标系中以 X- Y- Z- 指定的位置。

当局部坐标系设定时，后面的以绝对值方式 G90 指令的移动是在局部坐标系中的坐标值。用 G50/G52 指定新的零点，可以改变局部坐标系的位置。为了取消局部坐标系并在工件坐标系中指定坐标值，应使局部坐标系零点与工件坐标系零点一致。

Used to select the workpiece coordinate system or machine tool coordinate system.

Instruction format: G53 (G54/G55/G56/G57/G58/G59) (set as mode)

G53 machine tool coordinate system

G54 Workpiece coordinate system 1

G54.1/G54.48 Workpiece coordinate system 54.1/54.48

G55 Workpiece coordinate system 2

G56 Workpiece coordinate system 3

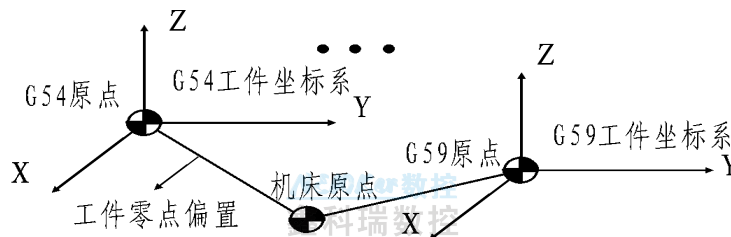
G57 Workpiece coordinate system 4

G58 Workpiece coordinate system 5

G59 Workpiece coordinate system 6

The G53 machine tool coordinate system refers to the coordinate system determined by the machine tool's first reference point (i.e., the machine tool's zero point). If the coordinate values in the machine tool coordinate system are modified, all workpiece coordinates will be offset accordingly. In general, it is recommended not to modify the G53 coordinate values, as G53 is the default coordinate system.

The offset of the workpiece coordinate system in the machine tool coordinate system for G54/G55/G56/G57/G58/G59 is determined by the setting of the coordinate system parameters.



Workpiece coordinate system selection (G54-G59)

For example: G00 G54 X50 Y60 Z70

Move quickly to the G00 coordinate system of the workpiece at X50, Y60, and Z70 coordinates under G54.

(2) Coordinate Instructions (G50/G52)

When programming in the workpiece coordinate system, a sub-coordinate system of the workpiece coordinate system can be set up to facilitate programming. G50 is the absolute coordinate setting of the workpiece coordinate system, and G52 is the relative coordinate setting of the local coordinate system.

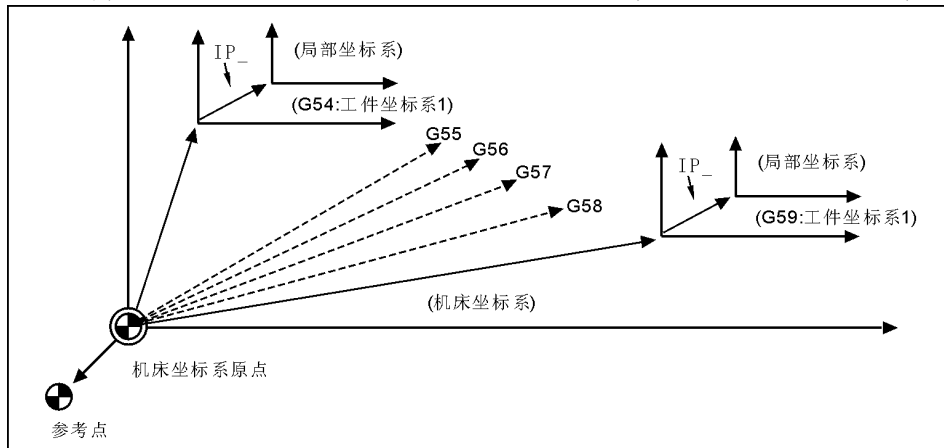
Instruction format: G50/G52 X-Y-Z; Set the coordinate system. (After setting, it becomes a mode)

G52 X0 Y0 Z0; Cancel the local coordinate system.

The G50/G52 command can be used to set the local coordinate system in either the machine tool coordinate system (G53) or the workpiece coordinate system (G54~G59). The origin of the local coordinate system is set at the position specified by X-Y-Z in either the machine tool coordinate system or the workpiece coordinate system.

When a local coordinate system is set, subsequent G90 absolute-value commands will operate using the coordinates within this system. By specifying new zero points with G50/G52 commands, the position of the local coordinate system can be adjusted. To cancel the local coordinate system and assign coordinates in the workpiece coordinate system, it is necessary to align the zero points of both systems.

局部坐标系设定不改变工件坐标系和机床坐标系。当用G50/G52 指令设定工件坐标系时，如果不是指令所有轴的坐标值的话，未指定坐标值的轴的局部坐标系不取消而保持不变。在G50/G52 程序段以后以绝对值方式指定运动指令。



(3) 编程方式指令 (G90/G91)

有两种方法指令刀具的移动：绝对值指令和增量值指令。在绝对值指令中，编程终点的坐标值；而在增量值指令中，编程移动距离。G90和G91 分别用来指明坐标字中用的是绝对编程或增量编程。

指令格式：G90	(模态，初态)	;绝对坐标编程方式。
G91	(模态)	;增量坐标编程方式。



注意：旋转轴绝对坐标编程时是就近计算执行，相对坐标编程是按编程计算执行；

(4) 平面选择指令 (G17/G18/G19)

指令格式：G17	(模态，初态)	;指定 XY 平面
G18	(模态)	;指定 ZX 平面
G19	(模态)	;指定 YZ 平面

用来指定圆弧插补的平面，不产生任何动作。

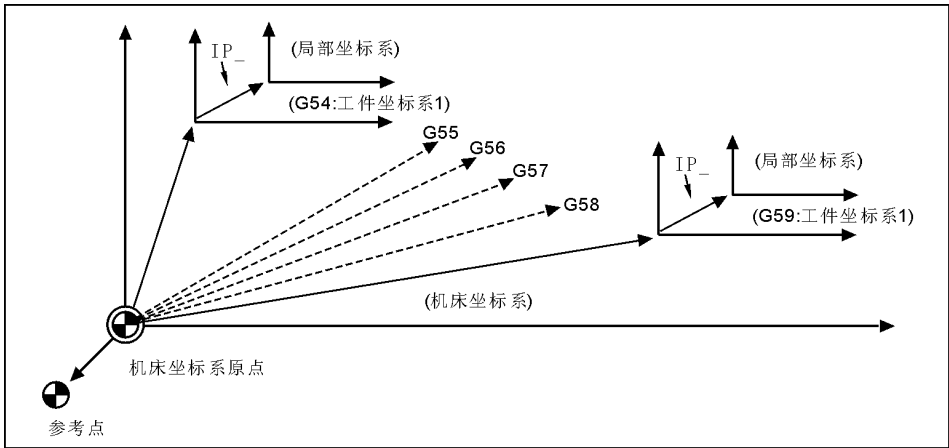
(5) 快速定位指令 (G00)

刀具以快速移动速度移动到用绝对值指令或增量值指令指定的工件坐标系中的位置。

以绝对值指令编程时编制终点的坐标值；

以增量值指令编程时编制刀具移动的距离。

The local coordinate system setting does not alter the workpiece coordinate system or the machine tool coordinate system. When using G50/G52 commands to set the workpiece coordinate system, if the command specifies coordinates for all axes, the local coordinate systems of unselected axes remain unchanged. Subsequent motion commands in the G50/G52 program segment must be specified using absolute values.



(3) Programming Instructions (G90/G91)

There are two types of commands to control tool movement: absolute and incremental. In absolute commands, the programming endpoint coordinates are specified; in incremental commands, the programmed travel distance is defined. The G90 and G91 commands respectively indicate whether absolute or incremental programming is used for coordinate values.

Instruction (Mode, initial format: G90 state)	; Absolute coordinate programming mode.
G91 (modality)	; Incremental coordinate programming mode.

□

Note: When the absolute coordinate programming of the rotation axis is calculated, the nearest calculation and execution are performed; when the relative coordinate programming is calculated, the programming is executed;

(4) Planar selection command (G17/G18/G19)

Instruction format: G17 (mode, initial state)	; Specify the XY plane
G18 (modality)	; Specify the ZX plane
G19 (modality)	; Specify the YZ plane

The plane used to specify the arc interpolation does not produce any action.

(5) Quick Positioning Instruction (G00)

The tool moves quickly to the position in the workpiece coordinate system specified by the absolute value instruction or the incremental value instruction.

Compile the coordinate value of the endpoint when programming with absolute value instructions;

The distance of tool movement is programmed when using incremental value instructions.

指令格式: G00 X-Y-Z-A-B-C-Xs-Ys- (模态, 初态)

说明: X, Y, Z, A, B, C, Xs, Ys 坐标字表示运行轴。其数据指明移动量和方向, 可以用增量编程或绝对编程。

G00 时各轴单独以各自设定的速度快速移动到终点, 互不影响。任何一轴到位自动停止运行, 另一轴继续移动直到指令位置。也可以联动方式: 通过杂类参数P9号D6=0设置成联动运行方式。

运行速度调用“速度参数”中所设定的 G00 速度值为依据。

(6) 直线插补指令 (G01)

用来单轴或二、三、四轴联动的插补加工切削。

指令格式: G01 X-Y-Z-A-B-C-Xs-Ys- F- (模态)

说明: X, Y, Z, A, B, C, Xs, Ys 坐标字表示运行轴, 其数据指明进给量和方向, 可以用增量编程或绝对编程。插补速度由程序中 F 字段指定。

G01 指令的F 进给速度可以通过面板上进给倍率上下调整(调整范围是0%~150%)。

G01 指令也可直接写成G1。

(7) 圆弧插补指令 (G02/G03)

用来在指定平面上, 进行 G02 顺圆(顺时针方向)或 G03 逆圆(逆时针方向)的圆弧插补切削。

指令格式: $G17 \begin{cases} G02 \\ G03 \end{cases} X_Y_ \begin{cases} I_J_ \\ R_ \end{cases} F_ ; XY \text{ 平面 (模态)}$

$G18 \begin{cases} G02 \\ G03 \end{cases} X_Z_ \begin{cases} I_K_ \\ R_ \end{cases} F_ ; ZX \text{ 平面 (模态)}$

$G19 \begin{cases} G02 \\ G03 \end{cases} Y_Z_ \begin{cases} J_K_ \\ R_ \end{cases} F_ ; YZ \text{ 平面 (模态)}$

说明: 圆弧插补必须明确插补平面, 坐标字中 X、Y、Z 是圆弧的终点坐标(增量或绝对编程), I、J、K 分别是 X、Y、Z 轴上起点(速度参数 P53 加 4 时表示从圆弧终点)到圆心的增量坐标值。也就是说以起点为零, 圆心在起点的正方向为正, 在起点的负方向为负。I J K 是描述圆心位置的坐标。也可不用 I、J、K 而直接用 R 编程, 当圆弧大于 180 度时 R 用负值, 整圆不能用 R 编程。

圆弧的走向以右手直角坐标系中从垂直于圆弧插补平面的正方向往负方向而言设定的。

在不同圆弧插补平面顺圆或逆圆走向如下图

Instruction format: G00 X-Y-Z-A-B-C-Xs-Ys- (mode, initial state)

Note: X, Y, Z, A, B, C, Xs, Ys coordinates indicate the running axis. The data indicates the amount and direction of movement, which can be programmed incrementally or absolutely.

At G00, each axis moves quickly to the end point at its own preset speed without affecting each other. When any axis reaches the position, it automatically stops running while the other axis continues moving until reaching the specified position. Alternatively, a linked operation mode can be set by configuring parameter P9 D6=0 to enable coordinated operation.

The operating speed is based on the G00 speed value set in the "Speed parameter".

(6) Linear Interpolation Command (G01)

Used for single axis or two, three, four axis linkage interpolation machining cutting.

Instruction format: G01 X-Y-Z-A-B-C-Xs-Ys-F- (mode)

Note: X, Y, Z, A, B, C, Xs, Ys coordinates indicate the running axis. Their data indicates the feed rate and direction. Incremental programming or absolute programming can be used. The interpolation speed is specified by the F field in the program.

The feed speed of G01 command F can be adjusted up and down by the feed multiplier on the panel, and the adjustment range is (0% ~ 150%).

The G01 instruction can also be written directly as G1.

(7) Circular Arc Interpolation Instruction (G02/G03)

Used to perform G02 clockwise (clockwise) or G03 counterclockwise (counterclockwise) arc interpolation and cutting on a specified plane.

指令格式: $G17 \begin{Bmatrix} G02 \\ G03 \end{Bmatrix} X_Y_ \begin{Bmatrix} I_J_ \\ R_ \end{Bmatrix} F_;$ XY plane (mode)

$G18 (G02) ^I-^K-);$ ZX plane (mode)

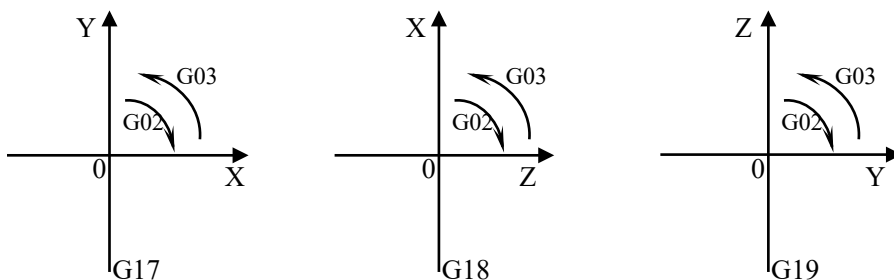
$\begin{Bmatrix} G03 \end{Bmatrix} X_Z_ \begin{Bmatrix} R_ \end{Bmatrix} F_$

$G19 \begin{Bmatrix} G02 \\ G03 \end{Bmatrix} Y_Z_ \begin{Bmatrix} J_K_ \\ R_ \end{Bmatrix} F_;$ YZ plane (mode)

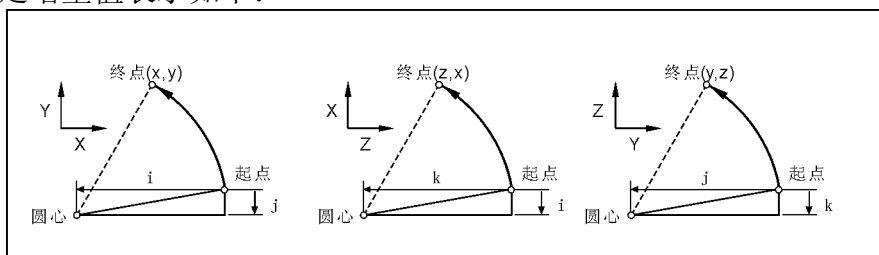
Note: Arc interpolation must specify the interpolation plane. In coordinate parameters, X, Y, and Z represent the endpoint coordinates of the arc (incremental or absolute programming). I, J, and K denote the incremental coordinates from the starting point (when P53 is set to 4, indicating the arc endpoint) to the center along the X, Y, and Z axes respectively. Specifically, with the starting point as zero, the positive direction of the center corresponds to the starting point's position, while the negative direction indicates the opposite position. I, J, and K describe the center's coordinates. Alternatively, R programming can be used directly instead of I, J, and K. When the arc exceeds 180 degrees, R should be set to a negative value. Full circles cannot be programmed using R programming.

The direction of the arc is defined in the right-handed Cartesian coordinate system from the positive direction to the negative direction perpendicular to the arc interpolation plane.

The different arc interpolation planes follow the clockwise or counterclockwise direction as shown in the figure below



用地址 I J K 指令的圆弧中心位置, I J K 后的数值是从起点向圆弧中心看的矢量分量, I J K 必须根据方向指定其符号正或负. I J K 可以省略, 当省略时终点与起点相同, 且中心用 I J K 指定时是 360 的圆弧整圆. 并且, 不管是 G90 还是 G91 总是增量值表示如下:



增强型圆弧功能(A/B 轴与 XYZ 插补圆弧):

格式: G177(G17.1)/G17.2 G02/G03 X-A/B-I-J-R- ;XA 平面

G188(G18.1)/G18.2 G02/G03 Z-A/B-K-I-R- ;ZA 平面

G199(G19.1)/G19.2 G02/G03 Y-A/B-J-K-R- ;YA 平面

G177(G17.1) 选 XA 平面; G188(G18.1) 选 ZA 平面; G199(G19.1) 选 YA 平面。G17.2 选 XB 平面; G18.2 选 ZB 平面; G19.2 选 YB 平面。

在 XA/XB 平面中用 IJ 指定圆心 (I 对应 X 轴; J 对应 A/B 轴);

在 ZA/ZB 平面中用 KI 指定圆心 (K 对应 Z 轴; I 对应 A/B 轴);

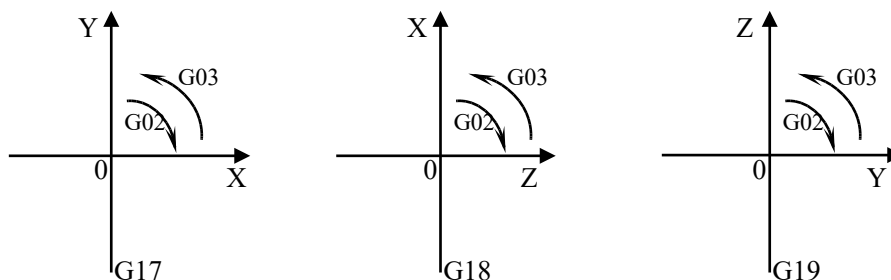
在 YA/YB 平面中用 JK 指定圆心 (J 对应 Y 轴; K 对应 A/B 轴)。

注意:

1, 当不使用 A/B 轴圆弧插补时, 必须回到 G17/G18/G19 平面;

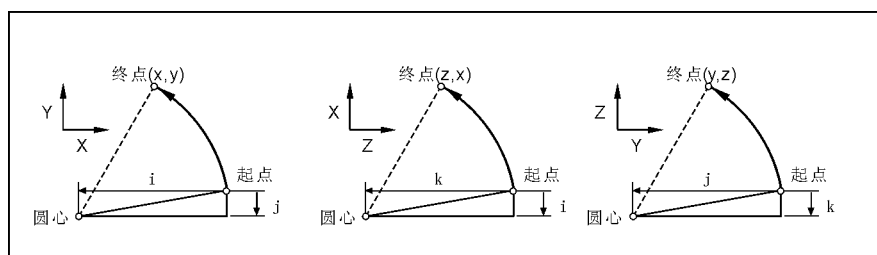
2, 当使用 A/B 轴圆弧插补指令时, A/B 轴必须设置为直线轴;

3, 用 R 指定半径, 当圆弧大于 180 度时 R 用负值表示, 整圆不能用 R 编程。



Use the address I J K to indicate the center position of the arc. The value after I J K is from the starting point to the center of the arc

The vector components I J K must be assigned positive or negative signs according to their direction. The components I J K may be omitted, in which case the endpoint coincides with the starting point, and when specified by I J K, it represents a full 360-degree arc. Additionally, whether using G90 or G91 commands, the incremental values are always expressed as follows:



Enhanced arc function (A/B axis and XYZ interpolated arc):

Format: G177 (G17.1)/G17.2 G02/G03 X-A/B-I-J-R-; XA
plane G188 (G18.1)/G18.2 G02/G03 Z-A/B-K-I-
R-; ZA plane G199 (G19.1)/G19.2 G02/G03 Y-A/
B-J-K-R-; YA plane

G177 (G17.1) Select the XA plane; G188 (G18.1) Select the ZA plane; G199 (G19.1) Select the YA plane. G17.2 Select the XB plane; G18.2 Select the ZB plane; G19.2 Select the YB plane.

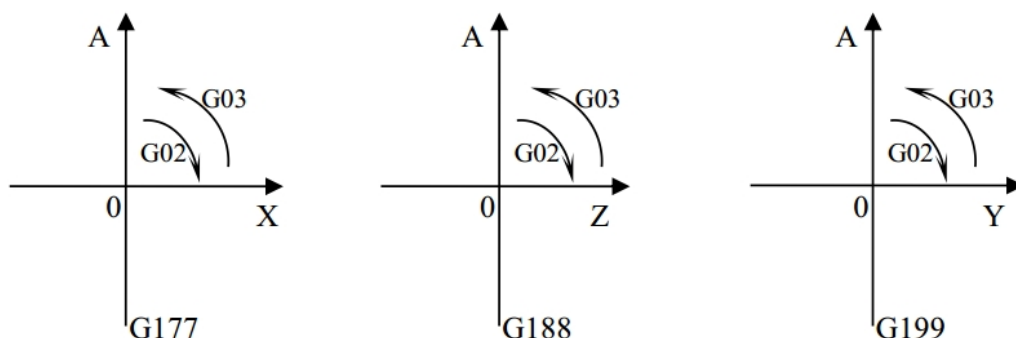
In the XA/XB plane, the center is specified by IJ (I corresponds to the X-axis; J corresponds to the A/B-axis);

In the ZA/ZB plane, the center is specified by KI (K corresponds to the Z-axis; I corresponds to the A/B axis);

In the YA/YB plane, the center of circle is specified by JK (J corresponds to the Y-axis; K corresponds to the A/B axis).

pay attention to :

1. When the A/B axis arc interpolation is not used, you must return to the G17/G18/G19 plane;
2. When using the A/B axis arc interpolation instruction, the A/B axis must be set as a straight axis;
3. Use R to specify the radius. When the arc is greater than 180 degrees, R is expressed with a negative value. The whole circle cannot be programmed with R.



举例：

1) 绝对编程

N0000 G00 X200 Y40 Z0 设起刀点绝对坐标位置
N0010 G90 G03 X160 Y40 I-20 J0 逆圆弧

F300

N0020 G02 X120 Y40 R20 顺圆弧
N0030 G02 X120 Y40 I20 J0 整圆
N0040 G26 M02 回起点结束

2) 相对编程

N0000 G91 G17 G03 X-40 Y0 R20 逆圆弧

F300

N0010 G02 X-40 Y0 R20 顺圆弧
N0020 G02 X0 Y0 I20 J0 整圆
N0030 G26 M02 回起点结束

(8) 螺旋线插补指令 (G02/G03)

螺旋移动的螺旋线插补可指定一个与圆弧插补轴同步移动的有关轴。指令方法只是简单地加上一个不是圆弧插补轴的移动轴，F 指令指定沿圆弧的进给速度。因此直线轴的进给速度为：

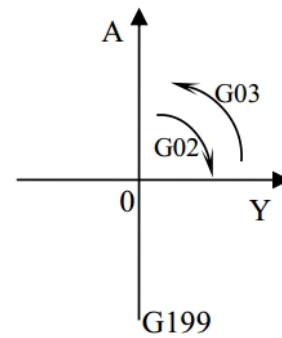
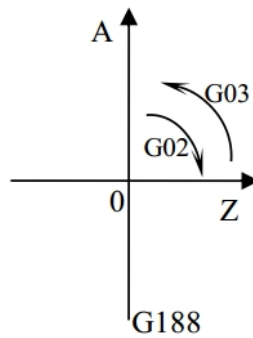
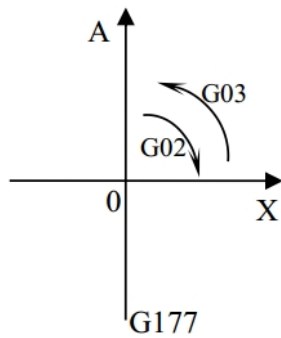
$$F \times \frac{\text{直线轴的长度}}{\text{圆弧的长度}}$$

指令格式：	$G17 \begin{Bmatrix} G02 \\ G03 \end{Bmatrix} \begin{Bmatrix} X \\ Y \end{Bmatrix} \begin{Bmatrix} I \\ J \end{Bmatrix} \begin{Bmatrix} R \\ R \end{Bmatrix} Z_{F_}$ ；XY 平面（模态）
	$G18 \begin{Bmatrix} G02 \\ G03 \end{Bmatrix} \begin{Bmatrix} X \\ Z \end{Bmatrix} \begin{Bmatrix} I \\ K \end{Bmatrix} \begin{Bmatrix} R \\ R \end{Bmatrix} Y_{F_}$ ；ZX 平面（模态）
	$G19 \begin{Bmatrix} G02 \\ G03 \end{Bmatrix} \begin{Bmatrix} Y \\ Z \end{Bmatrix} \begin{Bmatrix} J \\ K \end{Bmatrix} \begin{Bmatrix} R \\ R \end{Bmatrix} X_{F_}$ ；YZ 平面（模态）

螺旋线增强功能，可以编入螺距K(或J或I)用于多圈螺旋线加工：

G02(G03) X__ Y__ I__ J__ F__ Z__ K__ (G17, K螺距，单位毫米)；

G02(G03) X__ Z__ I__ K__ F__ Y__ J__ (G18, J螺距，单位毫米)；



give an example :

1) Absolute programming

N0000 G00 X200 Y40 Z0

N0010 G90 G03 X160 Y40 I-20 J0 F300

N0020 G02 X120 Y40 R20

N0030 G02 X120 Y40 I20 J0

N0040 G26 M02

Set the absolute coordinate position of the cutting point

The reverse arc
Smooth the arc
Round the whole thing
Return to the starting point

2) Relative programming

N0000 G91 G17 G03 X-40 Y0 R20 F300

N0010 G02 X-40 Y0 R20

N0020 G02 X0 Y0 I20 J0

N0030 G26 M02

The reverse arc

Smooth the arc
Round the whole thing
Return to the starting point

(8) Spiral Lead-in Instruction (G02/G03)

The helical interpolation of a spiral path allows specifying an additional axis that moves synchronously with the arc interpolation axis. The command method simply adds a movement axis other than the arc interpolation axis, while the F command specifies the feed rate along the arc. Consequently, the feed rate for the linear axis is:

$$F \times \frac{\text{直线轴的长度}}{\text{圆弧的长度}}$$

指令格式:	$G17 \begin{cases} G02 \\ G03 \end{cases} \begin{cases} X_Y_ \\ Z_F_ \end{cases} \begin{cases} I_J_ \\ R_ \end{cases}; \text{XY plane (mode)}$
	$G18 (G02) <\sup>I</sup>-<\sup>K</sup>-; \text{ZX plane (mode)}$
	$\begin{cases} \{ \\ \{G03\} \end{cases} \begin{cases} X_Z_ \\ \{R_ \} \end{cases} \begin{cases} Y_F_ \\ \{ \} \end{cases}$
	$G19 \begin{cases} G02 \\ G03 \end{cases} \begin{cases} Y_Z_ \\ X_F_ \end{cases} \begin{cases} J_K_ \\ R_ \end{cases}; \text{YZ plane (mode)}$

Spiral line enhancement function, which can be programmed with pitch K (or J or I) for multi-turn spiral processing:

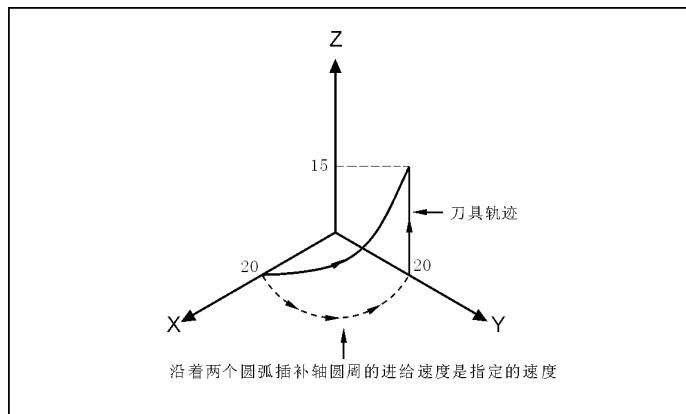
G02 (G03) X_Y_I_J_F_Z_K_ (G17, K pitch, unit mm);

G02 (G03) X_Z_I_K_F_Y_J_ (G18, J pitch, unit mm);

G02(G03) Y__ Z__ J__ K__ F__ X__ I__ (G19, I螺距, 单位毫米);
 G02(G03) X__ A__ I__ J__ F__ Z__ K__ (G177/XA平面, K螺距, 单位毫米);
 G02(G03) Z__ A__ K__ I__ F__ Y__ J__ (G188/ZA平面, J螺距, 单位毫米);
 G02(G03) Y__ A__ J__ K__ F__ X__ I__ (G199/YA平面, I螺距, 单位毫米);

刀具半径补偿只对圆弧进行, 在指令螺旋线插补的程序段中不能指令刀具偏置和刀具长度补偿。

举例:



```
N0001 G90 G17 G54
N0002 G01 X20 Y0 Z0 F600
N0003 G03 X0 Y20 R20 Z15 F180
.....
```

(9) 三维空间圆弧指令 G06

指令格式: G06 X__ Y__ Z__ I__ J__ K__ F__

指令功能: 如果不知道三维空间圆弧的圆心, 半径。但已知圆弧轮廓上的三个点的坐标, 则可使用三维空间圆弧指令 G06 功能, 通过始点和终点之间的中间点位置确定圆弧方向。

指令说明: G06 为模态 G 代码;

I: 圆弧所经过的中间点相对于起点的相对坐标值(X 向)(半径值表示, 带方向);

J: 圆弧所经过的中间点相对于起点相对坐标值(Y 向)(带方向);

K: 圆弧所经过的中间点相对于起点相对坐标值(Z 向)(带方向)。

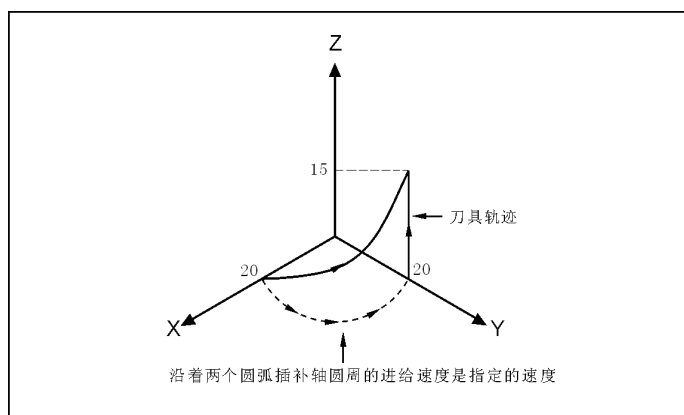
注意事项:

- 1), 中间点: 是指圆弧上除起点和终点之外的任意一点;
- 2), 当给出的三点共线时, 系统产生报警;
- 3), 当省略 I 时即认为 I=0, 当省略 K 时即认为 K=0, 当省略 J 时即认为 J=0; 当同时省略 I、J、K 时, 系统产生报警;
- 4), I、J、K 的意义类似于 G02/G03 指令中圆心坐标相对于起点坐标的位移

G02 (G03) Y_Z_J_K_F_X_I_ (G19, I pitch, unit mm);
 G02 (G03) X_A_I_J_F_Z_K_ (G177/XA plane, K pitch, unit mm);
 G02 (G03) Z_A_K_I_F_Y_J_ (G188/ZA plane, J pitch, unit mm);
 G02 (G03) Y_A_J_K_F_X_I_ (G199/YA plane, I pitch, unit mm);

Tool radius compensation is only for arcs, and tool offset and tool length compensation cannot be specified in the program segment of spiral interpolation.

give an example :



```
N0001 G90 G17 G54
N0002 G01 X20 Y0 Z0 F600
N0003 G03 X0 Y20 R20 Z15 F180
.....
```

(9) 3D Spatial Arc Command G06

Instruction format: G06 X_Y_Z_I_J_K_F_

Instruction function: If you do not know the center and radius of a 3d space arc, but know the coordinates of three points on the arc contour, you can use the 3d space arc instruction G06 function to determine the direction of the arc by the position of the midpoint between the starting point and the end point.

Instruction description: G06, is the mode G code;

I: The relative coordinate value (X direction) of the midpoint of the arc relative to the starting point (the radius value indicates the direction);

J: The relative coordinate value (Y direction) of the midpoint of the arc relative to the starting point;

K: Relative z-direction coordinate value (with direction) of the midpoint of the arc relative to the starting point.

Matters Need Attention :

1) The midpoint: refers to any point on the arc other than the starting point and the end point;

2) When the three points are given to be collinear, the system generates an alarm;

3) When I is omitted, I=0; when K is omitted, K=0; when J is omitted, J=0; when I, J and K are omitted at the same time, the system generates an alarm;

4) The meaning of I, J and K is similar to the displacement of the center coordinate relative to the starting point coordinate in the G02/G03 instruction

值 I、J、K;

5), G06 不能加工整圆。

6), 该指令数学计算量较大, 在总线系统上才能有效的运行, 在其它系统上可能会运行不流畅;

举例:

G0 X10 Y28 Z10

G06 X30 Y98 Z10 I5 J-6 K-5 F100

X130 Y198 Z120 I55 J-86 K-65

G0X0Z0

M02

(10) 延时指令 (G04)

用来在加工过程中因加工工艺需要, 延时一定时间再执行其它动作。

指令格式: G04 P_; 或
G04 X_; 或
G04 U_;

令功能: 执行该指令时, 各轴运动停止, 模态值保持, 延时指定的时间后, 再执行下一个程序段。

NC/OKer 数控
鑫科瑞数控

指令说明:

1. P_是以毫秒为单位指令暂停时间。单位: 毫秒。
2. X_、U_均是以秒为单位指令暂停时间。单位: 秒。
3. 实例:

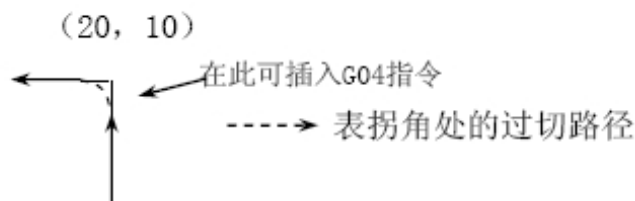
G04 X1; 表示程序暂停1秒。

G04 P1000; 表示程序暂停1秒。

G04 U1; 表示程序暂停1秒。

4. 特殊应用: 指令G04可看成为准确停指令, 如加工拐角类零件时, 在拐角处有时会出现过切现象, 如在拐角处加G04指令, 即可消除过切现象。

如下图所示:



例:

N150 G01 X20 Z10 F100;

N160 G04 P150; (可消除拐角处过切现象)

N170 G01 Y-10;

Values I, J, K;

5) , G06 cannot be processed into a full circle.

6) , The instruction has a large amount of mathematical calculation, and can only run effectively on the bus system, but may not run smoothly on other systems;

give an example :

```
G0 X10 Y28 Z10
G06 X30 Y98 Z10 I5 J-6 K-5 F100
X130 Y198 Z120 I55 J-86 K-65
G0X0Z0
M02
```

(10) Delay instruction (G04)

Used to delay other actions for a certain period of time during the processing process due to the processing requirements.

Instruction format: G04
P-; or G04 X_
_; or
G04 U ;

Function: When the instruction is executed, the movement of each axis stops, the mode value remains, and the next program segment is executed after a specified delay time.

Instruction description:

1.P__ is the instruction pause time in milliseconds. Unit: milliseconds.

2.X__ and U__ are the instruction pause time in seconds. Unit: seconds.

3. living example :

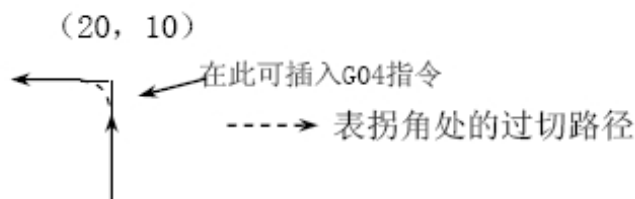
G04 X1; indicates that the program is suspended for 1 second.

G04 P1000; indicates that the program is suspended for 1 second.

G04 U1; indicates that the program is suspended for 1 second.

4. Special application: The instruction G04 can be regarded as an accurate stop instruction. For example, when machining corner parts, overcutting phenomenon may occur at the corner. If the G04 instruction is added at the corner, the overcutting phenomenon can be eliminated.

As shown in the figure below:



example :

```
N150 G01 X20 Z10 F100;
N160 G04 P150; (to eliminate overcut at corners)
N170 G01 Y-10;
```

.....

注：也可以设置加工参数P21号参数，消除过切现象。

(11) 镜像加工指令 (G11/G12)

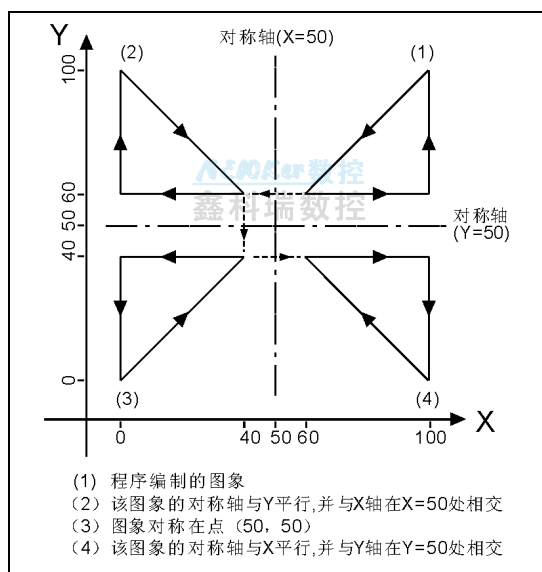
用于对称工件的加工，减少编程量。

指令格式：G11 X_ Y_ (Z_ X_) (Y_ Z_) (模态)
以 XYZ 指定直线的对称方向进给
G12 (模态，初态) ；取消镜像加工。

说明：镜像加工用改变轴进给方向实现在该轴上的对称点加工进给。要改变方向的轴由当前插补平面 G17 (X_ Y_)；G18 (Z_ X_)；G19 (Y_ Z_) 地址符指定。X_ Y_ Z_ 后的数值为当前工件坐标系下的坐标。

在可编程镜像方式中，返回参考点的指令G26、G61、G28、G30 等和改变坐标系的指令G52、G53/G59等不允许出现. 如果需要这些G 代码的任意一个必须在取消可编程镜像方式之后再指定。

举例：



子程序

KG11

N10 G00 G90 X60.0 Y60.0;

N20 G01 X100.0 F100;

N30 G01 Y100.0;

N40 G01 X60.0 Y60.0;

N50 M99;

主程序

N10 G00 G90;

.....

Note: The machining parameter P21 can also be set to eliminate the overcut phenomenon.

(11) Mirror Processing Command (G11/G12)

Used for processing symmetrical workpieces to reduce programming.

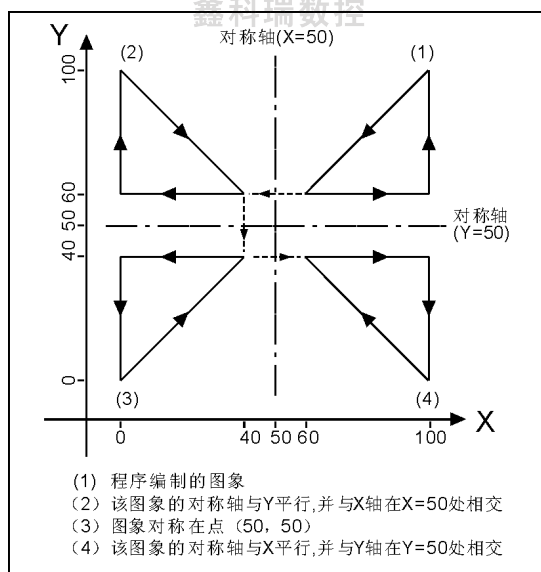
Instruction format: G11 X_Y_ (Z_X_) (Y_Z_) (mode)

Feed G12 (mode, initial state) in the direction of symmetry of the specified XYZ line; cancel mirror processing.

Note: Mirror machining achieves symmetrical point processing feed on the axis by altering the feed direction. The axis to be redirected is specified by the current interpolation planes: G17 (X_Y_), G18 (Z_X_), and G19 (Y_Z_). The values following X_Y_Z_ represent coordinates in the workpiece's current coordinate system.

In the programmable mirror mode, instructions such as G26, G61, G28, and G30 that return reference points, as well as commands like G52, G53/G59 that change the coordinate system, are not allowed. If any of these G codes is required, they must be specified after canceling the programmable mirror mode.

give an example :



subprogram

```
KG11
N10 G00 G90 X60.0 Y60.0;
N20 G01 X100.0 F100;
N30 G01 Y100.0;
N40 G01 X60.0 Y60.0;
N50 M99;
main program N10
G00 G90;
```

```

N20 M98 PKG11;
N30 G11 X50.0;
N40 M98 PKG11;
N50 G11 X50.0 Y50.0;
N60 M98 PKG11;
N70 G11 Y50.0;
N80 M98 PKG11;
N90 G12;
N100 M02

```

(12) 比例缩放指令 (G36/G37)

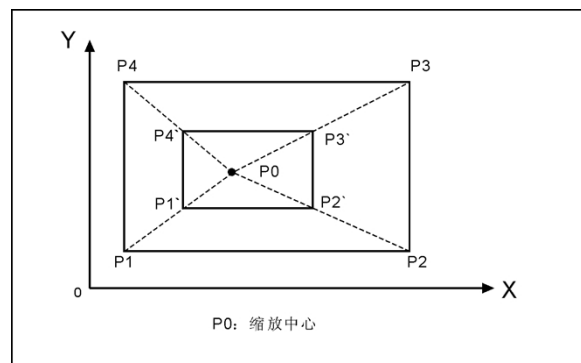
用于不改变加工程序的情况下，按比例缩小或放大加工出的工件。编程的形状被放大和缩小（比例缩放），用 X_、Y_ 和 Z_ 指定比例缩放中心。若某一轴未指定，则该轴不执行比例缩放。X_ Y_ Z_ 后的数值为当前工件坐标系下的坐标。

指令格式：G36 X_Y_Z_R_ （模态） ； 执行比例缩放
 G37 （模态，初态） ； 比例缩放取消

说明：地址符 R 后的数据为比例系数。

下图中当 P1' P2' P3' P4' 要放大到 P1P2P3P4 的图形时， $R=P0P4' / P0P4$ 。当 P1P2P3P4 要缩小到 P1' P2' P3' P4' 的图形时， $R=P0P4 / P0P4'$ 。即：图形放大 R 小于 1，图形缩小 R 大于 1。缺省值为 1。

图示：



比例缩放对刀具半径补偿值、刀具长度补偿值和刀具偏置值无效：

```
N20 M98 PKG11;  
N30 G11 X50.0;  
N40 M98 PKG11;  
N50 G11 X50.0 Y50.0;  
N60 M98 PKG11;  
N70 G11 Y50.0;  
N80 M98 PKG11;  
N90 G12;  
N100 M02
```

(12) Proportional Scaling Instructions (G36/G37)

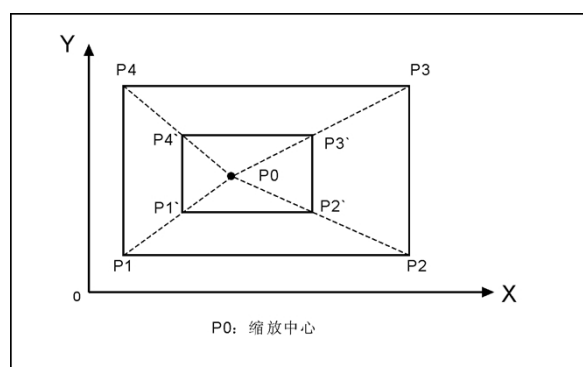
This process involves proportionally scaling (either enlarging or reducing) workpieces without altering the original machining program. The programmed shape undergoes proportional scaling, with X_, Y_, and Z_ coordinates specifying the scaling center. If no axis is specified, that axis remains unaffected by scaling. The values after X_Y_Z_ represent coordinates in the current workpiece's coordinate system.

Instruction format: G36 X_Y_Z_R_ ; Scale the exe-	(mode)
	cution ratio
G37 (modes, initial state)	; Zoom out

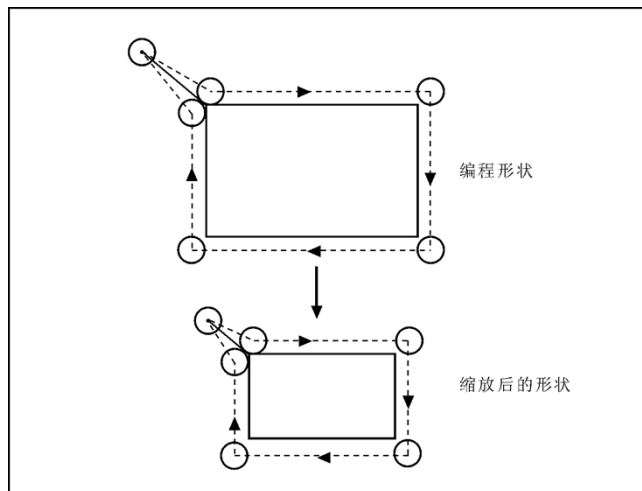
Note: The data after the address symbol R is the proportional coefficient.

In the diagram below, when scaling from P1 to P2 to P3 to P4, the ratio $R = P0P4 / P0P4'$. When scaling back from P1 to P2 to P3 to P4, the ratio $R = P0P4 / P0P4'$. This means: the scaling ratio is less than 1 for enlargement and greater than 1 for reduction. The default value is 1.

graphic :



Proportional scaling is invalid for tool radius compensation value, tool length compensation value and tool offset value:



(13) 坐标系旋转指令 (G68/G69)

编程形状能够旋转，用该旋转指令可将工件旋转某一指定的角度。另外，如果工件的形状由许多相同的图形组成，则可将图形单元编成子程序，然后用主程序的旋转指令调用。这样可简化编程，节省存储空间。

指令格式：G68 X- Y- R- （模态）；执行坐标系旋转
 G68 Z- X- R- （模态）；执行坐标系旋转
 G68 Y- Z- R- （模态）；执行坐标系旋转
 G69 （模态，初态）；坐标系旋转取消

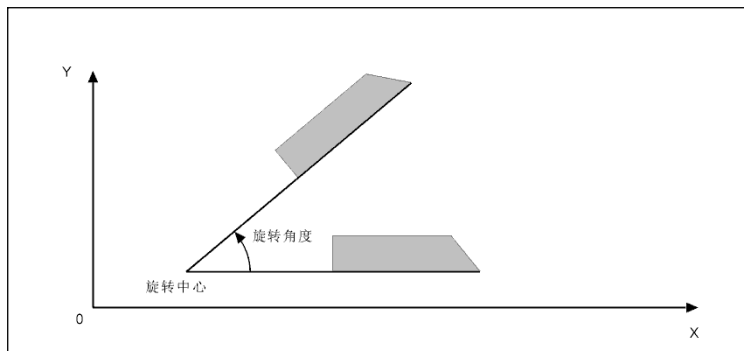
说明：

G68 后的 (G17) X-Y-或 (G18) Z-X-或 (G19) Y-Z-指定旋转中心

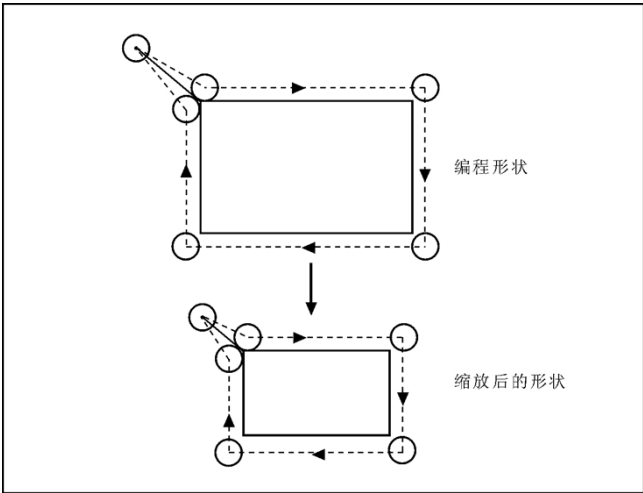
地址符 R 后的数据为旋转角度。X_ Y_ Z_后的数值为当前工件坐标系下的坐标。

R 为旋转角度，单位是 “°”；R 为正值时，表示逆时针旋转，R 为负值时，表示顺时针旋转。

图示：



在坐标系旋转代码G68 的程序段之前指定平面选择代码 (G17G18 或G19)，



(13) Coordinate System Rotation Command (G68/G69)

The program can rotate shapes by using rotation commands to turn the workpiece at a specified angle. Additionally, if a workpiece's shape consists of multiple identical graphic elements, these elements can be compiled into subroutines and then called using the main program's rotation command. This approach simplifies programming and saves storage space.

Instruction format: G68 X-Y-R- (mode); execute coordinate system rotation

G68 Z-X-R (mode); performs coordinate system rotation

G68 Y-Z-R (mode); performs coordinate system rotation

G69 (Mode, Initial State); Coordinate system rotation cancelled

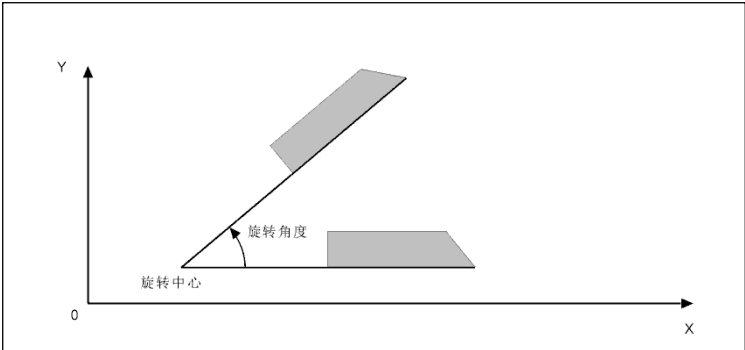
explain :

The X-Y or G18 Z-X or G19 Y-Z designation of the rotation center is after G68

The data after the address符 R is the rotation Angle. The values after X_Y_Z_ are the coordinates in the current workpiece coordinate system.

R is the rotation Angle, unit of "°"; R is positive for counterclockwise rotation and negative for clockwise rotation.

graphic :

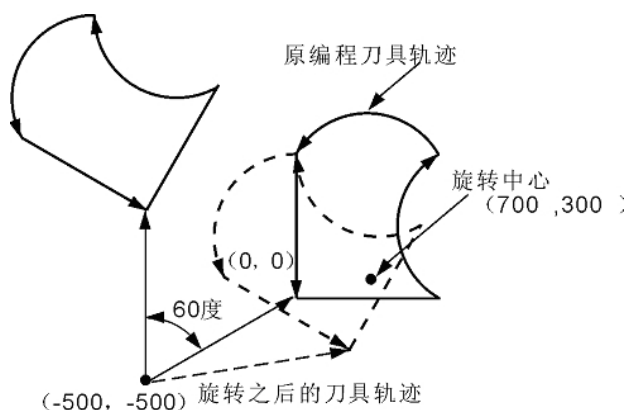


Before the program segment of coordinate system rotation code G68, specify the plane selection code (G17G18 or G19),

平面选择代码不能在坐标系旋转方式中指定。取消坐标系旋转方式的代码G69 必须单独占一程序段。在坐标系旋转之后，执行刀具半径补偿，刀具长度补偿，刀具偏置和其它补偿操作。

在坐标系旋转方式中，与返回参考点有关G 代码（G28、G26、G61、G30 等）和那些与坐标系有关的G 代码（G52、G54/G59）不能指定，如果需要这些G 代码，必须在取消坐标系旋转方式以后才能指令。坐标系旋转取消指令G69 以后的第一个移动指令必须用绝对值指定，如果用增量值指令，将不执行正确的移动。

举例：



```

N10 G01 X-500Y-500F2000 G17;
N20 G68 X700Y300R60;
N30 G90 G01 X0 Y0 F2000;
N40 G91 X1000
N50 G02 Y1000 R1000
N60 G03X-1000I-500J-500;
N70 G01Y-1000
N80 G69
N90 G90 X-500Y-500
N100 M02;

```

（14）回零点指令（G28-G288/G301-G308）

返回参考点指刀具沿着指定轴自动地移动到参考点。

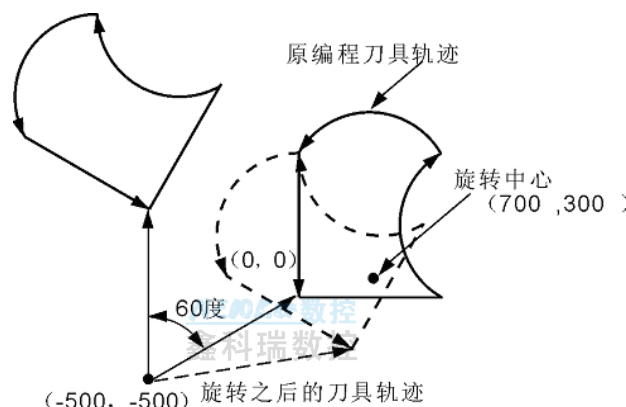
指令格式：G28 X(U)___ Z(W)___ Y(V)___ A___ B___； XZYAB通过中间点后再回参考点

G281	;X 轴回机床零点
G282	;Y 轴回机床零点
G283	;Z 轴回机床零点
G284	;A 轴回机床零点
G285	;B 轴回机床零点

The plane selection code cannot be specified in the coordinate system rotation mode. The code G69 for canceling the coordinate system rotation must occupy a separate program segment. After the coordinate system rotation, execute tool radius compensation, tool length compensation, tool offset, and other compensation operations.

In coordinate system rotation mode, G codes related to returning to reference points (e.g., G28, G26, G61, G30) and those associated with coordinate systems (G52, G54/G59) cannot be specified. If these G codes are required, they must be programmed after canceling the coordinate system rotation mode. The first movement command after canceling the coordinate system rotation mode with G69 must be specified using absolute values. If incremental value commands are used, the correct movement will not be executed.

give an example :



```
N10 G01 X-500Y-500F2000 G17;
N20 G68 X700Y300R60;
N30 G90 G01 X0 Y0 F2000;
N40 G91 X1000
N50 G02 Y1000 R1000
N60 G03X-1000I-500J-500;
N70 G01Y-1000
N80 G69
N90 G90 X-500Y-500
N100 M02;
```

(14) Return to zero instruction (G28-G288/G301-G308)

Return to reference point means that the tool automatically moves along the specified axis to the reference point.

Instruction format: G28 X (U) _ Z (W) _ Y (V) _ A _ B _ ; XZYAB returns to the reference point after passing through the midpoint

G281	; Return the X-axis to the machine zero point
G282	; Y-axis back to machine zero point
G283	; Z axis back to machine zero point
G284	; A axis back to machine zero point
G285	; B axis back to machine zero point

G286	;C 轴回机床零点
G287	;Xs 轴回机床零点
G288	;Ys 轴回机床零点
G301	;X 轴坐标回零
G302	;Y 轴坐标回零
G303	;Z 轴坐标回零
G304	;A 轴坐标回零
G305	;B 轴坐标回零
G306	;C 轴坐标回零
G307	;Xs 轴坐标回零
G308	;Ys 轴坐标回零

注意：执行该指令之前应该清除刀具半径补偿和刀具长度补偿。

(15) 刀具长度补偿指令 (G43/G44/G49)

设定编程时的刀具长度和实际使用的刀具长度之差值，用该功能补偿这个差值而不用修改程序。用G43 或G44 指定偏置方向，由输入的相应地址号H 代码从偏置表中选择刀具长度偏置值。

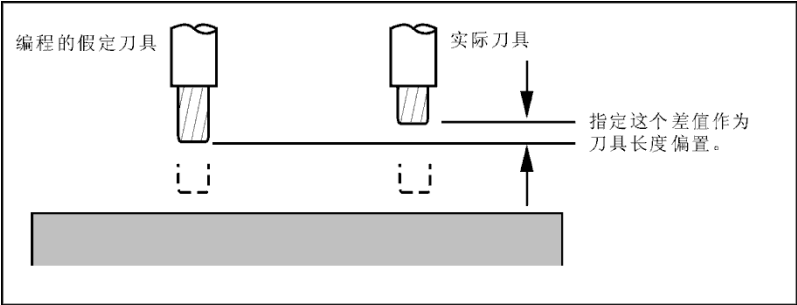
处于G17(XY)平面时，长度补偿H加在了Z轴上，所以刀具最好应装在Z轴上。
处于G18(ZX)平面时，长度补偿H加在了Y轴上，所以刀具最好应装在Y轴上。
处于G19(YZ)平面时，长度补偿H加在了X轴上，所以刀具最好应装在X轴上。

指令格式：

G43 H— ； 加一个刀具长度补偿参数值。
G44 H— ； 减一个刀具长度补偿参数值。
G49 或 H0 ； 取消刀具长度补偿。

说明：刀具长度补偿就是在 Z 轴或其他轴上加或减用 H 功能调用的刀具长度补偿参数值。调用号为 H1—H99（刀具最大数）。参数中的补偿数据都是以未进行补偿时的刀具位置作为起点。

图示：



例：N0000 G43 H2 X10 （H2 值为 5）
N0010 G44 H3 X20 （H3 值为 10）

G286	; C axis back to machine zero point
G287	; Xs axis back to machine zero point
G288	; Ys axis back machine zero point
G301	; The X-axis coordinate is reset to zero
G302	; Y-axis coordinates back to zero
G303	; Z-axis coordinates back to zero
G304	; The A-axis coordinate returns to zero
G305	; B-axis coordinates back to zero
G306	; C-axis coordinates back to zero
G307	; The Xs axis coordinate is reset to zero
G308	; Ys axis coordinates back to zero

Note: Tool radius compensation and tool length compensation should be cleared before executing this instruction.

(15) Tool length compensation command (G43/G44/G49)

The difference between the tool length specified during programming and the actual tool length used is compensated by this function without modifying the program. Use G43 or G44 to specify the offset direction, and select the tool length offset value from the offset table using the corresponding address code H input.

When in the G17 (XY) plane, the length compensation H is applied to the Z axis, so it is best to mount the tool on the Z axis.

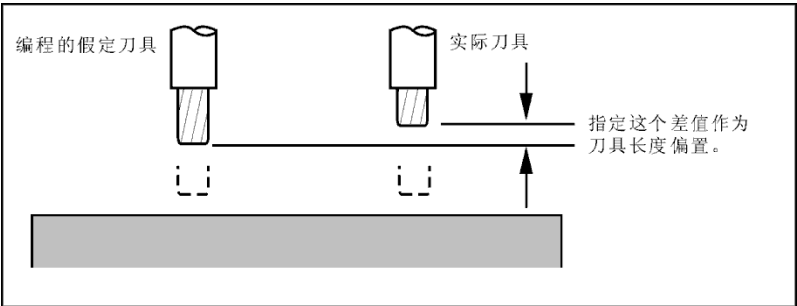
When in the G18 (ZX) plane, the length compensation H is applied to the Y axis, so it is best to mount the tool on the Y axis.

When in the G19 (YZ) plane, the length compensation H is applied to the X axis, so it is best to mount the tool on the X axis.

instruction format :
G43 H-; Add a tool length compensation parameter value.
G44 H-; subtract one tool length compensation parameter value.
G49 or H0; disable tool length compensation.

Note: Tool length compensation is the tool length compensation parameter value added or subtracted on the Z-axis or other axes using the H function. The call number is H1-H99 (maximum number of tools). The compensation data in the parameter is based on the tool position at the beginning without compensation.

graphic :



Example: N0000 G43 H2 X10 (H2
value is 5)
N0010 G44 H3 X20 (H3 value is
10)
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执行第一行程序时刀具伸长 5，执行第二个 G44 时，缩短 10。G49 不用参数，执行时回到补偿前的位置。

(16) 增减刀具半径指令 (G45/G46/G47/G48)

用于沟槽等工件的加工时，编程按工件图纸尺寸编制，指令行中加上这组指令之一。就可不同刀具半径时加工出符合图纸尺寸的产品。

指令格式	G45 T- D-	；增加一个刀偏值。
	G46 T- D-	；减少一个刀偏值。
	G47 T- D-	；增加两个刀偏值。
	G48 T- D-	；减少两个刀偏值。

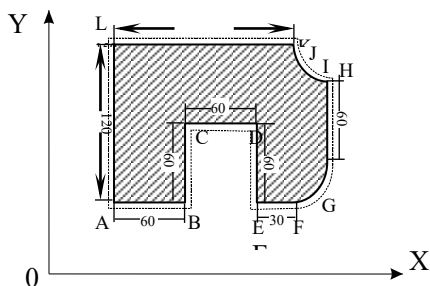
说明：增减刀具半径指令，是在执行中增加或减少一个或两个用 T 编号调用的刀具半径参数值。还要和 XY (G17) 平面上的 G00、G01、G02、G03 指令连用。不能和刀具半径补偿指令 (G41, G42) 混用。

对于 G00 和 G01 而言，是在给出轴的方向，分别增加或减少一或二个刀具半径。

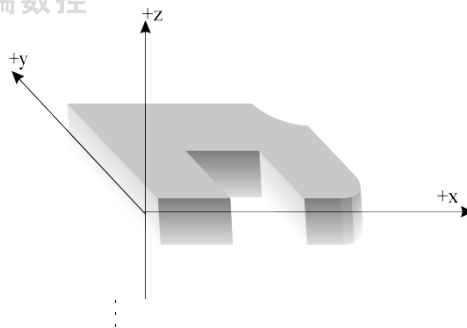
对于 G02 和 G03 则是在圆弧半径方向增加或减少一至二个刀具半径。

G45/G46/G47/G48 为非模态指令。

如下图：



鑫科瑞数控



编程序如下：

N0000 G01 Z-20 F400 G91	； Z 下刀
N0010 G46 T01 X55 Y55 D01 (T01)	； 点定位到 A 点，X Y 轴都减少一个刀具半径
N0020 G47 G01 X60 F200	； A-B X 轴增加两个刀具半径
N0030 Y60	； B-C 长度不变
N0040 G48 X60	； C-D X 轴减少两个刀具半径
N0050 Y-60	； D-E 长度不变
N0060 G45 X30	； E-F X 轴增加一个刀具半径
N0070 G45 G03 X30 Y30 R30	； F-G 逆圆半径增加一个刀具半径
N0080 G45 G01 Y60	； G-H Y 轴增加一个刀具半径
N0090 G46 X0	； H-I X 轴负方向走一个刀具半径

When the first line of program is executed, the tool lengthens by 5, and when the second G44 is executed, it shortens by 10. G49 has no parameters and returns to the position before compensation when executed.

(16) Add/subtract tool radius commands (G45/G46/G47/G48)

When used for the machining of workpieces such as grooves, the program is prepared according to the size of the workpiece drawing. One of these instructions is added to the instruction line, so that the product can be processed with different tool radii to meet the drawing size.

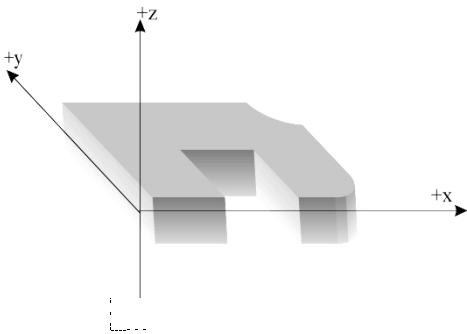
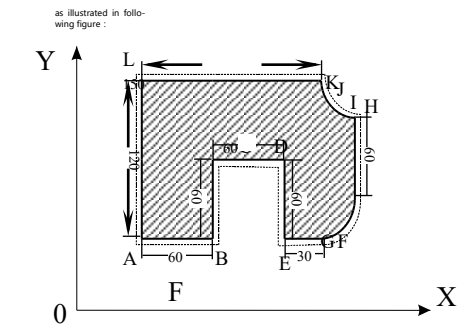
Instruction format: G45 T-D			; Add a tool offset.;
G46	T-	D-	Reduce a tool offset.
G47 T- D-			; Add two blade offsets.
G48 T- D-			; Reduce the two knife offsets.

Note: The tool radius add/subtract command is used to increase or decrease one or two tool radius parameters called by the T code during execution. It should be combined with G00, G01, G02, and G03 commands on the XY (G17) plane. It cannot be mixed with tool radius compensation commands (G41, G42).

For G00 and G01, the direction of the axis is given and one or two tool radii are added or subtracted respectively.

For G02 and G03, one or two tool radii are added or reduced in the direction of the arc radius.

G45/G46/G47/G48 are non-mode instructions.



The program is as follows:

N0000 G01 Z-20 F400 G91	; Z cuts the blade
	; The point is positioned at point A, and
N0010 G46 T01 X55 Y55 D01 (T01)	the X and Y axes are reduced by one
N0020 G47 G01 X60 F200	tool radius each; A-B X axis is increased
	by two tool radii
N0030 Y60	; B-C, the length remains unchanged
N0040 G48 X60	; C-D X axis reduces two tool radii
N0050 Y-60	; D-E, length unchanged
N0060 G45 X30	; Add a tool radius to the E-F X axis
N0070 G45 G03 X30 Y30 R30	; F-G The inverse circle radius is increased by a tool radius
N0080 G45 G01 Y60	; Add a tool radius to the G-Y axis
N0090 G46 X0	; H-I moves one tool radius in the negative direction of the X-axis

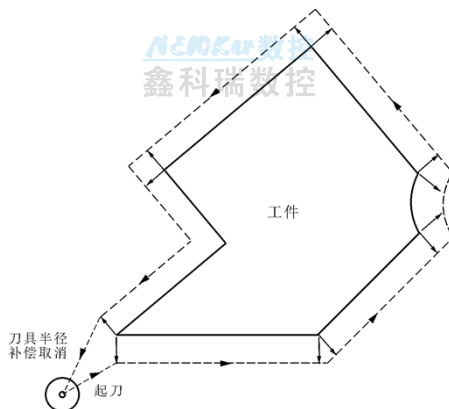
N0100 G46 G02 X-30 Y30 R30	; I-J 顺圆半径减少一个刀具半径
N0110 G45 G01 Y0	; J-K Y 轴正方向走一个刀具半径
N0120 G47 X-150	; K-L X 轴增加两个刀具半径
N0130 G47 Y-120	; L-M Y 轴增加两个刀具半径
N0140 G46 X-55 Y-55	; M-N X Y 轴都减少一个刀具半径
N0150 G26	; 程序回起点结束
N0151 M02	

上例指令的加(+)减(-)和刀具半径值的符号是同号相加,异号相减的关系,坐标轴数值可以是零。这时它的意义是用来指定坐标轴,运行的长度为一个或两个半径。方向由指令的加减和半径值的符号共同决定。

(17) 刀具半径补偿指令 (G40/G41/G42)

当刀具移动时,刀具轨迹可以偏移一个刀具半径。为了偏移一个刀具半径,CNC 首先建立长度等于刀具半径的偏置矢量,即起刀点。偏置矢量垂直于刀具轨迹。矢量的尾部在工件上而头部指向刀具中心。如果在起刀之后指定直线插补或圆弧插补,在加工期间刀具轨迹可以用偏置矢量的长度偏移。在加工结束时,为使刀具返回到开始位置,须取消刀具半径补偿方式。

图示:



指令格式: G40	(模态, 初态)	; 刀具半径补偿取消。
G41 T- D-	(模态)	; 刀具在工件左侧偏置的刀具半径补偿。
G42 T- D-	(模态)	; 刀具在工件右侧偏置的刀具半径补偿。

说明: G41/G42 中的左侧或右侧是指沿切削方向看,刀具在工件的左侧或右侧。刀具半径值用 T1-T99 调用。执行偏置是在 G41/G42 所在程序行开始。尖角处由系统自动产生过渡直线,使上一程序段终点的刀具半径偏移矢量过渡到下一

The + and - signs in the preceding instructions follow the rule of adding when same-sign values and subtracting when opposite-sign values, with coordinate axis values being zero. This mechanism specifies coordinate axes that operate over lengths equal to one or two radii. The direction is determined jointly by the instruction's sign and the radius value.

When a tool moves, its path may deviate by one tool radius. To achieve this, the CNC system first creates a bias vector equal in length to the tool radius, known as the tool start point. This bias vector is perpendicular to the tool's trajectory, with its tail resting on the workpiece and head pointing toward the tool center. If linear or arc interpolation is specified after tool start, the tool path will be adjusted by the vector's length during machining. To return the tool to its starting position at the end of processing, the tool radius compensation method must be disabled.

cutter

radius compensat-

ion .

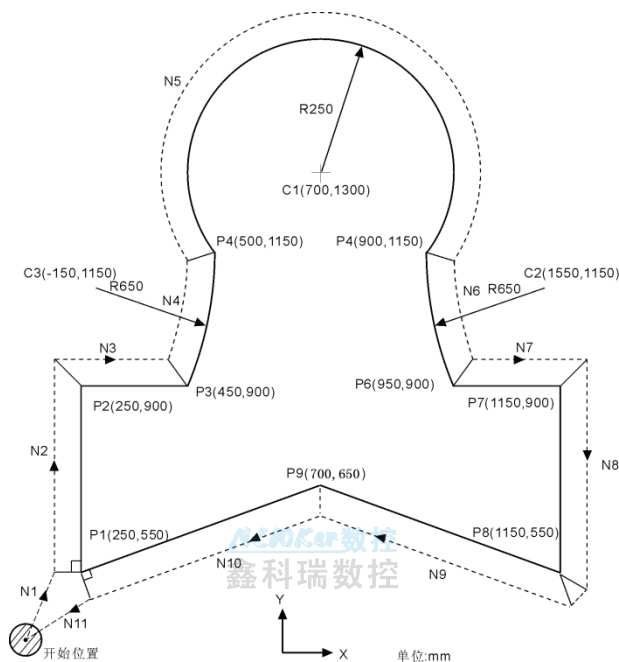
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程序段的起点的刀具半径偏移矢量、刀偏矢量是描述刀具偏移的数值和方向的方法。它的矢径为刀具半径。对于圆弧来说它的方向在半径方向。对于直线来说它的方向在直线的垂直方向。

刀具半径 C 补偿的建立和撤消方式分为 A 型与 B 型，可在刀具参数 P1、P2 中设置。具体方式见后图示。

注意：刀具半径补偿的建立和撤消必须在直线段进行。

例如：



G54 X0 Y0 Z0

G90 G17 G00 G41 T02 D3 X250.0 Y550.0

G01 Y900.0 F300

X450.0

G03 X500.0 Y1150.0 R650.0

G02 X900.0 Y1150.0 R-250.0

G03 X950.0 Y900.0 R650.0

G01 X1150.0

Y550.0

X700.0 Y650.0

X250.0 Y550.0

G00 G40 X0 Y0

；开始位置X0 Y0 Z0

；开始刀具半径补偿起刀，刀具用T02号刀的刀沿号D3 指定的距离偏移编程轨迹的右边。

；从P1 到P2 加工

；从P2 到P3 加工

；从P3 到P4 加工

；从P4 到P5 加工

；从P5 到P6 加工

；从P6 到P7 加工

；从P7 到P8 加工

；从P8 到P9 加工

；从P9 到P1 加工

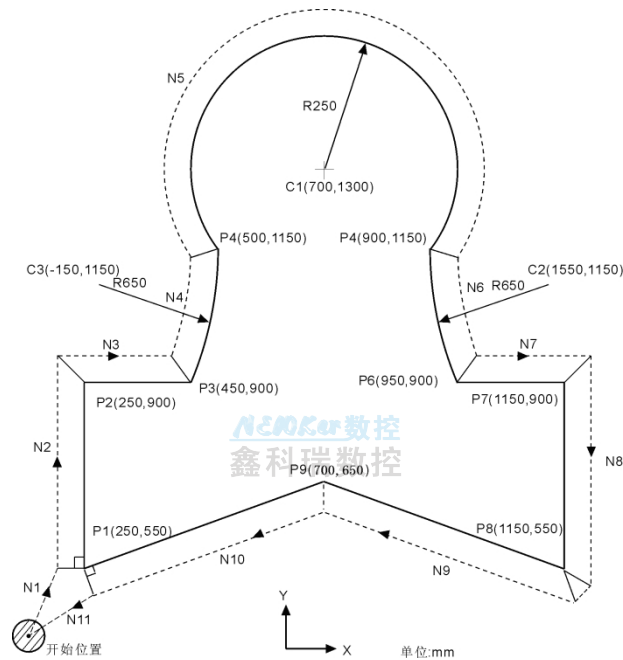
；取消偏置方式刀具返回到开始位置X0，Y0，Z0

The tool radius offset vector at the starting point of a program segment, known as the tool offset vector, is a method that describes both the magnitude and direction of tool movement. Its vector direction corresponds to the tool radius. For arcs, this direction aligns with the radius, while for straight lines, it points perpendicular to the line itself.

The establishment and cancellation of tool radius C compensation are divided into type A and Type B, which can be set in tool parameters P1 and P2. See the diagram below for details.

Note: The establishment and cancellation of tool radius compensation must be performed on straight segments.

for instance :



G54 X0 Y0 Z0 ; Start position X0 Y0 Z0

G90 G17 G00 G41 T02 D3 X250.0 Y550.0; Start tool radius compensation for cutting, tool

The edge number D3 of the tool T02 is offset to the right of the programmed path by a specified distance.

G01 Y900.0 F300	; Process from P1 to P2
X450.0	; From P2 to P3 processing
G03 X500.0 Y1150.0 R650.0	; From P3 to P4 processing
G02 X900.0 R-250.0	; From P4 to P5 processing
G03 X950.0 Y900.0 R650.0	; From P5 to P6 processing
G01 X1150.0	; From P6 to P7 processing
Y550.0	; From P7 to P8 processing
X700.0 Y650.0	; From P8 to P9 processing
X250.0 Y550.0	; From P9 to P1 processing
G00 G40 X0 Y0	; Cancel the biasing mode and return the tool

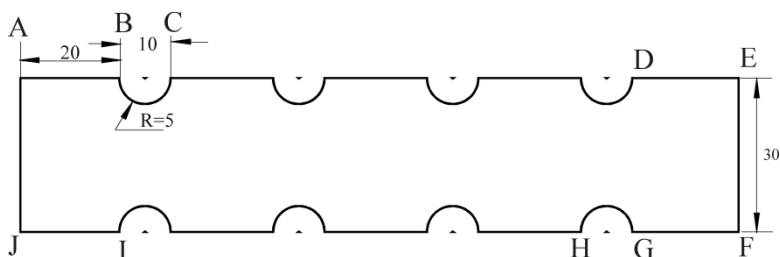
Start position X0, Y0, Z0

(18) 程序循环指令 (G22—G800)

G22 为程序循环指令，G800 是循环结束指令。二者必须配对使用，用于零件加工中需要反复加工的场所。L 为循环次数，取值范围为 1-99999。循环指令可以嵌套。

指令格式：G22 L-	；循环体开始
：	}
：	
：	
G800	；循环体结束

举例：



程序如下：

N0000 G17 G90 X0 Y0 F250 M03	；设置程序运行的开始状态
N0001 G91 G01 Z-10	
N0010 G22 L4	；循环体开始，指定循环体循环四次
N0020 G01 X20	；循环体内容：A-B 直线
N0030 G03 X10 I5 J0 Y0	；B-C 逆半圆
N0040 G800	；循环结束
N0050 G01 X20	；D-E 直线
N0060 Y-30	；E-F 直线
N0080 G22 L4	；循环体开始，指定循环体循环四次
N0090 G01 X-20	；F-G 直线
N0100 G03 X-10 I-5	；G-H 逆圆弧
N0110 G800	；循环结束
N0120 G01 X-20	；I-J 直线
N0130 G01 Y30	；J-A 直线
N0140 G26	；回程序起点
N0150 M02	；程序结束

(19) 准确定位/连续路径加工 (G60/G64)

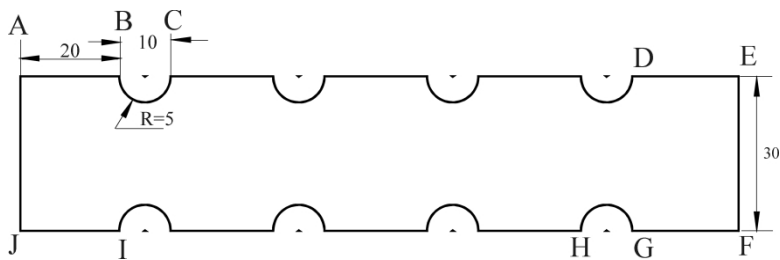
根据加工工艺的要求，可以通过 G60/G64 指令指定程序段之间的连接方式。

(18) Program loop instructions (G22--G800)

G22 is a program loop instruction, and G800 is a loop end instruction. These two instructions must be used in pairs for repeated processing in part machining. L is the number of loops, with a range of 1 to 99,999. Loop instructions can be nested.

Instruction format: G22 L-		
:	}	; The loop body begins; the
:		loop body; the loop body
:		ends
G800		

give an example :



The procedure is as follows:

```
N0000 G17 G90 X0 Y0 F250 M03; Set the starting state of the program
N0001 G91 G01 Z-10
N0010 G22 L4 ; The loop body starts, and the loop body is specified to
loop four times
N0020 G01 X20 ; Loop body content: A-B line
N0030 G03 X10 I5 J0 Y0 ; B-C, inverse semicircle
N0040 G800 ; loop end
N0050 G01 X20 ; D-E straight line
N0060 Y-30 ; E-F straight line
N0080 G22 L4 ; The loop body starts, and the loop body is specified to
loop four times
N0090 G01 X-20 ; F-G straight line
N0100 G03 X-10 I-5 ; G-H, inverse arc
N0110 G800 ; loop end
N0120 G01 X-20 ; I-J straight line
N0130 G01 Y30 ; J-A straight line
N0140 G26
; Return to the program starting point
N0150 M02 ; end of program
```

(19) Precise positioning/continuous path machining (G60/G64)

Depending on the requirements of the machining process, the G60/G64 instruction can be used to specify the connection between program segments.

指令格式: G60 ; 准确定位 (模态)
 G64 ; 连续路径加工 (模态, 初态)

(20) 固定循环宏定义指令 (G73、G80~G89)

用固定循环指令可以使频繁使用的加工程序变为更简便。

固定循环表(指令+500表示X方向进刀, +700表示Y方向进刀):

G代码	进给方式	在孔底的动作	退回	功能
G73/G573/G773	间歇进给	无	快速移动	高速深孔钻循环
G80/G580/G780	切削进给	无	无	取消固定循环
G81/G581/G781	切削进给	无	快速移动	钻孔, 点钻循环
G82/G582/G782	切削进给	停刀	快速移动	钻孔, 镗孔循环
G83/G583/G783	间歇进给	无	快速移动	深孔钻循环
G85/G585/G785	切削进给	无	切削进给	镗孔循环
G86/G586/G786	切削进给	主轴停	快速移动	镗孔循环
G87/G587/G787	切削进给	主轴停, 定向	快速移动	镗孔循环
G76/G576/G776	切削进给	主轴停, 定向	快速移动	镗孔循环
G89/G589/G789	切削进给	停刀	切削进给	镗孔循环
G74/G574/G774	切削进给	停刀-主轴正转	切削进给	左旋攻丝循环
G84/G584/G784	切削进给	停刀-主轴反转	切削进给	右旋攻丝循环

固定循环由6个顺序的动作组成

动作1 X轴和Y轴的定位

动作2 快速移动到R点

动作3 孔加工

动作4 在孔底的动作

动作5 返回到R点

动作6 快速移动到初始点

图示:

Instruction ; Precise positioning (modal-format: G60 ity)
G64 ; Continuous path machining (mode, initial state)

(20) Fixed Cycle Macro Definition Instructions (G73, G80~G89)

The use of fixed cycle instructions can make the frequently used machining program more convenient.

Fixed cycle table (instruction +500 indicates X direction feed, +700 indicates Y direction feed):

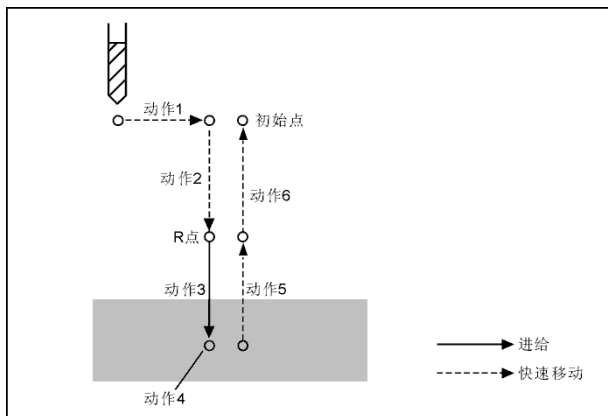
G code	Feed method	Action at the bottom of a hole	return	function
G73/G573/G773	index feed	not have	quick travel	High speed deep hole drilling cycle
G80/G580/G780	Cutting feed	not have	not have	Remove the fixed cycle
G81/G581/G781	Cutting feed	not have	quick travel	Borehole, point drill cycle
G82/G582/G782	Cutting feed	Stop the killing	quick travel	Bore, tap and bore cycle
G83/G583/G783	index feed	not have	quick travel	Deep hole drill cycles
G85/G585/G785	Cutting feed	not have	Cutting feed	Borehole circulation
G86/G586/G786	Cutting feed	The spindle is stopped	quick travel	Borehole circulation
G87/G587/G787	Cutting feed	Stop the spindle and orient	quick travel	Borehole circulation
G76/G576/G776	Cutting feed	Stop the spindle and orient	quick travel	Borehole circulation
G89/G589/G789	Cutting feed	Stop the killing	Cutting feed	Borehole circulation
G74/G574/G774	Cutting feed	Knife stop-spindle forward rotation	Cutting feed	Left turn screw cycle
G84/G584/G784	Cutting feed	Knife stop-spindle reverse	Cutting feed	Right turn screwing cycle

The fixed cycle consists of six sequential actions

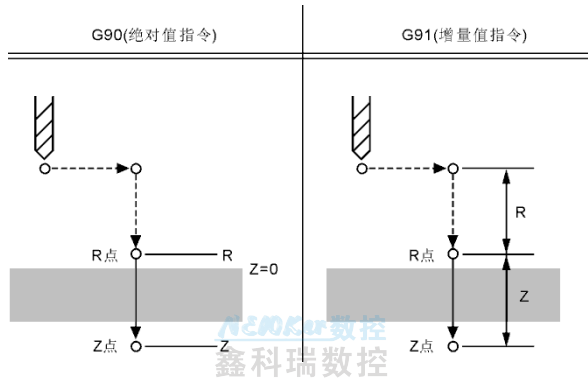
Action 1 Positioning
action of X and Y
axes 2 Quick move
to point R 3 Hole
machining

Action 4 Action at
the bottom of the
hole Action 5 Re-
turn to point R

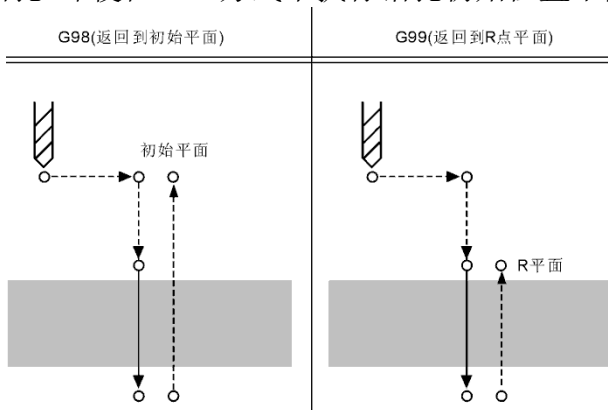
Action 6 Move quickly
to the starting point dia-
gram:



沿着钻孔轴的移动距离对 G90 和 G91 变化如下:

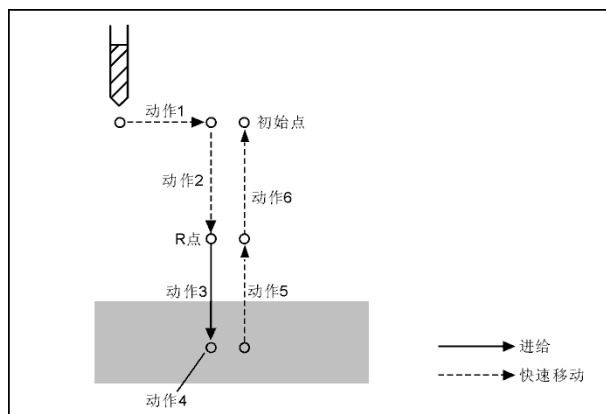


当刀具到达孔底后, 刀具可以返回到R 点平面或初始位置平面, 由G98 和G99 指定, 下图表示指定G98 或G99 时的刀具移动. 一般情况下, G99 用于第一次钻孔而G98 用于最后钻孔. 即使在G99 方式中执行钻孔初始位置平面也不变。

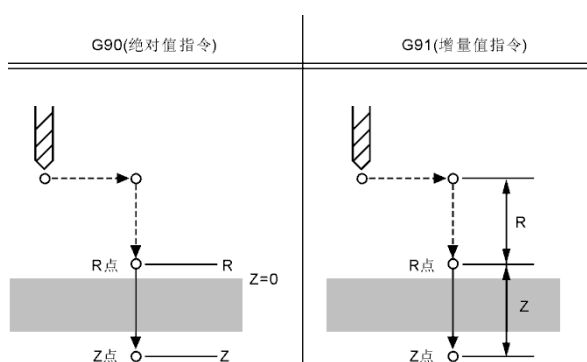


在L中指定重复次数, 在相同位置重复钻孔. L仅在被指定的程序段内有效. 重复次数L最大指令值=9999. 默认为L=1;

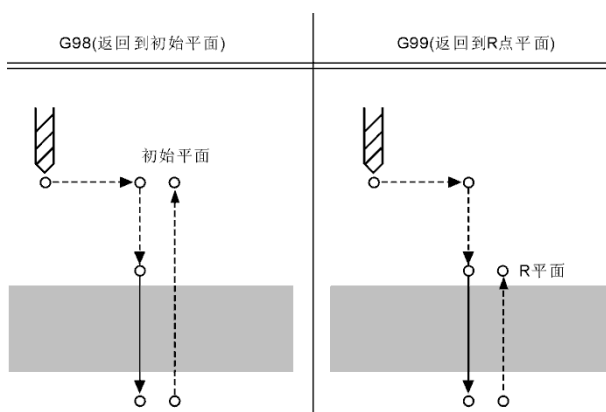
定位平面由G17 (XY) /G18 (ZX) /G19 (YZ) 决定, 另外速度参数P53圆弧间隙补偿再+2, 设置为2或者10。



The Changes of G90 and G91 Along the Moving Distance of the Drill Axis Are as Follows:



When the tool reaches the bottom of the hole, it can return to either the R-point plane or the initial position plane, as specified by G98 and G99 commands. The diagram below illustrates tool movement when G98 or G99 is activated. Typically, G99 is used for the initial drilling phase while G98 is employed for the final drilling. Notably, even when performing drilling in G99 mode, the initial position plane remains unchanged.



Specify the number of repetitions in L, and repeat drilling at the same location. L is only valid within the specified program segment. The maximum number of repetitions in L is 9999. The default value is L=1;

The positioning plane is determined by G17 (XY)/G18 (ZX)/G19 (YZ). In addition, the speed parameter P53 arc gap compensation is set to +2, and the setting is 2 or 10.

该循环执行高速深孔钻，它执行间歇切削进给直到孔的底部，同时从孔中排除切屑。

X-Y-: 孔位数据

R-：从初始点到 R 点的距离（G91）或坐标值

Q-: 每次切削深度

P-: 孔底的暂停时间, 单位: s

F-: 切削速度

L-: 重复次数

G73(G98)

Diagram illustrating the G73 (G98) drilling cycle. The cycle starts at the initial plane (初始平面) and moves down to the R point (R点). From the R point, the drill bit moves down to the Z point (Z点) in increments of depth d . The chip evacuation distance is q . The cycle repeats until the drill bit reaches the Z point. The diagram shows the drill bit moving down in increments of d and the chip evacuation distance q being maintained throughout the cycle.

G73(G99)

Diagram illustrating the G73 (G99) drilling cycle. The cycle starts at the initial plane (初始平面) and moves down to the R point (R点). From the R point, the drill bit moves down to the Z point (Z点) in increments of depth d . The chip evacuation distance is q . The cycle repeats until the drill bit reaches the Z point. The diagram shows the drill bit moving down in increments of d and the chip evacuation distance q being maintained throughout the cycle.

在指定 G73 之前用辅助功能旋转主轴 (M 代码)。

示例：

M3 S2000 ; 主轴开始旋转

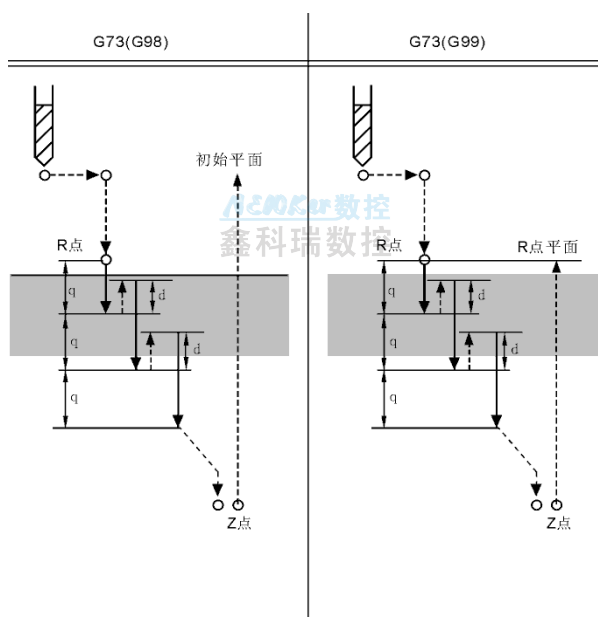
G90 G99 G73 X300 Y-250 Z-150 R-100 Q15 F120; 钻第1个孔, 返到R点

(21) High Speed Deep Hole Drilling Cycle (G73)

The cycle performs a high speed deep hole drill, which performs intermittent cutting feed until the bottom of the hole is reached while removing chips from the hole.

Instruction format: G73 X-Y-Z-R-Q-P-F-L-;
X-Y-: Hole position data
Z-: Distance from point R to bottom of hole (G91) or coordinate value (G90)
R-: Distance from the starting point to point R (G91) or coordinate value (G90)
Q-: Depth of cut per pass
P-: Pause time at the bottom of the hole,
unit: s
F-: cutting speed
L-: number of replication

graphic :



Note: The high-speed deep hole drilling cycle is performed with intermittent Z-axis feed. When using this cycle, chips can be easily discharged from the hole and a smaller retraction value can be set. This allows effective drilling. The tool rapid move retraction distance (d) is set in processing parameter P1.

Rotate the spindle using an auxiliary function (M code) before designating G73.

When a tool length offset (G43, G44, or G49) is specified in a fixed cycle, the offset is added while positioning to point R, and the tool radius offset is ignored.

instance :

M3 S2000 ; The spindle starts to rotate

G90 G99 G73 X300 Y-250 Z-150 R-100 Q15 F120; drill the first hole and return to point R

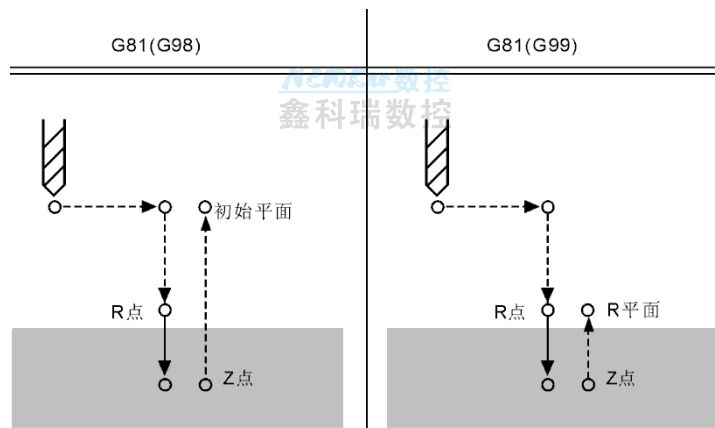
Y-550 ; 钻第2个孔, 然后返回到R点
Y-750 ; 钻第3个孔, 然后返回到R点
G98 Y-600 ; 钻第4个孔, 然后返回初始位置平面
G80
M30 ; 结束

(22) 钻孔循环, 点钻循环 (G81)

该循环用作正常钻孔, 切削进给执行到孔底。然后, 刀具从孔底快速移动退回。

指令格式: G81 X-Y-Z-R-P-F-L- ;
 X-Y-: 孔位数据
 Z-: 从 R 点到孔底的距离 (G91) 或坐标值 (G90)
 R-: 从初始点到 R 点的距离 (G91) 或坐标值 (G90)
 P-: 孔底的暂停时间, 单位: s
 F-: 切削速度
 L-: 重复次数

图示:



说明 在沿着 X 和 Y 轴定位以后快速移动到 R 点, 从 R 点到 Z 点执行钻孔加工, 然后刀具快速移动退回。

在指定G81之前用辅助功能M 代码旋转主轴

当在固定循环中指定刀具长度偏置 (G43、G44或G49) 时在定位到R点的同时加偏置, 刀具半径偏置被忽略。

示例:

M3 S2000 ; 主轴开始旋转
 G90 G99 G81 X300 Y-250 Z-150 R-100 F120
 ; 钻第1个孔, 然后返回到R点
 Y-550 ; 钻第2个孔, 然后返回到R点

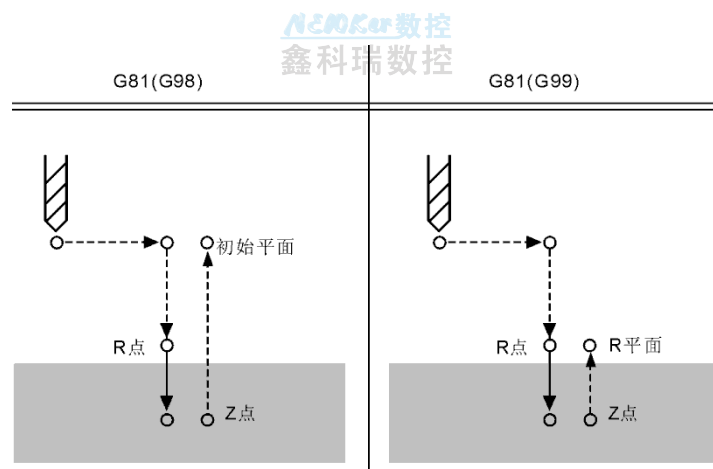
Y-550 ; Drill the second hole and return to point R
Y-750 ; Drill the third hole and return to point R
G98 Y-600 ; Drill the fourth hole and return to the initial position plane
G80
M30 ; finish

(22) Borehole cycle, point-to-borehole cycle (G81)

The cycle is used for normal drilling, and the cutting feed is executed to the bottom of the hole. The tool is then quickly moved back from the bottom of the hole.

Instruction format: G81 X-Y-Z-R-P-F-L-;
X-Y-: Hole position data
Z-: Distance from point R to hole bottom (G91) or coordinate value (G90)
R-: Distance from the starting point to point R (G91) or coordinate value (G90)
P-: Pause time at the bottom of the hole, unit: s
F-: cutting speed
L-: number of replication

graphic :



Description: After positioning along the X and Y axes, quickly move to point R, drill from point R to point Z, and then quickly move the tool back.

Rotate the spindle using the auxiliary function M code before specifying G81

When a tool length offset (G43, G44 or G49) is specified in a fixed cycle, the offset is added while positioning to point R, and the tool radius offset is ignored.

instance :

M3 S2000 ; The spindle starts to rotate
G90 G99 G81 X300 Y-250 Z-150 R-100 F120
; Drill the first hole and return to point R
Y-550 ; Drill the second hole and return to point R

G98 Y-750 ; 钻第3个孔, 然后返回初始位置平面

G80

M30 ; 结束

为了编程方便, 固定循环可以简化编程。

如: G90 G00X0Y0Z0M3;

G81 X10Y10F2;

G91X50L5;(循环5次, X方向5个孔, 间隔50);

Y60L5;(Y方向5个孔, 循环5次)

X10Y20L7;(斜线方向7个孔, 循环7次)

G90

(23) 钻孔循环, 镗镗循环 (G82)

该循环用作正常钻孔, 切削进给执行到孔底, 执行暂停。然后刀具从孔底快速移动退回。

指令格式: G82 X-Y-Z-R-P-F-L- ;

X-Y-: 孔位数据

Z-: 从 R 点到孔底的距离 (G91) 或坐标值 (G90)

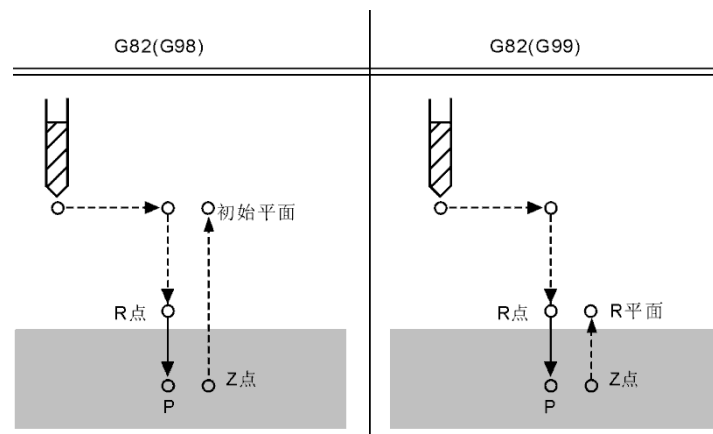
R-: 从初始点到 R 点的距离 (G91) 或坐标值 (G90)

P-: 孔底的暂停时间, 单位: s

F-: 切削速度

L-: 重复次数

图示:



说明: 沿着X和Y轴定位以后, 快速移动到R点。然后从R点到Z点执行钻孔加工。当到底孔底时, 执行暂停。然后刀具快速移动退回。

指定G82之前, 用辅助功能M代码旋转主轴。

当在固定循环中指定刀具长度偏置 (G43、G44或G49) 时在定位到R点的同时

G98 Y-750; drill the third hole and return to the initial position plane

G80

M30 ; finish

In order to facilitate programming, fixed loops can simplify programming.

For example: G90 G00X0Y0Z0M3;

G81 X10Y10F2;

G91X50L5; (cycled 5 times, 5 holes in the X direction, 50 intervals);

Y60L5; (5 holes in the Y direction, 5 cycles)

X10Y20L7; (7 holes in the diagonal direction, 7 cycles)

G90

(23) Drill Cycle, Counterboring Cycle (G82)

The cycle is used for normal drilling, the cutting feed is executed to the bottom of the hole, and a pause is performed. The tool is then rapidly moved back from the bottom of the hole.

Instruction format: G82 X-Y-Z-R-P-F-L-;

X-Y-: Hole position data

Z-: Distance from point R to bottom of hole (G91) or coordinate value (G90)

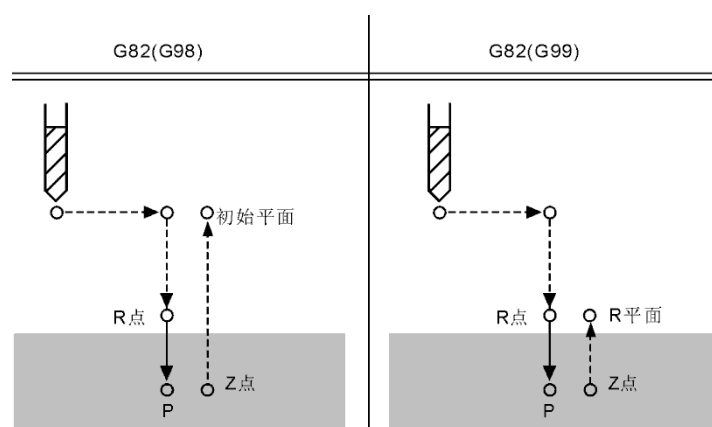
R-: Distance from the starting point to point R (G91) or coordinate value (G90)

P-: Pause time at the bottom of the hole, unit: s

F-: cutting speed

L-: number of replication

graphic :



Note: After positioning along the X and Y axes, quickly move to point R. Then drill from point R to point Z. When at the bottom of the hole, pause. Then the tool moves back quickly.

Rotate the spindle using auxiliary function M code before specifying G82.

When the tool length offset (G43, G44 or G49) is specified in a fixed cycle, it is positioned at point R

加偏置。刀具半径偏置被忽略。

示例：

```

M3 S2000          ; 主轴开始旋转
G90 G99 G82 X300 Y-250 Z-150 R-100 P1 F120
                  ; 钻第1个孔， 然后返回到R点
Y-550            ; 钻第2个孔， 然后返回到R点
G98 Y-750        ; 钻第3个孔， 然后返回初始位置平面
G80
M30              ; 结束
  
```

(24) 排屑钻孔循环 (G83)

执行间歇切削进给到孔的底部钻孔过程中从孔中排除切屑

指令格式：G83 X-Y-Z-R-Q-P-F-L- ；

X-Y-：孔位数据

Z-：从 R 点到孔底的距离 (G91) 或坐标值 (G90)

R-：从初始点到 R 点的距离 (G91) 或坐标值 (G90)

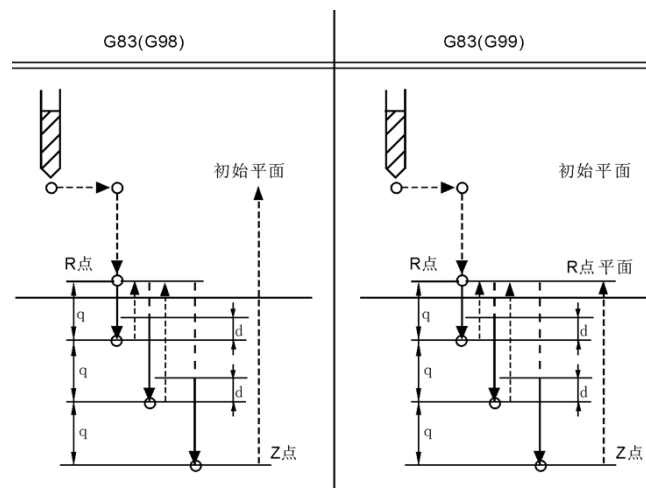
Q-：每次切削进给的切削深度

P-：孔底的暂停时间，单位：s

F-：切削速度

L-：重复次数

图示：



说明：Q 表示每次切削进给的切削深度，它必须用增量值指定。在第二次和以后的切削进给中，执行快速移动到上次钻孔结束之前的d 点再次执行，切削进给d在加工参数P2设定，在Q 中必须指定正值，负号被忽略。

指定G83之前，用辅助功能M 代码旋转主轴。

Add bias. Tool radius bias is ignored.

instance :

```
M3 S2000          ; The spindle starts to rotate
G90 G99 G82 X300 Y-250 Z-150 R-100 P1 F120
                  ; Drill the first hole and return to point R
Y-550             ; Drill the second hole and return to point R
G98 Y-750         ; Drill the third hole and return to the initial position plane
G80
M30               ; finish
```

(24) Drill Cuttings Circulation (G83)

Execution of intermittent cutting feed into the bottom of the hole to drill the hole and remove the chips from the hole

Instruction format: G83 X-Y-Z-R-Q-P-F-L-;

X-Y-: Hole position data

Z-: Distance from point R to hole bottom (G91) or coordinate value (G90)

R-: Distance from the starting point to point R (G91) or coordinate value (G90)

Q-: The depth of cut per pass

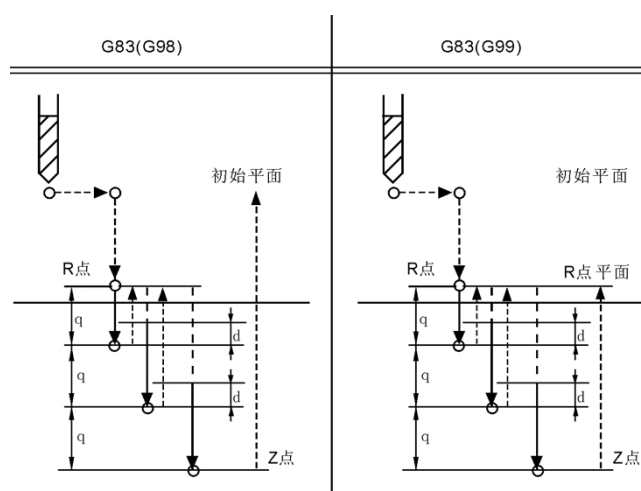
P-: Pause time at the bottom of the hole,

unit: s

F-: cutting speed

L-: number of replication

graphic :



Note: Q represents the cutting depth per feed, which must be specified as an incremental value. During the second and subsequent feed operations, perform a rapid move to point d before the previous drilling end to repeat the process. The cutting feed d is set in the machining parameter P2. A positive value must be specified in Q, and negative signs are ignored.

Rotate the spindle using the auxiliary function M code before specifying G83.

当在固定循环中指定刀具长度偏置（G43、G44 或G49）时，在定位到R点的同时加偏置， 刀具半径偏置被忽略。

示例：

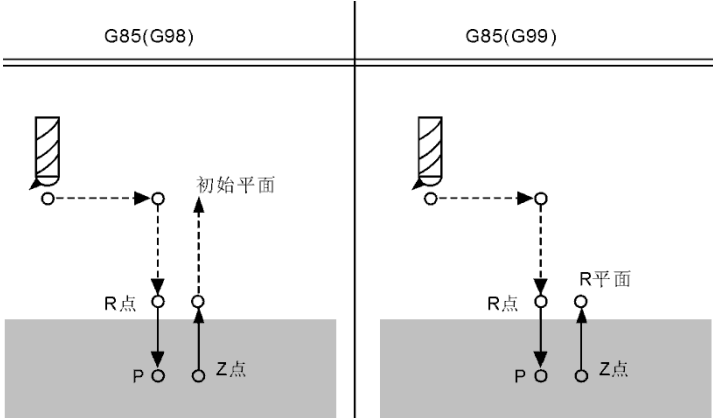
```
M3 S2000      ; 主轴开始旋转
G90 G99 G83 X300 Y-250 Z-150 R-100 Q15 F120 ; 钻第1个孔， 返到R点
Y-550        ; 钻第2个孔， 然后返回到R点
Y-750        ; 钻第3个孔， 然后返回到R点
G98 Y-600    ; 钻第4个孔， 然后返回初始位置平面
G80
M30          ; 结束
```

(25) 镗孔循环（G85）

G85指令用于镗孔。

指令格式：G85 X-Y-Z-R-P-F-L- ；
X-Y-：孔位数据
Z-：从 R 点到孔底的距离（G91）或坐标值（G90）
R-：从初始点到 R 点的距离（G91）或坐标值（G90）
P-：孔底的暂停时间，单位：s
F-：切削速度
L-：重复次数

图示：



说明：沿着X和Y轴定位以后，快速移动到R点，然后从R点到Z 点执行镗孔，当到达孔底时，执行切削进给，然后返回到R点。

在指定G85之前，用辅助功能M代码旋转主轴。

当在固定循环中指定刀具长度偏置（G43、G44 或G49）时在定位到R点的同时加偏置，刀具半径偏置被忽略。

示例：

When a tool length offset (G43, G44, or G49) is specified in a fixed cycle, the offset is added while positioning to point R, and the tool radius offset is ignored.

instance :

```
M3 S2000          ; The spindle starts to rotate
G90 G99 G83 X300 Y-250 Z-150 R-100 Q15 F120; drill the first hole
and return to point R Y-550      ; Drill the second hole and return
to point R
Y-750              ; Drill the third hole and return to point R
G98 Y-600          ; Drill the fourth hole and return to the initial position
```

plane

```
G80
M30          ; finish
(25) Borehole Circulation (G85)
```

The G85 command is used for boring.

Instruction format: G85 X-Y-Z-R-P-F-L-;

X-Y-: Hole position data

Z-: Distance from point R to hole bottom (G91) or coordinate value (G90)

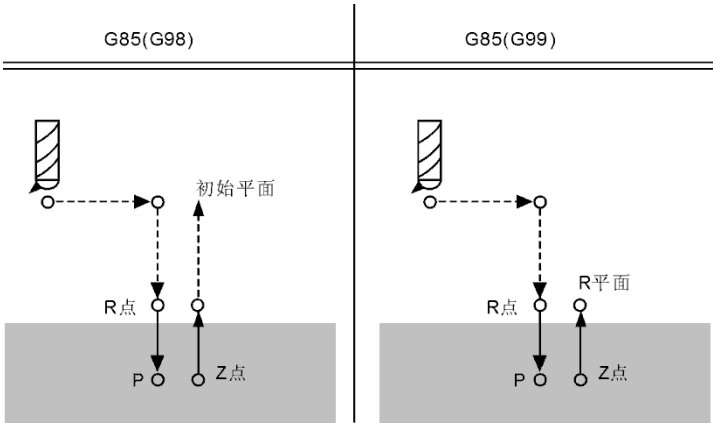
R-: Distance from the starting point to point R (G91) or coordinate value (G90)

P-: Pause time at the bottom of the hole, unit: s

F-: cutting speed

L-: number of replication

graphic :



Note: After positioning along the X and Y axes, quickly move to point R, then perform boring from point R to point Z. When reaching the bottom of the hole, perform cutting feed, and then return to point R.

Rotate the spindle using auxiliary function M code before designating G85.

When a tool length offset (G43, G44 or G49) is specified in a fixed cycle, the offset is added while positioning to point R, and the tool radius offset is ignored.

instance :

M3 S100 ; 主轴开始旋转
 G90 G99 G85 X300 Y-250 Z-150 R-120 F120 ; 镗第1个孔, 返到R点
 Y-550 ; 镗第2个孔, 然后返回到R点
 Y-750 ; 镗第3个孔, 然后返回到R点
 G98 Y-600 ; 镗第4个孔, 然后返回初始位置平面
 G80
 M30 ; 结束

(26) 镗孔循环 (G86)

G86指令用于镗孔.

指令格式: G86 X-Y-Z-R-P-F-L- ;

X-Y-: 孔位数据

Z-: 从 R 点到孔底的距离 (G91) 或坐标值 (G90)

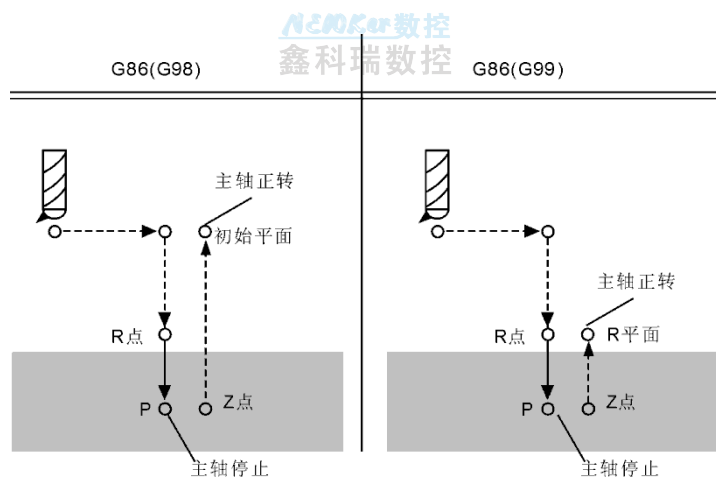
R-: 从初始点到 R 点的距离 (G91) 或坐标值 (G90)

P-: 孔底的暂停时间, 单位: s

F-: 切削速度

L-: 重复次数

图示:



说明: 沿着X和Y轴定位以后, 快速移动到R点, 然后从R点到Z点执行镗孔. 当主轴在孔底停止时, 刀具以快速移动退回.

在指定G86之前, 用辅助功能M代码旋转主轴.

当在固定循环中指定刀具长度偏置 (G43、G44 或G49) 时在定位到R点的同时加偏置, 刀具半径偏置被忽略.

示例:

M3 S2000 ; 主轴开始旋转
 G90 G99 G86 X300 Y-250 Z-150 R-100 F120 ; 镗第1个孔, 返到R点

M3 S100; the spindle starts to rotate

G90, G99, G85, X300, Y-250, Z-150, R-120, F120; bore the first hole and return to point R

Y-550 ; Bore the second hole and return to point R

Y-750 ; Bore the third hole and return to point R

G98 Y-600 ; Bore the fourth hole and return to the initial position plane
G80

M30 ; finish

(26) Borehole circulation (G86)

The G86 command is used for boring.

Instruction format: G86 X-Y-Z-R-P-F-L-;

X-Y-: Hole position data

Z-: Distance from point R to hole bottom (G91) or coordinate value (G90)

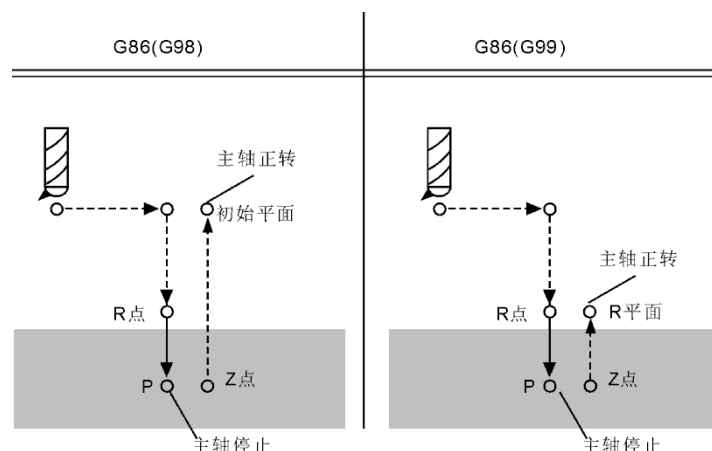
R-: Distance from the starting point to point R (G91) or coordinate value (G90)

P-: Pause time at the bottom of the hole, unit: s

F-: cutting speed

L-: number of replication

graphic :



Note: After locating along the X and Y axes, quickly move to point R and then perform a bore from point R to point Z. When the spindle stops at the bottom of the hole, the tool is rapidly moved back.

Rotate the spindle using the auxiliary function M code before designating G86.

When a tool length offset (G43, G44 or G49) is specified in a fixed cycle, the offset is added while positioning to point R, and the tool radius offset is ignored.

instance :

M3 S2000; the spindle starts to rotate

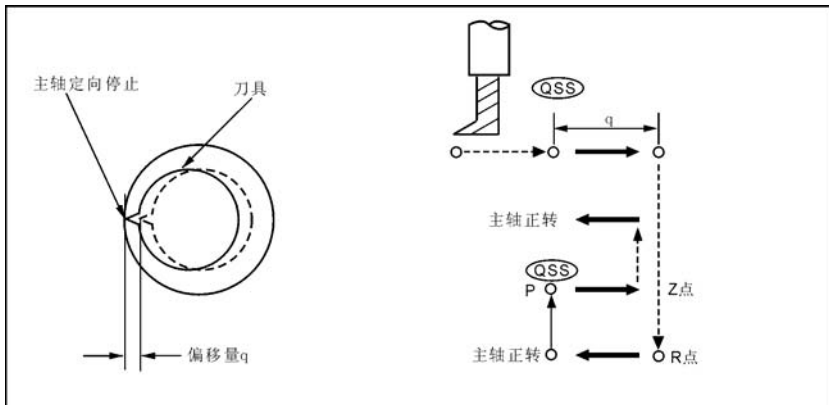
G90 G99 G86 X300 Y-250 Z-150 R-100 F120; bore the first hole and return to point R

Y-550 ; 镗第2个孔, 然后返回到R点
Y-750 ; 镗第3个孔, 然后返回到R点
G98 Y-600 ; 镗第4个孔, 然后返回初始位置平面
G80
M30 ; 结束

(27) 镗孔循环、背镗孔循环 (G87)

此指令只在加工中心系统上有效, 因为主轴需要定向停止功能, 所以前提条件为主轴必须具有定位功能(输出M61, 检测M22)。精镗循环镗削精密孔。当到达孔底时主轴停止, 切削刀具按加工参数中P4参数设定方向离开工件的被加工表面并返回。

指令格式: G87 X-Y-Z-R-Q-P-F-L- ;
X-Y-: 孔位数据
Z-: 从 R 点到孔底的距离 (G91) 或坐标值 (G90)
R-: 从初始点到 R 点的距离 (G91) 或坐标值 (G90)
Q-: 孔底的偏移量, 方向由加工参数 P4 设置
P-: 暂停时间, 单位: s
F-: 切削速度
L-: 重复次数



说明: 沿着X和Y轴定位以后, 主轴在固定的旋转位置上停止。刀具在刀尖的反方向(由加工参数中P4、P5参数设定)快速移动并在孔底R点定位。然后, 刀具在刀尖的方向上移动并且主轴正转。沿Z轴的正向镗孔直到Z点。在Z点, 主轴再次停在固定的旋转位置, 刀具在刀尖的反方向移动(由加工参数中P3、P5参数设定)。然后刀具返回到初始位置, 刀具在刀尖的方向上偏移, 主轴正转。执行下个程序段的加工。

在指定G87之前, 用辅助功能M代码旋转主轴。

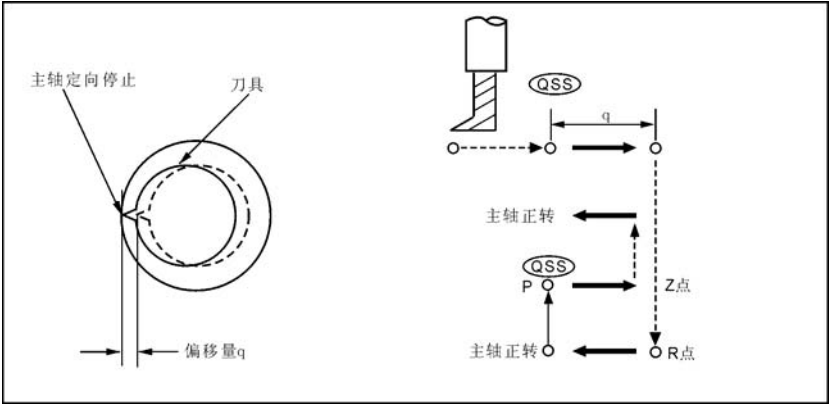
当在固定循环中指定刀具长度偏置(G43、G44或G49)时在定位到R点的同时加偏置, 刀具半径偏置被忽略。

Y-550 ; Bore the second hole and return to point R
Y-750 ; Bore the third hole and return to point R
G98 Y-600 ; Bore the fourth hole and return to the initial
position plane
G80
M30 ; finish

(27) Boring Cycle, Back Boring Cycle (G87)

This command is only valid on machining center systems, as the spindle requires orientation stop functionality. The prerequisite is that the spindle must have positioning capabilities (output M61, detection M22). Precision boring cycle for machining precision holes. When reaching the hole bottom, the spindle stops. The cutting tool then departs from the workpiece's machined surface in the direction specified by parameter P4 of the machining parameters and returns.

Instruction format: G87 X-Y-Z-R-Q-P-F-L-;
X-Y-: Hole position data
Z-: Distance from point R to bottom of hole (G91) or coordinate value (G90)
R-: Distance from the starting point to point R (G91) or coordinate value (G90)
Q-: The offset amount at the bottom of the hole, and the direction is set by the machining parameter P4
P-: Pause time, unit: s
F-: cutting speed
L-: number of replication



Procedure: After positioning along the X and Y axes, the spindle stops at a fixed rotation position. The tool rapidly moves in the opposite direction of the cutting edge (as set by parameters P4 and P5 in the machining parameters) to position at point R at the hole bottom. Subsequently, the tool moves in the cutting edge direction while the spindle rotates forward. Boring is performed along the Z-axis until reaching point Z. At Z-point, the spindle stops again at the fixed rotation position, and the tool moves in the opposite direction of the cutting edge (as set by parameters P3 and P5). The tool then returns to its initial position, shifts in the cutting edge direction, and rotates forward again. This completes the processing cycle for the next program segment.

Rotate the spindle using the auxiliary function M code before specifying G87.

When a tool length offset (G43, G44 or G49) is specified in a fixed cycle, the offset is added while positioning to point R, and the tool radius offset is ignored.

举例：

N10 M3 S100 主轴开始旋转

N20 G90 G99 G87 X300 Y-250 Z-150 R-100 Q5 ； 定位，镗1孔，然后返回到R点；孔底定向然后移动5mm

N30 Y-350 ； 定位，镗2孔，然后返回到R点

N40 Y-550 ； 定位，镗3孔，然后返回到R点

N50 Y-750 ； 定位，镗4孔，然后返回到R点

N60 G80

N70 M30

(28) 镗孔循环 (G89)

G89指令用于镗孔。

指令格式：G89 X-Y-Z-R-P-F-L- ；

X-Y-：孔位数据

Z-：从 R 点到孔底的距离 (G91) 或坐标值 (G90)

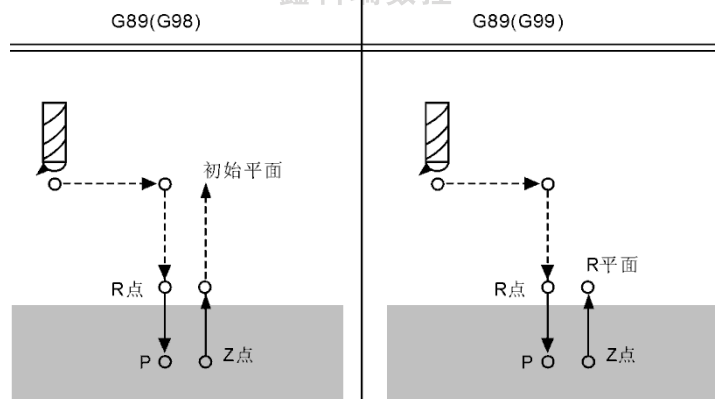
R-：从初始点到 R 点的距离 (G91) 或坐标值 (G90)

P-：暂停时间，单位：s

F-：切削速度

L-：重复次数

图示：



说明：除在孔底执行暂停，该循环与G85相同。

在指定G89之前，用辅助功能M代码旋转主轴。

当在固定循环中指定刀具长度偏置 (G43、G44或G49) 时在定位到R点的同时加偏置。刀具半径偏置被忽略。

示例：

M3 S100 ； 主轴开始旋转

G90 G99 G89 X300. Y-250. Z-150. R-120. P1000 F120.

； 镗第1个孔， 然后返回到R点在孔底停止1秒

give an example :

The N10 M3 S100 spindle starts to rotate

N20 G90 G99 G87 X300 Y-250 Z-150 R-100 Q5; Position, bore 1 hole, then return to point R; orient the bottom of the hole and move 5mm

N30 Y-350 ; Position, bore two holes, and return to point R

N40 Y-550 ; Position, bore 3 holes, and return to point R

N50 Y-750 ; Position, bore four holes, and return to point R

N60 G80

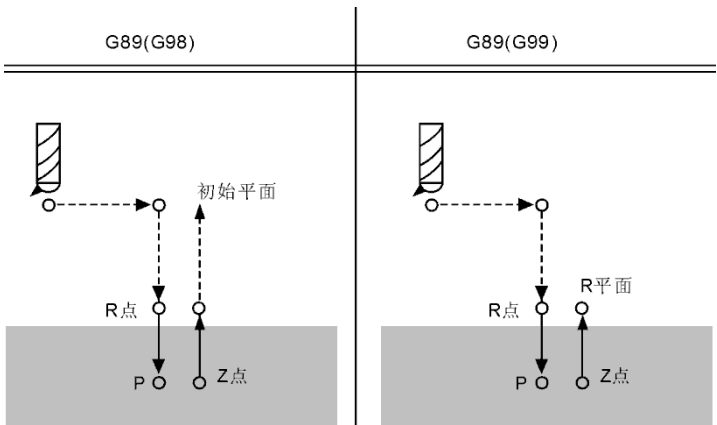
N70 M30

(28) Borehole Circulation (G89)

The G89 command is used for boring.

Instruction format: G89 X-Y-Z-R-P-F-L-;
X-Y-: Hole position data
Z-: Distance from point R to hole bottom (G91) or coordinate value (G90)
R-: Distance from the starting point to point R (G91) or coordinate value (G90)
P-: Pause time, unit: s
F-: cutting speed
L-: number of replication

graphic :



Note: Except for the pause at the bottom of the hole, this cycle is the same as G85.

Rotate the spindle using the auxiliary function M code before the specified G89.

When tool length offsets (G43, G44 or G49) are specified in a fixed cycle, the offsets are added while positioning to point R. Tool radius offsets are ignored.

instance :

M3 S100 ; The spindle starts to rotate

G90 G99 G89 X300. Y-250. Z-150. R-120. P1000 F120.

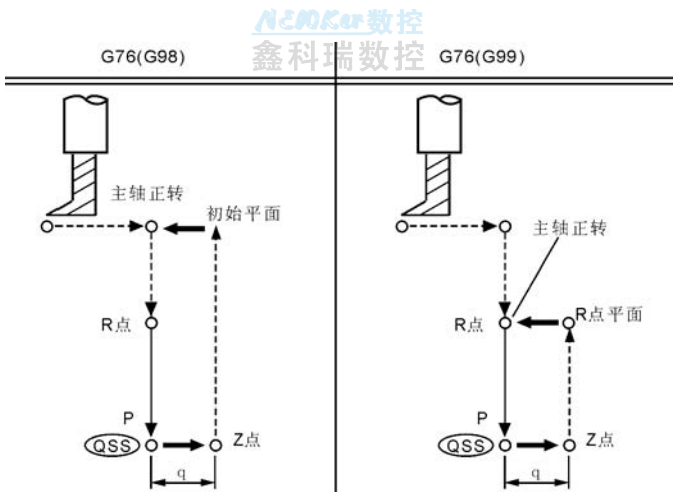
; Bore the first hole, then return to point R and stop at the bottom of the hole for 1 second

Y-550 ; 镗第2个孔, 然后返回到R点
G98 Y-750 ; 镗第3个孔, 然后返回初始位置平面
G80
M30 ; 结束

(29) 精镗循环 (G76)

此指令只在加工中心系统上有效, 因为主轴需要定向停止功能, 所以前提条件为主轴必须具有定位功能(输出M61, 检测M22)。精镗循环镗削精密孔。当到达孔底时主轴停止, 切削刀具按加工参数中P3参数设定方向离开工件的被加工表面并返回。

指令格式: G76 X-Y-Z-R-Q-P-F-L- ;
X-Y-: 孔位数据
Z-: 从 R 点到孔底的距离 (G91) 或坐标值 (G90)
R-: 从初始点到 R 点的距离 (G91) 或坐标值 (G90)
Q-: 孔底的偏移量, 方向由加工参数 P3 设置
P-: 暂停时间, 单位: s
F-: 切削速度
L-: 重复次数



说明: 当到达孔底时, 主轴在固定的旋转位置停止, 并且刀具以刀尖的相反方向移动退刀。这保证加工面不被破坏, 实现精密和有效的镗削加工。注: 主轴定向角度在加工参数P5中设置, 在指定G76 之前用辅助功能M代码旋转主轴。当在固定循环中指定刀具长度偏置 (G43、G44 或G49) 时在定位到R点的同时加偏置, 刀具半径偏置被忽略。

举例:

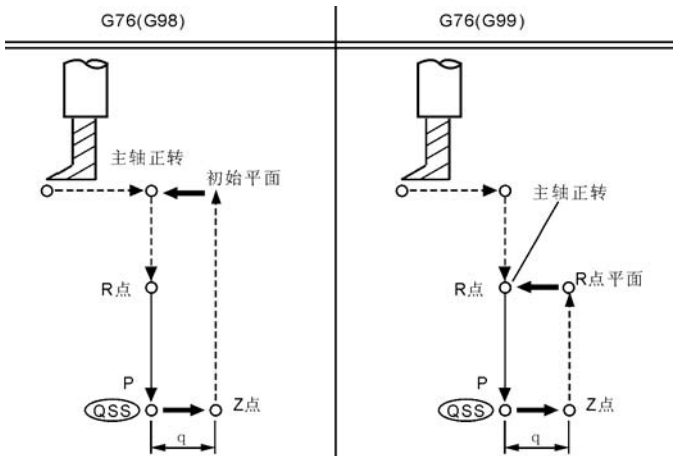
N10 M3 S100 主轴开始旋转
N20 G90 G99 G76 X300 Y-250 Z-150 R-100 Q5 ; 定位, 镗1孔, 然后返回到R点; 孔底定向然后移动5mm

Y-550 ; Bore the second hole and return to point R
G98 Y-750; Bore the third hole and return to the initial position
plane
G80
M30 ; finish

(29) Borehole circulation (G76)

This command is only valid on machining center systems, as the spindle requires orientation stop functionality. The prerequisite is that the spindle must have positioning capabilities (output M61, detection M22). Precision boring cycle for drilling precision holes: When reaching the hole bottom, the spindle stops. The cutting tool then departs from the workpiece's machined surface according to the direction specified in parameter P3 of the machining parameters and returns.

Instruction format: G76 X-Y-Z-R-Q-P-F-L-;
X-Y-: Hole position data
Z-: Distance from point R to hole bottom (G91) or coordinate value (G90)
R-: Distance from the starting point to point R (G91) or coordinate value (G90)
Q-: The offset of the hole bottom, and the direction is set by the machining parameter P3
P-: Pause time, unit: s
F-: cutting speed
L-: number of replication



Note: When reaching the bottom of the hole, the spindle stops at a fixed rotation position, and the tool moves backward in the opposite direction of the cutting edge. This ensures the workpiece surface remains undamaged, achieving precise and efficient boring operations.

Note: The spindle orientation angle is set in machining parameter P5, and the spindle rotation is performed using auxiliary function M codes before G76 commands. When tool length offsets (G43, G44, or G49) are specified in fixed cycles, the tool radius offset is ignored and offsets are applied simultaneously with positioning at point R.

give an example :

The N10 M3 S100 spindle starts to rotate
N20 G90 G99 G76 X300 Y-250 Z-150 R-100 Q5; position, bore 1
hole, then return to point R; orient the bottom of the hole and move
5mm

N30 Y-350 ; 定位, 镗2孔, 然后返回到R点
N40 Y-550 ; 定位, 镗3孔, 然后返回到R点
N50 Y-750 ; 定位, 镗4孔, 然后返回到R点
N60 G80
N70 M30

(30) 左旋啄式深孔攻丝循环(G74)

攻丝有二种方法: 跟踪主轴编码器方法 (P411=2, 主轴必须装配编码器) 和 Z 轴与主轴伺服插补方法 (P411=3), 此方式指令前主轴伺服要设置为位置控制方式如加指令 M881 (加工中心用 M881 指令, 控制主轴为位置控制方式)。

特别注意:

执行 G73/G84 指令停在中间或者攻丝过程断电时, 在 MDI 编辑状态下, 按“图形”键, 则自动生成 M180 攻丝回退代码, 按运行即可执行攻丝回退动作;

该循环执行左旋攻丝, 在左旋攻丝循环中, 每次进刀深度Q, 主轴反转回退距离d (加工参数P7设置) 再进刀, 排屑方式加工参数P6设置 (0表高速排屑回退d, 8表普通排屑回退到R), 当到达孔底时主轴顺时针旋转。

指令格式: G74 X-Y-Z-R-P-K(F)-Q-S-L- ;
X-Y-: 孔位数据
Z-: 从 R 点到孔底的距离 (G91) 或坐标值 (G90)
R-: 从初始点到 R 点的距离 (G91) 或坐标值 (G90)
Q-: 每次攻丝深度
P-: 暂停时间, 单位: s
K(F)-: 螺纹导程, E 英制
S-: 主轴转速
L-: 重复次数

图示:

N30 Y-350 ; Position, bore two holes,
and return to point R
N40 Y-550 ; Position, bore 3 holes, and
return to point R
N50 Y-750 ; Position, bore four holes,
and return to point R
N60 G80
N70 M30

(30) Left Rotary Deep Hole Tapping Cycle (G74)

There are two methods for thread cutting: the spindle encoder tracking method (P411=2, where the spindle must be equipped with an encoder) and the Z-axis and spindle servo interpolation method (P411=3). In this approach, the spindle servo must be set to position control mode before issuing commands, such as adding the instruction M881 (the M881 command is used in machining centers to control the spindle in position control mode). Special note:

When the G73/G84 instruction stops in the middle or the power is cut off during the tapping process, press the "graphics" key under the MDI editing state to automatically generate the M180 tapping back code, and press run to execute the tapping back action;

This cycle performs left-hand tapping. During each left-hand tapping cycle, the tool advances by depth Q, then the spindle reverses to retract by distance d (set in machining parameter P7), followed by another advance. The chip removal method is configured according to parameter P6 (Table 0: high-speed chip removal retraction d; Table 8: standard chip removal retraction to R). When reaching the hole bottom, the spindle rotates clockwise.

Instruction format: G74 X-Y-Z-R-P-K (F)-Q-S-L-;

X-Y-: Hole position data

Z-: Distance from point R to hole bottom (G91) or coordinate value (G90)

R-: Distance from the starting point to point R (G91) or coordinate value (G90)

Q-: Depth of each thread attack

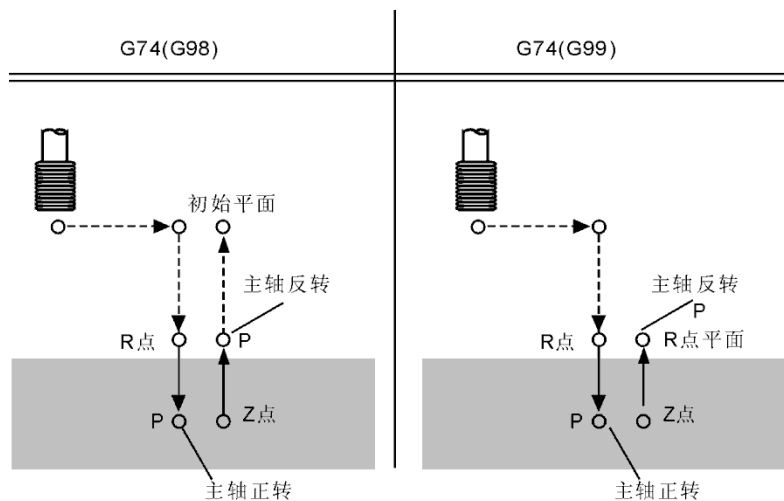
P-: Pause time, unit: s

K(F): Thread pitch, E in

S-: main shaft speed

L-: number of replication

graphic :



说明:

在加工螺纹时坐标轴速度由主轴转速和螺距确定，而与进给速度F值没有关系。系统自动将坐标轴速度限制在最大进给速度之内。

加工螺纹时，主轴倍率开关和进给轴倍率开关无效。

在每一螺纹加工程序段都应指定螺纹导程K值，否则，无法通过编译。

用主轴逆时针旋转执行攻丝。当到达孔底时，为了退回主轴顺时针旋转。该循环加工一个反螺纹。在左旋攻丝期间，进给暂停不停止机床，直到后退动作完成。

在指定G74之前使用辅助功能M代码使主轴逆时针旋转。当主轴为脉冲控制方式时，不需用辅助功能M代码旋转主轴。

当在固定循环中指定刀具长度偏置(G43, G44 或G49)时在定位到R点的同时加偏置。刀具半径偏置被忽略。

示例:

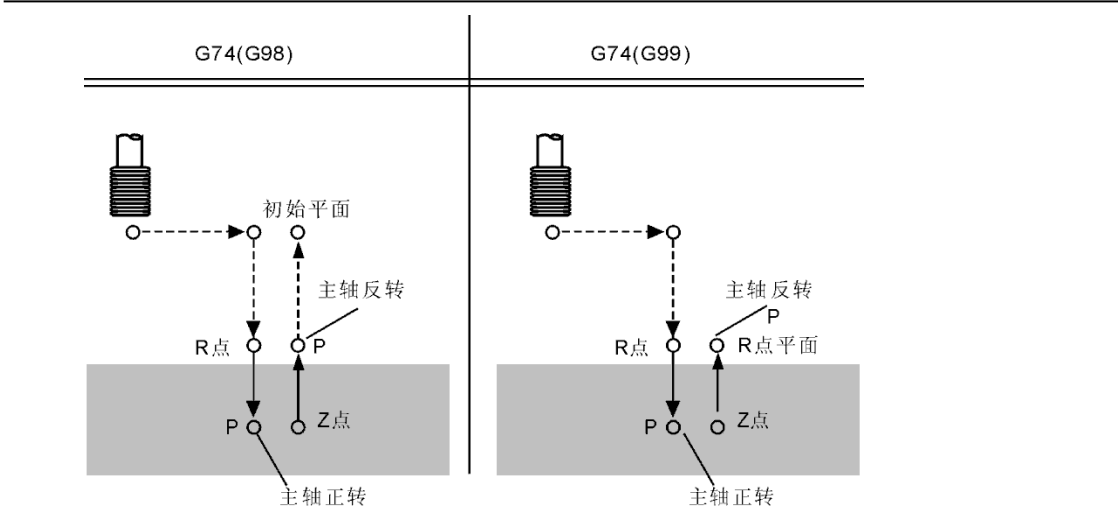
```

N10 M4 S100  主轴开始旋转
N20 G90 G99 G74 X300. Y-250. Z-150. R-100. K5 S100
                        ; 攻第1个孔，然后返回到R点
N30 Y-550. K5        ; 攻第2个孔，然后返回到R点
N40 G98 Y-750 K5      ; 攻第3个孔，然后返回初始位置平面
N50 G80
N60 M30            ; 结束
  
```

(31) 右旋琢式深孔攻丝循环(G84)

攻丝有二种方法跟踪主轴编码器方法(P411=2，主轴必须装配编码器)和Z轴与主轴伺服插补方法(P411=3)，此方式指令前主轴伺服要设置为位置控制方式如加指令M881(加工中心用M881指令，控制主轴为位置控制方式)。

特别注意:



explain :

In the process of thread processing, the coordinate axis speed is determined by spindle speed and pitch, and has nothing to do with feed speed F value. The system automatically limits the coordinate axis speed within the maximum feed speed.

During thread machining, the spindle ratio switch and feed axis ratio switch are invalid.

The thread pilot K value should be specified in each thread machining program segment, otherwise, it cannot be compiled.

Turn the spindle counterclockwise to execute the tap. When you reach the bottom of the hole, turn the spindle clockwise to retract it. Rotate. The cycle processes a counter-thread. During left-hand tapping, the feed pause does not stop the machine until the retraction action is completed.

Use auxiliary function M code to rotate the spindle counterclockwise before specifying G74. When the spindle is pulse-controlled, no rotation of the spindle is required using auxiliary function M code.

When a tool length offset (G43, G44 or G49) is specified in a fixed cycle, the offset is added while positioning to point R. The tool radius offset is ignored.

instance :

The N10 M4 S100 spindle starts to rotate

N20 G90 G99 G74 X300. Y-250. Z-150. R-100. K5 S100

; Attack the first hole and return to point R

N30 Y-550. K5; tap the second hole, then return to point R

N40 G98 Y-750 K5; tap the third hole, then return to the initial position plane

N50 G80

N60 M30 ; finish

(31) Right Rotation Twist Deep Hole Tapping Cycle (G84)

There are two ways to attack the thread: the method of tracking the spindle encoder (P411=2, the spindle must be equipped with an encoder) and the method of Z-axis and spindle servo interpolation (P411=3). In this way, the spindle servo must be set to position control mode before the instruction. For example, add the instruction M881 (the M881 instruction is used for machining centers to control the spindle in position control mode).

Special note:

执行 G73/G84 指令停在中间或者攻丝过程断电时，在 MDI 编辑状态下，按“图形”键，则自动生成 M180 攻丝回退代码，按运行即可执行攻丝回退动作；

在这个攻丝循环中，每次进刀深度Q,主轴反转回退距离d（加工参数P7设置）再进刀，排屑方式加工参数P6设置（0表高速排屑回退d, 8表普通排屑回退到R），当到达孔底时主轴以反方向旋转。

指令格式：G84 X-Y-Z-R-P-Q-K(F)-L-S- ；

X-Y-:孔位数据

Z-:从 R 点到孔底的距离 (G91) 或坐标值 (G90)

R-:从初始点到 R 点的距离 (G91) 或坐标值 (G90)

Q-:每次攻丝深度

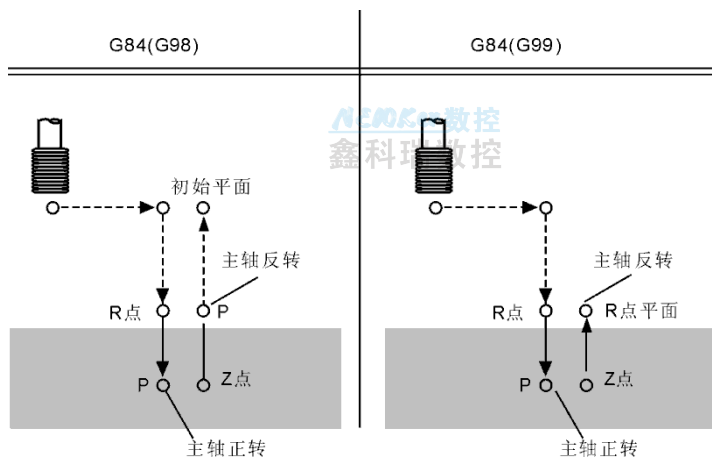
P-:暂停时间, 单位: s

K(F)-: 螺纹导程;E 英制;

S-:主轴转速

L-:重复次数

图示:



说明:

图中在加工螺纹时坐标轴速度由主轴转速和螺距确定，而与进给速度F值没有关系。系统自动将坐标轴速度限制在最大进给速度之内。

加工螺纹时，主轴倍率开关和进给轴倍率开关无效。

在每一螺纹加工程序段都应指定螺纹导程K值，否则，无法通过编译。

主轴顺时针旋转执行攻丝。当到达孔底时，为了回退，主轴以相反方向旋转。这个过程生成螺纹。在攻丝期间进给暂停不停止机床，直到返回动作完。

在指定G84 之前，用辅助功能M 代码旋转主轴。当主轴为脉冲控制方式时，不需辅助功能M 代码旋转主轴。

当在固定循环中指定刀具长度偏置 (G43, G44 或G49) 时在定位到R点的同时加偏置。刀具半径偏置被忽略。

When the G73/G84 instruction stops in the middle or the power is cut off during the tapping process, press the "graphics" key under the MDI editing state to automatically generate the M180 tapping back code, and press run to execute the tapping back action;

In this tapping cycle, each feed depth Q, the spindle reverses back and retreats a distance d (processing parameter P7 is set) before feeding again. The chip removal method is processed parameter P6 (0 table high-speed chip removal retreat d, 8 table normal chip removal to R). When reaching the bottom of the hole, the spindle rotates in the opposite direction.

Instruction format: G84 X-Y-Z-R-P-Q-K (F)-L-S-;

X-Y-: Hole position data

Z-: Distance from point R to hole bottom (G91) or coordinate value (G90)

R-: Distance from the starting point to point R (G91) or coordinate value (G90)

Q-: Depth of each thread attack

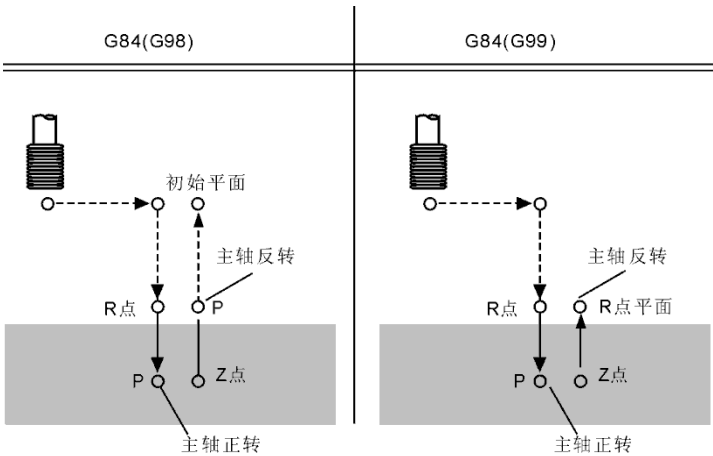
P-: Pause time, unit: s

K(F) -Thread pitch; E in imperial units;

S-: main shaft speed

L-: number of replication

graphic :



explain :

In the figure, the coordinate axis speed during thread processing is determined by spindle speed and pitch, and has nothing to do with feed speed F value. The system automatically limits the coordinate axis speed within the maximum feed speed.

During thread machining, the spindle multiplier switch and feed axis multiplier switch are invalid.

The thread pilot K value should be specified in each thread machining program segment, otherwise, it cannot be compiled.

The spindle rotates clockwise to perform tapping. When reaching the bottom of the hole, the spindle rotates in the opposite direction to retract. This process generates threads. During tapping, feed is paused without stopping the machine until the return action is complete.

Rotate the spindle using auxiliary function M code before specifying G84. When the spindle is pulse-controlled, no rotation of the spindle is required using auxiliary function M code.

When a tool length offset (G43, G44 or G49) is specified in a fixed cycle, the offset is added while positioning to point R. The tool radius offset is ignored.

举例:

```

N10 M3 S100  主轴开始旋转
N20 G90 G99 G84 X300 Y-250 Z-150 R-120 P300 K5 S100
           ; 攻第1个孔, 然后返回到R 点
N30 Y-550. K5      ; 攻第2个孔, 然后返回到R 点
N40 G98 Y-750 K5   ; 攻第3个孔,  然后返回初始位置平面
N50 G80
N60 M30           ;  结束

```

G74/G84 插补攻丝调试方法<主轴必须接编码器, Z 轴必须接位置反馈>

一, 轴类参数设置

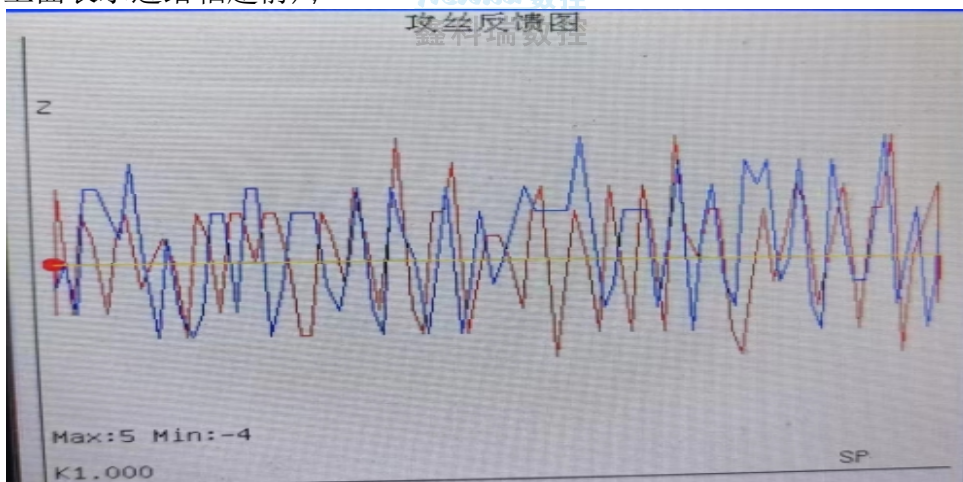
- <1> P607=1, 显示攻丝主轴与进给轴编码器反馈误差脉冲数;
- <2> P610=68, 开启反馈记录[+1 表直线;+2 表圆弧;+4 表攻丝;+64 显示图形]

二, 速度参数, 主轴与进给轴反向补偿量

- <1>P52, 攻丝时主轴反向间隙补偿(脉冲数)
- <2>P53, 攻丝主轴反转前 Z 轴提前提刀量(um) [10-5000]

三, 攻丝调试方法

- 1, 调整主轴和给进轴刚性, 尽量调整绝对误差<10(正值表进给轴超前);
- 2, 图形查看, 诊断界面按 F8 对直线/圆弧/攻丝进行图形换页显示;
- 3, 黄线为标准线, 红线为进刀线, 蓝线为提刀线, 尽量调整三线重合(在黄线上面表示进给轴超前);



四, 总线系统可以设置下面轴类参数(新方案攻丝)

- <1> P600=20200, 攻丝主轴位置控制增益[10001-30000]
- <2> P601=10800, 提刀主轴位置控制增益[10001-20000]
- <3> P602=20100, 刚性攻丝平滑处理时间常数
- <4> P603=5, 攻丝进给轴定位到位允差值(指令脉冲数) ;
- <5> P604=20, 攻丝提刀进给轴定位到位允差值(指令脉冲数);
- <6> P605, 插补攻丝进给轴最大定位误差;
- <7> P606, 插补攻丝主轴最大定位误差;

give an example :

The N10 M3 S100 spindle starts to rotate

N20 G90 G99 G84 X300 Y-250 Z-150 R-120 P300 K5 S100

; Attack the first hole and return to point R

N30 Y-550. K5; tap the second hole, then return to point R

N40 G98 Y-750 K5; tap the third hole, then return to the initial position plane

N50 G80

N60 M30 ; finish

G74/G84 Inserting and tapping debugging method <The spindle must be connected to the encoder, and the Z-axis must be connected to the position feedback> 1. Axis parameter setting

<1> P607=1, shows the number of feedback error pulses of the tapping spindle and feed axis encoder;

<2> P610=68, enable feedback recording [+1 for straight line; +2 for arc; +4 for thread attack; +64 for graphic display]

Second, speed parameters, spindle and feed axis reverse compensation

<1>P52, spindle backlash compensation (pulse number) during tapping

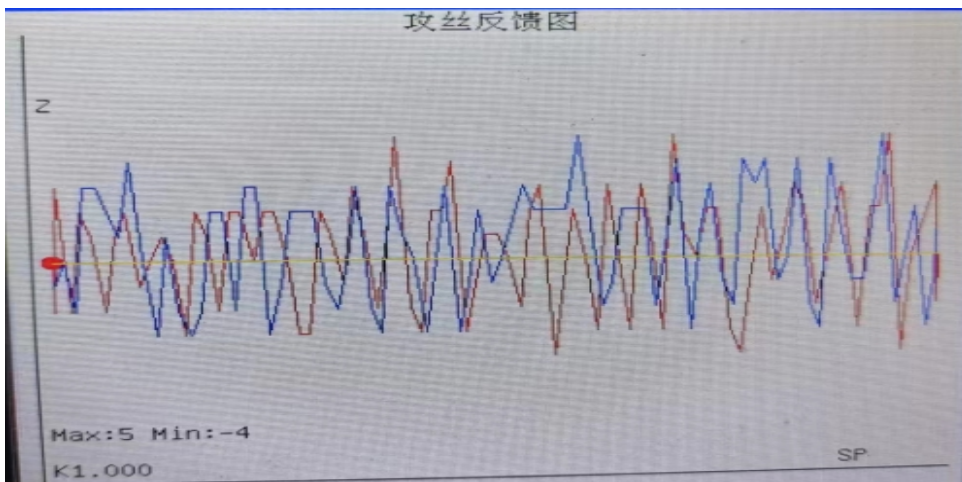
<2>P53, advance the Z-axis feed before the main spindle reversal (um) [10-5000]

Third, thread attack debugging method

1. Adjust the rigidity of the spindle and feed shaft, and try to adjust the absolute error to <10 (positive value indicates that the feed shaft is in advance);

2. For graphic viewing and diagnosis, press F8 to display the graphic page of straight line/arc/threading;

3. The yellow line is the standard line, the red line is the feed line, and the blue line is the lift line. Try to adjust the three lines to coincide (the yellow line above indicates that the feed axis is in advance);



IV. The bus system can set the following shaft parameters (new scheme tapping)

<1> P600=20200, tapping spindle position control gain [10001-30000]

<2> P601=10800, the control gain of the lifting tool spindle position [10001-20000]

<3> P602=20100, the constant of smooth treatment time for rigid thread attack

<4> P603=5, tapping feed axis positioning tolerance (instruction pulse number);

<5> P604=20, tapping feed axis positioning tolerance (instruction pulse number);

<6> P605, maximum positioning error of feed axis for plug-in tapping;

<7> P606, maximum positioning error of plug and tap spindle;

- 注意：(1) P600/P601 值越大刚性越好,效率越高,但容易超调;
 (2) 当有 XsYs 轴时,P600<20000; 没有 XsYs 轴时 P600>20000;
 (3) P603/P604 值越小精度越高,值越大效率越高;

(32) 固定循环取消 (G80)

该指令取消固定循环。

指令格式: G80 ;

说明: 取消所有的固定循环,执行正常的操作。

为了简化编程,固定循环添加如下功能。

例如: G90 G00X0Y0Z0M3;

G81 X10Y10F2;

G91X50L5;(循环5次,X方向5个孔,间隔50);

Y60L5;(Y方向5个孔,循环5次)

X10Y20L7;(斜线方向7个孔,循环7次)

G90

.....

另外: 该功能对极坐标G16增量编程极坐标角度也有效,

例如: G90 G00X0Y0Z0M3;

G16

G81 X10Y0F2; 指定初始位置: 极半径10,极角度0。

G91Y60L5;(逆时针隔60度1个孔,共5个孔,循环5次)

G90 G15

.....

(33) 极坐标指令 (G15/G16)

极坐标指令可以使坐标值用极坐标半径和角度输入,角度的正向是所选平面的第1 轴正向的逆时针转向,而负向是顺时针转向。半径可以用绝对值指令或增量值指令 (G90, G91), 角度用绝对值方式 (G90)。

指令格式:

G15 取消极坐标指令;

(G17/G18/G19) (G90/G91) G16 IP- ; 极坐标

指令

说明:

- 1、G17/G18/G19指定极坐标指令的平面选择。
- 2、G90 指定工件坐标系的零点作为极坐标系的原点从该点测量半径。
- 3、G91 指定当前位置作为极坐标系的原点从该点测量半径。
- 4、IP- 指定极坐标系选择平面的轴地址及其值:

- Note:(1) the Larger the Value of P600/P601, the Better the Rigidity and Efficiency, but It Is Easy to Overshoot;
(2) When there is an XsYs axis, P600 <20000; when there is no XsYs axis, P600> 20000;
(3) The smaller the value of P603/P604, the higher the precision; the larger the value, the higher the efficiency;
(32) Fixed-cycle cancellation (G80)

The instruction cancels the fixed loop.

Instruction format: G80;

Note: All fixed loops are removed and normal operations are performed. To simplify programming, the following functions are added to the fixed loop.

For example: G90 G00X0Y0Z0M3;

G81 X10Y10F2;
G91X50L5; (5 cycles, 5 holes in the X direction, spaced 50) and Y60L5; (5 holes in the Y direction, 5 cycles)
X10Y20L7; (7 holes in the diagonal direction, 7 cycles)

G90
.....

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In addition, this function is also effective for polar G16 incremental programming of polar angles,

For example: G90 G00X0Y0Z0M3;

G16
G81 X10Y0F2; Specify the initial position: pole radius 10, pole Angle 0.
G91Y60L5; (counterclockwise 60 degrees per hole, a total of 5 holes, cycle 5 times)
G90 G15
.....

(33) Polar Coordinate Instructions (G15/G16)

The polar coordinate command allows coordinates to be entered using the radius and angle of the polar coordinate system. The positive direction of the angle corresponds to the counterclockwise rotation of the first axis in the selected plane, while the negative direction corresponds to clockwise rotation. The radius can be specified using absolute value commands or incremental value commands (G90, G91), while angles are defined using absolute values (G90).

instruction format : G15 Take a negative coordinate instruction; (G17/G18/G19) (G90/G91) G16 IP-; Polar coordinate command
--

explain :

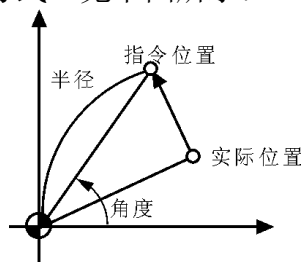
- 1、 Select the plane for the G17/G18/G19 specified polar coordinate command.
- 2、 The zero point of the G90 specified workpiece coordinate system is taken as the origin of the polar coordinate system and the radius is measured from that point.
- 3、 G91 Designates the current position as the origin of the polar coordinate system and measures the radius from that point.
- 4、 IP-Specify the axis address and value of the plane in the polar coordinate system:

第1轴：极坐标半径值

第2轴：极角度值

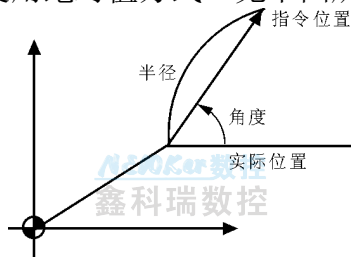
5、G90设定工件坐标系的零点作为极坐标系的原点：

用绝对值编程指令指定半径（零点和编程点之间的距离）。工件坐标系的零点设定为极坐标系的原点，当使用局部坐标系G52时，局部坐标系的原点变成极坐标的中心，角度用绝对值方式。见下图所示：

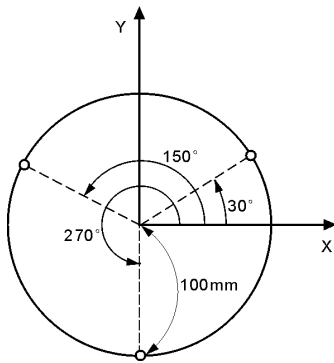


6、G91设定当前位置作为极坐标系的原点：

用增量值编程指令指定半径（当前位置和编程点之间的距离）。当前位置指定为极坐标系的原点。角度用绝对值方式。见下图所示：



举例：



G17 G90 G16 X0 Y0

；指定极坐标指令和选择XY平面设定工件坐标系的零点作为极坐标系的原点

G81 X100.0 Y30.0 Z-20.0 R-5.0 F200.0

；指定100mm 的距离和30 度的角度

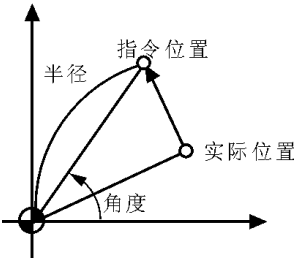
Y150.0 ；指定100mm 的距离和150 度的角度

Y270.0 ；指定100mm 的距离和270 度的角度

Axis 1: Polar radius value
Axis 2: Polar Angle value

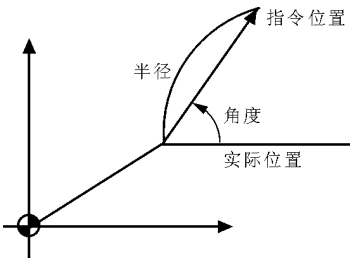
5、 The zero point of the workpiece coordinate system in G90 is set as the origin of the polar coordinate system:

Specify the radius (distance between the zero point and programming point) using absolute programming instructions. The zero point of the workpiece coordinate system is set as the origin of the polar coordinate system. When using the local coordinate system G52, the origin of the local coordinate system becomes the center of the polar coordinates, with angles expressed in absolute values. See the figure below:

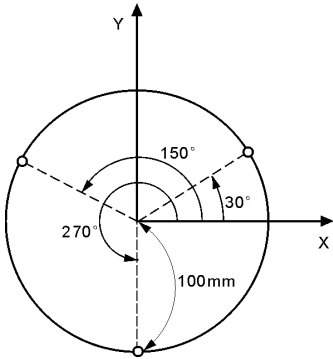


6、 The G91 sets the current position as the origin of the polar coordinate system:

Use incremental programming instructions to specify the radius (the distance between the current position and the programming point). The current position is designated as the origin of the polar coordinate system. The angle is expressed in absolute value. See the figure below:



give an example :



G17 G90 G16 X0 Y0
; Specify the polar coordinate instruction and select the XY plane to set the zero point of the workpiece coordinate system as the origin of the polar coordinate system

G81 X100.0 Y30.0 Z-20.0 R-5.0 F200.0

; Specify a distance of 100mm
and an Angle of 30 degrees Y150.0;
Specify a distance of 100mm and an Angle
of 150 degrees

Y270.0; specifies a distance of 100mm and an Angle of 270 degrees

G15 G80 ；取消极坐标指令

(34) 公英制转换指令（G20/G21）、进给方式（G94/G95）

公英制转换指令选择英制或公制输入。

指令格式：

G20 ； 英寸输入；
G21 ； 毫米输入；
G94 ； 分进给；
G95 ； 转进给；

说明：该G代码必须编在程序的开头，在设定坐标系之前，以单独程序段指定。在对英制/公制转换的G代码指定以后输入数据的单位切换到最小英制或公制输入增量单位，角度的数据输入的单位保持不变，在英制/公制转换之后改变下面值的单位制：

- 由F 代码指令的进给速度
- 位置指令
- 工件零点偏移值
- 刀具补偿值
- 手摇脉冲发生器的刻度单位
- 在增量进给中的移动距离

公英制转换指令 G20/G21 单位说明(参数中的单位必须重新设置)：

- 1) 英制G20的最小设定单位为0.0001inch,公制G21的最小设定单位为0.001mm;
- 2) 英制 G20 下所有与位置有关的指令单位为"inch";
- 3) 英制 G20 下进给速度 F 指令单位为"inch/min";
- 4) 英制 G20 下刀补和 G54-G59 工件系偏移值的单位为为"inch";
- 5) 英制 G20 下与距离(如软限位和补偿量等)有关参数的单位由公制 G21 的"mm"变成"0.1inch"或"um"变成"0.0001inch";
- 6) 英制G20下与速度有关参数的单位由公制G21的"mm/min"变成"0.1inch/min";
- 7) 英制 G20 下与加速度有关参数的单位由公制 G21 的"mm/min/s" 变成"0.1inch/min/s";
- 8) 英制 G20 下的手轮增量变成 0.0001inch,0.001inch,0.01inch;
- 9) 英制 G20 下的脉冲当量发生了变化,系统自动调整,系统电子齿轮按公制计算设置;
- 10) 英制 G20 下的坐标显示为小数点后四位数;
- 11) 在程序开头编写 G20 或 G21 指令,程序中间不应编写 G20 或 G21 指令;
- 12) 英制 G20 下的旋转轴,1 度对应 0.1inch;
- 13) 转换公英制模式时,用 MDI 执行 G20/G21 后,重启系统即可;
- 14) 系统公英制转换指令 G20/G21 执行后必须重新对刀;
- 15) 所以,只需要将参数配置文件 cncsystemcn (英文 cncsystemen) 的提示信息修改一下即可,如:有关速度的参数,将 "mm/min" 改成

G15 G80; Polar coordinate instruction is cancelled

(34) **Metric conversion instructions (G20/G21), feed mode (G94/G95)**

Select British or metric for input in the metric conversion instruction.

instruction format :

G20; Inch input;
G21; millimetre input;
G94; Feed;
G95; Feed transfer;

Note: The G code must be programmed at the beginning of the program and specified as a separate program segment before setting the coordinate system. After specifying the G code for imperial/metric conversion, switch the input data unit to either minimum imperial or metric increments. The unit of angle data input remains unchanged. After completing the imperial/metric conversion, adjust the unit system for the following values:

- The feed speed is specified by the F code instruction
- Location Directive
- Workpiece zero point offset value
- Tool compensation value
- The unit of measurement on a hand pulse generator
- The distance traveled in incremental feed

Instructions for metric conversion G20/G21 unit description (the unit in the parameter must be reset):

- 1) The minimum setting unit of the British G20 is 0.0001inch, and the minimum setting unit of the metric G21 is 0.001mm;
- 2) In the British G20, all positional instructions are in "inches";
- 3) In the British system, G20 feed speed F instruction unit is "inch/min";
- 4) In the British system, G20 tool cutting and G54-G59 workpiece offset values are in "inches";
- 5) Under the British G20, the unit of parameters related to distance (such as soft limit and compensation amount) is changed from "mm" in the metric G21 to "0.1inch", or "um" to "0.0001inch";
- 6) In the British G20, the unit of speed-related parameters changed from "mm/min" in the metric G21 to "0.1inch/min";
- 7) In the British G20, the unit of the parameter related to acceleration is changed from "mm/min/s" in the British G21 to "0.1inch/min/s";
- 8) Under the British G20, the increment of the handwheel becomes 0.0001inch, 0.001inch, 0.01inch;
- 9) The pulse equivalent under the British G20 has changed, the system automatically adjusts, and the electronic gear of the system is set according to the metric calculation;
- 10) The coordinates under the British G20 are shown in four decimal places;
- 11) Write G20 or G21 instructions at the beginning of the program, and do not write G20 or G21 instructions in the middle of the program;
- 12) The rotation axis under the British G20 is 1 degree corresponding to 0.1inch;
- 13) When switching to the metric mode, execute G20/G21 with MDI, and then restart the system;
- 14) After the system public metric conversion instruction G20/G21 is executed, the tool must be realigned;
- 15) Therefore, you only need to modify the prompt message of the parameter configuration file cncsystemcn (English cncsystemen), such as: For the speed parameter, change "mm/min" to

“0.1inch/min”；有关位置的参数，将“mm”或“度”改成“0.1inch”，将“um”改成“0.0001inch”。

(35) 返回程序起点指令 (G26/G261-G268)

用于返回程序起点。

指令格式：G26	； ZXY 均返回程序起点。
G261	； X 返回程序起点。
G262	； Y 返回程序起点。
G263	； Z 返回程序起点。
G264	； A 返回程序起点。
G265	； B 返回程序起点。
G266	； C 返回程序起点。
G267	； Xs 返回程序起点。
G268	； Ys 返回程序起点。

说明：G26 是按 G00 方式自动返回程序起点。

(36) 记忆当前点 (G25)

用于记忆当前坐标点位置。

指令格式：G25	； 记忆当前点 XYZABCXsYs 坐标
----------	-----------------------

(37) 返回程序记忆点指令 (G61/G611-G618)

用于返回程序G25记忆点。

指令格式：G61	； 返回 G25 记忆点的 XYZ 坐标。
G611	； 返回 G25 记忆点的 X 坐标。
G612	； 返回 G25 记忆点的 Y 坐标。
G613	； 返回 G25 记忆点的 Z 坐标。
G614	； 返回 G25 记忆点的 A 坐标。
G615	； 返回 G25 记忆点的 B 坐标。
G616	； 返回 G25 记忆点的 C 坐标。
G617	； 返回 G25 记忆点的 Xs 坐标。
G618	； 返回 G25 记忆点的 Ys 坐标。

说明：G61 是按 G00 方式自动返回记忆 G25 记忆的坐标点。

(38) 检测跳转指令 G31、G311

指令格式：G31 X_ Y_ Z_ A_ B_ C_ Xs_ Ys_ F_ P_

"0.1 inch/min"; for the parameters of relevant positions, change "mm" or "°" to "0.1 inch", and change "um" to "0.0001 inch".

(35) Return program starting instruction (G26/G261-G268)

Used to return to the program's starting point.

Instruction format: G26 ; ZXY both return the program starting point.
G261; X returns to the program starting point.
G262; Y returns to the starting point of the program.
G263; Z returns to the starting point of the program.
G264; A returns to the program starting point.
G265; B returns to the starting point of the program.
G266; C returns to the program starting point
G267; Xs returns to the program starting point
G268; Ys returns to the program starting point

Note: G26 is an automatic return to the starting point of the program in accordance with G00 mode.

(36) Memory Current Point (G25)

Used to remember the current coordinate point position.

Instruction format: G25; Remember the current point XYZABCXsYs coordinates
--

(37) Return program memory point

instruction (G61/G611-G618) is used to return program G25 memory point.

Instruction format: G61 Return the XYZ coordinates of the G25 memory point.
G611; returns the X-coordinate of the G25 memory point.
G612; Return to the Y-coordinate of the G25 memory point.
G613; return to the Z-coordinate of the G25 memory point.
G614; return to the A coordinate of the G25 memory point.
G615; return to the B coordinate of the G25 memory point.
G616; return to the C coordinate of the G25 memory point.
G617; return to the Xs coordinate of the G25 memory point.
G618; return to the Ys coordinate of the G25 memory point.

Note: G61 automatically returns to the memory G25 memory coordinates according to the G00 mode.

(38) Detect jump instructions G31 and G311

Instruction format: G31 X_Y_Z_A_B_C_Xs_Ys_F_P_

G311 X_ Y_ Z_ A_ B_ C_ Xs_ Ys_ F_ P_

G311 与 G31 的唯一不同在于如果没有检测到信号，G311 会报警提示，G31 不会报警提示而继续执行。

P 后的数据为：跳转行号(不输入时为下一行)加上输入点编号 X00-X39 再加上 1000 或再加上 2000，加上 1000 表示检测到该输入点有效时跳转；加上 2000 表示检测到该输入点无效时跳转。

例如：G31 X50 Z100 F100 P331022 表检测到输入点 X22 有效时就中止当前段而执行 N33 段。

G311 X50 Z100 F100 P2021 表检测到输入点 X21 无效时就中止当前段而执行下一段。若 X21 一直有效就报警。

注意：

<1>P32=28, P910=1 启动高速检测；

<2>XYZA 对应检测点必须为各轴对应的 X28/X27/X26/X25；

<3>G31/G311 检测时只能单独检测一个轴, 如:G31 X30 F100 P1010；

<4>G31/G311 指令之前必须使用 M4027 清 0；

(39) 自动倒斜角 I 和倒圆角 R 功能：

铣床默认为 G17 平面。

对于 G17 平面指令格式：

G01(G00) XI 自动倒斜角，下一段含坐标轴运行的程序段必须为 G01(G00) Y。

G01(G00) Y I 自动倒斜角，下一段含坐标轴运行的程序段必须为 G01(G00) X。

G01(G00) XR 自动倒圆角，下一段含坐标轴运行的程序段必须为 G01(G00) Y。

G01(G00) Y R 自动倒圆角，下一段含坐标轴运行的程序段必须为 G01(G00) X。

对于 G18 平面指令格式：

G01(G00) XI 自动倒斜角，下一段含坐标轴运行的程序段必须为 G01(G00) Z。

G01(G00) Z I 自动倒斜角，下一段含坐标轴运行的程序段必须为 G01(G00) X。

G01(G00) XR 自动倒圆角，下一段含坐标轴运行的程序段必须为 G01(G00) Z。

G01(G00) Z R 自动倒圆角，下一段含坐标轴运行的程序段必须为 G01(G00) X。

对于 G19 平面指令格式：

G01(G00) Y I 自动倒斜角，下一段含坐标轴运行的程序段必须为 G01(G00) Z。

G311 X_ Y_ Z_ A_ B_ C_ Xs_ Ys_ F_ P_

The only difference between G311 and G31 is that if no signal is detected, G311 will alarm, while G31 will continue to execute without alarm.

The data after P is: the line number (which defaults to the next line if not entered) plus the input point number X00-X39, then add 1000, or alternatively add 2000. Adding 1000 indicates a jump when the input point is detected as valid; adding 2000 indicates a jump when the input point is detected as invalid.

For example, G31 X50 Z100 F100 P331022, when the input point X 22 is detected as valid by the meter, the current section is terminated and the N33 section is executed.

When the G311 X50 Z100 F100 P2021 meter detects that the input point X21 is invalid, it terminates the current section and performs the next section. If X21 remains valid, an alarm is triggered.

pay attention to :

<1>P32=28, P910=1 start high speed detection;

<2>The corresponding detection points of XYZA must be X28/X27/X26/X25 corresponding to each axis;

<3>G31/G311 can only detect one axis at a time, such as: G31 X30 F 100 P1010;

<4>G31/G311 Before the instruction must be used, M4027 must be cleared 0;

(39) Automatic chamfer I and fillet R functions:

The milling bed is assumed to be G17 plane.

For G17 planar instruction format:

G01 (G00) X I automatically reverses the angle, and the next program segment containing the coordinate axis must be G01 (G00) Y.

G01 (G00) Y I automatically reverse the angle, and the next program segment containing the coordinate axis must be G01 (G00) X.

G01 (G00) X R automatically fillets corners. The next program segment containing the coordinate axis must be G01 (G00) Y.

G01 (G00) Y R automatically fillets corners. The next program segment containing the coordinate axis must be G01 (G00) X.

For the G18 plane instruction format:

G01 (G00) X I automatically reverses the angle, and the next program segment containing the coordinate axis must be G01 (G00) Z.

G01 (G00) Z I automatically reverses the angle, and the next program segment containing the coordinate axis must be G01 (G00) X.

G01 (G00) X R automatically fillets corners. The next program segment containing the coordinate axis must be G01 (G00) Z.

G01 (G00) Z R automatically fillets, and the next program segment containing the coordinate axis must be G01 (G00) X.

For the G19 plane instruction format:

G01 (G00) Y I automatically reverse the angle, and the next program segment containing the coordinate axis must be G01 (G00) Z.

G01 (G00) Z I 自动倒斜角，下一段含坐标轴运行的程序段必须为 G01 (G00) Y。

G01 (G00) Y R 自动倒圆角，下一段含坐标轴运行的程序段必须为 G01 (G00) Z。

G01 (G00) Z R 自动倒圆角，下一段含坐标轴运行的程序段必须为 G01 (G00) Y。

NCK数控
鑫科瑞数控**注意：**

- 1、地址 I 和 R 均以半径模式指定。该段直线和下一段直线运行的距离必须大于倒斜角的宽度 I 或倒圆角的半径 R，否则 系统会自动将倒斜角的宽度 I 或倒圆角的半径 R 减小到该段直线和下一段直线的最小运行距离。
- 2、相邻二条直线必须是 90° 。

举例：

```
N0 G54G0X-50Y-50Z20
N1 M03S500
N2 G01G42D01X0Y0F200
N3 G01Z-5
N4 X100I4
```

； 倒斜角 4x4

G01 (G00) Z I automatically reverses the angle, and the next program segment containing the coordinate axis must be G01 (G00) Y.

G01 (G00) Y R automatically fillets corners. The next program segment containing the coordinate axis must be G01 (G00) Z.

G01 (G00) Z R automatically fillets, and the next program segment containing the coordinate axis must be G01 (G00) Y.

NEOKer 数控
鑫科瑞数控

pay attention to :

- 1、 Both address I and R are specified in radius mode. The distance between the current straight line segment and the subsequent straight line segment must exceed either the width of the chamfer (I) or the radius of the fillet (R). If this condition is not met, the system will automatically reduce the width of the chamfer (I) or the radius of the fillet (R) to the minimum required distance between the two line segments.
- 2、 Two adjacent lines must be 90° apart.
give an example :

```
N0 G54G0X-50Y-50Z20
N1 M03S500
N2 G01G42D01X0Y0F200
N3 G01Z-5
N4 X100I4
```

; Inclined Angle 4x4

N5 Y40R6 ; 倒圆角 R6
 N6 X47R5 ; 倒圆角 R5
 N7 Y70I3 ; 倒斜角 3x3
 N8 X15
 N9 X0Y40
 N10 Y0
 N11 G0X-50Y-50G40
 N12 Z50
 N13 M30

(40) M 功能辅助指令

主 轴	M03 M04 M05 M203 M204 M205	主轴正转 主轴反转 主轴停止刹车 M7053/M7054 Pxxxx ; 主轴自 动正转/反转一段时间后自动 停止, 时间由参数 P 指定, 单 位为毫秒; 定时精度可达到 毫秒级。 例如: M7053 P2000; 表主轴正 转 2 秒钟后停止。	√	√ √ √	M203 L3/L4/L5/L6 S M204 L3/L4/L5/L6 S M205 L3/L4/L5/L6 S 第二到第六主轴正/ 反转/停止, S 指令转 速
冷 却	M08 M09	冷却开 冷却关	√	√ √	
卡 盘	M10 M11	松紧刀夹紧 松紧刀松开	√	√ √	
自定 义	M79 M78	自定义输出 7 开, 输出 M79 自定义输出 7 关, 输出 M73	√	√ √	双信号输出
润 滑	M32 M33	机床润滑开 机床润滑关	√	√ √	控制 M32 信号
吹 气	M59 M58	吹气开 吹气关	√	√ √	控制 M59 信号
自定 义 输 出	M61 M60	自定义 1 开 自定义 1 关		√ √	控制 M61 信号
	M63 M62	自定义 2 开 自定义 2 关		√ √	控制 M63 信号
	M65 M64	自定义 3 开 自定义 3 关		√ √	控制 M65 信号
	M67 M66	自定义 4 开 自定义 4 关		√ √	控制 M67 信号

N5 Y40R6 ; Reverse the fillet R6
N6 X47R5 ; Reverse the fillet R5
N7 Y70I3 ; Inclined Angle 3x3
N8 X15
N9 X0Y40
N10 Y0
N11 G0X-50Y-50G40
N12 Z50
N13 M30

(40) Function Assist Command M

principal axis	M03 M04 M05 M203 M204 M205	Main shaft forward rotation main shaft reverse main shaft stop brake M7053/M7054 Pxxxx; The spindle automatically stops after a period of automatic forward/reverse rotation, and the time is specified by parameter P, with the unit of millisecond; the timing accuracy can reach millisecond level. For example: M7053 P2000; the table spindle stops after 2 seconds of forward rotation.	√	√ √ √	M203 L3/L4/L5/L6 S M204 L3/L4/L5/L6 S M205 L3/L4/L5/L6 S indicates the forward/reverse/stop of the second to sixth spindles, and S indicates the speed
cold step back	M08 M09	Cool down Cooling guard	√	√ √	
block tray	M10 M11	Set the jaws of the screwdriver Loosen the wrench	√	√ √	
user-defined	M79 M78	Open custom output 7 and output M79 Custom output 7, output M73	√	√ √	Dual signal output
lubricating	M32 M33	Open the machine tool lubrication Machine tool lubrication	√	√ √	Control the M32 signal
scavenging	M59 M58	Turn on the air blowers Blow off the lid	√	√ √	Control the M59 signal
user-defined output	M61 M60	Customize 1 open Customization 1 concerns		√ √	Control the M61 signal
	M63 M62	Customize 2 open Customization 2		√ √	Control the M63 signal
	M65 M64	Customize 3 open Customization 3		√ √	Control the M65 signal
	M67 M66	Customize 4 open Customization 4		√ √	Control the M67 signal

	M69 M68	自定义 5 开 自定义 5 关		√ √	控制 M69 信号
	M71 M70	自定义 6 开 自定义 6 关		√ √	控制 M71 信号
	M75 M74	自定义 8 开 自定义 8 关		√ √	控制 M75 信号
自定义输入	M12 M13	检测 M12 有效 检测 M12 无效		√ √	条件成立转移, 如 M12 P120 即: 当 M12 信号有效时, 程序跳 转到 120 行执行
	M14 M15	检测 M14 有效 检测 M14 无效		√ √	
	M16 M17	检测 M16 有效 检测 M16 无效		√ √	
	M18 M19	检测 M18 有效 检测 M18 无效		√ √	
	M28 M29	检测 M28 有效 检测 M28 无效		√ √	
	M22 M23	检测 M22 有效 检测 M22 无效		√ √	
主轴 换档	M41 M42 M43 M44	第一档 第二档 第三档 第四档			M41/M44 输出 S01/S02 同时调整主 轴模拟电压
子 程 序	M97 M98 M99	程序跳转 子程序调用 子程序调用返回			L=1-99 P 为转移程序行号
程 序 控 制	M87 M188 M189 M00 M01 M02 M30 M20	工件数加 1 M188 进给保持手动手轮干预 坐标后不返回起点 M189 进给保持回起点 程序暂停 程序准停, 输入 X46 有效暂 停程序结束 M05、M09 程序结束 程序结束, 根据参数设置运 行次数自动重复运行程序, 用于考机			杂类参数 P10=0 设为 工件数不自动加 1 时, 指令 M87 工件数加 1

	M69 M68	Customize 5 open Customize 5		√ √	Control the M69 signal
	M71 M70	Customize 6 open Customization 6		√ √	Control the M71 signal
	M75 M74	Customize 8 open Customize 8		√ √	Control the M75 signal
Self-defined inputs	M12 M13	The M12 is valid for testing The M12 is invalid		√ √	Conditions for tr- ansfer, such as M12 P120: when the M 12 signal is valid, the program jumps to 120 and execu- tes the line
	M14 M15	The M14 was tested and found to be eff- ective The M14 is invalid		√ √	
	M16 M17	The M16 is valid for testing The M16 is invalid		√ √	
	M18 M19	The M18 is valid for testing The M18 is invalid		√ √	
	M28 M29	The M28 was tested and found to be eff- ective The M28 is invalid		√ √	
	M22 M23	The M22 was tested and found to be eff- ective The M22 is invalid		√ √	
Change the main axis	M41 M42 M43 M44	first gear second gear third gear Episode IV			M41/M44 outputs S01/S02 and adju- sts the spindle si- mulation voltage at the same time
subprogram	M97 M98 M99	Program jump subprogram call Subroutine call ret- urned			L=1-99 P is the transfer program line nu- mber
programmed control	M87 M188 M189 M00 M01 M02 M30 M20	Add 1 to the number of jobs M188 feed hold ha- ndwheel intervention coordinates without returning to the star- ting point M189 feed is maint- ained at the starting point program halt The program is sto- pped. Enter X46 to effectively pause the program and end it M05, M09 program end At the end of the program, the prog- ram will be automat- ically repeated acco- rding to the number of times set by par- ameters for computer testing			When the miscell- aneous parameter P10=0 is set to not automatically add 1 to the number of workpieces, the in- struction M87 adds 1 to the number of workpieces

主 轴 速 度	S SS SC/SD SG/SH	设定第一主轴速度 设定第二主轴转速 第三/四主轴转速 第五/六四主轴转速		√ √	S=0-99999 SS=0-99999
读 绝 对 式 电 机 位 置	M500 M501 M502 M503 M504 M505 M506 M507 M508	M500: 读取全部进给轴绝对电机位置并重置当前机床坐标。 M501-M508: 读取 XYZABCXsYs 轴绝对电机位置并重置当前机床坐标。			
清 工 件 坐 标	M313 M314 M315 M316 M317 M318 M319 M320	M313: 清零 Xs 轴工件坐标 M314: 清零 Ys 轴工件坐标 M315: 清零 A 轴工件坐标 M316: 清零 B 轴工件坐标 M317: 清零 X 轴工件坐标 M318: 清零 Y 轴工件坐标 M319: 清零 Z 轴工件坐标 M320: 清零所有轴工件坐标			
清 机 床 坐 标	M413 M414 M415 M416 M417 M418 M419 M420	M413: 清零 Xs 轴机床坐标 M414: 清零 Ys 轴机床坐标 M415: 清零 A 轴机床坐标 M416: 清零 B 轴机床坐标 M417: 清零 X 轴机床坐标 M418: 清零 Y 轴机床坐标 M419: 清零 Z 轴机床坐标 M420: 清零所有轴机床坐标			
总线 驱动 指令	M133 M135	M133: 驱动器以指定速度转动 (必须是 30 的倍数); 驱动参数 P30=0 恒转速, P30=1 往复摆动, 见轴类参数。 M135: 使能指定驱动器			例子: M133 X300; M135 Z0; 关闭 Z 轴驱动 M135 X1; 使能 X 轴驱动
总线 驱动 指令	M186	第 5 轴焊接跟随功能模式开启 第 5 轴焊接跟随功能模式关闭			为选配功能

Axis speed	s S S SC/ SD SG/ SH	Set the first spindle speed Set the second spindle speed and third/fourth spindle speed Fifth/sixth main shaft speed		√ √	S=0-99999 SS=0-99999
Read the absolute motor position	M500 M501 M502 M503 M504 M505 M506 M507 M508	M500: Read the absolute motor position of all feed axes and reset the current machine tool coordinates. M501-M508: Read the absolute motor position of the XYZABCXsYs axis and reset the current machine tool coordinates.			
Set the work coordinates	M313 M314 M315 M316 M317 M318 M319 M320	M313: Zero the Xs axis workpiece coordinates M314: Zero the Ys axis workpiece coordinates M315: Zero the A axis workpiece coordinates M316: Zero the B axis workpiece coordinates M317: Zero the X axis workpiece coordinates M318: Zero the Y axis workpiece coordinates M319: Zero the Z axis workpiece coordinates M320: Zero the coordinates of all shaft workpieces			
Clear the machine seat coordinates	M413 M414 M415 M416 M417 M418 M419 M420	M413: Zero the Xs axis machine tool coordinates M414: Zero the Ys axis machine tool coordinates M415: Zero the A axis machine tool coordinates M416: Zero the B axis machine tool coordinates M417: Zero the X axis machine tool coordinates M418: Zero the Y axis machine tool coordinates M419: Zero Z axis machine tool coordinates M420: Zero all axis machine tool coordinates			
Bus drive instruct	M133 M135	M133: The driver rotates at a specified speed (must be a multiple of 30); drive parameter P30=0 for constant rotation, P30=1 for reciprocating swing, see shaft parameters. M135: Enable the specified driver			example : M133 X300; M135 Z0; Turn off Z-axis drive M135 X1; Enable X-axis drive
highway drive instruct	M186	The fifth axis welding follow-up function mode is enabled The fifth axis welding follow-up function mode is turned off			Optional features

倍率 指令	M100 M101	倍率无效，固定为 100%； 倍率有效；			
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所有 M 指令输出，均为低电平有效。

所有 M 指令输入，均为低电平有效。

(41) 调用程序 M97、M98、M99

无条件跳转

M97 P 无条件跳转到P所指定的行号处运行，P4表示用4字段的数字指定程序转移主程序的入口行号（标记行号）。

子程序调用

在本系统中子程序应为一个独立的程序。

M98 P L 无条件调用子程序指令。P指定程序调用的子程序路径及名字，L指子程序调用次数地址。

M98指令可以省略不写，格式为：PP文件名，文件名可以是隐藏文件，隐藏文件首字符必须为“HIDEFILE”开头。如文件“HIDEFILE01”，此程序在程序区是不显示的，调用时可以用指令M98/G65 PHIDEFILE01或M98/G65 P*01或PP*01或PPHIDEFILE01。

例如：Psub/1390表子程序为tmp/NC/sub/1390

注：1、tmp/NC/为系统程序默认路径，sub为其下面的一个文件夹

2、子程序必须是一个单独的程序。

3、优盘中主程序调用优盘中子程序的方法：P[或P]。

例如：M98 P[A1234 表示调用优盘中的子程序A1234；

M98 P]SS12 表示调用优盘中的子程序SS12；

PP[FFDE 表示调用优盘中的子程序FFDE；

如果调用优盘中文件夹下的子程序，需要编上文件的路径。

子程序调用次数地址字符L前必须有空格。子程序运行结束，返回到主程序的下一个程序段。如果程序包含固定的顺序或多次重复的图形的话，这样的顺序或图形可以编成子程序在存储器中贮存以简化编程，子程序可以由主程序调用，被调用的子程序也可以调用另一个子程序。

M99子程序结束返回指令，子程序结束必须有此指令。

注意：

1) M99在主程序中同M02指令；

2) M99在主程序带P时同M97指令；

3) M99在子程序中返回主程序调用处下行；

4) M99在子程序中带P时返回主程序P程序行；

子程序可以调用其它子程序(嵌套)。不管主调程序的是主程序或子程序，子程序返回都是返回到主调程序行的下一个程序行。当主程序调用子程序时，它被认为是一级子程序。调用可以嵌套4级调用关系见图表。

multiplying power instruct	M 100	The multiplier is invalid and fixed at 100%; The multiplier is eff- ective;			
	M 101				

All M instruction outputs are valid at low level.

All M instruction inputs are valid at low level.

(41) Call the Programs M97, M98 and M99

unconditional jump

M97 P unconditionally jumps to the line number specified by P; P4 indicates the entry line number (marked line number) of the main program from which the program transfers using the number in the 4 field.

subprogram call

In this system, the subroutine should be a separate program.

M98 P L is an unconditional subroutine call instruction. P specifies the path and name of the subroutine called, and L specifies the address of the number of times the subroutine is called.

The M98 instruction can be omitted. Its format is: PP filename, where the filename may be a hidden file. Hidden files must start with "HIDEFILE". For example, if the file is named "HIDEFILE01", it will not appear in the program area. When calling this function, you can use the following instructions: M98/G65 PHIDEFILE01, M98/G65 P*01, PP*01, or PPHIDEFILE01.

For example, the subroutine Psub/1390 is tmp/NC/sub/1390

Note: 1. tmp/NC/ is the default path of the system program, and sub is a folder under it

2. A subroutine must be a separate program.

3 The method for the main program in a USB drive to call subroutines within it: P[or P]. For example: M98 P[A1234] indicates calling subroutine A1234 in the USB drive; M98 P]SS12 indicates calling subroutine SS12 in the USB drive; PP[FFDE indicates calling subroutine FFDE in the USB drive;

If you call a subroutine in a folder on the USB drive, you need to write the path of the file.

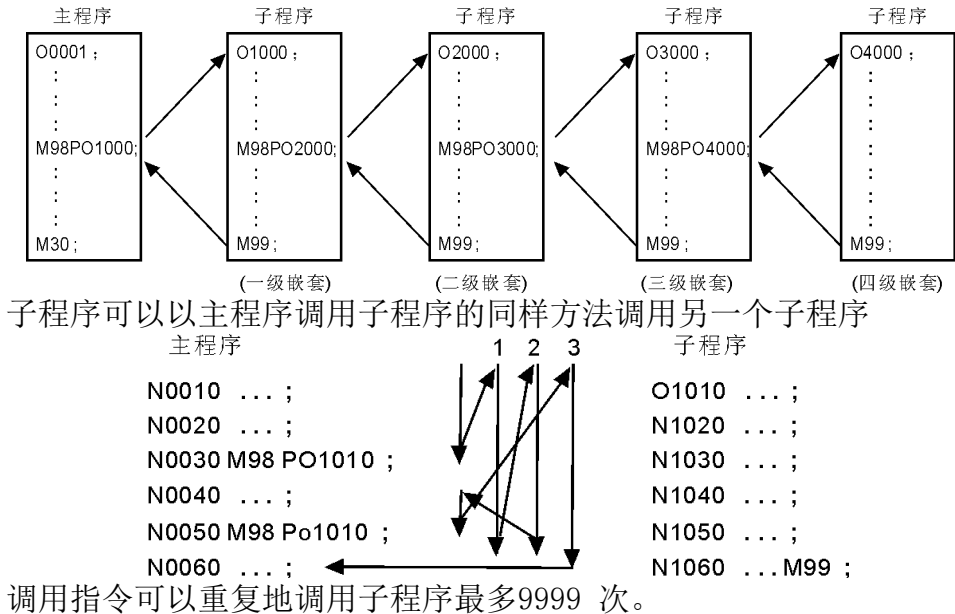
Subroutine calls must be preceded by a space character L. Upon completion of a subroutine execution, the program returns to the next segment of the main program. When a program contains fixed sequences or repetitive graphics, these elements can be stored as subroutines in memory to simplify programming. Subroutines can be called by the main program, and the called subroutine may also call another subroutine.

The M99 subroutine end return instruction must be present for the subroutine to end.

pay attention to :

- 1) M99 in the main program with M02 instruction;
- 2) M99 is the same as M97 instruction when P is in the main program;
- 3) M99 returns to the main program call point in the subroutine;
- 4) M99 returns to the main program line P when it carries P in the subroutine;

Subroutines can call other subroutines (nested). Whether the calling program is a main program or a subroutine, the return of a subroutine always goes to the next line of the program where the main program was called. When a main program calls a subroutine, it is considered a first-level subroutine. The call can be nested up to four levels. See the diagram for the call relationship.



条件等待、跳转指令

本系统中指令 M 代码有用于检测外部输入信号做为条件的情况，具体如下：

条件等待

M12 M13 指令用于检测输入信号 M12 的状态,程序行中 M12 表检测 M12 信号输入有效后执行下一行程序，M13 表检测 M12 信号输入无效后执行下一行程序。该指令独立一行。

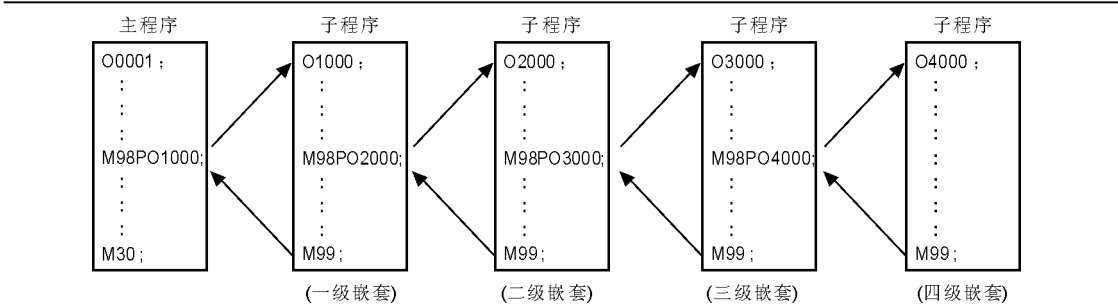
M14 M15 指令用于检测输入信号 M14 的状态，程序行中 M14 表检测 M14 信号输入有效后执行下一行程序，M15 表检测 M14 信号输入无效后执行下一行程序。该指令独立一行。

M16 M17 指令用于检测输入信号 M16 的状态，程序行中 M16 表检测 M16 信号输入有效后执行下一行程序，M17 表检测 M16 信号输入无效后执行下一行程序。该指令独立一行。

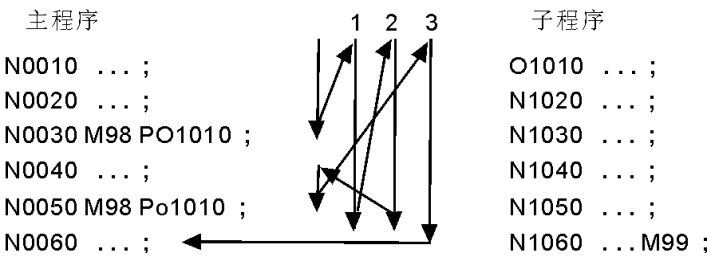
M18 M19 指令用于检测输入信号 M18 的状态，程序行中 M18 表检测 M18 信号输入有效后执行下一行程序，M19 表检测 M18 信号输入无效后执行下一行程序。该指令独立一行。

M22 M23 指令用于检测输入信号 M22 的状态，程序行中 M22 表检测 M22 信号输入有效后执行下一行程序，M23 表检测 M22 信号输入无效后执行下一行程序。该指令独立一行。

M28 M29 指令用于检测输入信号 M28 的状态，程序行中 M28 表检测 M28 信号输入有效后执行下一行程序，M29 表检测 M28 信号输入无效后执行下一行程序。该指令独立一行。



A subroutine can call another subroutine in the same way that a subroutine is called by a main program



The call instruction can call the subroutine up to 9,999 times.

Condition wait, jump instruction

In this system, the instruction M code is used to detect external input signals as conditions, as follows:

Conditions await

The M12 and M13 instructions are used to detect the status of the input signal M12. In the program line, the M12 table detects the validity of the M12 signal input and executes the next line of the program after it is valid, while the M13 table detects the invalidity of the M12 signal input and executes the next line of the program after it is invalid. This instruction is independent of a line.

The M14 and M15 instructions are used to detect the status of the input signal M14. In the program line, the M14 table detects the validity of the M14 signal input and executes the next line of the program. The M15 table detects the invalidity of the M14 signal input and executes the next line of the program. This instruction is independent of the line.

The M16 and M17 instructions are used to detect the status of the input signal M16. In the program line, the M16 table detects the validity of the M16 signal input and executes the next line of the program. The M17 table detects the invalidity of the M16 signal input and executes the next line of the program. This instruction is independent of a line.

The M18 and M19 instructions are used to detect the status of the input signal M18. In the program line, the M18 table detects the validity of the M18 signal input and executes the next line of the program. The M19 table detects the invalidity of the M18 signal input and executes the next line of the program. This instruction is independent of a line.

The M22 and M23 instructions are used to detect the status of the input signal M22. In the program line, the M22 table detects the validity of the M22 signal input and executes the next line of the program. The M23 table detects the invalidity of the M22 signal input and executes the next line of the program. This instruction is independent of a line.

The M28 and M29 instructions are used to detect the status of the input signal M28. In the program line, the M28 table detects the validity of the M28 signal input and executes the next line of the program. The M29 table detects the invalidity of the M28 signal input and executes the next line of the program. This instruction is independent of a line.

条件跳转

M12/M13/M14/M15/M16/M17/M22/M23/M28/M29 指令之后加上 Pxxxx (程序行号) 条件成立转移, 条件不成立则向下执行。

举例: 指令 M14 P0120

即当程序运行到该行时, 系统检测到 M14 输入信号有效时, 程序跳转到该程序第 120 行 (标记行) 执行, 若 M14 输入信号为无效则执行该程序下一行指令。

M186/M187 焊接跟随指令: 为总线系统选配功能 (杂类参数 P903 设置);

- (1) 第四轴为 A 轴旋转轴, 参数设置为全规算, 用于实现刀具摆动。
- (2) 第五轴为 C 轴旋转轴, 参数设置为 C 轴且为全规算, 用于实现刀具转动;

注意: 第五轴旋转的正方向设置应与 XYZ 坐标系符合右手坐标系原则。

(3) 程序段与段之间为了实现第五轴的旋转角度插入的一段第五轴运动, 按第五轴 G00 速度运行, 并按就近方向运动。

(4) 指令:

M186: 开启焊接跟随功能 (执行该指令后系统的界面显示 M186)

M187: 关闭焊接跟随功能 (执行该指令后系统的界面不显示 M186)

(5) 实例:

```
G54 G0X10Y60Z30
M3S200
M186           ;进入跟随模式
G1 Z50 F500   ;Z 轴提升
G1X10Y63F200  ;XY 轴刀尖对准
G1 Z0         ;Z 轴到焊点
X0Y-23
X10Y50
G2X-23.20Y130R90
M187           ;取消跟随模式
M5
M30
```

(42) F 进给速度功能指令

本系统切削的进给速度用 F 字段。它是模态, 实际运行速度为设置的速度乘以速度修调倍率。

F 用于指定进给指令 G01、G02、G03 的加工速度。

速度范围 F 为 0.01-15000mm/min, 进给速度为 Fx 速度修调, F 具有模态功能。

F 指令和移动指令在同一段时先执行 F 指令, 再执行移动指令。

(43) T/H/D 功能指令

Conditional jump

M12/M13/M14/M15/M16/M17/M22/M23/M28/M29 After the instruction, Pxxxx (program line number) is added. If the condition is true, transfer; if not, execute downward.

Example: Instruction M14 P0120

That is, when the program runs to this line, the system detects that the M14 input signal is valid, and the program jumps to the 120th line (marked line) of the program to execute. If the M14 input signal is invalid, the program executes the instruction of the next line.

M186/M187 welding follow instruction: optional function for the bus system (miscellaneous parameter P903 setting);

- (1) The fourth axis is the A-axis rotation axis, and the parameter is set to full specification calculation, which is used to realize tool swing.
- (2) The fifth axis is the rotation axis of the C-axis, and the parameter is set as the C-axis and full specification calculation, which is used to realize the rotation of the tool. Note: The positive direction setting of the rotation of the fifth axis should conform to the right-hand coordinate system principle in the XYZ axis coordinate system.
- (3) The program segment inserted between the segments to achieve the rotation Angle of the fifth axis is run at the G00 speed of the fifth axis and moves in the nearest direction.

(4) instruct :

M186: Enable welding follow function (the system interface displays M186 after executing this instruction)

M187: Disable the welding follow function (the system interface does not display M186 after executing this instruction)

(5) living example :

```
G54 G0X10Y60Z30
M3S200M186      ; Enter follow
                  mode
G1 Z50 F500      ; Z-axis lift
G1X10Y63F200     ; XY axis tool
                  tip aligned
G1 Z0            ; Z axis to the
                  weld point
X0Y-23
X10Y50
G2X-23. 20Y130R90
M187            ; Cancel the follow mode
M5
M30
```

(42) F Feed Speed Function Command

The feed rate for the system's cutting is given by the F field. It is a mode, and the actual running speed is the set speed multiplied by the speed adjustment multiplier.

F is used to specify the feed speed of G01, G02, and G03.

The speed range is F 0.01-15000mm/min, and the feed speed is Fx. The speed is adjusted, and F has the mode function.

F The F instruction is executed first and the move instruction is executed later in the same paragraph.

(43) T/H/D function instruction

本系统的 T/H/D 功能, 是对刀具半径或刀具长度进行补偿的参数调用功能, 为模态, 在程序中用编号调用。

其中 T01 到 T99 为刀具编号, 每把刀具有四个刀沿, H1 到 H99 为刀具长度补偿参数编号, D1 到 D99 为刀具半径补偿参数编号。

(44) S、SS、SC/SD/SG/SH 主轴速度功能指令

本系统提供二路主轴控制方式。

第一主轴转速用 S 指定, 第一主轴有二种档位控制方式:

(1) 第一种是四档主轴速度电气控制, 输出有级变速四位编码, M41-M44 指令控制, 对应输出 S01-S04 编码, 有级变速。用轴类参数 P48-P51 参数设置换档方式。

(2) 第二种是用四档+无级变速, M41-M44 指令控制, 对应输出 S01-S04 编码。用速度参数 P48-P51 设定其相应档位最高转速, 用轴类参数 P59-P63 参数设置换档方式。

无级变速, 其取值范围为 0-99999, 输出 0-10V 变频电压。输出电压为指定的 S_x 主轴修调 $\times 10V$ /最高速度。

第二主轴转速用 SS 指定, 最高速度由速度参数 P52 控制, 输出 0-10V 变频电压。

第三主轴转速用 SC 指定, 最高速度由速度参数 P52-1 控制, 输出 0-10V 变频电压, 与第一主轴共用一个模拟输出口, 第一主轴为全脉冲输出时此功能有效。

第四主轴转速用 SD 指定, 最高速度由速度参数 P52-2 控制, 输出 0-10V 变频电压, 与第二主轴共用一个模拟输出口, 第二主轴为全脉冲输出时此功能有效。

SC/SD/SG/SH 指定第三/四/五/六主轴的转速;

(45) 宏程序指令 (G65、G66、G67)

1、输入指令: WAT

等待输入口 X 有效或无效指令, 等待 X, Y, M 有效或无效指令。

格式: WAT+(-)X(Y/M)**+(-)X**+(-)X**+(-)X**+(-)X**+(-)X** P- D-

说明: “+” 表等待有效;

“-” 表等待无效;

Y 或 M 必须最多只能有一个, X 可以有任意多个。

“X” 表示输入口 X00-X55; 见系统 I/O 诊断;

P-: 等待时间 (ms), P0 表示一直等待;

D0: 报警, 按 ESC 退出运行, 按 Enter 继续等待;

D1: 报警并退出运行;

D2: 报警并继续运行;

D3: 不报警并退出运行;

D4: 不报警并继续运行;

The T/H/D function of this system is a parameter call function for compensation of tool radius or tool length. It is a mode and can be called by number in the program.

T01 to T99 are tool numbers, each tool has four tool edges, H1 to H99 are tool length compensation parameter numbers, D1 to D99 are tool radius compensation parameter numbers.

(44) S, SS, SC/SD/SG/SH spindle speed function commands

This system provides two spindle control modes.

The first spindle speed is designated by S. The first spindle has two types of gear control:

(1) The first type is the four-speed spindle speed electrical control, which outputs four-level variable speed four-bit encoding and M41-M44 command control. Corresponding output S01-S04 encoding is provided for stepless speed variation. The shifting mode is set by using spindle parameters P48-P51.

(2) The second type uses four gears + stepless transmission, which is controlled by M41-M44 instructions and corresponds to output S01-S04 codes. The maximum speed of the corresponding gear is set by speed parameters P48-P51, and the shifting mode is set by shaft parameters P59-P63.

Stepless speed regulation, whose value range is 0-99999, outputs 0-10V variable frequency voltage. The output voltage is specified as $S \times \text{spindle speed} \times 10V / \text{maximum speed}$.

The second spindle speed is designated by SS, and the maximum speed is controlled by the speed parameter P52, with an output of 0-10 V variable frequency voltage.

The third spindle speed is specified by SC, and the maximum speed is controlled by the speed parameter P52-1. The output frequency is 0-10V, and it shares an analog output port with the first spindle. This function is effective when the first spindle is fully pulse output.

The fourth spindle speed is specified by SD, and the maximum speed is controlled by the speed parameter P52-2. The output frequency voltage is 0-10V, and it shares an analog output port with the second spindle. This function is effective when the second spindle is fully pulse output.

SC/SD/SG/SH specifies the rotational speed of the third/four/fifth/sixth spindle;

(45) Macro Program Instructions (G65, G66, G67)

1 Input Command: WAT

Wait for the X input to be valid or invalid, wait for the X, Y, M to be valid or invalid.

Format: WAT+ (-) X (Y/M)** + (-) X** + (-) X** + (-) X** + (-) X** + (-) X**
P-D-

Note: "+" indicates that the wait is valid;

"-" The table is waiting for invalid;

There must be at most one Y or M, and X can have any number.

"X" indicates input ports X00-X55; see system I/O diagnosis;

P-: Waiting time (ms), P0 means waiting forever;

D0: Alarm, press ESC to exit operation, press Enter to continue waiting;

D1: Alarm and exit operation;

D2: Alarm and continue to run;

D3: Do not alarm and exit operation;

D4: Do not alarm and continue to run;

2、输出指令：OUT

设定输出口Y有效或无效指令

格式：OUT+(-)Y

说明：“+”表示输出有效；

“-”表示输出无效；

“Y”表示输出口Y00-Y31；见系统I/O诊断；

3、变量与赋值：=

宏变量根据变量号可以分成如下几种类型：

1) #0—#20 局部变量：局部变量只能用在宏程序中存储数据，例如运算结果，当断电时，局部变量被初始化为空；调用宏程序时自变量对局部变量赋值。

2) #21—#999 全局变量：全局变量在不同的宏程序中的意义相同。当断电时，变量#21—#100初始化为零；

变量#101—#999的数据保存即使断电也不丢失；

宏变量 #900—#999 可以在 PLC 中调用作 CMP 比较运算。

3) #1000— 系统变量：系统变量用于读取CNC运行时各种数据的变化。例如刀具的当前位置和补偿值。

特别注意：宏变量 #100—#204, #380—#390 已被系统功能占用，用户不要使用。

用户可使用的全局变量：#31—#99, #205—#379和#391—#999，其他全局变量已被系统占用或用作系统备用功能。

宏变量 #161—#165 用于记录和显示G31/G311指令跳转时的XYZAB机床坐标值。

#201：螺纹复合循环G176重复执行的次数(方式1)。

#202：螺纹复合循环G176重复执行时每次刀具沿Z轴的偏移距离(单位：mm)(方式1)。

#203：螺纹复合循环G176重复执行的次数(方式2)。

#204：螺纹复合循环G176重复执行时每次刀具沿Z轴的偏移距离(单位：mm)(方式2)。

4)，宏变量#1001—#1099对应车床T1—T99的X轴偏移值(单位为微米)。

宏变量#1401—#1499对应车床T1—T99的Z轴偏移值(单位为微米)。

可以读取其值，例如：#200=#1003；表将三号刀的X轴偏移值读入到宏变量#200中。

可以修改其值，例如：#1003=23000；表将三号刀的X轴偏移值改成23000微米。

#1003=#1003+50；表将三号刀的X轴偏移值增加50微米。

5) 输入输出变量：

#1800：X00-X07 (D0-D7)

#1801：X08-X15 (D0-D7)

#1802：X16-X23 (D0-D7)

#1802：X16-X23 (D0-D7)

#1803：X24-X31 (D0-D7)

#1804：X32-X39 (D0-D7)

2 Output Instruction: OUT

Set the instruction format
for output port Y to be
valid or invalid: OUT+ (-)
Y

Note: "+" indicates that the output is valid;

"-" indicates invalid output;

"Y" indicates the output ports Y00-Y31; see system I/O diagnosis;

3 Variable and Assignment: =

Macro variables can be divided into the following types according to variable number:

1) #0--#20 Local variables: Local variables can only be used in macro programs to store data, such as operation results. When the power is cut off, local variables are initialized to empty; when calling macro programs, self variables assign values to local variables.

2) #21--#999 Global variables: Global variables have the same meaning in different macro programs. When the power is cut off, variable #21--#100 is initialized to zero;

Variable #101-- #999 data is saved without loss even if the power is cut off;

Macro variables #900-- #999 can be invoked in the PLC as CMP comparison operations. 鑫科瑞数控

3) #1000- -System variables: System variables are used to read changes in various data during CNC operation, such as the current position of a tool and compensation values.

Special note: Macro variables #100-#204, #380-#390 have been occupied by system functions. Do not use them.

Global variables available to users: #31-- #99, #205-- #379 and #391-- #999. Other global variables have been occupied by the system or used as system backup functions.

Macro variables #161-- #165 are used to record and display the XYZAB machine tool coordinates when the G31/G311 instruction is jumped.

#201: The number of times the thread composite cycle G176 is repeated (mode 1).

#202: The offset distance of the tool along the Z-axis (unit: mm) when the thread composite cycle G176 is repeated (Mode 1).

#203: The number of times the thread composite cycle G176 is repeated (mode 2).

#204: The offset distance of the tool along the Z-axis (unit: mm) when the thread composite cycle G176 is repeated (Mode 2).

4) Macro variables #1001--#1099 correspond to the X-axis offsets (in microns) of lathes T1--T99. Macro variables #1401--#1499 correspond to the Z-axis offsets (in microns) of lathes T1--T99.

You can read its value, for example: #200=#1003; the table reads the X-axis offset of knife 3 into the macro variable #200.

You can modify its value, for example: #1003=23000; the table changes the X-axis offset value of No.3 knife to 23000 microns.

#1003= #1003 + 50; the table increases the X-axis offset value of No.3 knife by 50 microns.

5) Input/Output variables:

#1800: X00-X07 (D0-D7)

#1801: X08-X15 (D0-D7)

#1802: X16-X23 (D0-D7)

#1802: X16-X23 (D0-D7)

#1803: X24-X31 (D0-D7)

#1804: X32-X39 (D0-D7)

#1805: X40-X47 (D0-D7)

#1806: X60-X67 (D0-D7)

#1808: Y00-Y15 (D0-D15)

#1809: Y16-Y31 (D0-D15)

格式: #i=表达式;

6) 按用户程序格式保存当前全部全局宏变量

FILEMS (AABBCC) 或 FILEMS[AABBCC]

保存#21-#999到程序AABBCC文件;

要读取这些变量时, 只需要执行一下该文件, 或者将该文件作为子程序调用即可。

4、算术和逻辑运算

算术和逻辑运算表:

功能	格式	备注
定义	#i = #j	
加法	#i = #j + #k ;	
减法	#i = #j - #k ;	
乘法	#i = #j * #k ;	
除法	#i = #j / #k ;	
正弦	#i = SIN(#j) ;	
反正弦	#i = ASIN(#j) ;	
余弦	#i = COS(#j) ;	角度以度指定. 90度30分表示为90.5度。
反余弦	#i = ACOS(#j) ;	
正切	#i = TAN(#j) ;	
反正切	#i = ATAN(#j) ;	
平方根	#i = SQRT(#j) ;	
绝对值	#i = ABS(#j) ;	
舍入	#i = ROUND(#j) ;	
向下取整	#i = FIX(#j) ;	
向上取整	#i = FUP(#j) ;	
自然对数	#i = LN(#j) ;	
指数函数	#i = EXP(#j) ;	
或	#i = #j OR #k ;	逻辑运算一位一地按二进制数执行。
异或	#i = #j XOR #k ;	
与	#i = #j AND #k ;	

5、无条件转移: GOTO n

转移到标有顺序号n的程序段当指定1到99999以外的顺序号时出现报错, 可用表达式指定顺序号。例: GOTO 5, GOTO #100

6、条件转移: 1) IF (条件表达式) GOTO 或 THEN

#1805: X40-X47 (D0-D7)

#1806: X60-X67 (D0-D7)

#1808: Y00-Y15 (D0-D15)

#1809: Y16-Y31 (D0-D15)

Format: #i=expression;

6) Save all global macro variables in the current user program format
FILEMS (AABBCC) or FILEMS[AABBCC]

Save #21-#999 to program AABBCC file;

To read these variables, simply execute the file or call it as a subroutine.

Arithmetic and Logical Operations

Arithmetic and logical operation tables:

function	form	remarks
definition	#i = #j	
addition	#i = #j + #k ;	
subtraction	#i = #j - #k ;	
multiplication	#i = #j * #k ;	
division	#i = #j / #k ;	
Sine, arsine, cosine, inverse cosine, arctan- gent, arctangent	#i = SIN(#j) ; #i = ASIN(#j) ; #i = COS(#j) ; #i = ACOS(#j) ; #i = TAN(#j) ; #i = ATAN(#j) ;	The Angle is specif- ied in degrees. 90.30 is expressed as 90.5 degrees.
Round off the absolute value of square roots Round down round up to an integer natural logarithm exponential function	#i = SQRT(#j) ; #i = ABS(#j) ; #i = ROUND(#j) ; #i = FIX(#j) ; #i = FUP(#j) ; #i = LN(#j) ; #i = EXP(#j) ;	
Or yes or no	#i = #j OR #k ; #i = #j XOR #k ; #i = #j AND #k ;	Logical operations are performed bit by bit in binary numb- ers.

5 Unconditional Transfer: GOTO N

When the sequence number other than 1 to 99999 is specified, an error occurs when transferring to the program segment marked with sequence number n. You can specify the sequence number with an expression. Example: GOTO 5, GOTO #100

6 Condition transfer: 1) IF (condition expression) GOTO or THEN

2) IF (条件表达式)

<操作>

ELSE

<操作>

ENDIF

3) IF (条件表达式)

<操作>

ELIF

<操作>

ELIF

<操作>

ENDIF

如果指定的条件表达式满足时, 执行本条语句; 如果指定的条件表达式不满足, 执行下个程序段。

例: IF (#100 EQ 2) THEN #100=5

IF (#101 GT 2) GOTO 6

运算符含义:

EQ 等于=

NE 不等于

GT 大于>

GE 大于或等于>=

LT 小于<

LE 小于或等于<=

NCNOR数控
鑫科瑞数控

7、循环语句: WHILE (条件表达式) DO 1、2、3

在WHILE 后指定一个条件表达式。当指定条件满足时, 执行从DO 到END之间的程序。否则, 转到END后的程序段。循环嵌套最多3层。

例: WHILE (#100 LT 3) DO 1

.....

WHILE (#103 EQ 5) DO 2

.....

WHILE (#200 GE 20) DO 3

.....

END 3

.....

END 2

.....

END 1

8、非模态调用宏指令: G65

格式: G65 P- L- <A-B-C-..... 自变量传递数据>

2) IF (condition expression)

< operate >

ELSE

< operate >

ENDIF

3) IF (condition expression)

< operate >

ELIF

< operate >

ELIF

< operate >

ENDIF

If the specified conditional expression is satisfied, execute this statement; if the specified conditional expression is not satisfied, execute the next program segment.

Example: IF (#100 EQ 2) THEN #100=5

IF (#101 GT 2) GOTO 6

Operation meaning:

EQ equals

NE

does not

equal

GT

greater

than>

GE greater

than or equal

to or less than

LT less than

LE less than or equal to <=

N20Kc 数控

鑫科瑞数控

7、 The WHILE Statement: WHILE (condition Expression) DO 1, 2, 3

After WHILE, specify a conditional expression. When the specified condition is met, execute the program between DO and END. Otherwise, transfer to the program segment after END. The loop can be nested up to 3 levels.

Example: WHILE (#100 LT 3) DO 1

.....

WHILE (#103 EQ 5) DO 2

.....

WHILE (#200 GE 20) DO 3

.....

END 3

.....

END 2

.....

END 1

8. Non-Mode Calling Macro Instructions: G65

Format: G65 P-L- <A-B-C-..... Independent variable passes data>

P为宏程序名，L调用次数，A、B、C为自变量，自变量名对应如下：

#0→A、#1→B、#2→C、#3→D、#4→E、#5→F、#6→H、#7→I、#8→J、#9→K、 #10→M、
#11→Q、#12→R、#13→S、#14→T、#15→U、#16→V、#17→W、#18→X、
#19→Y、#20→Z。

特别注意：地址G、L、N、Q、P不能在自变量中使用。

例如：

主程序：9000

G00 X0 Z0

G65 P8000 L1 A5 B6

G0 X0 Z0

M30

宏程序：8000

N1 #2=#0+#1

N2 IF (#2 EQ 10) GOTO 4

N3 G00 X#2

N4 G00 Z#1

N5 M99 ; 返回

9、模态调用宏指令：G66、G67

G66为模态调用宏指令，G67取消模态。使用格式同G65。

例如：

主程序：9000

G00 X0 Z0

G66 P8000 L2 A5 B6

A8 B1

A9 B10

G67

M30

宏程序：8000

N1 #2=#0+#1

N2 IF (#2 EQ 10) GOTO 4

N3 G00 X#2

N4 G00 Z#1

N5 M99 ; 返回

10、信息提示对话框

格式：MSG(参数) 或 MSG[参数]；参数为信息字符串，暂停。

注意：该指令在一般 NC 程序(非宏程序)中也可以使用。

显示信息提示对话框后，系统自动切换到暂停状态。

格式：STAF(参数) 或 STAF[参数]；参数为信息字符串，不暂停。

11、自动生产加工程序

1>, 建立打开文件指令：

P is the macro program name, L is the number of calls, and A, B, C are the independent variables. The names of the independent variables correspond to the following:

#0→A、#1→B、#2→C、#3→D、#4→E、#5→F、#6→H、#7→I、#8→J、#9→K、 #10→M、
#11→Q、#12→R、#13→S、#14→T、#15→U、#16→V、#17→W、#18→X、#19→
Y、#20→Z。

Special note: Addresses G, L, N, Q, and P cannot be used in the independent variable.

for instance :

Main program: 9000

G00 X0 Z0
G65 P8000 L1 A5 B6
G0 X0 Z0
M30

Macro program: 8000

N1 #2=#0+#1
N2 IF (#2 EQ 10) GOTO 4
N3 G00 X#2
N4 G00 Z#1
N5 M99 ; return

9. Modal Call Macro Instructions: G66 and G67

G66 is a mode call macro instruction and G67 cancels the mode. The format is the same as G65.

for instance :

Main program: 9000

G00 X0 Z0
G66 P8000 L2 A5 B6
A8 B1
A9 B10
G67
M30

Macro program: 8000

N1 #2=#0+#1
N2 IF (#2 EQ 10) GOTO 4
N3 G00 X#2
N4 G00 Z#1
N5 M99 ; return

10. Information Prompt Dialog Box

Format: MSG (parameter) or MSG[parameter]; the parameter is an information string and is suspended.

Note: This instruction can also be used in general NC programs (non-macro programs).

After the message prompt dialog box is displayed, the system automatically switches to the pause state.

Format: STAF (parameter) or STAF[parameter]; The parameter is an information string and does not pause.

11. Automatic Production and Processing Program

1> Establish the file open command:

FILEON(参数) 或 FILEON[参数]

例如:

FILEON(AABBCC) 或 FILEON[AABBCC]

表示建立打开一个文件 AABBCC

2>, 关闭文件指令:

FILECE 表关闭当前打开的文件, 若没有该指令, 程序运行完毕后会自动关闭当前打开的文件。

3>, 向打开的文件中写入一串字符的指令:

FILEWD (参数) 或 FILEWD [参数]

例如:

FILEWD (G54G0X0Z0) 或 FILEWD [G54G0X0Z0]

表示打开的文件中写入一串字符 G54G0X0Z0

4>, 向打开的文件中写入当前进给轴的绝对坐标指令:

FILEWC

示例程序:

G0X0Z0

FILEON[AABBCC]

FILEWD [G54G0X0Z0]

G1X45Z89

FILEWC

G1X99Z76

FILEWC

FILECE

执行完该程序后则会在程序目录下生成一个文件 AABBCC, 其中的内容为:

G54G0X0Z0

X45Z89

X99Z76

(46) 自定义宏程序指令 (G101—G172, M880—M979/M6880—M6929)

1、G101—G170 是 G 宏程序, 对应宏程序 ProgramGxxx;

2、M880—M979 是 M 宏程序, 对应宏程序 ProgramUser0—ProgramUser49;

3、M6880—M6929 是 M 并行宏程序, 对应 M880—M979 功能, 与主程序并行执行, 但不能执行坐标移动指令, 只能执行 I/O 控制;

<1>G162 功能: 主轴模拟量控制准停到指令角度, 格式: G162 C- S- D- 主轴按 S 速度正转到 C 角度并执行 M63 锁紧主轴, D 越小精度越高。

<2>G163 功能: 输出点循环检测到位信号如果无信号就报错;

格式: G163 Q- M- K- X- D- R-

Q: 循环检测次数, 如 Q5;

FILEON (parameter) or FILEON[parameter]

for instance :

FILEON (AABBCC) or FILEON[AABBCC]

Indicates the opening of a file AABBCC

2>, Close file command:

FILECE closes the currently open file. If this instruction is not present, the program will automatically close the currently open file after the program is finished.

3. Write a string of characters to an open file:

FILEWD (parameter) or FILEWD [parameter]

for instance :

FILEWD (G54G0X0Z0) or FILEWD [G54G0X0Z0]

Write a string of characters G54G0X0Z0 into the open file

4> writes the absolute coordinate command of the current feed axis to the open file:

FILEWC

Example program:

G0X0Z0

FILEON[AABBCC]

FILEWD [G54G0X0Z0]

G1X45Z89

FILEWC

G1X99Z76

FILEWC

FILECE

After executing the program, a file AABBCC will be generated in the program directory, which contains:

G54G0X0Z0

X45Z89

X99Z76

(46) Custom macro program instructions (G101-G172, M880-M979/M6880-M6929)

1、 G101-G170 is a G macro program, corresponding to the macro program ProgramGxxx;

2、 M880-M979 is an M macro program, corresponding to the macro program ProgramUser0-ProgramUser49;

3、 M6880-M6929 is an M parallel macro program, corresponding to the function of M880-M979. It can be executed in parallel with the main program, but it cannot execute coordinate move instructions, and can only execute I/O control;

<1>G162 Function: spindle analog control stop to the command Angle, format: G162 C-S-D-spindle forward at S speed to C Angle and execute M63 to lock the spindle, the smaller the D, the higher the precision.

<2>G163 Function: Output point loop detection status signal is available. If no signal is available, error is reported;

Format: G163 Q-M-K-X-D-R

Q: Number of cycle tests, such as Q5;

M: 输出信号有效指令, 如 M61, 输出 Y00;

K: 输出信号无效指令, 如 K60, 关闭 Y00;

X: 检测输入点; 如 X01;

D: 输出信号有效后延时时间, 单位为秒;

R: 输出信号无效后延时时间, 单位为秒;

<3>G172 功能: 在非宏程序中调用宏程序, 非模态指令;

例如, 定义指令 G152 功能: 圆弧模式多孔钻孔加工循环。(必须将宏程序 ProgramG152 导入系统)。

格式: G152 Xx Yy Zz Rr Ii Aa Bb Hh Ff;

X:圆心的 X 坐标绝对值或增量值指定

Y:圆心的 Y 坐标绝对值或增量值指定

Z:孔深

R:快速趋近点坐标

F:切削进给速度

I:圆半径

A:第一孔的角度

B:增量角指定(负值时为顺时针)

H:孔数

宏程序 ProgramG152 的内容如下:

#80=#0

#81=#1

#82=#2

#83=#3

#84=#4

#85=#5

#86=#6

#87=#7

#88=#8

#89=#9

#90=#10

#91=#11

#92=#12

#93=#13

#94=#14

#95=#15

#96=#16

#97=#17

#98=#18

#99=#19

#100=#20

NCK数控
鑫科瑞数控

M: Output valid signal instruction, such as M61, output Y00;
K: Output invalid signal instruction, such as K60, turn off Y00;
X: Detection input point; e.g. X01;
D: Output signal effective delay time, unit is seconds;
R: The delay time after the output signal is invalid, in seconds;

<3>G172 Function: Call a macro program in a non-macro program, non-modal instruction;

For example, define the function of G152: Circular arc mode multi-hole drilling machining cycle. (Macro program ProgramG152 must be imported into the system).

Format: G152 Xx Yy Zz Rr li Aa Bb Hh Ff;

X: The absolute value or incremental value of the X coordinate of the center is specified

Y: The absolute value or incremental value of the Y coordinate of the center is specified

Z: hole depth

R: Coordinates of the point of rapid approach

F: Cutting feed speed

I: Radius of circle

A: Angle of the first hole

B: Incremental Angle designation (clockwise when negative)

H: hole count

The contents of the macro program ProgramG152 are as follows:

```
#80=#0
#81=#1
#82=#2
#83=#3
#84=#4
#85=#5
#86=#6
#87=#7
#88=#8
#89=#9
#90=#10
#91=#11
#92=#12
#93=#13
#94=#14
#95=#15
#96=#16
#97=#17
#98=#18
#99=#19
#100=#20
```

```
#30=#4003
#31=#4014
G90
IF[#30 EQ 90] GOTO 1
G53
#98=#5001+#98
#99=#5002+#99
N1 WHILE[#86 GT 0] DO 1
#35=#98+#87*COS[#80]
#36=#99+#87*SIN[#80]
G81X#35Y#36Z#100R#92F#85
#80=#80+#81
#86=#86-1
END 1
G#30 G#31 G80
M99
```

(47) 圆内凹槽粗铣G110/G111

指令格式: G99 G110/G111 X_ Y_ R_ Z_ I_ W_ Q_ K_ V_ E_ D_ F_

指令说明: 绝对式闭环系统才有此功能, G110-G142一样。

指令功能: 从圆心开始, 以螺旋线方式进行多次圆弧插补, 直至加工出编程尺寸的圆内凹槽。

指令说明: 相关指令请参照表3-2 固定循环的指令说明。G110: 逆时针圆内凹槽粗铣; G111: 顺时针圆内凹槽粗铣;

1. I: 圆内槽半径, I 应大于当前刀具的半径;
2. W: Z 轴方向首次切深, 是从R 基准面向下的距离, 应大于0 (若首次切深超过槽底位置, 则直接以槽底位置加工);
3. Q: Z 轴方向每次的切深增量;
4. K: 在XY 面内切削的宽度增量, 应小于刀具直径, 大于0;
5. V: 快速下刀时, 离末加工面的距离, 大于0;
6. E: 圆内凹槽粗铣循环加工余量。(大于等于0, 负数取绝对值);
7. D: 刀具半径序号, 取值范围为0 ~ 32, D0默认为0。根据给定的序号取出当前刀具半径值;

循环过程: (1) 快速定位到XY 平面的位置;

8. (2) 快速下至R 点平面;

9. (3) 以切削速度向下切削W 距离深度;

10. (4) 中心向外每次按K 值递增螺旋铣完半径为I 的圆面;

11. (5) Z 轴快速返回R 基准面;

```
#30=#4003
#31=#4014
G90
IF[#30 EQ 90] GOTO 1
G53
#98=#5001+#98
#99=#5002+#99 N1 WHILE
[#86 GT 0] DO 1
#35=#98+#87*COS[#80]
#36=#99+#87*SIN[#80] G81
X#35Y#36Z#100R#92F#85
#80=#80+#81
#86=#86-1
END 1
G#30 G#31 G80
M99
```

(47) Grind the Round Recess G110/G111

Instruction format: G99 G110/G111 X_Y_R_Z_I_W_Q_K_V_E_D_F_

Instruction: This function is only available in absolute closed-loop systems, and the same is true for G110-G142.

Instruction function: from the center of the circle, multiple arc interpolation is carried out in a spiral way until the circular groove with programmed size is processed.

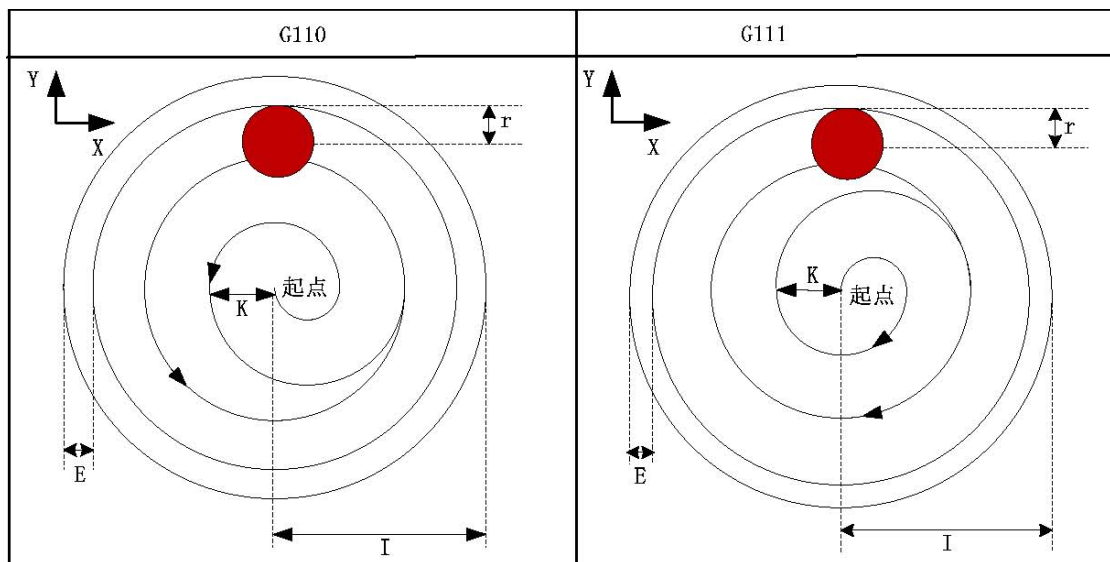
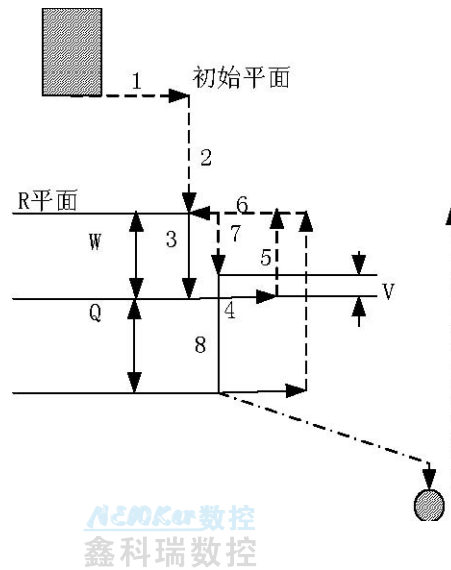
Instruction description: Refer to Table 3-2 for the instruction description of fixed cycle. G110: rough milling of counterclockwise circular groove; G111: rough milling of clockwise circular groove;

1. I: The radius of the circular groove, I should be greater than the radius of the current tool;
2. W: The first depth of cut in the Z direction is the distance from the R reference plane downward, which should be greater than 0 (if the first depth of cut exceeds the bottom position of the groove, the groove bottom position should be processed directly);
3. Q: The depth increment of each cut in the Z direction;
4. K: The width increment of cutting in XY plane should be less than the diameter of the tool and greater than 0;
5. V: When the blade is cut quickly, the distance from the last processing surface is greater than 0;
6. E: The machining allowance of the rough milling cycle in the circular groove. (Greater than or equal to 0, the absolute value of a negative number;
7. D: Tool radius serial number, the value range is 0 ~ 32, D0 is default 0. According to the given serial number, the current tool radius value is obtained;

Cycling process: (1) Quick positioning to the position in XY plane;

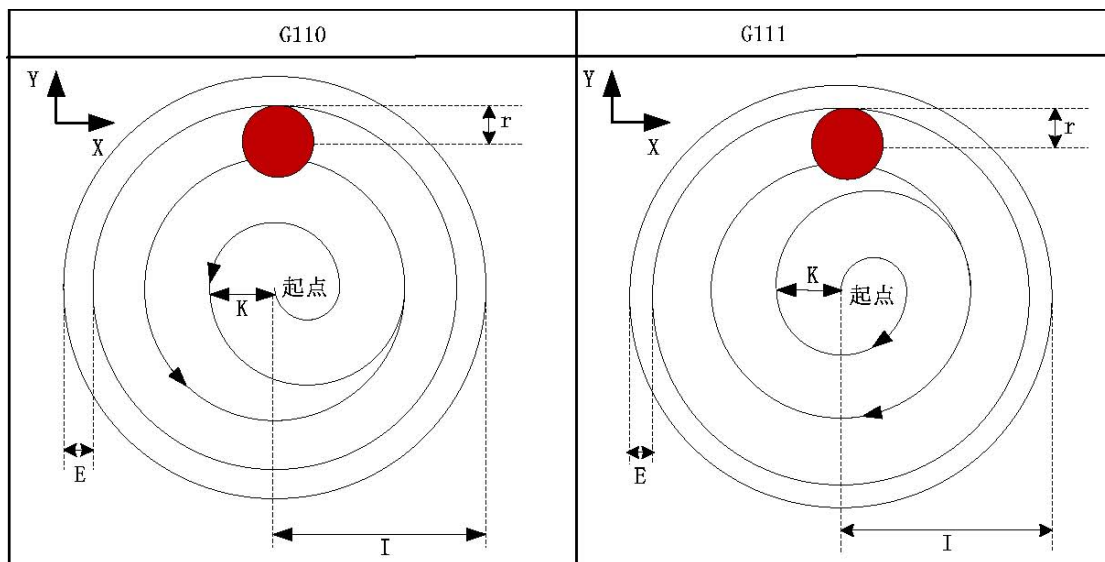
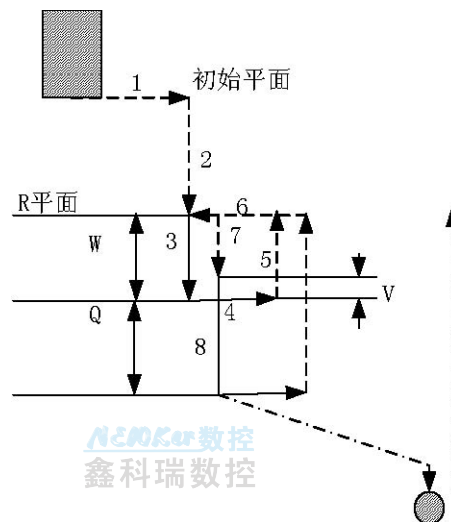
8. (2) Quickly descend to the plane of point R;
9. (3) The depth of downward cutting of W is cut at the cutting speed;
10. (4) The center is outward and each time the K value is increased, the circular surface with radius I is milled by helical milling;
11. (5) The Z-axis quickly returns to the R reference plane;

12. (6) X, Y 轴快速定位到圆心;
 13. (7) Z 轴快速下降至离末加工面V 的距离;
 14. (8) Z 轴向下切削 (Q+V) 的深度;
 15. (9) 循环 (4) ~ (8) 的动作, 直至加工完总切深的圆面;
 16. (10) 固定返回R点。
- (11) 返回XY 孔定位位置。指令轨迹:

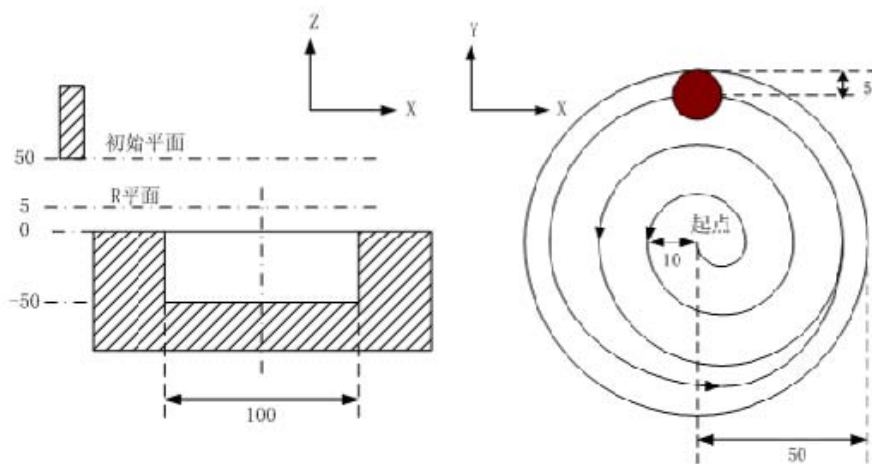


例：用固定循环G110 指令粗铣一个圆内凹槽，如下图所示：

12. (6) The X and Y axes are quickly positioned to the center;
13. (7) The Z-axis drops rapidly to the distance from the final processing surface V;
14. (8) The depth of downward cutting ($Q+V$) on the Z-axis;
15. (9) The action of cycle (4) ~ (8) is performed until the circular surface with the total cutting depth is processed;
16. (10) Fixed return to point R.
- (11) Return to XY hole positioning position. Instruction trajectory:



Example: Use the fixed cycle G110 command to rough mill a circular groove, as shown below:



G90 G00 X50 Y50 Z50; (G00 快速定位)

G99 G110 X25 Y25 R5 Z-50 I50 W20 Q10 K10 E0 V10 F800 D1; (进行圆内凹槽粗铣循环 D1=5)

G80 X50 Y50 Z50; (取消固定循环, 从R 点平面返回)

M30;

(48) 全圆内精铣循环 G112/G113

指令格式: G99 G112/G113 X_ Y_ R_ Z_ I_ J_ D_ F_

指令功能: 刀具以指定的半径值I 及方向在圆内部精铣一个整圆, 精铣完成后返回。

指令说明: 相关指令请参照表3-2 固定循环的指令说明。G112: 逆时针方向全圆内精铣循环。G113: 顺时针方向全圆内精铣循环。

I: 精铣圆半径, 取值范围-99999999~99999999×最小指令单位, 当为负值时取其绝对值。

J: 精铣起点与精铣圆圆心的距离, 取值范围 $J \leq I$ -刀具半径值, 当为负值时取其绝对值。

D: 刀具序号, 取值范围为1 ~ 32, D0 默认为0。根据给定的序号取出当前刀具半径。循环过程: (1) 快速定位到XY 平面的位置;

(2) 快速下至R 点平面;

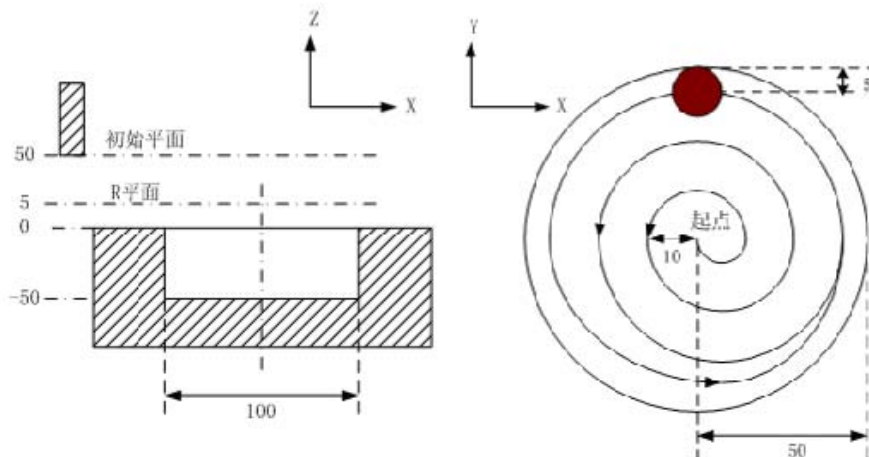
(3) 切削进给至孔底;

(4) 以过渡弧1 为轨迹进行圆弧插补;

(5) 以弧2, 弧3 为轨迹进行整圆插补;

(6) 以过渡弧4 为轨迹进行圆弧插补回到起点;

(7) 固定返回R点; 指令轨迹:



G90 G00 X50 Y50 Z50; (G00 Quick Positioning)

G99 G110 X25 Y25 R5 Z-50 I50 W20 Q10 K10 E0 V10 F800 D1; (perform rough milling cycle for circular internal groove D1=5)

G80 X50 Y50 Z50; (Remove fixed cycle and return from the R-point plane)

M30;

(48) Full Circle Internal Precision Milling Cycle G112/G113

Instruction format: G99 G112/G113 X_Y_R_Z_I_J_D_F_

Instruction function: The tool precision milling a whole circle inside the circle with the specified radius value I and direction, and returns after precision milling.

Instruction Description: Refer to Table 3-2 for instructions of fixed cycles. G112: Full circle internal precision milling cycle in counterclockwise direction. G113: Full circle internal precision milling cycle in clockwise direction.

I: Precision milling radius, the value range is -99999999~99999999× minimum instruction unit, when it is negative, take its absolute value.

J: The distance between the starting point of precision milling and the center of precision milling circle. The value range is $J \leq I$ - tool radius value. When it is negative, its absolute value should be taken.

D: Tool serial number, the value range is 1 ~ 32, D0 is default 0. According to the given serial number, the current tool radius is obtained. The cycle process: (1) Quick positioning to the XY plane position;

(2) Quickly descend to the plane of point R;

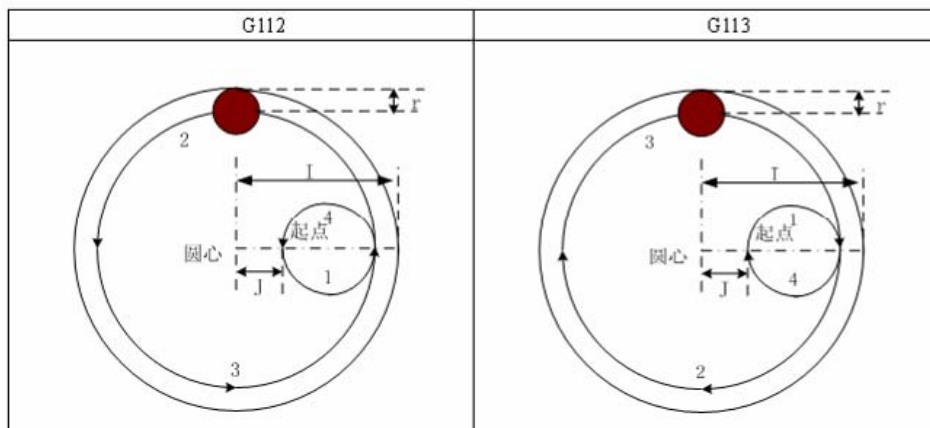
(3) Cutting feed to the bottom of the hole;

(4) Arc interpolation is performed with transition arc 1 as the trajectory;

(5) Use arc 2 and arc 3 as the trajectory for full circle interpolation;

(6) Use transition arc 4 as the trajectory to interpolate back to the starting point;

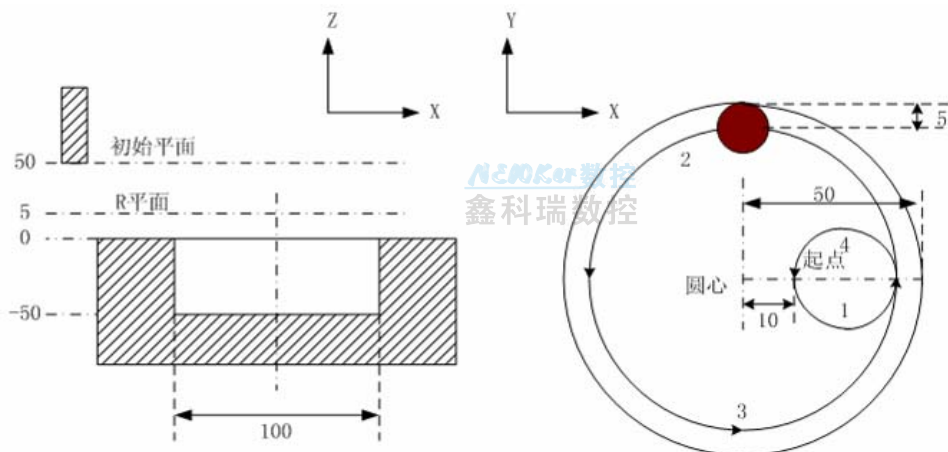
(7) Fixed return to point R; Instruction trajectory:



相关说明：

该循环下指令Q、P、L无效，但会保留Q、P的值作为固定循环模态数值保存。

例：用固定循环G112 指令，精铣如下图所示的已粗铣完的圆凹槽。



G90 G00 X50 Y50 Z50; (G00 快速定位)

G99 G112 X25 Y25 R5 Z-50 I50 J10 F800 D1; (开始固定循环，下到孔底进行圆内精铣循环D1=5)

G80 X50 Y50 Z50; (取消固定循环，从R 点平面返回)

M30;

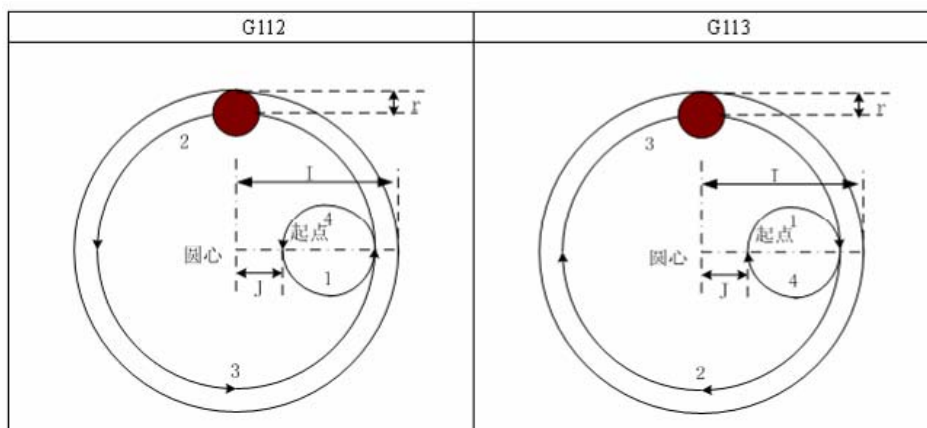
(49) 外圆精铣循环G114/G115

指令格式： G99 G114/G115 X_ Y_ R_ Z_ I_ J_ D_ F_

指令功能：刀具以指定的半径值及方向在圆外精铣一个整圆，精铣完成后返回。

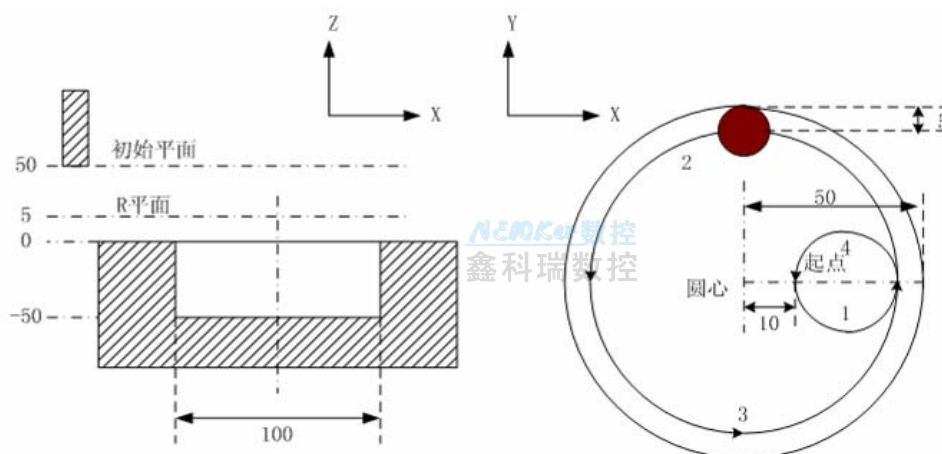
指令说明：相关指令请参照表3-2 固定循环的指令说明。G114：逆时针外圆精铣循环。G115：顺时针外圆精铣循环。

I：精铣圆半径，取值范围-99999999~99999999×最小指令单位，当为负值时取



Related notes:

In this cycle, the instructions Q, P, and L are invalid, but the values of Q and P are retained as fixed cycle mode values. Example: Use the fixed cycle G112 instruction to finish milling the rough-milled circular groove as shown in the figure below.



G90 G00 X50 Y50 Z50; (G00 Quick Positioning)

G99 G112 X25 Y25 R5 Z-50 I50 J10 F800 D1; (Start the fixed cycle, go down to the bottom of the hole for the inner circular precision milling cycle D1=5)

G80 X50 Y50 Z50; (Remove fixed cycle and return from the R-point plane)

M30;

(49) External Cylindrical Milling Cycle G114/G115

Instruction format: G99 G114/G115 X_Y_R_Z_I_J_D_F_

Instruction function: The tool will finish milling a whole circle outside the circle with the specified radius value and direction, and return after finishing milling.

Instruction Description: Refer to Table 3-2 for instructions of fixed cycles. G114: Counter clockwise external circle precision milling cycle. G115: clockwise external circle precision milling cycle.

I: Precision milling radius, the value range is -99999999~99999999× minimum instruction unit, when the value is negative, it is taken

其绝对值。

J: 精铣起点与精铣圆的距离, 取值范围 $-99999999 \sim 99999999 \times$ 最小指令单位, 且 $J \geq D$ (刀具半径), 当为负值时取其绝对值。

D: 刀具半径序号, 取值范围为 $0 \sim 32$, D0 默认为0。根据给定的序号取出当前刀具半径值。循环过程: (1) 快速定位到XY 平面的位置;

(2) 快速下至R 点平面;

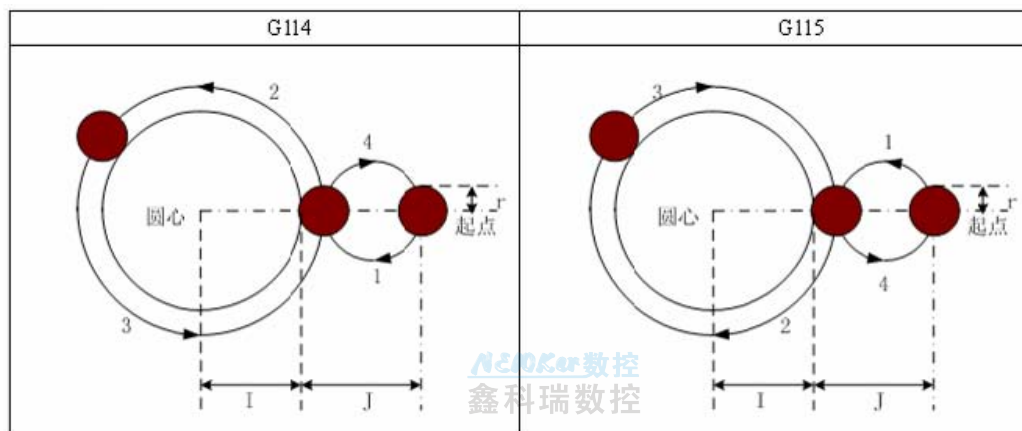
(3) 切削进给至孔底;

(4) 以过渡弧1 为轨迹进行圆弧插补;

(5) 以弧2, 弧3 为轨迹进行整圆插补;

(6) 以过渡弧4 为轨迹进行圆弧插补回到起点;

(7) 固定返回R点; 指令轨迹:



相关说明:

(1) 外圆精铣时, 过渡弧与精铣弧的插补方向不一致, 指令说明中的插补方向指精铣弧的插补方向。

(2) 该循环下指令Q、P、L无效, 但会保留Q、P的值作为固定循环模态数值保存。

例: 用固定循环G114 指令, 精铣如下图所示的已粗铣完的圆凸台。

Its absolute value.

J: The distance between the starting point of precision milling and the precision milling circle, the value range is $-99999999 \sim 99999999 \times$ minimum instruction unit, and $J \geq D$ (tool radius), when it is negative, take its absolute value.

D: Tool radius sequence number, the value range is $0 \sim 32$, D0 is default 0. The current tool radius value is obtained according to the given sequence number. The cycle process: (1) Quick positioning to the XY plane position;

(2) Quickly descend to the plane of point R;

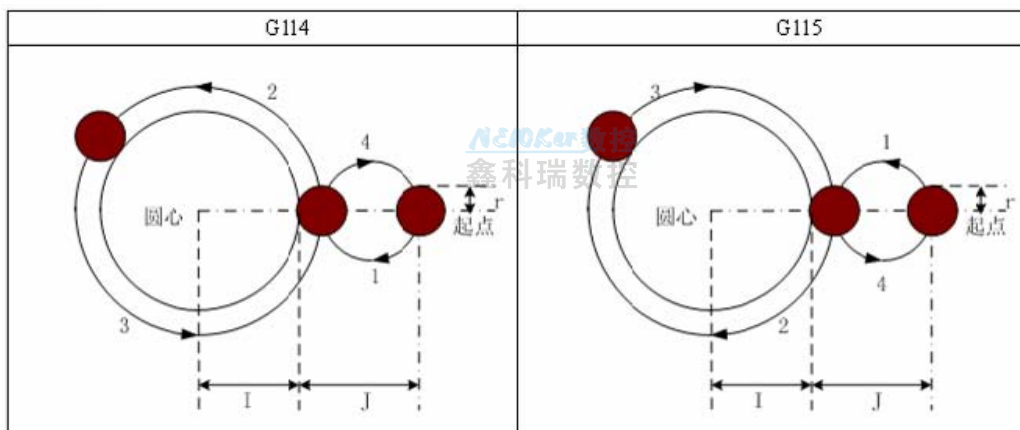
(3) Cutting feed to the bottom of the hole;

(4) Arc interpolation is performed with transition arc 1 as the trajectory;

(5) Use arc 2 and arc 3 as the trajectory for full circle interpolation;

(6) Use transition arc 4 as the trajectory to interpolate back to the starting point;

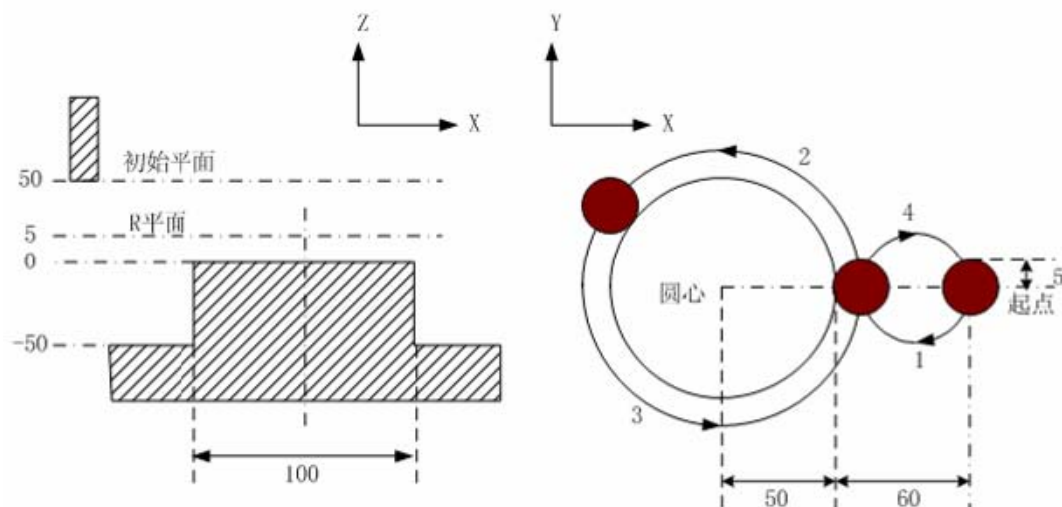
(7) Fixed return to point R; Instruction trajectory:



Related notes:

(1) During external circle precision milling, the direction of interpolation between transition arc and precision arc is not consistent. The direction of interpolation in the instruction specification refers to the direction of interpolation of precision arc.

(2) In this cycle, the instructions Q, P, and L are invalid, but the values of Q and P are retained as fixed cycle mode values. Example: Use the fixed cycle G114 instruction to finish milling the rough-milled circular boss as shown in the figure below.



G90 G00 X50 Y50 Z50; (G00 快速定位)

G99 G114 X25 Y25 R5 Z-50 I50 J60 F800 D1; (开始固定循环, 下到孔底进行外圆精铣循环 D1=5)

G80 X50 Y50 Z50; (取消固定循环, 从R 点平面返回) M30;

(50) 外圆粗铣循环G116/G117

指令格式: G99 G116/G117 X_ Y_ Z_ R_ I_ J_ W_ Q_ K_ C_ E_ D_ F_

指令功能: 由起点开始, 刀具以指定的半径值及方向进行全圆插补, 直至加工出编程尺寸的圆凸台。

指令说明: 相关指令请参照表3-2固定循环的指令说明。 G116: 逆时针外圆粗铣 G117: 顺时针外圆粗铣

I: 外粗铣圆半径 (应大于0, 负数取绝对值)

J: 毛坯工件半径 (应大于0, 负数取绝对值)

E: 外圆粗铣在XY平面的余量 (大于等于0, 负数取绝对值);

W: Z轴方向首次切深, 是从R基准面向下的距离, 应大于0, 负数取绝对值 (若首次切深超过槽底位置, 则直接以槽底位置加工)

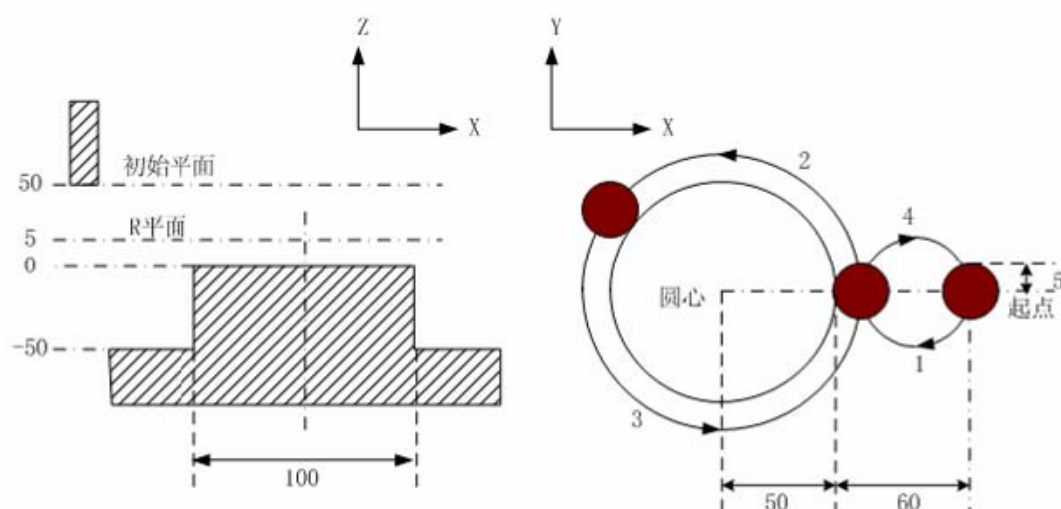
Q: Z轴方向每次的切深增量, 负数取绝对值

K: 在XY面内切削的宽度增量 (应小于刀具直径, 大于0, 负数取绝对值, 如果没指定K, 则默认K = 刀具半径D)

C: X方向首次进刀切屑量 (C值应大于等于刀具半径+2.0, 大于0时, X轴往正方向进刀, 工件在起点正方向处。小于0时, X轴往负方向进刀, 工件在起点负方向处,)

D: 刀具半径序号, 取值范围为0~32, D0默认0. 根据给定的序号取出当前刀具半径值

循环过程: (1) 快速定位到起点XY平面位置



G90 G00 X50 Y50 Z50; (G00 Quick Positioning)

G99 G114 X25 Y25 R5 Z-50 I50 J60 F800 D1; (Start the fixed cycle, go down to the bottom of the hole for the outer circle precision milling cycle D1=5)

G80 X50 Y50 Z50; (Remove the fixed cycle and return from the R-point plane) M30;

(50) Outer Circle Rough Milling Cycle G116/G117

Command format: G99 G116/G117 X_Y_Z_R_I_J_W_Q_K_C_E_D_F_

Instruction function: From the starting point, the tool will carry out full circle interpolation with a specified radius value and direction until the programming dimension of the circular boss is processed.

Instruction: Refer to Table 3-2 for instructions of fixed cycle. G116: Counter clockwise outer circle rough

Milling G117: Circular rough milling clockwise

I: Outer rough milling radius (should be greater than 0, negative values take absolute value)

J: radius of the rough workpiece (should be greater than 0, negative values are taken as absolute values)

E: The allowance of the outer circle rough milling in the XY plane (greater than or equal to 0, the absolute value of a negative number);

W: The first depth of cut in the Z direction is the distance from the R reference plane downward. It should be greater than 0, and the absolute value of a negative number (if the first depth of cut exceeds the bottom position of the groove, the groove bottom position shall be processed directly)

Q: The absolute value of the negative depth increment in the Z direction

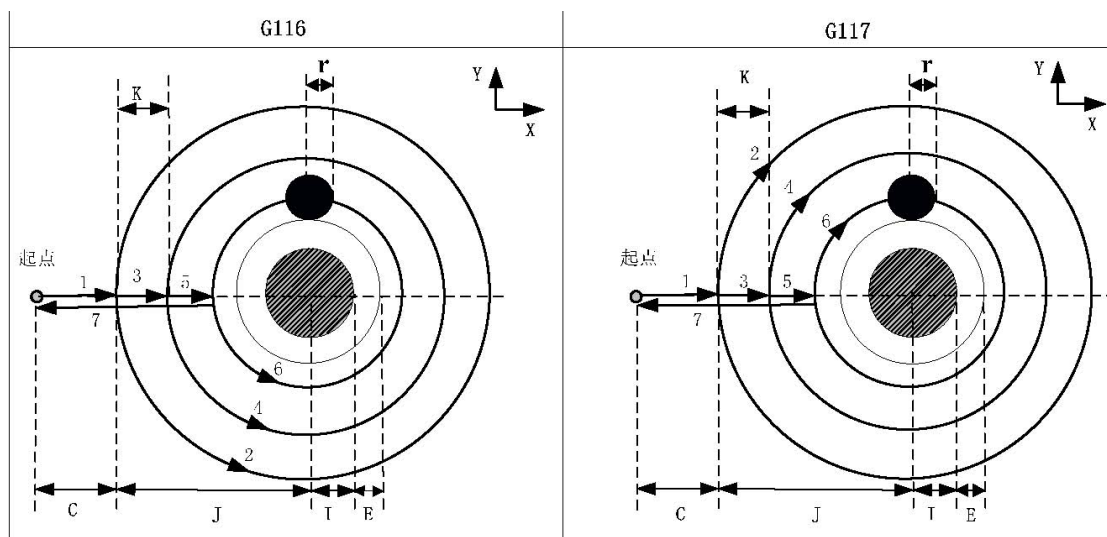
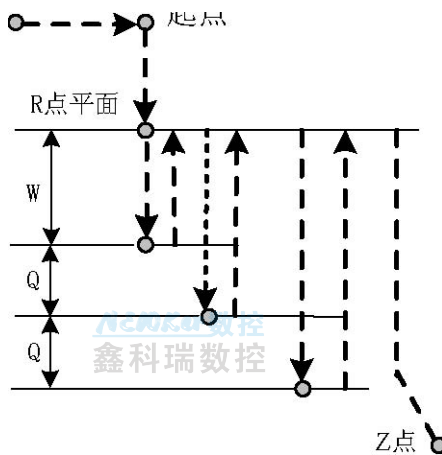
K: The width increment of cutting in the XY plane (should be less than the tool diameter, greater than 0, the absolute value of a negative number, if K is not specified, K = tool radius D by default)

C: The amount of chips cut during the first X-direction feed (The value of C should be greater than or equal to the tool radius +2.0. When it is greater than 0, the X-axis feeds in the positive direction, and the workpiece is at the starting point in the positive direction. When it is less than 0, the X-axis feeds in the negative direction, and the workpiece is at the starting point in the negative direction.)

D: Tool radius serial number, the value range is 0~32, D0 is default 0. The current tool radius value is obtained according to the given serial number

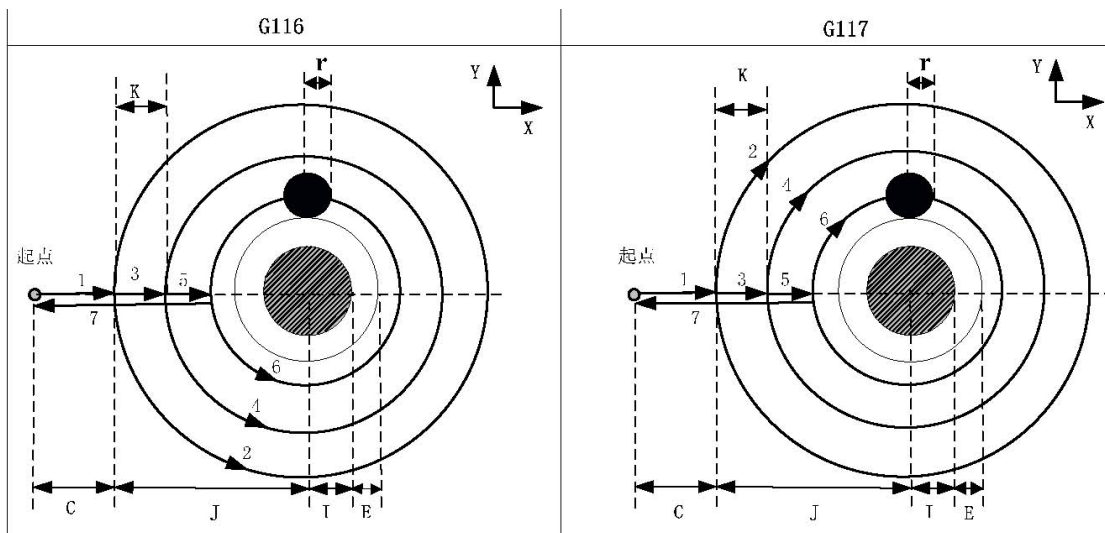
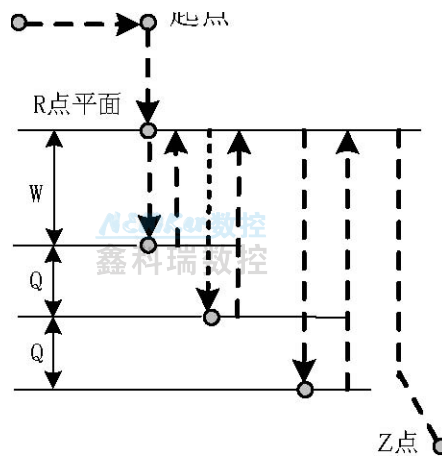
Cycle process: (1) Quickly locate the starting XY plane position

- (2) 快速下至R点平面。
- (3) 快速下至W距离深度
- (4) 以X轴首次进刀量C，直线1为轨迹进行直线插补进刀
- (5) 以圆弧2为轨迹进行整圆插补
- (6) 由外向中心每次按K值递增铣完半径为(I+E)的圆面 (7) Z轴快速定位到返回R基准面
- (8) 快速定位到起点位置XY平面 (9) Z轴快速下至(未加工面+Q)的距离深度
- (10) 循环5~9动作直到加工完总切深为Z的外圆粗铣圆面
- (11) 固定返回R点；
- (12) 返回XY起点位置



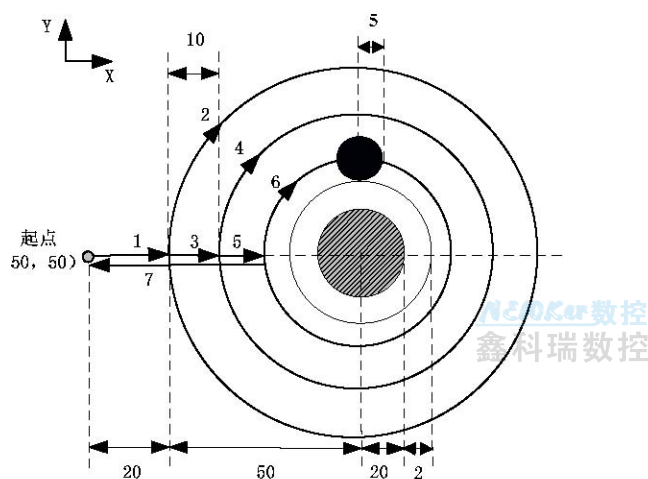
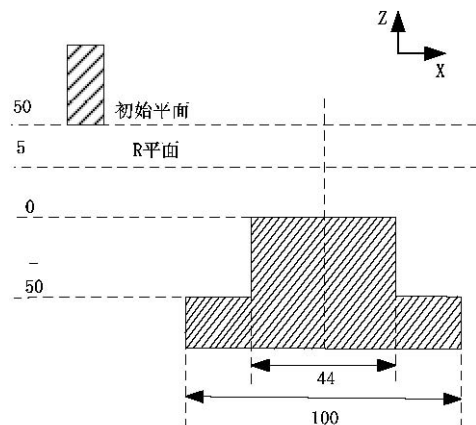
相关说明：该循环下指令P、L无效，但会保留P值作为固定循环模态数值保存。
 例：用固定循环G117 指令粗铣一个圆凸台，如下图所示：

- (2) Go Quickly to the R Plane.
- (3) Go quickly to the depth of W distance
- (4) Take the first feed amount C on the X axis as the trajectory of line 1 for linear interpolation
- (5) The arc 2 is used as the trajectory for full circle interpolation
- (6) From the outer to the center, each time the radius of the circle with K value increment is milled as (I+E)
- (7) The Z axis is quickly positioned to return to the R reference plane
- (8) The delivery is positioned on the XY plane of the starting point
- (9) and the distance depth of the Z axis is quickly lowered to (unprocessed surface +Q)
- (10) Repeat steps 5 to 9 until the outer circular surface with total cutting depth Z is milled
- (11) Fixed return to point R;
- (12) Return to XY starting position



Related note: The instruction P and L are invalid under this cycle, but the value of P is retained as a fixed cycle mode value.

Example: Use the fixed cycle G117 command to rough mill a circular boss, as shown in the following figure:



G90 G00 X0 Y0 Z50; (G00 快速定位)

G99 G117 X50 Y50 R5 Z-50 I20 J50 W20 Q10 K10 C20 E2 F800 D1; (进行外圆凸台粗铣循环 D1=5)

G80 X50 Y50 Z50; (取消固定循环, 从R 点平面返回)

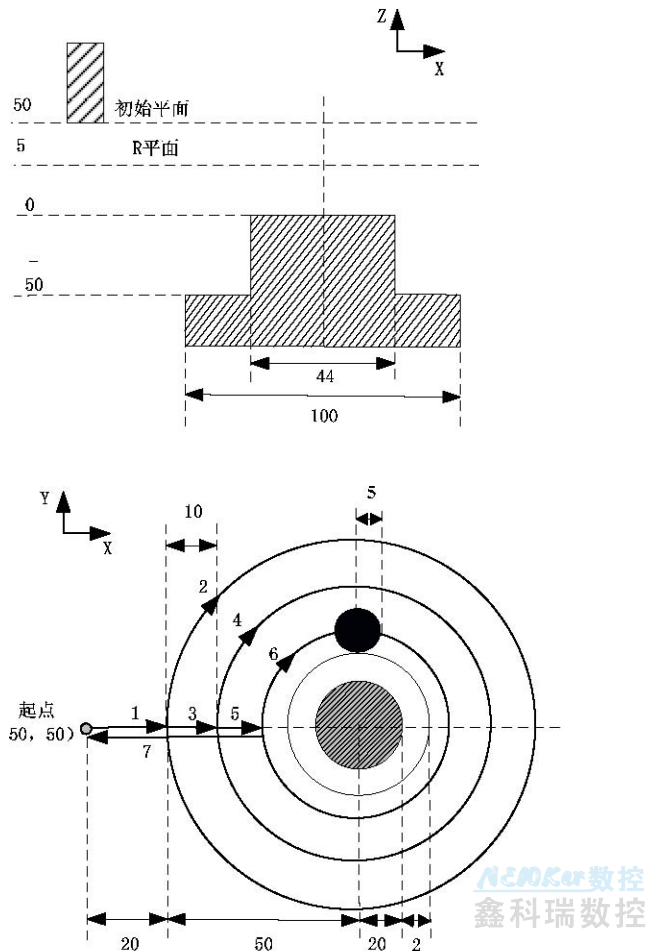
M30;

(51) 外矩形粗铣 G132/G133

指令格式: G99 G132/G133 X_ Y_ Z_ R_ I_ J_ A_ B_ W_ Q_ K_ C_ E_ D_ F_
指令功能: 从起点开始, 以指定的参数数据作直线切屑循环, 直至加工出编程尺寸的矩形凸台。

指令说明: 相关指令请参照表3-2固定循环的指令说明。 G132: 逆时针外矩形粗铣 G133: 顺时针外矩形粗铣 XY: 下刀点的XY平面坐标位置

I: 外粗铣矩形在X轴方向的宽度 (应大于0, 负数取绝对值)



G90 G00 X0 Y0 Z50; (G00 quick positioning)

G99 G117 X50 Y50 R5 Z-50 I20 J50 W20 Q10 K10 C20 E2 F800 D1;
(perform a rough milling cycle for the outer cylindrical boss D1=5)

G80 X50 Y50 Z50; (Remove fixed cycle and return from the R-point plane)

M30;

(51) External Rectangular Rough Milling G132/G133

Command format: G99 G132/G133 X_Y_Z_R_I_J_A_B_W_Q_K_C_E_D_F_

Instruction function: from the starting point, with the specified parameter data as a straight chip cycle, until the programming scale is processed

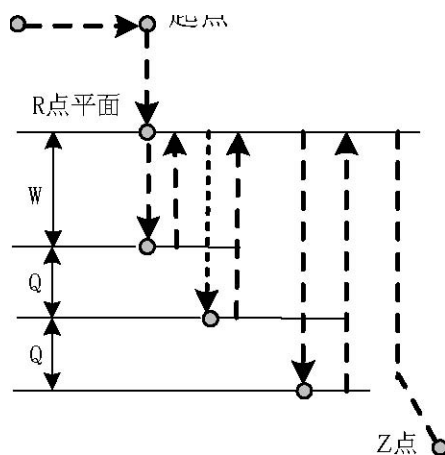
A Rectangular Protrusion an Inch in Size.

Instruction: Refer to the instruction for fixed cycle in Table 3-2 for relevant instructions. G132: counterclockwise outer rectangle

Coarse milling G133: clockwise outer rectangular coarse milling XY: XY plane coordinates of the lower cutting point

I: The width of the outer rough milling rectangle in the X-axis direction (should be greater than 0, and the absolute value of a negative number)

- J: 外粗铣矩形在Y轴方向的宽度 (应大于0, 负数取绝对值)
- A: 毛坯在X轴方向的宽度 (应大于0, 负数取绝对值)
- B: 毛坯在Y轴方向的宽度 (应大于0, 负数取绝对值)
- E: 外矩形粗铣在XY平面的余量 (应大于等于0, 负数取绝对值);
- W: Z轴方向首次切深, 是从R基准面向下的距离, 应大于0, 负数取绝对值 (若首次切深超过槽底位置, 则直接以槽底位置加工)
- Q: Z轴方向每次的切深增量, 负数取绝对值
- K: 在XY面内切削的宽度增量, (应小于刀具直径, 大于0, 负数取绝对值, 如果没指定K, 则默认K=刀具半径D)
- C: X方向首次进刀切屑量 (C值应大于等于刀具半径+2.0, 大于0时, X轴往正方向进刀, 工件在起点正方向处。小于0时, X轴往负方向进刀, 工件在起点负方向处)
- D: 刀具半径序号, 取值范围为1~32. 根据给定的序号取出当前刀具半径值。
- 循环过程: (1) 快速定位到XY平面起点位置
- (2) 快速下至R点平面
- (3) 首次快速下至W距离深度
- (4) 以X轴首次进刀量C, 直线1为轨迹进行直线插补进刀
- (5) 以轨迹2进行直线插补
- (6) 由外向中心每次按K值递增铣完长为 $(I+2E)$ 、宽为 $(J+2E)$ 的矩形
- (7) 铣矩形的圆弧转角 (8) Z轴快速返回R基准面 (9) X轴、Y轴快速定位到起点位置平面 (10) Z轴快速下至 (未加工面+Q) 的距离深度
- (11) 循环4~9动作直到加工完总切深为Z的外粗铣矩形面
- (12) 固定返回R点
- (13) 返回XY起点位置



J: The width of the outer rough milling rectangle in the Y-axis direction (should be greater than 0, and the absolute value of a negative number)

A: The width of the blank in the X direction (should be greater than 0, the absolute value of a negative number)

B: The width of the blank in the Y direction (should be greater than 0, the absolute value of a negative number)

E: The allowance of the outer rectangular rough milling in the XY plane (should be greater than or equal to 0, and the absolute value of a negative number);

W: The first depth of cut in the Z direction is the distance from the R reference plane downward. It should be greater than 0, and the absolute value of a negative number (if the first depth of cut exceeds the bottom position of the groove, it will be processed directly at the bottom position of the groove)

Q: The absolute value of the negative depth increment in the Z direction

K: The width increment of cutting in the XY plane (should be less than the tool diameter, greater than 0, and the absolute value of a negative number. If K is not specified, K= tool radius D by default)

C: The amount of chips cut during the first X-direction feed (The value of C should be greater than or equal to the tool radius + 2.0. When it is greater than 0, the X-axis feeds in the positive direction, and the workpiece is at the starting point in the positive direction. When it is less than 0, the X-axis feeds in the negative direction, and the workpiece is at the starting point in the negative direction)

D: Tool radius serial number, the value range is 1~32. According to the given serial number, the current tool radius value is obtained.

Cyclic process: (1) Quickly locate the starting position in XY plane

(2) Go quickly to the R plane

(3) Reach the depth of W distance for the first time quickly

(4) Take C as the first feed of the X-axis, and make linear interpolation for the straight path 1

(5) Straight interpolation is performed on trajectory 2

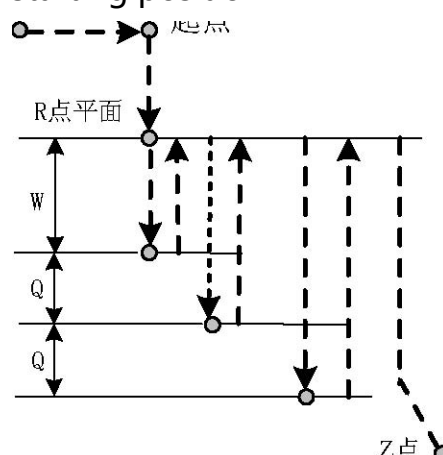
(6) From the outside to the center, each time a rectangle with length $(l+2E)$ and width $(J+2E)$ is milled according to the increasing K value

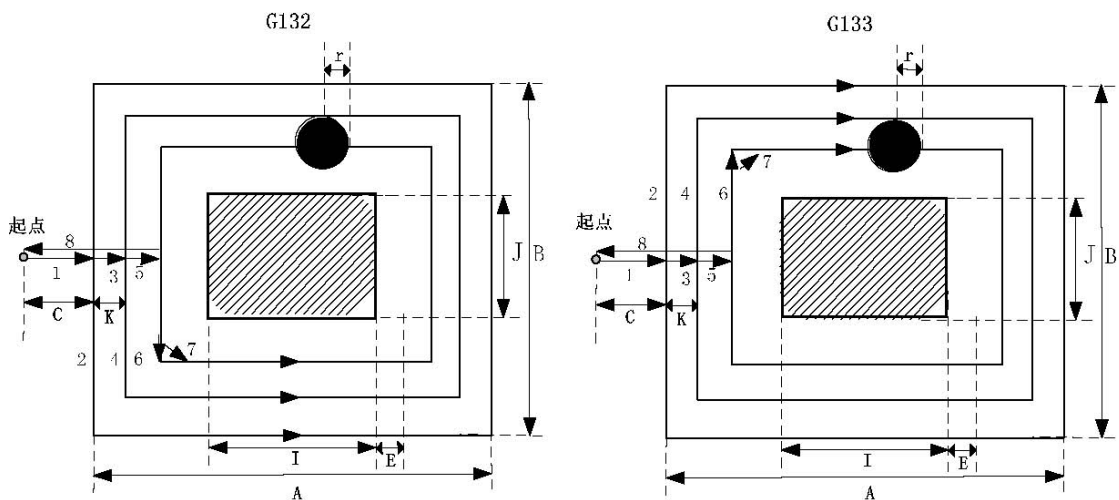
(7) Machining the arc corner of the rectangle (8) Z-axis quick return to the R reference plane (9) X-axis and Y-axis quick positioning to the starting plane (10) Z-axis quick down to the distance depth of (unprocessed surface +Q)

(11) Repeat steps 4 to 9 until the outer rough milling of the rectangular surface with total depth Z is completed

(12) Fixed return to point R

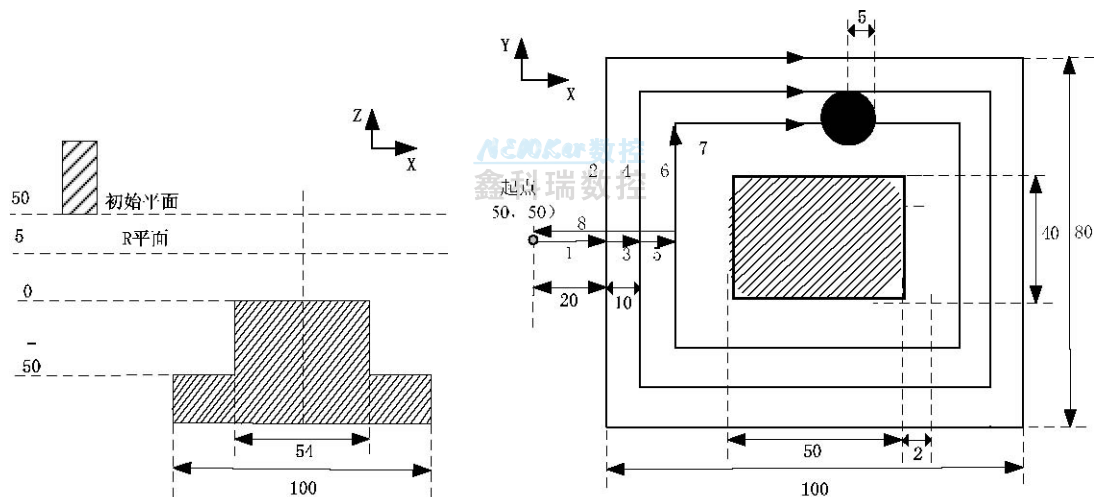
(13) Return to XY starting position





相关说明：该循环下指令P、L无效，但会保留P值作为固定循环模态数值保存。

例：用固定循环G133 指令粗铣一个外矩形凸台，如下图所示：



G90 G00 X0 Y0 Z50; (G00 快速定位)

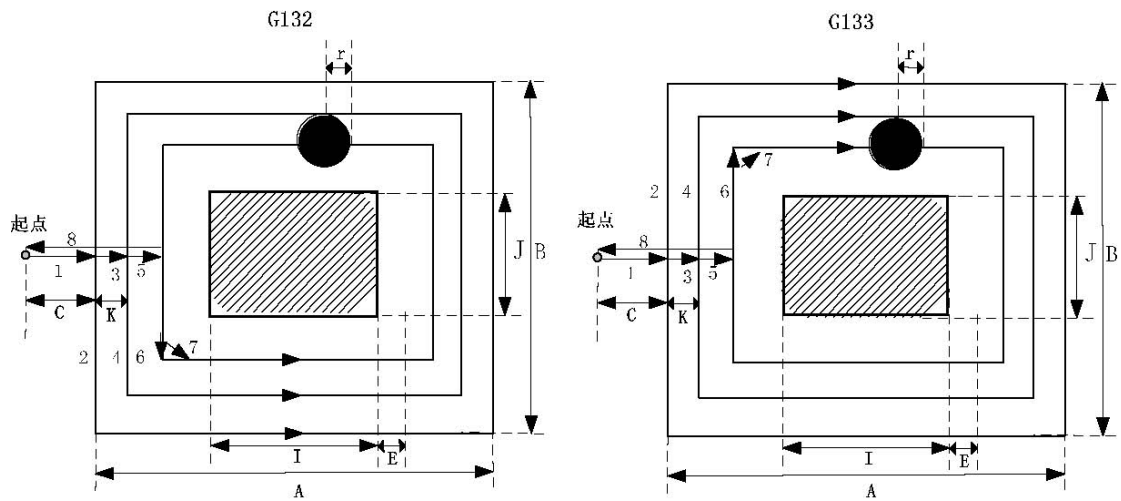
G99 G133 X50 Y50 R5 Z-50 I50 J40 A100 B80 W20 Q10 K10 C20 E2 U5 F800 D1; (进行矩形内凹槽粗铣循环D1=5)

G80 X50 Y50 Z50;

M30;

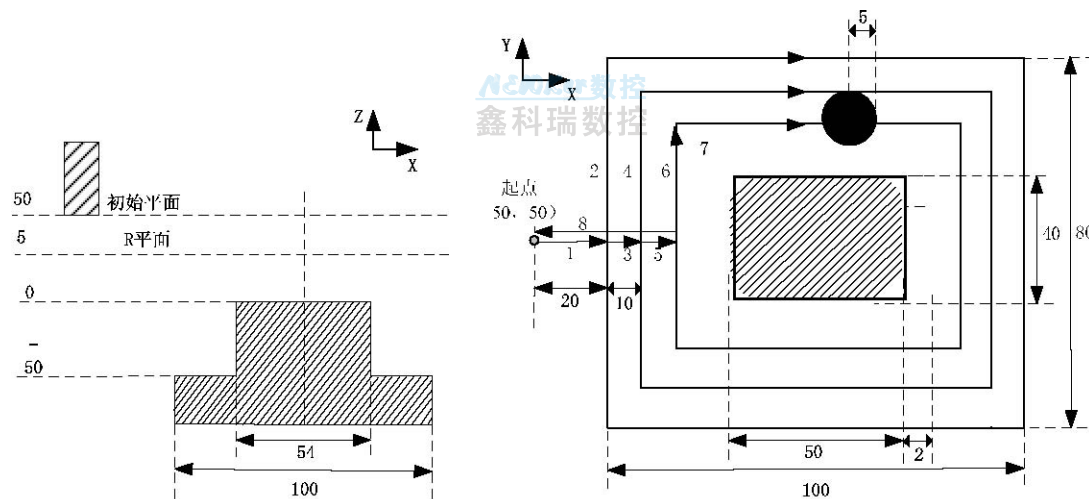
(52) 矩形凹槽粗铣G134/G135

指令格式：G99 G134/G135 X_Y_Z_R_I_J_K_W_Q_E_V_D_F_



Related note: The instruction P and L are invalid under this cycle, but the value of P is retained as a fixed cycle mode value.

Example: Use the fixed cycle G133 command to rough mill an external rectangular boss, as shown in the figure below:



G90 G00 X0 Y0 Z50; (G00 quick positioning)

G99 G133 X50 Y50 R5 Z-50 I50 J40 A100 B80 W20 Q10 K10 C20 E2 U5 F800 D1; (perform rectangular concave groove rough milling cycle D1=5)

G80 X50 Y50 Z50;

M30;

(52) Rough Milling of Rectangular Groove G134/G135

Instruction format: G99 G134/G135 X_Y_Z_R_I_J_K_W_Q_E_V_D_F_

指令功能: 从矩形中心开始, 以指定的参数数据作直线切削循环, 直至加工出编程尺寸的矩形凹槽。

指令说明 相关指令请参照表3-2 固定循环的指令说明。G134 : 逆时针矩形凹槽粗铣。G135 : 顺时针矩形凹槽粗铣。

I: 矩形凹槽在X 轴方向的宽度。

J: 矩形凹槽在Y 轴方向的宽度。

K: 在XY 面内切削的宽度增量, 应小于刀具直径, 大于0。

W: Z 轴方向首次切深, 是从R 基准面向下的距离, 应大于0 (若首次切深超过槽底位置, 则直接以槽底位置加工)。

Q: Z 轴方向每次的切深增量。

V: 快速下刀时, 离末加工面的距离, 大于0。

E: 矩形凹槽粗铣在XY 平面的余量 (应大于等于0, 负数取绝对值, 不指定时取数据参数5124 所设定的数值)。

D: 刀具半径序号, 取值范围为0 ~ 32, D0 默认为0。根据给定的序号取出当前刀具半径值。

循环过程: (1) 快速定位到XY 平面的位置;

(2) 快速下至R 点平面;

(3) (4) (5) (6) (7) (8) (9) (10)

(11) 指令轨迹:

以切削速度向下切削W 距离深度; 中心向外每次按K 值递增螺旋铣完矩形面; Z 轴快速返回R 基准面; X, Y 轴快速定位到矩形中心; Z 轴快速下降至离末加工面V 的距离; Z 轴向下切削 (Q+V) 的深度; 循环 (4) ~ (8) 的动作, 直至加工完总切深的圆面; 固定返回R点返回XY 孔定位位置。

Instruction function: Start from the center of the rectangle, make a straight cutting cycle with the specified parameter data, until the programming size of the rectangular groove is processed.

Instruction Description: Refer to Table 3-2 for instructions of fixed cycles. G134: Rough milling of counterclockwise rectangular groove. G135: Rough milling of clockwise rectangular groove.

I: The width of the rectangular groove in the X direction.

J: The width of the rectangular groove in the Y direction.

K: The width increment of cutting in XY plane should be less than the diameter of the tool and greater than 0.

W: The first depth of cut in the Z direction is the distance from the R reference plane downward, which should be greater than 0 (if the first depth of cut exceeds the bottom position of the groove, the groove bottom position should be processed directly).

Q: The depth increment of each cut in the Z direction.

V: When the blade is cut quickly, the distance from the last processing surface is greater than 0.

E: The allowance of rough milling of rectangular groove in XY plane (it should be greater than or equal to 0, the absolute value of negative number, and the value set by data parameter 5124 is taken if not specified).

D: Tool radius serial number, the value range is 0 ~ 32, D0 is default 0. The current tool radius value is selected according to the given serial number.

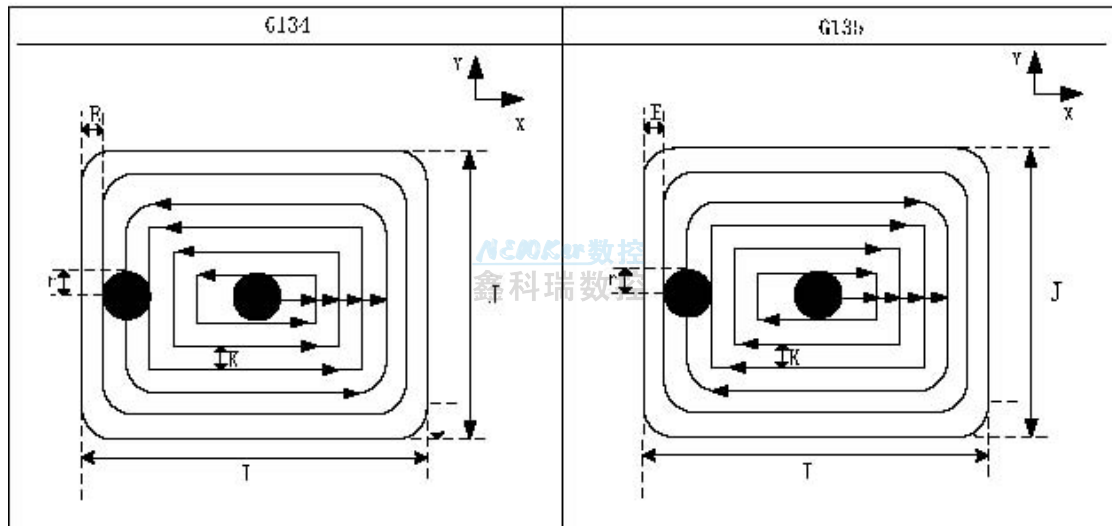
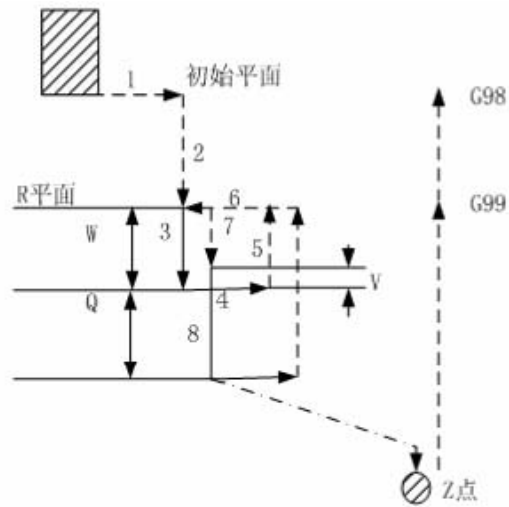
Cycling process: (1) Quick positioning to the position in XY plane;

(2) Quickly descend to the plane of point R;

(3) (4) (5) (6) (7) (8) (9) (10)

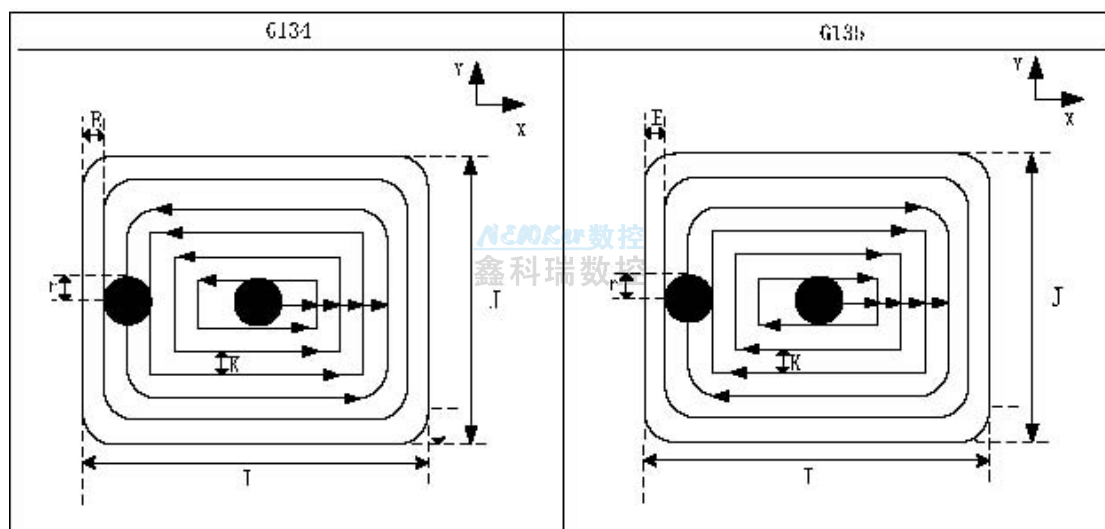
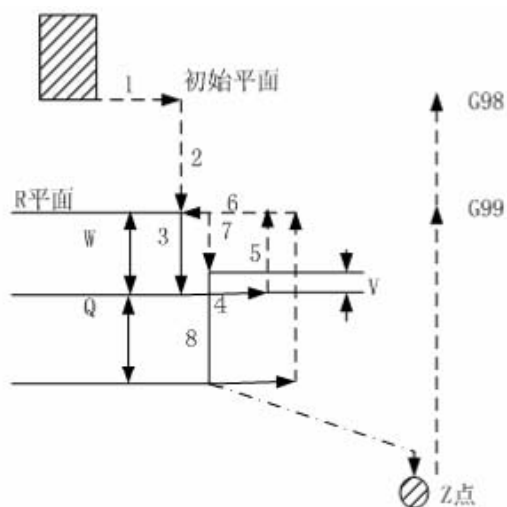
(11) Instruction trajectory:

Perform the following operations: Start by cutting downward at W distance depth with the cutting speed. Center outward, complete each rectangular surface via spiral milling with K value increments. Quickly return to the R reference plane along the Z-axis. Rapidly position the X and Y axes to the rectangular center. Lower the Z-axis to the distance from the final processed surface V. Cut downward with Z-axis depth (Q+V). Repeat cycles (4) to (8) until completing the circular surface machining. Finally, return to the XY hole positioning point via fixed return to R point.



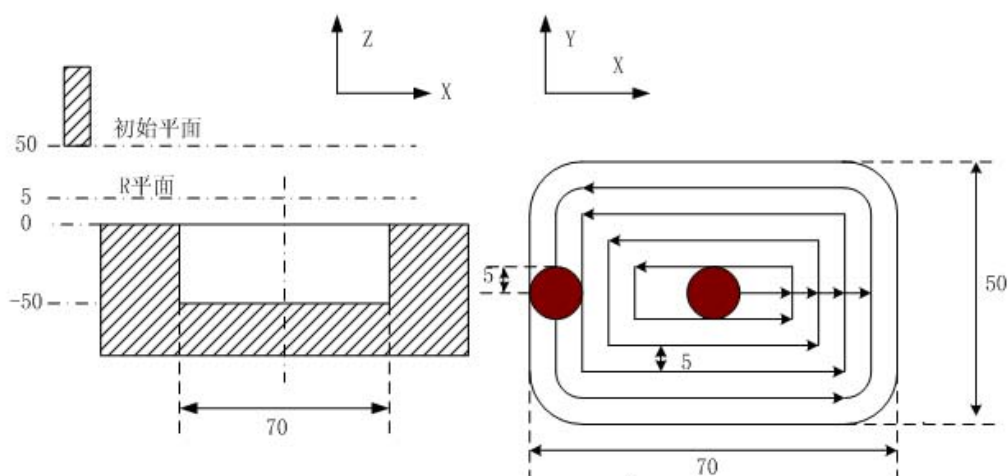
相关说明：该循环下指令P、L无效，但会保留P值作为固定循环模态数值保存。

例：用固定循环G134 指令粗铣一个矩形内凹槽，如下图所示：



Related Note: The Instruction P and L Are Invalid Under This Cycle, but the Value of P Is Retained as a Fixed Cycle Mode Value.

Example: Use the fixed cycle G134 command to rough mill a rectangular recessed groove, as shown in the figure below:



G90 G00 X50 Y50 Z50; (G00 快速定位)

G99 G134 X25 Y25 R5 Z-50 I70 J50 W20 Q10 K5 E0 V10 U10 F800 D1; (进行矩形内槽粗铣循环D1=5)

G80 X50 Y50 Z50; (取消固定循环, 从R 点平面返回)

M30;

(53) 矩形凹槽内精铣循环G136/G137

指令格式: G99 G136/G137 X_ Y_ Z_ J_ D_ K_ U_ F_;

指令功能: 刀具以指定的宽度及方向在矩形内部精铣, 精铣完成后返回。

指令说明: 相关指令请参照表3-2 固定循环的指令说明。G136: 逆时针矩形凹槽内精铣循环。G137: 顺时针矩形凹槽内精铣循环。

I: 矩形X 轴方向的宽度, 取值范围-99999999~99999999×最小指令单位。负数取其绝对值。

J: 矩形Y 轴方向的宽度, 取值范围-99999999~99999999×最小指令单位。负数取其绝对值。J的取值必须大于或者等于刀具半径D 值, 否则报警。

D: 刀具半径序号, 取值范围为0 ~ 32, D0 默认为0。根据给定的序号取出当前刀具半径值。

K: 精铣起点与矩形边X 轴方向的距离, 取值范围-99999999~99999999×最小指令单位。负数取其绝对值。

U: 转角圆弧半径, 省略则表示无转角圆弧过渡。U 省略或U=0 时且刀具半径>0, 则报警。循环过程: (1)快速定位到XY 平面的位置;

(2) 快速下至R 点平面;

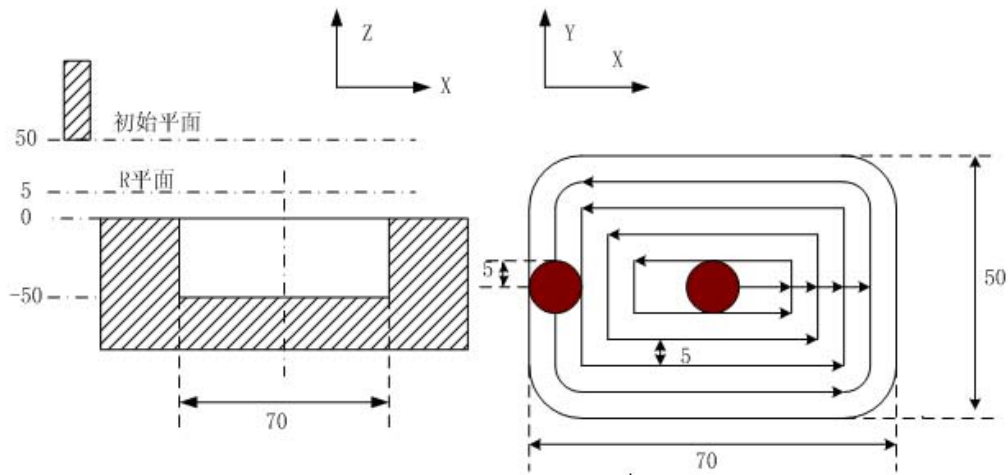
(3) 切削进给至孔底;

(4) 从起点以过渡弧1 为轨迹进行圆弧插补;

(5) 以2-3 为轨迹进行直线和圆弧插补;

(6) 以过渡弧4 为轨迹进行圆弧插补回到起点;

(7) 固定返回R点



G90 G00 X50 Y50 Z50; (G00 Quick Positioning)
 G99 G134 X25 Y25 R5 Z-50 I70 J50 W20 Q10 K5 E0 V10 U10 F800 D1;
 (perform rough milling cycles for rectangular recessed grooves D1=5)
 G80 X50 Y50 Z50; (Remove fixed cycle and return from the R-point
 plane)

M30;

(53) Precision Milling Cycle G136/G137 in Rectangular Groove

Instruction format: G99 G136/G137 X_Y_R_Z_I_J_D_K_U_F_;

Instruction function: The tool is precision-milled in the rectangle with the specified width and direction, and returns after precision-milling.

Instruction Description: Refer to Table 3-2 for instructions of fixed cycles. G136: Precision milling cycle in the counterclockwise rectangular groove. G137: Precision milling cycle in the clockwise rectangular groove.

I: The width in the X-axis direction of the rectangle, with a range of -99999999 to 99999999× minimum instruction unit. The absolute value of negative numbers is taken.

J: The width in the rectangular Y-axis direction, with a range of -99999999 to 99999999× minimum instruction unit. Negative numbers are taken as absolute values. The value of J must be greater than or equal to the tool radius D value, otherwise an alarm will be triggered.

D: Tool radius serial number, the value range is 0 ~ 32, D0 is default 0. The current tool radius value is selected according to the given serial number.

K: The distance between the starting point of precision milling and the rectangular edge in the X-axis direction. The value ranges from -99999999 to 99999999× minimum instruction unit. The absolute value of a negative number is taken.

U: Corner arc radius. If omitted, it indicates no corner arc transition. When U is omitted or U=0 and the tool radius>0, an alarm will be triggered. The cycle process: (1) Quickly locate the position in the XY plane;

(2) Quickly descend to the plane of point R;

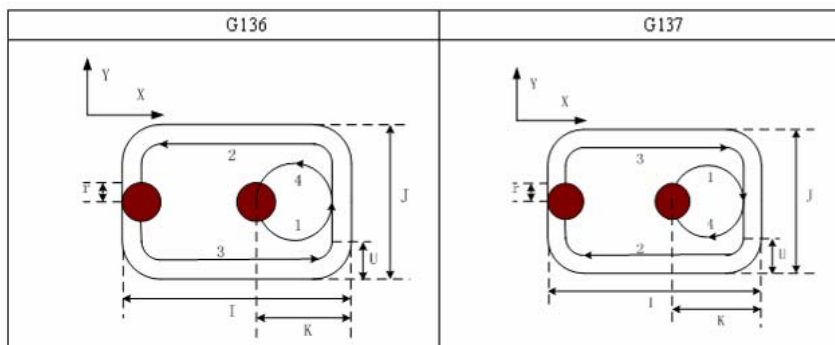
(3) Cutting feed to the bottom of the hole;

(4) Arc interpolation is performed from the starting point along the transition arc 1;

(5) Use 2-3 as the trajectory for straight line and arc interpolation;

(6) Use transition arc 4 as the trajectory to interpolate back to the starting point;

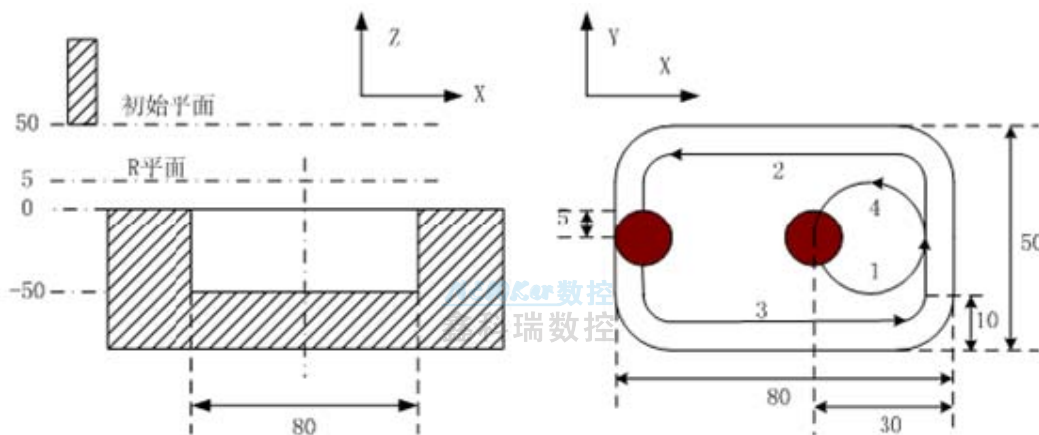
(7) Fixed return to point R



指令轨迹：相关说明：

该循环下指令Q、P、L无效，但会保留Q、P的值作为固定循环模态数值保存。

例：用固定循环G136 指令，精铣如下图所示的已粗铣完的矩形凹槽。



G90 G00 X50 Y50 Z50; (G00 快速定位)

G136 X25 Y25 R5 Z-50 I80 J50 K30 U10 F800 D1; (固定循环下到孔底进行矩形凹槽内精铣D1=5)

G80 X50 Y50 Z50; (取消固定循环，从R 点平面返回)

M30;

(54) 矩形外精铣循环G138/G139/G141/G142

指令格式：G99 G138/G139/G141/G142 X_ Y_ R_ Z_ I_ J_ D_ K_ U_ F_

指令功能：刀具以指定的宽度及方向在矩形外部精铣，精铣完成后返回。

指令说明：相关指令请参照表3-2 固定循环的指令说明。

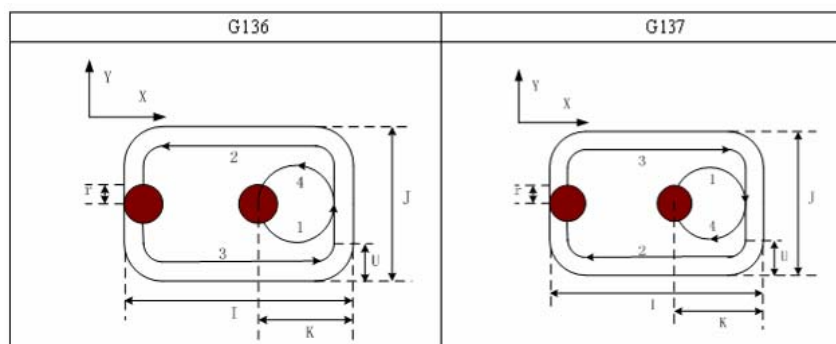
G138:逆时针矩形外精铣循环。

G139:顺时针矩形外精铣循环。

G141:横向长方形或正方形外精铣循环 (I>J)。

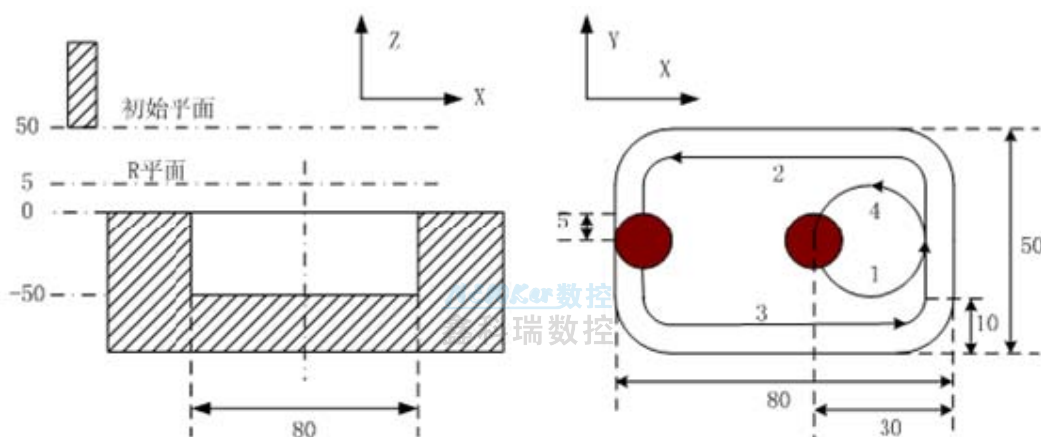
G142:竖向长方形或正方形外精铣循环 (I<J)。

I: 矩形X 轴方向的宽度，取值范围-99999999~99999999×最小指令单位。负数



Instruction trajectory: Related description:

In this cycle, the instructions Q, P, and L are invalid, but the values of Q and P are retained as fixed cycle mode values. Example: Use the fixed cycle G136 instruction to finish milling the rough-milled rectangular groove shown in the figure below.



G90 G00 X50 Y50 Z50; (G00 Quick Positioning)

G136 X25 Y25 R5 Z-50 I80 J50 K30 U10 F800 D1; (fixed cycle to the bottom of the hole for precision milling in the rectangular groove D1=5)

G80 X50 Y50 Z50; (Remove fixed cycle and return from the R-point plane)

M30;

(54) Rectangular outer milling cycles G138/G139/G141/G142

Instruction format: G99 G138/G139/G141/G142 X_Y_R_Z_I_J_D_K_U_F_

Instruction function: The tool is precision milled on the outside of the rectangle with the specified width and direction, and returns after the precision milling is completed.

Instruction: Refer to Table 3-2 for instructions of fixed cycles.

G138: counterclockwise rectangular outer milling cycle.

G139: clockwise rectangular outer milling cycle.

G141: Longitudinal rectangular or square outer milling cycle ($I > J$).

G142: Vertical rectangular or square outer milling cycle ($I < J$).

I: The width in the X-axis direction of the rectangle. The value ranges from -99999999 to 99999999 × minimum instruction unit. Negative number

取其绝对值。

J: 矩形Y 轴方向的宽度, 取值范围 $-99999999 \sim 99999999 \times$ 最小指令单位。负数取其绝对值。

D: 刀具半径序号, 取值范围为 $0 \sim 32$, D0 默认为0。根据给定的序号取出当前刀具半径值。

K: 精铣起点与矩形边X 轴方向的距离, 取值范围 $-99999999 \sim 99999999 \times$ 最小指令单位。负数取其绝对值。

U: 转角圆弧半径, 省略则表示无转角圆弧过渡。循环过程 (1) 快速定位到XY 平面的位置;

(2) 快速下至R 点平面;

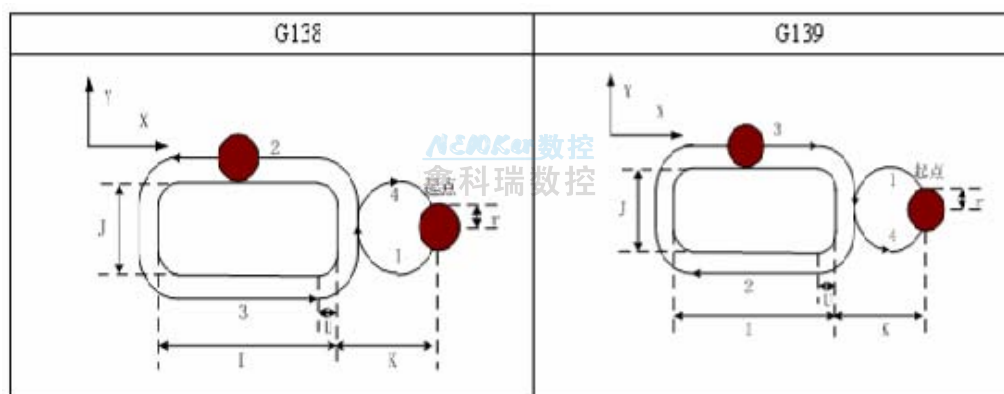
(3) 切削进给至孔底;

(4) 从起点以过渡弧1 为轨迹进行圆弧插补;

(5) 以2-3 为轨迹进行直线和圆弧插补;

(6) 以过渡弧4 为轨迹进行圆弧插补回到起点;

(7) 固定返回R点



相关说明:

(1) 矩形外精铣时, 过渡弧与精铣弧的插补方向不一致, 指令说明中的插补方向指精铣弧的插补方向。

(2) 该循环下指令Q、P、L无效, 但会保留Q, P的值作为固定循环模态数值保存。例: 用固定循环G138 指令, 精铣如下图所示的已粗铣完的矩形凸台。

Take the absolute value.

J: The width in the rectangular Y-axis direction, with a range of -99999999 to $99999999 \times$ minimum instruction unit. Negative numbers are taken as their absolute values.

D: Tool radius serial number, the value range is $0 \sim 32$, D0 is default 0. The current tool radius value is selected according to the given serial number.

K: The distance between the starting point of precision milling and the rectangular edge in the X-axis direction. The value ranges from -99999999 to $99999999 \times$ minimum instruction unit. For negative values, the absolute value is taken.

U: Angle radius of the arc, omitted to indicate no transition of the arc.

Cycle process: (1) Quick positioning to the position in the XY plane;

(2) Quickly descend to the plane of point R;

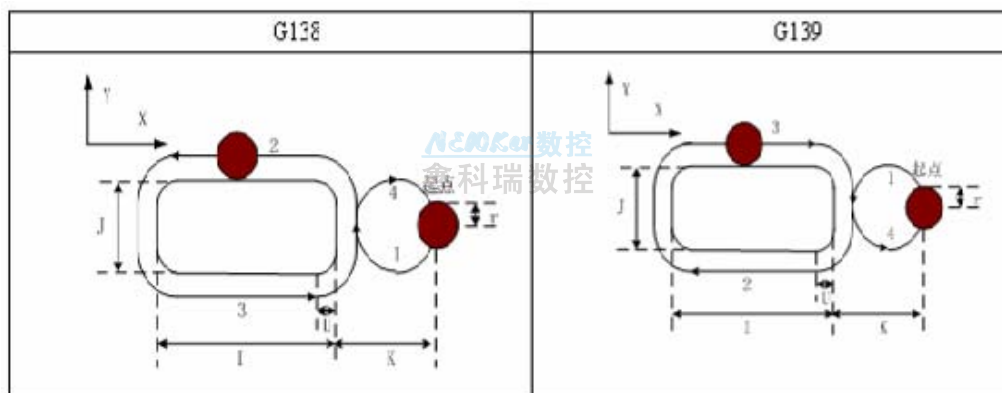
(3) Cutting feed to the bottom of the hole;

(4) Arc interpolation is performed from the starting point along the transition arc 1;

(5) Use 2-3 as the trajectory for straight line and arc interpolation;

(6) Use transition arc 4 as the trajectory to interpolate back to the starting point;

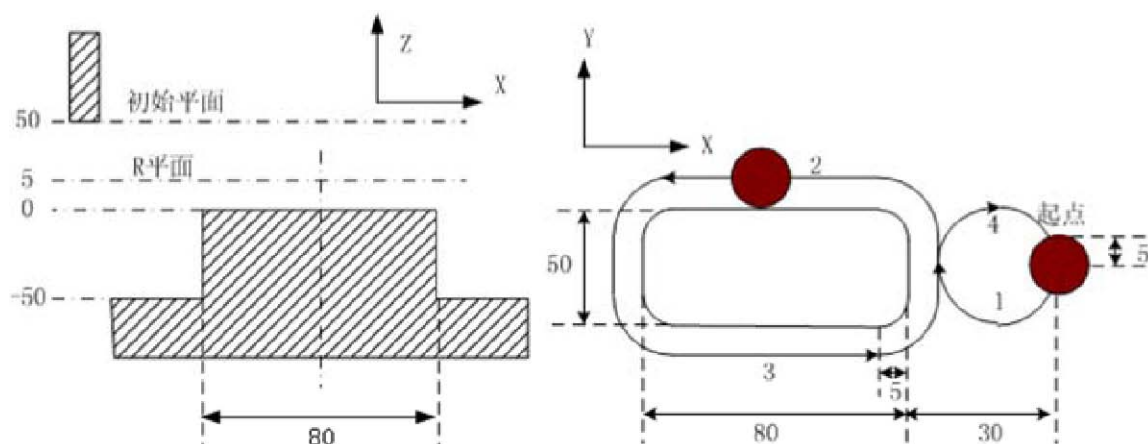
(7) Fixed return to point R



Related notes:

(1) When milling a rectangle outside, the direction of interpolation between the transition arc and the precision milling arc is not the same. The direction of interpolation in the instruction specification refers to the direction of interpolation between the precision milling arc.

(2) In this cycle, the instructions Q, P, and L are invalid, but the values of Q and P are retained as fixed cycle mode values. Example: Use the fixed cycle G138 instruction to finish milling the rough-milled rectangular boss as shown in the figure below.



G90 G00 X50 Y50 Z50; (G00 快速定位)

G99 G138 X25 Y25 R5 Z-50 I80 J50 K30 U5 F800 D1; (固定循环下到孔底进行矩形外精铣D1=5)

G80 X50 Y50 Z50; (取消固定循环, 从R 点平面返回)

M30;

(55) 可编程数据输入指令 G10/G710

可编程数据输入指令, 可以修改 G54-G59 工件坐标:

G10 Lxx 开启可编程数据输入
 Px Rxx 执行数据修改
 Px Ixx 执行数据修改
 X_Y_Z_A_ 设置指定工件坐标系的当前坐标值
 G710 关闭可编程数据输入

说明:

G10 表开启可编程数据输入, 为模态指令。

G710 表关闭可编程数据输入, 为模态指令。

L 地址字指定修改数据的类型, 54--59 分别对应 G54--G59 偏移量, 5401--5448 分别对应 G54.1--G54.48 偏移量, 为模态指令。

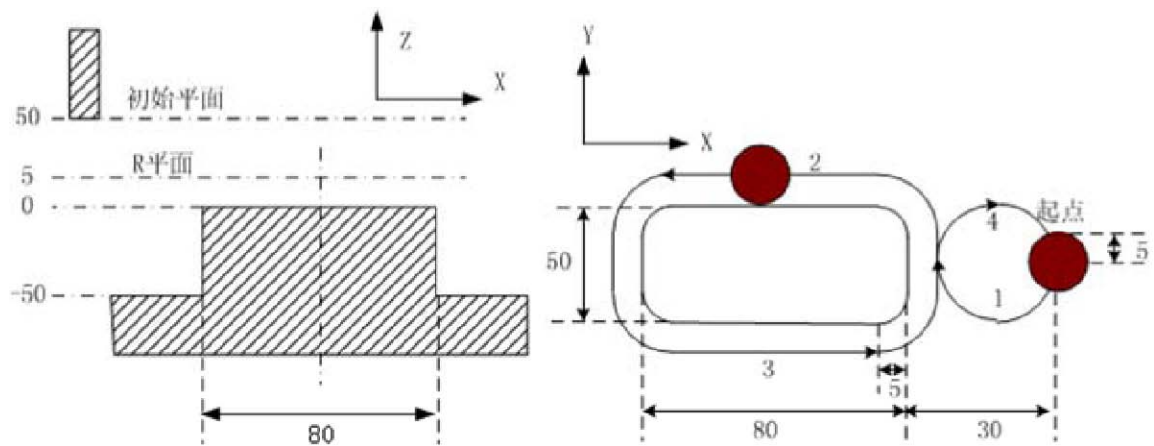
P 地址字指定修改数据对应的进给轴, 1--8 对应 8 个进给轴, 为模态指令。

R 地址字指定修改的数据值, 为绝对方式修改。

I 地址字指定修改的数据值, 为增量方式修改。

例如:

G10 L55 ;开启可编程数据输入, 修改 G55 工件坐标系偏移量
 P2 R6 ;G55 的 Y 轴工件坐标系偏移量设为 6 mm
 L56 X1Y2Z3 ;G56 的工件坐标设置为 X1Y2Z3
 P1 I1.5 ;G56 的 X 轴工件坐标系偏移量加上 1.5 mm
 G710 ;关闭可编程数据输入



G90 G00 X50 Y50 Z50; (G00 Quick Positioning)
 G99 G138 X25 Y25 R5 Z-50 I80 J50 K30 U5 F800 D1; (fixed cycle to the bottom of the hole for rectangular external milling D1=5)
 G80 X50 Y50 Z50; (Remove fixed cycle and return from the R-point plane)
 M30;

(55) Programmable Data Input Instruction G10/G710

Programmable data input command, which can modify the G54-G59 workpiece coordinates:

G10 Lxx enables programmable data input

Px Rxx performs data modification

Px lxx performs data modification

X_Y_Z_A_ sets the current coordinate value of the specified workpiece coordinate system

G710 Turn off programmable data input

explain :

The G10 table is opened for programmable data input and is a modal instruction.

The G710 table is closed to programmable data input and is a modal instruction.

L Address word specifies the type of modified data. 54-59 correspond to G54-G59 offsets respectively, and 5401-5448 correspond to G54.1-G54.48 offsets respectively, which are mode instructions.

The P address word specifies the feed axis corresponding to the modified data, and 1-8 correspond to 8 feed axes. It is a mode instruction. The R address word specifies the value of the modified data, which is modified in absolute mode.

I The address word specifies the modified data value in an incremental manner.

for instance :

G10 L55; Enable programmable data input to modify the G55 workpiece coordinate system offset

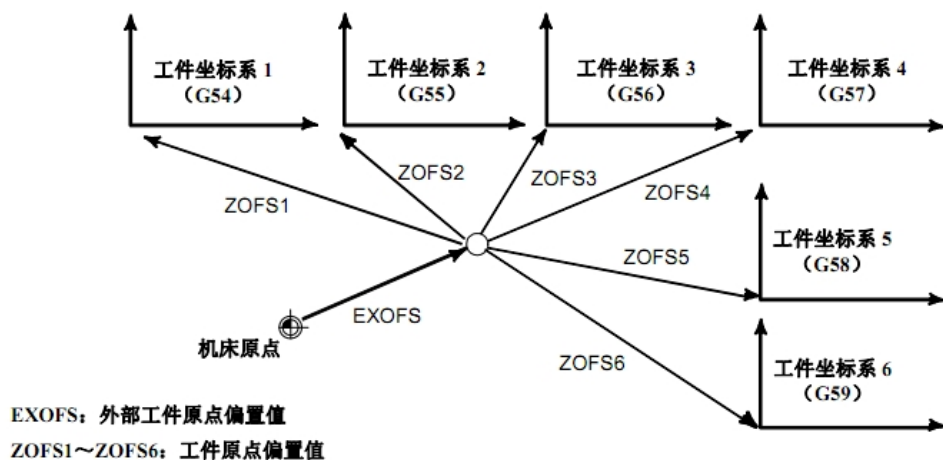
P2 R6 ; The Y-axis workpiece coordinate system offset of G55 is set to 6 mm

The workpiece coordinates of L56 X1Y2Z3; G56 are set to X1Y2Z3

P1 I1.5; the X-axis workpiece coordinate system offset of G56 is added by 1.5mm

G710 ; Turn off programmable data input

(56) 可编程数据输入指令 G10



<1>当调用 L2 时，表示设置外部工件坐标系或 G54—G59 工件坐标系的原点偏移值：

G10 L2 P_ X_ Y_ Z_ A_ B_ ...

地址字 P 为 0 表设置外部工件坐标系的原点偏移值，方法 1：P0 设置外部工件坐标系的原点偏移值；方法 2：在主界面在显示 G53 状态时按 F7 测量对刀设置外部工件坐标系的偏移值；方法 3：在参数坐标系状态，参数 P1-0 工件坐标系当前组[G53-G59]设置为 G53，后面的 P1-1/P1-8 参数表示各个进给轴在外部工件坐标系下的偏移值。

地址字 P 为 1—6 表设置工件坐标系 G54—G59 的原点偏移值。

当前为 G90 时按绝对方式设置，当前为 G91 时按增量方式设置。

调用功能 L2 时 G10 为非模态指令，每行必须编入 G10 L2 指令。

例如：

G10L2P1X0Y0Z0G90 表使 G54 工件坐标系与外部坐标系重合。

G10L2P3X11Y22Z33 表使 G56 工件坐标系与外部坐标系偏移 X11 毫米, Y22 毫米, Z33 毫米。

<2>当调用 L20 时，表示设置 G54.1—G54.48 工件坐标系的原点偏移值：

G10 L20 P_ X_ Y_ Z_ A_ B_ ...

地址字 P 为 1—48 表设置工件坐标系 G54.1—G54.48 的原点偏移值。

当前为 G90 时按绝对方式设置，当前为 G91 时按增量方式设置。

调用功能 L20 时 G10 为非模态指令，每行必须编入 G10 L20 指令。

例如：

G10L20P1X0Y0Z0G90 表使 G54.1 工件坐标系与外部坐标系重合。

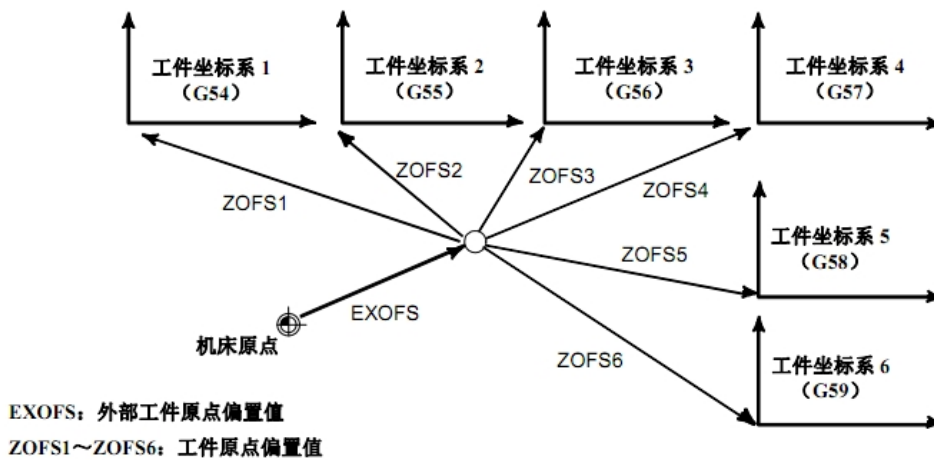
G10L20P3X11Y22Z33 表使 G54.3 工件坐标系与外部坐标系偏移 X11 毫米, Y22 毫米, Z33 毫米。

说明：当调用功能 L2 或 L20 时，指令兼容于发那科系统。

<3>外部坐标系的设置与修改

1. 在参数坐标系状态，选择 G53 修改偏移量；

(56) Programmable Data Input Instruction G10



<1>When calling L2, it indicates the offset value of the origin of the external workpiece coordinate system or G54-G59 workpiece coordinate system:

G10 L2 P_ X_ Y_ Z_ A_ B_ ...

Address parameter P is set to 0 to configure the origin offset value for the external workpiece coordinate system. Method 1: Set the origin offset value for the external workpiece coordinate system using P 0. Method 2: When the G53 status is displayed on the main interface, press F7 to measure and set the offset value for the external workpiece coordinate system during tool measurement. Method 3: In parameter coordinate system mode, set parameters P1-0 to the current workpiece coordinate system group [G53-G59] to G53. Subsequent parameters P1-1/P1-8 indicate the offset values for each feed axis in the external workpiece coordinate system.

Address word P is set to 1-6 to set the origin offset value of the workpiece coordinate system G54-G59.

When the current value is G90, it is set in absolute mode; when the current value is G91, it is set in incremental mode.

When the function L2 is called, G10 is a non-mode instruction, and each line must be programmed with G10 L2 instructions.

for instance :

G10L2P1X0Y0Z0G90 indicates that the G54 workpiece coordinate system coincides with the external coordinate system.

G10L2P3X11Y22Z33 indicates that the G56 workpiece coordinate system is offset from the external coordinate system by X11 mm, Y 22 mm, and Z33 mm.

<2>When L20 is called, it indicates that the origin offset value of G 54.1--G54.48 workpiece coordinate system is set:

G10 L20 P_ X_ Y_ Z_ A_ B_ ...

The address word P is set to 1--48, indicating the workpiece coordinate system G54. The origin offset value ranges from 1--G54.48. When the current mode is G90, it is set in absolute mode; when the current mode is G91, it is set in incremental mode.

When the function L20 is called, G10 is a non-modular instruction, and each line must be programmed with the G10 L20 instruction.

for instance :

G10L20P1X0Y0Z0G90 indicates G54.1 that the workpiece coordinate system coincides with the external coordinate system.

G10L20P3X11Y22Z33 indicates G54.3 workpiece coordinate system offset from external coordinate system by X11 mm, Y22 mm, and Z33 mm.

Note: When function L2 or L20 is called, the instruction is compatible with the FANUC system.

<3>Setting and modifying the external coordinate system

1. In the parameter coordinate system state, select G53 to modify the offset;

2. 在刀补状态，修改外部坐标；

(57) 椭圆指令 (G602/G603)

G17 G602/G603 X(U)_ Y(V)_ I_ J_ Q_

G18 G602/G603 Z(W)_ X(U)_ K_ I_ Q_

G19 G602/G603 Y(V)_ Z(W)_ J_ K_ Q_

I 表椭圆 X 方向半轴长(mm)；J 表椭圆 Y 方向半轴长(mm)；Q 表斜角(单位为度)；无论是直径编程还是半径编程，编程地址 I J K 均指半径值，单位为 mm。

G602 表加工顺时针椭圆， G603 表加工逆时针椭圆。

当 I J K 指定的半轴长均为正值时，表示椭圆对应的圆心角小于等于 180 度，当 I J K 指定的半轴长有一个或两个为负值时，表示椭圆对应的圆心角大于等于 180 度。

Q 表椭圆绕椭圆中心点转动的斜角，按右手坐标系逆时针方向转动为正，单位为度，不编写 Q 表没有转动的斜角。

指令 G602/G603 为模态指令，但 I J K Q 为非模态。

(58) 抛物线指令 (G702)

G17 G702 X(U)_ Y(V)_ R_ Q_ 对应抛物线方程为: $Y^2=2 \cdot R \cdot X$ (X 指半径值)

G18 G702 Z(W)_ X(U)_ R_ Q_ 对应抛物线方程为: $X^2=2 \cdot R \cdot Z$ (X 指半径值)

G19 G702 Y(V)_ Z(W)_ R_ Q_ 对应抛物线方程为: $Z^2=2 \cdot R \cdot Y$

R 表抛物线方程系数，含义表示抛物线焦点相对于抛物线顶点相对距离的两倍，有正负之分，单位为 mm。

Q 表抛物线曲线绕抛物线顶点转动的斜角，按右手坐标系逆时针方向转动为正，单位为度，不编写 Q 表没有转动的斜角。

指令 G702 为模态指令，但 R Q 为非模态。

注意: R 有正负值之分，正负值对应的抛物线曲线朝向相反，当 R 大于零时，抛物线开口朝右，当 R 小于零时，抛物线开口朝左。

抛物线示意图如下：

2. In the tool supplement state, modify the external coordinates;

(57) Ellipse Command (G602/G603)

G17 G602/G603 X(U)_ Y(V)_ I_ J_ Q_

G18 G602/G603 Z(W)_ X(U)_ K_ I_ Q_

G19 G602/G603 Y(V)_ Z(W)_ J_ K_ Q_

I represents the semi-major axis of the elliptical X direction (mm); J represents the semi-major axis of the elliptical Y direction (mm); Q represents the inclination angle (unit: degrees);

Whether it is diameter programming or radius programming, the programming address I J K refers to the radius value, and the unit is mm. G602 is for clockwise elliptical machining, and G603 is for counterclockwise elliptical machining.

When the semi-axes specified by I JK are all positive, it means that the corresponding central angle of the ellipse is less than or equal to 180 degrees; when one or two of the semi-axes specified by IJK are negative, it means that the corresponding central angle of the ellipse is greater than or equal to 180 degrees.

Q is the oblique angle by which an elliptic table rotates around the center point of the ellipse. The rotation in the counterclockwise direction according to the right-handed coordinate system is positive, and the unit is degrees. The oblique angle without rotation is not written in Q table.

Instructions G602/G603 are modal instructions, but I J K Q are non-modal.

(58) Parabolic Command (G702)

The corresponding parabolic equation for G17 G702 X (U)_Y (V)_R_ Q is: $Y^2 = 2 \cdot R \cdot X$ (where X is the radius value)

The corresponding parabolic equation for G18 G702 Z (W)_X(U)_R_ Q is: $X^2 = 2 \cdot R \cdot Z$ (where X is the radius value)

The corresponding parabolic equation for G19 G702 Y (V)_Z (W)_ R_ Q is $Z^2 = 2 \cdot R \cdot Y$

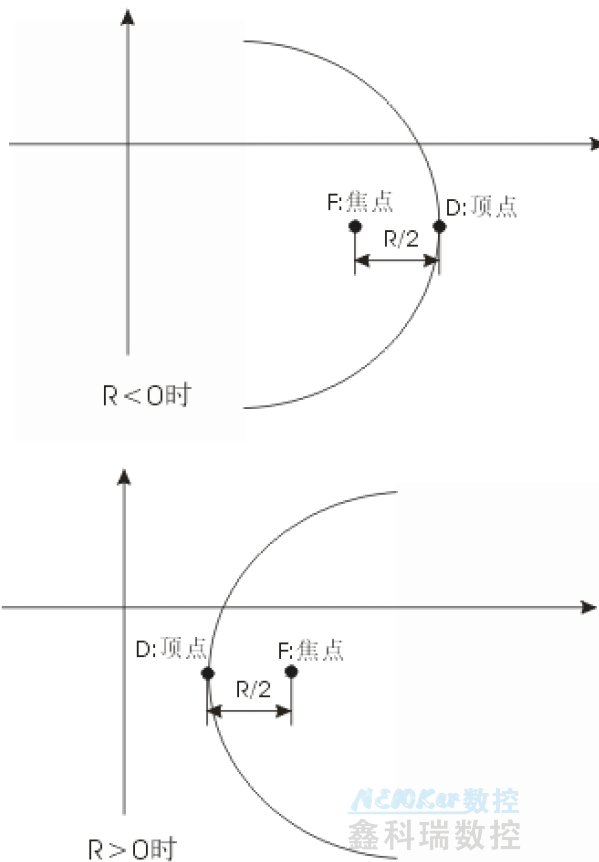
R is the coefficient of the parabola equation. The meaning is that it is twice the relative distance between the focus of the parabola and the vertex of the parabola. It has positive and negative values. The unit is mm.

Q is the rotation Angle of the parabola curve around the vertex of the parabola. If it rotates counterclockwise in the right-handed coordinate system, it is positive and measured in degrees. The rotation Angle without rotation is not written in Q table.

Instruction G702 is a modal instruction, but R Q is non-modal.

Note: R has positive and negative values, and the corresponding parabolic curve has opposite directions. When R is greater than zero, the parabolic curve opens to the right, and when R is less than zero, the parabolic curve opens to the left.

A diagram of the parabola is as follows:



(59) 样条曲线光顺处理指令 (M620/M621)

指令: M620 关闭样条曲线光顺处理插补

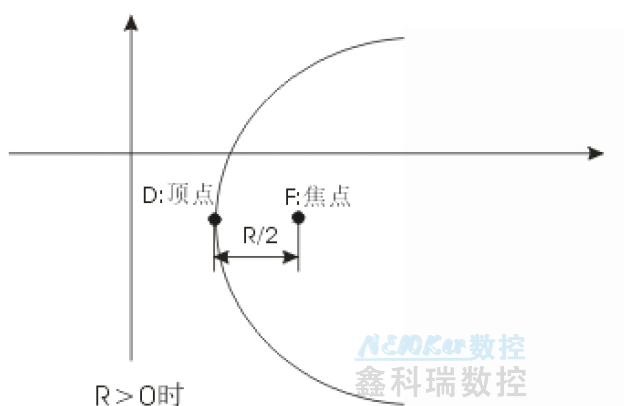
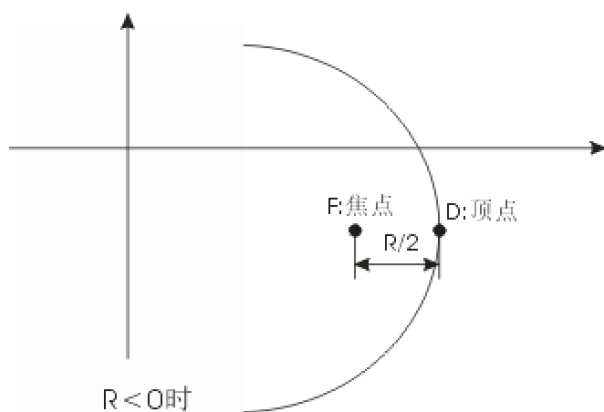
指令: M621 L_ 开启样条曲线光顺处理插补

只是针对连续的 G01 程序段进行处理, 对 G00/G02/G03 等程序段不作处理。
默认对于连续的 5 段以上的 G01 程序才进行光顺处理, 可以通过编写 L 指定最小光顺段数。

例如: 编入 M621 L2 表示即使是只有两段的 G01 程序, 也会进行光顺处理。

```
G0X0Y0Z0
M621 L2
G1X20Y60Z15
X150Y-96Z-36
X75Y-152Z69
```

(60) 倾斜姿态坐标系 G682/G680 (G68.2/G69)



(59) Spline Curve Smoothing Instruction (M620/M621)

Command: M620 turns off the smoothing interpolation of the spline curve

Command: M621 L_Enable smooth interpolation of spline curve light

It only processes continuous G01 program segments, and does not process G00/G02/G03 program segments. By default, smooth processing is performed for continuous G01 program segments of more than 5 segments. The minimum number of smooth segments can be specified by writing L.

For example, entry into M621 L2 means that even a G01 program with only two segments will be smooth.

```
G0X0Y0Z0
M621 L2
G1X20Y60Z15
X150Y-96Z-36
X75Y-152Z69
```

(60) Tilt axis coordinate system G682/G680 (G68.2/G69)

G680/G69 退出任意角度倾斜姿态坐标系状态

G682/G68.2 Q_ X_ Y_ Z_ I_ J_ K_ L8 进入倾斜姿态坐标系状态

Q_ 指定旋转顺序, 后面编 1--3 个数字, 不编 Q 默认 Q123; 例如:

Q123 表依次按绕 X 绕 Y 绕 Z 的顺序旋转;

Q3 表仅仅绕 Z 轴旋转;

Q32 表依次按绕 Z 绕 Y 的顺序旋转;

Q312 表依次按绕 Z 绕 X 绕 Y 的顺序旋转;

X_Y_Z_ 指定倾斜姿态坐标系坐标原点的位置, 以当前工件坐标系方式指定, 不指定则认为是当前位置;

I_J_K_ 指定绕 XYZ 轴的旋转角度(度), 不指定默认 0 度;

旋转原理: 将机床坐标系的原点平移到 X_ Y_ Z_ 指定位置, 再将该坐标系按 Q_ 指定顺序旋转 I_J_K_ 指定角度即可得到倾斜姿态坐标系。

附加功能: 当处于五轴 RTCP 模式时, 该行编入 L8 指令(类似发那科 G53.1 指令) 表该行会自动运行旋转轴, 以实现刀具轴线与倾斜坐标系的 Z 轴方向一致, 当前为 G0 状态就按 G0 速度运行, 否则按 G1 速度运行。

G533 开启法向进刀退刀指令; G530 关闭法向进刀退刀指令;

注意:

<1> 在 G682 时, 所有 XYZ 编程坐标是指在倾斜姿态坐标系下的坐标;

<2> 在非 RTCP 模式和 RTCP 模式下均可使用;

<3> 必须处于 M601/M602 使用 G533 指令才有效;

<4> G43.4/G49 指令可替代 M602/M600 指令;

G68.2/G69 指令可替代 G682/G680 指令;

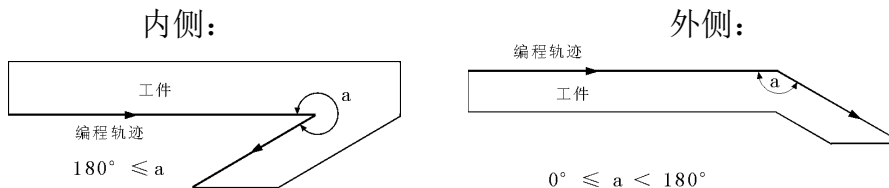
G53.1 与发那科使用方法一样。

2.5 刀具半径补偿 C 的介绍

刀具半径补偿 C 是指系统根据本段程序的上一段和下一段计算出半径补偿后的刀具轨迹。

2.5.1 内侧和外侧概念

当两段程序指令建立的刀具轨迹的夹角超过 180 时称该轨迹为内侧, 当夹角在 0 度和 180 度之间时称为外侧。见下图所示:



2.5.2 起刀时的刀具移动

G680/G69 exits the state of any Angle tilt attitude coordinate system
G682/G68.2 Q_X_Y_Z_I_J_K_L8 Enter the tilt attitude coordinate system state Q_ Specify the rotation order, followed by 1-3 digits, do not compile Q default Q123; for example:

Q123 The table rotates in the order of rotating around X, Y and Z axes;

Q3 The table only rotates around the Z-axis;

Q32 The table rotates in the order of rotating around the Z and Y axes;

Q312 The table rotates in the order of rotating around the Z, X and Y axes;

X_Y_Z_ specifies the position of the origin of the coordinate system in the inclined attitude, which is specified in the current workpiece coordinate system. If not specified, it is considered as the current position;

I_J_K_ specifies the rotation Angle around the XYZ axis (in degrees), not the default 0 degree;

Rotation principle: The origin of the machine tool coordinate system is moved to the specified position of X_Y_Z_, and then the coordinate system is rotated by the specified Angle of I_J_K_ according to the specified order of Q_ to obtain the inclined attitude coordinate system.

Additional functions: When in five-axis RTCP mode, this line is programmed with L8 instructions (similar to FANUC G53.1 instructions). The line will automatically rotate the axis to align the tool axis with the Z-axis direction of the tilting coordinate system. If currently in G0 state, it will operate at G0 speed; otherwise, it will operate at G1 speed.

G533 enables the feed advance/retreat command in the direction; G530 closes the feed advance/retreat command in the direction;

pay attention to :

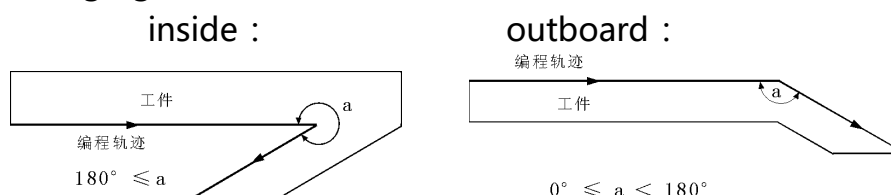
- <1> At G682, all XYZ programming coordinates refer to the coordinates in the oblique attitude coordinate system;
- <2> can be used in both non-RTCP mode and RTCP mode;
- <3> Must be in M601/M602 to use G533 command is valid;
- <4> G43.4/G49 The instruction can replace the M602/M600 instruction;
G68.2 /G69 instruction can replace the G682/G680 instruction; G53.1 is used in the same way as FANUC.

2.5 Introduction to Tool Radius Compensation C

Tool radius compensation C refers to the tool path calculated by the system based on the radius compensation of the previous and next sections of the program.

2.5.1 Internal and External Concepts

When the angle between two program instructions to establish a tool path exceeds 180, it is called the inside path, and when the angle is between 0 degrees and 180 degrees, it is called the outside path. See the following figure:



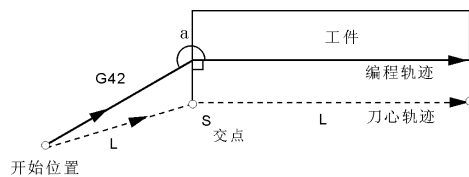
2.5.2 Movement of the Tool at the Time of Starting the Knife

由无刀具半径补偿建立刀具半径补偿。

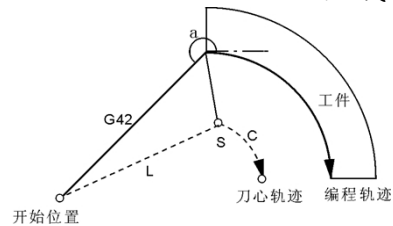
(1) 围绕拐角内侧的刀具移动 ($\alpha \geq 180$)

刀具中心将移至下段程序起点的刀具矢量半径顶点。

直线→直线



直线→圆弧

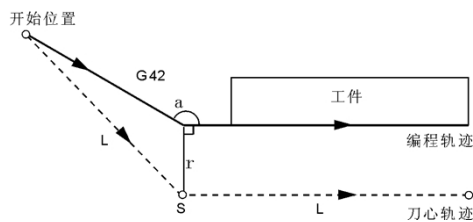


(2) 围绕钝角拐角外侧的刀具移动 ($90 \leq \alpha < 180$)

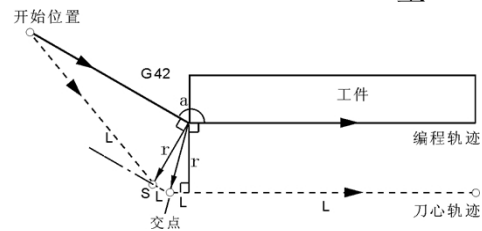
刀具中心将移至本段程序终点的刀具矢量半径顶点。

直线→直线

A型

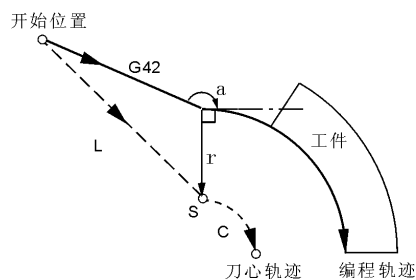


B型

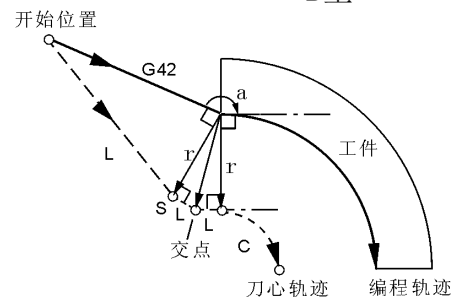


直线→圆弧

A型



B型



(3) 围绕锐角拐角外侧的刀具移动 ($\alpha < 90$)

刀具中心将移至本段程序终点的刀具矢量半径顶点。

直线→直线

A型

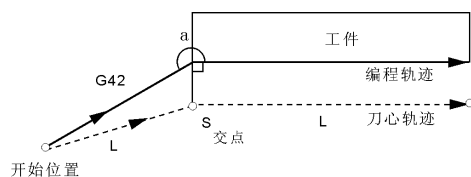
B型

The tool radius compensation is established by no tool radius compensation.

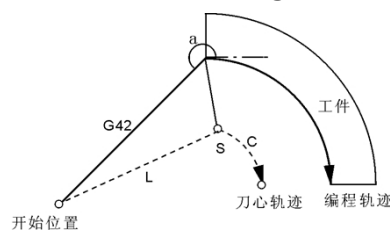
(1) Movement of the Tool Around the Inside of the Corner ($\alpha \geq 180$)

The tool center will be moved to the tool vector radius vertex at the starting point of the lower program.

Straight-> straight



Straight line-> Arc

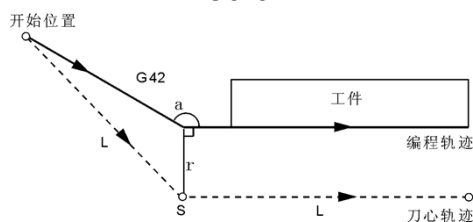


(2) Movement of the Tool Around the Outer Side of the Obtuse Angle ($90 \leq A < 180$)

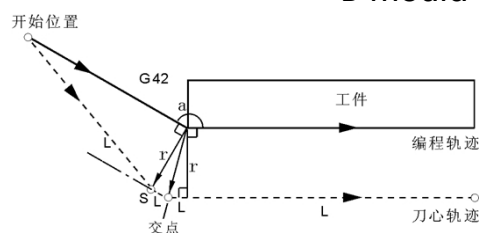
The tool center will move to the tool vector radius vertex at the end of this program.

Straight-> straight

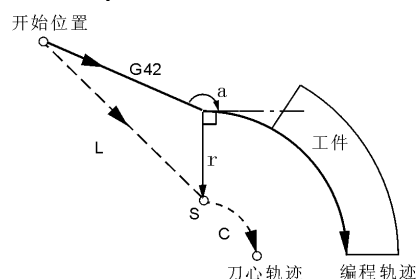
A mould



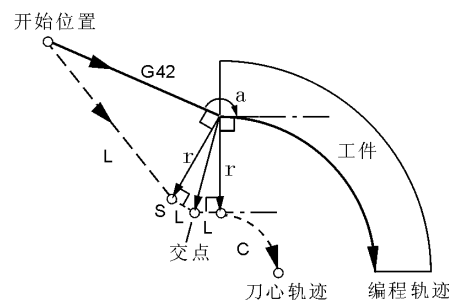
B mould



Straight
line
to
arc
A
type



B mould

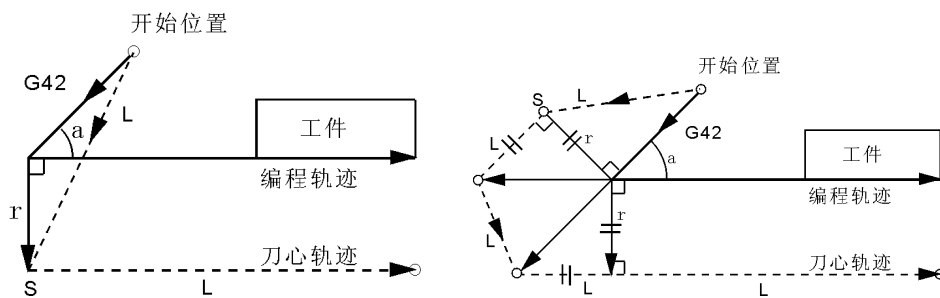


(3) Tool Movement Around the Outer Side of the Sharp Corner ($\alpha < 90$)

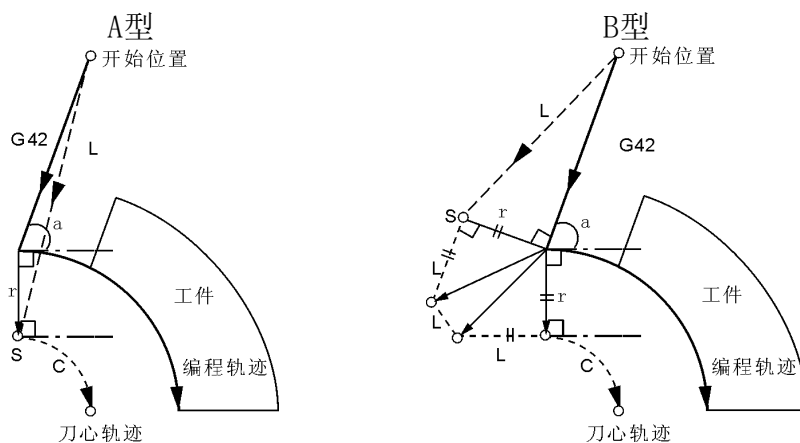
The tool center will move to the tool vector radius vertex at the end of this program segment. Line-> line

A mould

B mould



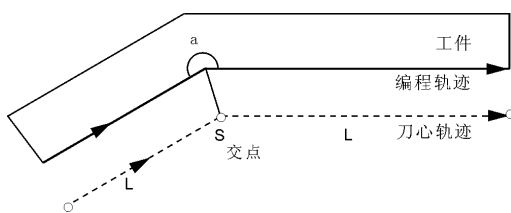
直线→圆弧



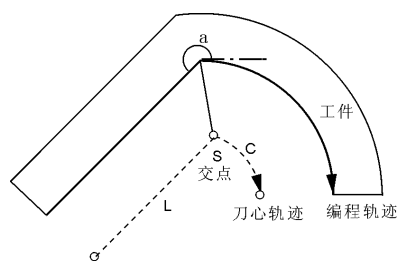
2.5.3 偏置方式中的的刀具移动

(1) 围绕拐角内侧的刀具移动 ($180 \leq \alpha$)

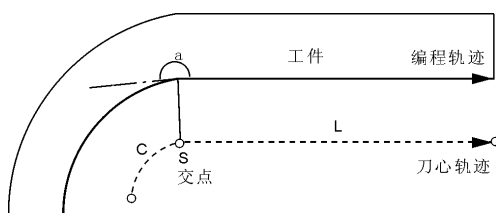
直线→直线



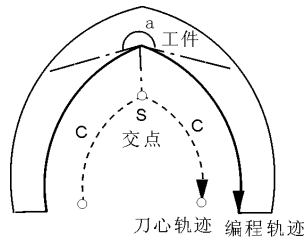
直线→圆弧

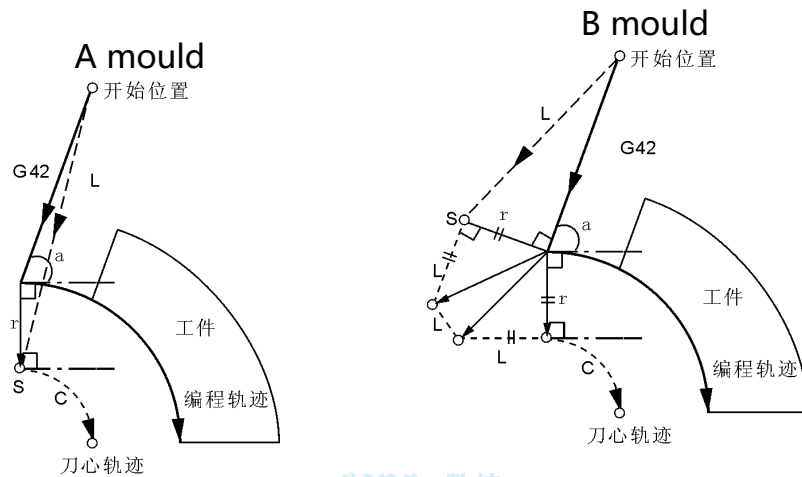
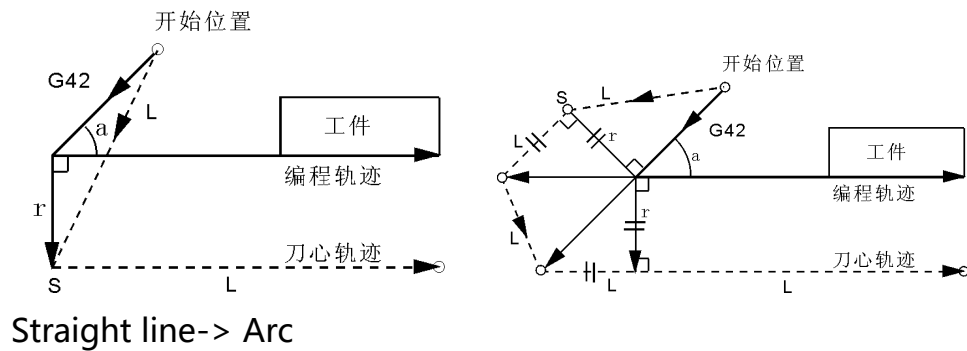


圆弧→ 直线



圆弧→圆弧

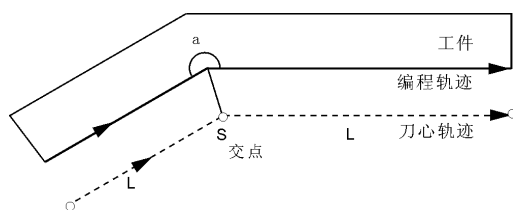




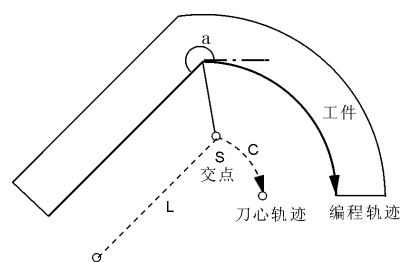
2.5.3 Tool Movement in a Biased Mode

- (1) The tool moves along the inner side of the corner ($180 \leq \alpha$)

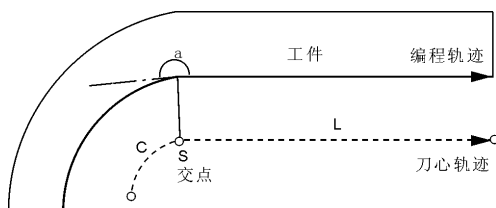
straight-> straight



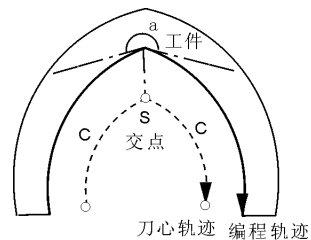
Straight line-> Arc

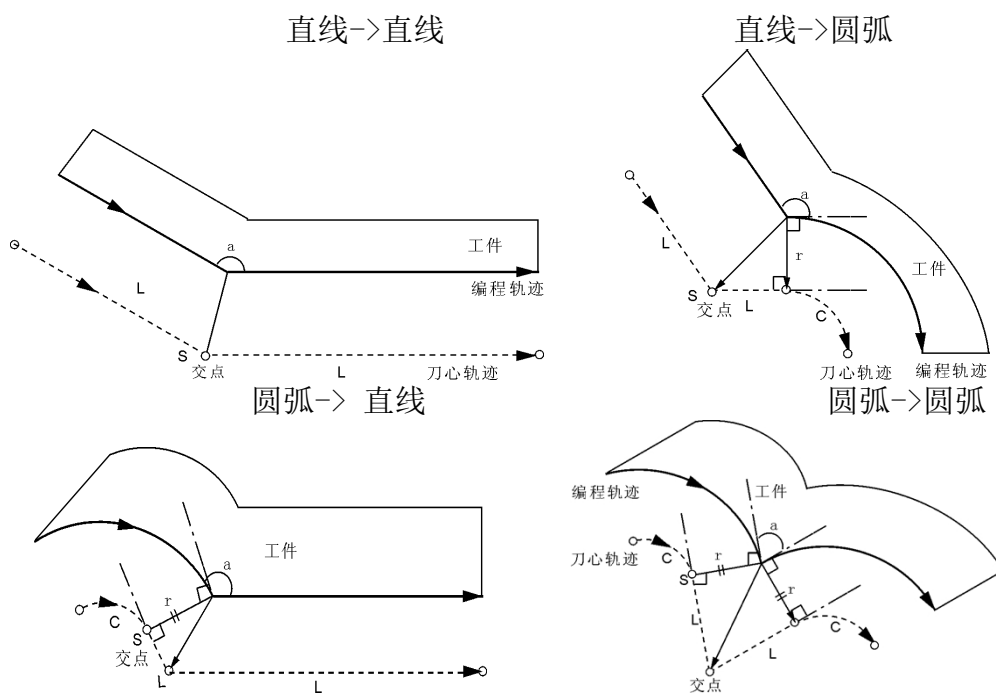
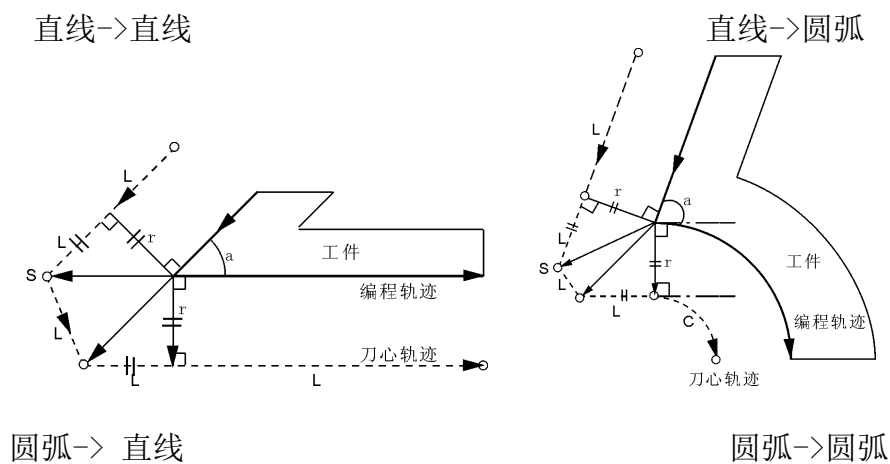


Circular arc-> straight line



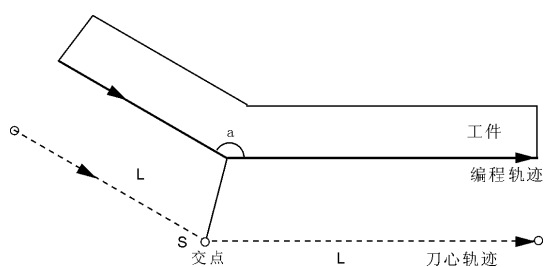
Arc-> arc



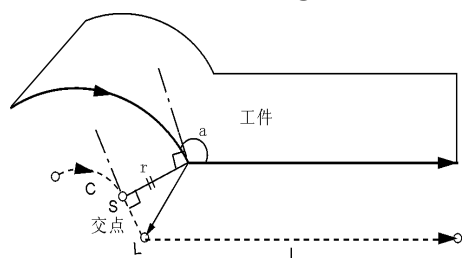
(2) 围绕钝角拐角外侧的刀具移动 ($90 \leq \alpha < 180$)(3) 围绕锐角拐角外侧的刀具移动 ($\alpha < 90$)

(2) Movement of the Tool Around the Outer Side of the Obtuse Angle ($90 \leq A < 180$)

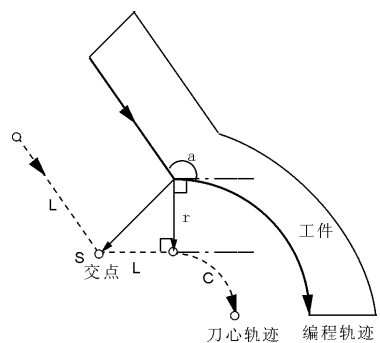
Straight line $\rightarrow$ straight line



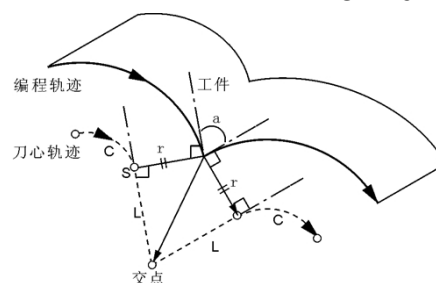
Circular arc $\rightarrow$ straight line



Straight line $\rightarrow$ Arc

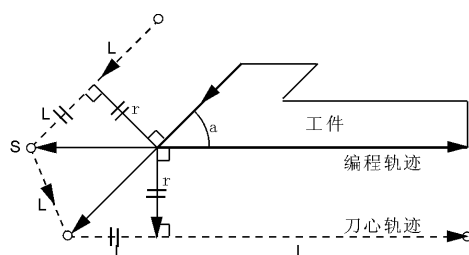


Arc $\rightarrow$ arc



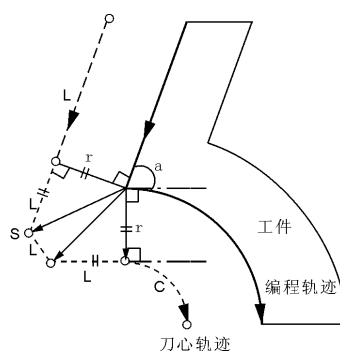
(3) Tool Movement Around the Outer Side of the Sharp Corner ($\alpha < 90$)

Straight line $\rightarrow$ straight line

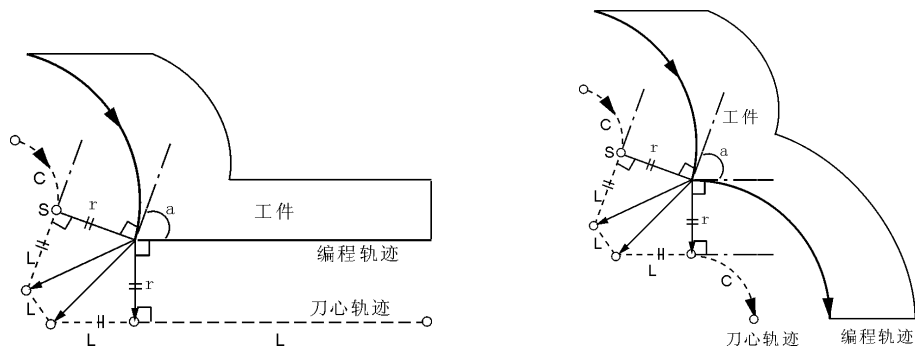


Circular arc $\rightarrow$ straight line

Straight line $\rightarrow$ Arc



Arc $\rightarrow$ arc

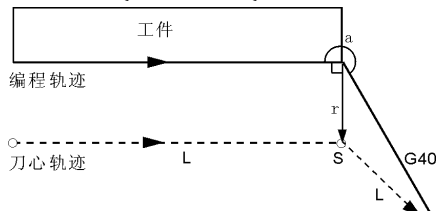


(4) 偏置取消方式中的的刀具移动

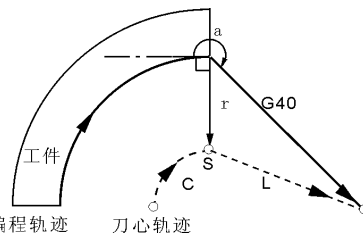
1、围绕拐角内侧的刀具移动 ($180 \leq \alpha$)

刀具中心将移至本段程序终点的刀具矢量半径顶点。

直线→直线



圆弧→直线

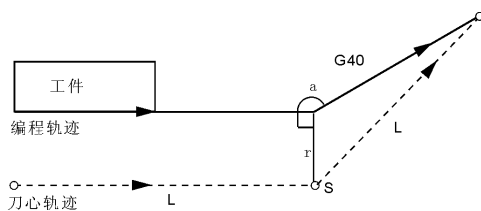


2、围绕钝角拐角外侧的刀具移动 ($90 \leq \alpha < 180$)

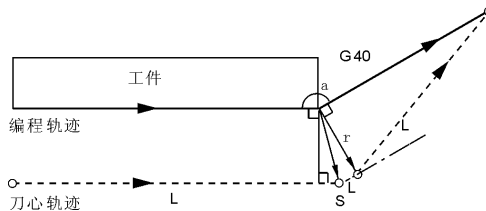
刀具中心将移至下段程序起点的刀具矢量半径顶点。

直线→直线

A型

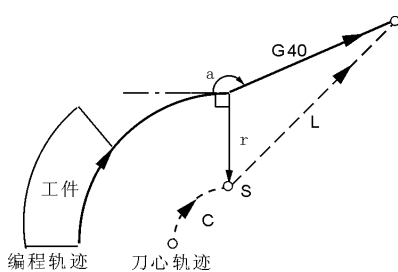


B型

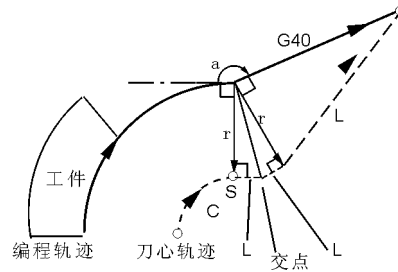


圆弧→直线

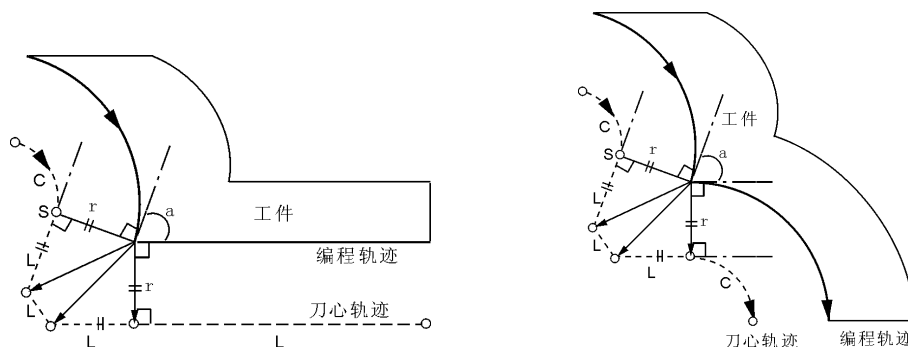
A型



B型



3、围绕锐角拐角外侧的刀具移动 ($\alpha < 90$)



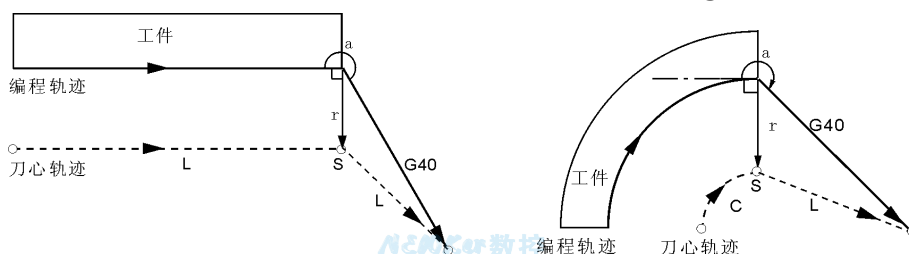
(4) Tool Movement in the Offset Cancellation Mode

1, Tool Movement Around the Inside of the Corner ($180 \leq \alpha$)

The tool center will move to the tool vector radius vertex at the end of this program section.

Straight line $\rightarrow$ straight line

Circular arc $\rightarrow$ straight line



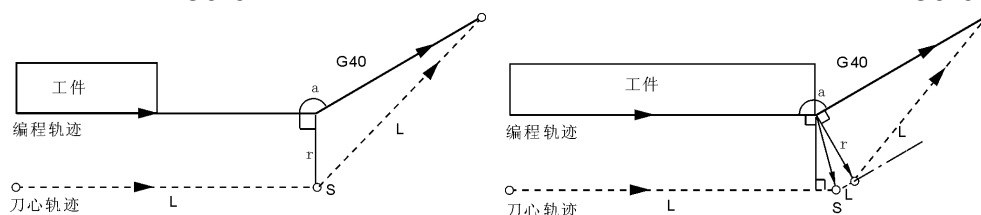
2. Tool Movement Around the Outer Side of Obtuse Angle ($90 \leq A < 180$)

The tool center will move to the tool vector radius vertex at the start of the next program.

Line $\rightarrow$ line

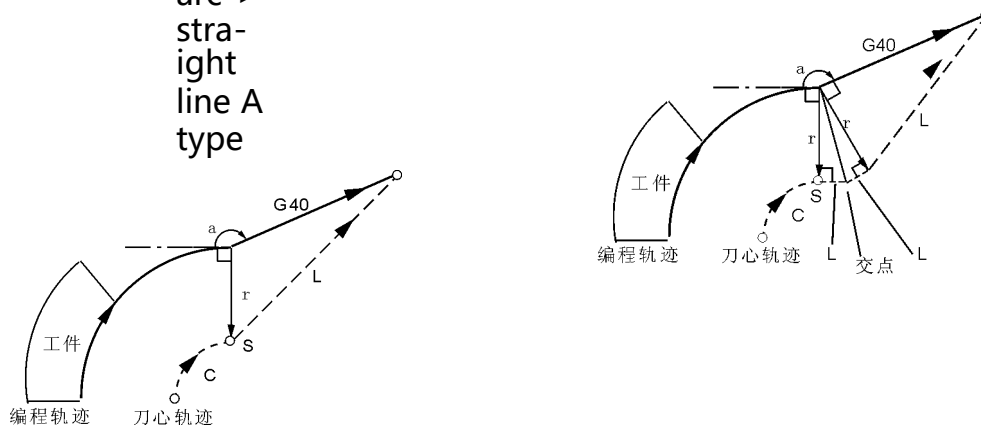
A mould

B mould



Circular
arc $\rightarrow$
stra-
ight
line A
type

B mould

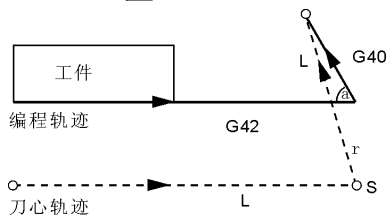


3、 Tool movement around the outer side of the sharp corner ($\alpha < 90$)

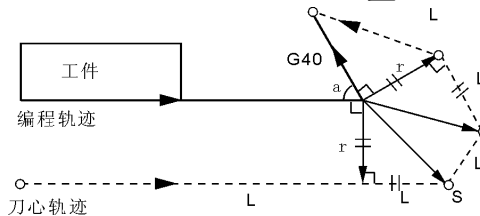
刀具中心将移至下段程序起点的刀具矢量半径顶点。

直线—>直线

A型

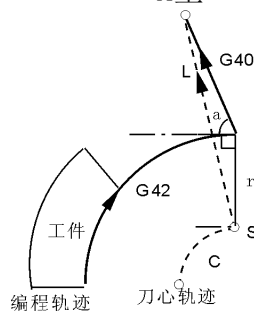


B型

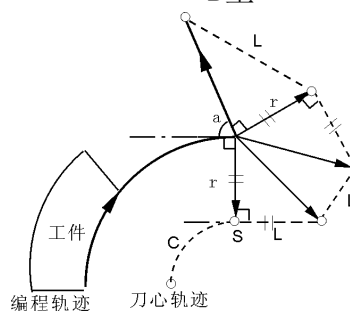


圆弧—>直线

A型



B型

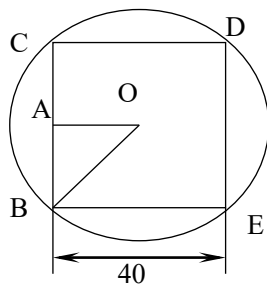


2.6 编程综合实例

NC980Ker数控
鑫科瑞数控

在实际编程时，必须根据图纸及工艺要求，选择正确的装夹方式和适合的刀具，并结合机床的实际工作性能选择正确的切削余量，现举例说明：

例 1：切方和切圆



从中心开始，中心坐标 G54 X0 Y0 Z50

刀具半径在参数 T01 中的 D01

N0 G54 G00 X0 Y0 Z50

；加工起点

N1 M03 S1000

；开主轴

N2 G00 Y-40 X-40 G90

；走出正方形外

N3 Z10

；

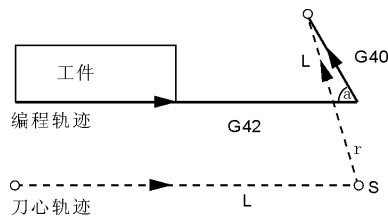
N4 G01 Z-10 F200

；Z轴下刀

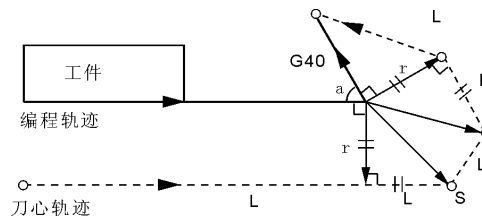
The tool center will move to the tool vector radius vertex at the start of the next program.

Line-> line

A mould

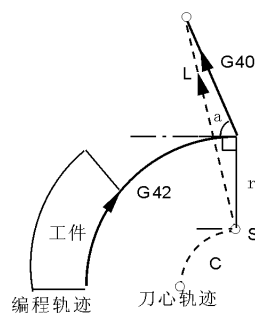


B mould

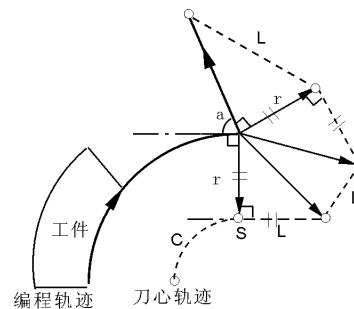


Circular

arc
->
straight
line
A
type



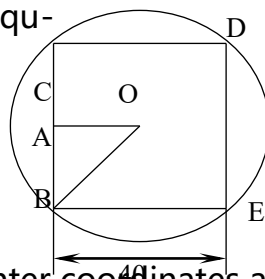
B mould



2.6 Programming Examples

In actual programming, we must choose the correct clamping method and suitable tools according to the drawings and process requirements, and select the correct cutting allowance according to the actual working performance of the machine tool. Now, we will give an example:

Example 1: Cutting a square and a circle



From the center, the center coordinates are G54 X0 Y0 Z50

The tool radius is D01 in parameter T01

N0 G54 G00 X0 Y0 Z50 ; Point of processing

N1 M03 S1000 ; Start the spindle

N2 G00 Y-40 X-40 G90 ; Out of the square

N3 Z10 ;

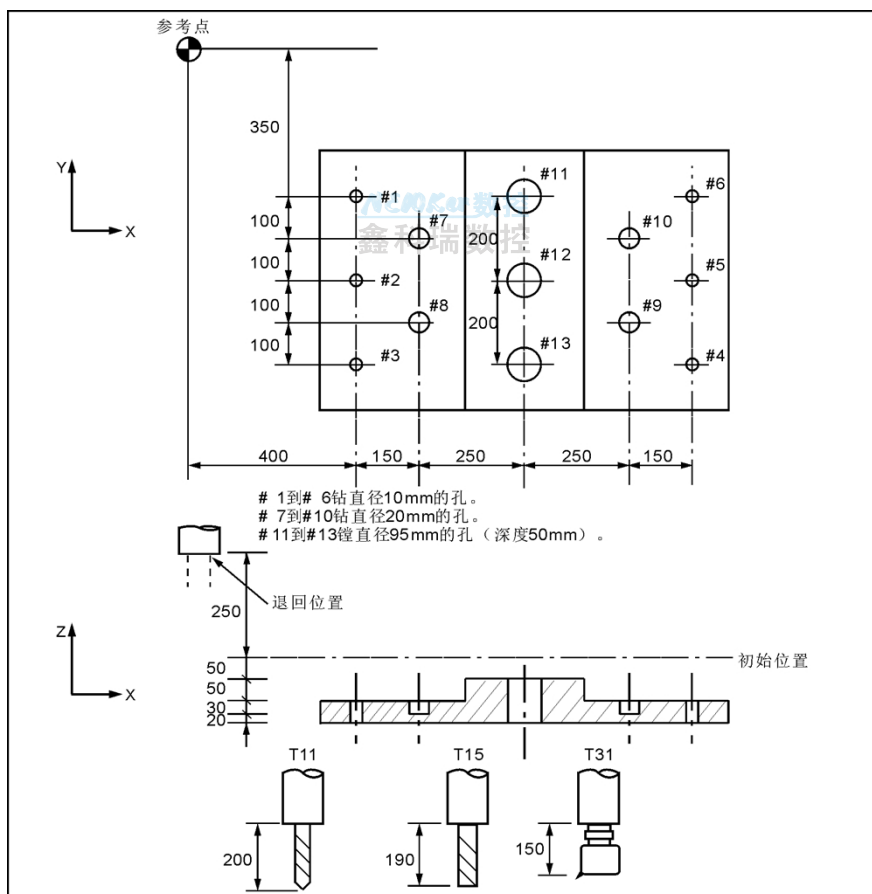
N4 G01 Z-10 F200 ; The Z-axis cuts down

```

N5 G01 G41 T01 D01 X-20 Y-20 F400 ; 走到 B 点
N6 Y20 ; 刀具在工件左侧切 BC 直线
N7 X20 ; C 点尖角过渡, 切 CD 直线
N8 Y-20 ; D 点尖角过渡, 切 DE 直线
N9 X-25 ; E 点尖角过渡, 切 EB 直线为了工件光
      洁度一致, 多走 5mm

N10 G00 X-40 Y-40 G40 ; 走出圆外
N11 G01 G41 X-20 Y-20 F500 ; 从 A-B 径向伸长 T01 值
N12 Z-20 F100 ; Z 轴下刀
N13 G91 G02 I20 J20 Y0 X0 ; 切圆
N14 G00 Z50 ; 提刀
N15 G40 G00 X0 Y0 ; 取消刀补
N16 M05 ; 关主轴
N17 M02 ; 程序结束
  
```

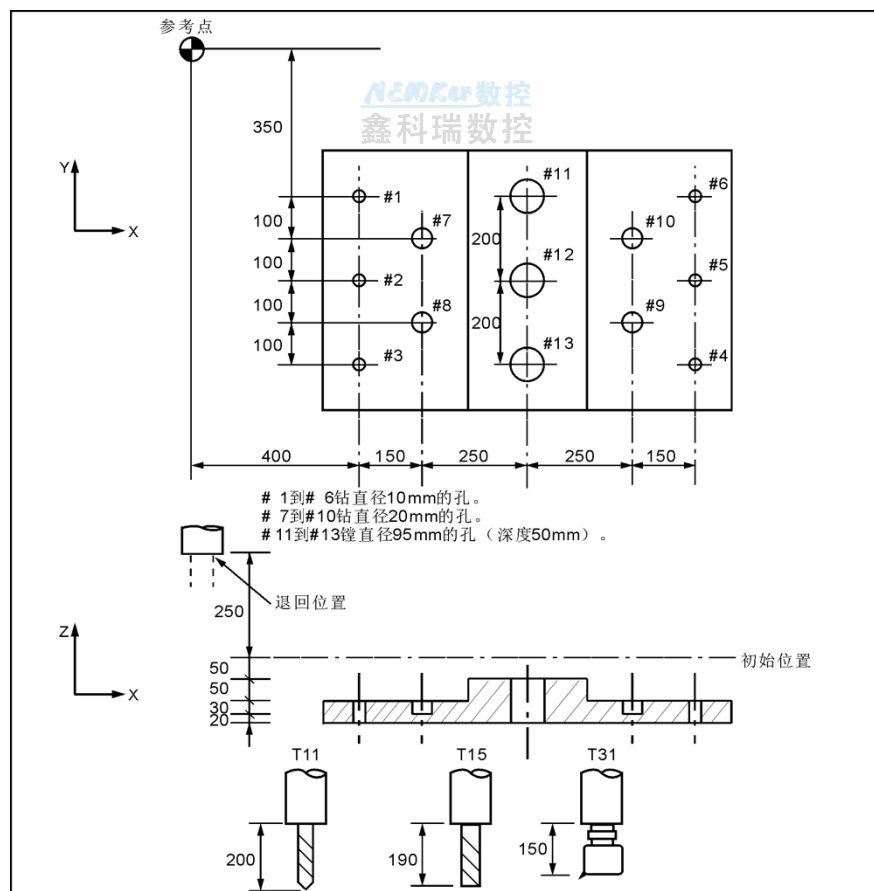
例 2:



假定使用了三把刀具 T11, T15, T31。其刀具长度补偿值分别为 200 (H1), 190 (H1), 150 (H1)。已经在刀具参数设置中输入。根据零件图的加工要求, 可以编写出程序如下:

N5 G01 G41 T01 D01 X-20 Y-20 F400; Walk to point B
N6 Y20
; The tool cuts the BC line on the left side of the workpiece
N7 X20 ; Point C is a sharp transition, cut the CD line
N8 Y-20 ; Point D is a sharp transition, cut the line DE
N9 X-25 ; E point sharp corner transition,
cut EB straight line for the workpiece smooth
The cleanliness is consistent and 5mm more
N10 G00 X-40 Y-40 G40 ; Go out of the circle
N11 G01 G41 X-20 Y-20 F500; radial elongation from A to B at T01 value
N12 Z-20 F100 ; The Z-axis cuts down
N13 G91 G02 I20 J20 Y0 X0 ; circle of contact
N14 G00 Z50 ; griffe
N15 G40 G00 X0 Y0 ; Eliminate knife subsidies
N16 M05 ; Turn the main shaft
N17 M02 ; end of program

Example 2:



Assuming the use of three cutting tools T11, T15, and T31. Their tool length compensation values are 200 (H1), 190 (H1), and 150 (H1) respectively. These parameters have been input into the tool parameter settings. Based on the machining requirements of the part drawing, the program can be written as follows:

N1 G54 X0 Y0 Z0	; 在参考点设置工件坐标
N2 G90 G00 Z250.0 T11	; 刀具交换
N3 G43 Z0 H1	; 初始位置刀具长度偏置
N4 S30 M3	; 主轴起动
N5 G99 G81 X400.0 Y-350.0 Z-153.0 R-97.0 F120	; 定位钻 1 孔
N6 Y-550.0	; 定位钻 2 孔并返回到 R 点位置
N7 G98 Y-750.0	; 定位钻 3 孔并返回到初始位置
N8 G99 X1200.0	; 定位钻 4 孔并返回到 R 点位置
N9 Y-550.0	; 定位钻 5 孔并返回到 R 点位置
N10 G98 Y-350.0	; 定位钻 6 孔并返回到初始位置
N11 G00 X0 Y0 M5	; 返回参考点主轴停止
N12 G49 Z250.0 T15	; 取消刀具长度偏置换刀
N13 G43 Z0 H1	; 初始位置刀具长度偏置
N14 S20 M3	; 主轴起动
N15 G99 G82 X550.0 Y-450.0 Z-130.0 R-97.0 F500	; 定位钻 7 孔返回到 R 点位置
N16 G98 Y-650.0	; 定位钻 8 孔返回到初始位置
N17 G99 X1050.0	; 定位钻 9 孔返回到 R 点位置
N18 G98 Y-450.0	; 定位钻 10 孔返回到初始位置
N19 G00 X0 Y0 M5	; 返回参考点主轴停止
N20 G49 Z250.0 T31	; 取消刀具长度偏置换刀
N21 G43 Z0 H1	; 初始位置刀具长度偏置
N22 S10 M3	; 主轴起动
N23 G85 G99 X800.0 Y-350.0 Z-153.0 R47.0 F150	; 定位镗 11 孔返回到 R 点位置
N24 G91 Y-200.0	; 定位镗 12 孔返回到 R 点位置
N25 Y-200.0	; 定位镗 13 孔返回到 R 点位置
N26 G90 G28 X0 Y0 M5	; 返回参考点主轴停止
N27 G49 Z0 G80	; 取消刀具长度偏置

N1 G54 X0 Y0 Z0	; Set the coordinates of the workpiece with reference points
N2 G90 G00 Z250.0 T11	; Tool change
N3 G43 Z0 H1	; Initial tool length offset position
N4 S30 M3	; Spindle starts
N5 G99 G81 X400.0 Y-350.0 Z-153.0 R-97.0 F120	; Position the drill 1 hole
N6 Y-550.0	; Position the drill 2 holes and return to point R
N7 G98 Y-750.0	; Position three holes and return to the initial position
N8 G99 X1200.0	; Position four holes and return to point R
N9 Y-550.0	; Position five holes and return to point R
N10 G98 Y-350.0	; Position six holes and return to the initial position
N11 G00 X0 Y0 M5	; Return to reference point spindle stop
N12 G49 Z250.0 T15	; Remove the tool length offset to replace the tool
N13 G43 Z0 H1	; Initial tool length offset position
N14 S20 M3	; Main shaft starts
N15 G99 G82 X550.0 Y-450.0 Z-130.0 R-97.0 F500	; Position the drill 7 back to point R
N16 G98 Y-650.0	; Position the drill and return to the initial position
N17 G99 X1050.0	; Position the drill 9 and return to point R
N18 G98 Y-450.0	; Position the drill 10 holes back to the initial position
N19 G00 X0 Y0 M5	; Return to reference point spindle stop
N20 G49 Z250.0 T31	; Remove the tool length offset to replace the tool
N21 G43 Z0 H1	; Initial tool length offset position
N22 S10 M3	; Main shaft starts
N23 G85 G99 X800.0 Y-350.0 Z-153.0 R47.0 F150	; Position boring 11 back to point R
N24 G91 Y-200.0	; Position the boring 12 holes back to the R point position
N25 Y-200.0	; Position the boring 13 hole and return to point R
N26 G90 G28 X0 Y0 M5	; Return to reference point spindle stop
N27 G49 Z0 G80	; Remove tool length bias

N28 M02

；程序停止

例 3：配高速冲床使用方法，本系统 3 个高速输入点；

<1>M22(X27) 检测+M295；

<2>M12(X29) 检测+M297；

<3>M18(X28)检测+M30；

T0101

G00 X0

G22 L50 ；循环次数

WAT+M30 ；冲头上抬到位，检测输入点 M18(X28)

OUT-M30

G0 U1

WAT+M297 ；冲头下冲到位，检测输入点 M12(X29)

OUT-M297

G800

M02

2.7 自动对刀仪使用方法

1、参数说明：

定义自动对刀仪功能的宏变量如下（对应杂类参数 P380—P392）：

#380：自动对刀初始定位点的 X 轴机床坐标值(mm)；

#381：自动对刀初始定位点的 Y 轴机床坐标值(mm)；

#382：自动对刀初始定位点及返回点的 Z 轴机床坐标值(mm)；

#383：自动对刀快速负向移动速度(mm/min)；

#384：自动对刀慢速正向移动速度(mm/min)；

#385：自动对刀后当前工件坐标系下工件上表面的 Z 轴坐标值(mm)；

#386：自动对刀快速到定位点的速度(mm/min)；

#387：自动对刀模式(1 表固定点对刀仪, 0 表浮动点对刀仪)。

#388：Z 轴最低机床坐标值(mm)；

#389：Z 轴落差值[即对刀仪上表面相对于工件上表面的高度](mm)；

#390：自动对刀总的检测次数(1—5)

#391：RTCP 刀具旋转中心到刀柄的距离(mm)

固定点对刀仪是指对刀仪放在一个固定的位置，每次对刀 XYZ 轴都会首先自动运行到该固定点；而浮动点对刀仪则直接向 Z 轴负向运行寻找对刀仪信号。

默认输入点 X25(常闭)作为自动对刀仪检测点输入。

2、指令：M880（对应 ProgramUser0）自动对刀指令。

3、自动对刀步骤：

a) 设置杂类参数 P380—P391；

N28 M02 ; program halt

Example 3: Use method of high-speed punch press, this system has 3 high-speed input points;

```
<M1> M22 (X27) detection + M295;
<M2> M12 (X29) detection + M297;
<M3> M18 (X28)
test + M30; T0101
G00 X0
G22 L50 ; cycle index
WAT+M30 ; Raise the nozzle to position, and detect input point M
18 (X28)
OUT-M30
G0 U1
WAT+M297 ; The punch is down and the input point M12 (X29) is
detected
OUT-M297
G800
M02
```

2.7 How to Use the Automatic Knife Sharpener

1、parameter declaration :

The macro variables defining the automatic tool setter function are as follows (corresponding to miscellaneous parameters P380-P392):

#380: Machine tool coordinate value (mm) of the initial X-axis positioning point of automatic tooling;

#381: Machine tool coordinate value of Y-axis initial positioning point of automatic tooling (mm);

#382: Machine tool coordinate values (mm) of Z-axis initial positioning point and return point for automatic tooling;

#383: Automatic tool cutting speed in negative direction (mm/min);

#384: Automatic tool slow forward movement speed (mm/min);

#385: The Z-axis coordinate value (mm) of the upper surface of the workpiece in the coordinate system of the current workpiece after automatic tooling;

#386: The speed of automatic tool cutting to the positioning point (mm/min);

#387: Automatic tooling mode (1 indicates fixed point tooling instrument, 0 indicates floating point tooling instrument).

#388: Minimum machine tool coordinate value on Z-axis (mm);

#389: Z-axis drop value [that is, the height of the upper surface of the knife instrument relative to the upper surface of the workpiece] (mm);

#390: Total number of automatic tooling tests (1-5)

#391: RTCP Distance between tool rotation center and handle (mm)

A fixed-point tool setter is one in which the tool setter is placed at a fixed position and the XYZ axis of the tool setter automatically moves to that fixed point each time; while a floating-point tool setter moves directly in the negative direction of the Z-axis to find the tool setter signal.

The default input point X25 (normally closed) is used as the input point for automatic tool setter detection.

2 Instruction: M880 (corresponding to ProgramUser0) automatic tool cutting instruction.

3 Automatic tooling step:

a) Set miscellaneous parameters P380-P391;

- b) 自动对刀：MDI 运行 M880 指令，自动设置当前刀具的刀具长度 H1, H2=H1+#391。

2.8 自动分中功能使用方法

1、X 轴分中：M884（对应 ProgramUser4）

- 1) 选择当前坐标系如 G54；
- 2) 手动移动 X 到工件负端；MDI 运行 M884 指令；
- 3) 手动移动 X 到工件正端；MDI 运行 M884 指令，自动分中 X 轴，并将工件中点设置为当前坐标系 X 轴原点，即自动设置当前坐标系如 G54 中 X 轴坐标偏移量；

2、Y 轴分中：M885（对应 ProgramUser5）

- 1) 选择当前坐标系如 G54；
- 2) 手动移动 Y 到工件负端；MDI 运行 M885 指令；
- 3) 手动移动 Y 到工件正端；MDI 运行 M885 指令，自动分中 Y 轴，并将工件中点设置为当前坐标系 Y 轴原点，即自动设置当前坐标系如 G54 中 Y 轴坐标偏移量。

3、内框 X 轴检测自动分中：M886（对应 ProgramUser6）

- 1) 选择当前坐标系如 G54；
- 2) 手动移动分中仪到工件内框中；
- 3) MDI 运行 M886 指令，自动检测输入点 X31，自动分中 X 轴，并将工件中点设置为当前坐标系 X 轴原点，并移动到 X 轴原点；

4、内框 Y 轴检测自动分中：M887（对应 ProgramUser7）

- 1) 选择当前坐标系如 G54；
- 2) 手动移动分中仪到工件内框中；
- 3) MDI 运行 M887 指令，自动检测输入点 X31，自动分中 Y 轴，并将工件中点设置为当前坐标系 Y 轴原点，并移动到 Y 轴原点；

5、外框 XY 对刀仪自动分中, G160/M893/M894（对应 ProgramUser13/14）：

要求手动移动对刀仪到工件中心, 设置工件坐标为 G54X0Y0Z0, 程序如下：

```
#300=600      ;X 方向工件大小
#301=400      ;Y 方向工件大小
#302=10       ;检测长度
#303=20       ;Z 方向抬起点；
#304=-10      ;Z 方向检测点；
G54 G0 X0 Y0 Z10 ;运行到工件中心
G160          ;调用外框分中程序
```

.....

Y31:检测仪使能；

X40: 检测仪准备好信号, 到位信号

X41:检测仪电池电压低；

X42:检测头与接收器之间有遮挡物；

6、内框 XY 对刀仪自动分中, G161/M893/M894（对应 ProgramUser13/14）：

- b) Automatic tool: MDI runs the M880 command, automatically sets the tool length of the current tool H1, $H2=H1+\#391$.

2.8 How to Use the Automatic Partitioning Function

- 1、 In the X-axis section: M884 (corresponding to ProgramUser4)
 - 1) Select the current coordinate system such as G54;
 - 2) Move X manually to the negative end of the workpiece; MDI runs the M884 command;
 - 3) Move X manually to the front end of the workpiece; MDI runs the M884 instruction to automatically center the X-axis and set the midpoint of the workpiece as the origin of the current coordinate system's X-axis, that is, automatically set the current coordinate system as the X-axis offset in G54;
- 2、 In the Y-axis section: M885 (corresponding to ProgramUser5)
 - 1) Select the current coordinate system, such as G54;
 - 2) Move Y manually to the negative end of the workpiece; MDI runs the M885 instruction;
 - 3) Move Y to the front end of the workpiece manually; MDI runs the M885 instruction to automatically center the Y-axis and set the midpoint of the workpiece as the origin of the current coordinate system's Y-axis, that is, automatically set the current coordinate system as the Y-axis offset in G54.
- 3、 Automatic centering of inner frame X-axis detection: M886 (corresponding to ProgramUser6)
 - 1) Select the current coordinate system, such as G54;
 - 2) Manual move the center instrument to the inner frame of the workpiece;
 - 3) MDI runs the M886 command to automatically detect the input point X31, automatically divide the X-axis, set the midpoint of the workpiece as the current coordinate system X-axis origin, and move to the X-axis origin;
- 4、 Automatic centering of inner frame Y-axis detection: M887 (corresponding to ProgramUser7)
 - 1) Select the current coordinate system, such as G54;
 - 2) Manual move the center instrument to the inner frame of the workpiece;
 - 3) MDI runs the M887 command, automatically detects the input point X31, automatically divides the Y-axis, sets the midpoint of the workpiece as the current coordinate system Y-axis origin, and moves to the Y-axis origin;
- 5、 In the automatic centering of the outer frame XY, G160/M893/M894 (corresponding to ProgramUser13/14): manually move the tool setter to the center of the workpiece and set the coordinates of the workpiece as G54X0Y0Z0. The program is as follows:


```
#300=600          ; X-direction workpiece size
#301=400          ; Y direction workpiece size
#302=10           ; Detection length
#303=20           ; Z-direction lifting point;
#304=-10          ; Z direction detection
point; G54 G0 X0 Y0 Z10; Run to
the center of the workpiece G
160          ; Calls the outer frame
subprogram
.....
Y31: Detector enabled;
X40: The detector is ready for the signal and the arrival signal
X41: The battery voltage of the detector is low;
X42: There is an obstacle between the detection head and the receiver;
```
- 6、 In the automatic centering of the inner frame XY, G161/M893/M894 (corresponding to ProgramUser13/14):

要求手动移动对刀仪到工件中心, 设置工件坐标为 G54X0Y0Z0, 程序如下:

```
#300=600      ;X 方向工件大小
#301=400      ;Y 方向工件大小
#302=10       ;检测长度
#303=20       ;Z 方向抬起点;
#304=-10      ;Z 方向检测点;
G54 G0 X0 Y0 Z10 ;运行到工件中心
G161         ;调用外框分中程序
.....
```

Y31:检测仪使能;

X40: 检测仪准备好信号, 到位信号

X41:检测仪电池电压低;

X42:检测头与接收器之间有遮挡物;

注意: 分中指令 M884-M887, M891, M892 必须先将外部坐标偏移量清为 0。

2.9 坐标轴随动功能使用方法

1、方法 1: 轴类参数 P414, 此方法可以不开启随动轴;

此参数设置 7/8/9 表 A 随 X/Y/Z; 17/18/19 表 C 随 X/Y/Z; 27/28/29 表 B 轴随 X/Y/Z; 14/15/16 表 Xs 随 X/Y/Z; 24/25/26 表 Ys 随 X/Y/Z; 23 表 Ys 随 Xs;

另外, 如果 D5..D20=1/表 A 随 X/Y/Z, C 随 X/Y/Z, B 随 X/Y/Z, Xs 随 X/Y/Z, Ys 随 X/Y/Z, Ys 随 Xs, 这样可以实现三轴同时 随动功能。

即 D5=1(+32)表 A 随 X; D6=1(+64)表 A 随 Y; D7=1(+128)表 A 随 Z; D8=1(+256)表 C 随 X; D9=1(+512)表 C 随 Y; D10=1(+1024)表 C 随 Z; D11=1(+2048)表 B 随 X; D12=1(+4096)表 B 随 Y; D13=1(+8192)表 B 随 Z; D14=1(+16384)表 Xs 随 X; D15=1(+32768)表 Xs 随 Y; D16=1(+65536)表 Xs 随 Z; D17=1(+131072)表 Ys 随 X; D18=1(+262144)表 Ys 随 Y; D19=1(+524288)表 Ys 随 Z; D20=1(+1048576)表 Ys 随 Xs。

2、方法 2: 此方法用指令指定哪个轴随动, 手动时不随动, 自动运行时随动, 坐标也随动。

M123: 开启 Y 轴自动时随动于 X 轴运动;

M124: 关闭 Y 轴随动于 X 轴运动;

M125: 开启 A 轴自动时随动于 X 轴运动;

M126: 关闭 A 轴随动于 X 轴运动。

注意: 随动功能开启后, 反向间隙补偿与螺距误差补偿是各轴补偿各轴的;

2.10 四轴/五轴系统 RTCP 功能

1、用户指令:

M601 开启五轴 RTCP 的一个旋转轴补偿功能。

The tool changer is required to be manually moved to the center of the workpiece and the workpiece coordinates are set as G54X0Y0Z0. The program is as follows:

```

#300=600      ; X-direction wo-
               rkpiece size
#301=400      ; Y direction wor-
               kpiece size
#302=10       ; Detection length
#303=20       ; Z-direction lift
               point;
#304=-10      ; Z direction dete-
               ction point;

G54 G0 X0 Y0 Z10; run to the center of the workpiece
G161          ; Calls the outer frame subprogram
.....

```

Y31: Detector enabled;
X40: The detector is ready for the signal and the arrival signal
X41: The battery voltage of the detector is low;
X42: There is an obstacle between the detection head and the receiver;
Note: The external coordinate offset must be cleared to 0 before the instruction M884-M887, M891, and M892 can be used.

2.9 Coordinate Axis Follower Function Usage

1、 Method 1: Shaft parameter P414, this method can not open the follower shaft;

This parameter setting: Table A follows X/Y/Z; Table C follows X/Y/Z; Table B follows X/Y/Z; Table B follows X/Y/Z; Table Xs follows X/Y/Z; Table Ys follows X/Y/Z; Table Ys follows Xs;

In addition, if D5..D20=1/ table A follows X/Y/Z, C follows X/Y/Z, B follows X/Y/Z, Xs follows X/Y/Z, Ys follows X/Y/Z, and Ys follows Xs, the three-axis simultaneous tracking function can be achieved.

D5=1 (+32) in Table A (X); D6=1 (+64) in Table A (Y); D7=1 (+128) in Table A (Z); D8=1 (+256) in Table C (X); D9=1 (+512) in Table C (Y); D10=1 (+1024) in Table C (Z); D11=1 (+2048) in Table B (X); D12=1 (+4096) in Table B (Y); D13=1 (+8192) in Table B (Z); D14=1 (+16384) in Table Xs (X); D15=1 (+32768) in Table Xs (Y); D16=1 (+65536) in Table Xs (Z); D17=1 (+131072) in Table Ys (X); D18=1 (+262144) in Table Ys (Y); D19=1 (+524288) in Table Ys (Z); D20=1 (+1048576) in Table Ys (Xs).

2、 Method 2: This method uses the instruction to specify which axis is followed, not followed when manual operation, followed when automatic operation, and coordinates are also followed.

M123: When Y-axis is automatically opened, it moves along with X-axis;

M124: Close the Y-axis follower to the X-axis movement;

M125: When the A-axis is automatically opened, it moves along with the X-axis;

M126: Close the A-axis follower to the X-axis movement.

Note: After the Sui dynamic function is enabled, the reverse clearance compensation and pitch error compensation are for each axis compensation;

2.10 Four-Axis/five-Axis System RTCP Function

1、 User Instructions:
M601, enables a rotational axis compensation function of the five-axis RTCP.

机床结构型式参数等于 10 时：只对旋转 B 轴进行补偿。

机床结构型式参数等于 11 时：只对旋转 A 轴进行补偿。

机床结构型式参数等于 12 时：只对旋转 A 轴进行补偿。

机床结构型式参数等于 13 时：只对旋转 A 轴进行补偿。

机床结构型式参数等于 14 时：只对旋转 A 轴进行补偿。

机床结构型式参数等于 15 时：只对旋转 B 轴进行补偿。

M602/G43.4 开启五轴 RTCP 的两个旋转轴补偿功能，对两个旋转轴均进行补偿。

M600/G49 关闭五轴 RTCP 功能。

在 M601 或 M602 状态下，手动移动 Z 轴时是沿刀具方向前进或后退，而不是单独机床坐标的 Z 轴移动。

辅助继电器 M1060：循环切换 M600 与 M602 状态通过编制梯形图用按键实现手动快捷切换 M600 与 M602 状态。

2、机床结构型式、坐标轴名称及方向定义规则：

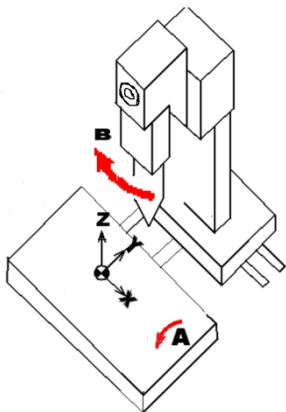
A 轴为绕直线轴 X 轴旋转的轴，旋转正方向满足右手坐标系规则。

B 轴为绕直线轴 Y 轴旋转的轴，旋转正方向满足右手坐标系规则。

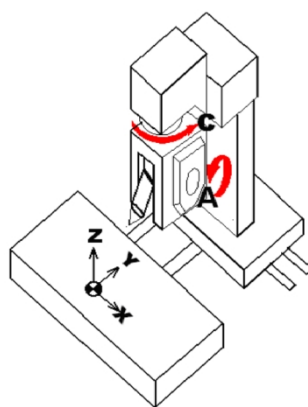
C 轴为绕直线轴 Z 轴旋转的轴，旋转正方向满足右手坐标系规则。

注意：

- 1) 直线轴的运动方向也是以刀具相对于工件的运动方向定义的；
- 2) 与刀具在一起的旋转轴旋转方向是以刀具相对于工件的旋转方向定义的；
- 3) 与工作台在一起的旋转轴旋转方向是以绕工作台自身的旋转方向定义的。



B 轴刀具-A 轴平台五轴机床(10)



AC 轴刀具回转型五轴机床(11)

When the structural type parameter of the machine tool is equal to 10, only the B-axis rotation compensation is performed.

When the structural type parameter of the machine tool is equal to 11, compensation is only performed for the rotating A axis.

When the structural type parameter of the machine tool is equal to 12, compensation is only performed on the rotating A axis.

When the structural type parameter of the machine tool is equal to 13, compensation is only performed on the rotating A axis.

When the structural type parameter of the machine tool is equal to 14, compensation is only performed on the rotating A axis.

When the structural type parameter of the machine tool is equal to 15, only the B-axis rotation compensation is performed.

M602/G43.4 Enable the two rotation axis compensation functions of the five-axis RTCP, and compensate both rotation axes.

M600/G49 Turn off the five-axis RTCP function.

In the M601 or M602 state, manual movement of the Z axis is forward or backward along the tool direction, rather than a separate machine coordinate Z axis movement.

Auxiliary relay M1060: The status of M600 and M602 is switched in a loop. The status of M600 and M602 can be manually switched quickly by programming ladder diagram and using keys.

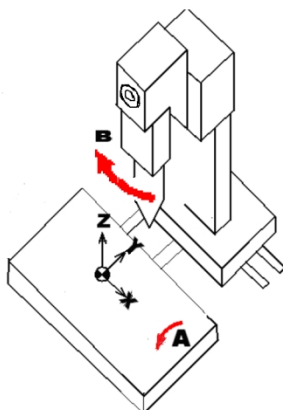
2, Machine Tool Structure Type, Coordinate Axis Name and Direction Definition Rules:

Axis A is the axis that rotates around the straight line axis X, and the rotation direction satisfies the right hand coordinate system rule.

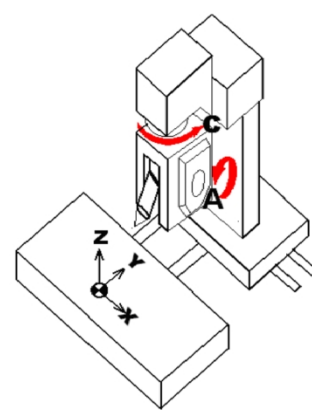
The B-axis is the axis that rotates around the straight-line axis Y-axis, and the rotation direction satisfies the right-hand coordinate system rule.

The C-axis is the axis that rotates around the straight-line axis Z-axis, and the rotation direction satisfies the right-hand coordinate system rule. Note:

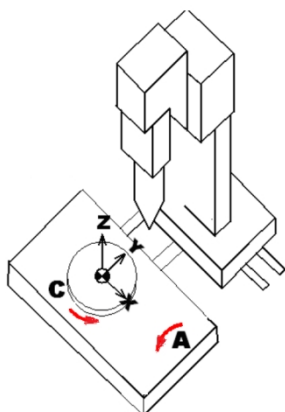
- 1) The direction of motion of the linear axis is also defined by the direction of motion of the tool relative to the workpiece;
- 2) The rotation direction of the rotating shaft associated with the tool is defined by the rotation direction of the tool relative to the workpiece;
- 3) The rotation direction of the rotating shaft associated with the worktable is defined by the rotation direction around the worktable itself.



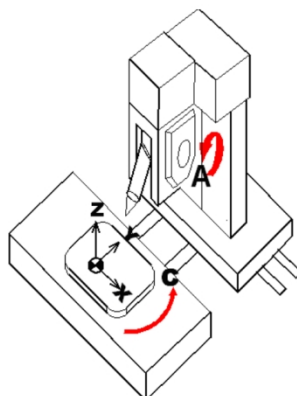
B-axis tool-A axis platform five-axis machine tool (10)



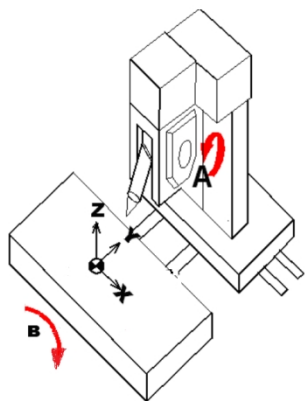
AC axis tool return five-axis machine tool (11)



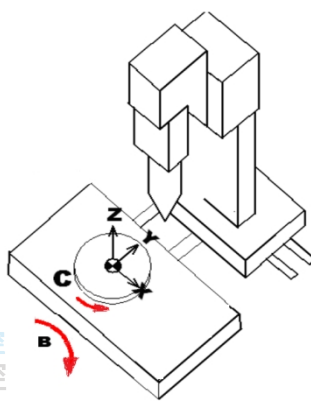
AC 轴刀具平台型五轴机床 (12)



A 轴刀具-C 轴平台五轴机床 (13)



A 轴刀具-B 轴平台五轴机床 (14)



BC 轴刀具平台型五轴机床 (15)



 鑫科瑞数控

3、各种五轴机床的旋转轴机床坐标零点位置定义规则：

B 轴刀具-A 轴平台混和型五轴数控机床 (10)：

方向与 Y 轴正方向一致的位置定义为 A 轴机床坐标零点。

刀尖与 Z 轴负方向一致的位置定义为 B 轴机床坐标零点。

AC 轴刀具回转型五轴数控机床 (11)：

刀尖与 Z 轴负方向一致的位置定义为 A 轴机床坐标零点。

方向与 Y 轴负方向一致的位置定义为 C 轴机床坐标零点。

AC 轴平台回转型五轴数控机床 (12)：

方向与 Y 轴正方向一致的位置定义为 A 轴机床坐标零点。

方向与 X 轴正方向一致的位置定义为 C 轴机床坐标零点。

A 轴刀具-C 轴平台混和型五轴数控机床 (13)：

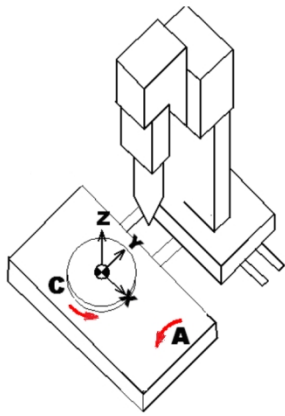
刀尖与 Z 轴负方向一致的位置定义为 A 轴机床坐标零点。

方向与 X 轴正方向一致的位置定义为 C 轴机床坐标零点。

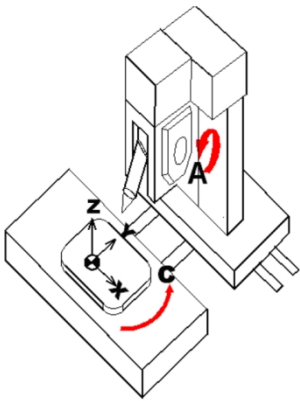
A 轴刀具-B 轴平台混和型五轴数控机床 (14)：

刀尖与 Z 轴负方向一致的位置定义为 A 轴机床坐标零点。

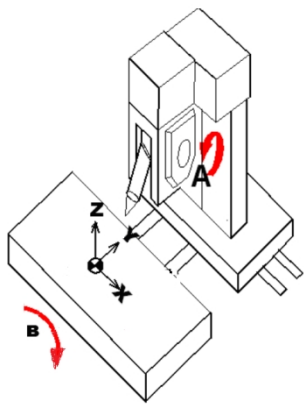
方向与 Z 轴正方向一致的位置定义为 B 轴机床坐标零点。



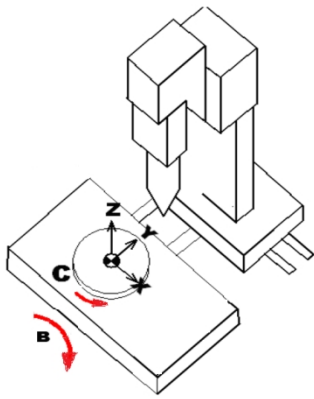
AC axis tooling platform type five axis machine tool (12)



Five-axis machine tool with A-axis tool and C-axis table (13)



Five-axis machine tool with A-axis tool and B-axis table (14)



BC axis tool platform type five axis machine tool (15)

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3, Definition Rules for the Zero Point Position of the Rotating Axis of Various Five-Axis Machine Tools:

B-axis tool-A axis platform hybrid five-axis CNC machine tool (10):

The position aligned with the positive direction of the Y-axis is defined as the A-axis machine coordinate zero point. The position aligned with the negative direction of the Z-axis is defined as the B-axis machine coordinate zero point.

AC axis tool return five-axis CNC machine tool (11):

The position where the tip is aligned with the negative direction of the Z-axis is defined as the A-axis machine coordinate zero point.

The position in the direction of the negative Y-axis is defined as the zero point of the C-axis machine tool.

AC axis platform rotary five-axis CNC machine tool (12):

The position in the positive direction of the Y-axis is defined as the A-axis machine coordinate zero point.

The position that is consistent with the positive direction of the X-axis is defined as the C-axis machine coordinate zero point.

A-axis tool-C-axis platform hybrid five-axis CNC machine tool (13):

The position where the tip of the knife is aligned with the negative direction of the Z axis is defined as the A axis machine coordinate zero point.

The position in the positive direction of the X-axis is defined as the C-axis machine coordinate zero point.

A-axis tool-B-axis platform hybrid five-axis CNC machine tool (14):

The position where the tip of the knife is aligned with the negative direction of the Z axis is defined as the A axis machine coordinate zero point.

The position that is consistent with the positive direction of the Z-axis is defined as the B-axis machine coordinate zero point.

BC 轴平台回转型五轴数控机床 (15):

方向与 Z 轴正方向一致的位置定义为 B 轴机床坐标零点。

方向与 X 轴正方向一致的位置定义为 C 轴机床坐标零点。

4、加工参数:

551, 五轴数控机床结构型式(10—99)

等于 10 时表示 B 轴刀具-A 轴平台混和型五轴数控机床;

等于 11 时表示 AC 轴刀具回转型五轴数控机床 (总线系统);

等于 12 时表示 AC 轴平台回转型五轴数控机床 (总线系统);

等于 13 时表示 A 轴刀具-C 轴平台混和型五轴数控机床 (总线系统);

等于 14 时表示 A 轴刀具-B 轴平台混和型五轴数控机床 (总线系统);

等于 15 时表示 BC 轴平台回转型五轴数控机床 (总线系统);

552, RTCP 补偿反向和坐标系重合 (反向 10/14:B+4, A+8; 11/12/13:A+4, C+8; 坐标系重合+16: [12A 轴, 13C 轴, 14B 轴] 对应编程和工件坐标系重合; +32: 12C 轴对应编程和工件坐标系重合)

类型 10/14(B+4; A+8): B 轴补偿反向+4; A 轴补偿反向+8;

类型 11/12/13(A+4; C+8): A 轴补偿反向+4; C 轴补偿反向+8;

类型 15(B+4; C+8): B 轴补偿反向+4; C 轴补偿反向+8;

+16 表示: 类型 12A 轴/13C 轴/14B/15B 轴的机床坐标对应编程坐标系和工件坐标系重合, 否则表该回转工作台轴的机床坐标为 0 时对应编程坐标系和工件坐标系重合;

+32 表示: 类型 12/15 回转工作台 C 轴的机床坐标对应编程坐标系和工件坐标系重合, 否则表 C 轴的机床坐标为 0 时对应编程坐标系和工件坐标系重合;

553, 五轴数控回转平台中心的机床坐标 1 (um)

机床结构型式参数等于 10 时: A 轴回转中心的 Y 轴的机床坐标。

机床结构型式参数等于 11 时: 无意义。

机床结构型式参数等于 12 时: A 轴回转中心的 Y 轴的机床坐标。

机床结构型式参数等于 13 时: C 轴回转中心的 X 轴的机床坐标。

机床结构型式参数等于 14 时: B 轴回转中心的 X 轴的机床坐标。

机床结构型式参数等于 15 时: B 轴回转中心的 X 轴的机床坐标。

554, 五轴数控回转平台中心的机床坐标 2 (um)

机床结构型式参数等于 10 时: A 轴回转中心的 Z 轴的机床坐标。

机床结构型式参数等于 11 时: 无意义。

机床结构型式参数等于 12 时: A 轴回转中心的 Z 轴的机床坐标。

机床结构型式参数等于 13 时: C 轴回转中心的 Y 轴的机床坐标。

机床结构型式参数等于 14 时: B 轴回转中心的 Z 轴的机床坐标。

机床结构型式参数等于 15 时: B 轴回转中心的 Z 轴的机床坐标。

555, 测量五轴数控回转平台中心时的刀长 (um)

机床结构型式参数等于 10 时: 测量 A 轴回转中心时的刀长。

机床结构型式参数等于 11 时: 无意义。

机床结构型式参数等于 12 时: 测量回转平台中心时的刀补值 H1

BC axis platform rotary five-axis CNC machine tool (15):

The position that is consistent with the positive direction of the Z-axis is defined as the B-axis machine coordinate zero point.

The position in the positive direction of the X-axis is defined as the C-axis machine coordinate zero point.

4. Processing Parameters:

551, five-axis CNC machine tool structure type (10-99)

When it is equal to 10, it means B-axis tool-A axis platform hybrid five-axis CNC machine tool;

When it is equal to 11, it is an AC axis tool return five-axis CNC machine tool (bus system);

When it is equal to 12, it means AC axis platform rotary five-axis CNC machine tool (bus system);

When it is equal to 13, it means that the A-axis tool-C-axis platform hybrid five-axis CNC machine tool (bus system);

When it is equal to 14, it indicates the A-axis tool-B-axis platform hybrid five-axis CNC machine tool (bus system);

when it is equal to 15, it indicates the BC-axis platform rotary five-axis CNC machine tool (bus system);

552. RTCP compensation for reverse and coordinate system alignment (reverse: 10/14: B+4, A+8; 11/12/13: A+4, C+8; coordinate system alignment +16: [12A-axis, 13C-axis, 14B-axis] corresponds to programming and workpiece coordinate system alignment; +32:12C-axis corresponds to programming and workpiece coordinate system alignment)

Type 10/14 (B+4; A+8): B-axis compensation reverse +4; A-axis compensation reverse +8;

Type 11/12/13 (A+4; C+8): A-axis compensation reverse +4; C-axis compensation reverse +8;

Type 15 (B+4; C+8): B-axis compensation reverse +4; C-axis compensation reverse +8;

+16 indicates: The machine tool coordinates of type 12A axis/13C axis/14B/15B axis correspond to the programming coordinate system and the workpiece coordinate system coincide, otherwise the machine tool coordinates of the rotary table axis are 0 when the programming coordinate system and the workpiece coordinate system coincide;

+32 indicates: Type 12/15, the machine tool coordinate corresponding to the rotary worktable C-axis coincides with the programming coordinate system and the workpiece coordinate system; otherwise, the machine tool coordinate corresponding to the rotary worktable C-axis is 0 when the programming coordinate system and the workpiece coordinate system coincide;

553, machine tool coordinates at the center of the five-axis CNC rotary table 1 (um)

When the structural type parameter of the machine tool is equal to 10: the machine tool coordinates on the Y-axis of the A-axis rotation center.

When the structural type parameter of the machine tool is equal to 11, it is meaningless.

When the structural type parameter of the machine tool is equal to 12: the machine tool coordinates on the Y-axis of the A-axis rotation center.

When the structural type parameter of the machine tool is equal to 13: the machine tool coordinate of the X axis of the C-axis rotation center.

When the structural type parameter of the machine tool is equal to 14, it is the machine tool coordinate of the X axis of the B-axis rotation center.

When the structural type parameter of the machine tool is equal to 15, it is the machine tool coordinate of the X axis of the B-axis rotation center.

554, machine tool coordinates 2 (um) at the center of the five-axis CNC rotary table

When the structural type parameter of the machine tool is equal to 10: the Z axis coordinate of the A axis rotation center.

When the structural type parameter of the machine tool is equal to 11, it is meaningless.

When the structural type parameter of the machine tool is equal to 12: the machine tool coordinates of the Z axis at the center of rotation of the A axis.

When the structural type parameter of the machine tool is equal to 13: the machine tool coordinates on the Y-axis of the C-axis rotation center.

When the structural type parameter of the machine tool is equal to 14, it is the machine tool coordinate of the Z axis of the B-axis rotation center.

When the structural type parameter of the machine tool is equal to 15, it is the machine tool coordinate of the Z axis of the B-axis rotation center.

55, measuring the tool length (um) when the center of the five-axis CNC rotary platform is measured

When the structural type parameter of the machine tool is equal to 10, measure the tool length when the A axis rotation center is turned.

When the structural type parameter of the machine tool is equal to 11, it is meaningless.

When the structural type parameter of the machine tool is equal to 12, the tool compensation value H1 is measured when the center of the rotary platform is measured

机床结构型式参数等于 13 时：测量 C 轴回转中心时的刀长。

机床结构型式参数等于 14 时：测量 B 轴回转中心时的刀长。

机床结构型式参数等于 15 时：测量回转平台中心时的刀补值 H1

556, AC/BC 轴平台回转型五轴机床的 C 轴回转中心 X/Y 轴机床坐标 (um)

机床结构型式参数等于 12 时此参数有效。

557, AC 轴刀具回转型的 AC 轴线偏移量 (um)

机床结构型式参数等于 11 时此参数有效在 AC 轴的机床坐标零点位置 A 轴的轴心线在 C 轴的轴心线前面时 (Y 轴方向)，偏移值 (Y 轴方向) 设为负值，反之设为正值。

558, AC 轴刀具回转型的 C 轴与主轴在 X 方向上的偏移量 (um)

机床结构型式参数等于 11 时此参数有效，在 AC 轴的机床坐标零点位置，C 轴的轴心线与主轴轴心线的偏移值 (X 轴方向)，可以设置为正负。

5、RTCP 对刀操作

在系统刀补界面按 ‘H’ 用于自动测量刀长：

注意：对刀过程中，C 轴要保持在 0 度位置，确保 A 轴正好是绕 X 旋转。

(1) 机床结构型式参数等于 10 时：

用于自动测量刀尖点到 B 轴回转中心的距离。

首先换刀选择刀具号，让刀尖点处于工件上某一标记点，按 ‘H’ 输入 1 设定的第一个点，然后将刀具提起来，转动一下 B 轴，转动角度不能太小，再慢慢移动 X 轴和 Z 轴，让刀尖点处于工件上的同一标记点，按 ‘H’ 输入 2 设定的第二个点，系统则自动计算出刀尖点到 B 轴回转中心的距离 (um)，并将其保存到当前光标处的 H2 中。系统进行 RTCP 的 B 轴补偿计算时会自动调用当前刀具号对应的 H2 值。

(2) 机床结构型式参数等于 11 或 13 或 14 时：

用于自动测量刀尖点到 A 轴回转中心的距离。

首先换刀选择刀具号，让刀尖点处于工件上某一标记点，按 ‘H’ 输入 1 设定的第一个点，然后将刀具提起来，转动一下 A 轴，转动角度不能太小，再慢慢移动 Y 轴和 Z 轴，让刀尖点处于工件上的同一标记点，按 ‘H’ 输入 2 设定的第二个点，系统则自动计算出刀尖点到 A 轴回转中心的距离 (um)，并将其保存到当前光标处的 H2 中。系统进行 RTCP 的 A 轴补偿计算时会自动调用当前刀具号对应的 H2 值。更换刀具时就需要进行该距离的测量。因为 H2 用于保存刀尖点到回转中心的距离，作为计算 RTCP 补偿的依据，所以 G43/G44 刀具长度补偿就不能调用 H2，只能使用每一把刀的 H1。当需要下刀或提刀修改 G54 的 Z 偏移值或每一把刀的 H1，不能修改每一把刀的 H2，注意 H2 反应的是真实的刀具长度值，必须与实际一致，否则补偿计算就不正确。程序中需要编入当前选那一把刀，系统才会调用相应刀具的 H2 值。刀具长度补偿 G43/G44 可以与 RTCP 补偿 M601/M602 指令同时使用。

如果对刀不准或者刀具磨损了，那么就需要同时对 H1 和 H2 进行增量修改：

在刀补界面修改长度补偿值时，在前面多输入 H 字符表修改 RTCP 的 H2 刀具长度值。

When the structural type parameter of the machine tool is equal to 13, measure the tool length when the C-axis rotation center is turned.

When the structural type parameter of the machine tool is equal to 14, measure the tool length when the B axis rotation center is measured.

When the structural type parameter of the machine tool is equal to 15, measure the tool compensation value H1 when the center of the rotary platform is measured

556, AC/BC axis platform rotary five-axis machine tool C-axis rotation center X/Y machine tool coordinates (um) the machine tool structure type parameter is equal to 12 when this parameter is valid.

557 AC axis tool return-to-center line offset (um)

When the machine tool structural type parameter is equal to 11, this parameter is valid. At the zero point position of the machine tool coordinate on the AC axis, when the A-axis axis line is in front of the C-axis axis line (Y-axis direction), the offset value (Y-axis direction) is set to a negative value, otherwise it is set to a positive value.

558, AC tool return type C axis and spindle offset in the X direction (um)

This parameter is valid when the machine tool structural type parameter equals 11. At the zero point position of the machine tool coordinate on the AC axis, the offset value (X direction) between the C-axis axis line and the spindle axis line can be set as positive or negative.

5、 RTCP, knife operation

Press "H" on the system tool supplement interface to automatically measure the tool length:

Note: During the cutting process, keep the C axis at 0 degrees and make sure that the A axis is rotating around X.

(1) When the structural type parameter of the machine tool is equal to 10:

Used to automatically measure the distance from the tip point to the B-axis rotation center.

First, switch to the tool number and position the tool tip at a designated mark on the workpiece. Press 'H' to input point 1 of the first setup. Lift the tool and rotate the B-axis slightly (avoid excessive rotation). Slowly adjust the X and Z axes until the tool tip aligns with the same mark. Press 'H' again to input point 2. The system automatically calculates the distance (um) from the tool tip to the B-axis rotation center and stores it in H2 at the current cursor position. During RTCP B-axis compensation calculations, the system automatically retrieves the corresponding H2 value from the current tool number.

(2) When the structural type parameter of the machine tool is equal to 11,13 or 14:

Used to automatically measure the distance from the tip point to the A-axis rotation center.

First, switch the tool and select its number. Position the tool tip at a designated mark on the workpiece. Press 'H' to input point 1 (the first preset position). Lift the tool and rotate the A-axis slightly (avoid excessive rotation), then gradually adjust the Y-axis and Z-axis until the tool tip aligns with the same mark. Press 'H' again to input point 2. The system automatically calculates the distance (um) from the tool tip to the A-axis rotation center and stores it in H2 at the current cursor position. During RTCP A-axis compensation calculations, the system automatically retrieves the corresponding H2 value for the selected tool. This measurement is essential when changing tools, as H2 stores the actual distance to the rotation center for RTCP compensation. Therefore, G43/G44 tool length compensation cannot use H2 but must employ each tool's H1. When modifying G54 Z offsets or individual tool H1 during tool lowering/raising operations, note that H2 reflects the actual tool length value which must match reality; otherwise, compensation calculations will be incorrect. The system only activates the appropriate H2 value based on the currently selected tool. Tool length compensation G43/G44 can be used concurrently with RTCP compensation commands M601/M602.

If the tool is not accurate or worn, then incremental modifications to both H1 and H2 are required:

When modifying the length compensation value in the tool supplement interface, input H character table to modify the RTCP H2 tool length value.

例如：按下 Enter 后在对话框输入 HE123.56 表按绝对方式修改 H2 为 123.56 毫米按下 Enter 后在对话框输入 H8.6 表按增量方式修改 H2 为原始值加上 8.6 毫米。

在系统主界面按 ‘R’ 用于自动测量回转平台中心的机床坐标值，即自动确定轴参数 553, 554, 555, 和 556 号：

(1) 机床结构型式参数等于 10 时：

自动测量 A 轴回转中心 Y(P553)和 Z(P554) 机床坐标。

首先换刀选择刀具号，将 B 轴的机床坐标走到零点，即 B 轴垂直于 X 轴，然后移动 Y 轴和 Z 轴使刀尖处于工件(装夹在 A 轴上)上某一标记点，按 ‘R’ 输入 1 设定第一个点，然后将刀具提起来，转动一下 A 轴，转动角度尽量大一点，再慢慢移动 Y 轴和 Z 轴，让刀尖点处于工件上的同一标记点，按 ‘R’ 输入 2 设定第二个点，依次方法设定第三个点，系统会自动计算出 A 轴回转中心的 Y 和 Z 机床坐标值，并且记录当前刀长到参数“测量五轴数控回转平台中心时的刀长(um)”中。刚刚装机或重新安装 A 轴时才需要测量 A 轴回转中心的 Y 和 Z 机床坐标值，更换刀具时不需要进行该操作。自动测量 A 轴回转中心之前必须先测量好刀尖点到 B 轴回转中心的距离才行。

参数“555, 测量五轴数控回转平台中心时的刀长(um)”记录的当前刀具的 H2 刀具长度值。

(2) 机床结构型式参数等于 11 时：

不需要该操作。

(3) 机床结构型式参数等于 12 时：

第一步：自动测量 A 轴回转中心 Y(P553)和 Z(P554) 机床坐标。

首先换刀选择刀具号，然后移动 Y 轴和 Z 轴使刀尖处于工件(装夹在 A 轴上)上某一标记点，按 ‘R’ 输入 1 设定第一个点，然后将刀具提起来，转动一下 A 轴，转动角度尽量大一点，再慢慢移动 Y 轴和 Z 轴，让刀尖点处于工件上的同一标记点，按 ‘R’ 输入 2 设定第二个点，依次方法设定第三个点，系统会自动计算出 A/B 轴回转中心的 Y 和 Z 机床坐标值。对于有多把刀具时，需要将长度补偿值 H1 设置好后，再进行该项操作。

参数“555, 测量五轴数控回转平台中心时的刀长(um)”记录的当前刀具的 H1 补偿值。如果不使用刀具长度补偿 G43 时，需要将长度补偿值 H1 清零。刚刚装机或重新安装 A 轴时才需要测量 A 轴回转中心的 Y 和 Z 机床坐标值，更换刀具时不需要进行该操作。

第二步：用于自动测量 C 轴回转中心的 X(P556) 机床坐标值。

首先换刀选择刀具号，将 A 轴的机床坐标走到零点，即 C 回转轴处于水平位置，然后移动 X 轴和 Y 轴使刀尖处于工件(装夹在 C 轴上)上某一标记点，按 ‘R’ 输入 A 设定第一个点，然后将刀具提起来，转动一下 C 轴，转动角度尽量大一点，再慢慢移动 X 轴和 Y 轴，让刀尖点处于工件上的同一标记点，按 ‘R’ 输入 B 设定第二个点，依次方法输入 C 设定第三个点，系统会自动计算出 C 轴回转中心的 X 机床坐标值(P556)。

刚刚装机或重新安装 C 轴时才需要测量 C 轴回转中心的 X 机床坐标值，更换

For example, press Enter and enter HE123.56 in the dialog box to modify H2 to 123.56 mm in absolute mode; press Enter and enter H8.6 in the dialog box to modify H2 to the original value plus 8.6 mm in incremental mode.

Press "R" on the main interface of the system to automatically measure the machine tool coordinates at the center of the rotary table, that is, automatically determine the axis parameters 553,554,555 and 556:

(1) When the structural type parameter of the machine tool is equal to 10:

Automatically measure the A-axis rotation center Y (P553) and Z (P554) machine tool coordinates.

First, switch the tool and select the tool number. Align the B-axis machine coordinates to the zero point (B-axis perpendicular to the X-axis). Then move the Y-axis and Z-axis to position the tool tip at a designated mark on the workpiece (clamped on the A-axis). Press 'R' to input 1 for setting the first point. Lift the tool, rotate the A-axis as wide as possible, then slowly adjust the Y-axis and Z-axis until the tool tip aligns with the same mark on the workpiece. Press 'R' again to set the second point, repeating this process for the third point. The system automatically calculates the Y and Z machine coordinates of the A-axis rotation center and records the current tool length in the parameter "Measured Tool Length (um) when measuring the five-axis CNC rotary platform center." This measurement is required only during initial setup or A-axis reinstallation; it's unnecessary when changing tools. Before automatic A-axis rotation center measurement, ensure the distance between the tool tip and B-axis rotation center has been accurately measured.

Parameter "555, measuring the tool length (um) when the center of the five-axis CNC rotary table is measured" records the current H2 tool length value of the tool.

(2) When the structural type parameter of the machine tool is equal to 11:

This operation is not required.

(3) When the structural type parameter of the machine tool is equal to 12:

Step 1: Automatically measure the machine tool coordinates of the A-axis rotation center Y (P553) and Z (P554).

First, select the tool number and move the tool along the Y and Z axes to position the tip at a designated mark on the workpiece (clamped on the A-axis). Press 'R' to input 1 for the first point. Lift the tool and rotate the A-axis as wide as possible, then slowly adjust the Y and Z axes until the tip aligns with the same mark on the workpiece. Press 'R' again to input 2 for the second point, repeating this process for each subsequent point. The system will automatically calculate the Y and Z coordinates for the A/B axis rotation center. When using multiple tools, set the length compensation value H1 before proceeding with this operation.

Parameter "555, Measure tool length (um) when centering the five-axis CNC rotary table" records the current H1 compensation value of the tool. If G43 tool length compensation is not used, the length compensation value H1 must be reset to zero. The measurement of Y and Z machine coordinates for A-axis rotation centering is only required when the machine is newly installed or the A-axis is reinstalled. This operation is not necessary when changing tools.

Step 2: Used to automatically measure the X (P556) machine tool coordinates of the C-axis rotation center.

First, switch the tool and select the tool number. Align the machine's A-axis coordinates to the zero point, positioning the C-axis horizontally. Then move the X and Y axes to align the tool tip with a designated mark on the workpiece (clamped on the C-axis). Press 'R' to set the first point. Raise the tool and rotate the C-axis as wide as possible. Slowly adjust the X and Y axes until the tool tip aligns with the same mark. Press 'R' again to set the second point, then repeat this process for the third point. The system will automatically calculate the machine's X-axis coordinates for the C-axis rotation center (P556).

The X coordinate value of the C-axis rotation center needs to be measured only when the C-axis is newly installed or reinstalled, and then replaced

刀具时不需要进行该操作对于有多把刀具时需要将长度补偿值 H1 设置好后在进行该项操作。

参数“555, 测量五轴数控回转平台中心时的刀长(um)”记录的当前刀具的 H1 补偿值。

(4) 机床结构型式参数等于 15 时:

第一步:自动测量 B 轴回转中心 X(P553)和 Z(P554)机床坐标。

首先换刀选择刀具号,然后移动 X 轴和 Z 轴使刀尖处于工件(装夹在 B 轴上)上某一标记点,按‘R’输入 1 设定第一个点,然后将刀具提起来,转动一下 B 轴,转动角度尽量大一点,再慢慢移动 X 轴和 Z 轴,让刀尖点处于工件上的同一标记点,按‘R’输入 2 设定第二个点,依次方法设定第三个点,系统会自动计算出 B 轴回转中心的 X 和 Z 机床坐标值。对于有多把刀具时,需要将长度补偿值 H1 设置好后,再进行该项操作。

参数“555, 测量五轴数控回转平台中心时的刀长(um)”记录的当前刀具的 H1 补偿值.如果不使用刀具长度补偿 G43 时,需要将长度补偿值 H1 清零。刚刚装机或重新安装 B 轴时才需要测量 B 轴回转中心的 X 和 Z 机床坐标值,更换刀具时不需要进行该操作。

第二步:用于自动测量 C 轴回转中心的 Y(P556)机床坐标值。

首先换刀选择刀具号,将 B 轴的机床坐标走到零点,即 C 回转轴处于水平位置,然后移动 X 轴和 Y 轴使刀尖处于工件(装夹在 C 轴上)上某一标记点,按‘R’输入 A 设定第一个点,然后将刀具提起来,转动一下 C 轴,转动角度尽量大一点,再慢慢移动 X 轴和 Y 轴,让刀尖点处于工件上的同一标记点,按‘R’输入 B 设定第二个点,依次方法输入 C 设定第三个点,系统会自动计算出 C 轴回转中心的 Y 机床坐标值(P556)。

刚刚装机或重新安装 C 轴时才需要测量 C 轴回转中心的 Y 机床坐标值,更换刀具时不需要进行该操作。对于有多把刀具时,需要将长度补偿值 H1 设置好后,在进行该项操作。

参数“555, 测量五轴数控回转平台中心时的刀长(um)”记录的当前刀具的 H1 补偿值。

注意 1:BC 类型 RTCP 补偿不使用 B 轴,只操作第二步测量 C 轴回转中心的 X 和 Y 机床坐标值即可;

注意 2: B 轴零点要注意,必须是当工作台处于水平时为 B 轴零点。而 C 轴零点可以任意设置;

注意 3: 刀补调用如果为发那科模式指令 G43 当 T1 号刀就调用 H1 刀补, T2 号刀调用 H2 刀补,以此类推;

(5) 机床结构型式参数等于 13 时:

用于自动测量 C 轴回转中心的 X(P553)和 Y(P554)机床坐标值。

首先换刀选择刀具号,将 A 轴的机床坐标走到零点,即 A 轴垂直于 Y 轴,然后移动 X 轴和 Y 轴使刀尖处于工件(装夹在 C 轴上)上某一标记点,按‘R’输入 1 设定第一个点,然后将刀具提起来,转动一下 C 轴,转动角度尽量大一点,再慢慢移动 X 轴和 Y 轴,让刀尖点处于工件上的同一标记点,按‘R’输入 2 设定第

This operation is not required for a single tool. When there are multiple tools, this operation should be performed after setting the length compensation value H1.

Parameter "555, measuring the tool length (um) when the center of the five-axis CNC rotary table is measured" records the current H1 compensation value of the tool.

(4) When the structural type parameter of the machine tool is equal to 15:

Step 1: Automatically measure the B-axis rotation center coordinates of the machine tool X (P553) and Z (P554).

First, select the tool number and move the tool along the X and Z axes to position the tip at a designated mark on the workpiece (clamped on the B-axis). Press 'R' to input 1 for setting the first point. Lift the tool and rotate the B-axis as wide as possible. Slowly adjust the X and Z axes until the tip aligns with the same mark on the workpiece. Press 'R' again to input 2 for setting the second point, repeating this process for the third point. The system will automatically calculate the B-axis rotation center coordinates for the X and Z axes. When using multiple tools, set the length compensation value H1 before proceeding with this operation.

Parameter "555, Measure tool length (um) when centering the five-axis CNC rotary table" records the current H1 compensation value of the tool. If G43 tool length compensation is not used, the H1 length compensation value must be reset to zero. The measurement of B-axis rotary center coordinates (X and Z) is only required when the machine is newly installed or the B-axis is reinstalled. This operation is not necessary when changing tools.

Step 2: Used to automatically measure the Y (P556) machine tool coordinates of the C-axis rotation center.

First, switch the tool and select its number. Align the machine's B-axis coordinates to the zero point (keeping the C-axis horizontal), then move the X and Y axes to position the cutting edge at a designated mark on the workpiece (clamped on the C-axis). Press 'R' to input A for setting the first point. Lift the tool and rotate the C-axis as wide as possible. Slowly adjust the X and Y axes until the cutting edge aligns with the same mark on the workpiece. Press 'R' again to input B for setting the second point. Repeat this process for C to set the third point. The system will automatically calculate the Y-coordinate values for the C-axis rotation center (P556).

The measurement of the Y coordinate value of the C-axis rotation center is required only when the C-axis has just been installed or reinstalled, and is not required when replacing a tool. For multiple tools, this operation should be performed after setting the length compensation value H1.

Parameter "555, measuring the tool length (um) when the center of the five-axis CNC rotary platform is measured" records the current H1 compensation value of the tool.

Note 1: BC type RTCP compensation does not use the B axis, only the second step to measure the C axis rotation center of the machine tool coordinates X and Y can be operated;

Note 2: The B-axis zero point must be set when the worktable is horizontal. The C-axis zero point can be set arbitrarily;

Note 3: Tool feed call If the tool feed is in the FANUC mode command G43, the T1 tool calls H1 tool feed, the T2 tool calls H2 tool feed, and so on;

(5) When the structural type parameter of the machine tool is equal to 13:

Used to automatically measure the X (P553) and Y (P554) machine tool coordinates of the C-axis rotation center.

First, switch the tool and select the tool number. Align the machine's A-axis coordinates to the zero position (perpendicular to the Y-axis). Then move the X and Y axes to position the cutting edge at a designated mark on the workpiece (clamped on the C-axis). Press 'R' to input 1 for the first point. Lift the tool and rotate the C-axis as wide as possible. Slowly adjust the X and Y axes until the cutting edge aligns with the same mark on the workpiece. Press 'R' again to input 2 for the second point

二个点,依次方法设定第三个点,系统会自动计算出C轴回转中心的X和Y机床坐标值,并且记录当前刀长到参数“测量五轴数控回转平台中心时的刀长(um)”中。

刚刚装机或重新安装C轴时才需要测量C轴回转中心的X和Y机床坐标值,更换刀具时不需要进行该操作。自动测量C轴回转中心之前必须先测量好刀尖点到A轴回转中心的距离才行。

参数“555,测量五轴数控回转平台中心时的刀长(um)”记录的当前刀具的H2刀具长度值。

(6) 机床结构型式参数等于14时:

用于自动测量B轴回转中心的X(P553)和Z(P554)机床坐标值。

首先换刀选择刀具号,将A轴的机床坐标走到零点,即A轴垂直于Y轴,然后移动X轴和Z轴使刀尖处于工件(装夹在B轴上)上某一标记点,按‘R’输入1设定第一个点,然后将刀具提起来,转动一下B轴,转动角度尽量大一点,再慢慢移动X轴和Z轴,让刀尖点处于工件上的同一标记点,按‘R’输入2设定第二个点,依次方法设定第三个点,系统会自动计算出B轴回转中心的X和Z机床坐标值,并且记录当前刀长到参数“测量五轴数控回转平台中心时的刀长(um)”中。

刚刚装机或重新安装B轴时才需要测量B轴回转中心的X和Z机床坐标值,更换刀具时不需要进行该操作。自动测量B轴回转中心之前必须先测量好刀尖点到A轴回转中心的距离才行。

参数“555,测量五轴数控回转平台中心时的刀长(um)”记录的当前刀具的H2刀具长度值。

机床结构型式参数等于10时需要注意如下几点(对于其他机床结构型式也有类似注意):

1>,在与Z轴负方向重合的地方定义为B轴机床坐标值为零,正方向就按右手坐标系规则。

2>,在与Z轴正方向重合的地方定义为A轴机床坐标值为零,正方向就按右手坐标系规则。

3>,在确定G54等工件坐标系的Z轴时将B轴的机床坐标走到零点,即B轴垂直于X轴后再进行。

4>,因为H2用于保存刀尖点到B轴回转中心的距离,作为计算RTCP的B轴旋转补偿的依据,所以G43/G44刀具长度补偿就不能调用H2,只能使用每一把刀的H1。

当需要下刀或提刀修改G54的Z偏移值或每一把刀的H1,不能修改每一把刀的H2,注意H2反应的是真实的刀具长度值,必须与实际一致,否则补偿计算就不正确。程序中需要编入当前选那把刀,系统才会调用相应刀具的H2值。刀具长度补偿G43/G44可以与RTCP补偿M601/M602指令同时使用。

如果刀具磨损了,程序中同时使用了长度补偿G43/G44和RTCP补偿M601/M602指令,那么就需要同时对H1和H2进行增量修改。

5>,在手动下执行MDI,输入M601或M602会进入RTCP状态,系统界面会有

Two points are set in sequence. The system will automatically calculate the X and Y machine tool coordinates of the C-axis rotation center, and record the current tool length to the parameter "tool length (um) when measuring the center of the five-axis CNC rotary platform".

The measurement of the X and Y machine coordinates at the C-axis rotation center is required only when the C-axis has just been installed or reinstalled. This operation is not required when replacing a tool. The distance between the tool tip point and the A-axis rotation center must be measured before automatically measuring the C-axis rotation center.

Parameter "555, measuring the tool length (um) when the center of the five-axis CNC rotary table is measured" records the current H2 tool length value of the tool.

(6) When the structural type parameter of the machine tool is equal to 14:

Used to automatically measure the X (P553) and Z (P554) machine tool coordinates at the B-axis rotation center.

First, switch the tool and select the tool number. Align the machine's A-axis coordinates to the zero position (perpendicular to the Y-axis), then move the X and Z axes to position the tool tip at a designated mark on the workpiece (clamped on the B-axis). Press 'R' to input 1 for setting the first point. Lift the tool, rotate the B-axis as wide as possible, then slowly adjust the X and Z axes until the tool tip aligns with the same mark on the workpiece. Press 'R' again to input 2 for setting the second point. Repeat this process for the third point. The system will automatically calculate the B-axis rotation center coordinates (X and Z) and record the current tool length in the parameter "Measured Tool Length (um) during five-axis CNC rotary table centering".

The measurement of B-axis rotation center X and Z machine tool coordinates is required only when the B-axis has just been installed or reinstalled. This operation is not required when replacing a tool. The distance between the tool tip point and the A-axis rotation center must be measured before automatically measuring the B-axis rotation center.

Parameter "555, measuring the tool length (um) when the center of the five-axis CNC rotary table is measured" records the current H2 tool length value of the tool.

When the structural type parameter of the machine tool is equal to 10, the following points should be noted (similar attention should also be paid to other machine tools).

1>, where the negative direction of the Z-axis is coincident, the B-axis machine coordinate value is defined as zero, and the positive direction follows the right-hand coordinate system rules.

2>, where the positive direction of the Z-axis is coincident, the A-axis machine coordinate value is defined as zero, and the positive direction follows the right-hand coordinate system rules.

3. When determining the Z-axis of the coordinate system of G54 and other workpieces, move the machine tool coordinates of the B-axis to the zero point, that is, after the B-axis is perpendicular to the X-axis.

4>, because H2 is used to store the distance between the tip of the knife and the B-axis rotation center, which serves as the basis for calculating the B-axis rotation compensation of RTCP, G43/G44 tool length compensation cannot call H2, but can only use H1 of each tool.

When modifying G54's Z offset or H1 for individual toolpaths, do not adjust the corresponding H2 values. Note that H2 reflects the actual tool length value, which must match the real measurement for accurate compensation calculation. The system will only retrieve the H2 value of the currently selected tool when programmed. Tool length compensation G43/G44 can be used in conjunction with RTCP compensation commands M601/M602.

If the knife tool is worn and the G43/G44 length compensation and RTCP M601/M602 instructions are used at the same time in the program, then incremental modifications to both H1 and H2 are required.

5>. When MDI is executed manually, input M601 or M602 to enter the RTCP state, and the system interface will be available

显示, 此时手动运行 B 轴或 A 轴, XYZ 轴的机床坐标会自动运行进行补偿。

6>, 调试时检查 RTCP 补偿方向的正确性:

首先调好 B 轴机床坐标值零点, 确保零点位置 Z 轴垂直于 X 轴, 然后自动测量刀尖点到 B 轴回转中心的距离, 自动测量 A 轴回转中心的 Y 和 Z 机床坐标值, 在手动下执行 MDI, 输入 M602 进入 RTCP 状态, 将刀尖对准工作台上一标记点, 再手动转动 B 轴, 此时 X 轴和 Z 轴会跟着移动, 若方向正确, 标记点会紧跟着刀尖移动, 否则表示若方向不正确, 修改 552 号参数。将刀尖对准工作台上一标记点, 再手动转动 A 轴, 此时 Y 轴和 Z 轴会跟着移动, 若方向正确, 标记点会紧跟着刀尖移动, 否则表示若方向不正确, 修改 552 号参数。

7>, 在 RTCP 状态下, 机床坐标是反应的真实坐标, 工件坐标则为虚拟坐标 (与程序编制的坐标一致)。

8>, 在确定工件坐标系 XYZ 时, 建议将旋转轴 ABC 轴处于机床坐标零点位置, 方便编程。

9>, 在编入 M601 或 M602 指令时最好先非 RTCP 模式下将 XYZAB 五轴定位到某一安全点, 例如:

```
G54 T1
M600
G0 X_ Y_ Z_ A_ B_
M601 或 M602
G0 X_ Y_ ...
G1 X_ A_ B_ ...
....
....
M600
M30
```

6、四轴系统的 RTCP 功能使用方法:

1>当有 XYZ 直线轴和一个绕 X 轴旋转的旋转轴 A 时 (当有 XYZ 直线轴和一个绕 Z 轴旋转的旋转轴 C 时, 可以将 Z 轴定义为 X 轴, X 轴定义为 Y 轴, Y 轴定义为 Z 轴, 旋转轴定义为 A 轴):

开启四个轴 XYZA 并将参数机床类型设为 14。

用 M601 开启 RTCP 功能, 用 M600 关闭 RTCP 功能。

在系统刀补界面按 ‘H’ 用于自动测量刀长。

自动测量刀尖点到 A 轴回转中心的距离: 首先换刀选择刀具号, 让刀尖点处于工件上某一标记点, 按 ‘H’ 输入 1 设定的第一个点, 然后将刀具提起来, 转动一下 A 轴, 转动角度不能太小, 再慢慢移动 Y 轴和 Z 轴, 让刀尖点处于工件上的同一标记点, 按 ‘H’ 输入 2 设定的第二个点, 系统则自动计算出刀尖点到 A 轴回转中心的距离 (um), 并将其保存到当前光标处的 H2 中。系统进行 RTCP 的 A 轴补偿计算时会自动调用当前刀具号对应的 H2 值。

2>当有 XYZ 直线轴和一个绕 Y 轴旋转的旋转轴 B 时:

开启五个轴 XYZAB 并将参数机床类型设为 10。

Display, at this time, manually run the B axis or A axis, the XYZ axis of the machine tool coordinates will automatically run for compensation.

6. Check the correctness of RTCP compensation direction during debugging:

First, calibrate the B-axis coordinate zero point to ensure the Z-axis is perpendicular to the X-axis. Then automatically measure the distance from the tool tip to the B-axis rotation center, and automatically determine the Y and Z coordinates of the A-axis rotation center. Manually execute MDI, input M602 to enter RTCP mode, and align the tool tip with a worktable marker. Manually rotate the B-axis; if the direction is correct, the marker will follow the tool tip's movement. If not, adjust parameter 552. Next, align the tool tip with another worktable marker and manually rotate the A-axis. If the direction is correct, the marker will follow the tool tip's movement. If not, adjust parameter 552 again.

7> In the RTCP state, the machine tool coordinates are the real coordinates reflected, and the workpiece coordinates are virtual coordinates (consistent with the programmed coordinates).

8) When determining the coordinate system XYZ of the workpiece, it is recommended that the rotation axis ABC be located at the zero point of the machine tool coordinate system for easy programming.

9. When entering the M601 or M602 command, it is best to first position the XYZAB five-axis in a safe point in non-RTCP mode, for example:

```
G54 T1
M600
G0 X_ Y_ Z_ A_ B_
M601 or M602
G0 X_ Y_ ....
G1 X_ A_ B_ ....
:::
M600
M30
```

6. How to Use the RTCP Function of the Four-Axis System:

1) When there is an XYZ linear axis and a rotating axis A that rotates around the X-axis (when there is an XYZ linear axis and a rotating axis C that rotates around the Z-axis, the Z-axis can be defined as the X-axis, the X-axis as the Y-axis, the Y-axis as the Z-axis, and the rotating axis as the A-axis):

Start four axes XYZA and set the parameter machine tool type to 14. Use M601 to enable RTCP function and use M600 to disable RTCP function.

Press 'H' on the system blade supplement interface to automatically measure the blade length.

Automatic measurement of the distance from the tool tip to the A-axis rotation center: First, switch tools and select the tool number. Position the tool tip at a designated mark on the workpiece. Press 'H' to input point 1 (the first preset position). Raise the tool and rotate the A-axis slightly upward. Slowly adjust the Y-axis and Z-axis until the tool tip aligns with the same mark. Press 'H' again to input point 2 (the second preset position). The system automatically calculates the distance (um) from the tool tip to the A-axis rotation center and stores it in H2 at the current cursor position. When performing RTCP A-axis compensation calculations, the system automatically retrieves the corresponding H2 value for the selected tool number.

2>When there is an XYZ linear shaft and a rotating shaft B that rotates around the Y-axis:

Start the five axes XYZAB and set the parameter machine tool type to 10.

因为实际没有 A 轴此时将 A 轴机床坐标清零然后通过编写 PLC 梯形图禁止 A 运行即可。

用 M601 开启 RTCP 功能，用 M600 关闭 RTCP 功能。

在系统刀补界面按 ‘H’ 用于自动测量刀长。

自动测量刀尖点到 B 轴回转中心的距离：首先换刀选择刀具号，让刀尖点处于工件上某一标记点，按 ‘H’ 输入 1 设定的第一个点，然后将刀具提起来，转动一下 B 轴，转动角度不能太小，再慢慢移动 X 轴和 Z 轴，让刀尖点处于工件上的同一标记点，按 ‘H’ 输入 2 设定的第二个点，系统则自动计算出刀尖点到 B 轴回转中心的距离 (um)，并将其保存到当前光标处的 H2 中。系统进行 RTCP 的 B 轴补偿计算时会自动调用当前刀具号对应的 H2 值。

7、RTCP 功能测试方法：

在手动 M602 状态下，让刀尖对准工作台上一点，转动 A 轴或 C 轴，XYZ 自动跟随运行，如果刀尖始终保持对准工作台上一点，说明系统 RTCP 工作正常。

Because there is no A-axis in fact, the A-axis machine tool coordinates are cleared, and then the A-axis operation can be prohibited by writing ladder diagram of PLC.

Use M601 to enable the RTCP function, and use M600 to disable the RTCP function.

Press 'H' on the system blade supplement interface to automatically measure the blade length.

Automatic measurement of the distance from the tool tip to the B-axis rotation center: First, switch tools and select the tool number. Position the tool tip at a designated mark on the workpiece. Press 'H' to input point 1 (the first preset position). Raise the tool and rotate the B-axis slightly upward. Slowly adjust the X and Z axes until the tool tip aligns with the same mark. Press 'H' again to set point 2. The system automatically calculates the distance (in micrometers) between the tool tip and the B-axis rotation center, storing this value in the H2 field at the current cursor position. When performing RTCP B-axis compensation calculations, the system automatically retrieves the corresponding H 2 value from the current tool number.

7、 RTCP Function Test Method:

In manual M602 state, make the tip of the knife point at a point on the worktable, rotate the A-axis or C-axis, and the XYZ automatically follows. If the tip of the knife always points at a point on the worktable, it indicates that the system RTCP works normally.

第三篇 操作篇

3.1 概述

使用本数控系统时，只要掌握系统参数、程序编辑、手动操作、自动运行几方面的操作内容，就可以很方便的进行操作了。以下内容将会针对详细说明。

3.2 操作面板

本系统面板分主面板和副面板。主面板用于参数设置和程序编辑，副面板用于对刀设置和加工操作。

3.3 键盘说明

3.3.1 倍率增减

(1) 快速倍率

快速倍率有 5%~100%共六档，通过快速倍率键调节，快速倍率对以下指令有效：G00、G26、G28、G61、G611-G618 固定循环的快速进给、手动快移进给。

(2) 进给倍率

进给倍率有 0%~150%共十六档，通过进给倍率键调节，进给倍率对以下指令有效：G01、G02、G03、固定循环的进给 F 有效；还对手动运行有效。

(3) 主轴倍率

主轴倍率有 5%~150%共十六档，通过主轴速度倍率键调节，主轴倍率对第一主轴速度 S 有效。

3.3.2 干预开关的使用

(1) 左则：正常加工。

(2) 中则：手动进给停止，自动进给暂停。返回左则时恢复正常。

(3) 右则：自动进给暂停、主轴暂停，返回中则时参见 (2)。当处于非坐标轴运行时主轴不受控制。

3.3.3 其它按键见下表

字母键	ABCDEFGHIJKLMNOPQRSTUVWXYZ 用于程序命令和参数设置；
数字键	0123456789.- 用于数据输入和参数设置；
符号键	#*[]()/%用于宏程序编辑；

The Third Part Is the Operation Part

3.1 Summary

When using this CNC system, as long as you master the system parameters, program editing, manual operation, automatic operation and other aspects of the operation content, you can easily operate. The following content will be detailed.

3.2 Operation Panel

The panel of the system is divided into main panel and auxiliary panel. The main panel is used for parameter setting and program editing, while the auxiliary panel is used for tool setting and machining operation.

3.3 Keyboard instructions

3.3.1 Multiplicative Rate Increase or Decrease

(1) Fast ratio

The rapid ratio has six grades from 5% to 100%. Through the adjustment of the rapid ratio key, the rapid ratio is effective for the following instructions: G00, G26, G28, G61, G611-G618 fixed cycle rapid feed and manual fast feed.

(2) Feed rate

The feed ratio has 16 steps from 0% to 150%, which can be adjusted by the feed ratio key. The feed ratio is effective for the following instructions: G01, G02, G03, and fixed cycle feed F, and also effective for manual operation.

(3) Shaft ratio

The spindle ratio has 16 steps from 5% to 150%, which can be adjusted by the spindle speed ratio key. The spindle ratio is effective for the first spindle speed S.

3.3.2 Use of Intervention Switches

(1) Left: normal processing.

(2) In the middle: manual feed stop, automatic feed pause. Return to the left side and normal.

(3) Right: Automatic feed pause, spindle pause, return to the middle of the time refer to (2). When in non-coordinate axis operation, the spindle is not controlled.

3.3.3 Other Keys Are Shown in the Table Below

letter key	ABCDEFGHIJKLMNOPQRSTUVWXYZ For program commands and parameter Settings;
numbering key	0123456789. -For data input and parameter setting;
symbolic key	#*[]()/=% Used for macro editing;

方向键	编程区的↑↓→← 用于光标移动;
+X、-X、 +Y、-Y、 +Z、-Z、+A、 -A、+B、-B	用于手动进给; +B、-B 用于 B\C\Xs\Ys 轴的正、负方向进给, 用“B”“C” “X”“Y”选择当前进给轴;
编辑键	“删除、上页、下页”用于程序编辑;
功能键	“复位” 自动运行程序复位结束程序运行并返回到程序头, 主轴等开关量关闭, 运行进给轴运行复位停止; “返回” 用于操作取消和功能返回; “确认” 用于功能选择和输入确认; “参数” 进入参数设置菜单; “程序” 进入程序编辑菜单; “刀补” 修正换刀误差; “手动” 手动功能选择; “自动” 自动功能选择; “手轮” 启动手轮功能; “诊断” 输入输出诊断; “单段” 选择自动执行方式, 加工中可中停; “单步” 用于手动单步(即增量)选择; “图形” 用于自动坐标或图形加工方式的选择; “空运行” 用于自动加工或自动空运行方式的选择;
控制键	“正转、反转、停止” 用于主轴的正转、反转、停止; “点动” 主轴正转点动; “冷却” 用于冷却液的启动和停止; “回零” 用于回机床零点; “松紧刀” 用于主轴松紧刀的卡紧和松开; “吹气” 用于主轴吹气的开或关; “润滑” 用于润滑的开关; “快速” 用于手动连续高、低速切换; “快速↑、快速↓” 用于 G00 或手动快速修调; “进给↑、进给↓” 用于进给速度修调; “主轴↑、主轴↓” 用于主轴转速修调; “启动” 执行自动加工程序; “暂停” 自动加工过程中暂停;
控制钮	紧急停止: 终止机床一切动作, 等待按键抬起; 干预开关: 在加工过程中暂停丝杠和主轴; 程序运行: 执行自动加工程序;

3.4 手动操作□

direction key	The ↑↓ → ← in the programming area is used for cursor movement;
+X、-X、 +Y、-Y、 +Z、-Z、+A、 -A、+B、-B	For manual feed; +B and -B are used for the positive and negative direction feed of the B,C,Xs,Ys axes, and "B", "C", "X", "Y" are used to select the current feed axis;
editing key	"Delete", "Previous page", "Next page" are used for program editing;
function key	"Reset" automatically runs the program reset ends the program runs and returns to the program head, spindle and other switches are closed, run feed axis run reset stop; "Return" is used for operation cancellation and function return; "Confirmation" is used for function selection and input confirmation; "Parameter" enters the parameter setting menu; The "Program" enters the program editing menu; "Knife supplement" to correct the error of changing the knife; "Manual" manual function selection; "Automatic" automatic function selection; "Handwheel" starts the handwheel function; "Diagnosis" input-output diagnosis; "Single segment" is selected to be executed automatically, and can be stopped during processing; "Single step" is used for manual single step (i.e. incremental) selection; "Graphics" is used to select the automatic coordinate or graphic processing mode; "Empty run" is used to select the mode of automatic processing or automatic empty run;
manipulating key	"Forward rotation, reverse rotation and stop" are used for the forward rotation, reverse rotation and stop of the spindle; "point movement" is the forward rotation point movement of the spindle; "Cooling" is used to start and stop the coolant; "Return to zero" is used to return the machine tool to zero; "Loose and tight knife" is used for the clamping and loosening of the spindle loose and tight knife; "blow" is used for the opening or closing of the spindle blow; "Lubrication" refers to the switch for lubrication; "Quick" is used for manual continuous high and low speed switching; "Quick ↑ and Quick ↓" are used for G00 or manual quick adjustment; "Feed up and feed down" are used to adjust the feed speed; "Up spindle, down spindle" is used for spindle speed adjustment; "Start" to execute the automatic processing program; "Pause" Pause in the machining process;
control button	Emergency stop: stop all movements of the machine tool and wait for the button to be lifted; Interference switch: pause the screw and spindle during processing; Program operation: execute automatic processing program;

3.4 Hand Operation ☐

系统采用一级菜单式操作，直观、方面、快捷，提示信息全面。
系统通电即进入系统主界面：

手动 连续		N00000		2016-06-06 06:06	
N1 G00 X0 Z0 N2 G00 X200 Z200 N3 G00 U-44		当前程序 99.TXT 指令代码 G53 T01H0D0			
○ X 168.000 ○ Y 168.000 ○ Z 168.000 ○ A 168.000 ○ B 168.000		状态 M05 M09 M10 不选停 M78 M33 M41 跳段 G00 X100% F2000 X100% S800 X100% SP000			
状态：正常		机床坐标 X 168.000 A 168.000 Y 168.000 B 168.000 Z 168.000			
		单件时间 0: 6 加工件数 0 主轴转速 0			
F1 相对坐标 F2 工件坐标 F3 综合坐标 F4 单步 F5 H D I F6 对刀 F7 设置坐标 F8 选择坐标					

屏幕第一行左面显示系统运行方式（可以是“手动连续”、“手动增量”、“自动连续”、“自动单段”、“自动空运行”、“手轮”），中间为加工程序的当前行，右面显示当前日期与时钟。

左上部为程序区，程序区第二行为当前程序行。

左下部为工件坐标显示区。

倒第二行为提示区，如报警提示等。

最后一行 F1-F8 功能键提示。

右边为当前程序名、指令代码、机床状态、机床坐标等的显示区。

菜单的操作：菜单的选择可通过功能键进行。例如：

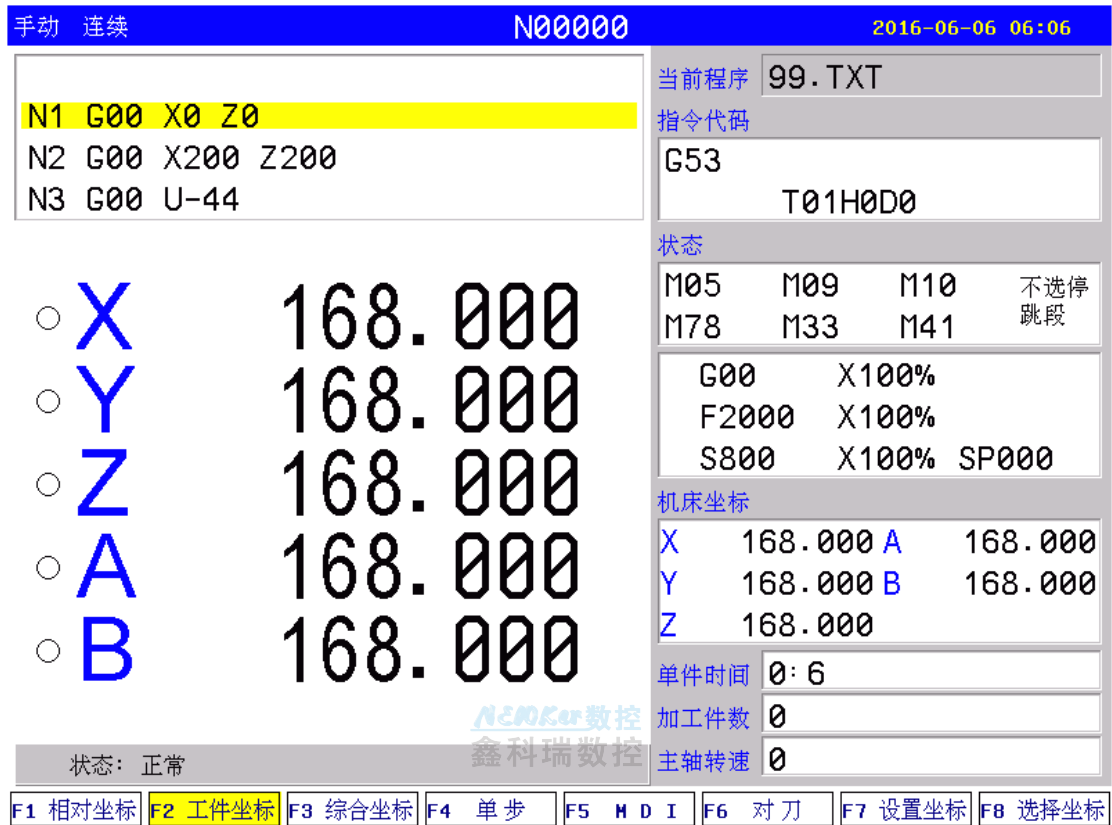
按“程序”则进入程序管理系统，可对程序进行编辑、修改、查错、删除、拷贝等工作。

按“参数”则进入参数管理系统，可对各类参数进行查阅、修改。

手动方式主要用于机床坐标的调整和机床辅助动作的启停，刀具的对刀，加工工件的起点确定，以及进行手动加工。手动方式不能进行程序加工。

3.4.1 手动方式按键操作

The system adopts the first-level menu operation, which is intuitive, convenient and fast, with comprehensive prompt information. The system enters the main interface immediately after power-on:



The left side of the first line of the screen shows the operation mode of the system (which can be "manual continuous", "manual incremental", "automatic continuous", "automatic single segment", "automatic empty run", "handwheel"), the middle part shows the current line of the processing program, and the right side shows the current date and clock.

The upper left is the program area, and the second behavior of the program area is the current program line.

The lower left part is the workpiece coordinate display area.

The second behavior indicates the area, such as alarm indication.

The last row of F1-F8 function keys is displayed.

On the right is the display area of the current program name, instruction code, machine tool status, machine tool coordinates, etc.

Menu operation: The menu can be selected by function keys. For example:

According to the "program", you can enter the program management system, and can edit, modify, check, delete, copy and other work on the program.

According to "parameter", you can enter the parameter management system, and check and modify all kinds of parameters.

Manual mode is mainly used for the adjustment of machine tool coordinates and the start and stop of machine tool auxiliary actions, tool alignment, determination of starting point of machining workpiece, and manual machining. Manual mode cannot be used for programmed machining.

3.4.1 Key Operation by Manual Mode

- (1) “F” 键：设定手动进给速度，以 mm/min 为单位，其输入范围为：1-30000mm/min。其输入方法参见参数中的数据输入方法。
- (2) “单步” 键：“手动连续”、“手动单步” 循环切换。
- (3) “S” 键：设定第一主轴转速。主轴转速范围为 0-99999，其最大值取决于速度参数状态中的 P48 号参数值。
- (4) “I” 键：在手动单步方式下修改单步值 I。
- (5) “进给↑” 键：当轴类参数 P1 号参数设置为 0 时，每按一次“进给↑” 键，进给速度倍率修调增加 10%。范围为 0—150%共 16 档。
- (6) “进给↓” 键：当轴类参数 P1 号参数设置为 0 时，每按一次“进给↓” 键，进给速度倍率修调减少 10%。范围为 0—150%共 16 档。
- (7) “快速↑” 键：每按一次“快速↑” 键，G00 或手动快速倍率修调增加 20%。范围为 5—100%共 6 档。
- (8) “快速↓” 键：每按一次“快速↓” 键，G00 或手动快速倍率修调减少 20%。范围为 5—100%共 6 档。
- (9) “主轴↑” 键：当轴类参数 P2 号参数设置为 0 时，每按一次“主轴↑” 键，第一主轴转速倍率增加 10%。范围为 5—150%共 16 档。
- (10) “主轴↓” 键：当轴类参数 P2 号参数设置为 0 时，每按一次“主轴↓” 键，第一主轴转速倍率减少 10%。范围为 5—150%共 16 档。
- (11) “手轮” 键：选择“手轮方式”，进行“0.001”、“0.01”、“0.1” 档位循环切换。
- (12) “润滑”： “润滑” 键为润滑开、关反复键，对应指令 M32、M33。
- (13) “F8 选择坐标”：按此键选择相应的坐标系（G53-G59），选择后屏幕右上角用相应的显示。
- (14) “F7 设置坐标”：在工件坐标系（G54-G59）状态时设置（G54-G59）的值，更新参数中的值；在机床坐标系（G53）状态时偏置所有刀具坐标，更新所有刀补值。
- (15) “自动” 键：选择“自动方式”。
- (16) “手动” 键：选择“手动方式”。
- (17) 主轴控制：“正转、反转、停止” 分别控制主轴正转、反转和停止，对应指令 M03、M04、M05。
- 按“点动” 键则输出点动信号，即按下时为主轴转动，抬起时主轴停止。
- (18) “冷却”：“冷却” 键为冷却液开、关反复键，对应指令 M08、M09。
- (19) 松紧刀：“松紧刀” 键为松紧刀开、关反复键，对应指令 M10、M11。
- (20) 手动速度：按“0”、“1”、“2”、“3”、“4”、“5”、“6”、“7”、“8”、“9” 进行“F30”、“F60”、“F120”、“F250”、“F500”、“F1000”、“F1500”、“F2000”、“F2500”、“F3000” 手动速度设置。
- (21) 吹气：“吹气” 键为吹气开、关反复键，对应指令 M59、M58。
- (22) 连续与单步的切换：按“单步” 键时手动连续方式与单步方式进行转换，若为单步时则显示单步值 I=XXXX.XXX。

- (1) "F" key: set the manual feed speed, in mm/min, its input range is: 1-30000mm/min. The input method can be found in the data input method of the parameter.
- (2) "Single step" key: "manual continuous" and "manual single step" cycle switch.
- (3) "S" key: Set the first spindle speed. The spindle speed ranges from 0 to 99,999, and the maximum value depends on the value of parameter P48 in the speed parameter state.
- (4) "I" key: Modify the single-step value I in manual single-step mode.
- (5) "Feed up" key: When parameter P1 is set to 0, each press of the "Feed up" key increases the feed speed multiplier by 10%. The range is 0 -150%, with a total of 16 steps.
- (6) "Feed Down" key: When parameter P1 is set to 0, each press of the "Feed Down" key reduces the feed speed multiplier by 10%. The range is 0-150%, with a total of 16 steps.
- (7) "Quick ↑" key: Each press of the "Quick ↑" key increases the G 00 or manual quick ratio adjustment by 20%. The range is 5-100%, a total of 6 steps.
- (8) "Quick ↓" key: Each press of the "Quick ↓" key reduces the G 00 or manual quick ratio adjustment by 20%. The range is 5-100%, a total of 6 steps.
- (9) "Spindle ↑" key: When parameter P2 is set to 0, each time the " spindle ↑" key is pressed, the first spindle speed multiplier increases by 10%. The range is 5-150%, with a total of 16 steps.
- (10) "Spindle ↓" key: When parameter P2 is set to 0, each time the " spindle ↓" key is pressed, the first spindle speed multiplier decreases by 10%. The range is 5-150%, a total of 16 steps.
- (11) Handwheel key: Select "handwheel mode" and cycle between "0.001", "0.01" and "0.1" gears.
- (12) "Lubrication": The "Lubrication" key is the repeated key to open and close lubrication, corresponding to the instructions M32 and M 33.
- (13) "F8 Select coordinate": Press this key to select the corresponding coordinate system (G53-G59), and select the corresponding display in the upper right corner of the screen.
- (14) "F7 Set coordinate": Set the value of (G54-G59) when the workpiece coordinate system is in state, update the value in the parameter; offset all tool coordinates and update all tool compensation values when the machine tool coordinate system is in state (G53).
- (15) "Auto" key: Select "Auto mode".
- (16) "Manual" key: Select "Manual mode".
- (17) Spindle control: "Forward, reverse and stop" respectively control the spindle forward, reverse and stop, corresponding to the instructions M03, M04 and M05.
Press the "point" key to output the point signal, that is, when pressed, the spindle turns, and when lifted, the spindle stops.
- (18) "Cooling": The "Cooling" key is the repeated key to open and close the coolant, corresponding to the instructions M08 and M09.
- (19) Looseness knife: The "Looseness knife" key is the repeatedly open and close key of the loosening knife, corresponding to the instructions M10 and M11.
- (20) Manual speed: Press "0", "1", "2", "3", "4", "5", "6", "7", "8", "9" to set "F30", "F60", "F120", "F250", "F500", "F1000", "F1500", "F2000", "F 2500", "F3000" manual speed.
- (21) Blow: The "blow" key is the repeatedly open and close button, corresponding to the instruction M59 and M58.
- (22) Continuous and single-step switching: When the "single-step" key is pressed, the manual continuous mode is converted to the single-step mode. If it is single-step, the single-step value I=XXXX.XXX is displayed.

(23) 回机床零点: 按“回零”键后, 再按“X”, 则 X 轴自动回机床零点; 若按“Y”则 Y 轴自动回机床零点; 若按“Z”则 Z 轴自动回机床零点; 若按“A”则 A 轴自动回机床零点; 若按“B”则 B 轴自动回机床零点; 若按“0”则先回 Z 轴, 再回 X、Y、A 轴、B 轴; 若按“返回”键则放弃。回机床零点执行速度为速度参数中的 P38-P47 号参数, 执行方向为轴类参数 P47 号参数决定。

在回机床零点过程中, 如果机床未安装行程开关或金属接近开关, 数控系统将检测不到回答信号而一直向前运行, 这时应及时按急停按钮以免撞车。

(24) 坐标进给: 直接按“+X、-X、+Y、-Y、+Z、-Z、+A、-A、+B、-B”键分别对应 X 轴和 Z 轴的正方向、负方向进给。

(25) 速度切换: 在连续进给时, 再按下“快速”键, 则立刻转化为系统速度参数的 P1、P2、P3 号参数对应的执行速度, 松开“快速”键则继续按原来的速度进给。若设定速度比参数中的速度高则按设定速度进给。

(26) 多坐标显示切换, 按“F1 相对坐标”“F2 绝对坐标”、“F3 综合坐标”键可在“绝对坐标”、“相对坐标”“综合坐标”三个界面切换

(27) 工件数清零: 按“删除”, 再按“确认”键, 加工工件数清零。

(28) 对刀基准: 按“T”键, 设置自动对刀的基准点 (Z 轴坐标点), 用于设置刀补参数中刀具长度补偿的基准。

(29) 自动对刀: 确认好对刀基准后, 装好刀具移动刀具到对刀点, 在“刀补状态”按“F5 对刀”键, 输入自动对刀的刀具号, 相应刀具的长度补偿值就确认好了。

(30) 自定义键“K1”: 开启/关闭 Y24。

(31) 自定义键“K2”: 开启/关闭 Y25。

(32) 自定义键“K3”: 开启/关闭 Y26。

(33) 增量坐标: 选择到“相对坐标”状态按“设置”键修改或设置为 0;

(34) 自动分中或自动找圆心操作, 切换到 G54—G59, 按“P/J”键。按提示操作, 系统自动计算值存入当前 G54—G59 的偏移值中。

注: 机床坐标清零: 在参数状态按“E”, 再按“确认”键。

3.4.2 手动连续

连续操作是以按键按下时间为依据, 按下则进给, 抬起则停止进给。用面板上“+X、-X、+Y、-Y、+Z、-Z、+A、-A、+B、-B”键在指定的轴和方向上进给, 进给速度为显示的速度 (F 的数据) 乘以速度修调倍率值。

连续执行时, 若按下“快速”键则速度切换为速度参数的 P1、P2、P3 号参数对应的执行速度。如果本身设定速度高于参数中的速度则速度不变, 进给速度为速度参数中的 P1、P2、P3 号参数值的速度乘以快速修调倍率值。

为了方便用户在手动功能进行单轴切削, 在手动中可设置手动速度。按“F”键提示 F 时, 键入速度值即可。

当进给超出运行轴的正负两个方向的硬限位点时, 则停止进给, 并提示已处

(23) Returning to the machine zero position: Press the "Zero" button, then press "X" to automatically return the X-axis to the machine zero position; press "Y" for the Y-axis; press "Z" for the Z-axis; press "A" for the A-axis; press "B" for the B-axis; press "0" to first return the Z-axis before moving to X, Y, A, and B axes; press the "Return" button to cancel. The execution speed for returning to the machine zero position is controlled by parameters P38-P47 in the speed settings, while the direction is determined by parameter P47 of the axis-related parameters.

During the return to the machine tool zero point process, if the machine tool is not installed with a travel switch or metal proximity switch, the CNC system will not be able to detect the response signal and will continue to run forward. In this case, the emergency stop button should be pressed in time to avoid collision.

(24) Coordinate feed: press the keys "+X, -X, +Y, -Y, +Z, -Z, +A, -A, +B, -B" respectively to feed in the positive direction and negative direction of the X-axis and Z-axis.

(25) Speed switching: When continuous feed is in operation, pressing the "Quick" button will immediately switch to the execution speed corresponding to parameters P1, P2, and P3 in the system speed parameters. Release the "Quick" button to resume the original feed speed. If the set speed exceeds the parameter speed, continue feeding at the set speed.

(26) Multi-coordinate display and switching: Press "F1 relative coordinate", "F2 absolute coordinate" and "F3 comprehensive coordinate" keys to switch between the three interfaces of "absolute coordinate", "relative coordinate" and "comprehensive coordinate"

(27) Zero workpiece number: press "Delete" and then press "Confirm" button to zero the number of processed workpieces.

(28) Tool reference: Press the "T" key to set the reference point (Z axis coordinate point) for automatic tool reference, which is used to set the reference for tool length compensation in the tool compensation parameters.

(29) Automatic tool alignment: After confirming the tool alignment reference, install the tool and move the tool to the tool alignment point. Press the "F5 tool alignment" key in the "tool compensation state", and input the tool number for automatic tool alignment. The length compensation value of the corresponding tool is confirmed.

(30) Custom key "K1": Turn Y24 on/off.

(31) Custom key "K2": Turn Y25 on/off.

(32) Custom key "K3": Turn Y26 on/off.

(33) Incremental coordinates: Select the "relative coordinates" state and press the "set" key to modify or set it to 0;

(34) For automatic center division or automatic center finding operation, switch to G54--G59 and press the "P/J" key. Follow the prompts and the system will automatically calculate the value and store it in the current G54--G59 offset value.

Note: Machine tool coordinate reset: press "E" in the parameter state, and then press "Confirm".

3.4.2 Manual Continuity

Continuous operation is based on the time of button press, feed when pressed, stop feed when lifted.

Use the "+" and "-" keys on the panel to feed in the specified axis and direction, and the feed speed is the displayed speed (F data) multiplied by the speed adjustment multiplier.

During continuous operation, pressing the "Quick" button will switch the speed to the execution speed corresponding to parameters P1, P2, and P3 in the speed settings. If the preset speed exceeds the parameter values, the speed remains unchanged, while the feed speed is calculated by multiplying the values of parameters P1, P2, and P3 by the Quick Speed Adjustment Factor.

In order to facilitate users to perform single axis cutting in manual mode, the manual speed can be set in manual mode. When the "F" key is pressed and the F prompt is displayed, enter the speed value.

When the feed exceeds the hard limit points in both positive and negative directions of the running axis, the feed is stopped and a prompt is given that it has been placed

于限位轴的方向上不能再进给，此时只能反方向进给。(下同)。

手动最大速度由速度参数的 P6 号参数限制，当设置速度高于此值时，则按此参数值运行。

杂类参数 P38=8 时，则“快速”叠加键为锁存开关键。

3.4.3 手动单步

此操作是以设定的单步值为依据，每按一次进给键坐标运行一个单步值。进入单步值操作后提示的“I=0010.000”表示单步值为 10mm，可以按“I”键修改单步值，按“确认”键确认。

其运行速度为显示速度(F 的数据)乘以速度修调倍率值。

3.4.4 回机床零点

在本系统中回零点有两种方式，即可有参考点开关也可设置浮动零点，具体方法如下：

有基点开关：

回参考点操作是使各坐标轴依次进给到机床上的基点开关位置。如当参数中进给轴回参考点方式为 0 时，坐标轴检测到基点信号后再返回检测零位脉冲信号时，将机床坐标数据自动置为 0。

数控机床断电后再次接通数控系统电源、超程报警解除及急停按钮解除以后，需要进行再次回参考点操作，以建立正确的机床坐标系。

说明：1、系统对每次开机时回零有要求可通过轴类参数 P45 号设置，可以是强制也可能是提示；

2、回零时检测信号的类型和方式可通过轴类参数 P46 号设置，即可只需检测参考点开关，也可检测参考点开关之后再检测电机 Z 脉冲信号（精度更高些），对于电机 Z 脉冲可正向检测，也可反向检测。

3、对于回零的方向可通过轴类参数 P47 号（位参数）设置，D2、D4 位分别设置 X、Z 轴，0 表正向，1 表负向。

4、对于 X、Z 同时回零时的先后顺序可通过轴类参数 P47 号（位参数）设置，D8 位为 0 表示先 X 后 Z，1 则反之。

5、对于参考点开关的类型可通过轴类参数 P48 号（位参数）设置，D0、D2 分别设置 X、Z 轴，0 表常开，1 表常闭。

6、对于需要检测电机 Z 脉冲时的检测最大长度可通过轴类参数 P30、P31 号参数设置，该参数值必须小电机一转脉冲。

7、对于回零后的偏移距离可通过轴类参数 P54-P58 号参数设置，即回零完成后快速移动坐标到该参数值。

无基点开关：

可设定浮动零点来确定，通过轴类参数 P39 启用相应轴浮动零点功能，并分

There is no further feed in the direction of the limit axis, and only reverse feed is possible. (same below).

The maximum manual speed is limited by the speed parameter P6. When the speed is set higher than this value, it will run according to this parameter value.

When the miscellaneous parameter P38=8, the "quick" overlay key is the latch key.

3.4.3 Manual Single Step

This operation is based on the preset single-step value. Each press of the feed button will move the coordinate by one single-step value. After entering the single-step value operation, the prompt "I=0010.000" indicates that the single-step value is 10mm. You can press the "I" key to modify the single-step value and press the "Confirm" key to confirm.

Its running speed is the display speed (F data) multiplied by the speed adjustment multiplier.

3.4.4 Return to Machine Zero

In this system, there are two ways to return to zero point, that is, there can be a reference point switch or set a floating zero point, the specific method is as follows:

A base point switch:

The return reference point operation feeds each coordinate axis to the machine tool's base point switch position in sequence. If the feed axis return reference point mode is set to 0 in the parameters, when the coordinate axis detects the base point signal and returns to detect the zero pulse signal, the machine tool's coordinate data will be automatically reset to 0.

After the power supply of the CNC system is turned on again after the power failure of the CNC machine tool, after the over-run alarm is removed and the emergency stop button is removed, it is necessary to perform the reference point operation again to establish the correct coordinate system of the machine tool.

Note: 1. The system requires zero return at each startup, which can be set through shaft parameter No. P45, which can be either mandatory or prompt;

2、 The type and method of the return zero detection signal can be set through the shaft parameter P46. Only the reference point switch can be detected, or the motor Z pulse signal can be detected after the reference point switch (with higher accuracy). The motor Z pulse can be detected in both forward and reverse directions.

3、 The direction of zero return can be set through the axis parameter P47 (bit parameter). D2 and D4 are respectively set for X and Z axes, where 0 indicates forward and 1 indicates negative.

4、 The sequence of X and Z returning to zero at the same time can be set by axis parameter P47 (bit parameter). D8 bit is 0 to indicate X first and then Z, and 1 to indicate the reverse.

5、 The type of reference point switch can be set through the shaft parameter P48 (bit parameter). D0 and D2 are respectively set for X and Z axes. 0 is normally on, and 1 is normally off.

6、 For the maximum detection length when the motor Z pulse needs to be detected, parameter P30 and P31 can be set. The value of this parameter must be smaller than the pulse of one rotation of the motor.

7、 The offset distance after zero return can be set by parameter P54-P58 of the shaft, that is, quickly move the coordinate to this parameter value after zero return.

Zero base point switch:

The floating zero point can be set to determine. The corresponding shaft floating zero point function is enabled through the shaft parameter P39, and divided

别设置 P40-P44 参数来确定 X/Y/Z/A/B/C/Xs/Ys 轴的浮动零点值, 确定机床的零点。

回参考点的操作方法如下:

在手动方式下, 按“回零”键, 系统提示 (X; Z; 0 (XZ)), 在对话框中要求选择哪轴回参考点。在图中输入 X 或 Z, 按 Enter 键则只将相应轴回参考点, 或输入 0 后, 按 Enter 键则按 X、Z 轴的顺序依次返回参考点。回参考点成功后, 在绝对坐标和相对坐标系中轴坐标前面的圆圈变为绿色, 否则回参考点失败。

回参考点过程中若要中止, 则按“停止”或“RESET”按钮则停止返回参考点动作。

特别注意: 每次启动系统后必须进行一次回参考点操作, 以确保机床加工精度。因意外断电或非正常关机后启动系统也必须进行回参考点操作, 否则可能造成故障。

3.4.5 手摇脉冲发生器 (手轮)

本系统中手轮可配置面板式和手持式手脉, 其通过杂类参数 P1 号设置。

面板式手轮操作:

按“手轮”键选择“手轮方式”。然后在字母键 X、Z 选择相应的轴, 再按“手轮”选择脉冲倍率。

NCK数控
鑫科瑞数控

手持手脉操作:

按“手轮”键选择“手轮方式”。然后操作手持手脉的轴选开关选择相应的轴, 操作手持手脉的轴、倍率开关选择脉冲倍率。

说明:

手轮主要用于对刀, 其运行速度与手轮每一格的进给量和旋转手轮的快慢相关, 若本系统在配用步进电机时, 进给速度不宜太快。

手摇脉冲发生器的转速要低于 200r/min (每转 100 脉冲的手轮)。

手轮方式在自动图形方式时无效, 只能在坐标运行方式下才有效。

3.4.6 工件坐标系

- 1、按“F8 选择坐标”键或“F5 MDI”选择相应的工件坐标系 (G54-G59);
- 2、移动要对刀的轴到指定位置;
- 3、按“F7 设置坐标”键, 输入对应工件坐标系的坐标。

3.5 自动操作

Do not set the P40-P44 parameters to determine the floating zero point value of the X/Y/Z/A/B/C/Xs/Ys axes to determine the zero point of the machine tool.

The operation method of the reference point is as follows:

In manual mode, press the "Zero Return" button. The system prompts (X; Z; 0 (XZ)). In the dialog box, select which axis should return to the reference point. Enter X or Z in the diagram and press Enter to return to the corresponding axis's reference point. Alternatively, enter 0 and press Enter to sequentially return to the reference points for both X and Z axes. After successful reference point return, the circle preceding the axis coordinates in absolute and relative coordinate systems will turn green. Failure to return to the reference point will result in a red circle instead.

If the return to reference point process is to be stopped, press the "STOP" or "RESET" button to stop the return to reference point action.

Special note: a reference point return operation must be performed after each system startup to ensure machining accuracy of the machine tool. A reference point return operation must also be performed after starting the system due to accidental power failure or abnormal shutdown, otherwise the system may fail.

3.4.5 Hand Pulse Generator (handwheel)

In this system, the handwheel can be configured with panel and handheld hand pulse, which is set by miscellaneous parameter P1.

Operation of panel handwheel:

Press the "Handwheel" key to select "Handwheel mode". Then select the corresponding axis by pressing the letter keys X and Z, and press the "Handwheel" key to select the pulse multiplier.

Hand pulse operation:

Press the "Handwheel" key to select "Handwheel mode". Then operate the axis selection switch of the handheld pulse meter to select the corresponding axis, and operate the axis and multiplier switch of the handheld pulse meter to select the pulse multiplier.

explain :

The handwheel is mainly used for tool cutting, and its running speed is related to the feed per grid of the handwheel and the speed of rotating the handwheel. If the system is equipped with a stepper motor, the feed speed should not be too fast.

The hand pulse generator should rotate at a speed of less than 200 r/min (100 pulses per turn of the handwheel).

The handwheel mode is invalid in automatic graphics mode and only works in coordinate run mode.

3.4.6 Workpiece Coordinate System

1. Press "F8 Select coordinate" key or "F5 MDI" to select the corresponding workpiece coordinate system (G54-G59);

2. The moving blade shaft should be moved to the specified position;

3. Press the "F7 Set coordinate" key and input the coordinates of the corresponding workpiece coordinate system.

3.5 Automata

自动是指对当前编辑的工件程序进行加工。本系统可以任意点起动,也可以任意行启动或任意刀具启动。任意行或任意点启动必须在程序中用绝对坐标编程。自动方式不能进行手动坐标的移动。

运行程序的选择: 在程序界面下,按“↑”“↓”将光标移动到需要执行的程序名上,按“F7 执行程序”键即选中需要自动运行的程序。

在手动方式按“自动”键进入自动方式。

多坐标显示切换,按“F1-F3”功能键可在“绝对坐标”、“相对坐标”“综合坐标”三个界面切换。

3.5.1 自动加工方式

“单段/连续”的选择:按“单段”键则单段/连续加工方式进行反复切换。

“连续运行”是指按“运行”按钮后,程序依次不断地执行各程序段(程序行),直到程序结束或有停止指令出现。

“单段运行”是在运行程序时只运行当前一行程序后即停止,等待按键操作或再次按“运行”按钮。即按一次“运行”按钮运行一段程序。

“坐标/图形”的选择:按“图形”键则坐标/图形执行方式进行反复切换。

“自动坐标”是指程序运行的坐标轴坐标以坐标数值方式显示。

“自动空运行”是指程序以空运行方式运行,坐标轴不移动。

“自动图形”是指程序运行的坐标轴以加工轨迹方式显示。

3.5.2 从程序任意行启动

A、实际行启动:

在自动加工方式下,按“-”键弹出下图对话框,输入行号,按Enter键确认,或者在自动坐标显示状态按“↑、↓”键选择程序行,即此行作为即将运行的当前行。当前行是指将要执行的程序行,显示在屏幕程序显示区的第一行。当按运行按钮时则从输入的实际行号开始运行。

注意:1、该行号为程序的实际行号,不是N地址符指定的行号。系统先自动以“速度参数”P8中的默认G01/G02/G03速度运行到该指定行的起点处,然后正常运行程序。

2、从实际行号加工时,缺省行号为上一次暂停程序时的行号,方便用户操作。

3、数控系统在坐标界面使用N查找过行号后再按“RESET”键就能返回到程序开头。

B、标记行启动:

系统具有从程序的某一标记行开始运行的功能。在自动加工方式下,按“N”

Automatic refers to the processing of the currently edited workpiece program. The system can be started at any point, line, or tool. Any line or point start must be programmed using absolute coordinates in the program. Automatic mode does not allow manual coordinate movement.

Selection of the running program: Under the program interface, press "↑" and "↓" to move the cursor to the name of the program to be executed. Press "F7 execute the program" to select the program to be automatically run.

Press the "Automatic" button manually to enter automatic mode.

Multi-coordinate display and switching: Press the function keys "F1-F3" to switch between the three interfaces of "absolute coordinate", "relative coordinate" and "comprehensive coordinate".

3.5.1 Automatic Processing Method

"Single/continuous" selection: Press the "single" key to repeatedly switch between single and continuous processing modes.

"Continuous run" means that after pressing the "run" button, the program executes each program segment (program line) continuously in turn until the end of the program or a stop instruction is issued.

"Single-line operation" is to run the program only one line at a time and then stop, waiting for a button operation or another "run" button. That is, press the "run" button to run a section of the program.

"Coordinate/Graphic" selection: Press the "Graphic" key to repeatedly switch between coordinate/Graphic execution modes. "Automatic coordinates" means that the coordinates of the coordinate axis on which the program runs are displayed in numerical coordinates.

"Auto run" means that the program runs in empty mode without moving the coordinate axes.

"Automatic graphics" refers to the program running coordinate axis is displayed in the form of a machining path.

3.5.2 Start from Any Line of the Program

A. Actual Start:

In automatic processing mode, pressing the "-" key will bring up the dialog box below. Enter the line number and press Enter to confirm. Alternatively, while in automatic coordinate display mode, press the "↑" and "↓" keys to select the program line that will be executed next. The currently selected line is displayed as the first line in the screen's program display area. When the run button is pressed, the program will execute from the line number you entered.

- Note:
1. The line number is the actual line number of the program, not the line number specified by the N address symbol. The system will automatically run at the default G01/G02/G03 speed in the "Speed parameter" P8 to the starting point of the specified line, and then run the program normally.
 2. In actual processing, the default line number is the line number when the program was paused last time, so as to facilitate user operation.
 3. In the coordinate interface of the CNC system, after using N to find the line number, press the "RESET" key to return to the beginning of the program.

B. Marking Line Initiation:

The system has the function of running from a certain marker line of the program. In automatic processing mode, press "N"

键，就会弹出一个对话框，在编辑框中输入开始运行的标记行号，按 Enter 键确认。当按运行按钮时则从输入的行号开始运行。

注意：该行号不是程序的实际行号，是 N 地址符指定的行号。系统先自动以“速度参数” P8 中的默认 G01/G02/G03 速度运行到该指定行的起点处，然后正常运行程序。

C、从某把刀启动

系统具有从程序的某一刀具开始运行的功能。在自动加工方式下，按“G”键，再输入开始运行的刀具号（只输入刀具号，不输入刀补号；如：0304，只输入“03”），按 Enter 键确认。当按运行按钮时则从输入的刀具号开始运行。

注意：该行号不是程序的实际行号，是 N 地址符指定的行号。系统先自动以“速度参数” P8 中的默认 G01/G02/G03 速度运行到该指定行的起点处，然后正常运行程序。

3.5.3 程序启动

程序启动必须按“自动”键切换为自动方式，有以下二种方式，程序自动开始执行。

- （1）面板按“启动”键。
- （2）接通外接运行 RUN 信号。

3.5.4 程序停止运行

程序停止运行有五种方式：

- （1）程序指令 M00、M01、M02、M30、M20。
- （2）按“单段”按键当前程序段运行完后停止。
- （3）干预开关置于中则或右则。
- （4）接通外接停止 HALT 信号或按“暂停”键，程序暂停。
- （5）按“RESET”键则中止程序的所有动作（如工作台、主轴、刀架等所有动作）

3.5.5 自动加工中的实时控制

（1）G00 快速修调：每按一次“快速↑”键，进给速度倍率修调增加 20%；每按一次“快速↓”键，进给速度倍率修调减少 20%。加工过程中根据不同情况可以任意地调整快速倍率。

（2）进给速度修调：当轴类参数 P1 号设置为 0 时，每按一次“进给↑”键，进给速度倍率修调增加 10%；每按一次“进给↓”键，进给速度倍率修调减少 10%。当轴类参数 P1 号设置为 1 时，将通过外接波段开关调节进给速度。进给速度修调分为 0%-150% 16 档，加工过程中根据不同情况可以任意地调整加工速

Press the key, and a dialog box will pop up. Enter the starting line number in the edit box and press Enter to confirm. When you press the run button, it will run from the line number you entered.

Note: The line number is not the actual line number of the program, but the line number specified by the N address. The system automatically starts with

The default G01/G02/G03 speed in the "Speed parameter" P 8 runs to the starting point of the specified line, and then the program runs normally.

C. Start from a Knife

The system operates by starting with a specific tool in the program. In automatic machining mode, press the "G" key and enter the tool number (only the tool number is required, not the tool supplement number; example: 0304, simply type "03"). Confirm by pressing Enter. When the run button is pressed, the system will execute operations starting from the entered tool number.

Note: The line number is not the actual line number of the program, but the line number specified by the N address. The system automatically starts with

The default G01/G02/G03 speed in the "Speed parameter" P 8 runs to the starting point of the specified line, and then the program runs normally.

3.5.3 The Program Starts

To start the program, you must switch to automatic mode by pressing the "Automatic" key. There are two ways to start the program automatically.

- (1) Press the "Start" button on the panel.
- (2) Connect the external RUN signal.

3.5.4 The Program Stops Running

There are five ways to stop a program from running:

- (1) Program instructions: M00, M01, M02, M30, M20.
- (2) Press the "single segment" button to stop the current program segment after it has run.
- (3) The intervention switch is placed in the middle or right.
- (4) Connect the external HALT signal or press the "pause" button to pause the program.
- (5) Press the "RESET" key to stop all actions of the program (such as worktable, spindle, tool holder and all other actions)

3.5.5 Real-Time Control in Automatic Processing

(1) G00 Fast Speed Adjustment: Press the "Fast ↑" key once to increase the feed speed ratio by 20%; press the "Fast ↓" key once to decrease the feed speed ratio by 20%. The fast speed ratio can be adjusted arbitrarily during processing according to different conditions.

(2) Feed Speed Adjustment: When parameter P1 is set to 0, each press of the "Feed↑" button increases the feed speed multiplier by 10%, while each press of the "Feed↓" button decreases it by 10%. When parameter P1 is set to 1, the feed speed adjustment is controlled via an external band switch. The feed speed adjustment ranges from 0% to 150% across 16 steps, allowing flexible adjustments during machining according to specific requirements

度。

(3) 主轴转速修调：当轴类参数 P2 号设置为 0 时，每按一次“主轴↑”键，主轴转速倍率修调增加 10%；每按一次“主轴↓”键，主轴转速倍率修调减少 10%。当轴类参数 P2 号设置为 1 时，将通过外接波段开关调节主轴转速。主轴转速修调分为 5%-150% 16 档，加工过程中根据不同情况可以任意地调整主轴转速。

(4) 加工中停：在连续方式加工过程中，如果按“单段”键当前段程序行执行完后将停止运行，等待操作。

(5) 加工暂停：可以将干预开关置于中则、右则或按面板上的“暂停”键，加工程序即可暂停。暂停后按“运行”按钮或“启动”键程序将继续运行；若此时按“复位”键，则退出自动加工运行状态，同时当前行指向加工程序第一行。

(6) 进给保持：在加工暂停（外接 HALT、按面板上的“暂停”键、按“手动”键或“单段”键）时，按“手动”键，自动进给保持，可以进行坐标调整等工作，再按“自动”，按“启动”键后自动运行到暂停点再继续运行。

(7) 退出加工：运行中或“暂停”时或“进给保持”时按“RESET”键。

(8) 加工过程中按“K”进入编辑当前行。

(9) 加工过程中或停止状态通过输入点启动内部宏程序。如 X0 有效时启动 ProgramUser0 程序，在 PLC 中加入：

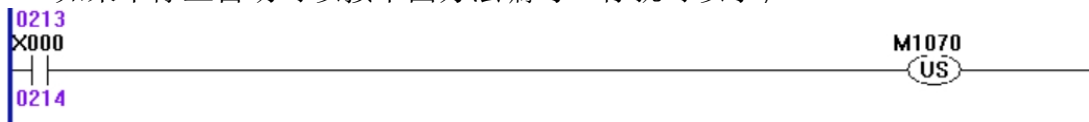


M309：中间寄存器，PLC 中不能有相同的 M309 出现；

M1014：复位当前程序；

M1070-M1079：对应 ProgramUser0-9 程序。

如果不停止自动可以按下面方法编写一行就可以了；



3.5.6 MDI 运行方式

在手动调整或自动坐标状态按“F5 MDI”键进入 MDI 执行方式。“MDI”方式是输入一段程序执行一段程序，在输入过程中按“返回”键则放弃退出，按“启动”键则马上执行当前输入程序段。

linear measure .

(3) Spindle Speed Adjustment: When parameter P2 is set to 0, each press of the "Spindle ↑" button increases the spindle speed multiplier by 10%, while pressing the "Spindle ↓" button decreases it by 10%. When parameter P2 is set to 1, external band switches are used to adjust the spindle speed. The speed adjustment ranges from 5% to 150% across 16 steps, allowing flexible adjustments during machining according to specific requirements.

(4) Processing stop: In the continuous processing process, if you press the "single segment" key, the operation will be stopped after the current segment program line is executed and the operation will be awaited.

(5) Process Pause: To pause the machining program, you can either set the control switch to the center position, right position, or press the "Pause" button on the panel. After pausing, pressing the "Run" button or "Start" key will resume the program. If you press the "Reset" button at this point, it will exit the automatic machining mode while moving the cursor to the first line of the current machining program.

(6) Feed Hold: When processing is suspended (external HALT, pressing the "Pause" button on the panel, pressing the "Manual" button or "Single Segment" button), press the "Manual" button to automatically hold feed. Coordinate adjustment and other tasks can be performed. Then press "Automatic", and after pressing the "Start" button, it will automatically run to the pause point and continue running.

(7) Exit processing: Press the "RESET" key when running or "pause" or "feed hold".

(8) Press "K" during processing to edit the current line.

(9) During processing or when the system is stopped, the internal macro program can be activated through input points. For example, if X0 is valid, the ProgramUser0 program will be activated. Add the following to the PLC:



M309: Intermediate register, there can be no same M309 in PLC;

M1014: Reset the current program;

M1070-M1079: Corresponding to ProgramUser0-9 program. If the automatic operation is not stopped, you can write one line according to the following method;



3.5.6 MDI Run Mode

To manually adjust or automatically coordinate the state, press the "F5 MDI" key to enter the MDI execution mode. The MDI mode allows you to input and execute a program segment. During input, pressing the "Return" key will abandon and exit the process, while pressing the "Start" key will immediately execute the current program segment.

3.5.7 手轮运行方式

在自动方式下按“手轮”，此时转动手轮程序将自动加工，其速度与程序 F 速度、进给修调倍率和转动手轮的快慢有关。此方式一般用于试加工运行程序。

注意：手轮运行时的加减速和最大速度可由速度参数的 P26-P33、P49 控制，当该参数设置为无效时，由采用默认的加减速和 G00 速度。

3.5.8 DNC 功能

在本系统中用户存储空间为 128Mbit（总线系统可以扩展 32Gbit），系统也可以用 DNC 功能加工。在本系统中 DNC 功能的实现可通过 RS232 和 USB 接口。

A、RS232-DNC 说明

1、将计算机与系统用专用通讯线连接好，在系统上设置好相应的通讯接口和速率。

2、在 PC 计算机上运行本系统专用通讯软件，设置好相应的通讯接口和速率。点击“发送 CNC 程序文件”，选择要联机加工的程序文件，进入正在发送程序文件状态。

3、在数控系统中进入程序文件界面下，按“L”键即可进入联机加工状态，此时界面的右上角显示“RS232--DNC”，在自动状态下按运行按钮即可进行联机加工。

4、在联机加工过程中要停止系统运行则旋转“干预开关”置中或右端，取消联机加工状态“停止”键或“RESET”键即可。

注意：1、使用串口传送文件时设置波特率的高低与使用环境（通讯线长度、现场环境）有关。

2、通讯线缆长度不大于 10 米。

3、在用户 PC 机一端需要使用本系统专用通讯软件方可传输程序。需特别注意的是将 PC 机和数控系统双方的传输速率设置一致，否则无法传输。

B、USB-DNC 说明：

USB-DNC 是通过优盘实现的，即将优盘外挂在系统上，选中优盘中的程序执行。

在程序界面下按“F6 打开优盘”键打开优盘，选中相应的程序后，按“F7 执行程序”键执行程序，按“自动”切换到自动状态，此时按“运行”键即可运行程序。

在联机加工过程中要停止系统运行则旋转“干预开关”置中或右端，取消联机加工状态“停止”键或“RESET”键即可。

注意：1、USB-DNC 过程中请不要拔动优盘，更不要取下优盘，则否会造成加工失败。

2、USB-DNC 完成后需从优盘界面回到系统程序界面。

3.5.7 Handwheel Operation Mode

In automatic mode, press the "handwheel". At this time, the handwheel rotation program will be automatically processed, and its speed is related to the program F speed, feed adjustment ratio and the speed of the handwheel rotation. This mode is generally used for trial processing of the program.

Note: The acceleration, deceleration and maximum speed of the handwheel can be controlled by the speed parameters P26-P33 and P49. When this parameter is set to invalid, the default addition/subtraction and G00 speeds are used.

3.5.8 DNC Function

In this system, the user storage space is 128Mbit (the bus system can be expanded to 32Gbit), and the system can also be processed with DNC function. In this system, the implementation of DNC function can be achieved through RS232 and USB interface.

A RS232-DNC description

1. Connect the computer and the system with a special communication line, and set up the corresponding communication interface and rate on the system.
2. Run the system-specific communication software on the PC, and set up the corresponding communication interface and speed. Click "Send CNC program file", select the program file to be processed online, and enter the status of sending the program file.
3. In the program file interface of the CNC system, press the "L" key to enter the online machining state. At this time, the upper right corner of the interface shows "RS232-DNC". In the automatic state, press the run button to carry out online machining.
4. If the system is to be stopped during online machining, rotate the "interference switch" in the middle or right end, and cancel the "stop" key or "RESET" key of online machining status.

Note: 1. The setting of baud rate when using serial port to transmit files is related to the use environment (communication line length, field environment).

2. The length of communication cable is not greater than 10 meters.

3. The system dedicated communication software is required to be used on the user's PC to transfer the program. It should be noted that the transmission rate between the PC and the CNC system should be set to the same, otherwise the transmission cannot be achieved.

B. USB-DNC Description:

USB-DNC is implemented through the USB drive, that is, the USB drive is attached to the system and the program in the USB drive is selected to be executed.

Under the program interface, press "F6 to open the USB drive" to open the USB drive. After selecting the corresponding program, press "F7 to execute the program" to execute the program. Press "Automatic" to switch to automatic mode, and then press "Run" to run the program.

In the process of online machining, if you want to stop the system operation, rotate the "interference switch" in the middle or right end, and cancel the "stop" key or "RESET" key of online machining state.

Note: 1. Do not pull out the USB-DNC process, and do not remove the USB, otherwise it will cause processing failure.

2. After USB-DNC is completed, you need to go back from the USB disk interface to the system program interface.

3、在 USB-DNC 时在执行程序前，最好选中程序后按“P”键对程序进行编译一遍，防止中途出现程序错误。

3.6 安全操作、报警提示

3.6.1 紧急停止

当发生紧急意外情况时可按下“紧急停止”按钮，系统将停止机床一切动作，系统复位屏幕显示“系统紧急停车”。等待按钮抬起。当杂类参数 P29 设置为有效时 M67 输出有效信号。

如果在加工过程中或在机床运动过程中按“紧急停止”按钮，系统坐标与机床位置可能发生变化，加工前需要重新确定系统坐标位置，最好进行回机床零点操作使显示坐标与机床位置一致。

系统可以外接急停 STOP 按钮，通过杂类参数 P27 设置按钮是常开或常闭。

3.6.2 系统复位

在系统运行的任何时候均可按“RESET”键中止当前操作，特别是在手、自动下按该键可中止工作台、主轴、刀架等所有机床动作，但对于坐标而言不会丢失，无需回零。

NCK数控
鑫科瑞数控

3.6.3 机床报警

机床出现报警时，屏幕以反白方式提示出错信息并闪烁，程序停止运行，中止坐标移动，此时应检查故障原因并排除故障后方可再运行。杂类参数当 P29 为 1 时 M67 输出信号有效。

- (1) X、Z 轴正向硬限位：X 或 Z 轴正方向正处于硬限位位置。
- (2) X、Z 轴负向硬限位：X 或 Z 轴负方向正处于硬限位位置。
- (3) 主轴、变频器报警：ALM1 机床变频器报警信号有效。
- (4) 无润滑油报警：ALM2 机床主轴报警信号有效。
- (5) XYZA 轴驱动器报警/X20：ALM 伺服驱动报警信号有效。在诊断状态按“B”键输出 INTH 信号复位伺服驱动。
- (6) BCXsYs 轴驱动器报警/X22：ALMB 伺服驱动报警信号有效
- (7) 门开关 5 号报警：M12 门开关报警信号有效。
- (8) 紧急停止：按下了急停按钮。
- (9) 没有检测到输入信号：指令 WAT 检测不到输入信号。

3.7 参数操作

在任何状态下，按“参数”键，进入参数设置状态。在本系统中参数分为“加

3、 In USB-DNC, before executing the program, it is best to select the program and press the "P" key to compile the program once to prevent errors from occurring in the middle.

3.6 Safety operation and alarm prompt

3.6.1 Emergency Stop

In case of emergency, press the "Emergency Stop" button. The system will stop all movements of the machine tool and reset the screen to display "System Emergency Stop". Wait until the button is lifted. When miscellaneous parameter P29 is set to effective, M67 outputs an effective signal.

If the "emergency stop" button is pressed during processing or during machine tool movement, the system stops

The position of the gauge and the machine tool may change. Before processing, the system coordinate position needs to be redefined. It is best to perform the return to the machine zero point operation to make the displayed coordinate consistent with the position of the machine tool.

The system can be connected with an emergency stop STOP button, which can be set as normally on or off through miscellaneous parameter P27.

3.6.2 System Reset

At any time during the operation of the system, you can press the "RESET" key to stop the current operation. Especially under manual and automatic conditions, pressing this key can stop all machine tool operations such as the worktable, spindle and tool holder, but the coordinates will not be lost and there is no need to return to zero.

3.6.3 Machine Tool Alarm

When the machine tool alarms, the screen will display the error information in reverse white and flash, and the program will stop running and coordinate movement will be stopped. At this time, the fault cause should be checked and the fault should be eliminated before running again. Miscellaneous parameters: When P29 is 1, M67 output signal is valid.

(1) X and Z axis positive hard limit: The X or Z axis is in the positive direction of the hard limit position.

(2) X, Z-axis negative hard limit: The X or Z-axis negative direction is in the hard limit position.

(3) Main shaft, frequency converter alarm: ALM1 machine tool frequency converter alarm signal is valid.

(4) No lubricating oil alarm: ALM2 machine tool spindle alarm signal is valid.

(5) XYZA axis driver alarm/X20: The ALM servo drive alarm signal is valid. In diagnostic mode, press the "B" key to output the INTN signal to reset the servo drive.

(6) BCXsYs Axis driver alarm/X22: ALMB servo drive alarm signal is valid

(7) Door switch No.5 alarm: M12 door switch alarm signal is valid.

(8) Emergency stop: The emergency stop button has been pressed.

(9) No input signal was detected: The WAT command did not detect the input signal.

3.7 Parameter Operations

In any state, press the "Parameter" key to enter the parameter setting state. In this system, parameters are divided into "add"

工参数”、“速度参数”、“轴类参数”、“刀具参数”、“杂类参数”、“坐标系”、“密码”七类。

手动连续

N00000

2016-06-06 06:06

1, 固定循环G73退刀	5.000
2, 固定循环G83退刀	5.000
3, 镗孔固定循环G76	1
4, 镗孔固定循环G87	1
5, 镗孔固定循环G76	0
6, 排屑攻丝G84G74排	0.000
7, 排屑攻丝G84G74退	0.000
17, 程序运行须主轴转	0
18, 指令M20运行次数	-1
21, G01/G02/G03段	0
22, G00后延时(ms)	0
23, 手轮平滑加减速常	70
200, 系统屏保时间	1

当前程序99.TXT

指令代码

G53

T01H0D0

状态

M05M09M10不选停

M78M33M41跳段

G00X100%

F2000X100%

S800X100%SP000

机床坐标

X168.000A168.000

Y168.000B168.000

Z168.000

单件时间0: 6

加工件数0

主轴转速0

状态: 正常

F1 加工参数

F2 速度参数

F3 轴类参数

F4 刀具参数

F5 杂类参数

F6 坐标系

F7 密码

F8 返回

进入参数界面后对应按“F1-F7”进入相应界面，按“↑”“↓”键选择参数号，按“确认”弹出对话框输入数据，输入完成后再“确认”键参数修改成功。
具体参数说明如下：

3. 7. 1 加工参数

- ★、1, 固定循环 G73 退刀量 d 的值(mm)
用于设置固定循环 G73 退刀量 d 的值(mm)，详见指令说明。
- ★、2, 固定循环 G83 退刀量 d 的值(mm)
用于设置固定循环 G73 退刀量 d 的值(mm)，详见指令说明。
- ★、3, 镗孔固定循环 G76 偏移量 Q 的方向
(以 G17 平面为例)(1 表+X,2 表-X,3 表+Y,4 表-Y)
用于设置固定循环 G76 偏移量 Q 的方向，详见指令说明。
- ★、4, 镗孔固定循环 G87 偏移量 Q 的方向
(以 G17 平面为例)(1 表+X,2 表-X,3 表+Y,4 表-Y)
用于设置固定循环 G87 偏移量 Q 的方向，详见指令说明。

There are seven categories: "working parameters", "speed parameters", "shaft parameters", "tool parameters", "miscellaneous parameters", "coordinate system" and "password".

手动 连续

N00000

2016-06-06 06:06

当前程序99.TXT

指令代码

G53

T01H0D0

状态

M05M09M10不选停

M78M33M41跳段

G00X100%

F2000X100%

S800X100%SP000

机床坐标

X168.000A168.000

Y168.000B168.000

Z168.000

单件时间0: 6

加工件数0

主轴转速0

1, 固定循环G73退刀5.000

2, 固定循环G83退刀5.000

3, 镗孔固定循环G761

4, 镗孔固定循环G871

5, 镗孔固定循环G760

6, 排屑攻丝G84G74排0.000

7, 排屑攻丝G84G74退0.000

17, 程序运行须主轴转0

18, 指令M20运行次数-1

21, G01/G02/G03段0

22, G00后延时(ms)0

23, 手轮平滑加减速常70

200, 系统屏保时间1

状态: 正常

F1 加工参数

F2 速度参数

F3 轴类参数

F4 刀具参数

F5 杂类参数

F6 坐标系

F7 密码

F8 返回

After entering the parameter interface, press "F1-F7" to enter the corresponding interface, press "↑" and "↓" keys to select the parameter number, press "Confirm" to pop up a dialog box to input data, and then press "Confirm" after the completion of input to confirm the successful modification of parameters.

The specific parameters are as follows:

3. 7. 1 Processing Parameters

- ★1. Value of the tool retraction length (mm) for fixed cycle G73
Used to set the value of the fixed cycle G73 and the tool retraction amount d (mm). See the instruction description for details.
- ★2, fixed cycle G83, value of the tool retraction d (mm)
Used to set the value of the fixed cycle G73 and the tool retraction amount d (mm). See the instruction description for details.
- ★3. Boring fixing cycle G76, and the direction of offset Q
(Using the G17 plane as an example) (Table 1+X, Table 2-X, Table 3+Y, Table 4-Y)
Used to set the direction of the offset Q for a fixed cycle G76. See instruction instructions for details.
- ★4. Boring fixing cycle G87, the direction of offset Q
(Using the G17 plane as an example) (Table 1 + X, Table 2-X, Table 3 + Y, Table 4-Y)
Used to set the direction of the offset Q for a fixed cycle G87. See instruction instructions for details.

★、5,镗孔固定循环 G76G87 的主轴定向停止角度(0.1 度)

用于设置固定循环 G76G87 的主轴定向停止角度, 详见指令说明。

★、6,排屑攻丝 G84G74 排屑方式(0 表高速排屑回退 d,8 表普通排屑回退到 R)

用于设置固定循环 G84G74 排屑方式, 0 表高速排屑回退 d,8 表普通排屑回退到 R。

★、7,排屑攻丝 G84G74 退刀量 d(mm)

用于设置固定循环 G84G74 退刀量 d, 详见指令说明。

★、8, 固定循环 G73G83 每次提刀前暂停时间(ms) [绝对值大于 5 有效, 设为负数时表提刀按 G01 方式, 否则按 G00 方式]

用于设置固定循环 G73G83 每次提刀前暂停时间, 大于 5 有效,设为负数时表提刀按 G01 方式,否则按 G00 方式。

★、17, 程序运行须主轴转动[1 表是, 0 表否]

用于设置主轴转动与程序运行互锁, 设置程序运行时是否检测主轴运行情况, 即该值设置为 1 时, 程序运行需要主轴转动; 当该值设置为 0 时, 程序运行不需检测主轴运转情况。

★、18, 指令 M20 运行次数

该参数用于设定当程序中使用 M20 指令时循环执行程序次数, 当此参数为负值时表示无限循环。

★、19, 当前加工件数

该参数用于显示或设置程序指令 M20 执行的次数, 即工件件数。

★、21, G01/G02/G03 段间延时 (ms) (>100 有效)

此参数设置 G01/G02/G03 程序段间的延时时间, 此参数可以解决加工拐角处(条 G01 之间)的过切问题。

★、22, G00 执行后延时 (ms) (>100 有效)

此参数设置执行 G00 后的延时时间。

★、23, 手轮平滑加减速常数[50—100]

用于设置使用手轮时平滑升降速时间常数, 设置范围 50-100, 值越小升降速越快, 但振动越大。

★、200, 系统屏保时间 (分钟)

当处于主要界面并且不处于对话框状态时等待加工参数 200 设定时间 (<2 分钟时不屏保) 后进入屏保, 屏保后任意键返回。

★、202, 系统内部参数

坐标轴显示(1 表正常, 4 表无 X, 8 表无 Y, 16 表无 Z, 32 只显示 X, 64 只显示 Y, 128 只显示 Z, 256 显示 XY)

内部参数, 可以设置坐标轴显示方式, 4 表示不显示 X, 8 表示不显示 Y, 16 表示不显示 Z, 32 表示只显示 X, 64 表示只显示 Y, 128 表示只显示 Z, 256 表示显示 XY。

★、203, 是否严格检查 G41/G42(23103490 表启用暂停键, 2131970 表严格检

★, 5, boring fixed cycle G76G87 spindle orientation stop Angle (0.1 degrees)

Used to set the spindle orientation stop angle for fixed cycle G76G87. See instruction instructions for details.

★6. Chip removal and tapping G84G74 Chip removal method (0 table high-speed chip removal return d, 8 table normal chip removal return to R)

Used to set the fixed cycle G84G74 chip removal mode. Table 0 high-speed chip removal retraction d, Table 8 normal chip removal retraction to R.

★, 7, chip removal and tapping G84G74 tool back clearance d (mm)

Used to set the fixed cycle G84G74 tool back amount d, see instruction description for details.

★8. Fixed cycle G73G83 Pause time before each tool lifting (ms) [Absolute value greater than 5 valid. If set as negative, the tool lifting follows G01 mode; otherwise, it follows G00 mode]

Used to set the pause time before each tool lifting of fixed cycle G 73G83. If the pause time is greater than 5, it is valid. If the pause time is negative, the tool lifting is in G01 mode; otherwise, it is in G00 mode.

★17. The program shall run when the spindle rotates [Table 1 is yes, Table 0 is no]

Used to set the interlock between spindle rotation and program operation, and set whether to detect spindle operation during program operation. That is, when this value is set to 1, the spindle rotation is required for program operation; when this value is set to 0, the spindle operation is not required for program operation.

★, 18, number of times the instruction M20 is run

This parameter is used to set the number of times the program is executed in a loop when the M20 instruction is used. If this parameter is negative, it indicates an infinite loop.

★19. Current number of processing pieces

This parameter is used to display or set the number of times the program instruction M20 is executed, that is, the number of workpieces.

★, 21, G01/G02/G03 inter-segment delay (ms) (>100 valid)

This parameter sets the delay time between G01/G02/G03 program segments. This parameter can solve the problem of overcutting at the eighth corner (between G01 bars).

★22, G00 execute delay (ms) (>100 valid)

This parameter sets the delay time after G00 is executed.

★, 23, smooth acceleration and deceleration constant of handwheel [50-100]

Used to set the smooth lifting speed time constant when using the handwheel. The setting range is 50-100. The smaller the value, the faster the lifting speed, but the greater the vibration.

★, 200, system screen time (minutes)

When you are in the main interface and not in the dialog box state, wait for the processing parameter 200 to be set for a time (<2 minutes without screen protection) before entering the screen protection. After the screen protection, return by any key.

★, 202, system internal parameters

The coordinate axis is displayed (1 for normal, 4 for no X, 8 for no Y, 16 for no Z, 32 only displays X, 64 only displays Y, 128 Shows Z, 256 Shows XY)

Internal parameters can set the display mode of coordinate axes. 4 means not to show X, 8 means not to show Y, 16 means not to show Z, 32 means only to show X, 64 means only to show Y, 128 means only to show Z, and 256 means to show XY.

★ 203, whether to strictly check G41/G42 (Table 23103490 enables the pause button, Table 2131970 strictly checks

查, 34818 表不严格检查, 6326274 表快速预读加工)

此参数为特殊参数, 2131970 表示严格检查 G41/G42 指令, 34818 表不严格检查 G41/G42 指令, 6326274 表快速预读 G41/G42 指令。

★、205, 内部参数 (+8 表插补攻丝要等待脉冲发完; 32 表三位小数; 96 四位小数; 160 表 HX-G 指令; +4096 表铣床循环不编坐标点就不执行)

内部参数, +8 表示插补攻丝要等待脉冲发完才执行下一条指令; 32 表示坐标轴显示为三位小数; 96 表示坐标轴显示四位小数; 160 表示 HX-G 特殊指令; +4096 表铣床循环指令不编坐标点时就不执行。

★、210, 启用图形显示区域 (8 表启用固定显示区域, +16 表按铣床模式显示图形)

P210 表示启用图形显示区域功能, 8 表示启用固定显示区域, +16 表示按铣床模式显示图形, P211-P216 表示 XYZ 区域设置。

★、211, 图形显示区域 X 轴负端

★、212, 图形显示区域 X 轴正端

★、213, 图形显示区域 Y 轴负端

★、214, 图形显示区域 Y 轴正端

★、215, 图形显示区域 Z 轴负端

★、216, 图形显示区域 Z 轴正端

★、230, 通过输入点选择执行程序 (+4+8+16+32+64+128 对应 X26-X31)

通过输入点选择执行程序, 如设置为+4+8=12, 则 X26 或者 X27 有效时选择程序 HIDEFILEX26 或 X26 或者 HIDEFILEX27 或 X27。

D2 位为 1 即 +4 表 X26 有效时执行 X26 程序或 HIDEFILEX26 程序;

D3 位为 1 即 +8 表 X27 有效时执行 X27 程序或 HIDEFILEX27 程序;

D4 位为 1 即 +16 表 X28 有效时执行 X28 程序或 HIDEFILEX28 程序;

D5 位为 1 即 +32 表 X29 有效时执行 X29 程序或 HIDEFILEX29 程序;

D6 位为 1 即 +64 表 X30 有效时执行 X30 程序或 HIDEFILEX30 程序;

D7 位为 1 即 +128 表 X31 有效时执行 X31 程序或 HIDEFILEX31 程序;

D8 位为 1 即 +256 表 X16 有效时执行 X16 程序或 HIDEFILEX16 程序;

D9 位为 1 即 +512 表 X17 有效时执行 X17 程序或 HIDEFILEX17 程序;

D10 位为 1 即+1024 表 X18 有效时执行 X18 程序或 HIDEFILEX18 程序;

D11 位为 1 即+2048 表 X19 有效时执行 X19 程序或 HIDEFILEX19 程序;

D12 位为 1 即+4096 表 X20 有效时执行 X20 程序或 HIDEFILEX20 程序;

D13 位为 1 即+8192 表 X21 有效时执行 X21 程序或 HIDEFILEX21 程序;

D14 位为 1 即+16384 表 X22 有效时执行 X22 程序或 HIDEFILEX22 程序;

D15 位为 1 即+32768 表 X23 有效时执行 X23 程序或 HIDEFILEX23 程序;

★、231, 删除键的操作方式[0 表向后删除, 1 表向前删除]

设置按“删除”键的方式, 0 表示向后删除, 1 表示向前删除。

★、232, 攻丝前是否检测主轴零位[18 表是, 0 表示否]

设置为 18 时表示攻丝前要检测主轴编码器零位开始攻丝, 设置为 0 时表示不检测编码器零位。

Check 34818, Table not strictly checked, 6326274 Table fast pre-reading processing)

This parameter is a special parameter. 2131970 indicates strict check of G41/G42 instructions, 34818 indicates not strict check of G41/G42 instructions, and 6326274 indicates fast pre-read of G41/G42 instructions.

★205, internal parameters (+8 indicates that thread insertion must wait for pulse completion; 32 indicates three decimal places; 96 indicates four decimal places; 160 indicates HX-G command; +4096 indicates that milling cycle will not execute if coordinate points are not programmed)

Internal parameters: +8 indicates that the thread insertion must wait until the pulse is completed before executing the next instruction; 32 indicates that the coordinate axis display uses three decimal digits; 96 indicates that the coordinate axis display uses four decimal digits; 160 indicates HX-G special instructions; +4096 means that the milling cycle instruction will not be executed if no coordinate points are programmed.

★210, enable the graphic display area (8 enables the fixed display area, +16 displays the graphics in milling mode)

P210 means to enable the function of graphic display area, 8 means to enable fixed display area, +16 means to display graphics in milling mode, P211-P216 means to set the XYZ area.

★, 211, the negative end of the X-axis in the graphical display area

★, 212, the positive end of the X-axis in the graphical display area

★, 213, the negative end of the Y-axis in the graphical display area

★, 214, the positive end of the Y-axis in the graphical display area

★, 215, graphics display area Z-axis negative end

★, 216, the graphic display area is on the positive end of the Z-axis

★230. Select the execution program by input points (+4, +8, +16, +32, +64, +128 corresponding to X26-X31)

Select the program by entering the input point. If it is set to +4+8=12, select the program HIDEFILEX26 or X26 or HIDEFILEX27 or X27 when X26 or X27 is valid.

If D2 is 1, that is, +4, X26 is valid and X26, program or HIDEFILEX26 program is executed;

If D3 is 1, that is, +8, and X27 is valid, execute X27, the program, or HIDEFILEX27 program;

If D4 is 1, that is, +16, X28 is valid and X28, program or HIDEFILEX28 program is executed;

If D5 is 1, that is, +32, X29 is valid and X29, program or HIDEFILEX29 program is executed;

When the D6 bit is 1 (i.e., +64), the X30 table is valid and the X30, program, or HIDEFILEX30 program is executed; when the D7 bit is 1 (i.e., +128), the X31 table is valid and the X31, program, or HIDEFILEX31 program is executed;

If the D8 bit is 1, that is, +256, execute X16, program or HIDEFILEX16 program when X16 is valid;

When the D9 bit is 1 (i.e., +512), the table X17 is valid and the X17 program or HIDEFILEX17 program is executed; when the D10 bit is 1 (i.e., +1024), the table X18 is valid and the X18 program or HIDEFILEX18 program is executed;

If D11 is 1, that is, +2048, execute X19 when X19 is valid, or the HIDEFILEX19 program;

If D12 is 1, that is, +4096, execute X20, program or HIDEFILEX20 program when X20 is valid;

When the D13 bit is 1 (i.e., +8192), the X21 table is valid and the X21 program or HIDEFILEX21 program is executed; when the D14 bit is 1 (i.e., +16384), the X22 table is valid and the X22 program or HIDEFILEX22 program is executed;

When the D15 bit is set to 1 (i.e., +32768), execute X23, program, or HIDEFILEX23 program when Table X23 is valid; ★. 231, delete key operation mode [0: backward deletion of Table, 1: forward deletion of Table]

Set the "Delete" key to 0 for backward deletion and 1 for forward deletion.

★232. Check whether the spindle zero position is detected before tapping [Table 18 is yes, Table 0 is no]

When set to 18, the spindle encoder zero position should be detected before tapping, and when set to 0, the encoder zero position should not be detected.

★、233, G06 圆弧示教功能[0 表关闭, 1 表开启]

设置为 1 时表示开启 G06 圆弧示教功能, 0 表示关闭此功能。

★、234, 程序运行回退功能[+8 表启动手轮反转回退]

设置为+8 时表示启动程序运行手轮回退功能, 正摇手轮时程序正向运行, 反摇手轮时程序回退运行。

★、235, 示教时自动生成注释[0 表是, 1 表否]

设置为 1 时表示教时自动生成指令注释功能, 0 表示关闭此功能。

★、236, 示教产生平面圆弧圆心类型[0 表示 G5/G6 IJK 格式, 1 表示 G2/G3 R 格式]

设置为 0 表示示教平面圆弧时采用 G5/G6 IJK 格式, 1 表示 G2/G3 R 格式。

★、237, 示教产生指令方式[0 表正常, 1 表示示教时每段直线加工行都产生 G01 代码]

设置为 1 时表示示教时每行产生 G01 代码。

★、238, 中间行示教时是否自动插入一行[0 表插入, 1 表修改]

设置为 0 时表示在中间行教时是插入一行, 1 表示修改此行。

★、239, 中间行示教时按 XYZ[0 表插入, 1 表修改]

设置为 0 时表示中间行示教按 XYZ 时是插入 XYZ 坐标字, 设置为 1 时表示是修改坐标字。

★、240, 示教时相同坐标记录方式[0 不记录, 1 记录]

设置示教时相同坐标时的记录方式, 设置为 0 时表示不记录坐标字, 设置为 1 时表示要记录坐标字。

★、307, M18xx/M28xx/WAT 等检测指令等待最长报警时间(ms) [≥10 有效]

例如: 此参数设置为 2000 时, 程序 WAT+X16, 检测 2 秒还没有检测到 X16 信号则报警。

★、400-402, 转换 DXF 文件要求[1 表排序, 4 表根据起刀点排序, 8 表不排序]

1) 当 P400 设为 1 时, 系统按照首尾相连的关系排序。

2) 当 P400 设为 4 时, 系统按照首尾相连的关系排序后根据前后两个端点与参数起刀点的距离来判断从哪一个端点下刀, 从距离近的端点下刀。

3) 当 P400 设为 8 时, 系统不会自动排序, 用 AutoCAD 绘图生成 DXF 文件时, 必须注意绘图的顺序, 因为 DXF 文件按绘图顺序保存数据, 所以生成的 G 代码加工顺序也是按 DXF 文件的数据保存顺序来执行的。

★、500, 自定义 G74 时对应的 ProgramG 编号[101-170 有效(101-150 非模态, 151-170 模态)]

设置自定义指令 G74 对应的宏程序编号, 如设置为 101 时表示 G74 指令执行宏程序 ProgramG101, 参数 P501-P504 功能类同。

★、501, 自定义 G81 时对应的 ProgramG 编号[101-170 有效(101-150 非模态, 151-170 模态)]

★、502, 自定义 G82 时对应的 ProgramG 编号[101-170 有效(101-150 非模态, 151-170 模态)]

★, 233, G06 arc teaching function [Table 0 is off, Table 1 is on]

Set to 1 to enable the G06 circular arc teaching function, and 0 to disable this function.

★, 234, program run back function [+8 Start handwheel reverse back]

When set to +8, the program starts to run the handwheel back function. When the handwheel is turned forward, the program runs forward; when the handwheel is turned backward, the program runs backward.

★, 235. Annotations are automatically generated during teaching [Table 0 is yes, Table 1 is no]

When set to 1, the instruction annotation function is automatically generated during teaching. When set to 0, this function is turned off.

★, 236, Teach generates a plane arc center type [0 indicates G5/G6 IJK format, 1 indicates G2/G3 R format]

Setting 0 indicates that the G5/G6 IJK format is used when teaching the plane arc, and setting 1 indicates that the G2/G3 R format is used.

★, 237, Teach generation instruction mode [0 indicates normal, 1 indicates G01 code is generated for each linear processing line during teaching]

When set to 1, the G01 code is generated per row during teaching.

★, 238, whether to automatically insert a line when teaching the middle line [0 for table insertion, 1 for table modification]

Setting it to 0 indicates that a line is inserted when teaching in the middle row, and setting it to 1 indicates that the line is modified.

★, 239, the middle line is taught by inserting according to XYZ[0 table, 1 table modification]

When set to 0, the middle line teaching is inserted according to XYZ when it is set to 1, and the coordinate word is inserted.

★, 240, the same coordinate recording mode is used during teaching [0 no record, 1 record]

Set the recording mode when the same coordinates are set for teaching. If 0 is set, the coordinate word is not recorded; if 1 is set, the coordinate word is recorded.

★, 307, M18xx/M28xx/WAT and other test instructions waiting longest alarm time (ms) [>=10 valid]

For example, if this parameter is set to 2000, the program WAT+X16 will detect no X16 signal within 2 seconds and alarm.

★400-402, Conversion of DXF files requires [Table 1 sorted, Table 4 sorted by starting point, Table 8 unsorted]

1) When P400 is set to 1, the system sorts in a head-to-tail relationship.

2) When P400 is set to 4, the system sorts the endpoints in a head-to-tail relationship and determines which endpoint to cut from based on the distance between the two endpoints and the parameter starting point, cutting from the nearest endpoint.

3) When P400 is set to 8, the system will not automatically sort. When drawing DXF files with AutoCAD, attention must be paid to the drawing order, because DXF files save data according to the drawing order, so the processing order of G code generated is also executed according to the data saving order of DXF files.

★When customizing G74, the corresponding ProgramG number [101-170 valid (101-150 non-modified, 151-170 modified)]

Set the macro program number corresponding to the G74 command. If set to 101, it indicates that the G74 command executes the macro program ProgramG101, and parameters P501-P504 have the same function.

★501, when customizing G81, the corresponding ProgramG number [101-170 valid (101-150 non-modal, 151-170 modal)]

★502, when customizing G82, the corresponding ProgramG number [101-170 valid (101-150 non-modal, 151-170 modal)]

★、503, 自定义 G83 时对应的 ProgramG 编号[101-170 有效(101-150 非模态, 151-170 模态)]

★、504, 自定义 G84 时对应的 ProgramG 编号[101-170 有效(101-150 非模态, 151-170 模态)]

★、551, 五轴数控机床结构型式(10--99)

[10 表 B 刀具-A 平台, 11 表 AC 刀具, 12 表 AC 平台, 13 表 A 刀具-C 平台, 14 表 A 刀具-B 平台; 15 表 BC 平台]

P551-P558 用于五轴数控机床 RTCP 功能参数设置, 详见相关指令及使用说明。

★、552, RTCP 补偿反向和坐标系重合 (反向 10/14:B+4, A+8; 11/12/13:A+4, C+8; 15:B+4, C+8; 坐标系重合+16: [12A 轴, 13C 轴, 14/15B 轴] 对应编程和工件坐标系重合; +32: 12/15C 轴对应编程和工件坐标系重合)

★、553, 五轴数控回转平台中心的机床坐标 1 (um)

(机床型式为 10/12:A 轴回转中心的 Y 坐标; 13:C 轴回转中心的 X 坐标; 14:B 轴回转中心的 X 坐标; 15:B 轴回转中心的 X 坐标)

★、554, 五轴数控回转平台中心的机床坐标 2 (um)

(机床型式为 10/12:A 轴回转中心的 Z 坐标; 13:C 轴回转中心的 Z 坐标; 14:B 轴回转中心的 Z 坐标; 15B 轴回转中心的 Z 坐标)

★、555, 测量五轴数控回转平台中心时的刀长 (um)

(机床型式为 10: 测量 A 轴回转中心刀长; 12/15: 测量五轴数控回转平台中心刀补值 H1; 13: 测量 C 轴回转中心刀长; 14: 测量 B 轴回转中心刀长)

★、556, AC 轴或 BC 轴平台回转型五轴机床的 C 轴回转中心 X 或 Y 轴机床坐标 (um) 机床型式为 12 或 15)

★、557, AC 轴刀具回转型的 AC 轴心线在 Y 方向的偏移量 (um) (机床型式为 11)

★、558, AC 轴刀具回转型的 C 轴与主轴在 X 方向的偏移量 (um) (机床型式为 11)

★、560, RTCP 超速时是否报警[1: 表示不报警自动降低运行速度; 0: 表示报警]

★、600-0, M7 指令最长等待时间[>10] (ms)

M7 指令最长等待时间, 如设置为 100 时表示等待时间为 1000ms。

★、601, X 轴 M7 (>10: 跟随误差小于该值; -6: 反馈脉冲不再变化; -8: 系统脉冲发完)

设置为-6 时表示 X 轴反馈脉冲不再变化时表示定位完成; 设置为-8 时表示 X 轴脉冲发完时定位完成; 设置为其它>10 值时表示跟随误差小于设置时定位完成。参数 P601-P608 表示 X/Y/Z/A/B/C/Xs/Ys 轴, 功能类同。

★、700, G31 没有检测到信号时是否报警[0: 表禁止进给, 1: 表报警, +1024: 表不报警]

设置 G31 没有检测到信号时报警方式, 0 表示禁止进给, 1 表示报警, +1024 时表示不报警。

★、701, 进给轴运动时每 2ms 执行一次 M880-M979[20000--20049 对应

★503, when customizing G83, the corresponding ProgramG number [101-170 valid (101-150 non-modal, 151-170 modal)]

★504, ProgramG number corresponding to custom G84 [101-170 valid (101-150 non-modified, 151-170 modified)]

★551, five-axis CNC machine tool structure type (10-99)

[Table 10: Table B Tool-A Platform, Table 11: Table AC Tool, Table 12: Table AC Platform, Table 13: Table A Tool-C Platform, Table 14: Table A Tool-B Platform; Table 15: Table BC Platform]

P551-P558 is used for RTCP function parameter setting of five-axis CNC machine tools. See related instructions and instructions for details.

★ 552. RTCP compensation for reverse and coordinate system alignment (Reverse 10/14: B+4, A+8; 11/12/13: A+4, C+8; 15: B+4, C+8; Coordinate system alignment +16: [12A-axis, 13C-axis, 14/15B-axis] corresponds to programming and workpiece coordinate system alignment; +32:12/15C-axis corresponds to programming and workpiece coordinate system alignment)

★, 553, machine tool coordinate 1 (um) at the center of the five-axis CNC rotary table

(Machine tool type: 10/12: Y-coordinate of the A-axis rotary center; 13: X-coordinate of the C-axis rotary center; 14: X-coordinate of the B-axis rotary center; 15: X-coordinate of the B-axis rotary center)

★, 554, machine tool coordinates 2 (um) at the center of the five-axis CNC rotary table

(Machine tool type: 10/12: Z-coordinate of A-axis rotary center; 13: Z-coordinate of C-axis rotary center; 14: Z-coordinate of B-axis rotary center; 15B: Z-coordinate of B-axis rotary center)

★555, measuring the tool length (um) when the center of the five-axis CNC rotary platform is measured

(Machine tool type: 10: measure the A-axis rotary center tool length; 12/15: measure the five-axis CNC rotary platform center tool compensation value H1; 13: measure the C-axis rotary center tool length; 14: measure the B-axis rotary center tool length)

★556, AC or BC axis platform rotary five-axis machine tool C-axis rotary center X or Y axis machine tool coordinates (um) Machine tool type 12 or 15)

★557, Y-directional offset of AC spindle centerline in return-to-tool mode (um) (machine tool type 11)

★ ★ 558, the X-axis offset between the AC tool return-type C-axis and spindle (inum) (machine type: 11)★ 560, whether to trigger an alarm when RTCP exceeds speed limit [1: indicates no alarm and automatic speed reduction; 0: triggers alarm]★ 600-0, maximum M7 command waiting time (>10) (ms)

M7 is the maximum waiting time. If set to 100, the waiting time is 1000ms.

★601, X-axis M7 (>10: the following error is less than this value; -6: the feedback pulse no longer changes; -8: the system pulse has been sent out)

When set to -6, the positioning is completed when the X-axis feedback pulse stops changing. When set to -8, positioning is completed upon completion of X-axis pulse transmission. For other values >10, positioning is achieved when the tracking error falls below the preset threshold. Parameters P601-P608 apply to the X/Y/Z/A/B/C/Xs/Ys axes, sharing similar functionality.

★, 700, G31 whether to alarm when no signal is detected [0: table prohibits feed, 1: table alarms, +1024: table does not alarm]

Set the alarm mode when no signal is detected in G31.0 means to prohibit feed, 1 means to alarm, and +1024 means not to alarm.

★, 701, the feed axis moves and performs M880-M979 every 2ms [20000-20049 corresponding

ProgramUser0--49]

表示坐标轴运动时每 2ms 执行一次宏程序，如设置为 20000 表示运行坐标轴时每 2ms 执行一次 ProgramUser0。

3. 7. 2 速度参数

★、1, X 轴 G00 速度 (mm/min)

自动运行时 X 轴的快移(G00 指令调用的速度)速度。最大值30000。单位mm/min, 参数 P1-P5 功能类同，分别是 X/Y/Z/A/B/C/Xs/Ys 轴的 G00 速度。

注：该值的大小与机床配置有关，设置不当可能造成设备故障或事故。

★、6, 手动最大进给速度 (mm/min)

设置手动状态下进给轴的最高运行速度限制。单位：mm/min。

参数 P6-1/P6-2/P6-3/P6-4/P6-5 分别是 X/Y/Z/A/B/C/Xs/Ys 的手动最大速度。

注意：该值设置建议为 G00 速度的 50%，以确保手动运行安全。

★、7, 自动最大进给速度 (mm/min)

自动运行的最高速度。最大值：30000。单位：mm/min。

为确保各轴能够运行的最大速度保证效率，可将此参数值高于 G00 速度，以确保在联动（限制的是合成速度）时各轴达到最大速度。

★、8, G01/G02/G03 缺省速度 (mm/min)

当程序中第一个插补指令（G01/G02/G03）没有给定速度时，在自动运行时该指令所调用的速度。最大值：5000。单位：mm/min。

★、9, 空运行速度 (mm/min)

程序空运行所执行的速度。最大值：30000。单位：mm/min。

★、10, 进给轴手动速度 (mm/min)

手动状态下进给轴速度设定值。设置范围：<手动最大进给速度。单位：mm/min。

注：手动状态按 F 键设置后自动刷新此参数值。

★、11, 主轴手动转速 (rpm)

手动状态下主轴速度设定值。单位：rpm。

注：手动状态按 S 键设置后自动刷新此参数值。

★、12, 进给轴运行起跳速度 (mm/min)

★、13, 进给轴速度最大突变量 (mm/min)

★、14, 限制 G1G2G3 单轴速度[1 表是, 0 表否]

用于设置是否限制插补时各轴的最高运行速度。当该参数设置为 1 表示限制，即插补时各轴最高运行速度将受参数限制，0 则反之。P15-P19 设置 X/Y/Z/A/B 轴的限制 G1G2G3 最高速度 (mm/min)。

★、20, X 轴加减速常数

X 轴升降速时间常数，值越大升降速越快。数值范围：1-99999。P20-P24 设置 X/Y/Z/A/B/C/Xs/Ys 轴的加减速常数。

ProgramUser0--49]

Indicates that the macro program is executed every 2ms when the coordinate axis moves. If set to 20000, it means that ProgramUser0 is executed every 2ms when the coordinate axis runs.

3. 7. 2 Parameter of Velocity

★, 1, X-axis G00 speed (mm/min)

The speed of the X-axis fast move (the speed called by the G00 command) during automatic operation. Maximum value: 30,000. Unit: mm/min. Parameters P1-P5 have similar functions, which are the G00 speeds of the X/Y/Z/A/B/C/Xs/Ys axes respectively.

Note: The value is related to the configuration of the machine tool, and improper setting may cause equipment failure or accident.

★, 6, maximum manual feed speed (mm/min)

Set the maximum feed speed limit in manual mode. Unit: mm/min.

Parameters P6-1/P6-2/P6-3/P6-4/P6-5 are the manual maximum speeds of X/Y/Z/A/B/C/Xs/Ys respectively.

Note: This value is recommended to be set at 50% of G00 speed to ensure safe manual operation.

★, 7, Automatic maximum feed speed (mm/min)

Maximum speed of automatic operation. Maximum value: 30,000. Unit: mm/min.

To ensure maximum speed of each axis to ensure efficiency, this parameter value can be higher than G00 speed to ensure that each axis reaches maximum speed when linkage (limited by synthesis speed) is achieved.

★, 8, G01/G02/G03 default speed (mm/min)

When the first interpolation instruction (G01/G02/G03) in the program does not give a speed, the speed called by this instruction is the maximum value during automatic operation. Maximum: 5000. Unit: mm/min.

★, 9, empty running speed (mm/min)

Maximum speed at which the program runs empty. Maximum: 30,000. Unit: mm/min.

★, 10, feed axis manual speed (mm/min)

Set the feed axis speed in manual mode. Setting range: <maximum feed speed in manual mode. Unit: mm/min.

Note: The manual status is automatically refreshed after setting this parameter value by F key.

★, 11, spindle manual speed (rpm)

Manual spindle speed setting. Unit: rpm.

Note: The manual status is automatically refreshed after being set to S key.

★, 12, feed axis running jump speed (mm/min)

★13. Maximum abrupt variation of feed axis speed (mm/min)

★14. Limit the G1G2G3 single-axis speed [1 indicates yes, 0 indicates no]

This parameter is used to set whether to limit the maximum operating speed of each axis during interpolation. When this parameter is set to 1, it indicates a restriction, meaning the maximum operating speed of each axis during interpolation will be limited by the parameter. If set to 0, the opposite applies. Parameters P15-P19 set the maximum speeds (mm/min) for G1, G2, G3 axes in X/Y/Z/A/B directions.

★, 20, X-axis acceleration and deceleration constant

X-axis acceleration and deceleration time constant. The larger the value, the faster the acceleration and deceleration. Value range: 1 -99999. P20-P24 Set the acceleration and deceleration constants of X/Y/Z/A/B/C/Xs/Ys axes.

注：该值的大小与机床配置有关，设置不当可能造成设备故障或事故。通常情况下负载越重参数值越小，配置步进时该值不能大于 15000。

★、25，自动运行加减速常数

用于设置自动运行升降速度常数，设置范围 1-500。当该项参数值设置在 1-500 范围内时自动运行升降速处理速度以本参数为准，否则按用户参数中各轴的升降加速度参数值为准。

本参数的设置主要为了区分手动和自动的升降速度，只有两种状态要求的升降速度差异较大时才设置此参数，通常情况下设为无效。

★、26，手轮加减速常数

用于设置使用手轮时升降速时间常数，设置范围 500-32000，值越大升降速越快。

★、27，手轮试机加减速常数

用于设置手轮触发自动运行时的升降速时间常数，设置范围 500-32000，值越大升降速越快。

注：该参数值低于 500 表示无效，各轴升降速按常规执行。

★、28，手轮试机 G00 速度 (mm/min)

用于设置手轮触发自动运行时的空运行速度，单位：mm/min。

注：该参数值低于 10 表示无效，按各轴 G00 执行。

★、29，手轮 X 轴限制速度 (mm/min)

用于设置手动状态X轴使用手脉时的最高速度，单位 mm/min。设置范围100～手动最大速度。P29-P33设置Y/Z/A/B/C/Xs/Ys轴的手轮限制速度 (mm/min)。

本参数设置的大小与机床配置有关系，建议此值不超过4000，以确保运行安全。

注意：当设定值大于100时方可生效，否则，不对最高速度进行限制。

★、33-1，螺纹加工 Z 轴加减速常数

用于设置螺纹加工时 Z 轴升降速时间常数，设置范围 300-90000，值越大升降速越快。

注：该参数设置与机床配置有关系，步进时可将该设置小一些，以确保运行安全。

该参数设置小于 300 表示无升降速，通常伺服为保证效率可设置为不需要升降速，即设置为 0。

★、33-2，螺纹加工 X 轴加减速常数[伺服驱动器设为 0]

用于设置螺纹加工时 X 轴升降速时间常数，设置范围 300-90000，值越大升降速越快。

注：该参数设置与机床配置有关系，步进时可将该设置小一些，以确保运行安全。

该参数设置小于 300 表示无升降速，通常伺服为保证效率可设置为不需要升降速，即设置为 0。

★、33-3，伺服电机螺纹循环 X 轴退尾速度倍率

用于设置伺服配置情况下螺纹加工时 X 退尾速度的快慢。单位：0.1 倍。

Note: The value is related to the configuration of the machine tool, and improper setting may cause equipment failure or accident. Under normal circumstances, the heavier the load, the smaller the parameter value. When configuring the step, this value should not be greater than 15000.

★, 25, automatically run the acceleration and deceleration constant

Used to set the automatic operation lifting speed constant, the setting range is 1-500. When this parameter value is set in the range of 1-500, the automatic operation lifting speed processing speed is based on this parameter, otherwise it is based on the user's parameters of each axis lifting acceleration parameters.

This parameter is mainly set to distinguish the lifting speed between manual and automatic. This parameter is only set when there is a large difference in the lifting speed required by the two states, and it is usually set as invalid.

★, 26, handwheel acceleration and deceleration constant

Used to set the lifting speed time constant when using the handwheel. The setting range is 500-32000. The larger the value, the faster the lifting speed.

★27. Test the acceleration and deceleration constant of the handwheel

Used to set the speed time constant when the handwheel is triggered to run automatically. The setting range is 500-32000, and the larger the value, the faster the speed of lifting and lowering.

Note: If the value of this parameter is lower than 500, it is invalid. The lifting speed of each axis is executed according to the conventional method.

★, 28, handwheel test machine G00 speed (mm/min)

Used to set the empty running speed when the handwheel is triggered to run automatically. Unit: mm/min.

Note: If the value of this parameter is less than 10, it is invalid and G 00 is executed for each axis.

★, 29, handwheel X-axis speed limit (mm/min)

Set the maximum speed of manual X-axis operation using palm print, unit: mm/min. The setting range is 100 to maximum manual speed. Set the handwheel limit speed (mm/min) for Y/Z/A/B/C/Xs/Ys axes on pages P29-P33.

The size of this parameter setting is related to the configuration of the machine tool. It is recommended that this value does not exceed 4000 to ensure safe operation.

Note: The maximum speed is not limited if the set value is greater than 100.

★33-1, Z-axis acceleration constant for thread machining

Used to set the time constant of Z-axis lifting speed during thread processing. The setting range is 300-90000. The larger the value, the faster the speed of descent.

Note: This parameter setting is related to the machine tool configuration. When stepping, you can set it smaller to ensure safe operation.

If this parameter is set to less than 300, it means that there is no speed rise or fall. Usually, the servo can be set to 0 to ensure efficiency.

★33-2, Thread processing X-axis acceleration and deceleration constant [servo driver set to 0]

Used to set the time constant of X-axis lifting speed during thread processing. The setting range is 300-90000, and the larger the value, the faster the speed of descent.

Note: This parameter setting is related to the machine tool configuration. When stepping, you can set it smaller to ensure safe operation.

If this parameter is set to less than 300, it means that there is no speed rise or fall. Usually, the servo can be set to 0 to ensure efficiency.

★33-3, servo motor thread cycle X-axis tail speed multiplier

Used to set the fastness of X tail-back speed during thread machining in servo configuration. Unit: 0.1 times.

★、34, 加减速型式

该参数用于设置各轴升降速的类型, 设置为 8 表示进给轴升降速为曲线型 (S 形), 0 表示直线型。

注: 该参数设置与驱动配置有关, 通常情况步进设置为曲线型, 伺服设置为直线型。

★、35, 曲线型的初始加减速常数

该参数用于设置曲线型加速度的初始加减速常数, 该值越大加速越快。

注: 该参数值大于 10 有效。

★、36, 曲线型的加减速二阶常数 [≥ 10]

该参数用于设置曲线型加速度的二阶加减速常数, 该值越大加速越快。

注: 该参数值大于 10 有效。

★、37, 曲线型的最大加减速常数 [≥ 500]

该参数用于设置曲线型加速度的最大加减速常数, 该值越大加速越快。

注: 该参数值大于 500 有效, 否则采用直线型的各轴加减速常数。

★、38, X 轴回零正向速度 (mm/min)

X 轴回参考点时正向碰参考点开关的运行速度。单位: mm/min。数值范围: 小于 X 轴 G00 速度。P38/P40/P42/P44/P46 分别设置 X/Y/Z/A/B/C/Xs/Ys 轴的回零正向速度 (mm/min)。

★、39, X 轴回零反向速度 (mm/min)

X 轴回参考点时, 脱开参考点开关后检测电机 Z 脉冲信号的速度。数值范围: 20-500。P39/P41/P43/P45/P47 分别设置 X/Y/Z/A/B/C/Xs/Ys 轴的回零反向速度 (mm/min)。

注意: 该参数值会影响回参考点精度, 值越小时精度会更高。此值一旦设定好了就不要再改动了, 否则也会影响到参考点位置。

★、41-0, C 轴回编码器零参考点前运行的距离 (1/1000 度)

系统主轴设置为 C 轴时, M800 指令 C 轴回编码器零位前的运行距离 (1/1000 度), P41 为 M800 指令 C 轴回编码器零位的速度 (mm/min)。

★、48-1, 恒线速控制下 G96 的主轴速度下限值 (rpm)

该参数用于设置执行 G96 恒线速控制时, 主轴速度下限值。

★、48, 主轴高档最高转速 (rpm)

用于设置主轴最高转速, 也就是对应 10V 指令电压时的转速。单位: r/min

注: 当有高低档、多档功能时此参数为主轴高档、第一档最高转速。

★、49, 主轴低档 (第二档) 最高转速 (rpm)

用于设置主轴低档 (第二档) 最高转速, 也就是对应 10V 指令电压时的转速。单位: r/min。

★、50, 主轴第三档最高转速 (rpm)

用于设置主轴第三档最高转速, 也就是对应 10V 指令电压时的转速。单位: r/min

★、51, 主轴第四档最高转速 (rpm)

用于设置主轴第四档最高转速, 也就是对应 10V 指令电压时的转速。单位:

★34. Acceleration and deceleration type

This parameter is used to set the type of the lifting speed of each axis. If set to 8, it indicates that the lifting speed of the feed axis is S-shaped, and 0 indicates that it is straight.

Note: This parameter setting is related to the drive configuration. In general, the step setting is curve type and the servo setting is straight line type.

★35, the initial acceleration and deceleration constant of the curve type

This parameter is used to set the initial acceleration and deceleration constant of the curved acceleration. The larger this value is, the faster the acceleration is.

Note: The parameter value is valid only if it is greater than 10.

★36, The second-order constant of the curve type acceleration and deceleration [≥ 10]

This parameter is used to set the second-order acceleration and deceleration constant of the curved acceleration. The larger this value is, the faster the acceleration is.

Note: The parameter value is valid only if it is greater than 10.

★, 37, maximum acceleration and deceleration constant of the curve type [≥ 500]

This parameter is used to set the maximum acceleration and deceleration constant of the curved acceleration. The larger this value is, the faster the acceleration is.

Note: The value of this parameter is valid only when it is greater than 500, otherwise the linear acceleration and deceleration constants of each axis are used.

★38, X-axis zero return forward speed (mm/min)

The forward speed of the reference point switch when the X-axis returns to the reference point. Unit: mm/min. Value range: Less than the G00 speed of the X-axis. P38/P40/P42/P44/P46 respectively set the forward speed of the X/Y/Z/A/B/C/Xs/Ys axis return to zero (mm/min).

★, 39, X-axis return zero reverse speed (mm/min)

When the X-axis returns to the reference point, the speed of the motor Z pulse signal is detected after disconnecting the reference point switch. Value range: 20-500. P39/P41/P43/P45/P47 respectively set the return zero reverse speed (mm/min) for the X/Y/Z/A/B/C/Xs/Ys axes.

Note: The value of this parameter affects the accuracy of the reference point. The smaller the value, the higher the accuracy. Once this value is set, do not change it, otherwise the reference point position will also be affected.

★, 41-0, Distance (1/1000 degree) traveled before the zero reference point of the C-axis return encoder

When the system spindle is set to the C-axis, M800 indicates the running distance (1/1000 degree) before the C-axis return encoder zero position, and P41 is the speed (mm/min) of the M800 command for the C-axis return encoder zero position.

★48-1, lower limit of spindle speed (rpm) for G96 under constant linear speed control

This parameter is used to set the lower limit of spindle speed when G96 constant line speed control is performed.

★48, maximum spindle speed (rpm)

Used to set the maximum spindle speed, that is, the speed corresponding to the 10V instruction voltage. Unit: r/min

Note: When there are high and low speed, multi-speed functions, this parameter is the spindle high speed, the first speed is the highest.

★49, spindle low gear (second gear) maximum speed (rpm)

Used to set the maximum speed of the spindle at low speed (second speed), that is, the speed corresponding to the 10V instruction voltage. Unit: r/min.

★50, maximum spindle speed (rpm) at third gear

Used to set the maximum speed of the spindle in the third gear, that is, the speed corresponding to the 10V instruction voltage. Unit: r/min

★51. Maximum spindle speed (rpm) at fourth gear

Used to set the maximum speed of the spindle in the fourth gear, that is, the speed corresponding to the 10V instruction voltage. Unit:

r/min

★、52, 第二主轴最高转速(rpm)

用于设置第二主轴最高转速, 也就是对应 10V 指令电压时的转速。单位: r/min。P52-1/P52-2 设置第三/四主轴最高转速(SC/SD 指令对应的主轴转速), P52-3/P52-4 设置第三/四主轴手动转速。

★、52-0, 主轴点动时固定转速(0.1rpm) [>2]

设置主轴点动时的固定转速, 设置值大于 2 时有效, 否则用手动时的转速。

★、52-5, 第一主轴最低转速(rpm)或最低电压(mV, 前面加-负号表电压) [S0 测试值]

方法 1: 第一主轴最低转速, 例如: 设置为 100 时表示 S0 时的转速为 100 rpm;

方法 2: 第一主轴最低电压, 例如: 设置为-100 时表示 S0 时输出 100mV。

★、52-6, 第一主轴最高转速(rpm)或最高电压(mV, 前面加-负号表电压) [S 最高转速测试值]

方法 1: 第一主轴最高转速, 例如: 设置为 3000 时表示输出 10V 时的转速为 3000 rpm;

方法 2: 第一主轴最高电压, 例如: 设置为-10000 时表示最高转速时输出 10V。

★、52-7, 第二主轴最低转速(rpm)或最低电压(mV, 前面加-负号表电压) [SS0 测试值]

方法 1: 第二主轴最低转速, 例如: 设置为 100 时表示 SS0 时的转速为 100 rpm;

方法 2: 第二主轴最低电压, 例如: 设置为-100 时表示 SS0 时输出 100mV。

★、52-8, 第二主轴最高转速(rpm)或最高电压(mV, 前面加-负号表电压) [SS 最高转速测试值]

方法 1: 第二主轴最高转速, 例如: 设置为 3000 时表示输出 10V 时的转速为 3000 rpm;

方法 2: 第二主轴最高电压, 例如: 设置为-10000 时表示最高转速时输出 10V。

★、53, 圆弧间隙补偿方式(0 表 A: 间隙值越大补偿速度越快, 8 表 B: 补偿速度由参数决定, +4 表: 圆弧编程 IJK 是终点到圆心的坐标, +256 螺旋线插补速度不超过 F)

设为 0 时, A 方式: 间隙值一次性补偿完, 执行圆弧反向间隙补偿时, 补偿轴的补偿速度受反向间隙值的影响, 反向间隙值越大补偿速度越快, 以保证刀具不发生暂停现象, 但补偿轴的补偿速度不会超过 10000mm/min;

设为+8 时, B 方式: 间隙值分段补偿, 执行圆弧反向间隙补偿时, 补偿轴的补偿速度由参数值决定;

+4 表时: 圆弧编程 IJK 是终点到圆心的坐标, 在原参数值上加上 4 表示: 圆弧指令 G02G03 的 IJK 编程相对于终点, 否则为相对于起点。

+256 时: 表示螺旋线加工时不超过设定 F 值。

★、54, 圆弧进行 B 型间隙补偿的速度(mm/min)

方式 B 间隙补偿速度。单位: r/min。P51-1 设置 B 间隙补偿的起跳速度

r/min

★52, maximum speed of the second spindle (rpm)

Set the maximum rotational speed of the second spindle, which corresponds to the 10V command voltage. Unit: r/min. P52-1/P52-2 Set the maximum rotational speed of the third/fourth spindles (spindle speeds corresponding to SC/SD commands), and P52-3/P52-4 Set the manual rotational speed of the third/fourth spindles.

★, 52-0, fixed speed (0.1 rpm) when spindle is moved [2]

Set the fixed speed of the spindle during point motion. The value is valid when it is greater than 2, otherwise use the speed of manual operation.

★52-5, Minimum spindle speed (rpm) or minimum voltage (mV, preceded by a minus sign for test voltage) [S0 test value]

Method 1: The lowest rotation speed of the first spindle, for example, is set to 100, indicating that the rotation speed of S0 is 100 rpm;

Method 2: The minimum voltage of the first spindle, for example, is set to -100 when output 100mV is S0.

★52-6, maximum spindle speed (rpm) or maximum voltage (mV, preceded by a minus sign for table voltage) [S maximum speed test value]

Method 1: The maximum speed of the first spindle, for example, if set to 3000, the speed at which output 10V is 3000 rpm;

Method 2: The maximum voltage of the first spindle, for example, is set to -10000 when the output is 10V at the highest speed.

★ 52-7, Minimum speed of the second spindle (rpm) or minimum voltage (mV, with a minus sign in front of the voltage) [SS0 test value]

Method 1: The lowest rotation speed of the second spindle, for example, is set to 100, indicating that the rotation speed of SS0 is 100 rpm;

Method 2: The minimum voltage of the second spindle, for example, is set to -100 when SS0 and output 100mV.

★52-8, maximum speed of the second spindle (rpm) or maximum voltage (mV, with a minus sign in front to indicate the voltage) [SS maximum speed test value]

Method 1: The maximum speed of the second spindle, for example, if set to 3000, the speed at which output 10V is 3000 rpm;

Method 2: The maximum voltage of the second spindle, for example, is set to -10000 when the output is 10V at the highest speed.

★ 53, arc gap compensation method (0 Table A: The larger the gap value, the faster the compensation speed; 8 Table B: The compensation speed is determined by parameters; +4 Table: Arc programming IJK is the coordinate from the end point to the center; +256 Spiral interpolation speed does not exceed F)

When set to 0, mode A: The gap value is compensated once. When performing arc reverse gap compensation, the compensation speed of the compensation shaft is affected by the reverse gap value. The larger the reverse gap value, the faster the compensation speed will be, so as to ensure that the tool does not pause, but the compensation speed of the compensation shaft will not exceed 10,000mm/min;

When set to +8, mode B: gap value is segmented compensation. When performing reverse arc gap compensation, the compensation speed of the compensation axis is determined by the parameter value;

+4 Time: Arc programming. IJK is the coordinate from the end point to the center of the circle. Adding 4 to the original parameter value indicates that the IJK programming of arc instruction G02G03 is relative to the end point, otherwise it is relative to the starting point.

+256: indicates that the spiral line processing does not exceed the set F value.

★54. Speed of arc B-type clearance compensation (mm/min)

Mode B Gap compensation speed. Unit: r/min. P51-1 Set the jump speed of B gap compensation

(mm/min) [>10]。P51-2 设置 B 间隙补偿的加减速常数 ((mm/min)/s) [>10]

★、55, 激活速度处理功能[1 表是, 0 表否]

用于设置系统对短直线是否需要预先进行速度处理, 如果设置为“1”, 则要进行速度处理, 设置为“0”则不处理。

★、55-1, 程序运行中反向降速时的终止速度(mm/min)

用于程序运行过程中要求反向前降速的终止速度, 单位: mm/min。值越大反向越快。

★、56, 硬限位速停临界速度(mm/min)

用于设置硬限位时电机开始减速的速度。伺服不需要减速, 一般设置为 1。

★、57, 手轮停止速度(mm/min) [>18]

用于设置手轮停止时的速度, 单位: mm/min。值越大停止越快。

★、58, 跟随攻丝提刀时主轴转速(rpm) [>1]

编码器跟随攻丝提刀时的主轴转速, 如: 设置为 100 时表示主轴转速降速到 100r/min 时, Z 轴开始反向提刀。

★、58-1, 攻丝时主轴反向间隙补偿(脉冲数)

跟随和刚性攻丝提刀前的主轴反向间隙补偿脉冲数, 如设置为 200 时表示 Z 轴提刀前主轴反转 200 个脉冲后 Z 轴开始反向提刀。

★、58-2, 攻丝主轴反转前 Z 轴提前提刀量(um) [10-5000]

跟随和刚性攻丝主轴反转前 Z 轴反向间隙补偿脉冲数, 如: 设置为 500 时表示 Z 轴提刀补偿 500um 主轴开始反转, 类似于 Z 轴反向间隙补偿。

★、58-3, 攻丝退刀时进给速度(mm/min) [≥ 60 有效]

刚性攻丝 Z 轴退刀速度, 要求比较高的时候使用, 设置 Z 轴退刀速度为参数固定速度。

★、60, 启动平滑处理功能(+4 手动;+8 手轮;+16 程序;P61-P67 值越大越平稳但精度越差)

<1>P60=28 表示手动/手轮/程序加工均开启平滑处理功能;

<2>平滑处理功能——通过 Savitzky-Golay 滤波器实现曲线平滑插值法对折线进行平滑曲线处理基于 Numpy.convolve 实现滑动平均滤波数据平滑处理, 再通过 log() 和 exp() 函数算法实现在曲线的波峰波谷时对数据帧数多的平滑滤波, 主要目的是为了解决直线与直线之间或者直线与圆弧之间产生的曲线噪声;

<3>由参数 P61-P68 设置平滑处理时间常数, 平滑处理时间常数值越大越平稳, 但直线与直线或者直线与圆弧拐角精度越差;

<4>增强平滑处理是将平滑处理出来的脉冲再进行平滑处理, 设置级数就是再重复平滑处理次数, 主要是解决一级平滑处理系数太大可能导致轮廓精度差问题

★、P61, 手动平滑处理时间常数[2-500]; 建议设为: 4

★、P62, 手轮平滑处理时间常数[2-500]; 建议设为: 4

★、P63, 程序平滑处理时间常数[2-500]; 建议设为: 6-12

★、P65, 手动增强平滑处理时间常数[2-500] 建议设为: 4

★、P66, 手轮增强平滑处理时间常数[2-500] 建议设为: 4

(mm/min) [$\geq$]10. P51-2 Set the acceleration and deceleration constants for B gap compensation ((mm/min)/s) [$\geq$]10★,55, activate speed processing function [1 table is, 0 table is]

Used to set whether the system needs to preprocess the speed of short lines. If set to "1", processing is required; if set to "0", no processing is required.

★55-1, termination speed (mm/min) during reverse deceleration of the program

The termination speed required for reverse deceleration during the operation of the program, unit: mm/min. The larger the value, the faster the reverse speed.

★56. Hard limit stop critical speed (mm/min)

Used to set the speed at which the motor starts to decelerate when a hard limit is set. The servo does not need to decelerate and is usually set to 1.

★, 57, handwheel stop speed (mm/min) [$\geq$]18

Used to set the speed at which the handwheel stops. Unit: mm/min. The larger the value, the faster the stop.

★58, spindle speed (rpm) when following the thread to lift the blade [$\geq$]

The spindle speed of the encoder is followed by the spindle speed of the screwdriver, such as: when set to 100, the spindle speed decreases to 100r/min, and the Z-axis starts to lift the tool in reverse.

★58-1, spindle backlash compensation (pulse number) during threading

Follow the number of backlash compensation pulses for the front spindle during rigid thread lifting and tool removal. For example, if it is set to 200, it indicates that the front spindle of the Z-axis lifts the tool in reverse after 200 pulses.

★58-2, advance the Z-axis tool feed before spindle reversal (um) [10-5000]

Follow the number of Z-axis backlash compensation pulses before the spindle reversal of the rigid thread, such as: set it to 500, indicating that the Z-axis tool lift compensation is 500um and the spindle starts to reverse, similar to the Z-axis backlash compensation.

★58-3, feed speed (mm/min) when tapping back [$\geq$ 60 valid]

Rigid thread Z axis retraction speed is used when the requirements are relatively high, and the Z axis retraction speed is set as a fixed parameter.

★60, start the smoothing processing function (+4 manual; +8 handwheel; +16 program; P61-P67 The larger the value, the more stable but the less accurate)

<1>P60=28 means that the smoothing processing function is enabled for manual/handwheel/program processing;

<2>Smoothing Function — Utilizes Savitzky-Golay filter for curve smoothing interpolation. The algorithm employs Numpy convolution to implement sliding average filtering, followed by log() and exp() functions to apply smoothing filter on data frames with higher frame counts during peak and trough detection. This process primarily addresses noise generated between straight lines or between straight lines and arcs.

<3>The smoothing processing time constant is set by parameters P61-P68. The larger the value of the smoothing processing time constant, the more stable it is, but the corner accuracy between a straight line and a straight line or between a straight line and an arc is worse;

<4>The enhanced smoothing process is to smooth the pulse after smoothing. The setting of the number of levels is the number of times of repeated smoothing, which is mainly used to solve the problem that the contour accuracy may be poor due to the large coefficient of the first smoothing process

★P61, manual smoothing time constant [2-500]; it is recommended to set it as: 4

★P62, smooth processing time constant of the handwheel [2-500]; it is recommended to set it as: 4

★, P63, program smoothing processing time constant [2-500]; it is recommended to set it as: 6-12

★, P65, manual enhancement smoothing processing time constant [2-500] It is recommended to set it as:

4

★P66, the time constant for handwheel smoothing enhancement [2-500] is recommended to be set as: 4

★、P67, 自动增强平滑处理时间常数[2-500] 建议设为: 12

★、P68, 增强平滑处理等级[1-9] (越大越平稳) 建议设为: 2-8

★、101-108, X 起跳速度(mm/min)

用于设定进给轴运行时加速的初始速度, 同理也是减速时的终止速度。也就是进给轴的运行速度低于此值时无升降速过程直接到达, 当运行速度高于此值时, 加速时从此值开始。P101-P108 表示 X/Y/Z/A/B/C/Xs/Ys 轴起跳速度(mm/min)[>1]。

注: 该值的大小与机床配置有关, 设置不当可能造成设备故障或事故。

通常情况下配步进驱动不大于 100, 伺服不大于 500。

★、111-118, X 轴速度最大突变量(mm/min)

用于设定多轴连续轨迹插补时速度发生变化的情况下, 为保证插补合成速度时各轴速度的突变最大增量, 也就是变化速度增量值大于此值时会有升(降)速过程, 若小于此值则直接到达。主要用于增加多轴连续轨迹插补的连贯性。P111-P118 表示 X/Y/Z/A/B/C/Xs/Ys 速度最大突变量(mm/min)[>1]。

例: 当此值为 300 时, 多轴连续轨迹插补中 X 轴速度从 F800 变为 F1600 时, 其变化过程应为 X 轴速度先从 F800 变为 1100, 再加速到 F1600。

★、200, 连贯运动是否对 G00 起作用[16 表是, 1 表否]

设置 G00 是否做预读处理功能, 设置为 16 时表示做预读连贯运动处理。

★、210, 螺纹加工主轴转速波动率的允许千分比[0 表不等待, >=2 有效]

当此参数设置>2 时表示加工螺纹之前等待主轴稳定后才开始加工, 如设置为 5 时表示主轴转速误差波动小于<5/1000 时开始加工螺纹。

★、211, 刚性攻丝进刀主轴初始速度(rpm)

P211-P218 设置刚性攻丝主轴和 Z 轴退刀初始速度/加速度/预留量/预留量。

★、212, 刚性攻丝退刀主轴初始速度(rpm)

★、213, 刚性攻丝进刀主轴始速度(rpm/S) [>1]

★、214, 刚性攻丝退刀主轴加减速(rpm/S) [>1]

★、215, 刚性攻丝进刀主轴预留量(1/1000 圈) [>2]

★、216, 刚性攻丝进刀主轴预留量(1/1000 圈) [>2]

★、217, 刚性攻丝退刀主轴预留量(1/1000 圈) [>2]

★、218, 刚性攻丝退刀主轴拖尾量(1/1000 圈) [>2]

★、230, 主轴 SP 指令 G00 定位速度(0.1rpm)

设置主轴定位时执行 SP 指令 G00 定位速度。

★、231, SP 方式[+4 按 F 执行, +8 按 G90/G91 执行, +16 由当前显示值发送脉冲, +8192 模拟量每秒钟发送一次, +16384 执行 M05 关模拟量]

+4 表执行 SP 指令时按当前 F 速度[度/分钟]定位 (在 G00 状态下依然按主轴定位速度参数执行), 否则按主轴定位速度参数执行。

+8 表执行 SP 指令时受当前 G90/G91 的控制, 为 G90 时表绝对会归算, 为 G91 时表增量不会归算, 否则会进行归算并按最近方向运行。

+16 表系统的 SP 数值显示由当前发送脉冲决定, 否则由编码器反馈决定显示值。

+8192 表主轴模拟量每 1 秒钟就重复发送一次。

★, P67, the time constant of automatic enhancement smoothing processing [2-500] is recommended to be set as: 12

★P68, enhance the smoothing processing level [1-9] (the higher, the smoother). It is recommended to set it to: 2-8

★, 101-108, X take-off speed (mm/min)

This parameter determines the initial speed during feed axis acceleration and serves as the termination speed for deceleration. Specifically, when the feed axis speed falls below this threshold, it reaches its target directly without acceleration/deceleration phases. Conversely, when the speed exceeds this value, acceleration begins at this threshold. Sections P101-P108 specify the jump speeds (mm/min) for X/Y/Z/A/B/C/Xs/Ys axes [1].

Note: The value is related to the configuration of the machine tool, and improper setting may cause equipment failure or accident.

Normally, the step drive is not greater than 100 and the servo is not greater than 500.

★111-118, maximum X-axis velocity variation (mm/min)

This parameter controls the maximum speed variation during multi-axis continuous trajectory interpolation. It ensures that when the interpolated speed reaches its maximum allowable change, the corresponding axis velocity will either accelerate or decelerate. If the actual speed change exceeds this threshold, a gradual acceleration/deceleration process occurs; otherwise, the system directly reaches the target speed. This primarily enhances the continuity of multi-axis trajectory interpolation. Sections P111-P118 specify the maximum speed variation parameters (mm/min) for X/Y/Z/A/B/C/Xs/Ys axes [>]

Example: When this value is 300, when the X-axis speed changes from F800 to F1600 in multi-axis continuous trajectory interpolation, the change process should be X-axis speed first changes from F800 to 1100, and then accelerates to F1600.

★200, whether the continuous motion plays a role in G00 [Table 16 is yes, Table 1 is no]

Set whether G00 performs the pre-reading function. If set to 16, the pre-reading coherent motion processing is performed.

★, 210, permissible thousand percentage rate of variation in spindle speed during thread machining [0 not waiting, ≥2 valid]

When this parameter is set to >2, the machining of the thread will be started after waiting for the spindle to stabilize. If it is set to 5, the machining of the thread will be started when the spindle speed error fluctuation is less than <5/1000.

★211, Initial spindle speed for rigid tapping (rpm)

P211-P218: Set the initial speed, acceleration, reserve amount, and reserve amount of the rigid tapping spindle and Z-axis tool withdrawal. ★ 212: Initial speed (rpm) of the rigid tapping spindle during tool withdrawal.

★, 213, spindle feed speed for rigid tapping [>] 1

★214, Rigid thread attack and retraction spindle acceleration and deceleration (rpm/S) [>]

★, 215, main shaft reserved capacity for rigid thread cutting (1/1000 turn) [>]2

★, 216, main shaft reserved capacity for rigid thread cutting (1/1000 turn) [>]2

★, 217, reserved allowance for rigid tapping and retraction of the spindle (1/1000 turn) [>]2

★, 218, Rigid thread attack retraction spindle tail length (1/1000 turn) [>]2

★230, spindle SP command G00 positioning speed (0.1 rpm)

Set the spindle positioning to execute SP command G00 positioning speed.

★ 231, SP mode [press F for +4, press G90/G91 for +8, send pulses from current display value for +16, send analog signal every second for +8192, execute M05 to turn off analog signal for +16384]

+4 When executing SP command, the table is positioned according to the current F speed [°/min] (the spindle positioning speed parameter is still executed in G00 state), otherwise the spindle positioning speed parameter is executed.

+8 When the table performs the SP instruction, it is controlled by the current G90/G91. When G90, the table absolute will be calculated; when G91, the table increment will not be calculated. Otherwise, the table will be calculated and run in the nearest direction.

+16 The SP value of the table system is determined by the current sending pulse, otherwise it is determined by the encoder feedback.

+8192 The table spindle analog signal is sent once every 1 second.

+16384 表执行 M05 主轴停止后模拟量也停止输出。

★、232, 插补方式主轴 SP 指令定位方向[0 表正向, 1 表负向, 其余表就近方向]

0 表示 SP 指令正方向定位, 1 表示 SP 指令负方向定位, 其余表就近方向定位。

★、233, 主轴回零方式(1 表脉冲控制, 2 表驱动器控制, 输出 M61 检测 M22, 16 或 32 表主轴准停后定位到参数 P234 角度, 16 表输出 M75 检测 M22)

主轴回零方式, 1 表示输出脉冲系统回零, 2 表示驱动器控制(输出 M61 检测 M22), 16 或 32 表示主轴准停后定位到参数 P234 角度, 16 表输出 M75 检测 M22。

★、234, 主轴定向停止角度(0.1 度)

★、235, 主轴回零速度(0.1 rpm)

3.7.3 轴类参数

★、1, 进给轴波段开关[1 表有, 0 表无]

该参数用于设置 X、Z 轴倍率调节采用的操作方式, 1 表示采用外接波段开关调节, 0 表示用面板的“进给↑”“进给↓”键调节。

说明: 当该参数设置为 1 时, 外接波段开关的接口为 CN11 插座的 VDK0(OFF)、VDK1(X100)、VDK2(X10)、VDK3(X1) 信号。

★、2, 主轴和快速波段开关[1 表主轴, 2 表 G00 快速, 0 表无]

该参数用于设置主轴倍率调节采用的操作方式, 1 表示采用外接波段开关调节, 0 表示用面板的“主轴↑”“主轴↓”键调节。

说明: 当该参数设置为 1 时, 外接波段开关的接口为 CN11 插座的 VDS0(HCOM)、VDS1(Z)、VDS2(Y)、VDS3(X) 信号。

★、3/4, X 轴负向/正向最大行程(mm)

设置 X 轴软限位的负方向/正方向的坐标值, 该值是以机床坐标系为基准的。P3-P12 设置 X/Y/Z/A/B/C/Xs/Ys 等轴的负方向/正方向软限位。

★、13, 主轴制动 Y12 信号模式[0 表主轴制动, 1 表主轴使能]

0 表示 M05 输出 Y12 主轴制动, 1 表示 M03/M04 时输出 Y12 主轴使能信号。

P13-1 为 0 时设置主轴制动为短信号, 1 设置主轴制动为长信号;

P13-2 设置主轴制动为短信号时的时间。

★、15, 软限位[D0=1 表主轴冷却自动启动; D2 轴 X; D3 轴 Y; D4 轴 Z; D5 轴 A; D6 轴 B; D9 轴 C; D10 轴 XS; D11 轴 YS; 1 表不限位; D12 MOVE; D14 换刀无限位]

设置软限位功能是否有效, 每轴单独设置。该参数为位参数, 其中 D2 位控制 X 轴, D4 位控制 Z 轴, 相应位设置为 1 表示该轴软限位功能无效, 0 则有效, D15=1 表相对坐标断电保存。

位参数数据格式(其它位参数均一样): D15D14……D1D0。

例: 若想设置 X 轴软限位功能无效则就该参数设置为: 00000100

★、16, X 轴反向间隙补偿(um)

用于设置 X 轴运动反向时, 传动机构存在的反向间隙值(半径值表示)。当该轴运动时涉及到反向的情况, 系统自动调用该值进行补偿。单位: μm (微米)。

P16-P20 设置 X/Y/Z/A/B/C/Xs/Ys 的反向间隙补偿。

+16384 After the execution of M05, the analog output is also stopped. ★, 232, interpolation mode spindle SP command positioning direction [0: positive table direction, 1: negative table direction, others: nearest direction]

0 1 indicates positive direction positioning of the SP command, 2 indicates negative direction positioning of the SP command, and the rest indicate near-direction positioning. ★ 233: Spindle zeroing method (1 indicates pulse control, 2 indicates driver control, output M61 detects M22,16 or 32 indicates spindle stop positioning to parameter P234 angle after stopping, 16 outputs M75 to detect M22).

The spindle reset modes are as follows: 1 indicates the output pulse system resets, 2 indicates driver control (output M61 detects M22),16 or 32 indicates the spindle stops and positions to parameter P234 angle after stopping, 16 outputs M75 detects M22. ★, 234, spindle directional stop angle (0.1 degrees)

★, 235, spindle return speed (0.1 rpm)

3. 7. 3 Axis Parameters

★1. Feed axis band switch [1 indicates yes, 0 indicates no]

This parameter is used to set the operation mode of X and Z axis ratio adjustment. 1 indicates that the external band switch is used for adjustment, and 0 indicates that the "feed up" and "feed down" keys on the panel are used for adjustment.

Note: When this parameter is set to 1, the interface of the external band switch is VDK0 (OFF), VDK1 (X100), VDK2 (X10), and VDK3 (X1) signals of the CN11 socket.

★, 2, spindle and fast band switch [1 for spindle, 2 for G00 fast band, 0 for none]

This parameter is used to set the operation mode of spindle ratio adjustment. 1 indicates the use of external band switch adjustment, and 0 indicates the use of "spindle ↑" and "spindle ↓" keys on the panel.

Note: When this parameter is set to 1, the external waveband switch is connected to the VDS0 (HCOM), VDS1 (Z), VDS2 (Y), and VDS3 (X) signals of the CN11 socket.

★, 3/4, maximum X-axis negative/positive stroke (mm)

Set the negative/positive direction coordinate values of the soft limit on the X-axis, which are based on the machine tool coordinate system. P3-P12 Set the negative/positive direction soft limits for axes such as X/Y/Z/A/B/C/Xs/Ys.

★13, spindle brake signal mode [0 indicates spindle brake, 1 indicates spindle enable]

0 Indicates that M05 outputs Y12 spindle brake, and 1 indicates that M03/M04 outputs Y12 spindle enable signal.

When P13-1 is 0, set the spindle brake as short signal, and 1 set the spindle brake as long signal;

P13-2 Time when the spindle brake is set to short signal.

★15, Soft Limit [D0=1: Spindle cooling automatically activated; D2: Axis X; D3: Axis Y; D4: Axis Z; D5: Axis A; D6: Axis B; D9: Axis C; D10: Axis XS; D11: Axis YS; 1: No Limitation; D12: Move; D14: Tool Change with Unlimited Limitation]

The effectiveness of the soft limit function is configured separately for each axis. This parameter is a bit parameter, where D2 controls the X-axis and D4 controls the Z-axis. Setting the corresponding bits to 1 indicates that the soft limit function for that axis is disabled, while setting it to 0 enables activation. When D15=1, the relative coordinate table is powered off and saved.

Bit parameter data format (all other bit parameters are the same): D15 D14...D1D0.

Example: If you want to disable the soft limit function of the X-axis, this parameter should be set to: 00000100

★, 16, X-axis backlash compensation (um)

When setting the reverse movement of the X-axis, the reverse clearance value (radius value) in the transmission mechanism is indicated. When the movement of this axis involves reverse movement, the system automatically calls this value for compensation. Unit: μm (micrometers).

P16-P20 Set the reverse gap compensation for X/Y/Z/A/B/C/Xs/Ys.

★、21, X 轴方向信号

用于改变 X 轴电机方向, 即当此值设为 0 时将 X 轴的移动方向等同于指令方向取反, 当此值为 1 时 X 轴移动方向等同于指令方向。

P21-P25 设置 X/Y/Z/A/B/C/Xs/Ys 的电机方向。

★、27/28, X 轴电子齿轮分子/分母

用于设置 X 轴电子齿轮的分子/分母, 设值范围为 1-999999。

P29-36 设置 X/Y/Z/A/B/C/Xs/Ys 的电子齿轮分子/分母。

★、37, XYZ 正限位(+L) [0 表常开, 1 表常闭]

用于设置正向硬限位开关的类型, 0 表常开, 1 表常闭。

★、38, XYZ 负限位(-L) [0 表常开, 1 表常闭]

用于设置负向硬限位开关的类型, 0 表常开, 1 表常闭。

★、39, 浮动零点及 C 轴坐标归算[D3 轴 X;D4 轴 C(Y);D5 轴 Z;D6 轴 A;D7 轴 B;D8 轴 C;D9 轴 Xs;D10 轴 Ys;1 表示浮动零]

设置浮动零点功能是否有效, 每轴单独设置。该参数为位参数, 其中 D3 位控制 X 轴, D5 位控制 Z 轴, 相应位设置为 1 表示该轴为浮动零点, 0 则表示有机零开关。

例: 若想设置 X 轴零点为浮动零点则就该参数设置为: 00001000

★、40, X 轴浮动零点时机床坐标设置

该参数设置当前机床停在的位置浮动零点 X 轴坐标值, 开机回零即回到该浮动零点位置。

当没用机械零点时就设置为浮动零点, 进行回零操作时系统将快速定位到该零点位置。

P40-P44 表示 X/Y/Z/A/B/C/Xs/Ys 轴浮动零点时机床坐标设置。

★、45, 进给轴回零要求[1 表不需要, 0 表提示, 8 表强制, 9 表超强制, 15 表所有轴强制, 31 表所有轴超强制]

用于设置系统对开机回参考点处理方式, 回参考点处理方式有6种:

设置为1时为不需要: 每次开机后系统不提示也不做任何限制。

设置为0时为提示方式: 每次开机后系统会弹出一个对话框提示操作者执行回参考点操作, 之后不做任何限制。

设置为8时为强制方式: 每次开机后运行程序系统会弹出一个对话框提示操作者执行回参考点操作, 并且若在自动运行前不执行回参考点操作, 系统将会提示“进给轴没有回参考点”并不执行程序。

设置为9时为超强制方式: 每次开机后进给轴移动系统会弹出一个对话框提示操作者执行回参考点操作, 不执行回参考点操作, 系统将会提示“进给轴没有回参考点”并不执行运动。

设置为15时所有轴为强制方式, 设置为31时所有轴为超强制方式。

★、46, 进给轴回零型式[0--反向检测零位; 1--反向不检测零位; 2--检测零位不反向; 3--不检测零位不反向]

用于设置各进给轴回参考点时检测参考开关和电机编码器的零脉冲信号的几种方法:

★, 21, X-axis direction signal

Used to change the direction of the X-axis motor. That is, when this value is set to 0, the movement direction of the X-axis is equal to the reverse of the instruction direction, and when this value is set to 1, the movement direction of the X-axis is equal to the instruction direction.

P21-P25 Set the motor direction of X/Y/Z/A/B/C/Xs/Ys.

★, 27/28, X-axis electron gear numerator/fraction

Used to set the numerator and denominator of the X-axis electronic gear. The value range is 1-999999.

P29-36 Set the electron gear numerator and denominator of X/Y/Z/A/B/C/Xs/Ys.

★, 37, XYZ positive limit (+L) [0 normally open, 1 normally closed]

The type used to set the forward hard limit switch is 0 for normally open and 1 for normally closed.

★, 38, XYZ negative limit (-L) [0 normally open, 1 normally closed]

The type used to set the negative hard limit switch is 0 for normally open and 1 for normally closed.

★39, Floating zero point and C-axis coordinate reduction [D3 axis X; D4 axis C (Y); D5 axis Z; D6 axis A; D7 axis B; D8 axis C; D9 axis Xs; D10 axis Ys; 1 indicates floating zero]

Whether the floating zero point function is set to be effective is set separately for each axis. This parameter is a bit parameter, where D3 bit controls the X-axis and D5 bit controls the Z-axis. If the corresponding bit is set to 1, it indicates that the axis has a floating zero point; if 0, it indicates that there is a mechanical zero point switch.

For example, if you want to set the X-axis zero point as a floating zero point, this parameter should be set to: 00001000

★40, X-axis floating zero point lathe coordinate setting

This parameter sets the floating zero point X axis coordinate value of the current machine tool stop position, and the machine will return to this floating zero point position when starting up.

When the mechanical zero point is not used, it is set as a floating zero point. When the zero return operation is performed, the system will quickly locate to this zero point position.

P40-P44 indicates the floating zero point timing of the X/Y/Z/A/B/C/Xs/Ys axes in the lathe coordinate setting.

★, 45, feed axis return to zero requirement [1 table does not need, 0 table indicates, 8 table forces, 9 table overforces, 15

Table All shafts are forced, 31 Table All shafts are overforced]

Used to set the system to the starting point return processing mode, there are 6 kinds of return reference point processing modes:

When set to 1, it is not required: the system does not prompt or restrict anything after each startup.

When set to 0, it is a prompt mode: after each startup, the system will pop up a dialog box to prompt the operator to perform the return to reference point operation, and then no restrictions are made.

When set to 8, it is a mandatory mode: after each startup, the program running system will pop up a dialog box prompting the operator to perform the return to reference point operation. If the return to reference point operation is not performed before automatic operation, the system will prompt "the feed axis has no return to reference point" and the program will not be executed.

When the setting is 9, it is an ultra-mandatory mode: after each startup, a dialog box will pop up on the feed axis moving system to prompt the operator to perform the return to reference point operation. If the return to reference point operation is not performed, the system will prompt "the feed axis has no return to reference point" and the movement will not be performed.

When set to 15, all axes are in forced mode, and when set to 31, all axes are in super forced mode.

★46, feed axis return zero mode [0--reverse detection of zero position; 1--reverse no detection of zero position; 2--detection of zero position no reverse; 3--no detection of zero position no reverse]

Several methods for detecting the zero pulse signal of the reference switch and motor encoder when setting each feed axis back to the reference point:

设值为 0 时: 回参考点时, 撞到参考点开关后, 反向运行检测到开关脱开后, 再检测电机编码器零脉冲信号。

设值为 1 时: 回参考点时, 撞到参考开点关后, 反向运行检测到开关脱开即可。

设值为 2 时: 回参考点时, 撞到参考开点关后, 继续向前运行检测到开关脱开后, 再检测电机编码器零脉冲信号;

设置为 3 时: 回参考点时, 撞到参考开点关后, 继续向前运行检测到开关脱开即可。

回参考点方式应根据设备的电路情况设置, 通常情况下建议设置为 0 或 2, 因为如果回参考点不检测电机编码器的零脉冲信号时, 精度的保证全靠参考点开关来保证, 相比电机编码器的零脉冲精度就会更差一些。

★、47, 回零反向位参数[D2 轴 X;D3 轴 Y;D4 轴 Z;D5 轴 A;D6 轴 B;D7 轴 C;D8 轴 Xs;D9 轴 Ys;0 表正向;1 表负向]

设置进给轴回零的运动方向和先后顺序, 每轴单独设置。该参数为位参数, 其中 D2-D9 位控制 X/Y/Z/A/B/C/Xs/Ys 轴的回零运动方向, 1 表负向, 0 表正向。

★、48, 回零开关常闭位参数[D0 轴 X;D1 轴 Y;D2 轴 Z;D3 轴 A;D4 轴 B;D5 轴 C;D14 轴 Xs;D15 轴 Ys;1 常闭;0 常开;D6=1 手轮/自动;D7=1 手动/自动;D18=1 回零后 G28 回坐标零]

设置进给轴零点开关的类型, 每轴单独设置。该参数为位参数, 其中 D0-D5 控制 X/Y/Z/A/B/C/Xs/Ys 轴, 1 表常闭, 0 表常开。D7 为 1 时手动/自动状态自动切换, D6 为 1 时手轮/自动状态自动切换。

例: 若 X、Z 轴均为常闭开关, 则该参数设置为: 000000101

★、49, X 轴检测零位长度

设定 X 轴回零点时开关脱开后, 在多大范围内检测电机编码器零脉冲信号。单位: 0.1mm。P49-P53 表示 X/Y/Z/A/B/C/Xs/Ys 轴检测零位长度。

注: 该参数值必须小于电机一转的距离, 否则可能造成回零点不对的情况。

★、54, X 轴回零后偏移量

用于设置 X 轴回参考点时, 当检测到伺服电机零脉冲信号后, 再偏移多少距离后才完成回参考点功能。单位: 0.01mm。设值范围 -99999~+99999。

P54-P58 表示 X/Y/Z/A/B/C/Xs/Ys 轴回零后的偏移量。

注: 回参考点后的偏移是以 G00 速度运动。

★、59, 主轴换档时主轴是否启动[1 表是, 0 表否, 64 表保持主轴状态]

用于设置主轴高低档位换档时启不启动主轴, 1 表示要启动, 0 则反之。

★、60, 主轴换档时电机转动速度(1/100rpm)

用于设置主轴高低档位换档时启动主轴的速度。

★、61, 主轴换档时转动方向[0 表正, 1 表反]

用于设置主轴高低档位换档时启动主轴的方向, 1 表示反转, 0 表示正转。

★、62, 主轴换档时停止时间(10ms)

用于设置主轴高低档位换档时启动主轴的时间。单位: 10ms。

★、63, 主轴换档时低速转动时间(10ms)

When the value is set to 0: when returning to the reference point, after hitting the reference point switch, reverse operation is detected and the switch is disengaged, then the zero pulse signal of the motor encoder is detected again.

When the value is set to 1, when returning to the reference point, if the reference opening is closed, the reverse operation can detect the switch is open.

When the value is set to 2: when returning to the reference point, after hitting the reference open switch, continue to run forward and detect the switch disengagement, then detect the zero pulse signal of the motor encoder;

When set to 3, when returning to the reference point, if the reference opening is closed, continue to run forward and detect the switch disengagement.

The return reference point mode should be set according to the circuit condition of the equipment. In general, it is recommended to set it as 0 or 2. Because if the return reference point does not detect the zero pulse signal of the motor encoder, the accuracy guarantee depends entirely on the reference point switch, which will be worse than the zero pulse accuracy of the motor encoder.

★ 47, Zero Reverse Position Parameters [D2 axis X; D3 axis Y; D4 axis Z; D5 axis A; D6 axis B; D7 axis C; D8 axis Xs; D9 axis Ys; 0 indicates forward; 1 indicates reverse]

Set the direction and sequence of feed axis return to zero, which is set separately for each axis. This parameter is a bit parameter, where D2-D9 bits control the return direction of X/Y/Z/A/B/C/Xs/Ys axes. 1 indicates negative direction, and 0 indicates positive direction.

★ 48, Zero reset switch normally closed position parameters [D0 axis X; D1 axis Y; D2 axis Z; D3 axis A; D4 axis B; D5 axis C; D14 axis Xs; D15 axis Ys; 1 normally closed; 0 normally open; D6=1 manualwheel/automatic; D7=1 manual/automatic; D18=1 G28 reset after zero]

The type of feed axis zero point switch is configured separately for each axis. This parameter is a bit parameter, where D0-D5 controls the X/Y/Z/A/B/C/Xs/Ys axes, with 1 representing normally closed and 0 representing normally open. When D7 is set to 1, the manual/automatic mode automatically switches; when D6 is set to 1, the handwheel/automatic mode automatically switches.

For example, if both the X and Z axes are normally closed switches, the parameter is set to: 000000101

★49, X-axis detection zero position length

When the switch is disengaged during X-axis zero point setting, detect the zero pulse signal of the motor encoder within a range of 0.1 mm. P49-P53 indicate the zero position length detected by the X/Y/Z/A/B/C/Xs/Ys axes.

Note: The value of this parameter must be less than the distance of one rotation of the motor, otherwise it may cause the zero point to be wrong. ★, 54, X-axis offset after zero return

When setting the X-axis return reference point, when the servo motor zero pulse signal is detected, how much distance should be offset before completing the return reference point function. Unit: 0.01 mm. Set value range: -99999~+99999.

P54-P58 indicates the offset after zero return of X/Y/Z/A/B/C/Xs/Ys axes.

Note: The offset after returning to the reference point is moved at G00 speed.

★59, whether the spindle starts when shifting gears [1 indicates yes, 0 indicates no, 64 indicates keeps the spindle status] is used to set whether the spindle starts when shifting gears between high and low gears. 1 indicates to start, and 0 indicates otherwise.

★, 60, motor rotation speed when the main shaft shifts (1/100rpm)

Used to set the speed at which the spindle starts when shifting between high and low gears.

★, 61, rotation direction when the main shaft shifts [0 indicates forward, 1 indicates reverse]

Used to set the direction of the spindle when shifting between high and low gears. 1 indicates reverse rotation, 0 indicates forward rotation. ★, 62, Stop time during spindle shifting (10ms)

Used to set the time when the spindle starts during high and low gear shifting. Unit: 10ms.

★, 63, low speed rotation time (10ms) when the spindle shifts gears

用于设置主轴高低档位换档时停止主轴（M05）的时间。单位：10ms。

★、64, 主轴停止延时时间(10ms)

设置撤消 M03/M04 信号与启动 M05 信号之间的延时时间。单位：10ms。

★、65, 检测主轴位置反馈[1 表是, 0 表否]

★、66, 主轴反转点动是否有效[8 表有效]

设置为 8 时表示手动按“反转”键为主轴反转点动, 否则为主轴反转。

★、68, 进给轴反向延时时间(ms)

设置坐标轴反向时的延时时间, 在要求高的情况下设置为 0。单位：ms。

★、69, 手动按键回零方式[1 表示全部回零, 0 表示选择回零]

★、80, XZ 轴是否为旋转轴

设置为“1”有效, “0”无效。D2Z 轴工件规算, D3X 轴工件规算, D4Z 轴机床规算, D5X 轴机床规算, D6Z 轴旋转轴, D7X 轴旋转轴。

★、91, 车床第三轴轴名设置[0 表 Y, 1 表 C, 32 表车铣复合四轴 XYZC/五轴 XYZAC]

0 表示第三轴显示 Y, 1 表示第三轴显示 C, 32 显示四轴 XYZC/XYZAC 五轴。

★、92, 车床 C 轴功能设定[0 表旋转轴, 1 表直线轴]

0 表示 C 第三轴为旋转轴, 1 表示第三轴为直线轴。

★、93, C 轴为旋转轴时工件坐标[0 表不归算, 1 表归算]

★、94, C 轴为旋转轴时机床坐标[0 表不归算, 1 表归算]

★、101, 第四轴功能设定[0 表旋转轴, 1 表直线轴]

0 表示 C 第四轴为旋转轴, 1 表示第四轴为直线轴。

★、102, 第四轴为旋转轴时的规算[工件坐标+1; 机床坐标+2]

1 表示规算工件坐标, 2 表示规算机床坐标, 3 表示规算工件坐标和机床坐标。

★、103, 坐标轴运行时自动松开对应输出点或辅助继电器[10000+Y 输出点编号, 20000+M 继电器编号]

设置坐标运行时的输出点, 大于 10000 表示设置 Y 点, 大于 20000 表示设置 M 继电器编号, 如: 设置为 10021 时表示输出 Y21。

★、104, 坐标轴运行时自动松开到位对应输入点或辅助继电器[10000+X 输入点编号, 20000+M 继电器编号, 30000+等待时间 ms]

设置坐标运行输出点时设置是否检测输入点, 大于 10000 时表示检测 X 点, 大于 20000 时表示检测 M 继电器, 大于 30000 时表示延时设置时间。

例如: 设置 20005 表示输出 Y 点后检测 X5 到位。

★、105, 自动松开夹紧轴(1 表 A 轴; 2 表 X 轴; 4 表 Y/C 轴; 8 表 Z 轴; 16 表 B 轴; 32 表 C 轴; 64 表 Xs 轴; 128 表 Ys 轴)

设置坐标运行输出点的坐标轴, D0-D7 表示 A/X/Y/Z/B/C/Xs/Ys 轴。

★、111, 第五轴功能设定[0 表旋转轴, 1 表直线轴]

0 表示 C 第五轴为旋转轴, 1 表示第五轴为直线轴。

★、112, 第五轴为旋转轴时的规算[工件坐标+1; 机床坐标+2]

1 表示规算工件坐标, 2 表示规算机床坐标, 3 表示规算工件坐标和机床坐

Used to set the time (M05) when the spindle stops during high and low gear shifting. Unit: 10ms.

★, 64, spindle stop delay time (10ms)

Set the delay time between the M03/M04 signal and the start of the M05 signal. Unit: 10ms.

★65. Check the spindle position feedback [Table 1: Yes, Table 0: No]

★66. Whether the main shaft reversal point action is effective [8 valid]

When set to 8, the "reverse" key is manually pressed for spindle reversal point movement; otherwise, the spindle reverses.

★, 68, feed axis reverse delay time (ms)

Set the delay time when the axis is reversed. Set it to 0 in high requirements. Unit: ms.

★, 69, manual button return to zero mode [1 indicates full return to zero, 0 indicates select return to zero]

★, 80, whether the XZ axis is the rotation axis

Set to "1" for valid and "0" for invalid. D2Z axis workpiece specification, D3X axis workpiece specification, D4Z axis machine tool specification, D5X axis machine tool specification, D6Z axis rotation shaft, D7X axis rotation shaft.

★, 91, lathe third axis name setting [0 table Y, 1 table C, 32 table lathe milling compound four-axis XYZC/five-axis XYZAC]

0 The third axis displays Y, 1 indicates the third axis displays C, 32 shows four-axis XYZC/XYZAC five-axis. ★, 92, lathe C-axis function settings [0 indicates rotary axis, 1 indicates linear axis]

0 Indicates that axis C is the rotation axis, and 1 indicates that axis C is the straight line axis.

★When the C-axis is the rotation axis, the coordinates of the workpiece are [0 unweighted, 1 weighted]

★When the C-axis is the rotation axis, the machine coordinate [0 indicates no calculation, 1 indicates calculation]

★101, fourth axis function setting [0 indicates rotation axis, 1 indicates linear axis]

0 Indicates that the fourth axis is a rotation axis, and 1 indicates that the fourth axis is a straight line axis.

★102. Calculation of the fourth axis as a rotation axis [workpiece coordinate + 1; machine tool coordinate + 2]

1 Table 2 shows the coordinate of the workpiece, Table 3 shows the coordinate of the machine tool, and Table 3 shows the coordinate of the workpiece and the machine tool.

★103, when the coordinate axis runs, the corresponding output point or auxiliary relay is automatically released [10000+Y output point number, 20000+M relay number]

Set the output point of the coordinate run. If it is greater than 10,000, it indicates that the Y point is set; if it is greater than 20,000, it indicates that the M relay number is set. For example, if it is set to 10021, it indicates that the output Y21 is set.

★104, when the coordinate axis runs, it automatically releases the corresponding input point or auxiliary relay [10000+X input point number, 20000+M relay number, 30000+waiting time ms]

When setting the coordinate running output point, set whether to detect the input point. If it is greater than 10000, it indicates that the X point is detected; if it is greater than 20000, it indicates that the M relay is detected; if it is greater than 30000, it indicates that the delay setting time is set.

For example, set 20005 to indicate that X5 is checked when Y point is output.

★ 105. Automatic release of clamping shafts (1 for the A-axis; 2 for the X-axis; 4 for the Y/C-axis; 8 for the Z-axis; 16 for the B-axis; 32 for the C-axis; 64 for the Xs-axis; 128 for the Ys-axis)

Set the coordinate axis of the running output point. D0-D7 represents the A/X/Y/Z/B/C/Xs/Ys axes.

★111, Fifth axis function setting [0 indicates rotation axis, 1 indicates linear axis]

0 Indicates that axis C is the rotation axis and axis 1 is the straight line axis.

★112. Calculation of the fifth axis as a rotation axis [workpiece coordinate +1; machine tool coordinate +2]

1 Table 2 shows the coordinate of the workpiece, Table 3 shows the coordinate of the machine tool, and Table 3 shows the coordinate of the workpiece and the machine tool

标。

★、113, 第五轴是否显示为 C 轴[0 表否, 1 表是]

1 表示第 5 轴显示成 C 轴。

★、200, 主轴编码器每转的脉冲数(编码器线数的 4 倍)

设置主轴编码器的线数, 如:4096 表示为 1024 线编码器。

★、200-1, 主轴编码器检测线数报警值(诊断值) [>10 有效]

★、200-2, 主轴每转四倍频脉冲数[主轴与编码器 1:1 时设置为 0, 不是 1:1 时设置>99 有效, 必须是整数, 否则不能用此功能]

★、201, 是否检测主轴位置反馈(1 表是, 0 表否)

1 表示要检测主轴编码器反馈, 0 表示不检测。

★、202, 主轴定向时检测角度

★、203, 主轴定向时角度检测误差

★、210, 插补攻丝坐标轴[91 表 X, 92 表 Y, 93 表 Z, 94 表 A, 95 表 B, 96 表 C, 97 表 Xs, 98 表 Ys]

Z 轴作为插补攻丝的主轴坐标轴, 如: 95 表示 Z 轴与第 5 轴插补攻丝。

★、210a, Y 轴攻丝坐标轴[91 表 X, 92 表 Y, 93 表 Z, 94 表 A, 95 表 B, 96 表 C, 97 表 Xs, 98 表 Ys]

Y 轴作为插补攻丝的主轴坐标轴, 如: 95 表示 Y 轴与第 5 轴插补攻丝。

★、210b, X 轴攻丝坐标轴[91 表 X, 92 表 Y, 93 表 Z, 94 表 A, 95 表 B, 96 表 C, 97 表 Xs, 98 表 Ys]

X 轴作为插补攻丝的主轴坐标轴, 如: 95 表示 X 轴与第 5 轴插补攻丝。

★、219, 第一主轴全脉冲共享轴(62[61-65 对应 XYZAB 轴]; 要求 P221=P29, P30=360000; M03 与+C 方向一致; 开机 M800 回零)

★、220, 第一主轴全脉冲控制通道(1 表模拟量输出, 正向 82, 反向 92)

第一主轴控制是否为全脉冲控制, 1 模拟量 0-10V 输出, 82 为全脉冲正方向旋转, 92 为全脉冲反方向旋转。

★、221, 第一主轴全脉冲控制每圈脉冲数,

第一主轴全脉冲每圈脉冲数, 即电机转一圈脉冲数。

★、222, 第一主轴全脉冲控制升降速(rpm/2ms)

第一主轴全脉冲电机升降速常数。

★、223, 第二主轴全脉冲控制通道(1 表模拟量输出, 正向 81-88 对应 XYZABCXSYS 轴, 反向 91-98 对应 XYZABCXSYS 轴)

第二主轴控制是否为全脉冲控制, 1 模拟量 0-10V 输出, 81-88 为全脉冲正方向旋转, 91-98 为全脉冲反方向旋转。

★、224, 第二主轴全脉冲控制每圈脉冲数,

第二主轴全脉冲每圈脉冲数, 即电机转一圈脉冲数。

★、225, 第二主轴全脉冲控制升降速(rpm/2ms)

第二主轴全脉冲电机升降速常数。

★、226, 双主轴跟随的转速分子[1 表无效, (1000+ 分子) 表示跟随第一主轴, -(1000+ 分子) 表示跟随第二主轴]

treetop .

★113. Is the fifth axis displayed as the C-axis [0 indicates no, 1 indicates yes]

1 Indicates that the fifth axis is displayed as the C axis.

★, 200, number of pulses per revolution of the spindle encoder (4 times the number of encoder lines)

Set the number of lines of the spindle encoder, for example: 4096 indicates a 1024-line encoder.

★200-1, spindle encoder detection line number alarm value (diagnostic value) [>]10 valid

★200-2, the number of quadrupled frequency pulses per rotation of the spindle [set to 0 when the spindle and encoder are 1:1, set to >99 when not 1:1, must be an integer, otherwise this function cannot be used]

★201, whether to detect spindle position feedback (Table 1 is yes, Table 0 is no)

1 Indicates that the spindle encoder feedback is to be detected, and 0 indicates that the spindle encoder feedback is not to be detected.

★202, Angle detected when spindle is oriented

★203. Angle detection error when spindle is oriented

★, 210, interpolation thread coordinate axes [91 table X, 92 table Y, 93 table Z, 94 table A, 95 table B, 96 table C, 97 table Xs, 98 table Ys]

The Z-axis serves as the main axis of the spindle for interpolation and tapping, such as: 95 indicates that the Z-axis is used for interpolation and tapping with the 5th axis.

★, 210a, Y-axis tapping coordinate axis [91 table X, 92 table Y, 93 table Z, 94 table A, 95 table B, 96 table C, Table Xs, Table Ys [97,98]

The Y-axis serves as the main axis coordinate of the plug-in thread, such as: 95 indicates that the Y-axis and the fifth axis are used for plug-in thread.

★210b, X-axis tapping coordinate axis [91 table X, 92 table Y, 93 table Z, 94 table A, 95 table B, 96 table C, Table Xs, Table Ys [97,98]

The X axis is the main axis of the spindle for interpolation and threading. For example, 95 indicates that the X axis and the fifth axis are used for interpolation and threading.

★ 219, First Main Pulse Shared Axis (62[61-65 corresponding to XYZAB axis]; requirements: P221 = P29, P30 = 360000; M03 direction consistent with +C; startup: M800 reset to zero)

★220, first spindle full pulse control channel (1 table analog output, forward 82, reverse 92)

Whether the first spindle control is full pulse control, 1 analog output 0-10V, 82 is full pulse forward rotation, 92 is full pulse reverse rotation.

★221, the number of pulses per revolution of the first spindle full pulse control,

The number of pulses per revolution of the first spindle, that is, the number of pulses per revolution of the motor.

★222. Full pulse control of the first spindle speed (rpm/2ms)

First, the full pulse motor speed constant of the first spindle.

★223, Second spindle full pulse control channel (1 table analog output, forward 81-88 corresponding to XYZABCXSYS axes, reverse 91-98 corresponding to XYZABCXSYS axes)

Second, whether the main shaft control is full pulse control. 1 analog output 0-10V, 81-88 is full pulse forward rotation, 91-98 is full pulse reverse rotation.

★224, the number of pulses per revolution of the second spindle full pulse control,

The second main shaft full pulse number of each revolution, that is, the number of pulses of the motor per revolution.

★, 225, second spindle full pulse control speed (rpm/2ms)

Second main shaft full pulse motor speed constant.

★226, the rotational speed of the dual spindle follower [1 indicates invalid, (1000+ molecules) indicates following the first spindle, - (1000+ molecules) indicates following the second spindle]

全脉冲主轴跟随分子，如：1002 表示第二主轴跟随第一主轴的分子为 2，-1002 表示第一主轴跟随第二主轴分子为 2。

★、227, 双主轴跟随的转速分母[1 表无效, (1000+分母) 表示跟随第一主轴, -(1000+分母) 表示跟随第二主轴]

全脉冲主轴跟随分母，如：1002 表示第二主轴跟随第一主轴的分母为 2，-1002 表示第一主轴跟随第二主轴分母为 2。

★、228, 第一主轴的启停是否同时控制第二主轴[0 表是, 1 表否]

第一主轴的启停是否同时控制第二主轴，0 表示同时控制。

★、229, 第二主轴的启停是否同时控制第一主轴[0 表是, 1 表否]

第二主轴的启停是否同时控制第一主轴，0 表示同时控制。

★、323, 旋转轴是否不按就近距离运动(+4:X 轴;+8:Y/C 轴;+16:Z 轴 ;+32:A 轴;+64:B 轴)

坐标设置为旋转轴时，是否不按就近距离运行，D2-D6 设置为 1 时表示 X/Y/Z/A/B 不按就近距离运行。

★、404, 主轴脉冲控制方向信号(0 表反向, 1 表正常)

第一主轴设置为全脉冲时的电机方式，0 表反向, 1 表正向。

★、405, 主轴是否使用电子齿轮(0 表是, 1 表否)

第一主轴设置为全脉冲时是否使用电子齿轮，0 表示要用电子齿轮。

P406-P408 表示使用电子齿轮时的低档/高档电子分子/分母。

★、411, 攻丝和共享轴控制方式[2 跟踪;3 插补;+8 共享轴根据编码器反馈位置再定位;+16 编码器反馈反向, 即 A/B 交换]

攻丝轴和共享轴控制方式，0 表示编码器跟随攻丝，4 表示主轴与攻丝轴插补攻丝，+8 表示共享轴根据编码器反馈位置再定位，+16 编码器反馈反向，即 A/B 交换。

★、412, 主轴齿数

第一主轴与编码器不是 1: 1 减速器的主轴齿数。

★、413, 编码器齿数

第一主轴与编码器不是 1: 1 时减速器的编码器齿数。

★、414, 随动轴(7/8/9, 17/18/19, 27/28/29/D5..D13=1/表 A 随 X/Y/Z, C 随 X/Y/Z, B 随 X/Y/Z;14/15/16/表 Xs 随 X/Y/Z;24/25/26/表 Ys 随 X/Y/Z;23 表 Ys 随 Xs)

随动轴就是设置一个或者几个坐标轴跟随着被随动坐标轴同时运动，7 (或者 D5=1) 表示 A 轴跟随 X 轴运动，8 (或者 D6=1) 表示 A 跟随 Y 轴运动，9 (或者 D7=1) 表示 A 跟随 Z 轴运动；17 (或者 D8=1) 表示 B 轴跟随 X 轴运动，18 (或者 D9=1) 表示 B 跟随 Y 轴运动，19 (或者 D10=1) 表示 B 跟随 Z 轴运动，27 (或者 D11=1) 表示 C 轴跟随 X 轴运动，28 (或者 D12=1) 表示 C 跟随 Y 轴运动，29 (或者 D13=1) 表示 C 跟随 Z 轴运动。

★、501, M123M125 随动指令超前 Z 轴距离[>9 或<-9 有效](um)

随动指令 M123M125 超前 Z 轴距离主要用于随动轴跟随 Z 轴运动的粗加工场合，P502 设置超前时粗加工余量，P503 设置粗加工最大进刀速度。

Full pulse spindle follow molecules, such as: 1002, indicating that the second spindle follows the first spindle molecule is 2, -1002 indicates that the first spindle follows the second spindle molecule is 2.

★227, the denominator of the rotational speed of the double spindle follower [1 table is invalid, (1000+ denominator) indicates following the first spindle, - (1000+ denominator) indicates following the second spindle]

The full pulse spindle follows the denominator, such as: 1002, indicating that the second spindle follows the first spindle with a denominator of 2, and -1002 indicates that the first spindle follows the second spindle with a denominator of 2.

★228. Is the start/stop of the first spindle controlled simultaneously with the second spindle? [0 for yes, 1 for no]

Whether the start/stop of the first spindle is controlled simultaneously with the second spindle, 0 indicates simultaneous control.

★229. Is the start/stop of the second spindle controlled simultaneously with the first spindle? [0 for yes, 1 for no]

Whether the start/stop of the second spindle is controlled simultaneously with the first spindle, 0 indicates simultaneous control.

★323. Whether the rotation axis does not move according to the nearest distance (+4: X-axis; +8: Y/C-axis; +16: Z-axis; +32: A-axis; +64: B-axis)

When the coordinate is set as the rotation axis, does not run according to the nearest distance? When D2-D6 is set to 1, X/Y/Z/A/B does not run according to the nearest distance.

★404, spindle pulse control direction signal (0 indicates reverse, 1 indicates normal)

The motor mode is set to full pulse for the first spindle, where 0 indicates reverse and 1 indicates forward.

★405. Whether the main shaft uses electronic gear (0 indicates yes, 1 indicates no)

Whether to use an electronic gear when the first spindle is set to full pulse, 0 indicates that an electronic gear is used.

P406-P408 indicates the low/high electronic numerator/fraction when using an electronic gear.

★411, thread attack and shared axis control mode [2 tracking; 3 interpolation; +8 shared axis repositioning based on encoder feedback position; +16 encoder feedback reverse, i.e. A/B swap]

The control mode of the tapping shaft and the shared shaft is 0 for the encoder to follow the tapping, 4 for the spindle and the tapping shaft to interpolate the tapping, +8 for the shared shaft to reposition according to the encoder feedback position, and +16 for the encoder feedback reverse, that is, A/B exchange.

★, 412, number of main shaft teeth

The first spindle and encoder are not 1:1 reduction of the spindle teeth number.

★, 413, encoder number of teeth

The number of encoder teeth in the reducer is not 1:1 when the first spindle and encoder are not.

★ 414, follower shaft (7/8/9, 17/18/19, 27/28/29/D5...D13=1; Table A: follower follows X/Y/Z; C follows X/Y/Z; B follows X/Y/Z; 14/15/16: Table Xs follows X/Y/Z; 24/25/26: Table Ys follows X/Y/Z; 23: Table Ys follows Xs)

A follower axis refers to one or more coordinate axes that move in synchronization with the driven axis. Specifically: 7 (or D5=1) indicates A-axis following X-axis movement; 8 (or D6=1) indicates A-axis following Y-axis movement; 9 (or D7=1) indicates A-axis following Z-axis movement. Similarly, 17 (or D8=1) denotes B-axis following X-axis movement, 18 (or D9=1) B-axis following Y-axis movement, and 19 (or D10=1) B-axis following Z-axis movement. Additionally, 27 (or D11=1) C-axis follows X-axis movement, 28 (or D12=1) C-axis follows Y-axis movement, and 29 (or D13=1) C-axis follows Z-axis movement.

★501, M123M125 Follow-up command advance Z-axis distance [>]9 or <-9 valid] (um)

The M123M125 advance Z-axis distance is mainly used for the rough machining of the follower shaft following the Z-axis movement. P 502 sets the rough machining allowance, and P503 sets the maximum feed speed for rough machining.

- ★、502, M123M125 随动指令超前时粗加工余量[半径值] (um)
- ★、503, M123M125 随动粗加工最大进刀速度[>9 有效] (mm/min)
- ★、600, 总线系统攻丝主轴位置控制增益[10001-30000] (增益值为后四位, 建议值 20200)

刚性攻丝时设置攻丝时的主轴位置控制增益。

- ★、601, 总线系统攻丝提刀时主轴位置控制增益[10001-20000] (增益值为后四位, 建议值 10800)

刚性攻丝时设置提刀时的主轴位置控制增益。

- ★、602, 总线系统刚性攻丝平滑处理时间常数[经典方式: 10002-10120, 全新方式: 20002-20120; 建议值 20100]

设置刚性攻丝平滑处理时间常数, 有经典方式和全新方式二种方式, 经典方式设置范围为 10002-10120, 全新方式设置范围为 20002-20120; 一般使用全新方式建议值 20100。

- ★、603, 总线系统攻丝进给轴定位到位允差值(进给轴指令脉冲数, 建议值 5)

刚性攻丝进给轴定位允许误差值, 如设置为 5 时表示误差值<5 时表示攻丝线到位。

- ★、604, 总线系统攻丝提刀进给轴定位到位允差值(进给轴指令脉冲数, 建议值 20)

刚性攻丝进给轴定位允许误差值, 如设置为 5 时表示误差值<5 时表示攻丝线到位。

- ★、605, 总线系统插补攻丝位置环控制时进给轴最大定位误差(指令脉冲数, 建议设置为 3 个螺距的脉冲数)

刚性攻丝进给轴最大允许误差值, 如设置为 3 时表示进给轴最大误差值设置为 3 个螺距的脉冲数。

- ★、606, 总线系统插补攻丝位置环控制时主轴最大定位误差(指令脉冲数, 建议设置为 3 个螺距的脉冲数)

刚性攻丝主轴最大允许误差值, 如设置为 3 时表示主轴最大误差值设置为 3 个螺距的脉冲数。

- ★、607, 显示攻丝误差脉冲数(1: 表示攻丝时攻丝轴反馈脉冲数与主轴编码器脉冲数的误差, 正值表示攻丝轴超前, 负值表示主轴超前)

1: 表示攻丝时攻丝轴反馈脉冲数与主轴编码器脉冲数的误差, 正值表示攻丝轴超前, 负值表示主轴超前, 可以通过此误差值的显示表示进给轴和主轴的刚性差异。

- ★、608, 精确显示攻丝误差脉冲数(1: 表示攻丝时等待脉冲发完, 可以精确显示攻丝误差但效率会降低, 所以调试完要设置为 0)

用于刚性攻丝调试时使用, 1: 表示攻丝时等待脉冲发完, 可以精确显示攻丝误差但效率会降低。

- ★、610, 开启位置反馈记录[+1 表直线; +2 表圆弧; +4 表攻丝; +8 表直线同步性; +16 表车方 G809 的 C 轴 A 轴同步性; +64 显示图形]

★502, M123M125 Coarse machining allowance [radius value] when the follow-up instruction is advanced (um)

★503, M123M125 Maximum feed speed for follow-up rough machining [>]9 valid (mm/min)

★, 600, bus system tapping spindle position control gain [10001-30000] (gain value is the last four digits, recommended value 20200)

Set the spindle position control gain during rigid tapping.

★ 601, spindle position control gain when threading the main shaft in the bus system [10001-20000] (the gain value is the last four digits, the recommended value is 10800)

Set the spindle position control gain when setting the tool lift during rigid tapping.

★ 602, bus system rigid thread smooth processing time constant [classic method: 10002-10120, new method: 20002-20120; recommended value: 20100]

Set the constant of smooth treatment time for rigid thread cutting. There are two ways: classical way and new way. The setting range of classical way is 10002-10120, and the setting range of new way is 20002-20120. Generally, the recommended value of new way is 20100.

★ 603, bus system tapping feed axis positioning tolerance (feed axis command pulse number, recommended value 5)

The allowable error value of the feed axis positioning for rigid tapping is set to 5, indicating that the error value is <5 and the tapping line is in place.

★ 604, bus system tapping feed axis positioning tolerance (feed axis command pulse number, recommended value 20)

The allowable error value of the feed axis positioning for rigid tapping is set to 5. When the error value is <5, the tapping line is in place.

★605, maximum positioning error of feed axis during bus system plug-in tapping position ring control (instruction pulse number, recommended to set the number of pulses for 3 pitch)

Maximum allowable error value of feed axis for rigid tapping, such as setting it to 3, means that the maximum error value of feed axis is set to 3 pulse numbers of pitch.

★ 606. Maximum spindle positioning error during bus system plug-in tapping position ring control (instruction pulse number, recommended to be set to 3 pulse numbers of pitch)

Maximum allowable error value of rigid tapping spindle, if set to 3, it means that the maximum error value of spindle is set to 3 pulse numbers of pitch.

★607, shows the tapping error pulse number (1: indicates the error between the tapping shaft feedback pulse number and the spindle encoder pulse number during tapping. A positive value indicates the tapping shaft is ahead, and a negative value indicates the spindle is ahead)

1: indicates the error between the feedback pulse number of the tapping shaft and the pulse number of the spindle encoder during tapping. A positive value indicates the lead of the tapping shaft, and a negative value indicates the lead of the spindle. The difference between the feed shaft and the spindle rigidity can be indicated by the display of this error value.

★ 608, accurately display the number of screwing error pulses (1: indicates that the waiting pulse is finished during screwing, which can accurately display the screwing error but reduces the efficiency, so it should be set to 0 after debugging)

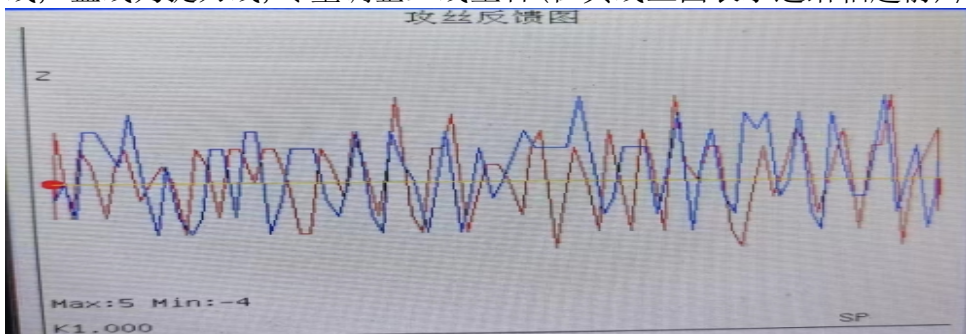
Used for rigid tapping debugging. 1: indicates that the pulse is waiting to be sent out during tapping, which can accurately display the tapping error but reduces the efficiency.

★ 610, Enable position feedback recording [+1 indicates straight line; +2 indicates arc; +4 indicates thread attack; +8 indicates straight line synchronization; +16 indicates synchronization between the C-axis and A-axis of G809 lathe; +64 displays graphics]

开启位置反馈记录功能,通过此功能可以显示 1 直线/+2 圆弧/+4 攻丝/+8 直线同步性/+16 车方 G809 的 C 轴 A 轴同步性的位置图形;指令运行完在“诊断界面”按“反馈图形”即显示位置反馈记录图形。

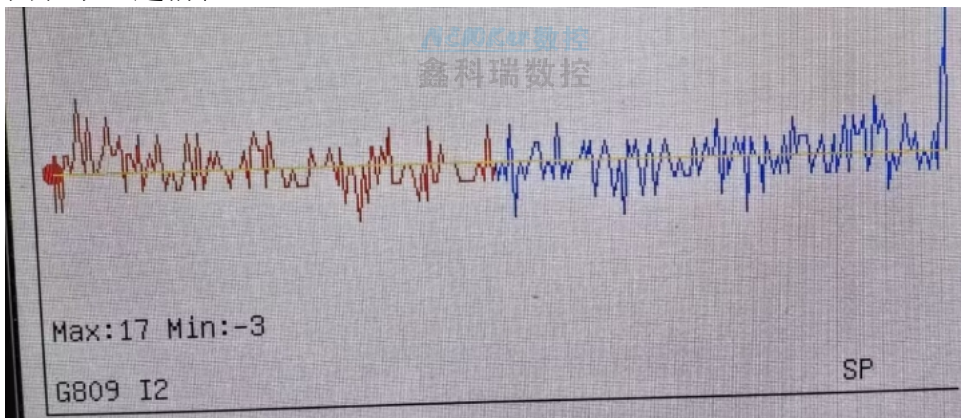
《1》刚性攻丝显示调试方法<P610=68>

- 1, 调整主轴和给进轴刚性, 要求绝对误差<10(正值表进给轴超前);
- 2, 黄线为标准线, 水平轴为主轴脉冲累计值, 垂直轴为同步误差;红线为进刀线, 蓝线为提刀线, 尽量调整三线重合(在黄线上面表示进给轴超前);



《2》飞方调试方法<P610=80>

- 1, 调整 C 轴和 A 轴刚性, 要求绝对误差<10(正值表 A 轴超前);
- 2, 黄线为标准线, 水平轴为主轴脉冲累计值, 垂直轴为同步误差, 在黄线上面表示 A 超前;



注意: 此功能要求系统必须接编码器反馈/攻丝 Z 轴反馈/飞方 A 轴反馈!!!

3. 7. 4 刀具参数

★、1, 半径 C 补偿的建立方式

设置半径 C 补偿的建立方式, 0 表 A 型, 1 表 B 型。

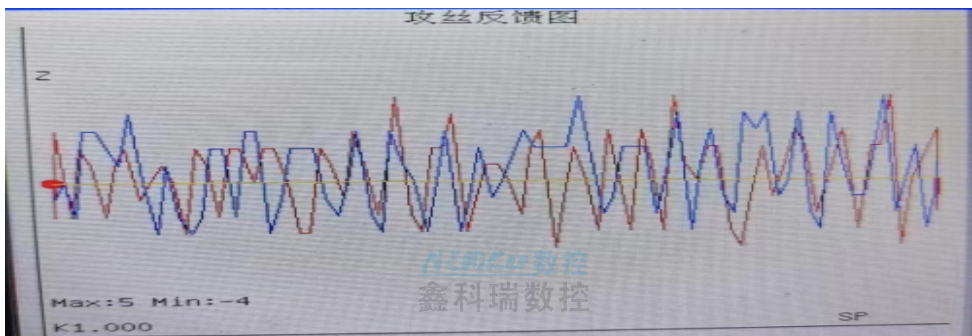
★、2, 半径 C 补偿的撤消方式

★、8, 刀具管理方式 (0 表使用 M06, 1 表 T 指令直接换刀, +32768 表开启刀具寿命管理)

刀具参数 P8=32768, 按二次 “刀补” 键即可进入刀具寿命管理界面;

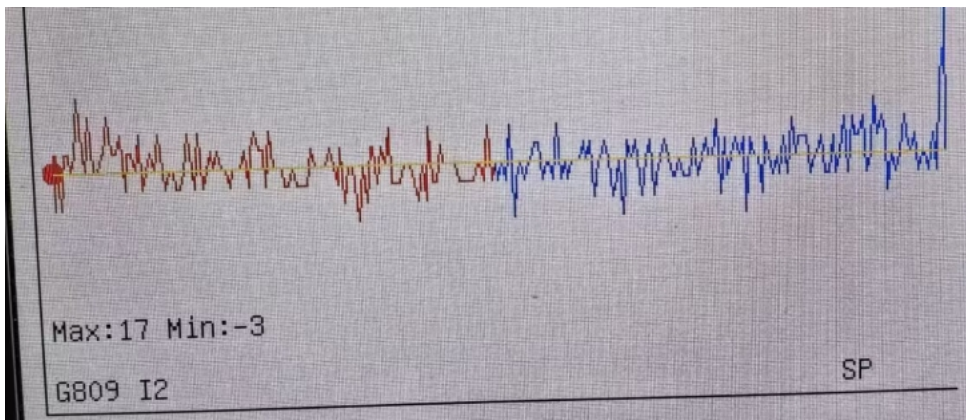
Enable the position feedback recording function. This feature displays the position graphics of G809's C-axis and A-axis synchronization: 1 straight line / +2 circular arc / +4 thread attack / +8 straight line synchronization / +16 wheel direction. After instruction execution, press "Feedback Graphics" in the "Diagnosis Interface" to display the position feedback record graphics.

1. Display debugging method for rigid thread attack <P610=68>
 1. Adjust the rigidity of the spindle and feed shaft, requiring absolute error <10 (positive value indicates that the feed shaft is in advance);
 2. The yellow line is the standard line, the horizontal axis is the main pulse cumulative value, and the vertical axis is the synchronous error; the red line is the feed line, and the blue line is the lift line. Try to adjust the three lines to coincide (the feed axis is in front of the yellow line);



<P610=80> Method 2: Feifang debugging method </P610=80>

1. Adjust the rigidity of C-axis and A-axis, requiring absolute error < 10 (positive value indicates A-axis overrunning);
2. The yellow line is the standard line, the horizontal axis is the main pulse cumulative value, and the vertical axis is the synchronous error. Above the yellow line, A leads;



Note: This function requires that the system must be connected to encoder feedback/threading Z axis feedback/flying A axis feedback!!!!

3. 7. 4 Tool Parameters

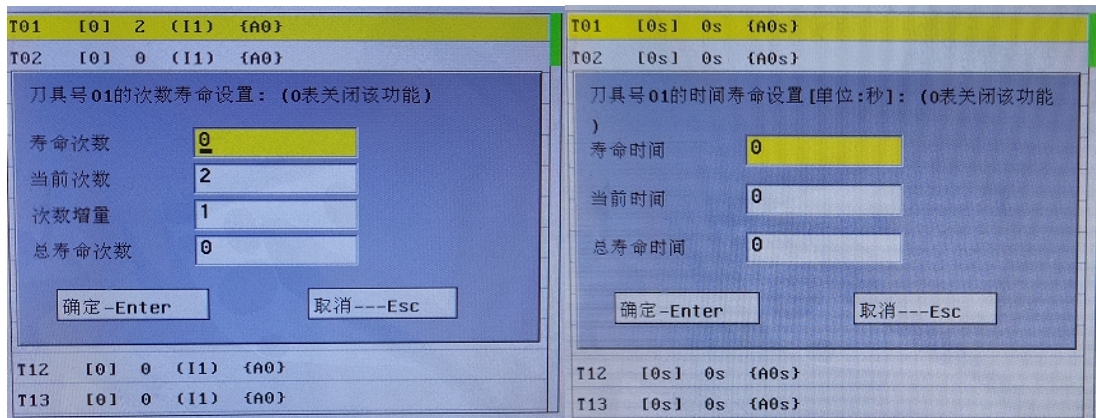
★1. The establishment of radius C compensation

The establishment mode of setting radius C compensation is 0 for Table A type and 1 for Table B type.

★2. Method of cancellation of radius C compensation

★ 8. Tool management mode (0 indicates the use of M06, 1 indicates the direct tool change with T command, +32768 indicates the opening of the tool Life management)

Tool parameter P8=32768, press the secondary "tool supplement" key to enter the tool life management interface;



在刀具寿命管理界面，有二个管理方案：

<1>按刀具使用次数：

刀具每换刀一次，“当前次数”自动加一个“次数增量”，如果到达设置的寿命次数，系统自动补偿所设置的“长度增量”和“半径增量”（0 表示不补偿）；如果到达设置的总寿命次数（0 表示不补报警）系统自动报警，报警后必须手动再次输入总寿命次数。

<2>按刀具使用时间：

“当前时间”自动按加工时间统计，如果到达设置的寿命时间，系统自动补偿所设置的“长度增量”和“半径增量”（0 表示不补偿）；如果到达设置的总寿命时间（0 表示不补报警）系统自动报警，报警后必须手动再次输入总寿命时间。

设置半径 C 补偿的撤消方式，0 表 A 型，1 表 B 型。

★、9, 基准刀具实际长度(mm) [绝对输入] (刀补界面按 ‘H’ 自动测量)

设置基准刀具实际长度(mm)，也可以在刀补界面按‘H’自动测量设置。

★、32, 刀位信号/WAT 信号滤波

D8=1 表对车床刀位检测信号进行 2ms 滤波；

D9=1 表对车床刀位检测信号进行 4ms 滤波；

D10=1 表对车床刀位检测信号进行 8ms 滤波，D8-D10 可同时设置为 1；

D11=1 表对 WAT 等待输入信号进行 2ms 滤波；

D12=1 表对 WAT 等待输入信号进行 4ms 滤波；

D13=1 表对 WAT 等待输入信号进行 8ms 滤波，D11-D13 可同时设置为 1；

3. 7. 5 杂类参数

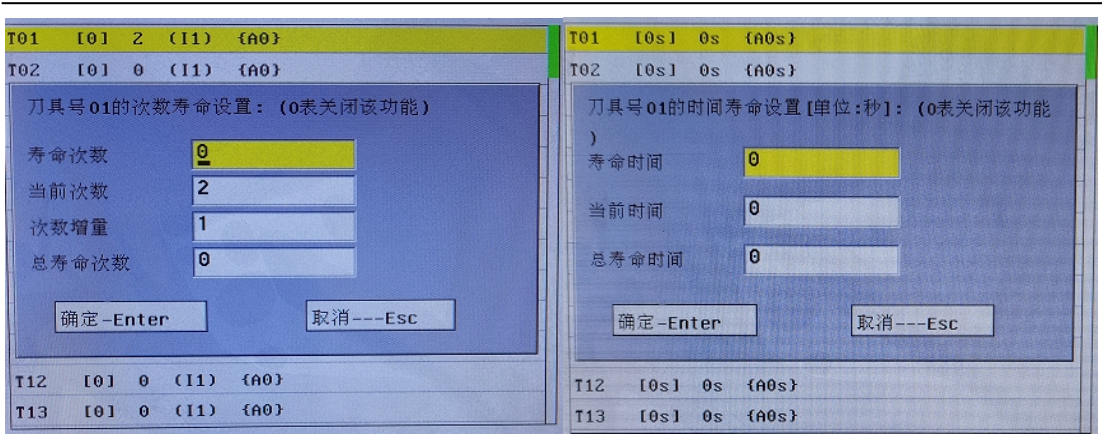
★、1、手轮型式

用于设置系统配置的手脉类型，1 表手持式,0 表面板式

注 当该参数设置为 1 时，手轮接口 CN11 此时不能使用波段开关进行主轴、进给修调，即“轴类参数”P1、P2 号参数只能设置为“0” A、X、Y、Z、X1、X10、X100、OFF 分别进行进给轴、档位的选择。

★、3、干预开关

用于设置系统面板上的干预开关是否起作用，88 表手动自动启用, 99 表自动



There are two management schemes in the tool life management interface:

<1>According to the number of times the tool is used;

Each time the tool is changed, the "current count" automatically increments by a "count increment". When the preset lifespan count is reached, the system automatically applies the set "length increment" and "radius increment" (0 indicates no adjustment). If the total lifespan count (0 indicates no alarm) is reached, the system triggers an automatic alarm. After the alarm occurs, the user must manually re-enter the total lifespan count.

<2>According to the tool use time;

The "Current Time" is automatically calculated based on processing duration. When the preset lifespan threshold is reached, the system automatically applies compensation to the set "Length Increment" and "Radius Increment" (0 indicates no compensation). If the total lifespan threshold is exceeded (0 indicates no alarm), the system triggers an automatic alert. After the alarm, users must manually re-enter the total lifespan value.

Set the cancellation mode of radius C compensation, 0 for Table A type and 1 for Table B type.

★9. Actual length of reference tool (mm) [Absolute input] (automatically measured by pressing 'H' on the tool supplement interface). Set the actual length of reference tool (mm), or set it automatically by pressing 'H9' on the tool supplement interface.

★, 32, tool position signal /WAT signal filter

D8=1 filters the lathe tool position detection signal for 2ms;

D9=1 filters the lathe tool position detection signal for 4ms;

D10=1 filters the lathe tool position detection signal for 8ms, and D8-D10 can be set to 1 at the same time;

D11=1 pairs the WAT to wait for the input signal to be filtered for 2 ms;

D12=1 indicates that the WAT waits for 4ms of input signal filtering;

D13=1 means that the WAT waits for 8ms filter input signal, and D11-D13 can be set to 1 at the same time;

3. 7. 5 Miscellaneous Parameters

★1. Handwheel type

Type of handheld pulse used to set up the system configuration: 1 for handheld, 0 for panel

Note: When this parameter is set to 1, the handwheel interface CN11 cannot use the band switch for spindle and feed adjustment at this time. That is, parameters P1 and P2 in the "Axis Parameters" can only be set to "0". A, X, Y, Z, X1, X10, X100, OFF are respectively used to select the feed axis and gear position.

★, 3. Intervention switch

Used to set whether the intervention switch on the system panel is working, 88 indicates manual automatic activation, 99 indicates automatic

启用,其余表关闭。

★、4、是否自动润滑

用于设置系统自动润滑功能是否有效,0 表有效,1 表无效。

注:自动润滑是根据运行时间来定的。

★、5、自动润滑时间(10ms)

用于设置每次润滑的时间,即 M32 保持有效的时间。单位:10ms。

★、6、自动润滑间隔时间(s)

用于设置每次润滑的间隔时间,即两次 M32 有效的间隔时间。单位:S。

★、7、门开关检测

用于设置系统是否要检测防护门开关信号,设置为 0 时表示无门开关;设置为 1 时表有门开关;为了安全建议设置检测。

注:1、对于门开关的检测是通过M12实现的。

2、当设置为检测后,当M12信号有效时,手动状态下可以运行坐标,自动侧会暂停程序。

★、8、门开关

用于设置门开的类型,0 表常开,1 表常闭

★、9、位参数

该参数为一个位参数,每一位都特定的功能,具体如下。

位参数数据格式(其它位参数均一样):D15D14……D1D0。

D0 位:无定义,默认为 1 不能改变。

D1 位:设置为 1 表 1 每次开机时加工件数自动清零,为 0 则保持。

D2 位:设置为 1 表示编辑程序时字母间自动插入空格,0 则不自动插入。

D3 位:无定义,默认为 0 不能改变。

D4 位:无定义,默认为 0 不能改变。

D5 位:设置为 1 表示按复位键时不停止主轴转动和冷却输出,0 则要停止。默认为 0。

D6 位:设置为 1 表示 G00 指令执行时各个轴按自己的速度独立运行,为非线性轨迹运行,即小坐标先到,0 则表示联动,同时到达。默认为 1。

D7 位:无定义,默认为 0 不能改变。

D8 位:设置为 1 表示关电时保存当前主轴松紧刀状态 M10/M11,系统重新上电后恢复该状态。0 则为上电时自动处于主轴紧刀状态。默认为 1。

D9 位:无定义,默认为 1 不能改变。

D10 位:设置为 1 时编程时行号自动排序。

D11 位:设置为 1 时,第一轴主轴模拟量输出从第一、第二主轴通道同时输出,第二主轴功能无效。

D12 位:设置为 1 时,屏蔽跳步功能,即带有程序段前加“/”跳转机能无效。

D13 位:设置为 1 时,屏蔽回零功能,手动按“回零”无效。

D14 位:设置为 1 时,屏蔽面板上的“启动”键功能,按“启动”无效。

D15 位:设置为 1 时,刀补显示相对值,否则刀补显示绝对值。

注:该位参数中有不能改变的位,设置时注意不能修改到,否则可能出现系

Enable, and disable the other tables.

★, 4. Whether it is automatically lubricated

Used to set whether the automatic lubrication function of the system is effective. 0 means effective, and 1 means invalid.

Note: Automatic lubrication is determined by the running time.

★, 5. Automatic lubrication time (10ms)

Used to set the time of each lubrication, i.e. the duration M32 remains active. Unit: 10ms.

★, 6. Automatic lubrication interval (s)

Used to set the interval time between lubrication, i.e. the interval time between two M32 operations. Unit: S.

★, 7. Door switch detection

Used to set whether the system should detect the door switch signal. If set to 0, there is no door switch; if set to 1, there is a door switch. For safety, it is recommended to set detection.

Note: 1. The detection of the door switch is realized by M12.

2 、 When set to detect, when the M12 signal is valid, the coordinates can be run in manual mode and the program will be suspended on the automatic side.

★, 8. Door switch

Used to set the type of door opening, 0 is normally open and 1 is normally closed

★, 9, bit parameters

This parameter is a bit parameter, each of which has a specific function as follows.

Bit parameter data format (all other bit parameters are the same): D15D14...D1D0.

D0 bit: undefined, default is 1 and cannot be changed.

D1: Set to 1 in Table 1. The number of processed parts is automatically reset to 0 each time the machine is started, and it remains at 0.

D2 bit: Set to 1 to automatically insert space between letters when editing, and set to 0 to not automatically insert.

D3 bit: undefined by default 0 and cannot be changed.

D4 bit: undefined, default is 0 and cannot be changed.

D5 position: Set to 1 to indicate that the spindle rotation and cooling output will not stop when the reset key is pressed, and 0 to stop. The default value is 0.

D6: Set to 1 indicates that each axis runs independently at its own speed when the G00 instruction is executed, which is a nonlinear trajectory running. That is, the small coordinate arrives first, and 0 indicates that the linkage arrives at the same time. The default value is 1.

D7 bit: undefined, default is 0 and cannot be changed.

D8 bit: Set to 1 indicates that the current spindle tension and tool state M10/M11 are saved when power is turned off, and this state will be restored after the system is re-powered. 0 indicates that the spindle is automatically in the tight tool state when power is turned on. The default value is 1.

D9 bit: undefined, default is 1 and cannot be changed.

D10 bit: When set to 1, the line number is automatically sorted during programming.

D11 bit: When set to 1, the first spindle analog output is output from both the first and second spindle channels, and the second spindle function is invalid.

D12 bit: When set to 1, the skip function is shielded, that is, the jump function with the program segment preceded by "/" is invalid.

D13 bit: When set to 1, the return to zero function is shielded and manual "return to zero" is invalid.

D14 bit: When set to 1, the "start" key function on the shielding panel is disabled and the "start" is invalid.

D15 position: When set to 1, the tool compensation display shows relative value; otherwise, the tool compensation display shows absolute value.

Note: There are non-modifiable bits in this bit parameter. Note that you should not modify them when setting, otherwise system may occur

统运行不正常。

★、10、加工件数自动统计

用于设置系统自动统计加工工件的件数，1 表有效，0 为表无效。

★、11、程序编辑序号增量

用于设置程序编辑是换行的行号增量。

★、12、系统内部参数[6 表自动进入, 7 表按键进入, 16 表按键快速进入, 8 表允许导入, 9 表清除]

开机显示 LOGO 界面设置，6 表示显示 LOGO 后自动进入主界面，7 表示按键后，16 表示开机自动读取电机位置按键快速进入主界面，8 表示可以导入 LOGO 界面文件，9 表示清除 LOGO 文件。

★、13、主轴转动与松紧刀是否互锁

用于设置主轴转动与松紧刀状态是否互锁：

0：表示松紧状态与主轴转动无任何联系；

1：表示主轴处于 M05 状态才能松紧刀；

8：表示主轴处于 M05 状态并且主轴编码器不能转动才能松紧刀；

16：表示检测转速互锁；

32：表示程序自动运行并且非暂停状态时松紧刀按键无效；

56：表示超完全互锁，程序自动运行并且非暂停状态时松紧刀按键无效。

本参数的设置跟机床设备的配置和用户使用情况有关，从安全角度考虑,建议设置为互锁。

★、14、自动下冷却按键是否有效

用于设置在自动状态下冷却按键是否有效。当本参数设置为 0 时表示无效；设置为 1 时表示有效。

本参数的设置与操作方式有关，当设置为有效时操作者可在自动运行时任意控制冷却的有效状态，满足加工工艺的要求。

★、17、XYZA 轴驱动器报警信号 (X20) [0 表常开, 1 表常闭]

用于设置系统检测 XYZA 伺服报警信号(CN5 的 12 脚 ALM)的类型,0 表常开,1 表常闭。

★、17-1、BCXsYs 轴驱动器报警信号 (X22) [0 表常开, 1 表常闭]

用于设置系统检测 BCXsYs 伺服报警信号 (CN6 的 10 脚 ALMB) 的类型，0 表常开,1 表常闭。

17-2、是否配置一体驱动器(驱动正常时输出 Y16 抱闸信号) [1 表有效, 0 表无效]

设置为 1 表示驱动器正常时输出驱动器 EN 使能 (如果驱动器使用内部使能时可以作抱闸信号 Y16(CN5-23), 0 表示不输出 EN 信号。

17-3、辅助继电器 M824--M831 功能 [0 表示 1--8 轴回零状态, 1 表 1--8 轴运行状态]

设置为 0 时辅助继电器 M824--M831 表示 1-8 轴的回零状态，设置为 1 时表示 1-8 轴的运行状态。

★、18、主轴变频伺服报警 1：ALM1 (X23)

The system is not working properly.

★10. Automatic statistics of the number of processed parts

Used to set the number of processed workpieces automatically by the system. 1 is valid and 0 is invalid.

★11. Increment the program editing serial number

Used to set the program editor to increment the line number by carriage return.

★ 12, system internal parameters [6 for automatic entry, 7 for key entry, 16 for quick key entry, 8 for import permission, 9 for clearing]

The startup display LOGO interface Settings: 6 indicates that the main interface is automatically entered after displaying the LOGO; 7 indicates that the key is pressed; 16 indicates that the motor position key is automatically read and the main interface is quickly entered; 8 indicates that the LOGO interface file can be imported; 9 indicates that the LOGO file is cleared.

★13. Whether the rotation of the spindle and the tightening of the knife are interlocked

Used to set whether the spindle rotation and the clamping state of the tool are interlocked:

0: indicates that the tension state has no connection with the rotation of the spindle;

1: indicates that the spindle is in the M05 state to loosen or tighten the tool;

8: indicates that the spindle is in the M05 state and the spindle encoder cannot rotate to loosen or tighten the tool;

16: Indicates that the rotation speed is interlocked;

32 : Indicates that the clamping knife button is invalid when the program is running automatically and not in a paused state;

56: Indicates full interlock. The program runs automatically and the clasp knife button is invalid when not in pause state.

The setting of this parameter is related to the configuration of machine tools and user usage. From the perspective of safety, it is recommended to set it as interlock.

★14. Whether the automatic cooling button is valid

Used to set whether the cooling button is valid in automatic mode. When this parameter is set to 0, it is invalid; when set to 1, it is valid.

The setting of this parameter is related to the operation mode. When set to effective, the operator can control the effective state of cooling at will during automatic operation to meet the requirements of processing technology.

★17. XYZA axis driver alarm signal (X20) [Table 0: normally open, Table 1: normally closed]

Used to set the type of XYZA servo alarm signal (12-pole ALM of CN5) for system detection. 0 means normally open, and 1 means normally closed.

★17-1, BCXsYs axis driver alarm signal (X22) [0 normally open, 1 normally closed]

Used to set the type of the BCXsYs servo alarm signal (CN6's 10-lead ALMB) for system detection. 0 is normally on and 1 is normally off.

17-2. Whether to configure an integrated drive (which outputs Y16 brake signal when the drive is normal) [Table 1: Valid, Table 0: Invalid]

Setting 1 indicates that the driver outputs the driver EN enable when the driver is normal (if the driver uses internal enable, it can be a brake signal Y16 (CN5-23), and 0 indicates that the EN signal is not output.

17-3, auxiliary relays M824-M831 function [0 indicates the return to zero state of axis 1-8, 1 indicates the running state of axis 1-8]

When set to 0, auxiliary relays M824-M831 indicate the zero return state of axis 1-8, and when set to 1, the running state of axis 1-8 is indicated.

★18. Spindle variable frequency servo alarm 1: ALM1 (X23)

用于设置系统检测机床报警信号（CN3 的 5 脚 ALM1）的类型，0 表常开,1 表常闭。

★、19、机床报警 2：ALM2 (X24)

用于设置系统检测机床报警信号（CN10 的 2 脚 ALM2）的类型，0 表常开,1 表常闭。

★、20、松紧刀控制信号

用于设置系统控制松紧刀是只需一个信号（单向阀）还是需两个信号（双向阀），该参数根机床配置有关。

当设置为 0 时，只是 M10 一个输出信号控制松紧刀松紧，当 M10 有效时，系统执行松紧刀紧动作，M10 无效时系统执行松紧刀松；

当设置为 1 时分别由 M71 和 M10 控制松紧刀的松和紧当继电器 M10 有效M71 无效时系统执行松紧刀紧动作，当 M71 有效、M10 无效时系统执行松紧刀松动作。即 M10 输出 M10，M11 输出 M71 信号。

是否检测松/紧是否到位。轴类参数P40-D1为1时表示要检测；0表示不检测。

若设置为检测，夹紧M10到位后会自动检测M14是否有效，当有效后才执行下一步动作；松开M11到位后会自动检测M12是否有效，当有效后才执行下一步动作。

★、22、外接松紧刀控制

用于设置系统是否外接控制松紧刀松紧的按钮（或脚踏开关），该信号为往复信号，即有效一次松紧刀的控制状态变化一次（有效一次松、再有效一次紧，往复方式）。当设置为 0 时表无外接松紧刀控制；当设置为 1 时表有外接松紧刀控制，该信号为 M16。

★、24、松紧刀 M10/M11 短信号时间

用于设置当松紧刀输出控制信号 M10、M71 为短信号时的保持时间，若该值为 0 则表示为长信号。单位：S（秒）。

★、26、面板急停[0 表常开,1 表常闭]

用于设置系统面板上的急停开关常开 0 或常闭 1，可通过此参数设置，为了安全建议设置为常闭。

★、27、手轮急停[0 表常开,1 表常闭]

用于设置系统 CN11-5 脚上的急停开关（手轮急停或外接急停）常开 0 或常闭 1，可通过此参数设置，为了安全建议设置为常闭。

★、28、运行输出(M69/Y04 运行,M65/Y02 停止)[0 表无效,1 表有效]

用于设置系统是否输出运行信号，即机床处于运行状态时，当该参数设置为 0 时，表无运行输出信号；当该参数设置为 1 时，表有运行输出信号，该信号为 M69，程序停止时输出 M65。

注：该信号输出可用来做机床状态指示，根据机床配置而定。

★、30、系统语言[1 表中文,0 mean English,9 表俄语,8 表其它语言]

用于设置系统界面是采用中文还是英文方式显示。1 时表示用中文方式；0 表示用英文方式，9 表示俄语，8 表示其它语言。

★、31、机床 I/O 逻辑控制器 PLC[1 表启动,0 表停止,32 表诊断时控制输出

Used to set the type of system detection machine tool alarm signal (CN3's 5-lead ALM1), 0 is normally open, 1 is normally closed.

★19. Machine tool alarm 2: ALM2 (X24)

Used to set the type of system detection machine tool alarm signal (CN10's 2 ALM2). 0 is normally on, 1 is normally closed.

★, 20. Looseness knife control signal

Whether a single signal (one-way valve) or two signals (two-way valve) is used to set the system control of the loosening knife depends on the configuration of the machine tool.

When set to 0, only the output signal of M10 controls the tightening and loosening of the knife. When M10 is valid, the system performs the tightening action of the knife; when M10 is invalid, the system performs the loosening of the knife;

When set to 1, the tension control of the clamping knife is managed by M71 and M10. When relay M10 is active and M71 is inactive, the system performs the clamping action. Conversely, when M71 is active and M10 is inactive, the system executes the release action. Specifically, M10 outputs M10 signals while M11 transmits M71 signals.

Whether to detect whether the looseness/tightness is in place. If P 40-D1 of shaft parameters is 1, it means to detect; if 0, it means not to detect.

If set to detect, after clamping M10 in place, it will automatically check whether M14 is valid, and the next action will be executed only when it is valid; after releasing M11 in place, it will automatically check whether M12 is valid, and the next action will be executed only when it is valid.

★22. External tightening knife control

This button (or foot pedal switch) is used to configure whether the system has an external control for the clamping knife. The signal operates in a reciprocating pattern, meaning each valid cycle (one release followed by one engagement) corresponds to a distinct control state change. When set to 0, the device lacks external clamping knife control; when set to 1, it enables external control via the signal M16.

★, 24, M10/M11 short signal time of the tightening knife

Used to set the holding time when the output control signals M10 and M71 of the loosening knife are short signals. If the value is 0, it indicates long signals. Unit: S (second).

★, 26, Panel emergency stop [0 indicates normally open, 1 indicates normally closed]

This parameter is used to set the on/off status of the emergency stop switch on the system panel as 0 or 1. For safety, it is recommended to set it as off.

★, 27, handwheel emergency stop [0 switch normally on, 1 switch normally off]

This parameter is used to set the on/off status of the emergency stop switch (manual or external) on the system CN11-5 foot. It can be set as 0 or 1. For safety, it is recommended to set it as closed.

★, 28, Operation output (M69/Y04 operation, M65/Y02 stop) [0 invalid, 1 valid]

Used to set whether the system outputs running signal, that is, when the machine tool is in running state, when this parameter is set to 0, there is no running output signal; when this parameter is set to 1, there is a running output signal, which is M69, and the program stops output M65.

Note: The signal output can be used to indicate the status of the machine tool, depending on the configuration of the machine tool.

★30, system language [1 indicates Chinese, 0 means English, 9 indicates Russian, 8 indicates other languages]

Used to set whether the system interface is displayed in Chinese or English. 1 indicates Chinese; 0 indicates English; 9 indicates Russian; 8 indicates other languages.

★ 31, machine tool I/O logic controller PLC [1 indicates start, 0 indicates stop, 32 indicates control output during diagnosis]

点与 PLC 同时有效, 64 表诊断时控制输出点有效与 PLC 无效]

用于设置系统内置 PMC 的运行状态, 1 表启动, 0 表停止, 32 表示诊断时与 PLC 同时控制输出点 (用于调试), 64 表示诊断控制输出 PLC 不控制 (用于调试)。

注: 该参数通常做为调试参数, 实际使用必须为启动, 否则机床的动作会异常。

★、32, 机床高速 I/O 逻辑控制器 PLC[0 表停止, 18 表启动高速, 28 表启动超高速]

用于设置系统内置 PLC 的运行时序, 1 表启动高速时序 (扫描周期为 8ms), 0 表停止高速时序 (扫描周期为 100ms), 28 表启动超高速时序 (扫描周期为 2ms)。

注: 该参数通常做为调试参数, 实际使用必须为启动, 否则机床的动作会异常。

★、33, 是否屏蔽外接停止按键[0 表不屏蔽, 64 表屏蔽]

设置为 64 时表示屏蔽外接的停止按键, 防止外部干扰暂停按钮。

★、35, 未回零手动软限位有效

用于设置系统在未回零状态下软限位功能是否有效, 1 表有效, 0 表无效。

注: 该参数的设置与安全操作习惯有关。

★、36, 系统时间[年-月-日-时-分]

用于设置系统日期和时间, 设置完成后系统将以此时间为基准, 根据内部时钟进行计时, 并显示在系统屏幕的右上角。

设置方法如下:

在该参数上, 按确认系统弹出对话框, 分别输入年、月、日、时、分几项进行设置, 每项之间用“-”隔开, 设置完成后按确认键; 如 2008 年 8 月 8 日 20:08, 则在对话框中输入 2008-08-08-20-08 按确认即可。

★、37, 串口通讯速率

用于设置系统串口的通讯速度, 值对应的波特率为: 0=7200; 1=9600; 2=14400; 3=19200; 4=38400; 5=57600; 6=115200

注: 1、该值越大通讯速度越快, 但会越不稳定。

2、通讯机的两端波特率必须设置为一致。

★、38, 手动快速键可锁存[8 表可锁存]

用于手动快速按键的使用方法, 此参数设为 8 时, 手动按“快速”键进行快速或慢速的转换, 快速时屏幕上有提示。

★、39, 40, 特殊参数

内部使用。

★、41, 当前参数备份

把当前参数定义为出厂参数, 用于设备调试完成后备份参数用, 便于以后的设备维护。

在此参数上连续按两次确认键即可完成备份。

★、42, 恢复备份参数

用于设置将备份参数设置为当前参数。当调试过程中出现参数调乱后, 用此参数把出厂参数设置为当前参数。

The control output point is valid when the PLC is active, and the 64-bit diagnostic test is valid for the PLC.

Used to set the running status of the built-in PMC in the system. Table 1 indicates start, Table 0 indicates stop, Table 32 indicates that the output points are controlled simultaneously with the PLC during diagnosis (for debugging), and Table 64 indicates that the PLC does not control the output during diagnosis (for debugging).

Note: This parameter is usually used as a debugging parameter. In actual use, it must be started, otherwise the machine tool will act abnormally.

★ 32, Machine tool high-speed I/O logic controller PLC[0 for stop, 18 for start high speed, 28 for start ultra high speed]

Used to set the operating timing of the built-in PLC. Table 1 starts the high-speed timing (scanning cycle is 8ms), Table 0 stops the high-speed timing (scanning cycle is 100ms), and Table 28 starts the ultra-high-speed timing (scanning cycle is 2ms).

Note: This parameter is usually used as a debugging parameter. In actual use, it must be started, otherwise the machine tool will act abnormally.

★, 33, whether to block the external stop button [0 means not blocked, 64 means blocked]

When set to 64, the stop button for external connection is shielded to prevent external interference from pausing the button.

★35. The manual soft limit is valid if it is not returned to zero

Used to set whether the soft limit function is effective when the system is not returned to zero. 1 indicates valid, 0 indicates invalid.

Note: The setting of this parameter is related to safe operation habits.

★, 36, system time [year-month-day-hour-minute]

Used to set the system date and time. After setting, the system will use this time as the reference, time according to the internal clock, and display in the upper right corner of the system screen.

Set it up as follows:

On this parameter, click the confirmation system dialog box to set the year, month, day, hour and minute respectively. Each item is separated by "-" and press the confirmation button after setting. For example, August 8, 200820:

08 Then enter 2008-08-08-20-08 in the dialog box and click OK.

★, 37, serial communication rate

Used to set the communication speed of the system serial port. The corresponding port rate is: 0=7200; 1=9600; 2=14400; 3=19200; 4=38400; 5=57600; 6=115200

Note: 1. The larger the value, the faster the communication speed, but it will be more unstable.

2. The baud rate at both ends of the communication machine must be set to be consistent.

★, 38, manual quick key can be locked [8 table can be locked]

Use the manual fast key. When this parameter is set to 8, press the "Fast" key manually to switch between fast and slow. A prompt is displayed on the screen when fast is selected.

★, 39,40, special parameters

For internal use only.

★ 41. Backup of current parameters

The current parameters are defined as factory parameters, and the backup parameters are used after equipment debugging is completed to facilitate future equipment maintenance.

Press the confirm button twice in succession on this parameter to complete the backup.

★, 42, Restore backup parameters

Used to set the backup parameters to the current parameters. When the parameters are out of order during debugging, use this parameter to set the factory parameters to the current parameters.

注意：执行该参数操作后，之前的参数将被覆盖。

★、50,从中间行启动是否先定点到上一段终点[8 表是,0 表否]

★、51, 从中间行启动回上一段终点时是否先回安全点(+4 表 Z, +8 表 X, +16 表进给保持, +128 表进给保持手轮运行时坐标不变, +256 表进给保持手动运行后不返回干预点)

设置从中间行启动回上一行坐标时是否先回安全点坐标, P52 设置回安全点的机床坐标。

+64 表进给保持时设置坐标偏移后对当前行立即生效;

+128 表进给保持时进入手轮模式,手轮运行坐标保持不变;

+256 表进给保持进行手动操作后,不自动返回原点,用于手动干预操作;

例如:

G00 Z0; 到工件附近

M00; 暂停后, 手动到工件表面, 下一行用 G50 设置起点坐标

G50 Z0; 切换到自动后按启动键, 不返回干预点直接执行此行

G0 Z50

★、120, 手动坐标按键反向(4 表立车 X/卧车 Z;8 表 Y/C;16 表立车 Z/卧车 X;32 表第 4 轴)

设置手动运行坐标时按键是否反向。

★、140, 当前用户坐标与参考点 1 比较(1001/2001/3001/4001 表 X/Y/Z/A, 小于参考点 M77 有效)

设置当前机床坐标与参考点 1 比较, 于小参考点时 M77 表效, 如: 1001 表示当 X 轴机床坐标小于参考点 1 时 M77 有效。此功能设置后 PLC 中的 M77 行要删除。P150-P153 为 X/Y/Z/A 的参考点 1 坐标。

★、141, 当前用户坐标与参考点 2 比较(1002/2002/3002/4002 表 X/Y/Z/A, 小于参考点 M78 有效)

设置当前用户坐标与参考点 2 比较, 于小参考点时 M77 表效, 如: 2002 表示当 Y 轴用户坐标小于参考点 2 时 M78 有效。此功能设置后 PLC 中的 M78 行要删除。P160-P163 为 X/Y/Z/A 的参考点 2 坐标。

★、200, X 轴反馈运行报警误差(pulse) [>1]

当系统配有坐标轴位置反馈功能时, 坐标轴运行中如果跟随误差大于此参数时将报警, 此参数大于 1 时有效。P200-P204 表示 X/Y/Z/A/B/C/Xs/Ys 轴。报警后可以在诊断界面按“G”清零指令位置和反馈位置并清除超差报警。

★、205, X 轴反馈停止报警误差(pulse) [>1]

当系统配有坐标轴位置反馈功能时, 坐标轴停止时如果跟随误差大于此参数时将报警, 此参数大于 1 时有效; P205-P209 表示 X/Y/Z/A/B/C/Xs/Ys 轴。报警后可以在诊断界面按“G”清零指令位置和反馈位置并清除超差报警。

★、210, X 轴反馈电子齿轮分子[自动计算输入: L 丝杠导程(um)M 编码器线数]

当系统配有坐标轴位置反馈功能时, 此参数为电子齿轮分子, 也可以输入丝

Note: After executing this parameter operation, previous parameters will be overwritten.

★50, whether to start from the middle line and first set the endpoint of the previous segment [8 indicates yes, 0 indicates no]

★51. When starting from the middle line to return to the end point of the previous section, does it first return to the safety point (+4 indicates Z, +8 indicates X, +16 indicates feed hold, +128 indicates coordinates remain unchanged during feed hold handwheel operation, +256 indicates no return to the intervention point after manual feed hold operation)

Set whether to return to the safety point coordinate first when starting back to the previous line coordinate from the middle line. P52 Set the machine tool coordinates to return to the safety point.

+64 When the feed is maintained, the coordinate offset is set and takes effect immediately on the current line; +128 When the feed is maintained, the handwheel mode is entered, and the running coordinate of the handwheel remains unchanged;

+256 The feed hold is maintained after manual operation, and does not automatically return to the origin. It is used for manual intervention. For example:....

G00 Z0; to the vicinity of the workpiece

M00; After pause, manually move to the surface of the workpiece, and set the starting point coordinates for the next line with G50

G50 Z0; After switching to automatic, press the start key and execute this line directly without returning to the intervention point

G0 Z50

★120, manual coordinate key reverse (4 for vertical lathe X/Z; 8 for Y/C; 16 for vertical lathe Z/X; 32 for the fourth axis)

Set whether the key is reversed when the coordinates are run manually.

★ 140, comparison between the current user coordinate and reference point 1 (1001/2001/3001/4001 in table X/Y/Z/A, valid if less than reference point M77)

Set the current machine tool coordinates and reference point 1. When comparing with the smaller reference point, M77 becomes active. For example, 1001 indicates that M77 is effective when the X-axis coordinate of the machine tool is less than reference point 1. After configuring this function, the M77 line in the PLC must be deleted. P 150-P153 represent the coordinates of reference point 1 for X/Y/Z/A.

★ 141. Comparison between the current user coordinate and reference point 2 (1002/2002/3002/4002 in Table X/Y/Z/A, valid if less than reference point M78)

Set the comparison between the current user coordinates and reference point 2. When the user coordinates are smaller than reference point 2, M77 becomes active. For example, 2002 indicates that M78 takes effect when the Y-axis user coordinates are less than reference point 2. After configuring this function, the M78 line in the PLC must be deleted. P160-P163 represent the coordinates of reference point 2 for X/Y/Z/A.

★, 200, X-axis feedback operation alarm error (pulse) [>]

When the system is equipped with coordinate axis position feedback functionality, an alarm will be triggered if the tracking error during axis operation exceeds this parameter. This parameter becomes effective when its value exceeds 1. The P200-P204 parameters represent the X/Y/Z/A/B/C/Xs/Ys axes. After triggering the alarm, users can execute the "G" reset command in the diagnostic interface to clear both the position and feedback positions while eliminating the out-of-tolerance alarm.

★, 205, X-axis feedback stop alarm error (pulse) [>]

When the system is equipped with coordinate axis position feedback function, an alarm will be triggered when the tracking error exceeds this parameter during axis stoppage. This parameter becomes effective only when its value is greater than 1. P205-P209 indicates the X/Y/Z/A/B/C/Xs/Ys axes. After triggering the alarm, the diagnostic interface can execute the "G" reset command to clear both the position and feedback positions while eliminating the out-of-tolerance alarm.

★210, X-axis feedback electronic gear molecule [Automatic calculation input: L lead screw pitch (um) M encoder line count]

When the system is equipped with coordinate axis position feedback function, this parameter is the electronic gear molecule, or you can also input silk

杠导程和编码器线数，如丝杠为 6mm，编码器为 2500 线时输入为：L6000M2500。
P210-P214 表示 X/Y/Z/A/B/C/Xs/Ys 轴。

特别注意：

- 1)、按 LxxxxMxxxx 方式时需要先设置系统中各个轴的电子齿轮分子和分母；
- 2)、当需要通过串口读取绝对编码器数据时必须按该方式设置。

★、215, X 轴反馈电子齿轮分母[自动计算输入：L 丝杠导程(um)M 编码器线数]

当系统配有坐标轴位置反馈功能时，此参数为电子齿轮分母，也可以输入丝杠导程和编码器线数，如丝杠为 6mm，编码器为 2500 线时输入为：L6000M2500。
P215-P219 表示 X/Y/Z/A/B/C/Xs/Ys 轴。

★、299, 设置电子齿轮时是否自动设置绝对值电机每圈对应坐标值的正负号
[0 表否, 1 表是]

1 表示设置电子齿轮时自动设置绝对值电机每圈对应坐标值的正负号。

★、300, 进给轴是否配绝对式编码器电机[X-D2, Y-D3, Z-D4, A-D5, B-D6, C-D7, 安川-D12, 0 表否, 1 表是, D14=1 偶校验, D15=1 无校验]

当系统配有配绝对式电机功能时，此参数为坐标轴选择是否配绝对式编码器电机。P301 为所配电机绝对编码器多圈低 16 位存储器地址（如：设为 92），P302 为绝对编码器单圈高 16 位存储器地址（如：设为 91），P303 为绝对编码器单圈低 16 位存储器地址（如：设为 90）。

特别注意：

系统通过 CN13 接口 RS485 通讯连接到驱动器，通讯协议如下：

- 1) RS485 通讯的驱动器站地址为：1-8 对应 X/Y/Z/A/B/C/Xs/Ys 轴。
- 2) RS485 通讯的波特率为 19200；数据位为 8 位；停止位为 1 位；格式为 RTU 格式；D14=0D15=0 奇校验，D14=1D15=0 偶校验，D14=0D15=1 无校验。

当系统配有配绝对式电机功能时，此参数为坐标轴选择是否配绝对式编码器电机。P301 为所配电机绝对编码器多圈低 16 位存储器地址（如：设为 92），P302 为绝对编码器单圈高 16 位存储器地址（如：设为 91），P303 为绝对编码器单圈低 16 位存储器地址（如：设为 90）。P303-1 设置为 1 时表示特殊单圈计算方法：单圈值=高 16 位*10000+低 16 位。

特殊驱动器地址设置方法如下：

- (1) P303=-999999(单圈 32 位/多圈 16 位), P301 多圈, P302 单圈;
- (2) P303=-999977(单圈 32 位/多圈 32 位), P301 多圈, P302 单圈;
- (3) P302=-999966(单圈 16 位/多圈 16 位), P301 多圈, P303 单圈;
- (4) P301 负值(多圈 32 位), 如: -5632 表多圈从 5632 读取 32 位数据;
- (5) P303=-999888, P302=-999888(有符号 32 位), P301 地址;
- (6) P303=-999222(有符号 64 位), P302 低 32 位, P301 高 32 位。

★、304, X 轴绝对编码器每圈脉冲数

当系统配有配绝对式电机功能时，此参数为 X 轴绝对编码器每圈脉冲数，17

The lead guide and encoder line numbers are as follows: For example, when the lead screw is 6mm and the encoder has 2500 lines, the input is: L6000M2500. P210-P214 indicates the X/Y/Z/A/B/C/Xs/Ys axes.

Special note:

1) When using the LxxxxMxxxx method, it is necessary to set the electronic gear numerator and denominator of each axis in the system first;

2) When the absolute encoder data needs to be read through the serial port, it must be set in this way.

★215, X-axis feedback electronic gear denominator [automatically calculated input: L lead screw thread (um) M encoder line count]

When the system is equipped with coordinate axis position feedback function, this parameter serves as the electronic gear denominator. You can also input the lead of the screw and encoder line count. For example, if the screw is 6mm and the encoder has 2500 lines, the input should be: L6000M2500. P215-P219 indicates the X/Y/Z/A/B/C/Xs/Ys axes.

★ 299. When setting the electronic gear, whether to automatically set the positive/negative sign of the absolute value of the coordinate value per revolution of the motor [0 indicates no, 1 indicates yes]

1 When setting the electronic gear, it automatically sets the positive and negative sign of the absolute value of the coordinate value per revolution of the motor.

★ 300, whether the feed axis is equipped with absolute encoder motor [X-D2, Y-D3, Z-D4, A-D5, B-D6, C-D7, Yaskawa-D12, 0 indicates no, 1 indicates yes, D14=1 for even check, D15=1 for no check]

When the system is equipped with absolute motor functionality, this parameter determines whether the coordinate axis motor should be configured with an absolute encoder. P301 represents the multi-turn low 16-bit memory address for the motor's absolute encoder (e.g., set to 92), while P302 denotes the single-turn high 16-bit memory address (e.g., set to 91). P303 indicates the single-turn configuration of the absolute encoder

Low 16-bit memory address (e.g., set to 90).

Special note:

The system is connected to the driver through the CN13 interface RS 485 communication, and the communication protocol is as follows:

1) The address of the RS485 communication driver station is: 1-8 corresponding to X/Y/Z/A/B/C/Xs/Ys axes.

2) The RS485 Communication Has a Baud Rate of 19200; 8 Bits for Data; 1 Bit for Stop; and a Format of RTU

Format; D14=0D15=0 odd check, D14=1D15=0 even check, D14=0D15=1 no check.

When the system is equipped with absolute motor functionality, this parameter determines whether to install an absolute encoder for the coordinate axis motor. P301 represents the multi-bit (low 16) memory address of the absolute encoder for the motor (e.g., set to 92), P302 denotes the single-bit (high 16) memory address (e.g., set to 91), and P 303 indicates the single-bit (low 16) memory address (e.g., set to 90). When P303-1 is set to 1, the special single-turn calculation method applies: Single-turn value = high 16 bits * 10000 + low 16 bits.

The special driver address is set as follows:

(1) P303= -999999 (32 bits for a single circle / 16 bits for multiple circles), P301 for multiple circles, P302 for a single circle;

(2) P303= -999977 (32 bits for a single circle/multi circle 32 bits), P301 multi circle, P302 single circle;

(3) P302= -999966 (16 bits per circle/multi circle 16 bits), P301 multi circle, P303 single circle;

(4) P301 negative value (multi-loop 32 bits), such as: -5632. The multi-loop meter reads 32 bits of data from 5632;

(5) P303= -999888, P302= -999888 (signed 32-bit), and P301 address;

(6) P303= -999222 (symbolic 64-bit), P302 low 32 bits, P301 high 32 bits.

★304, the number of pulses per revolution of the X-axis absolute encoder

When the system is equipped with absolute motor function, this parameter is the number of pulses per revolution of the X-axis absolute encoder, 17

位编码器驱动器此值设为 131072。P304-P308 表示 X/Y/Z/A/B/C/Xs/Ys 轴。

★、309, X 轴绝对编码器每圈对应坐标距离值 (nm)

当系统配有配绝对式电机功能时, 此参数 X 轴绝对编码器每圈对应坐标距离值 (nm), 单位为纳米, 设为负数时表示坐标计数方向取反。如: X 轴 4mm 丝杠直联时此值设为: 4000000 或 -4000000。P309-P313 表示 X/Y/Z/A/B/C/Xs/Ys 轴。

★、314, X 轴绝对编码器多圈偏移[输入 E 清零多圈]

当系统配有配绝对式电机功能时, 此参数设置 X 轴绝对式多圈偏移量, 输入 E 绝对式多圈数据清零, 用于解决多圈值太大导致机床坐标太大或溢出等问题。P314-P318 表示 X/Y/Z/A/B/C/Xs/Ys 轴。

★、319, 485 绝对值/Ethercat 总线驱动器反馈显示 16 位地址[231/232 负载 0.1%/282 转速 P//283 转速 R//284 电流 A//435 转矩 0.1% (NM)]

设置系统显示驱动器反馈状态:

231/232: 显示电机负载, 单位 0.1%;

282: 显示电机转速, 单位 P/s (Ethercat);

283: 显示电机转速, 单位 r/min; (Modbus485)

284: 显示电机电流, 单位 A (Modbus485);

435: 显示电机转矩, 单位 0.1% (Ethercat)/NM (Modbus485);

★、320, X 轴绝对编码器每圈对应坐标距离值 (nm)

当系统配有配绝对式电机功能时, 此参数 X 轴绝对编码器每圈对应坐标距离值分母, 即 X 轴绝对编码器每圈对应坐标距离值 = P309/P320。如: X 轴 4mm 丝杠减速比为 3/1 时 P309=4000000, P320=3。P320-P324 表示 X/Y/Z/A/B/C/Xs/Ys 轴。

系统的网络功能:

★、350, 网络[0 表关, 1 表开, 8 表自动, 12 表 OPC<TCP: 系统为 Server 服务器>, 40 表 OPC<UDP>, +512 表冲压工艺采用网络通讯, +1024 表冲压工艺采用 CAN 通讯]

设置是否开启网络功能, 设置为 0 表示关闭网络功能, 设置为 1 表示开启网络功能, 设置为 8 表示开机自动启动网络, 40 表示开启 OPC 功能。通过 P350-P354 设置网络功能。P350a 选择局域网通讯 OPC 功能网口, 0: 选网口 1, 1: 选网口 2

★、350a, Ethercat 驱动器网口选择[0: 选网口 1, 1: 选网口 2]

设置为 0 表示 Ethercat 驱动器通讯选择网口 1, 设置为 1 表示 Ethercat 驱动器通讯选择网口 2。

★、350b, 通讯 OPC 功能网口选择[0: 选网口 1, 1: 选网口 2]

设置为 0 表示系统 OPC 功能通讯选择网口 1, 设置为 1 表示系统 OPC 功能通讯选择网口 2。

★、351, 网关 IP 地址

任意设置, 最好局域网内前三个数一样, 如 192.168.1.1。

★、352, 本机 IP 地址

本机器人系统的 IP 地址, 最好局域网内前三个数一样, 如 192.168.1.103

★、353, 本机 MAC 地址

The encoder driver value is set to 131072. P304-P308 indicates the X/Y/Z/A/B/C/Xs/Ys axes.

★309, X-axis absolute encoder corresponding coordinate distance value per circle (nm)

When the system is equipped with absolute motor functionality, the parameter X-axis absolute encoder corresponds to a coordinate distance value (nm) per revolution. The unit is nanometers, and setting it to a negative value indicates an inverted direction for coordinate counting. For example, when connecting a 4mm screw rod directly, this value should be set to either 4,000,000 or -4,000,000. P309-P313 denotes the X/Y/Z/A/B/C/Xs/Ys axes.

★, 314, X-axis absolute encoder multi-turn offset [Input E reset multi-turn]

When the system is equipped with absolute motor function, this parameter sets the X-axis absolute multi-turn offset. Enter the E-axis absolute multi-turn data to reset, which helps resolve issues like excessive multi-turn values causing machine tool coordinates to become too large or overflow. P314-P318 indicates the X/Y/Z/A/B/C/Xs/Ys axes.

★ 319,485 absolute value / Ethercat bus driver feedback shows 16-bit address [231/232 load 0.1%/282 speed P/283 speed R/284 current A/435 torque 0.1% (NM)]

Set the system to display the drive feedback status:

231/232: Display motor load, unit 0.1%;

282: Display the motor speed, unit P/s (Ethercat);

283: Display motor speed, unit r/min; (Modbus485)

284: Display motor current, unit A (Modbus485);

435 Display motor torque, unit: 0.1% (Ethercat)/NM (Modbus485); ★, 320, X-axis absolute encoder per revolution corresponding to coordinate distance value (nm)

When the system is equipped with absolute motor functionality, parameter X-axis absolute encoder corresponds to the denominator of each revolution's coordinate distance value. Specifically, the ratio between each revolution's coordinate distance value and the absolute encoder is calculated as P309/P320. For example: When the screw reducer ratio for the X-axis is 3/1 (with a 4mm feed), P309 equals 4,000,000 and P320 equals 3. The range from P320 to P324 represents the X/Y/Z/A/B/C/Xs/Ys axis configuration.

System network functions:

★350, network [0 for off, 1 for on, 8 for automatic, 12 for OPC <TCP: system as Server>, 40 for OPC <UDP>, +512 for stamping process using network communication, +1024 for stamping process using CAN communication]

Set the network function to either 0 (disabled) or 1 (enabled). Setting 8 enables automatic network startup upon device activation, while 40 activates OPC functionality. Network configuration is performed via P350-P354 terminals. For P350a models, select the LAN communication OPC port: 0 for Port 1 and 1 for Port 2

★350a, Ethercat, driver network port selection [0: Select Network Port 1, 1: Select Network Port 2]

Setting to 0 indicates that the Ethercat driver communication selects network port 1, and setting to 1 indicates that the Ethercat driver communication selects network port 2.

★350b, communication OPC function network port selection [0: select network port 1, 1: select network port 2]

Setting to 0 indicates that the system OPC function communication selects network port 1, and setting to 1 indicates that the system OPC function communication selects network port 2.

★, 351, gateway IP address

Set it arbitrarily, preferably the first three numbers in the LAN are the same, such as 192.168.1.1.

★, 352, IP address of this machine

The IP address of the robot system is preferably the first three numbers in the LAN, such as 192.168.1.103 ★, 353, and the MAC address of the machine

本机器人系统的物理地址，任意设置，确保网内每一台不一样，如 255:255:255:1

★、354, 网络服务器 IP 地址

服务器 IP 地址，就是电脑的 IP 地址，最好前三个数一样，如 192.168.1.13

★、P361-P366 设置 WIFI 功能，通过系统 USB 实现网络 WIFI 功能：USB 网卡型号采用 EP-N8508GS 和 BL-LW05-AR5 两种网卡均可，设置方法：

- 1、在安卓手机上安装 ES 浏览器；
- 2、打开安卓手机的 ES 浏览器的网络-->远程管理器-->设置；
- 3、设置端口号设为 3721 或 2121 或 2221 三种均可；
- 4、设置根目录，选择手机的内部存储；
- 5、设置管理账号，采用匿名；
- 6、设置编码为 GBK 模式(否则有中文时会乱码)；
- 7、打开远程管理器(上面显示的 IP 地址输入到参数 P361)。

如果现场没有 WIFI 信号，可以手机热点信号来构成局域网，设置方法：

- 1、打开安卓手机 ES 浏览器的网络-->网络管理器-->创建网络热点；
- 2、使用自定义热点，设置好热点用户名和热点密码；
- 3、将该热点用户名和密码输入到参数 P362 和 P363；
- 4、退出远程管理器，打开远程管理器(上面显示的 IP 地址输入到参数 P561)。

在系统客户端进入 FTP 服务器文件夹的操作方法：

1、在程序目录界面下按“N”键即可进入。(如果为复合键则按“Shift”键后再按“N”键或按住快速叠加键后再按“N”键)。

2、进入 FTP 服务器文件夹后，按上面同样的操作按“N”键(或按 F6 键)即可退出 FTP 服务器文件夹。

使用 FTP 传输文件的优点是：在系统上可以看到 PC 机或者手机上文件夹中的内容，并且可以自由选择想要传输的文件，使用起来更加方便。可以采用匿名登陆 FTP 服务器不设置参数 P365, P366),也可采用 FTP 的用户名和密码，FTP 的用户名和密码采用原来有线网络的 FTP 用户名和密码。

当系统使用 WiFi 功能时只能与安卓手机进行程序文件上传和下传；当系统使用网络上网时只能与电脑进行程序文件上传和下传。

当系统同时连接上无线 WIFI 和有线以太网时，打开 FTP 服务器时，无线 WIFI 优先，系统中的参数如下：

★、361, WiFi 热点的 FTP 服务器 IP 地址[根据手机中 FTP 服务器的值输入，如:192.168.2.206]

★、362, WiFi 热点用户名

★、363, WiFi 热点密码

The physical address of this machine person system can be set arbitrarily to ensure that each network is not the same, such as 255:255:255:1

★, 354, IP address of the web server

Server IP address, which is the IP address of the computer, preferably the first three numbers are the same, such as 192.168.1.13

★P361-P366 Set the WIFI function and realize the network WIFI function through the system USB: USB, the network card model can be either EP-N8508GS or BL-LW05-AR5. The setting method is as follows:

1. Install ES browser on Android phone;
2. Open the network of the ES browser on the Android phone--> Remote Manager--> Settings;
3. Set the port number to 3721,2121 or 2221;
4. Set the root directory and select the internal storage of the phone;
5. Set up a management account and use anonymity;
6. Set the encoding mode to GKB (otherwise there will be garbled Chinese);
7. Open the remote manager (enter the IP address shown above into parameter P361).

If there is no WIFI signal on the site, you can use the mobile phone hotspot signal to form a local area network. The setting method is:

1. Open the network of ES browser on Android phone--> Network Manager--> Create a network hotspot;
2. Use the custom hotspot, set the hotspot user name and password;
3. Enter the hot spot user name and password into parameter P362 and P363;

4. Exit the Remote Manager and open the Remote Manager (enter the IP address shown above into parameter P561). The method to enter the FTP server folder from the system client is as follows:

1. Press the "N" key in the program directory interface to enter. (If it is a composite key, press "Shift" and then "N", or press "N" while holding down the fast overlay key).
2. After entering the FTP server folder, press the "N" key (or press F6) to exit the FTP server folder in the same way.

The advantages of using FTP for file transfer include: The system allows users to view folder contents on PCs or mobile devices, enabling flexible selection of files for transfer and enhanced operational convenience. Users can either log into the FTP server anonymously (without setting parameters P365 or P366) or use existing wired network credentials for both FTP and TP accounts.

When the system uses WiFi function, it can only upload and download program files with Android phones; when the system uses Internet access, it can only upload and download program files with computers.

When the system is connected to both wireless WIFI and wired Ethernet, when the FTP server is opened, the wireless WIFI takes priority. The parameters in the system are as follows:

★361, WiFi hotspot FTP, server IP address [Enter the value of the FTP server in the phone, such as: 192.168.2.206]

★, 362, WiFi hotspot user name

★, 363, WiFi hotspot password

★、364, WiFi 热点[8 表开启, 其余表关]

★、365, FTP 用户名

★、366, FTP 密码

P380-P391 为自动对刀专用参数, 详见相关功能说明。

★、380, 自动对刀初始定位点的 X 轴机床坐标值 (mm)

★、381, 自动对刀初始定位点的 Y 轴机床坐标值 (mm)

★、382, 自动对刀初始定位点及返回点的 Z 轴机床坐标值 (mm)

★、383, 自动对刀快速负向移动的速度 (mm/min)

★、384, 自动对刀慢速正向移动的速度 (mm/min)

★、385, 自动对刀后当前工件坐标系下工件上表面的 Z 轴坐标值 (mm)

★、386, 自动对刀快速定位点的速度 (mm/min)

★、387, 自动对刀模式 (1 表固定对刀仪, 0 表浮动对刀仪)

★、388, Z 轴最低机床坐标值 (mm)

★、389, Z 轴落差值[即对刀仪上表面相对于工件上表面的高度] (mm)

★、390, 自动对刀总的检测次数 (1—5)

★、391, RTCP 刀具旋转中心到刀柄的距离 (mm)

★、400, 坐标轴名字显示配置 [xxxxx2] (123456 分别表示 XYZABC, 如 123462 显示 XYZAC 轴, 124502 显示 XYAB 轴)

设置系统坐标名字显示顺序。

★、401, 进给轴脉冲端口配置 [xxxxxxxx0] (12345678 分别表示 XYZABC78, 如 123478560 第 7/8/5/6 轴脉冲从 B/C/XS/YS 口输出)

设置系统脉冲输出口顺序。

★、501, 界面显示颜色反转 [1 表正常 (白底), 8 表反转 (黑底), 128 表显示繁体字]

设置系统界面显示颜色与字体。

★、591, 一键定义步进

把当前参数设置为出厂默认步进电机参数, 用于机床安装的是步进装置。
该参数主要是对电机特性对部分速度参数进行了调整。

该操作通常在刚开始调试机床时。

★、592, 一键定义伺服

把当前参数设置为出厂默认伺服电机参数, 用于机床安装的是伺服装置。
该参数主要是对电机特性对部分速度参数进行了调整。

该操作通常在刚开始调试机床时。

★、600, 控制方式 (4000-4099 表 etherCAT, 6000 表 Modbus, 6001 表 1-6 轴 Modbus, 6002 表 3 轴 Modbus), 设为 4003;

★、601, etherCAT 位置环增益系数, 设为: 40;

★、602, etherCAT 螺纹加工位置环增益系数, 设为: 40;

★、603, etherCAT 定位到位允差值 (电机编码器脉冲数), 设为: 10005;

★, 364, WiFi hotspot [8 tables on, the rest off]

★, 365, FTP user name

★, 366, FTP password

P380-P391 are special parameters for automatic tool alignment. See the relevant function description for details.

★, 380, automatic tool initial positioning point X axis machine coordinate value (mm)

★381, machine tool coordinate value (mm) of the initial Y-axis tool positioning point

★, 382, Z axis machine tool coordinate values (mm) of the initial positioning point and return point of automatic tooling

★, 383, speed of automatic tool cutting and rapid negative movement (mm/min)

★, 384, speed of automatic tool slow forward movement (mm/min)

★★ 385, Z-axis coordinate value (mm) on the upper surface of the workpiece in the current workpiece coordinate system after automatic tool alignment ★ 386, speed of the quick positioning point for automatic tool alignment (mm/min)

★, 387, automatic tooling mode (1 indicates fixed tooling instrument, 0 indicates floating tooling instrument)

★, 388, Z-axis minimum machine tool coordinate value (mm)

★389, Z-axis drop value [i.e., the height of the upper surface of the knife gauge relative to the upper surface of the workpiece] (mm) ★ 390, total automatic tooling detection count (1-5)

★391, RTCP distance between tool rotation center and handle (mm)

★400, coordinate axis name display configuration [xxxxx2] (123456 respectively represent XYZABC, such as 123462 shows XYZAC axis, 124502 shows XYAB axis)

Set the order in which system coordinate names are displayed.

★ 401, feed axis pulse port configuration [xxxxxxxx0] (12345678 respectively represent XYZABC78, such as 123478560 where the 7th/8th/5th/6th axis pulses are output from B/C/XS/YS ports)

Set the sequence of system pulse output ports.

★, 501, the interface display color reversal [Table 1 normal (white background), Table 8 reversed (black background), Table 128 display traditional characters]

Set the color and font of the system interface display.

★, 591, one-click step definition

Set the current parameter as the factory default stepper motor parameter, which is used for the stepper device installed in the machine tool. This parameter mainly adjusts some speed parameters according to the characteristics of the motor.

This operation is usually done at the beginning of machine tool debugging.

★, 592, one-click definition of servo

Set the current parameters to the factory default servo motor parameters, which are used for machine tool installation. This parameter is mainly used to adjust some speed parameters of motor characteristics.

This operation is usually done at the beginning of machine tool debugging.

★600, control mode (4000-4099 for etherCAT, 6000 for Modbus, 6001 for 1-6 axis Modbus, 6002 for 3-axis Modbus), set to 4003;

★, 601, etherCAT position loop gain coefficient, set as: 40;

★602, etherCAT thread processing position ring gain coefficient is set as: 40;

★603, etherCAT positioning tolerance value (motor encoder pulse number) is set as: 10005;

对于采用周期性位置控制方式(0x6060 等于 8)设置值应大于 10000 才行(此时, 系统的位置环增益系数不再起任何作用);

对于采用 PID 位置环控制(0x6060 等于 9)设置值不需要加上任何数值, 设置值小于等于 800 即可(此时, 系统的位置环增益系数会起作用)。

该参数可以实时在线修改, 对于迈信伺服, 会将 0x6060 的值显示在数码管上, 如 OP8rn 或 OP9rn 。

etherCAT 总线, 对于将位置环 PID 控制设在系统上时要修改如下两个参数:

P601, etherCAT 位置环增益系数;

P602, etherCAT 螺纹加工位置环增益系数;

注意: 建议采用周期性位置控制方式(0x6060 等于 8, P603 设置值应大于 10000)更好。

P603=1-9999 表示周期性速度控制, 无 0x6081/6083/6084PDO 映射;

P603=10001-12000 表示周期性位置控制, 无 0x6081/6083/6084PDO 映射;

P603=20001-29999 表示周期性速度控制, 有 0x6081/6083/6084PDO 映射;

P603=30001-32000 表示周期性位置控制, 有 0x6081/6083/6084PDO 映射;

注意: 0x6081 是前馈速度, 0x6083 是加速度, 0x6084 是减速度, 系统最好都设成不映射, 采用周期速度控制时, 驱动器升降速必须最快, 否则会振荡;

《1》Ethercat 光栅尺全闭环控制方法:

杂类参数设置:

P601=40; etherCAT 位置环增益系数

P602=40; etherCAT 螺纹加工位置环增益系数

P603=5; etherCAT 闭环速度控制, 允差值 5

P631=1310720; X 轴开启闭环光栅尺功能, 报警误差值 1310720(电机 10 圈)

P641=131072; X 轴闭环光栅尺反馈脉冲电子齿轮分子[光栅尺反馈脉冲数 × 分子 / 分母 = 电机编码器反馈脉冲数, 负值表示反馈方向取反, P641-P658 相同]

P651=10000; X 轴闭环光栅尺反馈脉冲电子齿轮分母

举例:

电机编码器 1 圈 131072, 丝杆 10mm, 光栅尺精度 1μ, 减速器 1:1

分子/分母 = 电机编码器反馈脉冲数 × 光栅尺精度 / [减速比 × 丝杆距离]

= 131072 × 0.001 / 10 = 131072 / 10000

光栅尺 AB 指令 = 光栅尺反馈脉冲数 × 光栅尺反馈脉冲电子齿轮分子 / 光栅尺反馈脉冲电子齿轮分母

《2》Ethercat 光栅尺 AB 信号补偿功能

杂类参数设置:

For the periodic position control mode (0x6060 equals 8), the set value should be greater than 10000 (at this time, the system's position ring gain coefficient no longer plays any role);

For PID position loop control (0x6060 equals 9), no value needs to be added to the set value. The set value should be less than or equal to 800 (in this case, the system's position loop gain coefficient will take effect).

This parameter can be modified online in real time. For Maixin servo, the value of 0x6060 will be displayed on the digital tube, such as OP8rn or OP9rn.

EtherCAT bus. When the position loop PID control is set on the system, the following two parameters should be modified:

P601, etherCAT position loop gain coefficient;

P602, etherCAT thread processing position ring gain coefficient;

Note: It is recommended to use periodic position control (0x6060 equals 8, and P603 should be set to greater than 10000) for better results.

P603=1-9999, indicating periodic speed control, no 0x6081/6083/6084PDO mapping;

P603=10001-12000 indicates periodic position control, no 0x6081/6083/6084PDO mapping;

P603=20001-29999 indicates periodic speed control, with 0x6081/6083/6084PDO mapping;

P603=30001-32000 indicates periodic position control, with 0x6081/6083/6084PDO mapping;

Note: 0x6081 is the feedforward speed, 0x6083 is the acceleration, and 0x6084 is the deceleration. It is best to set all of them without mapping. When using periodic speed control, the driver must have the fastest speed increase and decrease, otherwise it will oscillate;

1. Full closed-loop control method of Ethercat grating ruler:

Miscellaneous parameter Settings:

P601=40; etherCAT position loop gain coefficient

P602=40; etherCAT thread processing position ring gain coefficient

P603=5; etherCAT closed-loop speed control, tolerance value 5

P631=1310720; The X-axis closed-loop optical scale function is activated, with an alarm error value of 1310720 (equivalent to 10 full rotations of the motor). P641=131072; The feedback pulse from the X-axis closed-loop optical scale electronic gear molecule [Calculation formula: Feedback pulse count of optical scale × Molecular count / Molecular count = Feedback pulse count of motor encoder. Negative values indicate reversed feedback direction. P641-P658 are identical].

P651=10000; X-axis closed-loop grating scale feedback pulse electronic gear denominator

give an example :

Motor encoder 1 circle 131072, screw rod 10mm, grating ruler precision 1u, reducer 1:1

Molecular/fraction = motor encoder feedback pulse number * grating ruler precision / [reduction ratio * screw rod distance] = 131072*0.001/10=131072/10000

Ruler AB instruction = Ruler feedback pulse number * Ruler feedback pulse electron gear numerator / Ruler feedback pulse electron gear denominator

2. Ethercat grating scale AB signal compensation function

Miscellaneous parameter Settings:

P601=40; etherCAT 位置环增益系数

P602=40; etherCAT 螺纹加工位置环增益系数

P603=10005; etherCAT 闭环位置控制, 允差值 5

P631=-888888; X 轴开启光栅尺 AB 信号补偿功能

P641=131072; X 轴闭环光栅尺反馈脉冲电子齿轮分子[光栅尺反馈脉冲数 x 分子 / 分母=电机编码器反馈脉冲数, 负值表示反馈方向取反, P641-P658 相同]

P651=10000; X 轴闭环光栅尺反馈脉冲电子齿轮分母

举例: 电机编码器一圈 131072, 如 AB 反馈 10000 个脉冲;

电机编码器反馈脉冲数 [补偿量] = 光栅尺反馈脉冲数 x 分子 / 分母
= 10000 * 131072 / 10000 = 131072;

★、610, etherCAT 电子齿轮共用分母, 设为: 10000;

★、611-618, etherCAT 的 J1-J6/Xs/Ys 轴电子齿轮分子, 设为: 131072;

★、620, etherCAT 站号设置[>=10 生效], 设为: 12345678;

★、621, etherCAT 驱动(1 迈信\之山\英威腾\埃斯顿\新时达\信捷\台达\雷赛\禾川\汇川\合信, 2 山洋\松下 A5, 3 恩普, 4 研控\图科\松下 A6, 5 图科双驱, 8 个 6 清能德创, 7 帧正\8 个 7+3 欧瑞+4 东菱\步科+6 迪维讯, 8 久同\韦德)

★、621a, EtherCAT 主轴驱动类型(前二位数表示第 1/2 主轴控制模式[9: 闭环速度控制模式, 1: 开环位置控制模式], 后二位数表示第 1/2 主轴驱动类型)

★、622-627, etherCAT 的 X/Y/Z/A/B/C 轴碰撞阈值

★、631-738, X/Y/Z/A/B/C/Xs/Ys 轴开启闭环光栅尺功能允许报警误差值[>2 表示开启闭环功能]

★、641-648, X/Y/Z/A/B/C/Xs/Ys 轴闭环光栅尺反馈脉冲电子齿轮分子[光栅尺反馈脉冲数 x 分子 / 分母=电机编码器反馈脉冲数, 负值表示反馈方向取反, P641-P658 相同]

★、651-658, X/Y/Z/A/B/C/Xs/Ys 轴闭环光栅尺反馈脉冲电子齿轮分母

★、700, EtherCAT 开机延时时间[>2 有效] (*50ms)

本系统对有些 Ethercat 驱动器有些特殊要求, 开机需要延时才能正常开机。如: 设置为 800 时表示开机延时 800*50=40000ms 后正常启动 Ethercat 驱动器。

★、900, 自定义加工参数对话框[1 表无效, 4 表部分, 12 表全部, 4108 表按页进入, +256 表 US0-54 为用户坐标, +8192 表冷却和快速键交换]

设置自定义加工参数对话框功能。4 表示显示左半部分界面, 12 表示全部界面, 4108 表示的按页进入界面, +256 表示用户坐标可以用 US 指令表示。

★、901, 回零轴顺序(>9) [5-9 位数, 前 8 位为 1/2/3/4/5/5/6/7/8 表 X/Y/Z/A/B/C/Xs/Ys, 最后位为 0]

设置自动回零时各个坐标轴的回零顺序, 如设置为 32145780 时, 回零顺序为 ZXYABXsYs 轴。

P601=40; etherCAT position loop gain coefficient

P602=40; etherCAT thread processing position ring gain coefficient

P603=10005; etherCAT closed-loop position control, tolerance value 5

P631=-888888; the X-axis open grating ruler AB signal compensation function is enabled

P641=131072; X-axis closed-loop linear encoder feedback pulse electronic gear molecular [linear encoder feedback pulse count x molecular numerator/denominator = motor encoder feedback pulse count, negative value indicates reversed feedback direction, P641-P658 identical] P651=10000; X-axis closed-loop linear encoder feedback pulse electronic gear denominator

For example, a motor encoder has 131072 pulses per revolution, such as AB feedback with 10000 pulses;

The feedback pulse count of the encoder [recovery amount] = light gate scale feedback pulse count × frequency / frequency

$$=10000*131072/10000=131072; \star,$$

610, etherCAT electronic gear shares the denominator, set as: 10000;

★In 611-618, the etherCAT J1-J6/Xs/Ys axis electron gear molecules are set as: 131072;

★620, etherCAT station number setting ≤10 effective, set as: 12345678;

★621, etherCAT drive (1: Moxin, Zhishan, Invt, Estun, Xinshida, Xinja, Delta, Raisa, Hechuan, Inovance, Hexin; 2: Shanyang, Panasonic A5; 3: Enpu; 4: R&D Control, Tuke, Panasonic A6; 5: Tuke Dual Drive; 8: Qingneng Dechuang; 7: Frame Zheng; 8: 7+3 Ourei +4 Dongling BuKe +6 Diversign; 8: Jiutong, Weide)

★621a, EtherCAT spindle drive type (the first two digits indicate the control mode of the first/second spindle [9: closed-loop speed control mode, 1: open-loop position control mode], and the last two digits indicate the drive type of the first/second spindle)

★, 622-627, etherCAT's X/Y/Z/A/B/C axis collision thresholds

★631-738, X/Y/Z/A/B/C/Xs/Ys axis open closed-loop grating scale function allows alarm error value [>]2 indicates open closed-loop function

★641-648, X/Y/Z/A/B/C/Xs/Ys, axis closed-loop grating scale feedback pulse electronic gear molecules [grating scale feedback pulse count x molecules denominator = motor encoder feedback pulse count, negative value indicates reversed feedback direction, P641-P658 same]

★651-658, X/Y/Z/A/B/C/Xs/Ys axis closed-loop grating scale feedback pulse electronic gear denominator ★, 700, EtherCAT startup delay time [>]2 valid (*50ms)

This system has some special requirements for some Ethercat drivers. The normal startup requires a delay. For example, if the setting is 800, the startup delay is 800*50=40000ms, and then the Ethercat driver starts normally.

★, 900, Custom machining parameters dialog [1 table invalid, 4 table partial, 12 table all, 4108 table page entry, +256 table US0-54 as user coordinates, +8192 table cooling and quick key switch]

Function of setting custom machining parameters dialog box. 4 indicates the display of the left half of the interface, 12 indicates the full interface, 4108 indicates the page entry interface, and +256 indicates that user coordinates can be expressed by US instructions.

★901, return zero axis sequence (>9) [5-9 digit, the first 8 digits are 1/2/3/4/5/6/7/8 X/Y/Z/A/B/C/Xs/Ys, and the last digit is 0]

Set the zero return sequence of each coordinate axis when setting automatic zero return. If set to 32145780, the zero return sequence is ZXYABXsYs axis.

★、903, 内部参数[2 表启用 SD 卡功能, +16 表第 5 轴焊接跟随加工功能(优先), +32 表第 4 轴焊接跟随加工功能]

内部参数, 特殊功能, 使用方法请联系系统制造厂家。

★、904, 内部参数[4 表文件解密, 8 表文件加密]

内部参数, 特殊功能, 使用方法请联系系统制造厂家。

★、910, X 轴-M18 (X28)/Y 轴-M22 (X27)/Z 轴-M24 (X26)/A 轴-M28 (X25) 是否为 G31/G311 的高速输入[1 表是]

注意:

<1>P32=28, P910=1 启动高速检测;

<2>XYZA 对应检测点必须为各轴对应的 X28/X27/X26/X25;

<3>G31/G311 检测时只能单独检测一个轴, 如: G31 X30 F100 P1010;

<4>G31/G311 指令之前必须使用 M4027 清 0;

★、911, 是否启动 M18 为示教, M24 为中点记录, M28 为终点记录功能[1 启动, 0 不启动]

设置示教功能外接记录按钮, 1 表示启动此功能。

★、912, 按复位键是否复位输出接口[1 复位, 0 不复位]

1 表示按复位键时复位输出 Y 信号。

★、913, 辅助继电器 M768-M783 是否作为扩展自定义报警[16 启动, 0 不启动]

★、915, 是否显示当前通道数的标志(设为 85 表开启)

85 表示开启双通道系统显示通道数标志。

★、916, 设置显示当前通道数(设为 1--9 有效)

设置双通道系统当前通道数。

★、994, 数控系统出厂编号

★、995, 机床出厂编号

★、996, 自定义出厂编号

3. 7. 6 坐标系

本参数有多坐标系功能, 即6个工件坐标系和一个机床坐标系G53, 。一个加工程序可设置一个工件坐标系, 也可设置多个工件坐标系, 工件坐标系可以用移动它的原点来改变。即参数中坐标系的值是其自身坐标原点(零点)在机床坐标系中的坐标值。

用G54 到G59可以设置54个工件坐标系, 在坐标系设置界面可以修改54个工件坐标系的原点在机床坐标系中的坐标值。

工件坐标系 G54--G59 整体的修改, 在参数的坐标系偏移值修改对话框中:

1) 输入 Axxx 表示所有坐标系该轴整体偏移, 例如: 输入 A12.5 表示当前的修改项所有坐标系整体偏移 12.5mm。

2) 输入 E 表示清零, 输入 Exxx 表示输入绝对偏移值。

3) 输入 EA 表示所有坐标系该轴整体清零, 输入 EAxxx 表示所有坐标系该轴

★, 903, internal parameters [2 table enables SD card function, +16 table fifth axis welding follow-up processing function (priority), +32 table fourth axis welding follow-up processing function]

Internal parameters, special functions, use methods contact the system manufacturer.

★, 904, internal parameters [Table 4 for file decryption, Table 8 for file encryption]

Internal parameters, special functions, use methods contact the system manufacturer.

★Whether 910, X-axis-M18 (X28), Y-axis-M22 (X27), Z-axis-M24 (X26), and A-axis-M28 (X25) are high-speed inputs for G31/G311 [Table 1 is shown]

pay attention to :

<1>P32=28, P910=1 start high speed detection;

<2>The corresponding detection points of XYZA must be X28/X27/X26/X25 corresponding to each axis;

<3>G31/G311 can only detect one axis at a time, such as: G31 X30 F100 P1010;

<4>G31/G311 Before the instruction must be used, M4027 must be cleared 0;

★911: Whether to start M18 for teaching, M24 for midpoint recording, and M28 for endpoint recording function [1 starts, 0 does not start]

Set the teaching function and external recording button. 1 indicates to start this function.

★, 912, press the reset button to reset the output interface [1 reset, 0 not reset]

1 Indicates that the reset output Y signal is generated when the reset button is pressed.

★, 913, whether the auxiliary relays M768-M783 are used as extended custom alarms [16 start, 0 not start]

★, 915, whether to display the flag of the current channel number (set to 85 to enable)

85 Indicates the opening of the dual-channel system display channel number sign.

★, 916, set to display the number of current channels (set to 1-9 valid)

Set the current number of channels in the dual-channel system.

★, 994, CNC system factory number

★, 995, machine tool factory number

★, 996, customize the factory number

3. 7. 6 Coordinate System

This parameter system features multi-coordinate systems, including six workpiece coordinate systems and one machine tool coordinate system (G53). A single machining program can be configured with either a single or multiple workpiece coordinate systems, which can be adjusted by moving their origin points. Specifically, the values in these parameters represent the coordinates of their respective workpiece origin points (zero points) within the machine tool coordinate system.

From G54 to G59, you can set 54 workpiece coordinate systems. In the coordinate system setting interface, you can modify the coordinates of the origin of the 54 workpiece coordinate systems in the machine tool coordinate system.

The modification of the workpiece coordinate system G54--G59 is performed in the dialog box of parameter coordinate system offset value modification:

1) Enter Axxx to indicate the overall offset of all coordinate systems along this axis. For example, enter A12.5 to indicate that the overall offset of all coordinate systems for the current modification is 12.5mm.

2) Enter E to indicate zero, and enter Exxx to indicate absolute offset value.

3) Enter EA to indicate that the axis of all coordinate systems is cleared, and enter EAxxx to indicate the axis of all coordinate systems

输入该绝对偏移值。

- 4) 在 X 轴的任意坐标系下, 输入 EALL 表示所有坐标系所有轴全部偏移值清零。
系统有 54 个工作坐标系: G54 到 G59、G54.1-G54.48。

参数列表:

1-0, 工件坐标系当前组[G54-G59]

此参数指明 1-1 至 1-8 显示的是哪一组工件坐标[G54-G59]

1-1, 用户坐标系 G54-G59 的 X(mm) [增量输入, E 开头表绝对]

1-2, 用户坐标系 G54-G59 的 Y(mm) [增量输入, E 开头表绝对]

1-3, 用户坐标系 G54-G59 的 Z(mm) [增量输入, E 开头表绝对]

1-4, 用户坐标系 G54-G59 的 A(度) [增量输入, E 开头表绝对]

1-5, 用户坐标系 G54-G59 的 B(度) [增量输入, E 开头表绝对]

1-6, 用户坐标系 G54-G59 的 C(度) [增量输入, E 开头表绝对]

1-7, 用户坐标系 G54-G59 的 XS(mm) [增量输入, E 开头表绝对]

1-8, 用户坐标系 G54-G59 的 YS(mm) [增量输入, E 开头表绝对]

2-0, 工件坐标系当前组[G54.1-G54.48]

此参数指明 2-1 至 2-8 显示的是哪一组工件坐标[G54.1-G54.48]

2-1, 用户坐标系 G54.1-G54.48 的 X(mm) [增量输入, E 开头表绝对]

2-2, 用户坐标系 G54.1-G54.48 的 Y(mm) [增量输入, E 开头表绝对]

2-3, 用户坐标系 G54.1-G54.48 的 Z(mm) [增量输入, E 开头表绝对]

2-4, 用户坐标系 G54.1-G54.48 的 A(度) [增量输入, E 开头表绝对]

2-5, 用户坐标系 G54.1-G54.48 的 B(度) [增量输入, E 开头表绝对]

2-6, 用户坐标系 G54.1-G54.48 的 C(度) [增量输入, E 开头表绝对]

2-7, 用户坐标系 G54.1-G54.48 的 XS(mm) [增量输入, E 开头表绝对]

2-8, 用户坐标系 G54.1-G54.48 的 YS(mm) [增量输入, E 开头表绝对]

1, 用户坐标系 G54 的 X(mm) [增量输入, E 开头表绝对]

2, 用户坐标系 G54 的 Y(mm) [增量输入, E 开头表绝对]

3, 用户坐标系 G54 的 Z(mm) [增量输入, E 开头表绝对]

4, 用户坐标系 G54 的 A(度) [增量输入, E 开头表绝对]

5, 用户坐标系 G54 的 B(度) [增量输入, E 开头表绝对]

6, 用户坐标系 G54 的 C(度) [增量输入, E 开头表绝对]

7, 用户坐标系 G54 的 XS(mm) [增量输入, E 开头表绝对]

8, 用户坐标系 G54 的 YS(mm) [增量输入, E 开头表绝对]

11, 用户坐标系 G55 的 X(mm) [增量输入, E 开头表绝对]

12, 用户坐标系 G55 的 Y(mm) [增量输入, E 开头表绝对]

13, 用户坐标系 G55 的 Z(mm) [增量输入, E 开头表绝对]

14, 用户坐标系 G55 的 A(度) [增量输入, E 开头表绝对]

15, 用户坐标系 G55 的 B(度) [增量输入, E 开头表绝对]

16, 用户坐标系 G55 的 C(度) [增量输入, E 开头表绝对]

17, 用户坐标系 G55 的 XS(mm) [增量输入, E 开头表绝对]

Enter this absolute offset value.

- 4) Under any coordinate system on the X-axis, input EALL indicates that all axis offset values in all coordinate systems are reset to zero. The system has 54 working coordinate systems: G54 to G59, G54.1 to G54.48.

Parameter list:

1-0, workpiece coordinate system current group [G54-G59]

This parameter specifies which set of workpiece coordinates [G54-G59] is displayed from 1-1 to 1-8

1-1, user coordinate system G54-G59 X (mm) [incremental input, E open table absolute]

1-2, Y (mm) in user coordinate system G54-G59 [incremental input, E start table absolute]

1-3, user coordinate system G54-G59 Z (mm) [incremental input, E start table absolute]

1-4, user coordinate system G54-G59 A (°) [incremental input, E start table absolute]

1-5, user coordinate system G54-G59 B (°) [incremental input, E start table absolute]

1-6, user coordinate system G54-G59 C (°) [incremental input, E start table absolute]

1-7, user coordinate system G54-G59 XS (mm) [incremental input, E open table absolute]

1-8, User coordinate system G54-G59 YS (mm) [Incremental input, E open table absolute]

2-0, workpiece coordinate system current group [G54.1-G54.48]

This parameter specifies which set of workpiece coordinates [G54.1-G54.48] is displayed from 2-1 to 2-8. The user coordinate system G54.1-G54.48 has an X (mm) [incremental input, E start table absolute].

2-2, user coordinate system G54.1-G54.48 Y (mm) [incremental input, E open table absolute]

2-3, user coordinate system G54.1-G54.48 Z (mm) [incremental input, E start table absolute]

2-4, user coordinate system G54.1-G54.48 A (°) [incremental input, E start table absolute]

2-5, user coordinate system G54.1-G54.48 B (°) [incremental input, E start table absolute]

2-6, user coordinate system G54.1-G54.48, C (°) [incremental input, E starting table absolute] 2-7, user coordinate system G54.1-G54.48, XS (mm) [incremental input, E starting table absolute]

2-8, User coordinate system G54.1-G54.48 YS (mm) [Incremental input, E open table absolute]

1. User coordinate system G54 X (mm) [Incremental input, E open table absolute]

2. User coordinate system G54 Y (mm) [Incremental input, E open table absolute]

3. User coordinate system G54 Z (mm) [Incremental input, E open table absolute]

4. User coordinate system G54 A (°) [Incremental input, E start table absolute]

5. User coordinate system G54 B (°) [Incremental input, E start table absolute]

6. User coordinate system G54 C (°) [Incremental input, E start table absolute]

7. User coordinate system G54 XS (mm) [Incremental input, E open table absolute]

8. User coordinate system G54 YS (mm) [Incremental input, E open table absolute]

11, user coordinate system G55 X (mm) [incremental input, E open table absolute]

12. User coordinate system G55 Y (mm) [incremental input, E start table absolute]

13. User coordinate system G55 Z (mm) [Incremental input, E start table absolute]

14. User coordinate system G55 A (°) [incremental input, E start table absolute]

15. User coordinate system G55 B (°) [incremental input, E start table absolute]

16. User coordinate system G55 C (°) [incremental input, E start table absolute]

17. User coordinate system G55 XS (mm) [Incremental input, E open table absolute]

- 18, 用户坐标系 G55 的 YS(mm) [增量输入, E 开头表绝对]
- 21, 用户坐标系 G56 的 X(mm) [增量输入, E 开头表绝对]
- 22, 用户坐标系 G56 的 Y(mm) [增量输入, E 开头表绝对]
- 23, 用户坐标系 G56 的 Z(mm) [增量输入, E 开头表绝对]
- 24, 用户坐标系 G56 的 A(度) [增量输入, E 开头表绝对]
- 25, 用户坐标系 G56 的 B(度) [增量输入, E 开头表绝对]
- 26, 用户坐标系 G56 的 C(度) [增量输入, E 开头表绝对]
- 27, 用户坐标系 G56 的 XS(mm) [增量输入, E 开头表绝对]
- 28, 用户坐标系 G56 的 YS(mm) [增量输入, E 开头表绝对]
- 31, 用户坐标系 G57 的 X(mm) [增量输入, E 开头表绝对]
- 32, 用户坐标系 G57 的 Y(mm) [增量输入, E 开头表绝对]
- 33, 用户坐标系 G57 的 Z(mm) [增量输入, E 开头表绝对]
- 34, 用户坐标系 G57 的 A(度) [增量输入, E 开头表绝对]
- 35, 用户坐标系 G57 的 B(度) [增量输入, E 开头表绝对]
- 36, 用户坐标系 G57 的 C(度) [增量输入, E 开头表绝对]
- 37, 用户坐标系 G57 的 XS(mm) [增量输入, E 开头表绝对]
- 38, 用户坐标系 G57 的 YS(mm) [增量输入, E 开头表绝对]
- 41, 用户坐标系 G58 的 X(mm) [增量输入, E 开头表绝对]
- 42, 用户坐标系 G58 的 Y(mm) [增量输入, E 开头表绝对]
- 43, 用户坐标系 G58 的 Z(mm) [增量输入, E 开头表绝对]
- 44, 用户坐标系 G58 的 A(度) [增量输入, E 开头表绝对]
- 45, 用户坐标系 G58 的 B(度) [增量输入, E 开头表绝对]
- 46, 用户坐标系 G58 的 C(度) [增量输入, E 开头表绝对]
- 47, 用户坐标系 G58 的 XS(mm) [增量输入, E 开头表绝对]
- 48, 用户坐标系 G58 的 YS(mm) [增量输入, E 开头表绝对]
- 51, 用户坐标系 G59 的 X(mm) [增量输入, E 开头表绝对]
- 52, 用户坐标系 G59 的 Y(mm) [增量输入, E 开头表绝对]
- 53, 用户坐标系 G59 的 Z(mm) [增量输入, E 开头表绝对]
- 54, 用户坐标系 G59 的 A(度) [增量输入, E 开头表绝对]
- 55, 用户坐标系 G59 的 B(度) [增量输入, E 开头表绝对]
- 56, 用户坐标系 G59 的 C(度) [增量输入, E 开头表绝对]
- 57, 用户坐标系 G59 的 XS(mm) [增量输入, E 开头表绝对]
- 58, 用户坐标系 G59 的 YS(mm) [增量输入, E 开头表绝对]
- 61, 用户坐标系 G54.1 的 X(mm) [增量输入, E 开头表绝对]
- 62, 用户坐标系 G54.1 的 Y(mm) [增量输入, E 开头表绝对]
- 63, 用户坐标系 G54.1 的 Z(mm) [增量输入, E 开头表绝对]
- 64, 用户坐标系 G54.1 的 A(度) [增量输入, E 开头表绝对]
- 65, 用户坐标系 G54.1 的 B(度) [增量输入, E 开头表绝对]
- 66, 用户坐标系 G54.1 的 C(度) [增量输入, E 开头表绝对]
- 67, 用户坐标系 G54.1 的 XS(mm) [增量输入, E 开头表绝对]

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- 18. User coordinate system G55 YS (mm) [Incremental input, E open header absolute]
 - 21, User coordinate system G56 X (mm) [incremental input, E open table absolute]
 - 22, user coordinate system G56 Y (mm) [incremental input, E open table absolute]
 - 23. User coordinate system G56 Z (mm) [Incremental input, E start table absolute]
 - 24. User coordinate system G56 A (°) [Incremental input, E start table absolute]
 - 25. User coordinate system G56 B (°) [Incremental input, E start table absolute]
 - 26. User coordinate system G56 C (°) [Incremental input, E start table absolute]
 - 27, User coordinate system G56 XS (mm) [Incremental input, E open table absolute]
 - 28. User coordinate system G56 YS (mm) [Incremental input, E open table absolute]
 - 31, User coordinate system G57 X (mm) [Incremental input, E open table absolute]
 - 32, User coordinate system G57 Y (mm) [incremental input, E open table absolute]
 - 33, user coordinate system G57 Z (mm) [incremental input, E open table absolute]
 - 34, user coordinate system G57 A (°) [incremental input, E start table absolute]
 - 35, User coordinate system G57 B (°) [Incremental input, E start table absolute]
 - 36, User coordinate system G57 C (°) [Incremental input, E start table absolute]
 - 37, User coordinate system G57 XS (mm) [Incremental input, E open table absolute]
 - 38, User coordinate system G57 YS (mm) [Incremental input, E open table absolute]
 - 41, User coordinate system G58 X (mm) [incremental input, E open table absolute]
 - 42, User coordinate system G58 Y (mm) [Incremental input, E open table absolute]
 - 43, User coordinate system G58 Z (mm) [incremental input, E start table absolute]
 - 44, user coordinate system G58 A (°) [incremental input, E start table absolute]
 - 45, user coordinate system G58 B (°) [incremental input, E start table absolute]
 - 46, user coordinate system G58 C (degree) [incremental input, E start table absolute]
 - 47, User coordinate system G58 XS (mm) [Incremental input, E open table absolute]
 - 48, User coordinate system G58 YS (mm) [Incremental input, E open table absolute]
 - 51, User coordinate system G59 X (mm) [Incremental input, E open table absolute]
 - 52, User coordinate system G59 Y (mm) [Incremental input, E start table absolute]
 - 53, User coordinate system G59 Z (mm) [Incremental input, E start table absolute]
 - 54, User coordinate system G59 A (°) [Incremental input, E start table absolute]
 - 55, user coordinate system G59 B (°) [incremental input, E start table absolute]
 - 56, user coordinate system G59 C (degree) [incremental input, E start table absolute]
 - 57, User coordinate system G59 XS (mm) [Incremental input, E open table absolute]
 - 58, User coordinate system G59 YS (mm) [Incremental input, E open table absolute]
 - 61, User coordinate system G54.1 X (mm) [Incremental input, E start table absolute]
 - 62, User coordinate system G54.1 Y (mm) [Incremental input, E start table absolute]
 - 63, User coordinate system G54.1 Z (mm) [Incremental input, E start table absolute]
 - 64, User coordinate system G54.1 A (°) [Incremental input, E start table absolute]
 - 65, User coordinate system G54.1 B (°) [Incremental input, E start table absolute]
 - 66, user coordinate system G54.1 C (°) [incremental input, E start table absolute]
 - 67, User coordinate system G54.1 XS (mm) [Incremental input, E start table absolute]

- 68, 用户坐标系 G54.1 的 YS(mm) [增量输入, E 开头表绝对]
- 71, 用户坐标系 G54.2 的 X(mm) [增量输入, E 开头表绝对]
- 72, 用户坐标系 G54.2 的 Y(mm) [增量输入, E 开头表绝对]
- 73, 用户坐标系 G54.2 的 Z(mm) [增量输入, E 开头表绝对]
- 74, 用户坐标系 G54.2 的 A(度) [增量输入, E 开头表绝对]
- 75, 用户坐标系 G54.2 的 B(度) [增量输入, E 开头表绝对]
- 76, 用户坐标系 G54.2 的 C(度) [增量输入, E 开头表绝对]
- 77, 用户坐标系 G54.2 的 XS(mm) [增量输入, E 开头表绝对]
- 78, 用户坐标系 G54.2 的 YS(mm) [增量输入, E 开头表绝对]
- 81, 用户坐标系 G54.3 的 X(mm) [增量输入, E 开头表绝对]
- 82, 用户坐标系 G54.3 的 Y(mm) [增量输入, E 开头表绝对]
- 83, 用户坐标系 G54.3 的 Z(mm) [增量输入, E 开头表绝对]
- 84, 用户坐标系 G54.3 的 A(度) [增量输入, E 开头表绝对]
- 85, 用户坐标系 G54.3 的 B(度) [增量输入, E 开头表绝对]
- 86, 用户坐标系 G54.3 的 C(度) [增量输入, E 开头表绝对]
- 87, 用户坐标系 G54.3 的 XS(mm) [增量输入, E 开头表绝对]
- 88, 用户坐标系 G54.3 的 YS(mm) [增量输入, E 开头表绝对]
- 91, 用户坐标系 G54.4 的 X(mm) [增量输入, E 开头表绝对]
- 92, 用户坐标系 G54.4 的 Y(mm) [增量输入, E 开头表绝对]
- 93, 用户坐标系 G54.4 的 Z(mm) [增量输入, E 开头表绝对]
- 94, 用户坐标系 G54.4 的 A(度) [增量输入, E 开头表绝对]
- 95, 用户坐标系 G54.4 的 B(度) [增量输入, E 开头表绝对]
- 96, 用户坐标系 G54.4 的 C(度) [增量输入, E 开头表绝对]
- 97, 用户坐标系 G54.4 的 XS(mm) [增量输入, E 开头表绝对]
- 98, 用户坐标系 G54.4 的 YS(mm) [增量输入, E 开头表绝对]
- 101, 用户坐标系 G54.5 的 X(mm) [增量输入, E 开头表绝对]
- 102, 用户坐标系 G54.5 的 Y(mm) [增量输入, E 开头表绝对]
- 103, 用户坐标系 G54.5 的 Z(mm) [增量输入, E 开头表绝对]
- 104, 用户坐标系 G54.5 的 A(度) [增量输入, E 开头表绝对]
- 105, 用户坐标系 G54.5 的 B(度) [增量输入, E 开头表绝对]
- 106, 用户坐标系 G54.5 的 C(度) [增量输入, E 开头表绝对]
- 107, 用户坐标系 G54.5 的 XS(mm) [增量输入, E 开头表绝对]
- 108, 用户坐标系 G54.5 的 YS(mm) [增量输入, E 开头表绝对]
- 111, 用户坐标系 G54.6 的 X(mm) [增量输入, E 开头表绝对]
- 112, 用户坐标系 G54.6 的 Y(mm) [增量输入, E 开头表绝对]
- 113, 用户坐标系 G54.6 的 Z(mm) [增量输入, E 开头表绝对]
- 114, 用户坐标系 G54.6 的 A(度) [增量输入, E 开头表绝对]
- 115, 用户坐标系 G54.6 的 B(度) [增量输入, E 开头表绝对]
- 116, 用户坐标系 G54.6 的 C(度) [增量输入, E 开头表绝对]
- 117, 用户坐标系 G54.6 的 XS(mm) [增量输入, E 开头表绝对]

68, User coordinate system G54.1 YS (mm) [Incremental input, E start table absolute]

71, User coordinate system G54.2 X (mm) [Incremental input, E start table absolute]

72, User coordinate system G54.2 Y (mm) [Incremental input, E start table absolute]

73, User coordinate system G54.2 Z (mm) [Incremental input, E start table absolute]

74, user coordinate system G54.2 A (°) [incremental input, E start table absolute]

75, User coordinate system G54.2 B (°) [Incremental input, E start table absolute]

76, User coordinate system G54.2 C (°) [Incremental input, E start table absolute]

77, User coordinate system G54.2 XS (mm) [Incremental input, E start table absolute]

78, User coordinate system G54.2 YS (mm) [Incremental input, E start table absolute]

81, User coordinate system G54.3 X (mm) [Incremental input, E start table absolute]

82, User coordinate system G54.3 Y (mm) [Incremental input, E start table absolute]

83, User coordinate system G54.3 Z (mm) [Incremental input, E start table absolute]

84, user coordinate system G54.3 A (°) [incremental input, E start table absolute]

85, user coordinate system G54.3 B (°) [incremental input, E start table absolute]

86, User coordinate system G54.3 C (°) [Incremental input, E start table absolute]

87, User coordinate system G54.3 XS (mm) [Incremental input, E start table absolute]

88, User coordinate system G54.3 YS (mm) [Incremental input, E start table absolute]

91, User coordinate system G54.4 X (mm) [Incremental input, E start table absolute]

92, User coordinate system G54.4 Y (mm) [Incremental input, E start table absolute]

93, User coordinate system G54.4 Z (mm) [Incremental input, E start table absolute]

94, user coordinate system G54.4 A (°) [incremental input, E start table absolute]

95, User coordinate system G54.4 B (°) [Incremental input, E start table absolute]

96, user coordinate system G54.4 C (°) [incremental input, E start table absolute]

97, User coordinate system G54.4 XS (mm) [Incremental input, E start table absolute]

98, User coordinate system G54.4 YS (mm) [Incremental input, E start table absolute]

101, user coordinate system G54.5 X (mm) [incremental input, E start table absolute]

102, user coordinate system G54.5 Y (mm) [incremental input, E start table absolute]

103, user coordinate system G54.5 Z (mm) [incremental input, E start table absolute]

104, user coordinate system G54.5 A (°) [incremental input, E start table absolute]

105, user coordinate system G54.5 B (°) [incremental input, E start table absolute]

106, user coordinate system G54.5's C (degree) [incremental input, E start table absolute]

107, User coordinate system G54.5 XS (mm) [Incremental input, E start table absolute]

108, User coordinate system G54.5 YS (mm) [Incremental input, E start table absolute]

111, user coordinate system G54.6 X (mm) [incremental input, E open table absolute]

112, user coordinate system G54.6 Y (mm) [incremental input, E start table absolute]

113, user coordinate system G54.6 Z (mm) [incremental input, E start table absolute]

114, user coordinate system G54.6 A (°) [incremental input, E start table absolute]

115, user coordinate system G54.6 B (°) [incremental input, E start table absolute]

116, user coordinate system G54.6 C (°) [incremental input, E start table absolute]

117, User coordinate system G54.6 XS (mm) [Incremental input, E open table absolute]

- 118, 用户坐标系 G54.6 的 YS(mm) [增量输入, E 开头表绝对]
- 121, 用户坐标系 G54.7 的 X(mm) [增量输入, E 开头表绝对]
- 122, 用户坐标系 G54.7 的 Y(mm) [增量输入, E 开头表绝对]
- 123, 用户坐标系 G54.7 的 Z(mm) [增量输入, E 开头表绝对]
- 124, 用户坐标系 G54.7 的 A(度) [增量输入, E 开头表绝对]
- 125, 用户坐标系 G54.7 的 B(度) [增量输入, E 开头表绝对]
- 126, 用户坐标系 G54.7 的 C(度) [增量输入, E 开头表绝对]
- 127, 用户坐标系 G54.7 的 XS(mm) [增量输入, E 开头表绝对]
- 128, 用户坐标系 G54.7 的 YS(mm) [增量输入, E 开头表绝对]
- 131, 用户坐标系 G54.8 的 X(mm) [增量输入, E 开头表绝对]
- 132, 用户坐标系 G54.8 的 Y(mm) [增量输入, E 开头表绝对]
- 133, 用户坐标系 G54.8 的 Z(mm) [增量输入, E 开头表绝对]
- 134, 用户坐标系 G54.8 的 A(度) [增量输入, E 开头表绝对]
- 135, 用户坐标系 G54.8 的 B(度) [增量输入, E 开头表绝对]
- 136, 用户坐标系 G54.8 的 C(度) [增量输入, E 开头表绝对]
- 137, 用户坐标系 G54.8 的 XS(mm) [增量输入, E 开头表绝对]
- 138, 用户坐标系 G54.8 的 YS(mm) [增量输入, E 开头表绝对]
- 141, 用户坐标系 G54.9 的 X(mm) [增量输入, E 开头表绝对]
- 142, 用户坐标系 G54.9 的 Y(mm) [增量输入, E 开头表绝对]
- 143, 用户坐标系 G54.9 的 Z(mm) [增量输入, E 开头表绝对]
- 144, 用户坐标系 G54.9 的 A(度) [增量输入, E 开头表绝对]
- 145, 用户坐标系 G54.9 的 B(度) [增量输入, E 开头表绝对]
- 146, 用户坐标系 G54.9 的 C(度) [增量输入, E 开头表绝对]
- 147, 用户坐标系 G54.9 的 XS(mm) [增量输入, E 开头表绝对]
- 148, 用户坐标系 G54.9 的 YS(mm) [增量输入, E 开头表绝对]
- 151, 用户坐标系 G54.10 的 X(mm) [增量输入, E 开头表绝对]
- 152, 用户坐标系 G54.10 的 Y(mm) [增量输入, E 开头表绝对]
- 153, 用户坐标系 G54.10 的 Z(mm) [增量输入, E 开头表绝对]
- 154, 用户坐标系 G54.10 的 A(度) [增量输入, E 开头表绝对]
- 155, 用户坐标系 G54.10 的 B(度) [增量输入, E 开头表绝对]
- 156, 用户坐标系 G54.10 的 C(度) [增量输入, E 开头表绝对]
- 157, 用户坐标系 G54.10 的 XS(mm) [增量输入, E 开头表绝对]
- 158, 用户坐标系 G54.10 的 YS(mm) [增量输入, E 开头表绝对]

操作说明:

1、工件坐标系的建立

工件坐标系的建立通常在手动状态下, 其操作步骤如下:

- (a) 按“F8 选择坐标”或“F5 MDI”键选择相应的工件坐标系 (G54-G59)。
- (b) 用手动将机床开到一定的位置 (方便测量的一个位置), 测量出该点相对对应工件坐标系零点的位置

- 118, User coordinate system G54.6 YS (mm) [Incremental input, E start table absolute]
 - 121, User coordinate system G54.7 X (mm) [Incremental input, E start table absolute]
 - 122, user coordinate system G54.7 Y (mm) [incremental input, E start table absolute]
 - 123, user coordinate system G54.7 Z (mm) [incremental input, E start table absolute]
 - 124, user coordinate system G54.7 A (°) [incremental input, E start table absolute]
 - 125, user coordinate system G54.7 B (°) [incremental input, E start table absolute]
 - 126, user coordinate system G54.7 of C (degree) [incremental input, E start table absolute]
 - 127, User coordinate system G54.7 XS (mm) [Incremental input, E start table absolute]
 - 128, User coordinate system G54.7 YS (mm) [Incremental input, E start table absolute]
 - 131, User coordinate system G54.8 X (mm) [Incremental input, E start table absolute]
 - 132, user coordinate system G54.8 Y (mm) [incremental input, E start table absolute]
 - 133, user coordinate system G54.8 Z (mm) [incremental input, E start table absolute]
 - 134, user coordinate system G54.8 A (°) [incremental input, E start table absolute]
 - 135, user coordinate system G54.8 B (°) [incremental input, E start table absolute]
 - 136, user coordinate system G54.8 of C (degree) [incremental input, E start table absolute]
 - 137, User coordinate system G54.8 XS (mm) [Incremental input, E start table absolute]
 - 138, User coordinate system G54.8 YS (mm) [Incremental input, E start table absolute]
 - 141, User coordinate system G54.9 X (mm) [Incremental input, E open table absolute]
 - 142, user coordinate system G54.9 Y (mm) [incremental input, E start table absolute]
 - 143, user coordinate system G54.9 Z (mm) [incremental input, E start table absolute]
 - 144, user coordinate system G54.9 A (°) [incremental input, E start table absolute]
 - 145, user coordinate system G54.9 B (°) [incremental input, E start table absolute]
 - 146, user coordinate system G54.9 C (°) [incremental input, E start table absolute]
 - 147, User coordinate system G54.9 XS (mm) [Incremental input, E start table absolute]
 - 148, User coordinate system G54.9 YS (mm) [Incremental input, E start table absolute]
 - 151, User coordinate system G54.10 X (mm) [Incremental input, E start table absolute]
 - 152, user coordinate system G54.10 Y (mm) [incremental input, E start table absolute]
 - 153, User coordinate system G54.10 Z (mm) [Incremental input, E start table absolute]
 - 154, user coordinate system G54.10 A (°) [incremental input, E start table absolute]
 - 155, user coordinate system G54.10 B (°) [incremental input, E start table absolute]
 - 156, user coordinate system G54.10 C (°) [incremental input, E start table absolute]
 - 157, User coordinate system G54.10 XS (mm) [Incremental input, E start table absolute]
 - 158, User coordinate system G54.10 YS (mm) [Incremental input, E start table absolute]
- operation declaration :

1. Establishment of Workpiece Coordinate System

The establishment of the workpiece coordinate system is usually in manual mode, and the operation steps are as follows:

- (a) Select the corresponding workpiece coordinate system (G54-G59) by pressing "F8 Select Coordinate" or "F5 MDI".
- (b) Manually open the machine tool to a certain position (a convenient position for measurement), and measure the position of this point relative to the zero point of the corresponding workpiece coordinate system

(c) 按“F7 设置坐标”键，按“X”键确认，再将上属测量出 X 轴的值输入对话框中，按“确认”键确认。

(d) 按“F7 设置坐标”键，按“Y”键确认，再将上属测量出 Y 轴的值输入对话框中，按“确认”键确认。

(e) 按“F7 设置坐标”键，按“Z”键，再将上属测量出 Z 轴的值输入对话框中，按“确认”键确认。

此时相就应的工件坐标系就设置好，进入参数中的坐标系界面就会看到对应参数中有相应的数值，该值即为该工件坐标系零与机床坐标系零的偏置值。

2、工件坐标系的偏移设置

若工件坐标系在手动中设置好后根据加工情况还需进一步调整值，此时需到参数的坐标界面。操作方法如下：

在坐标系参数界面，按上下光标键选中对应工件坐标系的相应轴，按确认键，系统即弹出对话框，此时直接输入需要的偏移量即可（即增量设置，如：往负向偏 10mm，即输入-10），按确认键修改完成。

说明：1、参数修改完成后，系统坐标界面立即刷新对应工件坐标系的坐标值。

2、坐标系界面内，参数后方括弧内的值为偏移修调的总量，方便操作者查看偏移修调的情况。

NCK数控
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3. 7. 7 密码

在本系统中为防止参数被意外修改，保证机床正常运行，系统采用了参数的分级权限设置。分为“数控厂家”、“机床厂家”、“用户厂家”三级权限。其中“数控厂家”权限为设置系统的功能，属系统内部参数；“机床厂家”权限为设置与机床设备配置和机械指标以及安全有关的参数；“用户厂家”权限为设置与加工工艺、性能等参数以及加工程序等有关的参数。

在本系统中三级权限初始情况为：“数控厂家”权限已启用，“机床厂家”、“用户厂家”权限均未启用。若想要启用该权限功能，必须先用初始口令，启用权限功能，再设置相应的新的权限口令，达到限权的目的。需要注意的是初始口令只能使用一次，设置新口令后初始口令将作废，请务必记住新口令。

参数列表：

- 1, 是否开启数控厂家密码保护
- 2, 是否开启机床厂家密码保护
- 3, 是否开启用户厂家密码保护
- 4、修订数控厂家保护密码
- 5、修订机床厂家保护密码
- 6、修订用户厂家保护密码
- 7、系统使用时间
- 8、软件版次

(c) Press the "F7 Set coordinate" key, press the "X" key to confirm, then input the value measured by the upper part on the X-axis into the dialog box, and press the "Confirm" key to confirm.

(d) Press the "F7 Set coordinate" key, press the "Y" key to confirm, and then input the Y-axis value measured by the upper part into the dialog box, and press the "Confirm" key to confirm.

(e) Press the "F7 Set coordinate" key, press the "Z" key, and then input the Z-axis value measured by the upper part into the dialog box, and press the "OK" key to confirm.

At this time, the corresponding workpiece coordinate system is set up. When entering the coordinate system interface in the parameters, you will see the corresponding value in the corresponding parameters, which is the offset value between the zero of the workpiece coordinate system and the zero of the machine tool coordinate system.

2. Offset Setting of Workpiece Coordinate System

If the coordinate system of the workpiece is set manually and needs to be further adjusted according to the processing situation, you need to go to the coordinate interface of parameters. The operation method is as follows:

In the coordinate system parameter interface, press the up and down cursor keys to select the corresponding axis of the workpiece coordinate system. Press the confirm key, and a dialog box will pop up. At this time, you can directly input the required offset (that is, incremental setting, such as: offset 10mm in the negative direction, that is, input -10), and press the confirm key to complete the modification.

Note: 1. After parameter modification is completed, the system coordinate interface will refresh the coordinate values of the corresponding workpiece coordinate system immediately.

2. In the coordinate system interface, the value in the parentheses behind the parameter is the total amount of offset adjustment, which is convenient for the operator to view the offset adjustment.

3. 7. 7 Password

To prevent accidental parameter modifications and ensure proper machine operation, this system implements a hierarchical permission structure for parameters. The three-tier authorization system consists of "CNC Manufacturer", "Machine Tool Manufacturer", and "User Manufacturer" levels. Specifically: The "CNC Manufacturer" level manages system configuration functions as internal parameters; the "Machine Tool Manufacturer" level handles parameters related to equipment specifications, mechanical performance metrics, and safety protocols; while the "User Manufacturer" level controls parameters concerning machining processes, operational performance characteristics, and program configurations.

In this system, the default three-tier authorization configuration is as follows: The "CNC Manufacturer" privilege is enabled, while both the "Machine Tool Manufacturer" and "User Manufacturer" privileges remain disabled. To activate these privileges, users must first use the initial password to enable the privilege function, then set up new privilege passwords to enforce access controls. It's important to note that the initial password is only valid once. After setting a new password, the initial one becomes invalid. Please remember your new password carefully.

Parameter list:

1. Whether to open the password protection of CNC manufacturer
2. Whether the machine tool manufacturer password protection is enabled
3. Whether to enable user manufacturer password protection
4. Revise the protection password of CNC manufacturer
5. Revise the protection password of the machine tool manufacturer
6. Revise the user manufacturer protection password
7. System usage time
8. Software version

参数说明:

★、1，是否开启数控厂家密码保护
系统内部参数不能操作

★、2，是否开启机床厂家密码保护

用于设置是否开启机床厂家密码保护，该参数启动后与机床配置相关的参数将会被屏蔽起来，无法操作。若需要修改需关闭该参数才能看到并修改。

该参数初始权限密码为“NEWNEW”，设置时需先输入密码，打开参数才能修改。

★、3，是否开启用户厂家密码保护

用于设置是否开启用户厂家密码保护，该参数启动后可将与加工相关的参数将屏蔽起来，无法操作若需要修改需关闭该参数才能看到并修改；或对程序进行保护，需密码才能编辑。

该参数初始权限密码为“KERKER”，设置时需先输入密码，打开参数才能修改。

★、4，修订数控厂家保护密码

★、5，修订机床厂家保护密码

★、6，修订用户厂家保护密码

上述三个参数用于修改新密码，再新改新密码时必须先输入老密码。

鑫科瑞数控

★、7，系统使用时间

系统开机使用时间，用户不能修改。

★、8，软件版次

用于观察系统软件版本号。

3.8 刀补参数设置

在任意界面按“刀补”键进入刀补界面，在刀补界面下包括“半径补偿”、“长度补偿”、“全部清零”、“当前清零”、“对刀”、“刀套表”、“设置”七项功能，分别按“F1-F7”键可进入该功能界面。退出该功能界面，按“返回”键则返回主菜单界面。

3.8.1 半径补偿

在刀补界面按“F1 半径补偿”键进入刀具半径补偿界面。该参数用于设置采用刀具的半径值。

设置方法：按“↑”“↓”键将光标移动到相应的刀具号上按“确认”键，

parameter declaration :

★1. Whether to open the password protection of CNC manufacturer Internal system parameters cannot be operated

★2. Whether the machine tool manufacturer password protection is enabled

Used to set whether to enable the machine tool manufacturer password protection. After this parameter is enabled, the parameters related to machine tool configuration will be blocked and cannot be operated. If you need to modify, you must disable this parameter to see and modify.

The initial permission password of this parameter is "NEWNEW". When setting, you need to input the password first and open the parameter before modification.

★3. Whether to enable user manufacturer password protection

Used to set whether to enable user manufacturer password protection. After this parameter is enabled, the processing-related parameters will be shielded and cannot be operated. If you need to modify, you must disable this parameter to see and modify. Or the program can be protected, and you must edit with a password.

The initial permission password of this parameter is "KERKER". When setting, you need to input the password first and open the parameter before modification.

★4. Revise the protection password of CNC manufacturer

★5. Revise the protection password of the machine tool manufacturer

★6. Revise the user manufacturer protection password

The above three parameters are used to modify the new password. When you change the new password, you must first enter the old password.

★, 7, system usage time

Users cannot modify the system startup time.

★8. Software version

Used to observe the version number of the system software.

3.8 Adjust Parameter Settings

To access the tool compensation interface, press the "Tool Compensation" button on any screen. This interface contains seven functions: Radius Compensation, Length Compensation, Full Reset, Current Reset, Tool Alignment, Tool Sleeve Table, and Settings. Press the "F1-F7" keys to activate each function. To exit the interface, press the "Return" key to return to the main menu.

3. 8. 1 Radius Compensation

Press the "F1 radius compensation" key in the tool supplement interface to enter the tool radius compensation interface. This parameter is used to set the radius value of the tool.

Setting method: Move the cursor to the corresponding tool number by pressing the "↑" and "↓" keys, and press the "Confirm" key,

系统弹出对话框，在对话框中输入相应的刀具半径值（绝对值输入）后按“确认”即可。

手动 连续		N00000		2016-06-06 06:06																																																					
在主界面按 T 键设置长度补偿对刀基准: No				当前程序 99.TXT																																																					
<table border="1"> <tr><td>T01</td><td>H:</td><td>0.000</td><td>[0.000]</td></tr> <tr><td>T02</td><td>H:</td><td>0.000</td><td>[0.000]</td></tr> <tr><td>T03</td><td>H:</td><td>0.000</td><td>[0.000]</td></tr> <tr><td>T04</td><td>H:</td><td>0.000</td><td>[0.000]</td></tr> <tr><td>T05</td><td>H:</td><td>0.000</td><td>[0.000]</td></tr> <tr><td>T06</td><td>H:</td><td>0.000</td><td>[0.000]</td></tr> <tr><td>T07</td><td>H:</td><td>0.000</td><td>[0.000]</td></tr> <tr><td>T08</td><td>H:</td><td>0.000</td><td>[0.000]</td></tr> <tr><td>H09</td><td>H:</td><td>0.010</td><td>[0.000]</td></tr> <tr><td>H10</td><td>H:</td><td>0.000</td><td>[0.000]</td></tr> <tr><td>H11</td><td>H:</td><td>0.000</td><td>[0.000]</td></tr> <tr><td>H12</td><td>H:</td><td>0.000</td><td>[0.000]</td></tr> <tr><td>H13</td><td>H:</td><td>0.000</td><td>[0.000]</td></tr> </table>				T01	H:	0.000	[0.000]	T02	H:	0.000	[0.000]	T03	H:	0.000	[0.000]	T04	H:	0.000	[0.000]	T05	H:	0.000	[0.000]	T06	H:	0.000	[0.000]	T07	H:	0.000	[0.000]	T08	H:	0.000	[0.000]	H09	H:	0.010	[0.000]	H10	H:	0.000	[0.000]	H11	H:	0.000	[0.000]	H12	H:	0.000	[0.000]	H13	H:	0.000	[0.000]	指令代码 G53 T01H0D0	
T01	H:	0.000	[0.000]																																																						
T02	H:	0.000	[0.000]																																																						
T03	H:	0.000	[0.000]																																																						
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T06	H:	0.000	[0.000]																																																						
T07	H:	0.000	[0.000]																																																						
T08	H:	0.000	[0.000]																																																						
H09	H:	0.010	[0.000]																																																						
H10	H:	0.000	[0.000]																																																						
H11	H:	0.000	[0.000]																																																						
H12	H:	0.000	[0.000]																																																						
H13	H:	0.000	[0.000]																																																						
状态				M05 M09 M10 不选停 M78 M33 M41 跳段																																																					
				G00 X100% F2000 X100% S800 X100% SP000																																																					
机床坐标				X 168.000 A 168.000 Y 168.000 B 168.000 Z 168.000																																																					
单件时间				0: 6																																																					
加工件数				0																																																					
主轴转速				0																																																					
状态: 正常																																																									
F1 半径补偿 F2 长度补偿 F3 全部清零 F4 当前清零 F5 对刀 F6 刀套表 F7 设置 F8 返回																																																									

3. 8. 2 长度刀补

在刀补界面按“F2 长度补偿”键进入刀具长度补偿界面。该参数用于修调采用刀具的长度值或重新设置采用刀具有长度值。

修调刀具的长度补偿值设置方法:

按“↑”“↓”键将光标移动到相应的刀具号上按“确认”键，系统弹出对话框，再输入需要的值（该输入为增量输入，若要减小 0.05mm，即输入-0.05）后按“确认”键即可。

自动对刀的方法:

将机床移动到方便测量对应刀具坐标的位置，按“↑”“↓”键将光标移动到相应的刀具号上按“F5 对刀”键，系统弹出对话框，再按“确认”键即可，相应刀具的长度补偿值就确认好了。

初始化刀具半径和长度补偿值的方法:

The system will pop up a dialog box. Enter the corresponding tool radius value (absolute value) in the dialog box and press "OK".

手动 连续		N00000		2016-06-06 06:06	
在主界面按 T 键设置长度补偿对刀基准: No				当前程序 99.TXT	
指令代码				G53	
				T01H0D0	
状态				M05 M09 M10 不选停	
				M78 M33 M41 跳段	
G00 X100%					
F2000 X100%					
S800 X100% SP000					
机床坐标					
X 168.000 A 168.000					
Y 168.000 B 168.000					
Z 168.000					
单件时间 0: 6					
加工件数 0					
主轴转速 0					
状态: 正常					
F1 半径补偿	F2 长度补偿	F3 全部清零	F4 当前清零	F5 对刀	F6 刀套表
F7 设置	F8 返回				

3. 8. 2 Length Knife Cuts

Press the "F2 Length Compensation" key in the tool compensation interface to enter the tool length compensation interface. This parameter is used to adjust the length value of the tool or reset the length value of the tool.

Setting method of length compensation value for tuning tool:

Move the cursor to the corresponding tool number by pressing the "↑" and "↓" keys. Press the "Confirm" key, and the dialog box will pop up. Then input the required value (this input is incremental input. If you want to reduce 0.05mm, that is, input-0.05), and then press the "Confirm" key.

Automatic alignment method:

Move the machine tool to a position convenient for measuring the corresponding tool coordinates. Press the "↑" and "↓" keys to move the cursor to the corresponding tool number and press the "F5 tool" key. The system will pop up a dialog box, and then press the "Confirm" key. The length compensation value of the corresponding tool is confirmed.

Method to initialize tool radius and length compensation values:

若想初始所有或当前刀具的半径和长度补偿值为 0 可按“F3”（全部清零）或“F4”（当前清零）。

特别注意：长度补偿值可以为正值或负值，半径补偿值只能为正值。

3.8.3 刀套表

在刀补界面按“F6 刀套表”键进入设置界面。该参数用于设置采用刀具半径补偿时刀套类型。

设置方法：按“↑”“↓”键将光标移动到相应的刀具号上按“确认”键，系统弹出对话框，在对话框中输入相应的刀套型代码后按“确认”即可。

若想初始所有刀具的刀套类型为本把刀具的刀套代码可按“F1 初始化”键即可。

3.8.4 设置刀具数量

在刀补界面按“F7 设置”键可在系统弹出对话框中设置系统管理的刀具总数，本系统可设置 99 把刀具。

3.9 螺补参数设置

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在参数界面下再按两次“参数”键即进入螺补界面。在此界面下可设置螺距补偿。

螺补用于螺距误差自动补偿，补偿由于丝杠的螺距误差对机床传动精度的影响。系统采用存贮式螺距误差补偿：调试时以机床零点为起点，测出丝杠误差曲线，根据误差曲线拟出修正曲线，把修正曲线值输入到修正参数表中，在自动运行中系统按照修正表进行补偿。

To initialize the radius and length compensation values of all or current tools to 0, press "F3" (all cleared) or "F4" (current cleared).

Special note: The length compensation value can be positive or negative, and the radius compensation value can only be positive.

3. 8. 3 A Knife Set

Press "F6, Tool Kit Table" key in the tool compensation interface to enter the setting interface. This parameter is used to set the tool kit type when using tool radius compensation.

Setting method: Move the cursor to the corresponding tool number by pressing the "↑" and "↓" keys, press the "Confirm" key, the system will pop up a dialog box, enter the corresponding tool sleeve type code in the dialog box and press "Confirm".

If you want to initialize the type of tool holder for all tools, the tool holder code for the tool in the main tool can be set by pressing the "F1, initialize" key.

3. 8. 4 Set the Number of Tools

Press "F7 Settings" on the tool supplement interface to set the total number of tools in system management in the dialog box that pops up. This system can set 99 tools.

3.9 Adjustment of Screw Parameters

Press the "Parameter" key twice under the parameter interface to enter the screw compensation interface. In this interface, you can set the screw pitch compensation.

The screw pitch compensation system automatically compensates for pitch errors in the lead screw, addressing their impact on machine tool transmission accuracy. The system employs a stored pitch error compensation mechanism: During calibration, it starts from the machine's zero point to measure the lead screw error curve. Based on this measurement, the system generates a correction curve and inputs its values into a correction parameter table. During automatic operation, the system applies compensation according to the pre-programmed correction parameters.

手动连续

N00000

2016-06-06 06:06

基本参数 [绝对方式]

1, 备用: 0

2, 负向检测点数: 3

3, 正向检测点数: 2

4, 补偿倍率: 1.000

5, 补偿点距um: 80000

检测点 机床坐标mm 误差值um

1)<160.000> 0

2)<80.000> 0

3)<0.000> 0

4)<-80.000> 0

5)<-160.000> 0

6)<-240.000> 0

当前程序99.TXT

指令代码

G53

T01H0D0

状态

M05M09M10不选停

M78M33M41跳段

G00X100%

F2000X100%

S800X100%SP000

机床坐标

X168.000A168.000

Y168.000B168.000

Z168.000

单件时间0: 6

加工件数0

主轴转速0

状态: 正常

F1 X-轴

F2 Y-轴

F3 Z-轴

F4 A-轴

F5 B-轴

F6 检测程序

F7 清零

F8 返回

螺补界面鑫科瑞数控

螺补参数是以轴为单位进行设置存贮的，X、Y、Z、A、B 轴分开设置，通过按“F1”、“F2”、“F3”、“F4”、“F5”切换，每个轴的螺补界面有两个区（基本参数和补偿量设置区），它们的切换是通过“→”“←”键来移动光标实现的。

基本参数设置：

在基本参数使用“↑”“↓”键可以选择当前设置的基本参数。当按下“确认”键后,会弹出输入各轴螺距误差补偿的基本参数对话框,输入螺补的基本信息。

补偿量设置：

在补偿量设置区，就会显示各轴的螺距误差补偿点和补偿量。使用“↑”“↓”键及翻页键可以选择当前补偿点,当按下“确认”键后,会弹出输入补偿量对话框,输入当前补偿点的值。

激光干涉仪进行螺补测量方法：

- (a) 系统设置好检测点点数和间距等螺补基本参数；
- (b) 系统自动生成检测运行程序；
- (c) 执行检测程序,用激光干涉仪测量螺补，激光干涉仪产生.rtl(或者.RTL)检测文件；
- (d) 导入激光干涉仪生成的.rtl(或者.RTL)检测文件；
- (e) 系统会自动计算出螺补值和反向间隙值；

手动连续

N00000

2016-06-06 06:06

基本参数 [绝对方式]

1, 备用: 0

2, 负向检测点数: 3

3, 正向检测点数: 2

4, 补偿倍率: 1.000

5, 补偿点距um: 80000

检测点 机床坐标mm 误差值um

1)<160.000> 0

2)<80.000> 0

3)<0.000> 0

4)<-80.000> 0

5)<-160.000> 0

6)<-240.000> 0

当前程序 99.TXT

指令代码

G53

T01H0D0

状态

M05 M09 M10 不选停

M78 M33 M41 跳段

G00 X100%

F2000 X100%

S800 X100% SP000

机床坐标

X 168.000 A 168.000

Y 168.000 B 168.000

Z 168.000

单件时间 0: 6

加工件数 0

主轴转速 0

状态: 正常

F1 X-轴 F2 Y-轴 F3 Z-轴 F4 A-轴 F5 B-轴 F6 检测程序 F7 清零 F8 返回

Screw patch interface

The screw compensation parameters are configured and stored per axis, with separate settings for X, Y, Z, A, and B axes. These parameters can be switched using the "F1", "F2", "F3", "F4", and "F5" keys. Each axis's compensation interface contains two sections: the basic parameter section and the compensation value setting section. The cursor movement between these sections is achieved by pressing the "→" and "←" keys.

Basic parameter Settings:

The basic parameters can be selected by pressing the "↑" and "↓" keys. After pressing the "Confirm" key, a dialog box for inputting the basic parameters of pitch error compensation for each axis will pop up, and the basic information of the pitch error compensation can be input.

Compensation setting:

In the compensation setting area, the pitch error compensation points and compensation values of each axis will be displayed. The "↑", "↓", and page-turning keys can be used to select the current compensation point. When the "Confirm" key is pressed, a dialog box for inputting compensation values will pop up, and the value of the current compensation point can be entered.

Laser interferometer method for screw repair measurement:

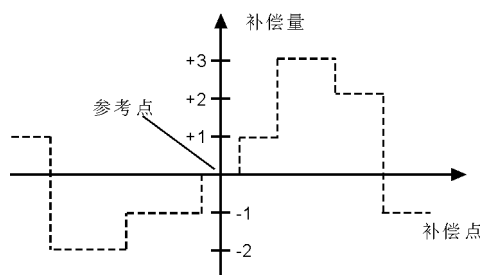
- (a) Set the basic parameters of the system, such as the number and spacing of detection points;
- (b) The system automatically generates detection running programs;
- (c) Perform the detection procedure, measure the screw repair with a laser interferometer, and generate a RTL (or. RTL) detection file;
- (d) Import the.rtl (or RTL) detection file generated by the laser interferometer;
- (e) The system will automatically calculate the screw value and reverse gap value;

自动检测程序生成:

即自动生成用激光干涉仪检测丝杠螺补的程序。进入螺补界面, 设置好基本参数, 按检测程序弹出对话框, 按确定生成对应的螺补检测程序。

补偿点个数可自由设定, 每轴的补偿点最多可达 300 点。各轴螺距误差补偿的基本参数包括如下方面:

17. 备用.
18. 负方向检测点数.
19. 正方向检测点数.
20. 补偿倍率.
21. 补偿点间距 (um) .



系统自动根据基本参数计算出各轴螺距误差补偿点的位置, 各轴螺距误差补偿点时等间距分布的, 用户可以输入各点的补偿值 (本系统要求输入绝对值, 相对于零点的误差)。

螺距误差补偿的补偿点是等间距分布的, 把此间距设定在各轴上。

示例:

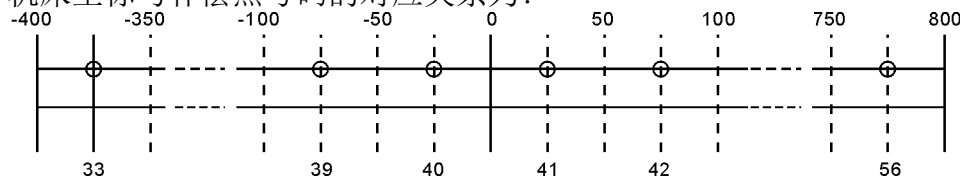
例 1: 直线轴的情况: 当行程为-400mm~+800mm, 螺距误差补偿点间距为 50mm, 参考点的补偿号码为 40, 则可以求出负方向最远端的补偿点号为:

参考点补偿号-机床负方向行程长度/补偿点间距+1=40-400/50+1=33。

正方向最远端的补偿点号为:

参考点补偿号+机床正方向行程长度/补偿点间距=40+800/50=56。

机床坐标与补偿点号码的对应关系为:



在 0 的位置输出补偿量

参数设置如下:

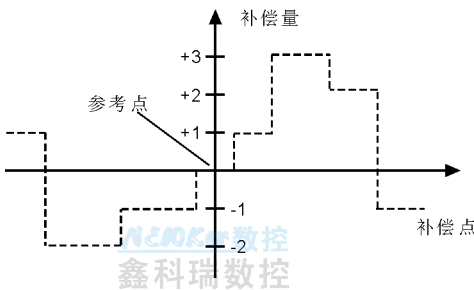
- 参考点的补偿点号: 40
- 负方向最远一端的补偿点号: 33
- 正方向最远一端的补偿点号: 56
- 补偿倍率: 1
- 补偿点间距: 50000

Automated detection program generated:

That is, automatically generate the program for laser interferometer detection of screw thread repair. Enter the screw thread repair interface, set the basic parameters, click the dialog box pop up by the detection program, and click OK to generate the corresponding screw thread repair detection program.

The number of compensation points can be freely set, and the maximum number of compensation points per shaft can reach 300. The basic parameters of pitch error compensation for each shaft include the following aspects:

- 17. reserve .
- 18. Negative direction detection points.
- 19. Positive direction detection points.
- 20. Compensation ratio.
- 21. Compensation point spacing (um).



The system automatically calculates the position of each shaft pitch error compensation point according to the basic parameters. Each shaft pitch error compensation point is evenly distributed, and users can input the compensation value of each point (the system requires the absolute value, the error relative to the zero point).

The compensation points of the pitch error compensation are evenly distributed, and this spacing is set on each axis.

instance :

Example 1: The case of straight shaft: when the stroke is-400mm ~ +800mm, the pitch error compensation point spacing is 50mm,

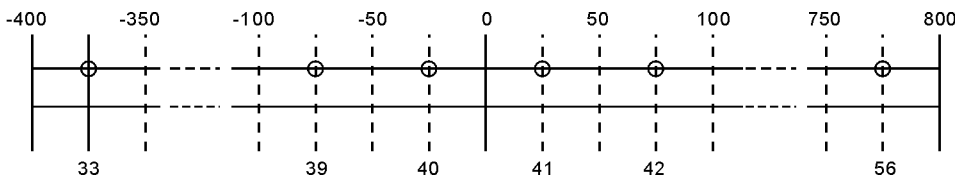
If the reference point compensation number is 40, then the compensation point number at the farthest end of the negative direction can be calculated as:

Reference point compensation number-machine tool negative direction stroke length/compensation point spacing +1=40-400/50+1=33.

The compensation point number at the farthest end of the positive direction is:

Reference point compensation number + machine tool forward stroke length / compensation point spacing =40 +800/50=56.

The corresponding relationship between the machine tool coordinate and the compensation point number is:



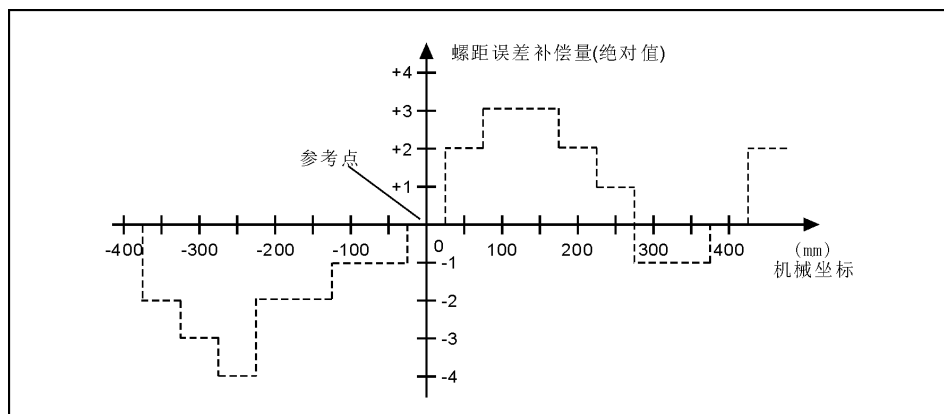
The compensation quantity parameters output at the O position are set as follows:

- Reference point compensation point number: 40
- The compensation point number at the farthest end of the negative direction is 33
- The compensation point number at the farthest end of the positive direction is 56
- Compensation ratio: 1
- Compensation point spacing: 50,000

在对应的补偿点输出补偿量：

号	3	3	3	3	3	3	3	4	4	4	4	4	4	4	4	4	5
码	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	6
补偿量	+	+	+	-	0	-	0	-	+	+	0	-	-	-	0	+	+
	2	1	1	2		1	0	1	2	1	0	1	1	2	0	1	2
																	1

补偿点与补偿量的对照图如下：

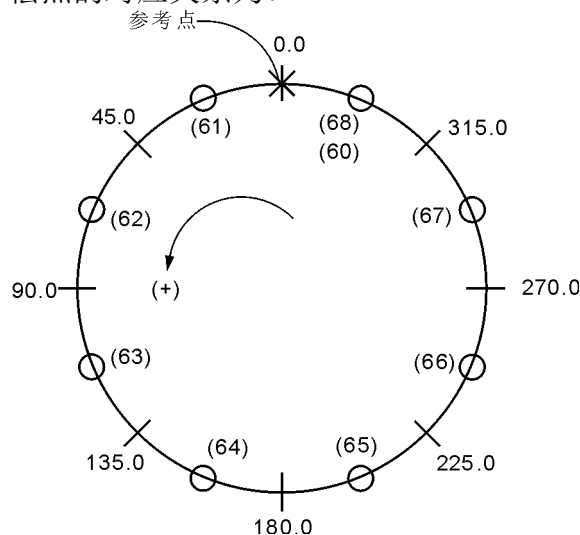


例 2: 当为回转轴时：若每转的移动量为 360 度，轴距误差补偿点的间距为 45 度，参考点的补偿点号码为 60，在回转轴情况下，负方向最远端的补偿点号码通常与参考点的补偿点号码相同。

正方向最远端的补偿点号码为：

参考点的补偿点号+每转的移动量/补偿点的间距=60+360/45=68。

机床坐标与补偿点的对应关系为：



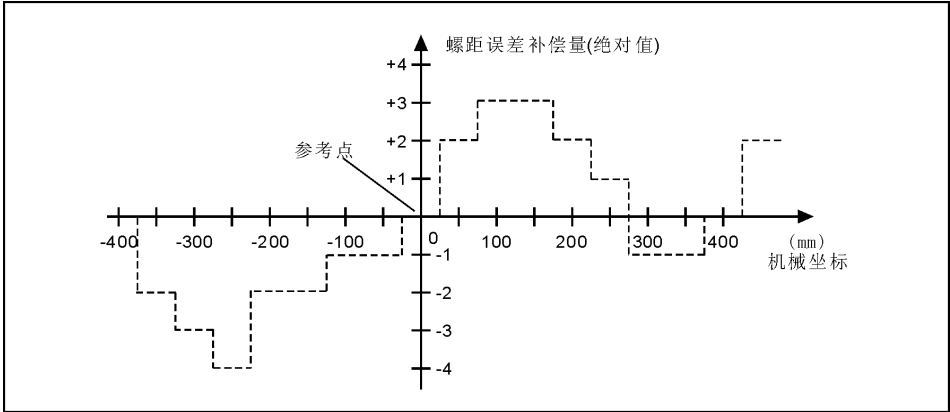
说明：在画圈的位置输出补偿量。若从 61 到 68 的补偿量之和不为零时，因累积每转的螺距误差会出现位置偏移，所以补偿点 60 与 68 应输入相同的补偿量。

在对应的补偿点输出补偿量：

Output compensation amount at the corresponding compensation point:

number	3	3	3	3	3	3	3	4	4	4	4	4	4	4	4	4	5
	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	6
compensation rate	+	+	+	-	0	-	0	-	+	+	0	-	-	-	0	+	+
	2	1	1	2	0	1	0	1	2	1	0	1	1	2	0	1	2

The comparison diagram of compensation point and compensation quantity is as follows:



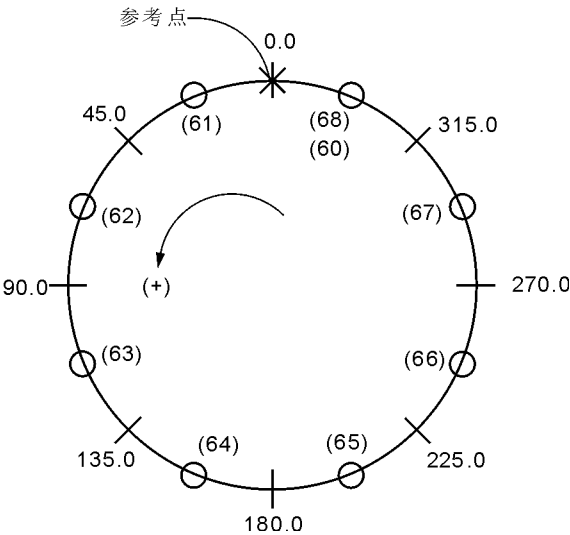
Example 2: When it is a rotary shaft: If the movement per rotation is 360 degrees, the spacing of the shaft distance error compensation points is 45 degrees, and the compensation point number of the reference point is 60, in the case of a rotary shaft, the compensation point number at the farthest end in the negative direction is usually

The reference point has the same compensation point number.

The number of the compensation point at the farthest end in the positive direction is:

Reference point compensation point number + moving amount per turn / spacing of compensation point = 60+360/45=68.

The corresponding relationship between machine tool coordinates and compensation points is:

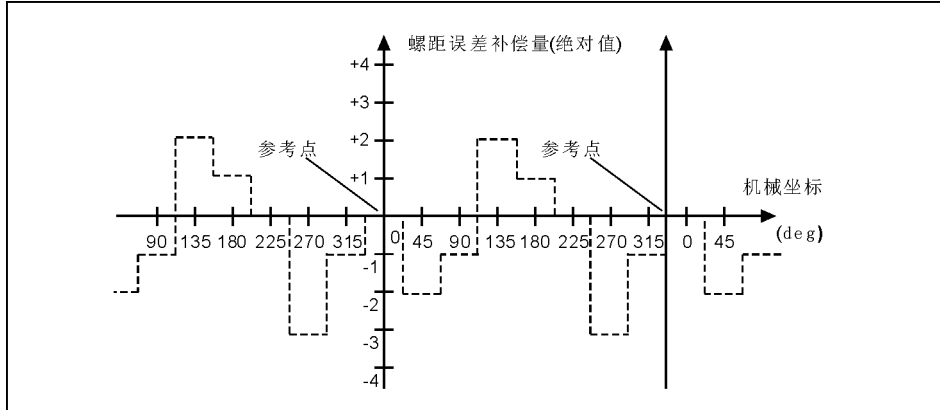


Note: The compensation amount is output at the position where the circle is drawn. If the sum of the compensation amounts from 61 to 68 is not zero, the position offset will occur due to the accumulated pitch error per revolution. Therefore, compensation amounts should be input at both compensation points 60 and 68.

Output the compensation amount at the corresponding compensation point:

补偿点号	60	61	62	63	64	65	66	67	68
设定补偿量	+1	-2	+1	+3	-1	-1	-3	+2	+1

补偿点与补偿量的对照图如下：



3.10 诊断

在参数界面下再按“参数”键或直接“诊断”键即进入诊断界面。在此界面下可设置螺距补偿。

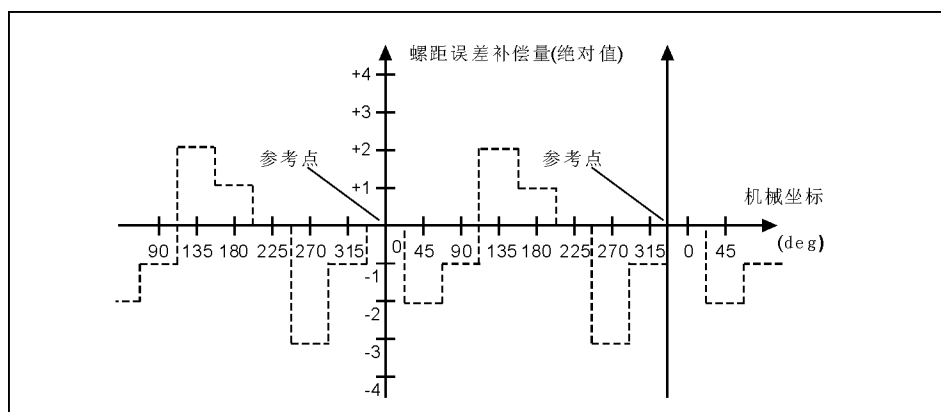
系统诊断界面（输入信号）

在“诊断”界面下可观察系统所有输入、输出状态和机床报警信息。

输入、输出状态按“F2 I/O”键和“上页”“下页”键查看，报警信息按“F4 报警”键查看。

Compensation point number	60	61	62	63	64	65	66	67	68
Set the compensation amount	+1	-2	+1	+3	-1	-1	-3	+2	+1

The comparison diagram of compensation point and compensation quantity is as follows:



3.10 Diagnose

Under the parameter interface, press the "parameter" key or directly press the "diagnosis" key to enter the diagnosis interface. In this interface, pitch compensation can be set.

System diagnostic interface (input signal)

Under the "diagnosis" interface, you can observe the status of all input and output ports of the system and machine tool alarm information.

The status of input and output ports can be viewed by pressing "F2 I/O" key and "Page up" and "Page down" keys. The alarm information can be viewed by pressing "F4 Alarm" key.

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X24	X25	X26	X27	X28	X29	X30	X31	X32	X33	X34	X35		
ALM2	M28	M24	M22	M18	M12	M14	M16						
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当前程序 99.TXT 指令代码 G53 T01H0D0 状态 M05 M09 M10 不选停 M78 M33 M41 跳段 G00 X100% F2000 X100% S800 X100% SP000 机床坐标 X 168.000 A 168.000 Y 168.000 B 168.000 Z 168.000 单件时间 0: 6 加工件数 0 主轴转速 0													
状态: 正常 F2 I/O F3 梯图状态 F4 报警 F5 梯图编辑 F6 复位驱动 F8 返回													

手动 连续		N00000								2016-06-06 06:06																																																																																																																																	
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系统诊断界面输出信号查看界面 科瑞数控

在输入输出诊断界面，上排数字 0 或 1 表示状态，1 表有效，0 表无效。

手动 连续		N00000								2016-06-06 06:06																																																																																																																																	
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System diagnostic interface output signal view interface

In the input/output diagnostic interface, the upper row number 0 or 1 indicates the status, 1 is valid, 0 is invalid.

手动连续

N00000

2016-06-06 06:06

输出点

0	0	0	0	0	0	0	0
Y00	Y01	Y02	Y03	Y04	Y05	Y06	Y07
M61	M63	M65	M67	M69	M71	M73	M75
0	0	0	0	0	0	0	0
Y08	Y09	Y10	Y11	Y12	Y13	Y14	Y15
M32	M79	M10	M08	M05	M04	M03	M75
0	0	0	0	0	0	0	1
Y16	Y17	Y18	Y19	Y20	Y21	Y22	Y23
	INTH	+T	-T	S04	S03	S02	S01
0	0	0	0	0	0	0	0
Y24	Y25	Y26	Y27	Y28	Y29	Y30	Y31

当前程序

99.TXT

指令代码

G53

T01H0D0

状态

M05M09M10不选停

M78M33M41跳段

G00X100%

F2000X100%

S800X100%SP000

机床坐标

X168.000A168.000

Y168.000B168.000

Z168.000

单件时间

0: 6

加工件数

0

主轴转速

0

状态: 正常

F2I/O

F3梯形状态

F4报警

F5梯形编辑

F6复位驱动

F8返回

鑫科瑞数控

系统诊断界面报警信息界面

该界面第一行显示的为当前报警信息，以下为历史报警信息，总共记录 10 条，超过的自动清除，只显示最近发生的 10 项报警信息。

- 161 -

手动连续

N00000

2016-06-06 06:06

输出点

0	0	0	0	0	0	0	0
Y00	Y01	Y02	Y03	Y04	Y05	Y06	Y07
M61	M63	M65	M67	M69	M71	M73	M59
0	0	0	0	0	0	0	0
Y08	Y09	Y10	Y11	Y12	Y13	Y14	Y15
M32	M79	M10	M08	M05	M04	M03	M75
0	0	0	0	0	0	0	1
Y16	Y17	Y18	Y19	Y20	Y21	Y22	Y23
	INTH	+T	-T	S04	S03	S02	S01
0	0	0	0	0	0	0	0
Y24	Y25	Y26	Y27	Y28	Y29	Y30	Y31

当前程序

99.TXT

指令代码

G53

T01H0D0

状态

M05M09M10不选停

M78M33M41跳段

G00X100%

F2000X100%

S800X100%SP000

机床坐标

X168.000A168.000

Y168.000B168.000

Z168.000

单件时间

0:6

加工件数

0

主轴转速

0

状态: 正常

F2I/O

F3梯形状态

F4报警

F5梯形编辑

F6复位驱动

F8返回

System diagnostic interface alarm information interface

The first line of the interface displays the current alarm information, and the following is the historical alarm information. A total of 10 items are recorded, and the excess is automatically cleared, and only the last 10 alarm information is displayed.

手动 连续		N00000		2016-06-06 06:06	
清除历史报警按删除键， 刷新报警信息按R键				当前程序 99.TXT	
主轴编码器线数检测：0 [第1编码器]				指令代码	
1, [当前报警]：无				G53	
2, [历史报警]：无				T01H0D0	
				状态	
				M05 M09 M10 不选停	
				M78 M33 M41 跳段	
				G00 X100%	
				F2000 X100%	
				S800 X100% SP000	
				机床坐标	
				X 168.000 A 168.000	
				Y 168.000 B 168.000	
				Z 168.000	
				单件时间 0: 6	
				加工件数 0	
				主轴转速 0	
状态：正常					
	F2 I/O	F3 梯图状态	F4 报警	F5 梯图编辑	F6 复位驱动
					F8 返回

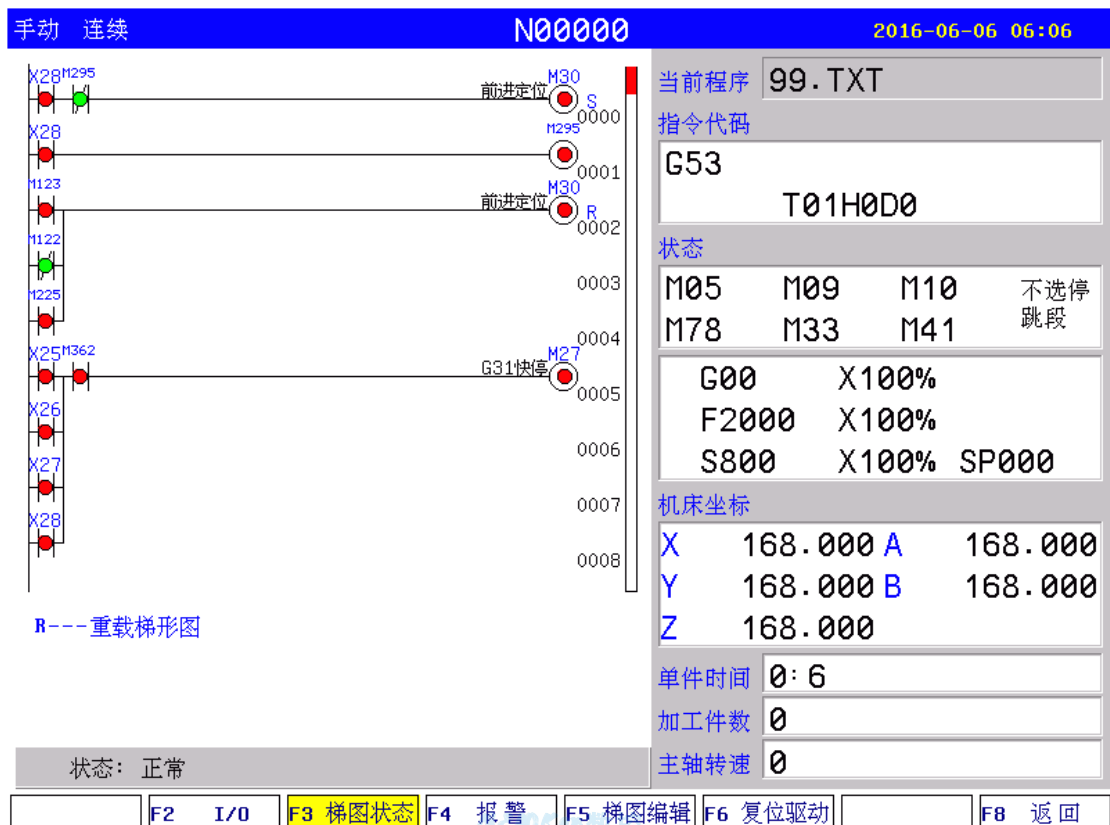
系统 PLC 查看和编辑界面

按“F3”查看 PLC 梯形图实时诊断界面，按“F5”编辑 PLC 梯形图，编辑梯形图或诊断梯形图时，按“S”键为搜索功能。当在梯形图编辑界面下修改梯形图并保存后，重新启动修改后的梯图才会执行，若不需要重新启动而立即执行，则在查看 PLC 梯形图实时诊断界面按“R”键即可。

手动 连续		N00000		2016-06-06 06:06	
清除历史报警按删除键， 刷新报警信息按R键				当前程序 99.TXT	
主轴编码器线数检测： 0 [第1编码器]				指令代码	
1, [当前报警]： 无				G53	
2, [历史报警]： 无				T01H0D0	
				状态	
				M05 M09 M10 不选停	
				M78 M33 M41 跳段	
				G00 X100%	
				F2000 X100%	
				S800 X100% SP000	
				机床坐标	
				X 168.000 A 168.000	
				Y 168.000 B 168.000	
				Z 168.000	
				单件时间 0: 6	
				加工件数 0	
				主轴转速 0	
状态： 正常					
	F2 I/O	F3 梯图状态	F4 报警	F5 梯图编辑	F6 复位驱动
					F8 返回

System PLC view and edit interface

Press "F3" to access the real-time diagnostic interface for PLC ladder diagrams. Press "F5" to edit PLC ladder diagrams. When editing ladder diagrams or diagnostic diagrams, press the "S" key to activate the search function. After modifying and saving the ladder diagram in the editing interface, the updated configuration will only take effect upon system restart. To execute changes immediately without restarting, simply press the "R" key in the real-time diagnostic interface for PLC ladder diagrams.



鑫科瑞数控

3.11 编程操作

在任何菜单下，按“程序”键，进入程序编制状态(如下图)。

程序管理采用文件管理方式，本系统可存贮 32M 字节容量程序，对于程序个数无限制。程序编辑采用全屏幕操作方式。

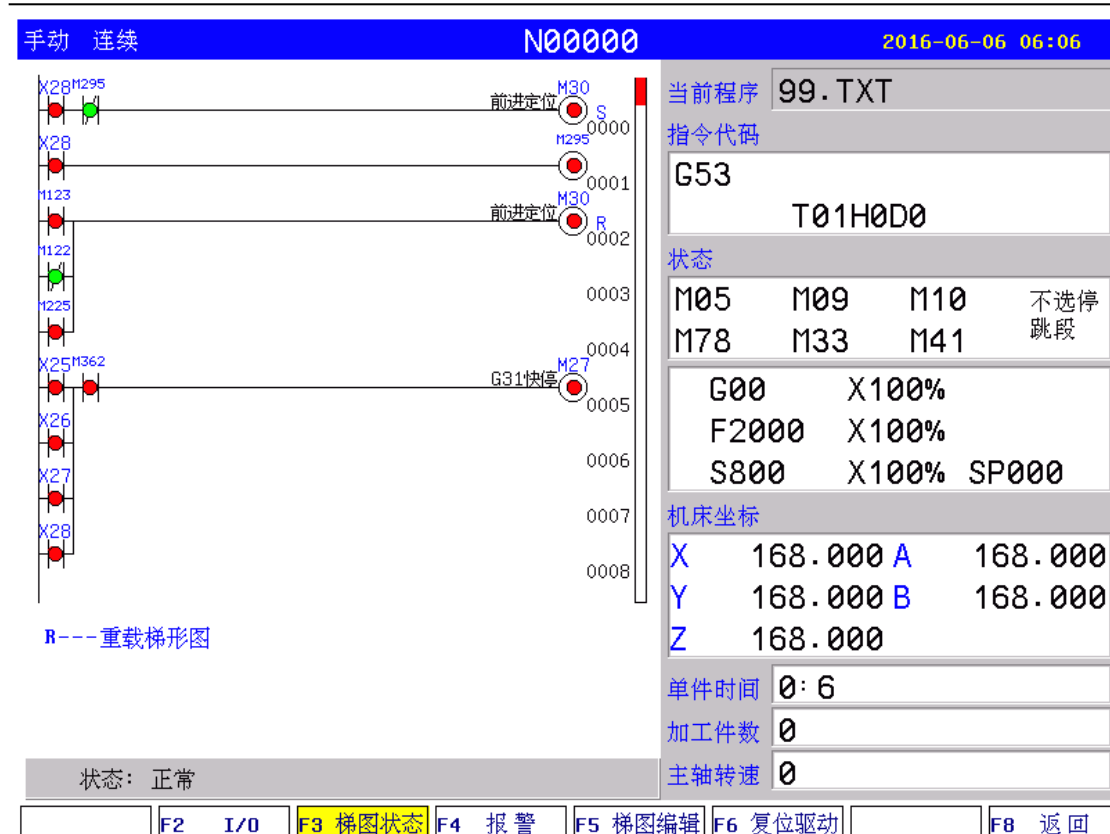
程序编辑界面中部为程序目录区，显示内存中所有程序名和文件夹，蓝色显示”为当前程序或文件夹；可通过“↑、↓、上页、下页”键任意选择程序，并以反白方式显示，按“确认”键即进入相应操作状态。

3.11.1 编辑

选择“新建查找”功能后，系统弹出对话框输入程序名，若输入系统已存在的文件名，则将原文件调出，若输入的文件名不存在，则建立一个新文件。

程序名可以是数字、字母或混用，长度为 100 位。

本系统程序名不允许重名，新建好一个程序名或选中一个程序名后按“确认”键进入程序编辑界面。



3.11 Programming Operations

Under any menu, press the "program" key to enter the programming state (as shown below).

The program management adopts the file management mode. The system can store 32M byte capacity programs, and there is no limit to the number of programs. The program editing adopts the full-screen operation mode.

In the middle of the program editing interface is the program directory area, which shows all program names and folders in memory. The blue "" indicates the current program or folder. You can select any program by pressing the keys "↑, ↓, previous page, and next page", and the program is displayed in white reverse. Press the "Confirm" key to enter the corresponding operation state.

3.11.1 Edit

After selecting the "Create search" function, a dialog box will pop up to input the program name. If the system has an existing file name, the original file will be called up; if the file name does not exist, a new file will be created.

The program name can be a number, a letter, or a mix of letters and numbers, and can be up to 100 characters long.

The program name of this system is not allowed to be duplicated. After creating a new program name or selecting a program name, press the "Confirm" key to enter the program editing interface.

手动 连续		N00000		2016-06-06 06:06						
系统中的文件夹/文件列表 当前目录: /NC		当前程序 99.TXT								
<table><tr><td>00</td></tr><tr><td>001</td></tr><tr><td>99-1</td></tr><tr><td>99.TXT</td></tr><tr><td>990TDI-1</td></tr></table>		00	001	99-1	99.TXT	990TDI-1	指令代码			
00										
001										
99-1										
99.TXT										
990TDI-1										
		G53 T01H0D0								
		状态								
		M05 M09 M10 不选停 M78 M33 M41 跳段								
		G00 X100% F2000 X100% S800 X100% SP000								
		机床坐标								
		X 168.000 A 168.000 Y 168.000 B 168.000 Z 168.000								
		单件时间 0: 6								
		加工件数 0								
		主轴转速 0								
编译--P; 串口接收--R, 发送--T, 联机--L		状态: 正常								
F1 新建查找	F2 拷贝	F3 更名	F4 信息	F5 上一级	F6 打开优盘	F7 执行程序	F8 返回			

鑫科瑞数控

文件名和程序内容的中文输入法:

程序目录界面下的“新建/查找”,“拷贝”,“更名”,“拷入系统”,“拷入优盘”可以通过全拼拼音输入法输入汉字;按系统提示信息可以转换输入法。在程序编辑界面按系统提示信息也可以输入汉字。

手动 连续		N00000		2016-06-06 06:06													
系统中的文件夹/文件列表 当前目录: /NC		当前程序 99.TXT															
<table border="1"> <tr><td>00</td></tr> <tr><td>001</td></tr> <tr><td>99-1</td></tr> <tr><td>99.TXT</td></tr> <tr><td>990TDI-1</td></tr> </table>		00	001	99-1	99.TXT	990TDI-1	指令代码										
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		G53															
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		状态															
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M05	M09	M10	不选停														
M78	M33	M41	跳段														
		<table border="1"> <tr> <td>G00</td> <td>X100%</td> </tr> <tr> <td>F2000</td> <td>X100%</td> </tr> <tr> <td>S800</td> <td>X100% SP000</td> </tr> </table>				G00	X100%	F2000	X100%	S800	X100% SP000						
G00	X100%																
F2000	X100%																
S800	X100% SP000																
		机床坐标															
		<table border="1"> <tr> <td>X</td> <td>168.000</td> <td>A</td> <td>168.000</td> </tr> <tr> <td>Y</td> <td>168.000</td> <td>B</td> <td>168.000</td> </tr> <tr> <td>Z</td> <td>168.000</td> <td></td> <td></td> </tr> </table>				X	168.000	A	168.000	Y	168.000	B	168.000	Z	168.000		
X	168.000	A	168.000														
Y	168.000	B	168.000														
Z	168.000																
编译--P; 串口接收--R, 发送--T, 联机--L		单件时间 0: 6															
		加工件数 0															
		主轴转速 0															
状态: 正常																	
F1 新建查找		F2 拷贝	F3 更名	F4 信息	F5 上一级	F6 打开优盘	F7 执行程序	F8 返回									

Chinese input method for file name and program content:

In the program directory interface, "New/Find", "Copy", "Renew", "Copy to system", and "Copy to USB" can be used to input Chinese characters through the full Pinyin input method. The input method can be converted according to the system prompts. In the program editing interface, Chinese characters can also be input according to the system prompts.

手动 连续		N00000		2016-06-06 06:06	
编辑程序: /NC/99.TXT			当前程序 99.TXT		
<pre> N1 G00 X0 Z0 N2 G00 X200 Z200 N3 G00 U-44 N4 W-5 N5 G01 U4 W-2 F300 N6 W-11 N7 U-4 W-2 N8 W-5 N9 U10 W-10 N10 W-6 N11 G02 U-6 W-9 I12 K-9 F5 N12 U10 W-15 I25 K0 F300 </pre>			指令代码 G53 T01H0D0		
第1行 / 共28行			状态 M05 M09 M10 不选停 M78 M33 M41 跳段		
			G00 X100% F2000 X100% S800 X100% SP000		
			机床坐标 X 168.000 A 168.000 Y 168.000 B 168.000 Z 168.000		
			单件时间 0: 6 加工件数 0 主轴转速 0		
状态: 正常					
F1 编译	F2 首尾行	F3 示教	F4 定位	F6 行删除	F7 > >
F8 中文					

在编辑状态下屏幕左上角提示所编辑的程序名; 左面为程序内容, 右面为机床状态信息区, 在编辑状态可进行以下操作:

(1) 当前光标定位:

按“↑↓←→”键可将光标移到程序内容的任意位置。

按“上页”键光标到上页。

按“下页”键光标到下页。

(2) 字符修改: 在光标处删除要改的字符, 再直接输入新字符。

(3) 字符插入: 在光标处直接输入新字符。当输入为字母时, 字母前面自动产生空格。如果想输入一个空格, 则先输入一个字母, 再删除这个字母即可。

(4) 字符删除: 光标处直接按“删除”键。

(5) 行间插入: 直接按“确认”键, 若光标在行首则在当前行之前插入一行, 否则在当前行之后插入一行。

(6) “快速”叠加键操作:

第一功能页:

A、“F1”: 编译当前程序。

B、“F2”: 到程序第一行或最后一行。

C、“F3”: 示教功能, 进入手轮状态; 此时可以按相应坐标轴字母 X/Y/Z/A/B/C/ “F4” (所有坐标轴) 读取当然刀具当前工件坐标系的坐标。

手动连续

N00000

2016-06-06 06:06

编辑程序: /NC/99.TXT

N1 G00 X0 Z0

N2 G00 X200 Z200

N3 G00 U-44

N4 W-5

N5 G01 U4 W-2 F300

N6 W-11

N7 U-4 W-2

N8 W-5

N9 U10 W-10

N10 W-6

N11 G02 U-6 W-9 I12 K-9 F5

N12 U10 W-15 I25 K0 F300

第1行 / 共28行

当前程序99.TXT

指令代码

G53

T01H0D0

状态

M05M09M10不选停

M78M33M41跳段

G00X100%

F2000X100%

S800X100%SP000

机床坐标

X168.000A168.000

Y168.000B168.000

Z168.000

单件时间0: 6

加工件数0

主轴转速0

状态: 正常

F1 编译F2 首尾行F3 示教F4 定位F6 行删除F7 > >F8 中文

In the editing state, the name of the edited program is displayed in the upper left corner of the screen; the content of the program is on the left, and the status information area of the machine tool is on the right. In the editing state, the following operations can be performed:

- (1) Current cursor position:
Press the "↑↓←→" keys to move the cursor to any position in the program.
Press the "Previous page" key to move the cursor to the previous page.
Press the "Next Page" key to move to the next page.
- (2) Character modification: Delete the character to be changed at the cursor and then directly enter the new character.
- (3) Character insertion: Type a new character directly at the cursor. When typing a letter, a space is automatically generated before the letter. If you want to type a space, type a letter first and then delete it.
- (4) Character deletion: press the Delete key directly at the cursor.
- (5) Line insertion: Press the "OK" key directly. If the cursor is at the beginning of the line, insert a line before the current line; otherwise, insert a line after the current line.
- (6) "Quick" overlay button operation:

First function page:
A, "F1": Compile the current program.
B, "F2": to the first or last line of the program.
C, "F3": Teaching function, enter the hand rotation state; at this time, you can press the corresponding coordinate axis letters X/Y/Z/A/B/C/F4 (all coordinate axes) to read the coordinates of the current workpiece coordinate system of the tool.

示教功能可以通过杂类参数 P911 启动外接开关 M18、M24、M28 记录坐标点。

D、“F4”：定位到指定行。

E、“F5”：无操作。

F、“F6”：删除当前行。

G、“F7”：第一、二功能页选择。

H、“F8”：汉字与字符输入转换。

第二功能页：

A、“F1”：删除程序块。

B、“F2”：拷贝指定程序块。

C、“F3”：程序排序。

D、“F4”：查找指定字符。

E、“F5”：替换指定字符。

F、“F6”：全部内容替换指定字符。

G、“F7”：第一、二功能页选择。

H、“F8”：汉字与字符输入转换。

3.11.2 拷贝

在程序主界面，按“↑”“↓”键，选定要拷贝的源程序，之后按“F2”键，在系统弹出的对话框中输入新的程序名，即复制一个不同程序名而内容完全相同的程序，以便修改、备份和更名。

拷贝新文件名后如果加上“[”，会自动删除原文件中的乱字符。

3.11.3 删除

在程序主界面，按“↑”“↓”键，选定要删除的程序，之后按“删除”键则该程序删除。

按“E”输入机床厂家密码，则删除内存中或者U盘中的全部程序文件。

删除操作需谨慎，一旦删除后程序将无法恢复。

3.11.4 更名

在程序主界面，按“↑”“↓”键，选定要更名的源程序，之后按“F3”键，在系统弹出的对话框中输入新的程序名，便将当前程序更名为新输入的程序名。

3.11.5 信息

在程序主界面，按“↑”“↓”键，选定要查看的程序，之后按“F4”键，在系统弹出的对话框将当前程序的大小和系统剩余空间大小显示出来。

3.11.6 程序检查

在程序主界面，按“↑”“↓”键，选定要检查的程序按“P”键，系统自动对程序格式及语法进行检查。若发现错误自动提示。

The teaching function can start the external switches M18, M24 and M28 to record the coordinate points through miscellaneous parameters P 911.

D, "F4": Position to the specified line.

E, "F5": no operation.

F, "F6": Delete the current line.

G, "F7": Select the first and second function pages.

H, "F8": Chinese characters and character input conversion.

Second function page:

A, "F1": Delete the program block.

B, "F2": copy the specified program block.

C, "F3": program sorting.

D, "F4": Look for a specified character.

E, "F5": Replace specified characters.

F, "F6": Replace all content with a specified character.

G, "F7": Select the first and second function pages.

H, "F8": Chinese characters and character input conversion.

3.11.2 copy

In the main interface of the program, press "↑" and "↓" keys to select the source program to be copied. Then press "F2" key. Enter a new program name in the dialog box that pops up in the system. That is, copy a program with the same content but different program names, so as to modify, backup and rename it.

If you add "[" to the copied new file name, the wild character in the original file will be automatically deleted.

3.11.3 Delete

In the main interface of the program, press "↑" and "↓" keys to select the program to be deleted, and then press "Delete" key to delete the program.

Enter the machine tool manufacturer's password as "E" to delete all program files in memory or on the U disk.

Deletion should be done carefully, once deleted the program will not be able to recover.

3.11.4 Rename

In the main interface of the program, press "↑" and "↓" keys to select the source program to be renamed. Then press "F3" key. In the dialog box displayed by the system, enter the new program name, and the current program will be renamed to the new program name.

3.11.5 Information

In the main interface of the program, press "↑" and "↓" keys to select the program to be viewed. Then press "F4" key. The dialog box displayed by the system will show the size of the current program and the remaining space size of the system.

3.11.6 Program Checkout

In the main interface of the program, press "↑" and "↓" keys to select the program to be checked. Press "P" key, and the system will automatically check the format and grammar of the program. If errors are found, the system will prompt automatically.

3.11.7 文件夹管理

在本系统中可建立文件，程序主界面按“F1”键输入文件名之后输入“。”即建立一个文件夹，其名称后的个“夹”标示。

光标移动到文件名上按“确认”键即将文件夹打开，又可在其里面新建程序文件和文件夹。

在下一级文件夹向上一级文件夹换按“F5”即可。

要删除文件夹，直接将光标移动到文件夹名上按“删除”即可。

3.11.8 选择自动运行程序

在程序主界面，按“↑”“↓”键，选定要执行的程序按“F7”键，系统自动选择该程序并切换到主界面（进入程序前的主界面）。

3.11.9 程序通讯

系统可采用 RS232 串口方式进行程序文件的传递。

发送

发送是将系统上选定的程序发送到另一台系统，或发送到计算机长期保存下来，在程序主界面，按“↑”“↓”键，选定要发送的程序按“T”键，发送过程中若要中断发送，可以按“复位”键。

接收

接收是将接收另外一台系统上选定的程序，或接收计算机上的程序（必须是文本文件），在程序主界面，按“R”在对话框中输入接收后的文件名，按“确认”键即可，在接收过程中若要中断，可以按“复位”键。

- 注意：
- 1、在用户 PC 机端需要使用本系统专用通讯软件方可传输程序。
 - 2、需特别注意的是将 PC 机和数控系统双方的传输速率设置一致，否则无法传输。
 - 3、PC 机与设备联接的串口号应与数控系统设置一致。
 - 4、RS232 通讯线的长度不得大于 10 米。
 - 5、PC 机主编辑的程序必须是文本文件格式。

3.11.10 U 盘管理

在本系统中可通过 USB 接口用优盘进行与 PC 机或其他系统的程序交换和参数文件及系统软件的升级与备份。

注意：使用优盘时文件夹名中不能有空格符号。

当将优盘插入 USB 接口后，在程序主界面下按“F6 打开优盘”即可进入优

3.11.7 File Folder Management

In this system, you can create a file. The main interface of the program is "F1" to enter the file name and then enter "." to create a folder. The "File" label after the name indicates.

Move the cursor to the file name and press the "OK" button to open the folder, and you can create new program files and folders in it.

Press F5 to switch from the lower level folder to the upper level folder.

To delete a folder, simply move the cursor over the name of the folder and press Delete.

3.11.8 Select the Program to Run Automatically

In the main interface of the program, press "↑" and "↓" keys to select the program to be executed. Press "F7" key, and the system will automatically select the program and switch to the main interface (the main interface before entering the program).

3.11.9 Programme Communication

The system can use RS232 serial port for program file transfer.
transmit by radio

Send is to send the selected program on one system to another system, or to send it to a computer for long-term storage. On the main interface of the program, press "↑" and "↓" keys to select the program to be sent, press "T" key, and if you want to interrupt the sending process, you can press "reset" key.

receive

Acceptance is to receive a selected program on another system, or to receive a program on the computer (must be a text file). In the main interface of the program, press "R" to enter the name of the received file in the dialog box, and press the "Confirm" key. If you want to interrupt the acceptance process, you can press the "Reset" key.

Note: 1. The system dedicated communication software is required on the user's PC to transfer the program.

2、 It should be noted that the transmission rate between PC and CNC system should be set in the same way, otherwise it cannot be transmitted.

3、 The serial port number connected to the PC and equipment should be consistent with the CNC system.

4、 The length of RS232 communication line shall not be greater than 10 meters.

5、 The program edited by the PC must be in text file format.

3.11.10 U Disk Management

In this system, USB can be used to exchange programs with PC or other systems through USB interface, and upgrade and backup of parameter files and system software.

Note: Do not use spaces in folder names when using USB drives.

When the USB drive is inserted into the USB port, press "F6 to open the USB drive" under the main interface of the program to enter the USB drive

盘管理界面。再按“F6 打开优盘”键可退回到系统界面。

手动 连续		N00000		2016-06-06 06:06	
优盘中的文件夹/文件列表 当前目录: /USB		当前程序 99.TXT 指令代码 G53 T01H0D0 状态 M05 M09 M10 不选停 M78 M33 M41 跳段 G00 X100% F2000 X100% S800 X100% SP000 机床坐标 X 168.000 A 168.000 Y 168.000 B 168.000 Z 168.000 单件时间 0: 6 加工件数 0 主轴转速 0			
00 001 99-1 99.TXT 990TDI-1 CURBMP0.BMP CURBMP1.BMP CURBMP10.BMP CURBMP11.BMP CURBMP12.BMP CURBMP13.BMP CURBMP14.BMP		编译--P; 程式全部导入--R, 全部导出--T, 联机--L 状态: 正常			
F1 导出参数	F2 导入参数	F3 拷入优盘	F4 拷入系统	F5 上一级	F6 退出优盘
F7 执行程序	F8 返回				

A、加工程序管理

优盘中的程序文件或文件夹拷入系统:

将优盘插入后，在程序主界面按下“F6 打开优盘”键进入 U 盘目录，通过按“↑”“↓”键移动光标选中要拷入系统的程序文件或文件夹，此时按“F4 拷入系统”键，则会弹出要求输入文件名的对话框，键入程序文件或文件夹名并按“确认”键。即将 U 盘上的当前文件或文件夹拷入系统中，系统自动切换到程序主界面，若键入的程序文件或文件夹名已存在会即弹出是否覆盖原程序文件或文件夹的信息。

如果 U 盘上的程序全部拷入系统则按“R”键。

系统中的程序文件或文件夹拷入优盘:

先应在系统程序主界面按“↑”“↓”键移动光标选中需要拷贝的程序文件或文件名，再按“F6”键，进入优盘界面，再按“F3 拷入优盘”键则会弹出要求输入文件名的对话框，键入程序文件或文件夹名并按“确认”键，即将系统当前文件或文件夹拷入 U 盘中。若键入的程序文件或文件夹名已存在会即弹出是否覆盖原程序文件或文件夹的信息。

如果系统中的文件全部拷入 U 盘则按“T”键。

Disk management interface. Press "F6 to open the USB drive" key to return to the system interface.

手动 连续		N00000		2016-06-06 06:06																											
优盘中的文件夹/文件列表 当前目录: /USB			当前程序 99.TXT 指令代码 G53 T01H0D0																												
<table border="1"> <tr><td>00</td></tr> <tr><td>001</td></tr> <tr><td>99-1</td></tr> <tr><td>99.TXT</td></tr> <tr><td>990TDI-1</td></tr> <tr><td>CURBMP0.BMP</td></tr> <tr><td>CURBMP1.BMP</td></tr> <tr><td>CURBMP10.BMP</td></tr> <tr><td>CURBMP11.BMP</td></tr> <tr><td>CURBMP12.BMP</td></tr> <tr><td>CURBMP13.BMP</td></tr> <tr><td>CURBMP14.BMP</td></tr> </table>			00	001	99-1	99.TXT	990TDI-1	CURBMP0.BMP	CURBMP1.BMP	CURBMP10.BMP	CURBMP11.BMP	CURBMP12.BMP	CURBMP13.BMP	CURBMP14.BMP	状态 <table border="1"> <tr> <td>M05</td> <td>M09</td> <td>M10</td> <td>不选停</td> </tr> <tr> <td>M78</td> <td>M33</td> <td>M41</td> <td>跳段</td> </tr> </table> <table border="1"> <tr> <td>G00</td> <td>X100%</td> </tr> <tr> <td>F2000</td> <td>X100%</td> </tr> <tr> <td>S800</td> <td>X100% SP000</td> </tr> </table>			M05	M09	M10	不选停	M78	M33	M41	跳段	G00	X100%	F2000	X100%	S800	X100% SP000
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Y	168.000	B	168.000																												
Z	168.000																														
状态: 正常			单件时间 0: 6 加工件数 0 主轴转速 0																												
<table border="1"> <tr> <td>F1 导出参数</td> <td>F2 导入参数</td> <td>F3 拷贝优盘</td> <td>F4 拷贝系统</td> <td>F5 上一级</td> <td>F6 退出优盘</td> <td>F7 执行程序</td> <td>F8 返回</td> </tr> </table>						F1 导出参数	F2 导入参数	F3 拷贝优盘	F4 拷贝系统	F5 上一级	F6 退出优盘	F7 执行程序	F8 返回																		
F1 导出参数	F2 导入参数	F3 拷贝优盘	F4 拷贝系统	F5 上一级	F6 退出优盘	F7 执行程序	F8 返回																								

A. Process management

Copy program files or folders from the USB drive to the system:

To copy files from a USB drive to the system, insert the device and press the "F6" key on the main interface to access the USB directory. Use the arrow keys (↑↓) to navigate the selection bar, then click "F4 Copy to System" to open the file dialog. Enter the desired filename or folder path and confirm. The system will automatically switch back to the main interface after copying. If the target path already exists, you'll see an option to overwrite the existing file or folder.

If all the programs on the U disk are copied into the system, press the "R" key.

Copy program files or folders in the system to a USB drive:

To copy a program file or folder, first navigate the system's main interface using the ↑/↓ keys to select the target location. Press F6 to enter the USB drive interface, then press F3 "Copy to USB" to open the dialog box asking for a filename. Enter the desired name and click Confirm to transfer the contents to the USB drive. If the target path already exists, the system will prompt you to choose whether to overwrite the original file or folder.

If all the files in the system are copied to the U disk, press the "T" key.

注意：拔出优盘前，必须返回到显示系统程序文件目录的界面下（退出优盘）。否则优盘中刚刚被拷入的数据可能会丢失。

使用优盘时文件夹名中不能有空格符号。

B、用优盘进行参数、系统软件的管理

在本系统中可以通过优盘接口进行参数文件、系统软件的传送，从而达到升级、更新或备份的目的。其具体操作方法如下：

用优盘进行参数文件、系统软件的导入（升级、更新）系统：

先将优盘插入系统USB接口后，在面板上按“程序”进入程序主界面，按“F6 打开优盘”键，则显示优盘中的文件。通过按光标键上下移动选中要导入系统的文件夹并按确认打开，在能看到需要导入的文件后，按“F2导入参数”键，输入密码后，按“确认”键，等待几秒则导入参数成功。导入成功后，按“F6退出优盘”键，退出优盘。重启系统，系统自动重载新的文件，以达升级参数的目的。

注：需要导入的参数必须在电脑中单独建一个文件夹，然后拷到优盘的根目录下下来导入，以防止操作失误损坏系统文件系统。

优盘进行参数文件导出备份

先将优盘插入系统 USB 接口后，在面板上按“程序”进入程序主界面，按“F6 打开优盘”键，则显示优盘中的文件。按“F1 导出参数”键并输入密码后，按“确认”键，等待几秒则导出参数成功。即将该系统所有的参数导出到优盘里。按“F6 退出优盘”键，退出优盘。

注：导出参数用的优盘最好是个空盘，方便文件整理（参数文件有几十个之多），或在电脑上先建一个文件夹，导出前先打开文件夹，将参数导在文件夹中。

3. 11. 11 DXF 文件转换成 G 代码

在程序文件目录下选中 DXF 文件，按“-”号键则将其转换为 G 代码文件，该 G 代码文件以 .CNC 作为文件扩展名。要求导入系统的 DXF 文件必须以 .DXF 或 .dxf 作为文件扩展名。使用时参见加工参数 P400-P402。

在产生 G 代码文件时，根据当前文件目录下是否有 HEADDXF.TXT 和 ENDDXF.TXT 文件（或 headdxf.txt 和 enddxf.txt 文件）自动产生相应头尾代码。

注意：头文件和尾文件必须处于当前文件目录下。

3. 11. 12 FTP 服务器文件夹操作方法

如果系统连接上了 Internet 网络或连接上了 WiFi 情况下，可以进入 FTP 服务器文件夹。

Note: Before you can remove the USB drive, you must return to the interface that displays the system program file directory (exit the USB drive). Otherwise, the data that has just been copied into the USB drive may be lost.

Do not use spaces in folder names when using USB drives.

B. Use USB to Manage Parameters and System Software

In this system, the parameter files and system software can be transferred through the USB interface, so as to achieve the purpose of upgrade, update or backup. The specific operation method is as follows:

Import (upgrade, update) system parameter files and system software using USB drive:

First, insert the USB drive into the system's USB port. On the panel, press "Program" to enter the main interface. Press the "F6 Open USB Drive" key to display files on the drive. Use the cursor keys to navigate and select the folder you want to import. After selecting it, confirm the import by pressing "F2 Import Parameters". Enter your password and click "Confirm". Wait a few seconds for successful parameter import. Finally, press "F6 Close USB Drive" to exit. Restart the system, which will automatically load new files to update parameters.

Note: The parameters to be imported must be created in a separate folder on the computer, and then copied to the root directory of the USB drive to import, so as to prevent system file system damage due to operational errors.

The USB drive exports the parameter file for backup

First, insert the USB drive into the system's USB port. After connecting, press "Program" on the panel to enter the main interface. Press the "F6 Open USB Drive" key to display files from the drive. Then press "F1 Export Parameters" and enter your password. After confirming, wait a few seconds for successful parameter export. This will transfer all system parameters to the USB drive. Finally, press "F6 Exit USB Drive" to complete the process.

Note: The USB drive used for exporting parameters is preferably an empty disk, which is convenient for file organization (there are dozens of parameter files), or you can build a folder on the computer first. Before export, open the folder and export the parameters in the folder.

3.11.11 The DXF File Is Converted to G Code

Select the DXF file in the program directory. Press the "-" key to convert it into a G code file, which will be saved as a.CNC extension. Import files must have.DXF or.dxf extensions. Refer to machining parameters P400-P402 for usage instructions.

When generating G code documents, the corresponding header and footer codes are automatically generated according to whether there are HEADDXF.TXT and ENDDXF.TXT files (or headdxf.txt and enddxf.txt files) in the current document directory.

Note: Header and footer files must be in the current directory.

3.11.12 FTP Server Folder Operation Method

If the system is connected to the Internet or WiFi, you can access the FTP server folder.

1、在程序目录界面下按“N”键即可进入。(如果为复合键则按“Shift”键后再按“N”键或按住快速叠加键后再按“N”键)。

2、进入 FTP 服务器文件夹后，按上面同样的操作按“N”键(或按 F6 键)即可退出 FTP 服务器文件夹。

使用 FTP 传输文件的优点是：在系统上可以看到 PC 机文件夹中的内容，并且可以自由选择想要传输的文件，使用操作起来更加方便。

NCNOR数控
鑫科瑞数控

第四篇 连 接 篇

1、 Press the "N" key in the program directory interface to enter.
(If it is a composite key, press the "Shift" key and then press the "N" key
or press the "N" key while holding the fast overlay key).

2、 After entering the FTP server folder, press the "N" key (or press
F6) to exit the FTP server folder in the same way.

The advantage of using FTP to transfer files is that the contents of
PC folders can be seen on the system, and the files you want to transfer
can be freely selected, so it is more convenient to use.

NEOKer 数控
鑫科瑞数控

Part IV. Connectivity

4.1 系统电气特性

- 高性能工业级 32 位 ARM+DSP+FPGA
- 128M（总线系统可以扩展到 32G）用户存储空间
- 800x600 TFT LCD 亮度自适应、LED 背光液晶显示器
- 轻触式薄膜面板
- RS232 通讯接口
- USB 接口
- 高抗干扰开关电源
- 双路主轴变频调速
- 手摇脉冲发生器
- 波段开关修调进给速度、主轴速度

4.2 系统技术指标

- 控制轴数：XYZABCXsYs 八轴
- 联动轴数：直线 XYZABCXsYs 八轴，圆弧 XYZ 任意二轴
- 脉冲当量：0.001mm
- 最高速度：240m/min
- 加工速度：0.01-30000mm/min
- 最小输入单位：0.001mm
- 编程尺寸范围：± 99999.999mm
- 编程代码：符合 ISO-840 国际标准
- 编程坐标系定义：符合 ISO-841 国际标准
- 平均无故障时间(MTBF)：大于 6000 小时

4.3 系统使用环境：

- 电源：交流 220V (+10%，-15%)，频率 50Hz±1%
- 电源功率≤100W
- 运行温度 5~45℃，相对湿度 40-80%
- 储运温度 0~55℃，相对湿度小于 90% (40℃)

4.4 系统安装连接

首先应检查数控系统、驱动电源、伺服电机、光电编码器、电动刀架等需要安装的硬件是否齐备、完好无损以及是否匹配。

数控系统安装必须牢固，四周应保留一定的空间，保证空气流通，面板安放

4.1 System Electrical Characteristics

High performance industrial grade 32-bit ARM+DSP+FPGA
128M (the bus system can be expanded to 32G) user storage space
800x600 TFT LCD with adaptive brightness and LED backlight
Touch-sensitive thin film panel
RS232 communication interface
。 USB joggle
High anti-interference switching power supply
Two-way spindle variable frequency speed control
Hand-cranked pulse generator
Band switch adjust feed speed, spindle speed

4.2 System Technical Indicators

Number of control axes: XYZABCXsYs eight axes
Number of linkage axes: eight axes for straight lines XYZABCXsYs, and any two axes for arcs XYZ
Pulse equivalent: 0.001mm
Maximum speed: 240m/min
Processing speed: 0.01-30000mm/min
Minimum input unit: 0.001mm
Programming size range: ± 99999.999 mm
Programming code: Complies with ISO-840 international standard
Programming coordinate system definition: Complies with ISO-841 international standard
Mean time between failures (MTBF): greater than 6000 hours

4.3 System Usage Environment:

Power supply: AC 220V (+10%, -15%), frequency 50Hz $\pm$ 1%
。 Power supply power $\leq$ 100W
Operating temperature 5~45°C, relative humidity 40-80%
Storage temperature 0~55°C, relative humidity less than 90% (40°C)

4.4 System Installation and Connection

First of all, it is necessary to check whether the hardware to be installed such as CNC system, drive power supply, servo motor, photoelectric encoder, electric knife holder, etc. are complete, intact and matched.

The installation of the CNC system must be firm, and a certain space should be reserved around it to ensure air circulation and panel placement

位置应既便于操作又能避开加工铁屑等烫伤。

强、弱电应分开，数控系统及驱动电源应尽可能与机床强电分开，各种信号线应尽可能远离交流接触器以减少干扰，光电编码器、限位、零点信号最好不通过强电箱直接与数控系统相连；电源线必须严格正确接地。

将各种插头插接牢固，上紧固定螺钉。禁止带电插拔驱动、电机和各种信号线插头。

安装数控系统时，面板不要被硬物、利器等弄伤，若需油漆时应将数控系统取下以免弄脏面板。

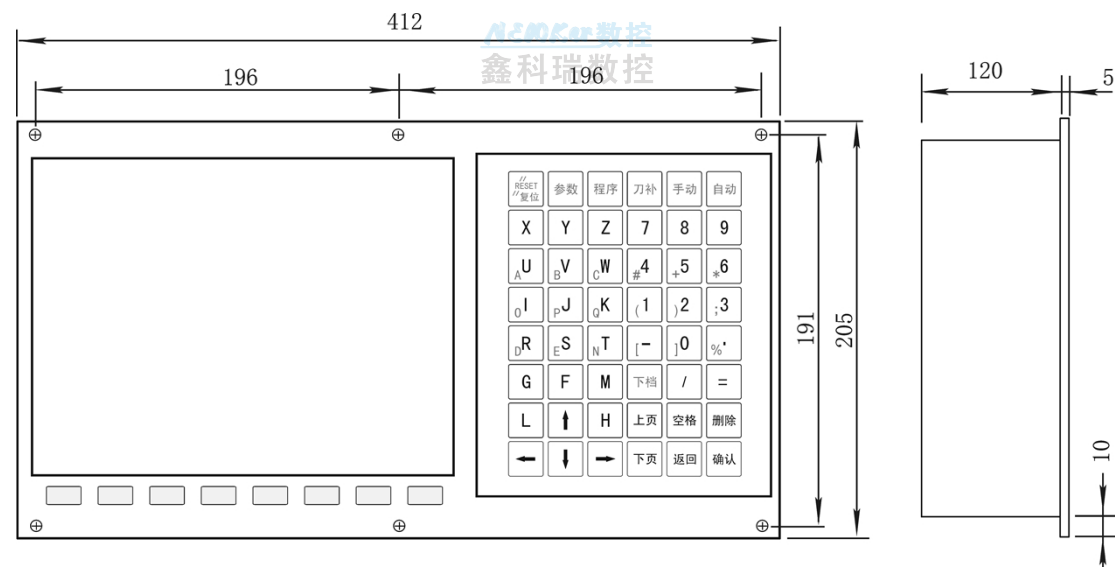
数控系统周围应无强磁、强电干扰源，尽量远离易燃、易爆物品和各种危险品。

注意事项：

- 1、必须安装在避雷等保护良好的电柜内
- 2、必须安装牢固，防止振动松脱
- 3、不可安装在易燃物体上面或附近，防止火灾

4.5 系统安装尺寸

第一种：横式主面板 (412x205x120)



AB 型副面板 (A 型带手轮；B 型带进给波段开关)：

The location should be convenient for operation and avoid scalding from processing iron chips.

Strong and weak power should be separated. CNC system and drive power supply should be separated from the machine tool's strong power as far as possible. All signal lines should be kept as far away as possible from AC contactors to reduce interference. Photoelectric encoders, limit switches and zero-point signals are best not directly connected to the CNC system through the strong power box. Power lines must be strictly and correctly grounded.

Plug all kinds of plugs firmly and tighten the fixing screws. Do not plug or unplug the driver, motor and signal line plugs while they are live.

When installing the CNC system, do not damage the panel with hard objects, tools, etc. If you need to paint, remove the CNC system to avoid dirtying the panel.

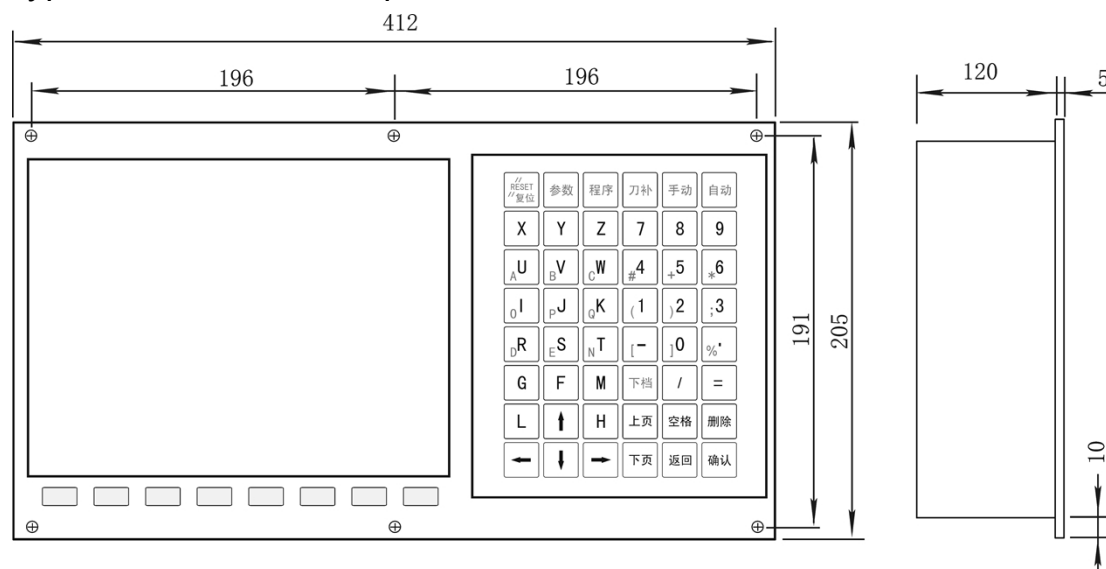
There should be no strong magnetic and electric interference sources around the CNC system, and it should be as far away as possible from inflammable and explosive articles and all kinds of dangerous goods.

Matters Need Attention :

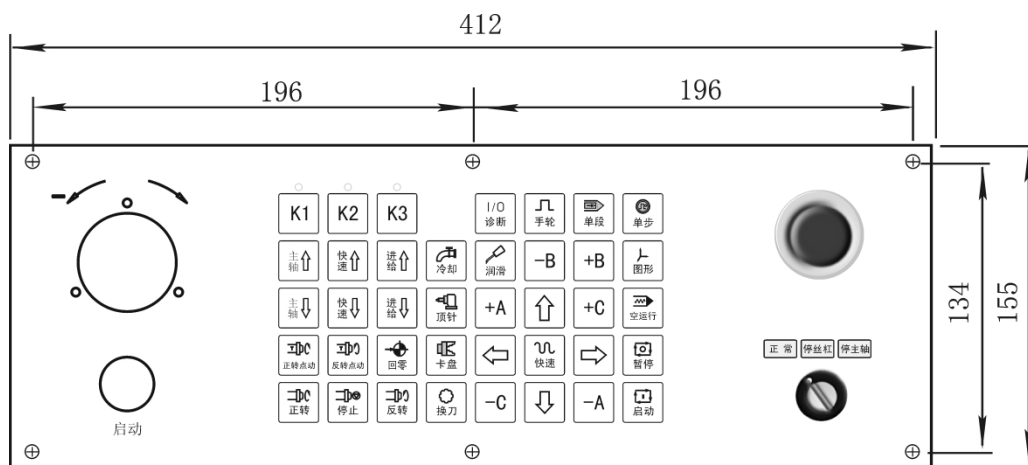
- 1、 It must be installed in a well-protected electrical cabinet such as a lightning arrester
- 2、 It must be installed firmly to prevent vibration and loosening
- 3、 Do not install on or near flammable objects to prevent fire

4.5 System Installation Dimensions

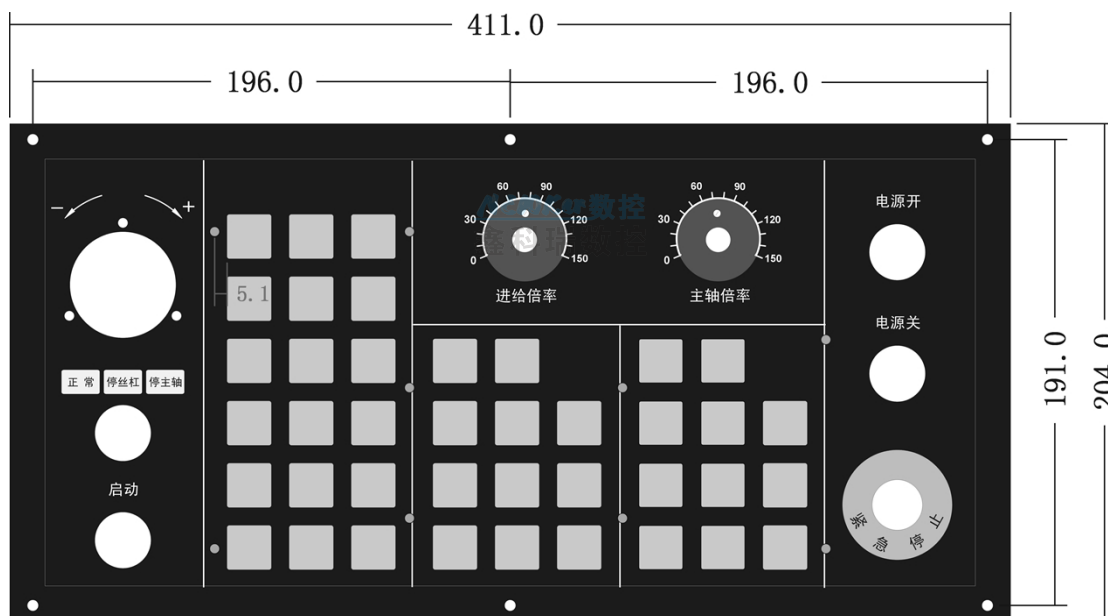
Type 1: Horizontal main panel (412x205x120)



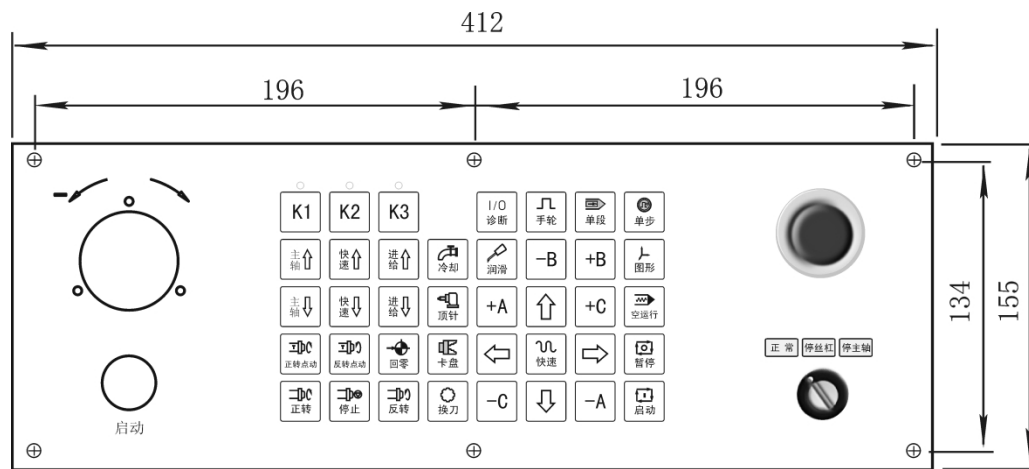
AB type auxiliary panel (A type with handwheel; B type with feed band switch):



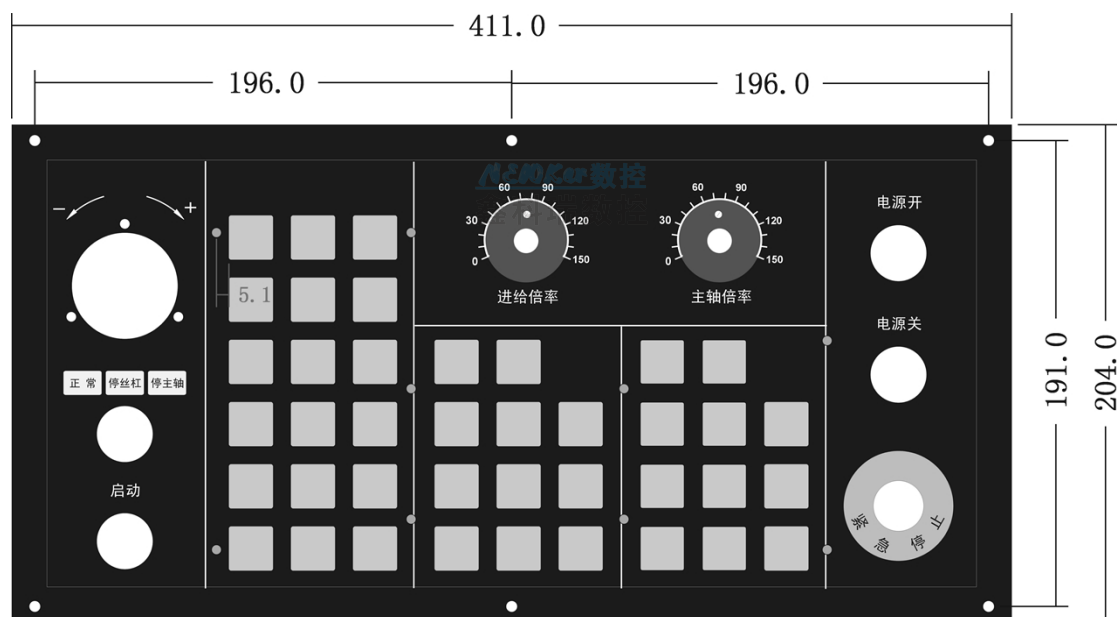
C 型副面板:



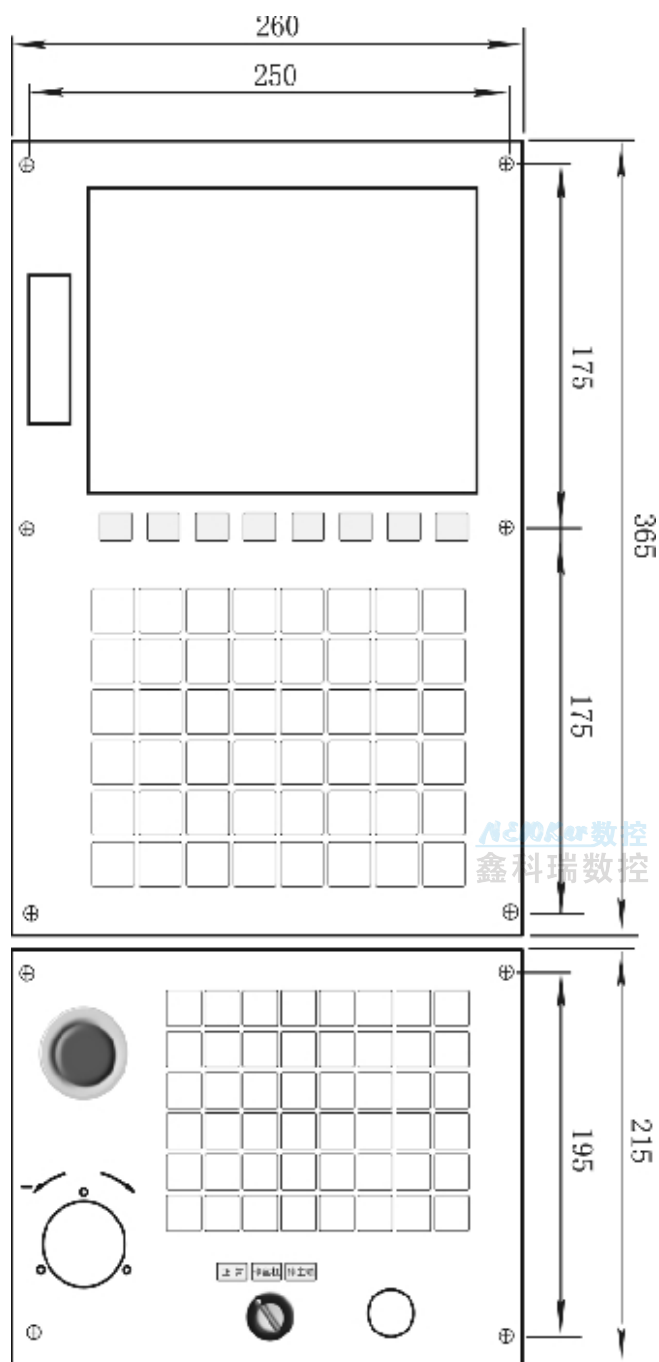
第二种:立式安装尺寸(260x265x120+215x265x120)



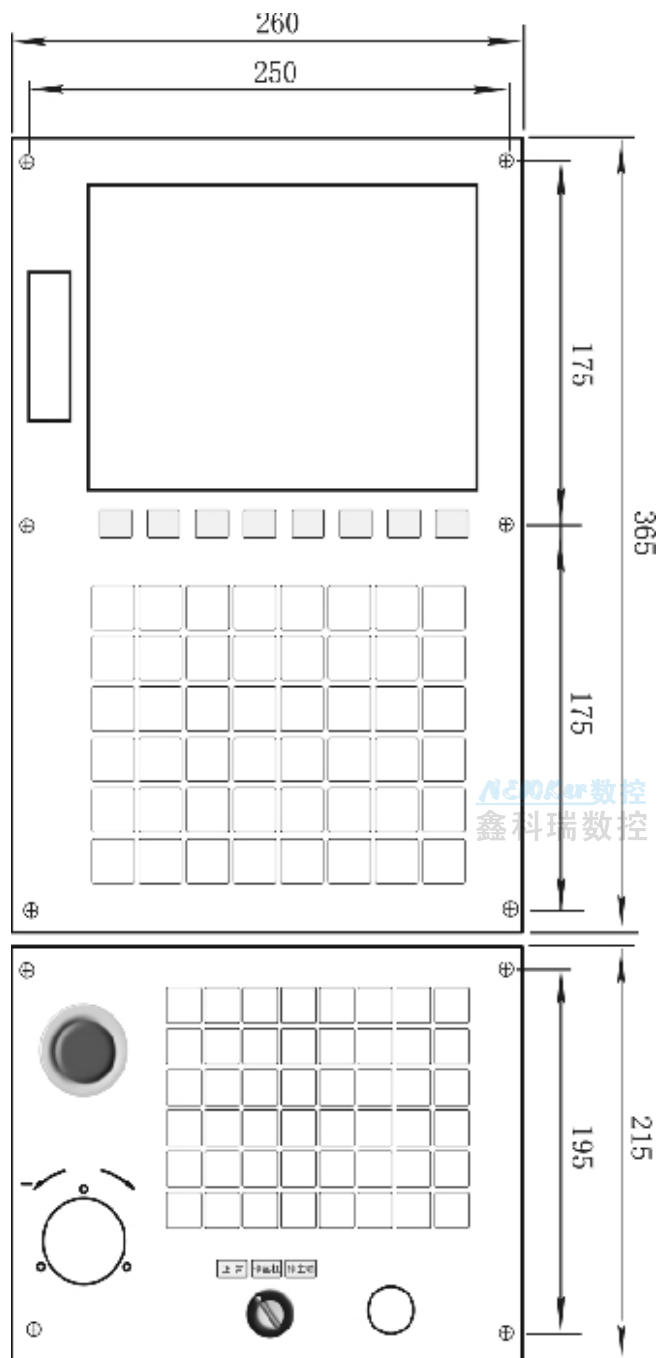
Type C subpanel:



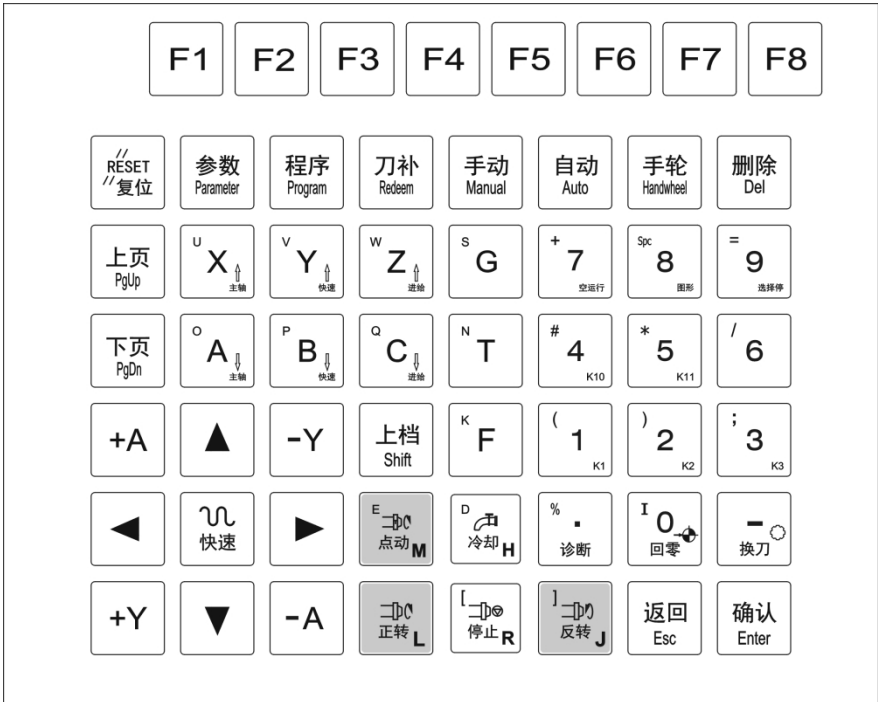
Type 2: Vertical installation dimensions (260x265x120+215x265x120)



第三种:简易立式安装尺寸(260x265x120)



Third: simple vertical installation size (260x265x120)

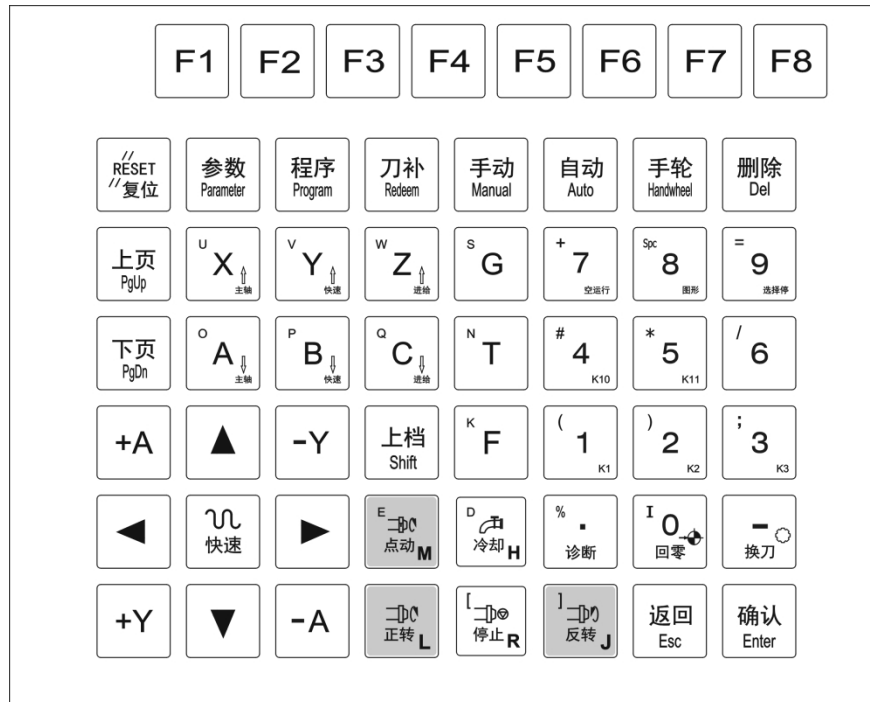


简易立式面板操作说明:

1. 二档按键: 按“上档”键后,再按一下则为上档小字母或字符键;
2. 三档按键: 字母/数字/字符键是编辑中有效,在手动/自动中的功能键如下;

- 鑫科瑞数控
- <1> U X (主轴), V Y (快速), W Z (进给) 主轴/快速/进给倍率上升调整;
- <2> O A (主轴), P B (快速), Q C (进给) 主轴/快速/进给倍率下降调整;
- <3> + 7 (空运行), Spc 8 (图形), = 9 (选择停) 自动状态空运行/图形/选择停功能键;
- <4> # 4 (K10), * 5 (K11), / 6 卡盘/顶针/手动单步或自动跳段功能键;
- <5> 正转 L, 停止 R, 反转 J 主轴正转/停止/反转功能键;
- <6> (1 (K1),) 2 (K2), ; 3 (K3) Y24/Y25/Y26 手动键(指令 OUT+M306/M307/M308);
- <7> 启动/暂停键:按一下“上档”再按“快速”为启动;直接按“0”为暂停;

第四种:横式主面板(420x260x120)

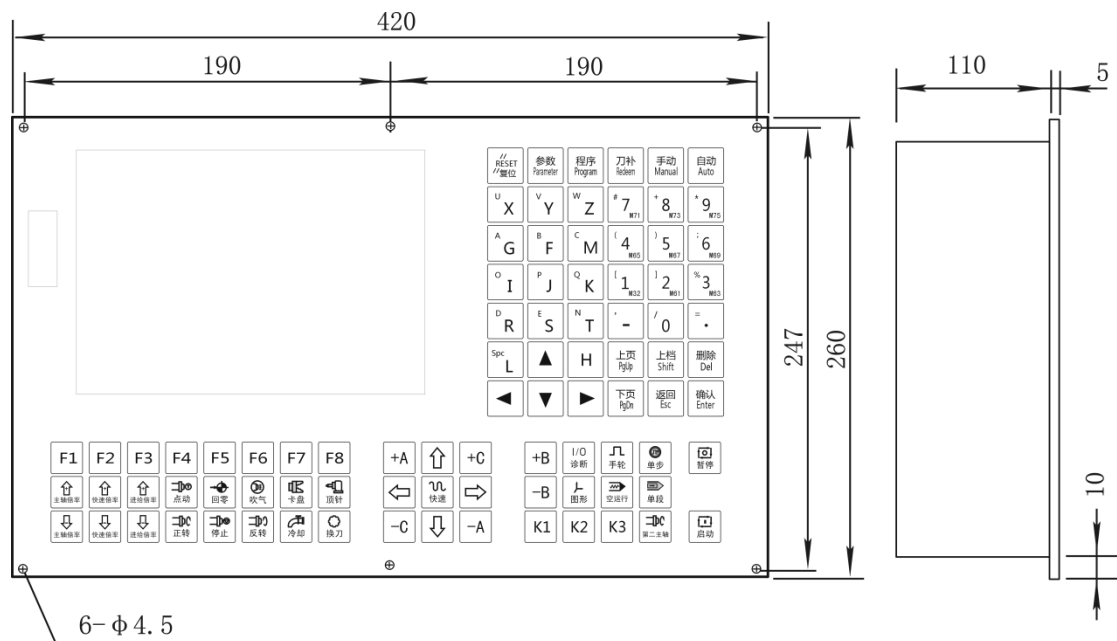


Simple vertical panel operation instructions:

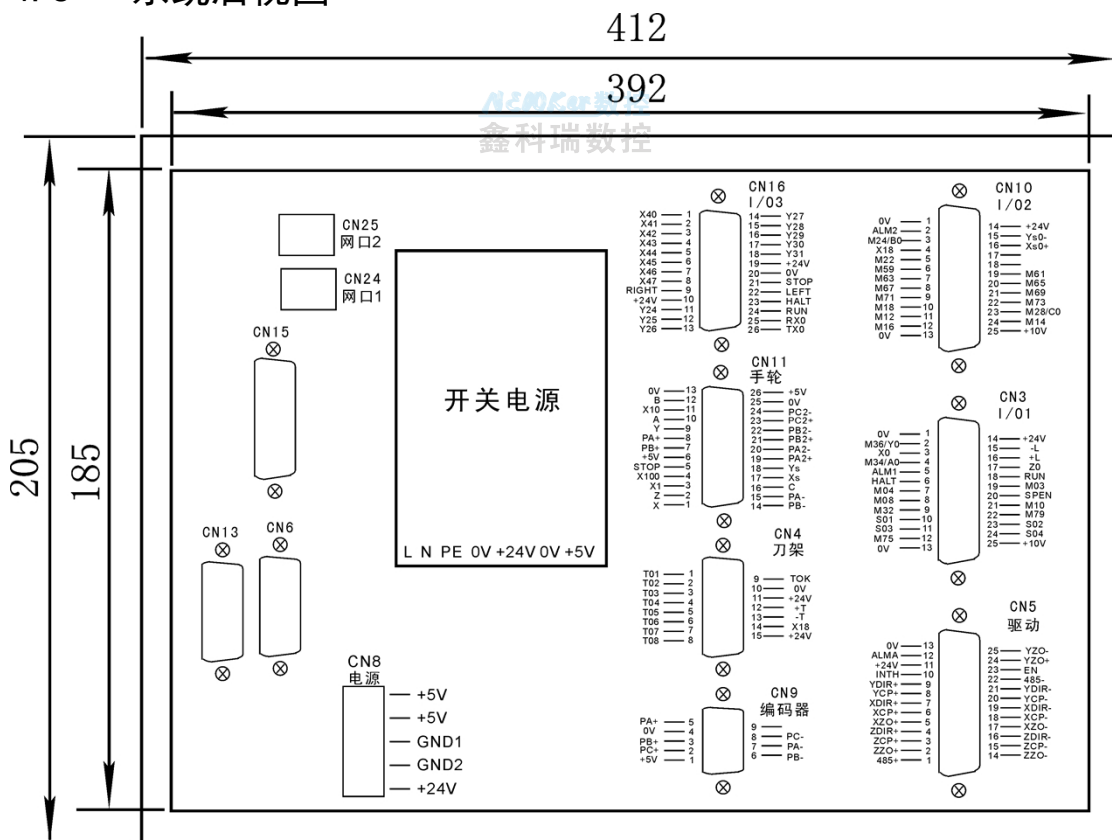
1. Two-way key: press the "up" key, and then press it again to be the small upper case letter or character key;
2. Three key keys: The letter/number/character keys are valid in editing. The function keys in manual/automatic mode are as follows;

- <1> handle/quick feed ratio upward;
- <2> work feed reduction adjustment;
- <3> /graphics/select stop function keys;
- <4> Card plate/thimble Manual single step (K10, K11, K12) function key;
- <5> Manual/stop/reverse function key;
- <6> Manual key (command OUT+M306/M307/M308);
- <7> : press "Up" and then press "Quick" to start; press "0" directly to pause;

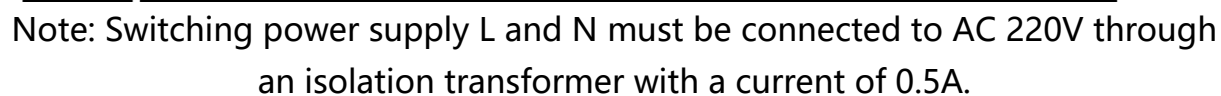
Type 4: Horizontal main panel (420x260x120)



4.6 系统后视图



注：开关电源 L、N 必须经过隔离变压器接入交流 220V，电流 0.5A。



4.7 接口连接图

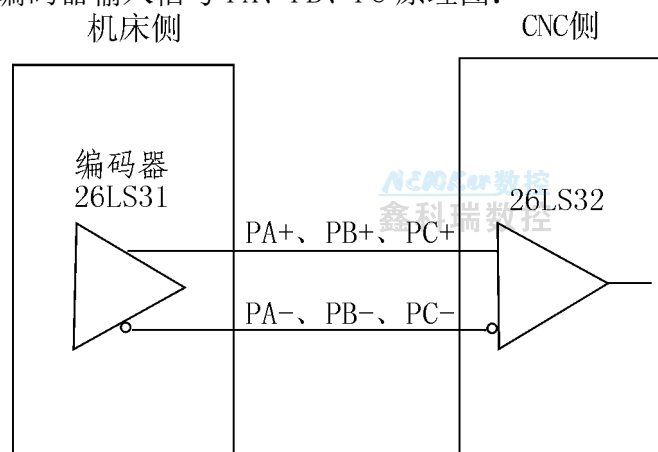
4.7.1 主轴编码器插座 CN9 的连接 DB9 针座

CN9 主轴编码器信号 DB9 针座				
信号	脚号	I/O	功能	有效电平
+5V/0V	1/4	OUT	电源	
PA+/PA-	5/7	IN	A 信号	5V
PB+/PB-	3/6	IN	B 信号	5V
PC+/0C-	2/8	IN	Z 同步信号	5V

注意：

- 1、编码器输出信号采用长线输出方式, 电源为+5V。
- 2、信号线必须采用双绞屏蔽电缆线, 长度不得超过 20m。

编码器输入信号 PA、PB、PC 原理图：



4.7.2 机床电器 CN3 的连接 DB25 孔座

CN3 I/O1 机床信号 DB25 孔座				
信号	脚号	I/O	功能	有效电平
0V	1	OUT	电源地	0V
+24V	14	OUT	24 电源	+24V
M36/Y0	2	IN	Y 轴回零信号	0V
X0	3	IN	X 轴回零信号	0V
Z0	17	IN	Z 轴回零信号	0V
-L	15	IN	机床正限位信号	0V
+L	16	IN	机床负限位信号	0V
M34/A0	4	IN	A 轴回零信号	0V
ALM1	5	IN	主轴驱动器报警	0V

4.7 Interface connection diagram

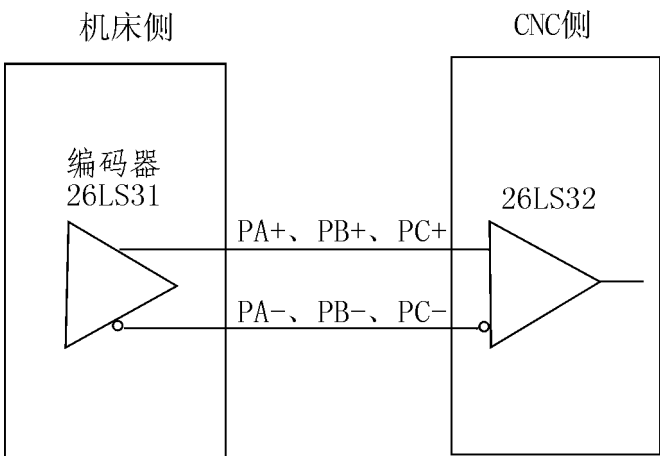
4.7.1 The Main Shaft Encoder Socket CN9 Is Connected to the DB 9 Pin Socket

CN9 spindle encoder signal DB9 needle holder				
signal	subscript	I / O	function	Effective level
+5V/0V	1/4	OUTPUT	source	
PA+/PA-	5/7	I ₊	A signal	5V
PB+/PB-	3/6	I ₊	B signal	5V
PC+/0C-	2/8	I ₊	Z synch	5V

pay attention to :

1. The Encoder Output Signal Adopts Long Line Output Mode, and the Power Supply Is +5V.
- 2 The Signal Line Must Be a Double Shielded Cable, and the Length Must Not Exceed 20m.

Schematic diagram of encoder input signals PA, PB and PC:



4.7.2 Machine Tool Electrical CN3 Is Connected to DB25 Socket

CN3 I/O1, machine tool signal DB25 socket				
signal	subscript	I / O	function	Effective level
0V	1	OUTPUT	Power supply	0V
+24V	14	OUTPUT	24 source	+24V
M36/Y0	2	I ₊	Y-axis zero return signal	0V
X0	3	I ₊	X-axis zero return signal	0V
Z0	17	I ₊	Z-axis zero return signal	0V
-L	15	I ₊	Machine tool positive limit signal	0V
+L	16	I ₊	Machine tool negative limit signal	0V
M34/A0	4	I ₊	A zero return signal on the A axis	0V
ALM1	5	I ₊	The spindle driver alarms	0V

HALT	6	IN	外接暂停按钮信号	0V
RUN	18	IN	外接运行按钮信号	0V
M03	19	OUT	主轴正转	0V
M04	7	OUT	主轴反转	0V
SPEN	20	OUT	主轴停止/主轴使能	0V
M08	8	OUT	冷却	0V
M10	21	OUT	主轴松紧刀	0V
M32	9	OUT	机床润滑	0V
M79	22	OUT	自定义输出 7	0V
S01	10	OUT	主轴第 1 档	0V
S02	23	OUT	主轴第 2 档	0V
M203/S3	11	OUT	第二主轴正转/3 档	0V
M204/S4	24	OUT	第二主轴反转/4 档	0V
M75	12	OUT	自定义输出 8	0V
+10V	25	OUT	第一主轴变频	0~10V
0V	13	OUT	变频地	0V

机床电器 CN10 的连接 DB25 孔座

说明：2028G 主板 CN10-15. 16 为第 7. 8 轴回零信号, 4 脚为 X18 输入备用, 17. 18 脚为空;

CN10 I/O2 机床信号 DB25 孔座				
信号	脚号	I/O	功能	有效电平
0V	1	OUT	电源地	0V
+24V	14	OUT	24 电源	+24V
SPCP+ (X17/Xs0)	15	OUT	主轴脉冲正	0V
SPCP- (X16/Ys0)	16	OUT	主轴脉冲负	
SPDIR+/空	17	OUT	主轴方向正	0V
SPDIR-/空	18	OUT	主轴方向负	
ALM (X18)	4	IN	驱动器报警	0V
ALM2	2	IN	无润滑油报警	0V
M24/B0	3	IN	自定义输入 7 或 B 轴回零信号	0V
M22	5	IN	M01 准停输入	0V
M59	6	OUT	吹气输出	0V
M61	19	OUT	自定义输出 1	0V
M63	7	OUT	自定义输出 2	0V
M65	20	OUT	自定义输出 3	0V
M67	8	OUT	自定义输出 4	0V

HALT	6	IN	External override button signal	0V
RUN	18	IN	External operation button signal	0V
M03	19	OUT	The spindle rotates forward	0V
M04	7	OUT	The spindle reverses	0V
SPEN	20	OUT	Spindle stopped/enabled	0V
M08	8	OUT	cooling	0V
M10	21	OUT	Shaft tension knife	0V
M32	9	OUT	Machine tool lubrication	0V
M79	22	OUT	Custom output 7	0V
S01	10	OUT	Main shaft 1 gear	0V
S02	23	OUT	Second gear of the spindle	0V
M203/S3	11	OUT	Second spindle forward rotation/3 gear	0V
M204/S4	24	OUT	Second spindle reverse/4 gear	0V
M75	12	OUT	Custom output 8	0V
+10V	25	OUT	First, the main shaft is variable frequency	0~10V
0V	13	OUT	Variable frequency ground	0V

Machine tool electrical CN10 is connected to DB25 socket

Note: The 2028G motherboard CN10-15.16 is the return zero signal of the 7.8th axis, pin 4 is the X18 input backup, and pin 17.18 is empty;

CN10 I/O2, machine tool signal DB25 socket				
sig nal	subscript	I / O	funct- ion	Effective level
0V	1	OUT	Power supply	0V
+24V	14	OUT	24 source	+24V
SPCP+ (X17/ Xs0)	15	OUT	The spindle pulse is positive	0V
SPCP- (X16/ Ys0)	16	OUT	The spindle pulse is negative	
SPDIR+/ empty	17	OUT	The spindle direction is correct	0V
SPDIR-/ empty	18	OUT	The spindle direction is negative	
ALM (X18)	4	I	Drive alarm	0V
ALM2	2	I	No lubricating oil alarm	0V
M24/ B0	3	I	Customize the input 7 or B axis zero return signal	0V
X22	5	I	M01 Stop input	0V
X59	6	OUT	Blow air out	0V
X61	19	OUT	Custom output 1	0V
X63	7	OUT	Custom output 2	0V
X65	20	OUT	Custom output 3	0V
X67	8	OUT	Custom output 4	0V

M69	21	OUT	自定义输出 5	0V
M71	9	OUT	自定义输出 6	0V
M73	22	OUT	自定义输出 7	0V
M18	10	IN	自定义输入 1	0V
M28/C0	23	IN	自定义输入 2 或 C 轴回零信号	0V
M12	11	IN	自定义输入 3	0V
M14	24	IN	自定义输入 4	0V
M16	12	IN	自定义输入 5	0V
+10V	25	OUT	第二主轴变频	0~10V
0V	13	OUT	变频地	0V

机床电器 CN16 的连接 DB26 孔座

CN16		I/O3 机床信号 DB26 孔座		
信号	脚号	I/O	功能	有效电平
0V	20	OUT	电源地	0V
+24V	10、19	OUT	24 电源	+24V
X40	1	IN	备用输入 0	0V
X41	2	IN	备用输入 1	0V
X42	3	IN	备用输入 2	0V
X43	4	IN	备用输入 3	0V
X44	5	IN	备用输入 4	0V
X45	6	IN	备用输入 5	0V
X46	7	IN	备用输入 6	0V
X47	8	IN	备用输入 7	0V
RIGHT	9	IN	右干预开关	0V
Y24	11	OUT	备用输出 0	0V
Y25	12	OUT	备用输出 1	0V
Y26	13	OUT	备用输出 2	0V
Y27	14	OUT	备用输出 3	0V
Y28	15	OUT	备用输出 4	0V
Y29	16	OUT	备用输出 5	0V
Y30	17	OUT	备用输出 6	0V
Y31	18	OUT	备用输出 7	0V
STOP	21	IN	外接急停	0V
LEFT	22	IN	左干预开关	0V
HALT	23	IN	外接停止	0V
RUN	24	IN	外接运行	0V

M69	21	OUT	Custom output 5	0V
M71	9	OUT	Custom output 6	0V
M73	22	OUT	Custom output 7	0V
M18	10	IN	Custom input 1	0V
M28/C0	23	IN	Customize the input 2 or C axis zero return signal	0V
M12	11	IN	Custom input 3	0V
M14	24	IN	Custom input 4	0V
M16	12	IN	Custom input 5	0V
+10V	25	OUT	Second main shaft frequency conversion	0~10V
0V	13	OUT	Variable frequency ground	0V

Machine tool electrical CN16 is connected to DB26 socket

CN16 I/O3, machine tool signal DB26 socket				
signal	subscript		function	Effective level
0V	20		Power supply	0V
+24V	10、19		24 source	+24V
X40	1		Backup input 0	0V
X41	2		Backup input 1	0V
X42	3		Backup input 2	0V
X43	4		Backup input 3	0V
X44	5		Backup input 4	0V
X45	6		Backup input 5	0V
X46	7		Backup input 6	0V
X47	8		Backup input 7	0V
RIGHT	9		Right intervention switch	0V
Y24	11		Backup output 0	0V
Y25	12		Backup output 1	0V
Y26	13		Backup output 2	0V
Y27	14		Backup output 3	0V
Y28	15		Backup output 4	0V
Y29	16		Backup output 5	0V
Y30	17		Backup output 6	0V
Y31	18		Backup output 7	0V
STOP	21		External emergency stop	0V
LEFT	22		Left intervention switch	0V
HALT	23		External connections are discontinued	0V
RUN	24		External operation	0V

TXD	25	OUT	RS232 发送	
RXD	26	IN	RS232 接收	

CN4 的连接 DB15 孔座

CN4 I/O4 信号 DB15 孔座				
信号	脚号	I/O	功能	有效电平
0V	10	OUT	电源地	0V
+24V	11、15	OUT	24 电源	+24V
Y18	12	OUT	+T 备用输出	0V
Y19	13	OUT	-T 备用输出	0V
X18	14	IN	X18 备用输入	0V
X00	1	IN	T1 备用输入	0V
X01	2	IN	T2 备用输入	0V
X02	3	IN	T3 备用输入	0V
X03	4	IN	T4 备用输入	0V
X04	5	IN	T5 备用输入	0V
X05	6	IN	T6 备用输入	0V
X06	7	IN	T7 备用输入	0V
X07	8	IN	T8 备用输入	0V
X21	9	IN	TOK 备用输入	0V

4.7.3 X/Y/Z 驱动信号插座 CN5 的连接 DB25 针座

CN5 伺服驱动信号 DB25 针座				
信号	脚号	I/O	功能	有效电平
XCP+/-	6/18	OUT	X 轴脉冲信号	5V
XDIR+/-	7/19	OUT	X 轴方向信号	5V
YCP+/-	8/20	OUT	Y 轴脉冲信号	5V
YDIR+/-	9/21	OUT	Y 轴方向信号	5V
XZO+/-	5/17	IN	X 轴电机零位	5V
ZCP+/-	3/15	OUT	Z 轴脉冲信号	5V
ZDIR+/-	4/16	OUT	Z 轴方向信号	5V
ZZO+/-	2/14	IN	Z 轴电机零位	5V
YZO+/-	24/25	IN	Y 轴电机零位	5V
ALM/0V	12/13	OUT	ALM	0V
+24V/EN	11/23	OUT	+24V/使能信号	24V
INTH	10	OUT	伺服报警清除	0V
RS485+/-	1/22		Modbus485 信号	

TXD	25	OUT	Send RS232	
RXD	26	IN	RS232 receiver	

CN4 is connected to the DB15 socket

CN4 I/O4 signal DB15 socket				
signal	subscript	I / O	function	Effective level
0V	10	OUT	Power supply	0V
+24V	11、15	OUT	24 source	+24V
Y18	12	OUT	+T Backup output	0V
Y19	13	OUT	-T Standby output	0V
X18	14	I	X18 alternate input	0V
X00	1	I	T1 Backup input	0V
X01	2	I	T2 alternate input	0V
X02	3	I	T3 alternate input	0V
X03	4	I	T4 alternate input	0V
X04	5	I	T5 Backup input	0V
X05	6	I	T6 Backup input	0V
X06	7	I	T7 Backup input	0V
X07	8	I	T8 Backup input	0V
X21	9	I	TOK alternate input	0V

4.7.3 The X/Y/Z drive signal socket CN5 is connected to the DB25 socket

CN5 servo drive signal DB25 socket				
sig-nal	subscript	I / O	function	Effective level
XCP+/-	6/18	OUT	X-axis pulse signal	5V
XDIR+/-	7/19	OUT	X-axis direction signal	5V
YCP+/-	8/20	OUT	Y-axis pulse signal	5V
YDIR+/-	9/21	OUT	Y-axis signal	5V
XZ0+/-	5/17	I	Zero position of X-axis motor	5V
ZCP+/-	3/15	OUT	Z-axis pulse signal	5V
ZDIR+/-	4/16	OUT	Z-axis direction signal	5V
ZZ0+/-	2/14	I	Zero position of Z-axis motor	5V
YZ0+/-	24/25	I	Y-axis motor zero position	5V
ALM/0V	12/13	OUT	ALM	0V
+24V/EN	11/23	OUT	+24V/enable signal	24V
INTH	10	OUT	Servo alarm cleared	0V
RS485+/-	1/22		Modbus485 signal	

A/B 驱动信号插座 CN6 的连接 DB15 孔座

CN6		伺服驱动信号 DB15 孔座		
信号	脚号	I/O	功能	有效电平
ACP+/-	1/9	OUT	A 轴脉冲信号	5V
ADIR+/-	2/10	OUT	A 轴方向信号	5V
BCP+/-	4/12	OUT	B 轴脉冲信号	5V
BDIR+/-	5/13	OUT	B 轴方向信号	5V
AZO+/-	3/11	IN	A 轴电机零位	5V
BZO+/-	6/14	IN	B 轴电机零位	5V
ALM/OV	7/8	IN	伺服报警/OV	0V
+24V	15	OUT	+24V 电源	24V

2028G 主板 A/B/C 驱动信号插座 CN6 的连接 DB26 孔座

CN6		伺服驱动信号 DB15 孔座		
信号	脚号	I/O	功能	有效电平
ACP+/-	9/18	OUT	A 轴脉冲信号	5V
ADIR+/-	8/17	OUT	A 轴方向信号	5V
AZO+/-	7/16	IN	A 轴电机零位	5V
BCP+/-	6/15	OUT	B 轴脉冲信号	5V
BDIR+/-	5/14	OUT	B 轴方向信号	5V
+24V/EN	4/13	OUT	+24V/使能信号	5V
CCP+/-	3/12	OUT	C 轴脉冲信号	5V
CDIR+/-	2/11	OUT	C 轴方向信号	5V
XsCP+/-	26/25	OUT	Xs 轴脉冲信号	5V
XsDIR+/-	24/23	OUT	Ys 轴方向信号	5V
YsCP+/-	22/21	OUT	Xs 轴脉冲信号	5V
YsDIR+/-	20/19	OUT	Ys 轴方向信号	5V
ALMB/OV	10/1	IN	伺服报警/OV	0V

注意：

- 1、驱动信号线必须采用双绞屏蔽电缆线，长度不得超过 20m。
- 2、报警信号 ALM 常开常闭输入方式由“杂类参数”P17 号决定。

伺服报警信号原理图：

The A/B drive signal socket CN6 is connected to the DB15 socket

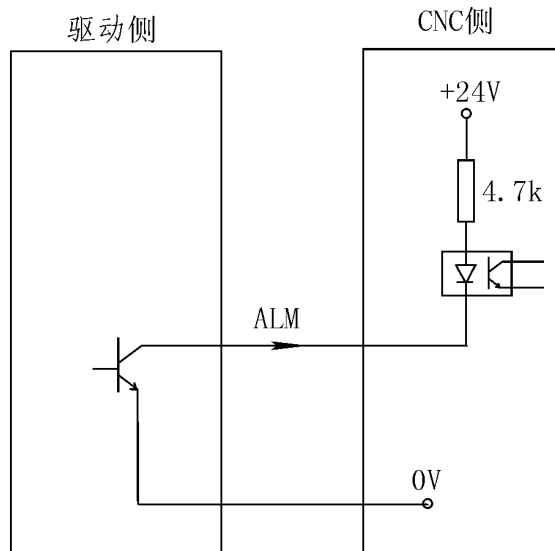
CN6 servo drive signal DB15 socket				
sig-nal	subscript	I / O	func-tion	Effective level
ACP+/-	1/9	OUT	An A-axis pulse signal	5V
ADIR+/-	2/10	OUT	Signal in the A axis direction	5V
BCP+/-	4/12	OUT	B-axis pulse signal	5V
BDIR+/-	5/13	OUT	B-axis signal	5V
AZO+/-	3/11	+	Axis A motor zero position	5V
BZO+/-	6/14	+	B-axis motor zero position	5V
ALM/0V	7/8	+	Servo alarm / 0V	0V
+24V	15	OUT	+24V power supply	24V

The 2028G motherboard A/B/C drive signal socket CN6 is connected to the DB26 socket

CN6 servo drive signal DB15 socket				
sig-nal	subscript	I / O	func-tion	Effective level
ACP+/-	9/18	OUT	An A-axis pulse signal	5V
ADIR+/-	8/17	OUT	Signal in the A-axis direction	5V
AZ0+/-	7/16	+	Axis A motor zero position	5V
BCP+/-	6/15	OUT	B-axis pulse signal	5V
BDIR+/-	5/14	OUT	B-axis signal	5V
+24V/EN	4/13	OUT	+24V/enable signal	5V
CCP+/-	3/12	OUT	C-axis pulse signal	5V
CDIR+/-	2/11	OUT	C-axis direction signal	5V
XsCP+/-	26/25	OUT	Xs axis pulse signal	5V
XsDIR+/-	24/23	OUT	Y-axis signal	5V
YsCP+/-	22/21	OUT	Xs axis pulse signal	5V
YsDIR+/-	20/19	OUT	Y-axis signal	5V
ALMB/0V	10/1	+	Servo alarm / 0V	0V

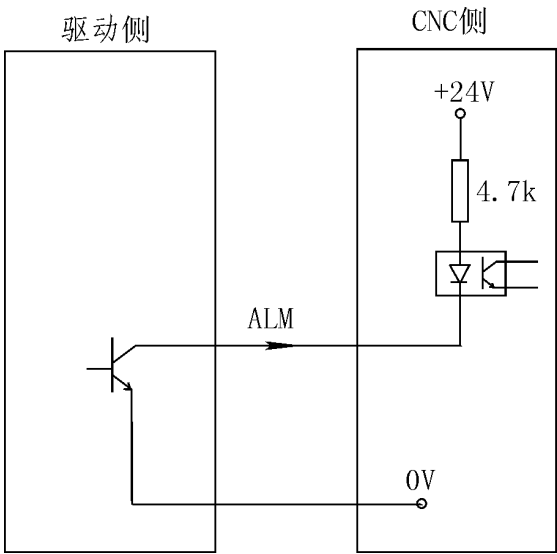
pay attention to :

- 1、The Drive Signal Line Must Be a Double Shielded Cable, and the Length Must Not Exceed 20m.
- 2、The Alarm Signal ALM Normally Open and Closed Input Mode Is Determined by "Miscellaneous Parameters" P17.
- Servo alarm signal schematic:



4.7.4 外接开关、电子手轮插座 CN11 的连接 DB15 (总线 DB26) 针座

CN11 电子手轮信号 DB15 (DB26) 针座				
信号	脚号	I/O	功能	有效电平
0V	13	OUT	电源地	0V
+5V	6	OUT	5V 电源	+5V
PA+	8	IN	A 信号正	5V
PA-	15	IN	A 信号负	
PB+	7	IN	B 信号正	5V
PB-	14	IN	B 信号负	
STOP	5	IN	外接急停信号	0V
H_B	12	IN	B 轴选择信号	0V
H_X100	4	IN	100 档位信号	0V
H_X10	11	IN	10 档位信号	0V
H_X1	3	IN	1 档位信号	0V
H_A/HALT	10	IN	A 轴选择信号/停止信号	0V
H_Z	2	IN	Z 轴选择信号	0V
H_Y/RUN	9	IN	Y 轴选择信号/运行信号	0V
H_X	1	IN	X 轴选择信号	0V
H_C	16	IN	C 轴选择信号	0V
H_Xs	17	IN	Xs 轴选择信号	0V
H_Ys	18	IN	Ys 轴选择信号	0V
Z2_A+/-	19/20	OUT	第二编码器 A 信号	
Z2_B+/-	21/22	OUT	第二编码器 B 信号	
Z2_C+/-	23/24	OUT	第二编码器 C 信号	
0V/+5V	25/26	OUT	电源	



4.7.4 Connect the External Switch and Electronic Handwheel Socket CN11 to the DB15 (bus DB26) Pin Seat

CN11 electronic handwheel signal DB15 (DB26) needle holder				
signal	subscript	I/O	function	Effective level
0V	13	OUT	Power supply	0V
+5V	6	OUT	5V power supply	+5V
PA+	8	IN	A signal is positive	5V
PA-	15	IN	A signal is negative	
PB+	7	IN	B Signal positive	5V
PB-	14	IN	B Signal negative	
STOP	5	IN	External emergency stop signal	0V
H_B	12	IN	B-axis selection signal	0V
H_X100	4	IN	100 Gate signal	0V
H_X10	11	IN	10 Gate signal	0V
H_X1	3	IN	1 Gate signal	0V
H_A/HALT	10	IN	A shaft selection signal/stop signal	0V
H_Z	2	IN	Z-axis selection signal	0V
H_Y/RUN	9	IN	Y-axis selection signal/run signal	0V
H_X	1	IN	X-axis selection signal	0V
H_C	16	IN	C-axis selection signal	0V
H_Xs	17	IN	Xs axis selection signal	0V
H_Ys	18	IN	Y-axis selection signal	0V
Z2_A+/-	19/20	OUT	Second encoder A signal	
Z2_B+/-	21/22	OUT	Second encoder B signal	
Z2_C+/-	23/24	OUT	Second encoder C signal	
0V/+5V	25/26	OUT	source	

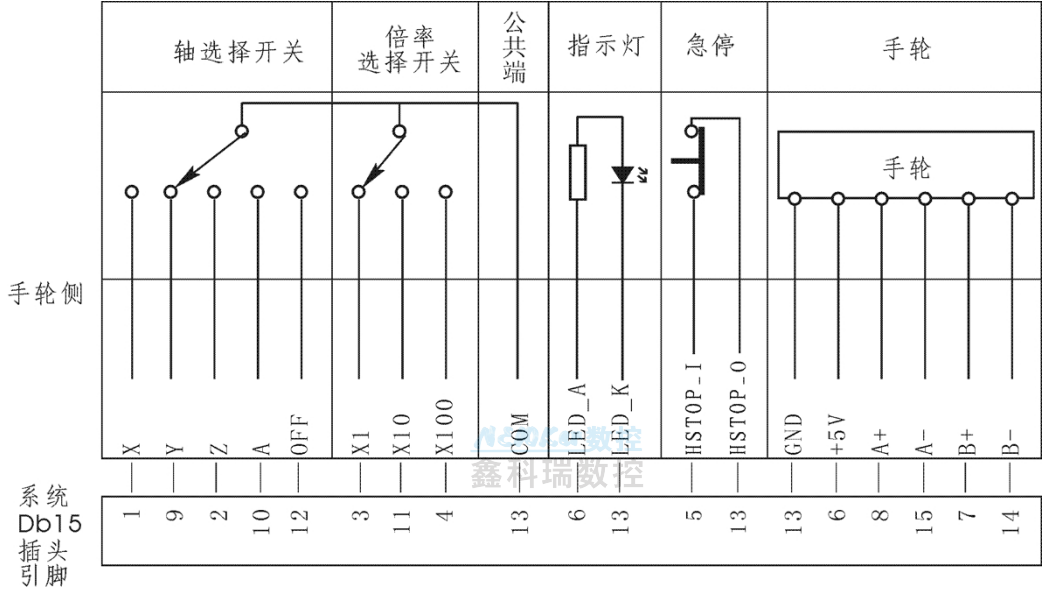
注意：所有信号内部为+5V 电源，切勿接入高于 5V 的电压。

4.7.4.1 电子手轮（手摇脉冲发生器）的使用

当“杂类参数”P1 号设置为 1 时，可以接标准的悬挂式手轮，A、X、Y、Z、X1、X10、X100 分别进行进给轴、档位的选择。

PA+、PB-、PA+、PA-分别为手轮正交脉冲 A、B 输入信号。

手轮接线示意图如下



注意：

- 1、手轮输出信号采用长线输出方式，电源为+5V。
- 2、如果采用的是电压输出方式，只需接 PA+、PB+即可。
- 3、手摇脉冲发生器无需确认按钮，如果已有此按钮，用线将开关两端短路即可。
- 4、所有信号内部为+5V 电源，切勿接入高于 5V 的电压。

手轮输入信号 PA、PB 原理图：

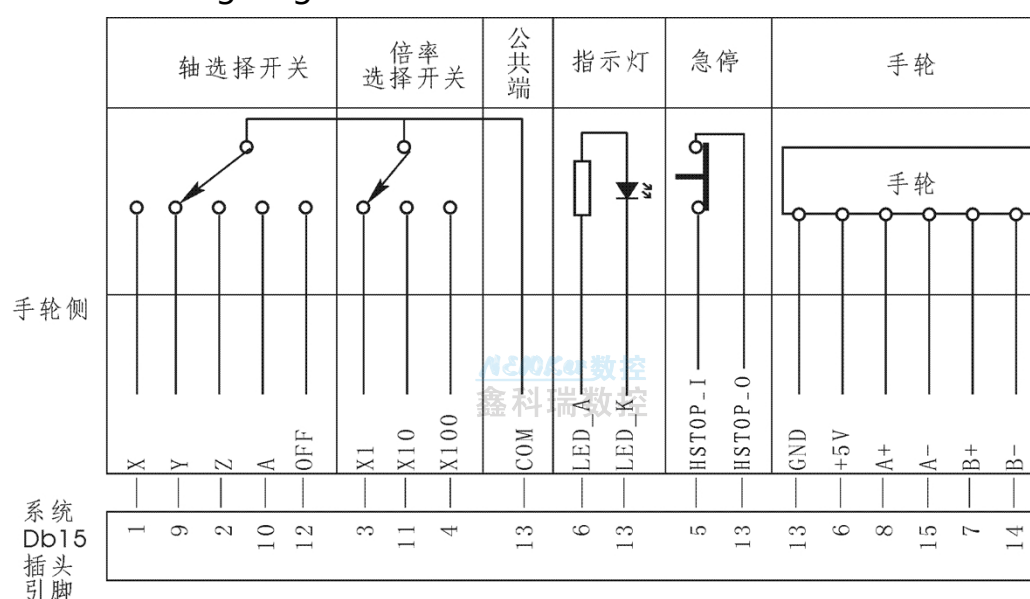
Note: All signals are powered by +5V internally. Do not connect to a voltage higher than 5V.

4.7.4.1 Use of Electronic Handwheel (hand-Cranked Pulse Generator)

When the "miscellaneous parameters" P1 is set to 1, a standard suspended handwheel can be connected, and A, X, Y, Z, X1, X10, X100 are respectively selected for feed axis and gear selection.

PA+, PB-, PA+, PA- are the input signals of handwheel orthogonal pulse A and B respectively.

The wiring diagram of the handwheel is as follows



pay attention to :

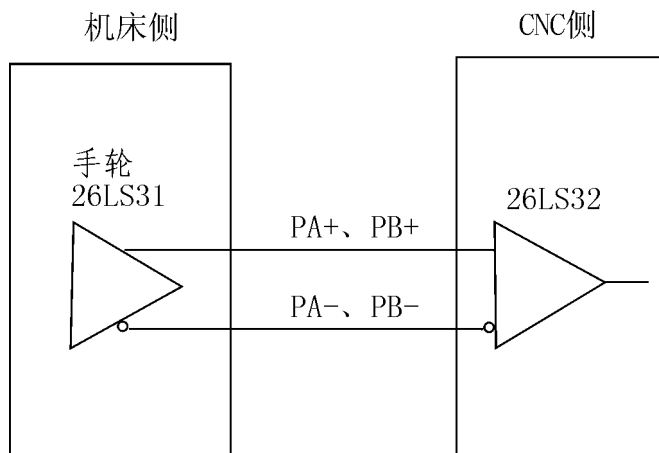
1、 The output signal of the handwheel adopts long line output mode, and the power supply is +5V.

2、 If the voltage output mode is used, only PA+ and PB+ need to be connected.

3. The hand-cranked pulse generator does not need to confirm the button. If there is such a button, short circuit the two ends of the switch with a wire.

4. All signals are powered by +5V internally. Do not connect to a voltage higher than 5V.

Schematic diagram of handwheel input signal PA and PB:



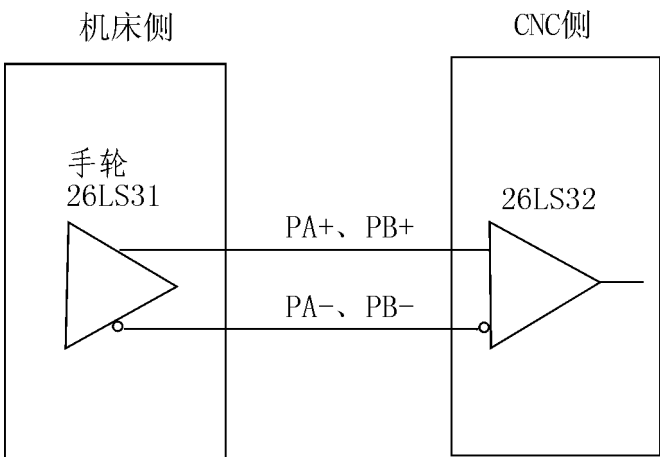
4.7.4.2 副面板后面接线端子波段开关的使用

信号	CN15 孔座	I/O	功能	有效电平
0V	29、39	OUT	电源地	0V
+5V	11、26	OUT	5V 电源	+5V
PA+	41	IN	A 信号正	5V
PA-	12	IN	A 信号负	
PB+	27	IN	B 信号正	5V
PB-	42	IN	B 信号负	
STOP	44	IN	外接急停信号	0V
KRUN	14	IN	运行按钮	0V
KLEFT	15	IN	干预开关左信号	0V
KRIGHT	30	IN	干预开关右信号	0V
DS0	8	IN	主轴波段 1 位	0V
DS1	38	IN	主轴波段 2 位	0V
DS2	23	IN	主轴波段 3 位	0V
DS3	37	IN	主轴波段 4 位	0V
DK0	7	IN	进给轴波段 1 位	0V
DK1	22	IN	进给轴波段 2 位	0V
DK2	9	IN	进给轴波段 3 位	0V
DK3	24	IN	进给轴波段 4 位	0V
LRUN	43	OUT	程序启动输出信号	0V

当“轴类参数”P1 号参数和 P2 号参数均设置为“1”时，可以使用波段开关。DS0、DS1、DS2、DS3 为主轴转速修调档位输入信号，四位控制共 16 档。DK0、DK1、DK2、DK3 为进给速度修调档位输入信号，四位控制共 16 档。

注意：所有信号内部为+5V 电源，切勿接入高于 5V 的电压。

4.7.4.3 外接急停按钮的使用



4.7.4.2 Use of Terminal Band Switch on the Back of the Auxiliary Panel

signal	CN15 seat	I/O	function	Effective level
0V	29、39	OUT	Power supply	0V
+5V	11、26	OUT	5V power supply	+5V
PA+	41	IN	A signal is positive	5V
PA-	12	IN	A signal is negative	
PB+	27	IN	B Signal positive	5V
PB-	42	IN	B Signal negative	
STOP	44	IN	External emergency stop signal	0V
KRUN	14	IN	Run button	0V
KLEFT	15	IN	Intercept the left signal of the switch	0V
KRIGHT	30	IN	Intercept the right signal on the switch	0V
DS0	8	IN	The main shaft band is 1 bit	0V
DS1	38	IN	The spindle band is 2 bits	0V
DS2	23	IN	The spindle band is 3 bits	0V
DS3	37	IN	The spindle band is 4 bits	0V
DK0	7	IN	Advance the shaft band 1 position	0V
DK1	22	IN	Set the feed shaft band to 2	0V
DK2	9	IN	Set the feed shaft band to 3	0V
DK3	24	IN	Four bits for the feed axis band	0V
LRUN	43	OUT	The program starts to output the signal	0V

When both parameter P1 and P2 in the "Axis Parameters" section are set to 1, the band switch can be activated. The four-position control units DS0, DS1, DS2, and DS3 provide 16 speed adjustment signals for spindle rotation, while the four-position control units DK0, DK1, DK2, and DK3 supply 16 feed rate adjustment signals.

Note: All signals are powered by +5V internally. Do not connect to a voltage higher than 5V.

4.7.4.3 Use of External Emergency Stop Button

STOP 信号为外接急停按钮输入信号，“杂类参数” P27 号参数可以选择外接急停开关是常开还是常闭。

4.7.5 坐标轴位置反馈信号插座 CN13 的连接 DB26 针座

CN13 位置反馈编码器或光栅尺信号 DB26 针座				
信号	脚号	I/O	功能	有效电平
+5V/0V	9/10	OUT	+5V/0V	
+24V/0V	26/23	OUT	+24V/0V	
DX+/DX-	25/24		RS485+/RS485-	
XA+/-	18/8	IN	X 轴 A 信号	5V
XB+/-	17/7	IN	X 轴 B 信号	5V
YA+/-	14/4	IN	Y 轴 A 信号	5V
YB+/-	13/3	IN	Y 轴 B 信号	5V
ZA+/-	16/6	IN	Z 轴 A 信号	5V
ZB+/-	15/5	IN	Z 轴 B 信号	5V
AA+/-	12/2	IN	A 轴 A 信号	5V
AB+/-	11/1	IN	A 轴 B 信号	5V
BA+/-	22/21	IN	B 轴 A 信号	5V
BB+/-	20/19	IN	B 轴 B 信号	5V

“杂类参数” P200 至 P220 设置此功能。报警后可以在诊断界面按“G”清零指令位置和反馈位置并清除超差报警。

注意：

- 1、编码器或光栅尺输出信号采用长线输出方式, 电源为+5V。
- 2、信号线必须采用双绞屏蔽电缆线, 长度不得超过 20m。
- 3、此功能为选配功能。

4.7.6 总线系统总线网口插座 CN22 的 DB15 连接孔座

335G 主板 CN22 总线网口信号 DB15 孔座(6 对线)				
信号	脚号	I/O	功能	有效电平
TXA-/TXA+	1/2 (1 对)		总线 A 信号	
TXB-/TXB+	3/4 (2 对)		总线 B 信号	
+24V/0V	5/6 (3 对)	OUT	+24V/0V	电源
+24V/0V	7/8 (4 对)	OUT	+24V/0V	
+5V/0V	11/12 (5 对)	OUT	+5V/0V	
485+/485-	10/15 (6 对)		485+/485-	

2026G/535G 主板 CN22 总线网口信号 DB15 孔座(6 对线)				
信号	脚号	I/O	功能	有效电平

STOP signal is the input signal for the external emergency stop button. The parameter P27 in "Miscellaneous parameters" can be selected to choose whether the external emergency stop switch is normally on or off.

4.7.5 Connect the Coordinate Axis Position Feedback Signal Socket CN13 to the DB26 Pin

CN13 position feedback encoder or grating scale signal DB26 needle holder				
signal	subscript	I / 0	func-tion	Effective level
+5V/0V	9/10	0 U T	+5V/0V	
+24V/0V	26/23	0 U T	+24V/0V	
DX+/DX-	25/24		RS485+/RS485-	
XA+/-	18/8	I *	X-axis A signal	5V
XB+/-	17/7	I *	X-axis B signal	5V
YA+/-	14/4	I *	Y-axis A signal	5V
YB+/-	13/3	I *	Y-axis B signal	5V
ZA+/-	16/6	I *	Z-axis A signal	5V
ZB+/-	15/5	I *	Z-axis B signal	5V
AA+/-	12/2	I *	Axis A signal	5V
AB+/-	11/1	I *	Axis A signal B	5V
BA+/-	22/21	I *	B-axis A signal	5V
BB+/-	20/19	I *	B-axis B signal	5V

The "Miscellaneous Parameters" P200 to P220 are set to enable this function. After the alarm, the "G" reset instruction position and feedback position can be pressed in the diagnostic interface to clear the out-of-tolerance alarm.

pay attention to :

1. The output signal of encoder or linear scale adopts long line output mode, and the power supply is +5V.
- 2 The signal line must be a double shielded cable, and the length must not exceed 20m.
- 3 This function is optional.

4.7.6 Bus System Bus Network Port CN22 DB15, Connection Socket

335G motherboard, CN22 bus network port signal, DB15 socket (6 pairs of wires)				
signal	subscript	I/O	function	Effective level
TXA-/TXA+	1/2 (1 pair)		Bus A signal	
TXB-/TXB+	3 / 4 (2 pairs)		Bus B signal	
+24V/0V	5 / 6 (3 pairs)	OUT	+24V/0V	source
+24V/0V	7 / 8 (4 pairs)	OUT	+24V/0V	
+5V/0V	11 / 12 (5 pairs)	OUT	+5V/0V	
485+/485-	15 October (6 pairs)		485+/485-	

2026G/535G motherboard CN22 bus network port signal DB15 socket (6 pairs of wires)				
signal	subscript	I/O	function	Effective level

T422+/T422-	1/2 (1 对)		总线 A 信号	
R422+/R422-	3/4 (2 对)		总线 B 信号	
X22/X20	11/6 (3 对)	IN	X22/X20	
+5V/EN	12/7 (4 对)	OUT	+5V/EN	
+24V/0V	13/8 (5 对)	OUT	+24V/0V	电源
+24V/0V	14/9 (6 对)	OUT	+24V/0V	

4.7.7 总线系统驱动脉冲信号插座 CN21 的 DB26 连接孔座

CN21 驱动脉冲信号 DB26 孔座				
信号	脚号	I/O	功能	有效电平
DX+/DX-	1/10		RS485+/RS485-	
ACP+/ACP-	2/3	OUT	A 轴脉冲信号	
ADIR+/ADIR-	4/5	OUT	A 轴方向信号	
XSCP+/XSCP-	6/7	OUT	XS 轴脉冲信号	
XSDIR+/XSDIR-	8/9	OUT	XS 轴方向信号	
XCP+/XCP-	11/12	OUT	X 轴脉冲信号	
XDIR+/XDIR-	13/14	OUT	X 轴方向信号	
ZCP+/ZCP-	15/16	OUT	Z 轴脉冲信号	
ZDIR+/ZDIR-	17/18	OUT	Z 轴方向信号	
C/YCP+/-	19/20	OUT	Y 轴脉冲信号	
C/YDIR+/-	21/22	OUT	Y 轴方向信号	
SP/YSCP+/-	23/24	OUT	YS 轴脉冲信号	
SP/YSDIR+/-	25/26	OUT	YS 轴方向信号	

注意：使用 RS485 功能时（接 1，10 脚），系统 0V 必须与驱动信号 0V 相连。

4.7.8 总线系统网口孔座 CN24, CN25

CN24 用于 Ethercat 驱动器网口使用，CN25 用于 OPC 功能；当系统接脉冲式驱动器时 CN24 可用于 OPC 功能。通过杂类参数 P350a 参数选择。

4.8 机床电器安装原理

4.8.1 限位：以 XYZ 轴正向限位+L 为例（-L 相同）

采用 NPN 型霍尔接近开关方式

T422+/T422-	1/2 (1 pair)		Bus A signal	
R422+/R422-	3 / 4 (2 pairs)		Bus B signal	
X22/X20	11 / 6 (3 pairs)	IN	X22/X20	
+5V/EN	12 / 7 (4 pairs)	OUT	+5V/EN	
+24V/0V	13 / 8 (5 pairs)	OUT	+24V/0V	source
+24V/0V	14 / 9 (6 pairs)	OUT	+24V/0V	

4.7.7 The Bus System Drives the Pulse Signal Socket CN21's DB26,
Which Connects to the Socket

CN21 Drive pulse signal DB26 socket				
signal	subscript	I/O	function	Effective level
DX+/DX-	1/10		RS485+/RS485-	
ACP+/ACP-	2/3	OUT	An A-axis pulse signal	
ADIR+/ADIR-	4/5	OUT	Signal in the A-axis direction	
XSCP+/XSCP-	6/7	OUT	XS-axis pulse signal	
XSDIR+/XSDIR-	8/9	OUT	XS axis direction signal	
XCP+/XCP-	11/12	OUT	X-axis pulse signal	
XDIR+/XDIR-	13/14	OUT	X-axis direction signal	
ZCP+/ZCP-	15/16	OUT	Z-axis pulse signal	
ZDIR+/ZDIR-	17/18	OUT	Z-axis direction signal	
C/YCP+/-	19/20	OUT	Y-axis pulse signal	
C/YDIR+/-	21/22	OUT	Y-axis signal	
SP/YSCP+/-	23/24	OUT	YS axis pulse signal	
SP/YSDIR+/-	25/26	OUT	YS axis signal	

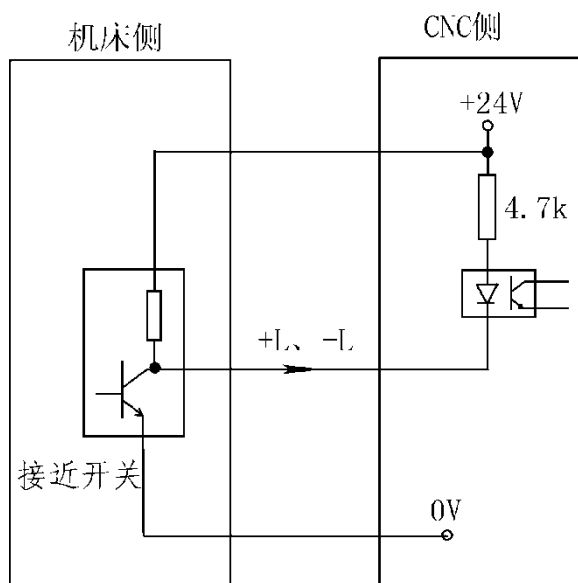
Note: When RS485 is used, the system 0V must be connected to the drive signal 0V when functioning (1, 10 feet).

4.7.8 Bus System Network Port Seat CN24, CN25

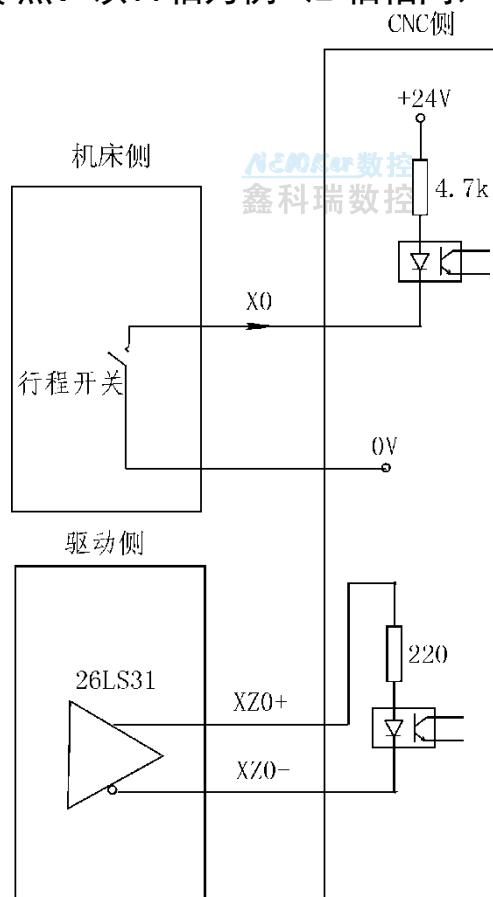
CN24 is used for Ethercat driver network ports, and CN25 is used for OPC functions. When the system is connected to a pulse driver, CN 24 can be used for OPC functions. This is selected through the miscellaneous parameter P350a.

4.8 Installation principle of machine tool electrical equipment

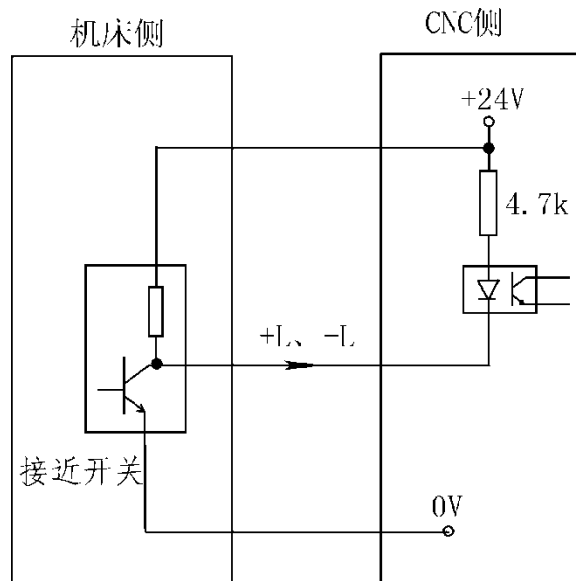
4.8.1 Limit: Take XYZ positive limit +L as an example (-L, the same) and use NPN type Hall approach switch



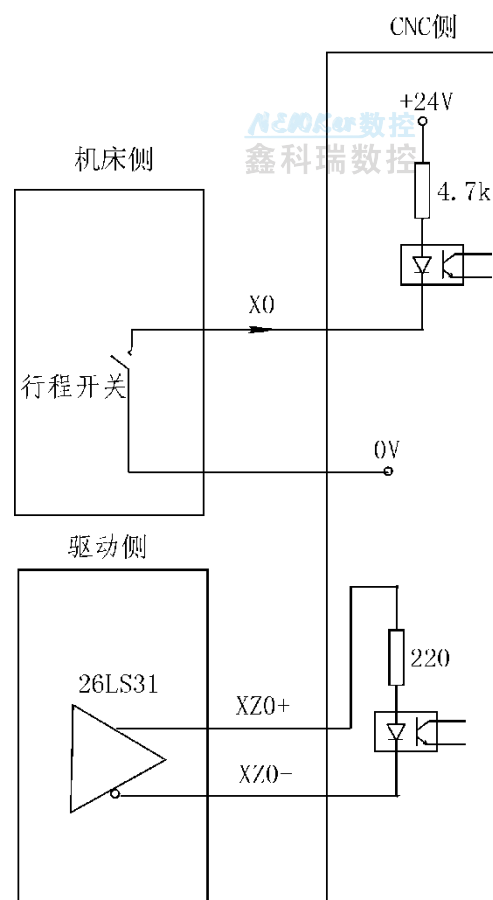
4.8.2 机床零点：以X轴为例（Z轴相同）



回零信号原理图：



4.8.2 Machine Tool Zero Point: Take X Axis as an Example (Z Axis Is the Same)



Return zero signal schematic:

在系统设置浮动零点功能无效的情况下回零需要检测接近开关信号和电机 Z 脉冲信号。即“轴类参数”的 P46 设置为“00000000”。

对回零功能要求由“轴类参数”的 P45 设置：

本系统对开机回参考点处理方式有六种：

设置为1时为不需要：每次开机后系统不提示也不做任何限制。

设置为0时为提示方式：每次开机后系统会弹出一个对话框提示操作者执行回参考点操作，之后不做任何限制。

设置为8时为强制方式：每次开机后运行程序系统会弹出一个对话框提示操作者执行回参考点操作，并且若在自动运行前不执行回参考点操作，系统将会提示“进给轴没有回参考点”并不执行程序。

设置为9时为超强制方式：每次开机后进给轴移动系统会弹出一个对话框提示操作者执行回参考点操作，不执行回参考点操作，系统将会提示“进给轴没有回参考点”并不执行运动。

回零检测信号的方式由“轴类参数”的 P46 设置：

设置为 0 时：回参考点时，撞到参考点开关后，反向运行检测到开关脱开后，再检测电机编码器零脉冲信号。

设置为 1 时：回参考点时，撞到参考开点关后，反向运行检测到开关脱开即可。

设置为 2 时：回参考点时，撞到参考开点关后，继续向前运行检测到开关脱开后，再检测电机编码器零脉冲信号；

设置为其余值时：回参考点时，撞到参考开点关后，继续向前运行检测到开关脱开即可。

回参考点方式应根据设备的电路情况设置，通常情况下建议设置为 0 或 2，因为如果回参考点不检测电机编码器的零脉冲信号时，精度的保证全靠参考点开关来保证，相比电机编码器的零脉冲精度就会更差一些。

回零的运行方向和先后顺序由“轴类参数”的 P47 设置：

每轴单独设置。该参数为位参数，其中 D2 位控制 X 轴的回零运动方向，D4 位控制 Z 轴的回零运动方向，1 表负向，0 表正向；D8 位控制 X、Z 回零的先后顺序，1 表示先 Z 后 X，0 则反之。

零点开关的类型由“轴类参数”的 P48 设置：

每轴单独设置。该参数为位参数，其中 D0 位控制 X 轴，D2 位控制 Z 轴，1 表常闭，0 表常开。

回零时检测电机 Z 脉冲运行长度由“轴类参数”的 P49-P53 设置：

回零完成后的偏移量由“轴类参数”的 P54-58 设置：

本参数具体设置值与机床参考点安装位置和机床坐标系设置有关，根据设备而定。

注：回参考点后的偏移是以 G00 速度运动。

注意：

1、减速开关也可以使用 NPN 霍尔开关。

2、安装减速开关时必须考虑减速的行程长度，要求 $\leq 25\text{mm}$ 。

When the floating zero point function is invalid in the system setting, the zero return requires detection of the proximity switch signal and motor Z pulse signal. That is, "Axis parameters" P46 is set to "00000000".

The zero return function requirement is set by P45 in "Axis Parameters":

There are six ways to process the reference point when starting up this system:

When set to 1, it is not required: the system does not prompt or restrict anything after each startup.

When set to 0, it is a prompt mode: after each startup, the system will pop up a dialog box to prompt the operator to perform the return to reference point operation, and then no restrictions are made.

When set to 8, it is a mandatory mode: after each startup, the program running system will pop up a dialog box to prompt the operator to perform the return to reference point operation. If the return to reference point operation is not performed before automatic operation, the system will prompt "the feed axis has no return to reference point" and not execute the program.

When the setting is 9, it is an ultra-mandatory mode: after each startup, a dialog box will pop up on the feed axis moving system to prompt the operator to perform the return to reference point operation. If the return to reference point operation is not performed, the system will prompt "the feed axis has no return to reference point" and the movement will not be performed.

The return-to-zero detection signal is set by the "shaft parameters" at P46:

When the value is set to 0: when returning to the reference point, after hitting the reference point switch, the reverse operation detects the switch is disengaged, and then the zero pulse signal of the motor encoder is detected.

When the value is set to 1, when returning to the reference point, if the reference opening is closed, the reverse operation can detect the switch is open.

When the value is set to 2: when returning to the reference point, after hitting the reference open switch, continue to run forward and detect the switch disengagement, then detect the zero pulse signal of the motor encoder;

When setting the other value: When returning to the reference point, if the reference opening is closed, continue to run forward and detect the switch disengagement.

The return reference point mode should be set according to the circuit condition of the equipment. In general, it is recommended to set it as 0 or 2. Because if the return reference point does not detect the zero pulse signal of the motor encoder, the accuracy guarantee depends entirely on the reference point switch, which will be worse than the zero pulse accuracy of the motor encoder.

The running direction and sequence of return zero are set by "Axis Parameters" on P47:

Each axis is set independently. This parameter is a bit parameter, where the D2 bit controls the return-to-zero direction of the X-axis movement, and the D4 bit controls the return-to-zero direction of the Z-axis (1 indicates negative direction, 0 indicates positive direction). The D8 bit controls the sequence of X and Z return-to-zero operations: 1 indicates Z before X, and 0 indicates the reverse order.

The type of zero-point switch is set by P48 in "Axis Parameters":

Each axis is set separately. This parameter is a bit parameter, where D0 bit controls the X-axis, D2 bit controls the Z-axis, and 1 Table is normally closed, 0 Table is normally open.

The pulse running length of motor Z detected at zero is set by "shaft parameters" P49-P53;

The offset after return zero is set by "Axis Parameters" P54-58;

The specific setting value of this parameter is related to the installation position of the machine tool reference point and the setting of the machine tool coordinate system, which depends on the equipment.

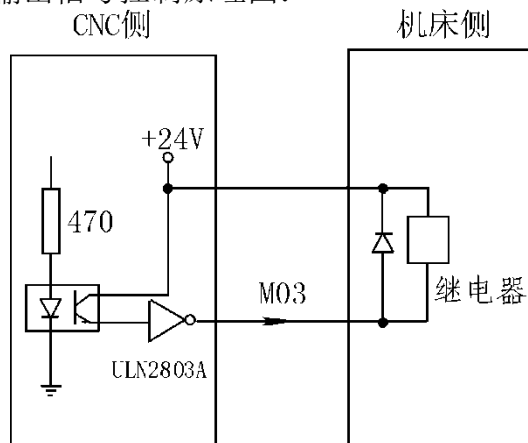
Note: The offset after returning to the reference point is moved at G00 speed.

pay attention to :

- 1、 The deceleration switch can also use an NPN Hall switch.
- 2、 When installing the deceleration switch, the travel length of deceleration must be considered, which should be $\leq 25\text{mm}$.

4.8.3 开关量控制信号：以 M03 为例（M04、M05、M08、M10、M32、M79、M75、M59、M61、M63、M65、M67、M69、M71、M73、S01-S04 相同）

M03 输出信号控制原理图：



如图所示，系统发 M03 时与 24V 组成回路，中间继电器动作，其一组常开触点接主轴正转交流接触器线包回路，使其动作即主轴正转。

所有输出信号均为低电平 0V 有效。

注意：

1、所用继电器等电感负载时，必须并接二极管吸收反向电流以免损坏系统，如果使用的是电磁接触器还要加阻容消火花电路。

2、芯片 ULN2803A 对应输出接口：

1)、U28 对应 M59、M61、M63、M65、M67、M69、M71、M73

2)、U29 对应 M03、M04、M05、M08、M10、M79、M32、M75

3)、U30 对应 +T、-T、S01、S02、S03、S04、LRUN、INTH

3、自定义信号 M71/M70、M73/M72 用作卡盘顶针输出信号时不能用作其它自定义信号。

4、自定义信号 M65、M67、M69 用作运行报警输出信号时不能用作其它自定义信号。M65 程序停止输出、M69 程序运行输出、M67 报警输出。

4.8.3.1 系统主轴控制（M03/M04/M05）

系统输出两路主轴（第一、第二主轴）控制信号，相关参数如下：

轴类参数

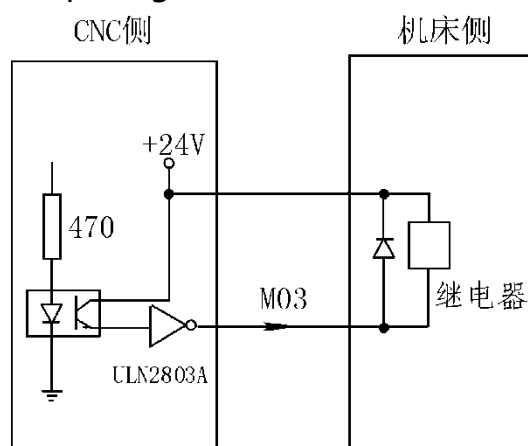
P13-2 号参数：设置主轴制动所用的时间，即 M05 输出的保持时间，制动时间设置越短主轴刹车越快，制动时间设置越长主轴刹车越慢。单位：10ms。

P13-1 号参数：设置主轴制动信号是长信号 1 或短信号 0

P65 号参数：用于设置系统是否检测主轴位置反馈信号，即主轴编码器信号。

4.8.3 Switching quantity control signals: Taking M03 as an example (M04, M05, M08, M10, M32, M79, M75, M59, M61, M63, M65, M67, M69, M71, M73, S01-S04 are the same)

M03 output signal control schematic:



As shown in the figure, when the system sends M03, it forms a circuit with 24V. The intermediate relay is operated, and one set of normally open contacts are connected to the main shaft forward AC contactor coil circuit, so that the main shaft rotates forward when it is operated.

All output signals are valid at low level 0V.

pay attention to :

1、 When the relay and other inductive loads are used, diodes must be connected in parallel to absorb reverse current to avoid system damage. If the electromagnetic contactor is used, a resistive capacitive spark elimination circuit should be added.

2、 Chip ULN2803A corresponds to output interface:

1) U28 corresponds to M59, M61, M63, M65, M67, M69, M71, and M73

2) U29 corresponds to M03, M04, M05, M08, M10, M79, M32, and M75

3) U30 corresponds to +T, -T, S01, S02, S03, S04, LRUN, and INTH

3、 Custom signals M71/M70 and M73/M72 cannot be used as other custom signals when used as chuck pin output signals.

4、 Custom signals M65, M67, and M69 cannot be used as other custom signals when serving as operational alarm output signals. M65 is programmed to stop output, M69 is programmed to run output, and M67 is programmed to alarm output.

4.8.3.1 System spindle control (M03/M04/M05)

The system outputs two spindle control signals (first and second spindles), and the relevant parameters are as follows:

Axis parameters

Parameter P13-2: Set the time used for spindle braking, that is, the holding time output by M05. The shorter the braking time is set, the faster the spindle brake will be, and the longer the braking time is set, the slower the spindle brake will be. Unit: 10ms.

Parameter P13-1: Set the spindle brake signal to long signal 1 or short signal 0

Parameter P65: Used to set whether the system detects the spindle position feedback signal, that is, the spindle encoder signal.

当参数值设为 1 时表示检测；当参数值设为 0 时表示不检测。

P200, 主轴编码器每转的脉冲数(编码器线数的 4 倍)

P200-1, 主轴编码器检测线数报警值(诊断值) [>10 有效]

P59 号参数: 主轴换档时主轴是否启动[1 表是, 0 表否]。

P60 号参数: 主轴换档时电机转动速度(1/100rpm)。

P61 号参数: 主轴换档时转动方向[0 表正, 1 表反]。

P62 号参数: 主轴换档时停止时间(10ms)。

P63 号参数: 主轴换档时低速转动时间(10ms)。

P64 号参数: 主轴停止延时时间(10ms)。

P220, 第一主轴全脉冲控制通道(正向 82, 反向 92)

P221, 第一主轴全脉冲控制每圈脉冲数

P222, 第一主轴全脉冲控制升降速(rpm/2ms)

P223, 第二主轴全脉冲控制通道(正向 84A 轴/85B 轴/88Ys 轴, 反向/94A 轴/95B 轴/98Ys 轴)

P224, 第二主轴全脉冲控制每圈脉冲数

P225, 第二主轴全脉冲控制升降速(rpm/2ms)

P226, 第二主轴跟随第一主轴的转速分子[1:表无效, 1000+分子:表示跟随第一主轴]

P227, 第二主轴跟随第一主轴的转速分母[1:表无效, 1000+分母:表示跟随第一主轴]

P228, 第一主轴的启停是否同时控制第二主轴[0 表是, 1 表否]

速度参数

P48 号参数: 用于设置主轴最高转速, 也就是对应 10V 指令电压时的转速。
单位: r/min

注: 当有高低档、多档功能时此参数为主轴高档、第一档最高转速。

P49 号参数: 用于设置主轴低档(第二档)最高转速或为第二主轴的最高转速, 也就是对应 10V 指令电压时的转速。单位: r/min

P50 号参数: 用于设置主轴第三档最高转速, 也就是对应 10V 指令电压时的转速。单位: r/min

P51 号参数: 用于设置主轴第四档最高转速, 也就是对应 10V 指令电压时的转速。单位: r/min

P52 号参数: 用于设置第二主轴最高转速, 也就是对应 10V 指令电压时的转速。单位: r/min

杂类参数

P13 号参数: 用于设置主轴转动与卡盘状态是否互锁: 设置为 0 时表示卡盘的松、紧状态与主轴转动无任何联系; 当此参数设置为 1 时表示卡盘的松、紧状态与主轴转动之间有互锁关系, 也就是只有卡盘在夹紧状态下主轴才可以转动。主轴转动时顶针不能开或关。

本参数的设置跟机床设备的配置和用户使用情况有关, 从安全角度考虑, 建议设置为互锁。

When the parameter value is set to 1, detection is performed; when the parameter value is set to 0, detection is not performed.

P200, number of pulses per revolution of the spindle encoder (four times the number of encoder lines)

P200-1, spindle encoder detection line count alarm value (diagnostic value) [$>$]10 valid

Parameter P59: Whether the spindle starts when shifting gears [1 indicates yes, 0 indicates no].

Parameter P60: Motor rotation speed (1/100rpm) when the spindle shifts gears.

Parameter P61: rotation direction when the spindle shifts gears [0 for forward, 1 for reverse].

Parameter P62: Stop time when the spindle shifts gears (10ms).

Parameter P63: Low speed rotation time (10ms) when the spindle shifts gears.

Parameter P64: Spindle stop delay time (10ms).

P220, first spindle full pulse control channel (forward 82, reverse 92)

P221, the number of pulses per revolution of the first spindle is fully controlled

P222, first spindle full pulse control speed (rpm/2ms)

P223, second spindle full pulse control channel (forward 84A axis / 85B axis / 88Ys axis, reverse / 94A axis / 95B axis / 98Ys axis)

P224, the number of pulses per revolution of the second spindle full pulse control

P225, second spindle full pulse control lift speed (rpm/2ms)

P226, the rotational speed of the second spindle relative to the first spindle [1: invalid table, 1000+ molecules: indicates following the first spindle]

P227, the denominator of the second spindle's rotational speed relative to the first spindle [1: invalid table, 1000 + denominator: indicates following the first spindle]

P228, whether the start/stop of the first spindle is controlled simultaneously with the second spindle [0 for yes, 1 for no]

parameter of velocity

Parameter P48: Used to set the maximum spindle speed, that is, the speed corresponding to the 10V instruction voltage. Unit: r/min

Note: When there are high and low speed, multi-speed functions, this parameter is the spindle high speed, the first speed is the highest.

Parameter P49: Used to set the maximum speed of the spindle low gear (second gear) or the maximum speed of the second spindle, that is, the speed corresponding to the 10V instruction voltage. Unit: r/min

Parameter P50: Used to set the highest speed of the third gear of the spindle, that is, the speed corresponding to the 10V instruction voltage. Unit: r/min

Parameter P51: Used to set the highest speed of the fourth gear of the spindle, that is, the speed corresponding to the 10V instruction voltage. Unit: r/min

Parameter P52: Used to set the maximum speed of the second spindle, that is, the speed corresponding to the 10V instruction voltage. Unit: r/min

Miscellaneous parameters

Parameter P13: Controls spindle rotation interlock with chuck status. Setting to 0 indicates no correlation between chuck clamping state and spindle rotation. When set to 1, the system enforces an interlock mechanism where spindle rotation is restricted until the chuck is securely clamped. This parameter ensures the ejector pin remains locked during spindle operation.

The setting of this parameter is related to the configuration of the machine tool equipment and user usage. From the perspective of safety, it is recommended to set it as interlock.

4.8.3.2 系统润滑控制 (M32/M33)

本系统可以通过“杂类参数”P4号设置自动润滑功能。设置此功能后“杂类参数”P6号参数为润滑间隔时间（单位：秒）；“杂类参数”P5号参数为润滑时间（单位：ms）。

信号通过 M32 输出控制。

4.8.3.3 松紧刀控制 (M10/M11)

本系统的松紧刀控制相关参数如下：

杂类参数

P13号参数：用于设置主轴转动与松紧刀状态是否互锁：设置为0时表示松紧刀的松、紧状态与主轴转动无任何联系；设置为1时表示松紧刀的松、紧状态与主轴转动之间有互锁关系，也就是只有松紧刀在夹紧状态下主轴才可以转动。

P20号参数：用于设置系统控制松紧刀是只需一个信号（单向阀）还是需两个信号（双向阀），该参数根机床配置有关。

当设置为0时，只是M10一个输出信号控制松紧刀松紧，当M10有效时，系统执行松紧刀紧动作，M10无效时系统执行松紧刀松；

当设置为1时分别由M10和M71控制松紧刀的松和紧当继电器M10有效M71无效时系统执行松紧刀紧动作，当M71有效、M10无效时系统执行松紧刀松动作。即M10时输出M10、M78时输出M71。

P22号参数：用于设置系统是否外接控制松紧刀松紧的按钮（或脚踏开关），该信号为往复信号，即有效一次松紧刀的控制状态变化一次（有效一次松、再有效一次紧，往复方式）。当设置为0时表无外接松紧刀控制；当设置为1时表有外接松紧刀控制，该信号为M16。

P24号参数：用于设置当松紧刀输出控制信号M10、M71为短信号时的保持时间，若该值为0则表示为长信号。单位：S（秒）。

注意：M16为多功能信号，使用时只能选一种功能使用。

4.8.4 系统报警信号：ALM、ALM1、ALM2、门报警 M12、急停

杂类参数

P7号参数：用于设置系统是否要检测防护门开关信号，设置为0时表示无门开关；设置为1时表有门开关；为了安全建议设置检测。

P8号参数：用于设置门开关的类型，0表常开，1表常闭

P17号参数：用于设置系统检测伺服报警信号（CN5的12脚ALM）的类型，0表常开，1表常闭。

P18号参数：用于设置系统检测主轴驱动报警信号（CN3的5脚ALM1）的类型，0表常开，1表常闭。

P19号参数：用于设置系统检测机床报警信号（CN19的2脚ALM2）的类型，0

4.8.3.2 System Lubrication Control (M32/M33)

The system can set the automatic lubrication function through parameter P4 in Miscellaneous Parameters. After setting this function, parameter P6 in Miscellaneous Parameters is the lubrication interval time (unit: seconds), and parameter P5 in Miscellaneous Parameters is the lubrication time (unit: ms).

The signal is controlled by the M32 output.

4.8.3.3 Loose and Tight Knife Control (M10/M11)

The control parameters of the loosening knife in this system are as follows:

Miscellaneous parameters

Parameter P13: Configures the interlock between spindle rotation and tool clamping status. When set to 0, the tool's clamping state remains independent of spindle rotation. When set to 1, the tool's clamping status triggers an interlock mechanism, meaning the spindle can only rotate when the tool is securely clamped.

Parameter P20: Used to set whether the system control knife is only one signal (one-way valve) or two signals (two-way valve), which is related to the machine tool configuration.

When set to 0, only the output signal of M10 controls the tightening and loosening of the knife. When M10 is valid, the system performs the tightening action of the knife; when M10 is invalid, the system performs the loosening of the knife;

When set to 1, the tension control of the clamping knife is managed by M10 and M71. The system performs clamping actions when M10 is active and M71 inactive. Conversely, clamping actions are executed when M71 is active and M10 inactive. Specifically, M10 outputs M10 and M78 during operation, while M71 is activated when M10 is inactive.

Parameter P22: Controls whether the system is connected to an external control lever (or foot pedal switch) for operating the clamping knife. This signal operates in a reciprocating pattern, meaning each valid cycle (one effective release followed by one effective engagement) corresponds to a complete control state change. When set to 0, the system lacks external clamping knife control; when set to 1, it enables external control via M16 signal.

Parameter P24: Used to set the holding time when the output control signals M10 and M71 of the tightening knife are short signals. If this value is 0, it means long signals. Unit: S (second).

Note: M16 is a multi-functional signal, and only one function can be selected when using it.

4.8.4 System Alarm Signal: ALM, ALM1, ALM2, Door Alarm M12, Emergency Stop

Miscellaneous parameters

Parameter P7: Used to set whether the system should detect the opening and closing signal of the protective door. If set to 0, there is no door switch; if set to 1, there is a door switch. For safety, it is recommended to set detection.

Parameter P8: Used to set the type of door switch, 0 is normally open, 1 is normally closed

Parameter P17: Used to set the type of servo alarm signal (12 ALM pins of CN5) detected by the system. 0

Table is normally open, 1 Table is normally closed.

Parameter P18: Used to set the type of alarm signal (CN3's 5-lead ALM1) detected by the system for spindle drive. 0 means normally open, and 1 means normally closed.

Parameter P19: Used to set the type of system detection machine tool alarm signal (2 ALM2 pins of CN19), 0

表常开, 1 表常闭。

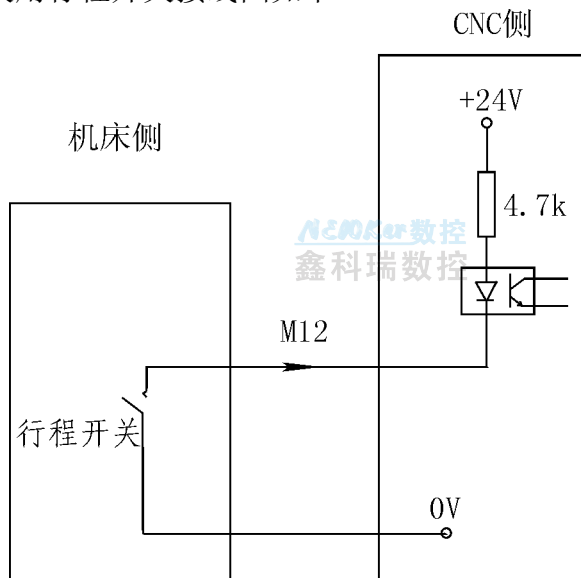
P27 号参数 用于设置系统 CN11 上的 STOP 急停开关（手轮急停或外接急停）常开或常闭，可通过此参数设置，为了安全建议设置为常闭。

急停 STOP：当发生紧急意外情况时可按下“紧急停止”按钮，系统将停止机床一切动作，系统复位屏幕显示“紧急停车”。等待按钮抬起。当杂类参数 P29 号有效后，M67 输出信号有效（报警输出）。该信号输出可用来做机床安全保护（切断电源等）或状态指示。

当出现机床报警时若杂类参数 P29 号有效后，M67 输出信号有效。

4.8.5 自定义输入信号 M12（M14、M16、M18、M28、M22、M24/B0、X40/X47、外接运行 RUN、外接暂停 HALT、外接急停 STOP 相同）

M12 开关用行程开关接线图如下



M12 开关也可以使用 NPN 型霍尔元件作检测开关，见下图

Table is normally open, 1 Table is normally closed.

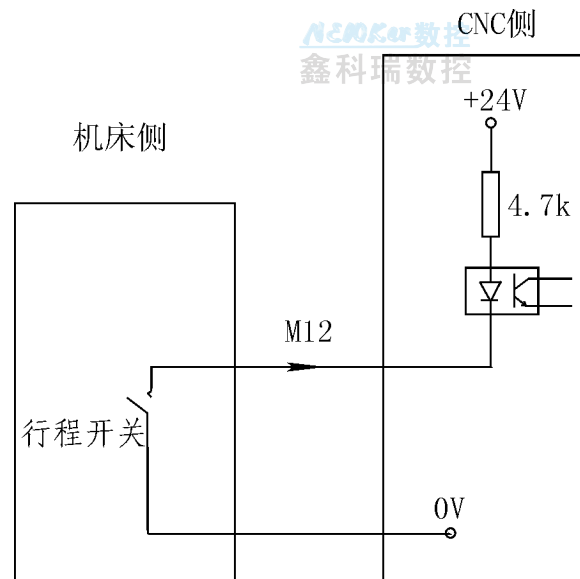
Parameter P27: Used to set the STOP and emergency stop switch (handwheel emergency stop or external emergency stop) on the system CN11 to be normally open or normally closed. This parameter can be set. For safety, it is recommended to set it as normally closed.

Emergency Stop: In case of emergency, press the "Emergency Stop" button to halt all machine operations. The system will reset and display "Emergency Shutdown" on the screen. Wait until the button is released. When parameter P29 becomes active, the M67 output signal becomes effective (alarm output). This signal can be used for machine safety protection (e.g., power disconnection) or status indication.

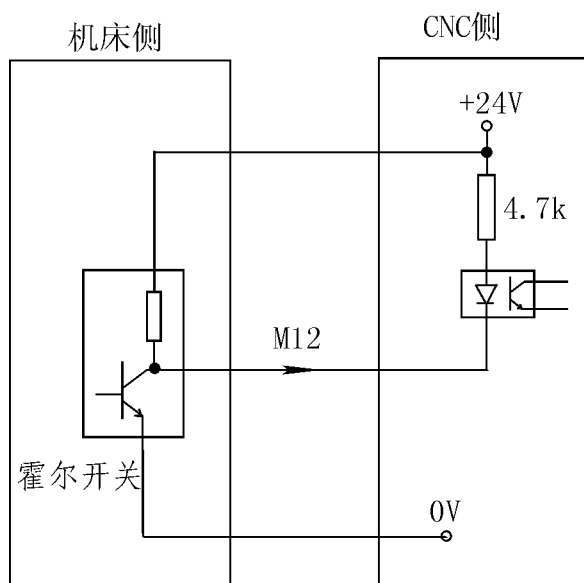
When the machine tool alarm occurs, if the miscellaneous parameter P29 is valid, the output signal of M67 is valid.

4.8.5 Custom input signals: M12 (also applicable to M14, M16, M18, M28, M22, M24/B0, X40/X47, RUN, HALT, STOP for external operation, external pause, and emergency stop)

The wiring diagram of the M12 switch is shown below



The M12 switch can also be used with an NPN type Hall element as a detection switch, as shown in the figure below

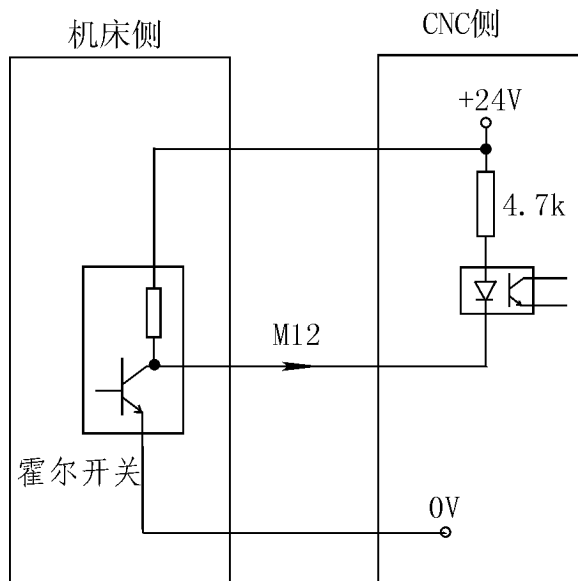


注意：（输入/输出配置见系统 I/O 诊断画面）

- 1、M12、M14、M16、M18、M28 均是多功能信号，在使用时只能用其中之一。
- 2、所有输入信号均为低电平 0V 有效。
- 3、通过 PLC 可编程输入 56 点 X00-Y47, X60-X67。注意功能不能与原来的 PLC 及系统功能有重复使用的地方。
- 4、所有输入点中 X32-X39、X60-X67 内部为+5V 电源（内部为常闭），使用时切勿接入高于+5V 的电压。

4.9 机床电器板

本公司生产有机床电器板，可以选配，见下图。



Note:(Input/output configuration is shown in the system I/O diagnostic screen)

1、 M12, M14, M16, M18 and M28 are all multi-purpose signals, only one of which can be used in use.

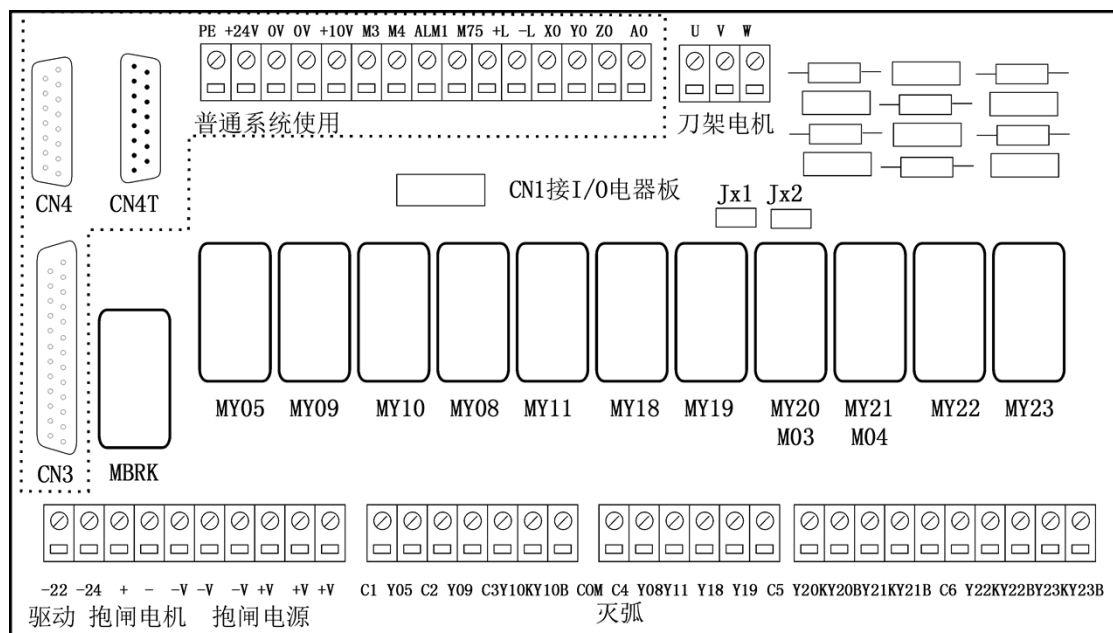
2、 All input signals are valid at low level 0V.

3、 Through the PLC, 56 programmable inputs are available: X00-Y 47 and X60-X67. Note that the functions cannot be reused in places where they are duplicated with the original PLC and system functions.

4、 In all input points, X32-X39 and X60-X67 are powered by +5V (internal normally closed). Do not connect the voltage higher than +5V when using.

4.9 Machine Tool Electrical Board

Our company produces electrical board for machine tools, which can be selected and equipped. See the figure below.

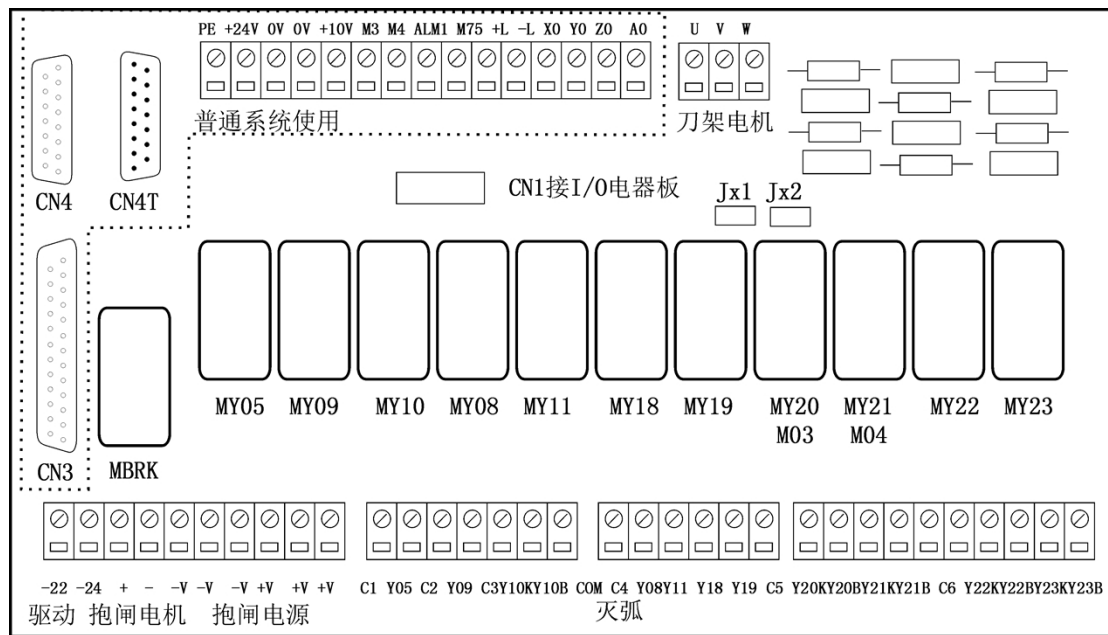


本公司生产有机床继电器板，可以选配，见图。

- 1、CN1 接继电器板 CN1, CN1-1 至-11 脚控制继电器:
Y05, Y09, Y10, Y08, Y11, Y18, Y19, Y20, Y21, Y22, Y23 对应:
M71, M79, M10, M32, M08, +T, -T, S04, S03, S02, S01。
- 2、_22 接驱动器 CN3-22 脚, _24 接驱动器 CN3-24 脚。
- 3、-, +接抱闸电机。
- 4、-V, +V 外接抱闸电源。
- 5、PE 接线端子必须牢靠接到机床接地排, 以防止干扰。
- 6、虚线部分为普通系统使用: CN3、CN4 接系统 CN3、CN4; CN4T 接电动刀架信号插座;
- 7、跳针 JX1/JX2 选择 MY20/MY21 继电器是控制 M03/M04 还是控制 S04/S03。
- 8、C1 是 M71/Y05 公共瑞; C2 是 M79/Y09 公共瑞; C3 是 M10/Y10 公共瑞;
C4 是 M32/Y08, M08/Y11, +T/Y18, -T/Y19 公共瑞;
C5 是 S04 (M03), S03 (M04) 公共瑞; C5 是 S02/Y22, S01/Y23 公共瑞;

4.10 总线系统 I/O 机床电器板

《1》 335G 接口板:

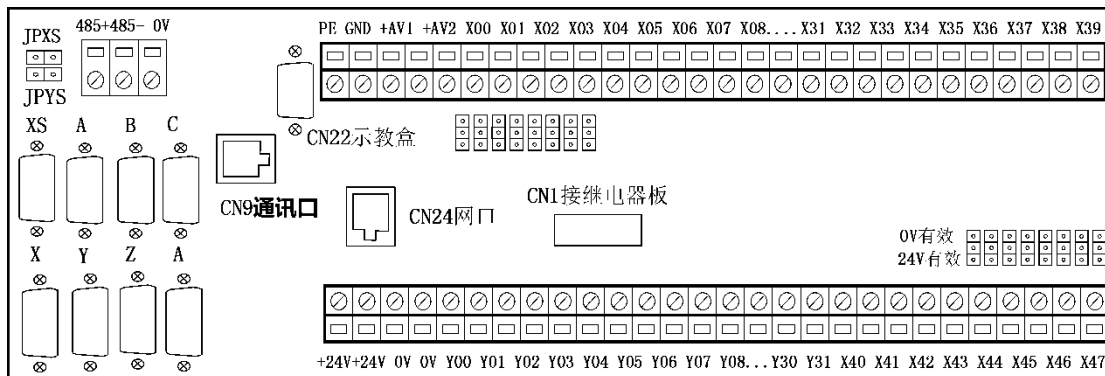


Our company produces machine tool relay board, which can be selected and equipped, as shown in the figure.

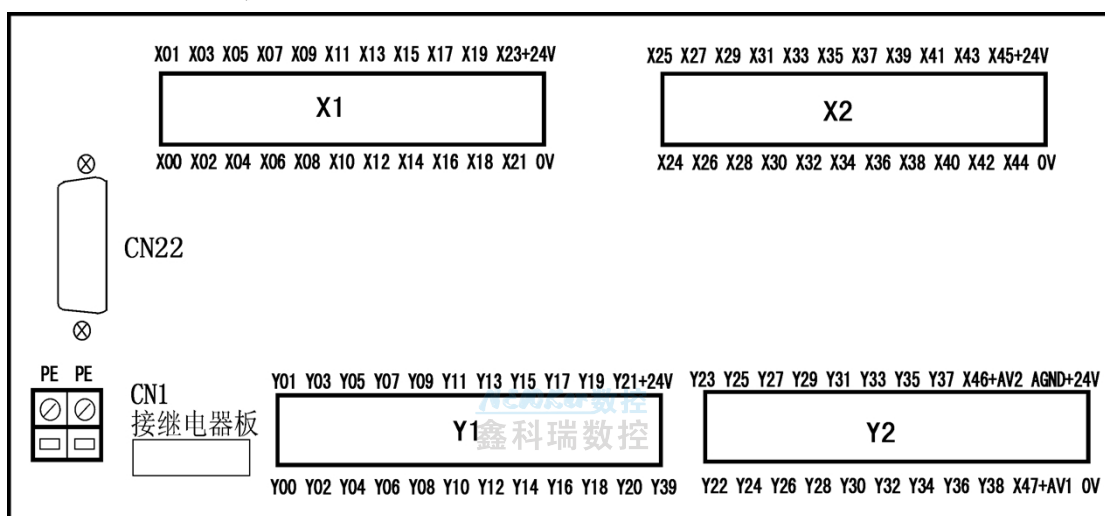
- 1、 CN1 Relay board CN1, CN1-1 to CN1-11 control relays:
Y05, Y09, Y10, Y08, Y11, Y18, Y19, Y20, Y21, Y22, Y23 correspond to:
M71, M79, M10, M32, M08, +T, -T, S04, S03, S02, S01。
- 2、 _22 Connect the driver CN3-22 foot, _24 Connect the driver CN 3-24 foot.
- 3、 - + Connect the brake motor.
- 4、 -V, +V external brake power supply.
- 5、 PE terminal must be firmly connected to the machine tool grounding panel to prevent interference.
- 6 The dashed line part is for the use of ordinary system: CN3 and CN4 are connected to the system CN3 and CN4; CN4T is connected to the signal socket of the electric knife frame;
- 7、 JX1/JX2 select MY20/MY21 relay to control M03/M04 or S04/S03.
- 8、 C1 is M71/Y05 public rite; C2 is M79/Y09 public rite; C3 is M 10/Y10 public rite; C4 is M32/Y08, M08/Y11, +T/Y18, -T/Y19 public rites;
C5 is the common reference for S04 (M03) and S03 (M04); C5 is the common reference for S02/Y22 and S01/Y23;

4.10 Bus System I/O Machine Tool Electrical Board

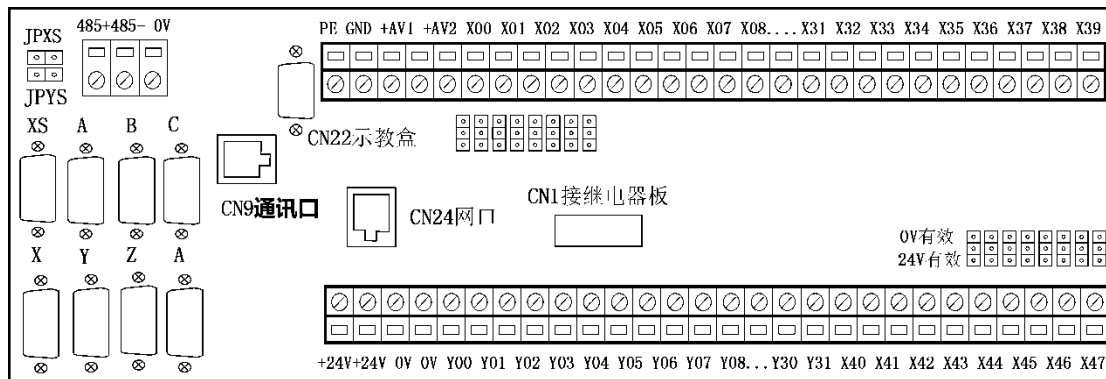
- 1) 335G interface board;



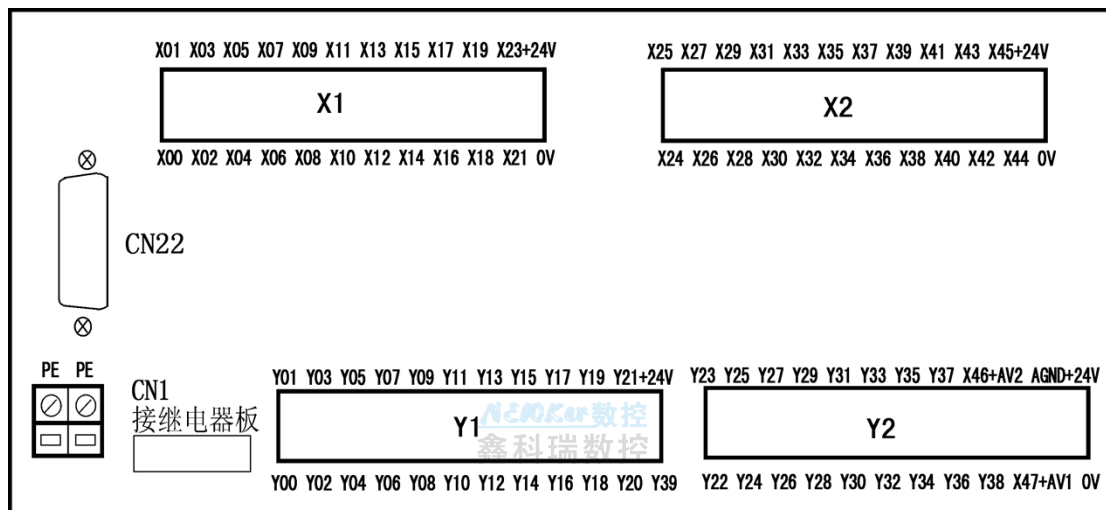
《2》2026G 接口板;



- 1) CN22 接机器人示教盒;
 - 2) CN24 接 POWERlink 总线驱动器;
 - 3) CN9 接 RS485 通讯口;
 - 4) CJ1, CJ2 接输入输出信号;
 - 5) CJ3, CJ4 接标准 RS485 通讯;
 - 6) XYZABC/XS/YS 接脉冲型伺服驱动器;
 - 7) JPXS、JPYS 跳针, 当接脉冲型驱动器时, 如果报警信号是常闭信号, 当没有 XS 坐标轴或没有坐标 YS 轴时对应的 JPXS、JPYS 要插上跳针 (即有 XS/YS 轴时要取掉对应的跳针);
 - 8) +AV1, +AV2 是二路模拟量 0-10V 输出信号, 0V 是信号地;
 - 9) CN1 接继电器板 CN1, CN1-1 至-11 脚控制继电器:
Y05, Y09, Y10, Y08, Y11, Y18, Y19, Y20, Y21, Y22, Y23 对应:
M71, M79, M10, M32, M08, +T, -T, S04, S03, S02, S01;
- 特别注意:



"2" 2026G interface board;



- 1) CN22 Robot teaching box;
 - 2) CN24 Connect to POWERlink bus driver;
 - 3) CN9 connects to RS485 communication port;
 - 4) CJ1 and CJ2 receive input/output signals;
 - 5) CJ3 and CJ4 are connected to standard RS485 communication;
 - 6) XYZABC/XS/YS pulse-connected servo driver;
 - 7) JPXS and JPYS jump pins: When connected to a pulse-type driver, if the alarm signal is normally closed, the corresponding JPXS and JPYS should be inserted with jump pins (i.e., remove the corresponding jump pins when there is XS/YS axis);
 - 8) +AV1 and +AV2 are two analog 0-10V output signals, and 0V is the signal ground;
 - 9) CN1 Relay board CN1, CN1-1 to CN1-11 control relays:
Y05, Y09, Y10, Y08, Y11, Y18, Y19, Y20, Y21, Y22, Y23 correspond to:
M71, M79, M10, M32, M08, +T, -T, S04, S03, S02, S01;
- Special note:

PE 接线端子必须牢固连接到接地排，以防止干扰。

1、输入

X0-X07 为电动刀架 T01-T08；

X20 为 XYZA 驱动器报警 ALM；

X21 为电动刀架到位信号 TOK；

X22 为 B/C/XS/YS 驱动器报警 ALMB；

X23 为主轴变频器报警；

X24 无润滑油报警 ALM2；

X09 为负限位开关信号-L；

X10 为正限位开关信号+L；

X08/X11/X12/X13/X26/X25/X17/X16 为 A/Y/X/Z/B/C/XS/YS 轴回零信号

A0/Y0/X0/Z0/B0/C0/XS0/YS0；

X27/M22； X28/M18； X29/M12； X30/M14； X31/M16；

X14/X15/X18/X19/X40-X47 为用户自定义输入点；

2、输出

Y16 为驱动使能信号 EN, 也是总线系统配一体驱动电机抱闸输出信号；

Y17 为驱动报警清除信号 INTH；

Y30 (SPEN) 为主轴伺服使能信号；

Y31 (SPEN2) 作为第二主轴伺服的使能信号；

Y00/M61； Y01/M63； Y02/M65； Y03/M67； Y04/M69； Y05/M71；

Y06/M73； Y07/M59； Y08/M32； Y09/M79； Y10/M10； Y11/M08；

Y12/M05； Y13/M04； Y14/M03； Y15/M75； Y16/EN； Y17/INTH；

Y18/+T； Y19/-T； Y20/S04； Y21/S03； Y22/S02； Y23/S01；

Y28/M203； Y29/M204； Y30/SPEN； Y31/SPEN2；

Y24-Y27 为用户自定义输出点；

4. 10. 1 脉冲型驱动器插座 XYZABC/XS/YS 的 DB15 连接针头

总线系统接脉冲型伺服驱动器接线表：

XYZABC/XS/YS 脉冲型驱动器 DB15 针头 (7 对)				301 驱动	302 驱动
信号	脚号	I/O	功能	DB25 针头	DB25 针头
CP+/CP-	1/9 (1 对)	OUT	脉冲信号	6/18	6/18
DIR+/DIR-	2/10 (2 对)	OUT	方向信号	7/19	7/19
EN/INTH	3/4 (3 对)	OUT	使能/报警清除	23/10	302G:23/10 302H:14/==
+24V/0V	11/13 (4 对)	OUT	24V 电源/0V	11/21	302G:11/21 302H:2/25

PE terminal must be securely connected to the grounding bar to prevent interference.

1、import

X0-X07 is the electric knife holder T01-T08;

X20 is the XYZA driver alarm ALM;

X21 is the electric knife holder position signal TOK;

X22 is the alarm ALMB for B/C/XS/YS driver;

X23 is the alarm of spindle frequency converter;

X24 No lubricating oil alarm ALM2;

X09 is the negative limit switch signal-L;

X10 is the positive limit switch signal +L;

X08/X11/X12/X13/X26/X25/X17/X16 are the A/Y/X/Z/B/C/XS/YS axis return signals A0/Y0/X0/Z0/B0/C0/XS0/YS0;

X27/M22; X28/M18; X29/M12; X30/M14; X31/M16;

X14/X15/X18/X19/X40-X47 are user-defined input points;

2、output

Y16 is the drive enable signal EN, which is also the output signal of the drive motor brake in the bus system;

Y17 is the drive alarm clear signal INTH;

Y30 (SPEN) is the spindle servo enable signal;

Y31 (SPEN2) as the enabling signal of the second spindle servo;

Y00/M61; Y01/M63; Y02/M65; Y03/M67; Y04/M69; Y05/M71;

Y06/M73; Y07/M59; Y08/M32; Y09/M79; Y10/M10; Y11/M08;

Y12/M05; Y13/M04; Y14/M03; Y15/M75; Y16/EN; Y17/INTH;

Y18/+T; Y19/-T; Y20/S04; Y21/S03; Y22/S02; Y23/S01;

Y28/M203; Y29/M204; Y30/SPEN; Y31/SPEN2;

Y24-Y27 are user-defined output points;

4.10.1 Pulse-type driver socket XYZABC/XS/YS DB15 connector bus system connected to pulse-type servo driver wiring list:

XYZABC/XS/YS pulse driver DB15 needle (7 pairs)				301 drive	302 drive
signal	subscript	I/O	function	DB25 needle	DB25 needle
CP+/CP-	1 / 9 (1 pair)	OUT	pulsing signal	6/18	6/18
DIR+/DIR-	2 / 10 (2 pairs)	OUT	Directional signal	7/19	7/19
EN/INTH	3 / 4 (3 pairs)	OUT	Enable/alarm clearance	23/10	302G:23/10 302H:14/==
+24V/0V	11 / 13 (4 pairs)	OUT	24V power / 0V	11/21	302G:11/21 302H:2/25

BP+/BP-	15/8 (5 对)	IN	编码器反馈 B	4/16	302G: 不焊 302H: 8/1
AP+/AP-	14/7 (6 对)	IN	编码器反馈 A	3/15	302G: 不焊 302H: 21/20
ALMB-/ALMB+	6/5 (7 对)	IN	驱动报警常闭	13/12	13/12
ALMK+/ALMK-	12/13 (7 对)	IN	驱动报警常开		
485+/485-				CN2-5/6	5/17
BK+/BK-				22/24	22/24
第二通道脉冲型驱动器 DB15 针头 (4 对)					
CP+/CP-	1/9 (1 对)	OUT	脉冲信号		3/15
DIR+/DIR-	2/10 (2 对)	OUT	方向信号		4/16
BP+/BP-	15/8 (3 对)	IN	编码器反馈 B		302G: 不焊 302H: 23/9
AP+/AP-	14/7 (4 对)	IN	编码器反馈 A		302G: 不焊 302H: 11/10
ALMB-/ALMB+	6/5 (短接)	IN	驱动报警常闭		



4. 10. 2 驱动器 Powerlink 总线网口插座 CN24 的连接

此插座连接到 Powerlink 伺服驱动器的总线网口插座。

CN24 驱动器总线网口信号插座 (2 对)				
信号	脚号	I/O	功能	对应标准网口
TXA-/TXA+	1/2	IN	A 信号	2/1
TXB-/TXB+	3/6	IN	B 信号	3/6

4. 10. 3 RS485 通讯口 CN9 的连接

CN9 RS485 通讯口 (2 对)				
信号	脚号	I/O	功能	
485+/485-	4/5		信号	
0V	1		信号地	

注意：信号线必须采用双绞屏蔽电缆线。

BP+/BP-	15 / 8 (5 pairs)	IN	Encoder feedback B	4/16	302G: No welding 302H: 8/1
AP+/AP-	14 / 7 (6 pairs)	IN	Encoder feedback A	3/15	302G: No welding 302H: 21/20
ALMB-/ALMB+	6 / 5 (7 pairs)	IN	Drive alarm normally closed	13/12	13/12
ALMK+/ALMK-	12 / 13 (7 pairs)	IN	Drive alarm is normally on		
485+/485-				CN2-5/6	5/17
BK+/BK-				22/24	22/24
DB15 needle (4 pairs) for second channel pulse driver					
CP+/CP-	1 / 9 (1 pair)	OUT	pulsing signal		3/15
DIR+/DIR-	2 / 10 (2 pairs)	OUT	Directional signal		4/16
BP+/BP-	15 / 8 (3 pairs)	IN	Encoder feedback B		302G: No welding 302H: 23/9
AP+/AP-	14 / 7 (4 pairs)	IN	Encoder feedback A		302G: No welding 302H: 11/10
ALMB-/ALMB+	6 / 5 (short circuit)	IN	Drive alarm normally closed		

4.10.2 Drive Power Link Bus Network Port Connection CN24

This Socket Is Connected to the Bus Network Port Socket of the Powerlink Servo Drive.

CN24 Driver bus network port signal socket (2 pairs)				
sig-nal	subscript	I / 0	function	Correspond to the standard network port
TXA-/TXA+	1/2	1	A signal	2/1
TXB-/TXB+	3/6	1	B signal	3/6

4.10.3 Connect the RS485 Communication Port CN9

CN9 RS485 communication port (2 pairs)				
signal	subscript	I / 0	function	
485+/485-	4/5		signal	
0 V	1		Signal ground	

Note: Signal lines must be double shielded cable.

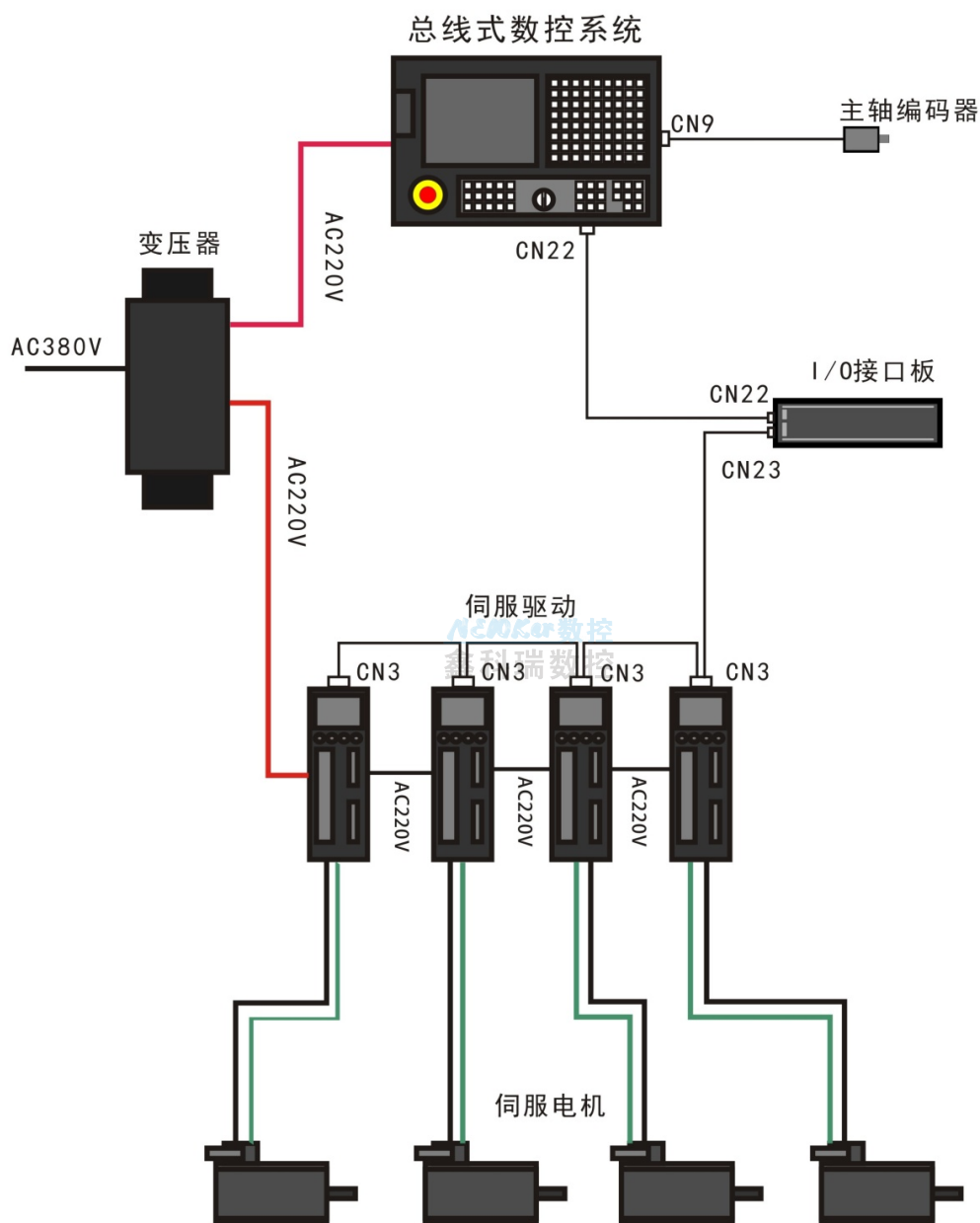
4.11 总线系统电路连接图

NEOKer数控
鑫科瑞数控

4.11 Bus System Circuit Diagram

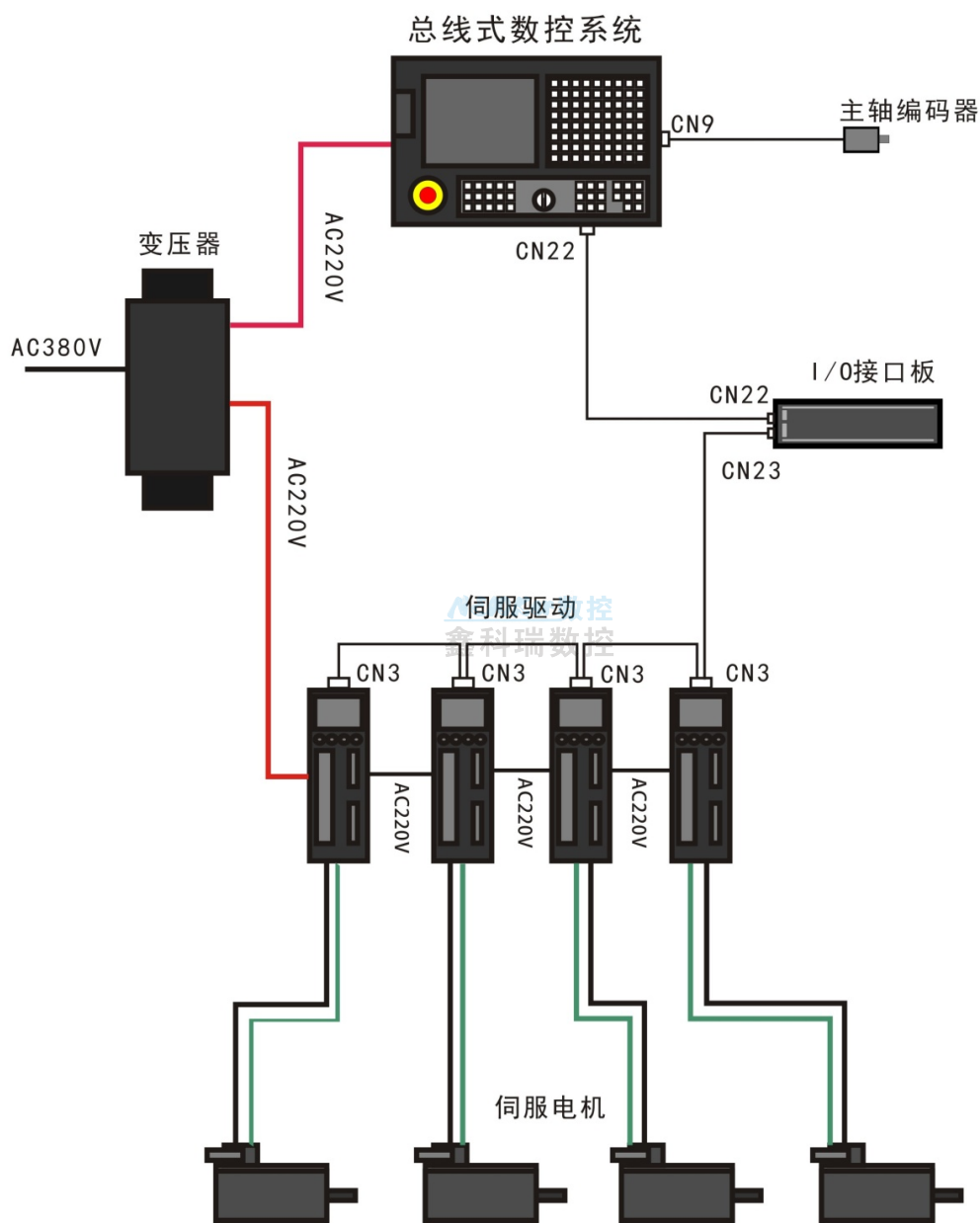
NEOKer 数控
鑫科瑞数控

总线式系统驱动连接图



第五篇 加工中心篇

总线式系统驱动连接图



Part V: Processing Centers

5.1 输入输出接口

I/O	信号	斗笠式刀库加工中心	机械手刀库加工中心
X00	T01	刀库电机过载检测	刀库电机过载检测;
X01	T02	主轴松刀检测	主轴松刀检测
X02	T03	润滑油位检测	润滑油位检测
X03	T04	冷却缺液检测	冷却缺液检测
X04	T05	主轴紧刀检测	主轴紧刀检测
X05	T06	刀库前进检测	
X06	T07	刀库后退检测 要求: 启动主轴时刀库必须后退到位	
X07	T08	刀库计数(0V 有效 CN4-8)	刀库计数(0V 有效 CN4-8)
X08	M36/A0	A 轴回零信号	A 轴回零信号
X23	ALM1	主轴驱动器报警	主轴驱动器报警
X24	ALM2	冷却电机过载检测	冷却电机过载检测
X25	M28/C0	C 轴回零信号	C 轴回零信号
X26	M24/B0	B 轴回零信号	B 轴回零信号
X27	M22	主轴定位完成/回零到位检测	主轴定位完成/回零到位检测
X29	M12	机床门开关检测 要求: 杂类参数 P7=1	机床门开关检测 要求: 杂类参数 P7=1
X30	M14	压缩空气压力检测报警	压缩空气压力检测报警
X31	M16	主轴远程松刀按钮 要求: 杂类参数 P22=1	主轴远程松刀按钮 要求: 杂类参数 P22=1
X40		排屑电机过载检测报警	排屑电机过载检测报警
X41			刀库定位检测, 如果刀库没有定位信号可以接到 T08
X42			刀库机械手停止检测: (机械手扣刀、机械手抓刀换刀、机械手归位)
X43			1、机械手原点信号, 换刀前检测此信号为无效 2、伺服刀库夹紧松开到位信号
X44			刀套抬起检测 要求: 换刀前必须抬起到位
X45			刀套落下检测

5.1 Inputoutput Interface

I/O	signal	Hat-shaped knife store machining center	Mechanical tool magazine machining center
X00	T01	Overload detection of knife library motor	Overload detection of knife library motor;
X01	T02	Shaft loosening detection	Shaft loosening detection
X02	T03	Lubrication level detection	Lubrication level detection
X03	T04	Cooling and fluid loss detection	Cooling and fluid loss detection
X04	T05	Spindle tight knife detection	Spindle tight knife detection
X05	T06	Knife library advance detection	
X06	T07	Knife library backstop detection Requirement: The tool magazine must be retracted to the right position when starting the spindle	
X07	T08	Knife count (0V valid CN4-8)	Knife count (0V valid CN4-8)
X08	M36/A0	A zero return signal on the A axis	A zero return signal on the A axis
X23	ALM1	The spindle driver alarms	The spindle driver alarms
X24	ALM2	Cool motor overload detection	Cool motor overload detection
X25	M28/C0	C-axis zero return signal	C-axis zero return signal
X26	M24/B0	B-axis zero return signal	B-axis zero return signal
X27	M22	Main shaft positioning is complete/ zero return is detected	Axis positioning is complete/zero return is detected
X29	M12	Machine door switch detection Request: Miscellaneous parameter P7=1	Machine door switch detection Request: Miscellaneous parameter P7=1
X30	M14	Compressed air pressure detection alarm	Air pressure detection alarm
X31	M16	Main shaft remote tool release button requirements: Miscellaneous parameter P22=1	Main shaft remote tool release button requirements: Miscellaneous parameter P22=1
X40		Overload detection alarm of dust removal motor	Overload detection alarm of dust removal motor
X41			Tool magazine positioning detection. If the tool magazine has no positioning signal, it can be connected to T08
X42			The knife library manipulator stops detection: (Mechanical hand knife, mechanical hand grab and change the knife, mechanical hand return)
X43			1、The mechanical tool origin signal is invalid before tool change 2. Servo tool magazine clamping release signal
X44			Raise the knife handle to detect Requirement: The blade must be lifted in place before changing
X45			Knife sleeve drop detection

X46		准停指令 M01 输入	准停指令 M01 输入
X47		主轴风机过载检测	主轴风机过载检测
Y00	M61	主轴回零输出, 检测回零到位 M22	主轴回零输出, 检测回零到位 M22
Y01	M63	刀库正转控制输出, 检测 T08 刀位计数;	刀库正转控制输出, 检测 T08 刀位计数;
Y02	M65	黄指示灯 (停止指示灯)	黄指示灯 (停止指示灯)
Y03	M67	红指示灯 (故障指示灯)	红指示灯 (故障指示灯)
Y04	M69	绿指示灯 (运行指示灯)	绿指示灯 (运行指示灯)
Y05	M71	刀库反转控制输出, 检测 T08 刀位计数;	刀库反转控制输出, 检测 T08 刀位计数;
Y06	M73	刀库前进控制输出, 检测前进到位 T06	
Y07	M59	M59 吹气	M59 吹气
Y08	M32	润滑控制输出	润滑控制输出
Y10	M10	松紧刀控制输出: 主轴松刀: 输出 M10, 检测松刀到位 T02; 主轴紧刀: 撤消 M10, 检测紧刀到位 T05;	松紧刀控制输出: 主轴松刀: 输出 M10, 检测松刀到位 T02; 主轴紧刀: 撤消 M10, 检测紧刀到位 T05;
Y11	M08	冷却控制输出	冷却控制输出
Y12	M05	M05	M05
Y13	M04	主轴反转控制输出	主轴反转控制输出
Y14	M03	主轴正转控制输出	主轴正转控制输出
Y15	M75	主轴控制模式切换输出	主轴控制模式切换输出
Y24		刀库后退控制输出, 检测后退到位 T07	刀库机械手电机控制输出: (机械手扣刀、机械手抓刀换刀、机械手归位)
Y25			伺服刀库夹紧松开输出
Y26			刀套抬起控制输出, 检测抬起到位 X44
Y27			刀套落下控制输出, 检测倒下到位 X45, 刀库定位检测 X41 必须到位;
Y28			
Y29		刚性攻丝提刀输出信号	刚性攻丝提刀输出信号
Y30			
Y31	SPEN2	主轴伺服使能输出信号	主轴伺服使能输出信号

X46		Enter the stop instruction M01	Enter the stop instruction M01
X47		Overload detection of main fan	Overload detection of main fan
Y00	M61	The spindle returns to zero output and detects that the return to zero is in place M22	The spindle returns to zero output and detects that the return to zero is in place M22
Y01	M63	The tool library controls the output in forward rotation and detects the T08 tool position count;	The tool library controls the output in reverse, and detects the T08 tool position count;
Y02	M65	Yellow indicator light (stop indicator light)	Yellow indicator light (stop indicator light)
Y03	M67	Red indicator light (fault indicator)	Red indicator light (fault indicator)
Y04	M69	Green indicator light (operating indicator light)	Green indicator light (operating indicator light)
Y05	M71	Turnaround control output of the tool library, and detect the T08 tool position count;	The tool column reverses the control output and detects the T08 tool position count;
Y06	M73	The knife library forward control output detects the forward position T06	
Y07	M59	M59 blew	M59 blew
Y08	M32	Lubrication control output	Lubrication control output
Y10	M10	Looseness control output: Main shaft loosening: output M10, detect the loosening of T02; Master shaft tight knife: remove M10 and check the tight knife is in place T05;	Looseness control output: Main shaft loosening: output M10, detect the loosening of T02; Master shaft tight knife: remove M10 and check the tight knife is in place T05;
Y11	M08	Cooling controls output	Cooling controls output
Y12	M05	M05	M05
Y13	M04	Shaft reversal control output	Shaft reversal control output
Y14	M03	Main shaft forward rotation control output	Main shaft forward rotation control output
Y15	M75	Switch output of spindle control mode	Switch output of spindle control mode
Y24		The knife library retracts to control the output and detects that it is in place T07	Motor control output of knife library manipulator: (manipulator clamping knife, manipulator grabbing and changing knife, manipulator returning)
Y25			Release the output of the servo tool magazine
Y26			The knife handle lifts to control the output and detects that it is in place at X44
Y27			The knife sleeve is lowered to control the output, and the detection of X45 for the reverse position and X41 for the positioning of the tool magazine must be in place;
Y28			
Y29		Hardening screw drive output signal	Hardening screw drive output signal
Y30			
Y31	SPEN2	The spindle servo enables the output signal	The spindle servo enables the output signal

5.2 加工中心参数

（一）轴类参数

57, 主轴编码器每转的脉冲数(编码器线数的 4 倍)

57-1, 主轴编码器检测线数报警值(诊断值)[>10 有效]

57-2, 主轴每转四倍频脉冲数[主轴与编码器 1:1 时设置为 0, 不是 1:1 时设置>99 有效]

P400, 换刀时是否检测主轴位置反馈(1 表是, 0 表否)

P401, 主轴定向时检测角度

P402, 主轴定向时角度检测误差

P403, 主轴脉冲控制方向信号(0 表反向, 1 表正常)

P404, 主轴是否使用电子齿轮(0 表是, 1 表否)

P405, 主轴低档电子齿轮的分子(1-32767)

P406, 主轴低档电子齿轮的分母(1-32767)

P407, 主轴高档电子齿轮的分子(1-32767)

P408, 主轴高档电子齿轮的分母(1-32767)

（二）刀具参数

刀具总数: 在刀补状态按“F7 设置”进行设置。

P10, 是否启动选刀及换刀功能[0 表否, 1 表是]

P11, 回换刀点轴[3 表 ZX; 4 表 ZY; 5 表 ZXY; 其余表 Z]

P12, 换刀时进给轴回第几参考点[1, 2, 3, 4 代表相应的参考点, 其余值表不运动]

P13, 换刀前是否检测主轴定向和进给轴回换刀点[0 表不检测, 1 表检测, 16 表换刀时先主轴定向]

P14, 换刀时主轴是否定向[0 表不定向, 1 表定向]

P15, 刀库类型[0 表机械手, 1 表排刀库, 2 表斗笠式]

16, 特殊刀库[0 标准, 16 特殊斗笠, 32 伺服刀库, 64 特殊机械手, 128 排刀库, +2 换刀后停主轴, +8 同动打刀, +1024 表 M06 与 M106 交换, +2048 表 M06 与 T 顺序执行, +16384 伺服刀库不回零]

说明: +2 表加工中心换刀后停止主轴运转;

+4 表屏蔽 K5 键;

+8 表采用同动打刀机械手刀库;

+16 表特殊的简易斗笠式刀库, 采用换刀程序 ProgramTool 控制;

+32 表刀库采用伺服电机驱动;

+64 表采用特殊机械手刀库, 采用换刀程序 ProgramM6 控制;

+128 表当刀库类型参数设为 1 时(既不是机械手又不是斗笠式刀库), 通过换刀程序 ProgramTool 来执行 T 指令;

+256 表机械手刀库有刀库前进和后退动作;

+512 表采用特殊刀盘, 刀盘的选刀转动采用换刀程序 ProgramTool 控制;

+1024 表 M06 与 M106 功能交换;

+2048 表 M06 与 T 指令同一行时按编程顺序执行(兼容发那科系统),

5.2 Processing Center Parameters

(1) Axis Parameters

57, Number of pulses per revolution of the spindle encoder (4 times the number of encoder lines) 57-1, Alarm value for the number of detection lines of the spindle encoder (diagnostic value) [>]>10 valid] 57-2, Four times the pulse count per revolution of the spindle [set to 0 when the spindle and encoder are 1:1, set to>99 when not 1:1 valid]

P400, whether the spindle position feedback is detected when changing the tool (1 table yes, 0 table no) P401, detect the Angle when the spindle is oriented

P402, Angle detection error when spindle is oriented

P403, spindle pulse control direction signal (0 indicates reverse, 1 indicates normal)

P404, whether the spindle uses electronic gear (0 indicates yes, 1 indicates no)

P405, molecular components of the spindle low-grade electronic gear (1-32767)

P406, denominator of the low-grade electronic gear of the spindle (1-32767)

P407, molecular components for high-end spindle electronic gears (1-32767)

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P408, denominator of high-grade electronic gear spindle (1-32767)

(2) Tool Parameters

Total number of tools: Set it in "F7 Settings" when the tool is in supplementary state. P10: Whether to start tool selection and tool changing function [0 for no, 1 for yes]

P11, return tool point axis [Table 3: ZX; Table 4: ZY; Table 5: ZXY; all tables Z]

P12: Which reference point does the feed axis return to during tool change? [1,2,3,4 represent corresponding reference points, other values remain stationary] P13: Does spindle orientation and feed axis return to tool change point get checked before tool change? [0 indicates no check, 1 indicates check, 16 indicates spindle orientation first during tool change]

P14, whether the spindle is oriented when changing the tool [0 indicates not oriented, 1 indicates oriented] P15, tool magazine type [0 indicates manipulator, 1 indicates tool magazine, 2 indicates conical bucket]

16, Special Tool Magazine [0 Standard, 16 Special Hat, 32 Servo Tool Magazine, 64 Special Robotic Arm, 128 Tool Magazine, +2 Stop spindle after tool change, +8 simultaneous tool striking, +1024 M06/M106 tool swap, +2048 M06/T sequential execution, +16384 Servo Tool Magazine non-zero return]

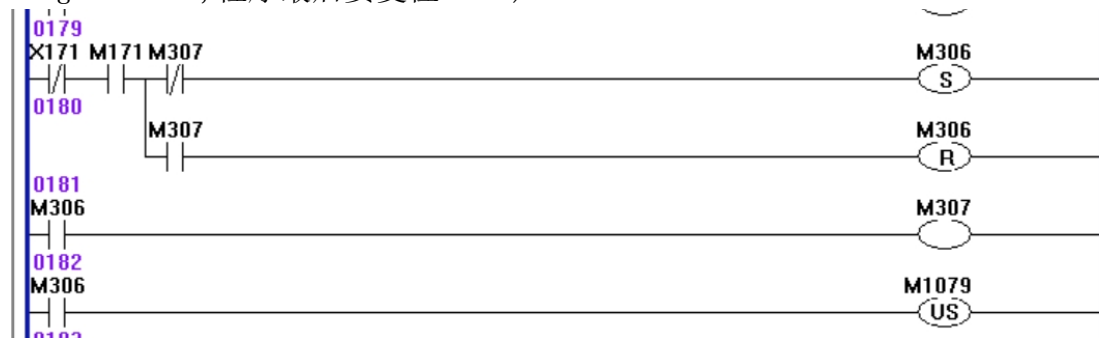
Note: +2 Stop the spindle operation after tool change in the machining center;

- +4 Shielded K5 key;
 - +8 The table adopts the same dynamic knife mechanical hand knife library;
 - +16 A special simple conical knife storage, controlled by the program tool ProgramTool;
 - +32 The table knife library is driven by servo motor;
 - +64 The table adopts a special mechanical knife tool library, which is controlled by the tool changing program ProgramM 6;
 - +128 When the type parameter of the blade library is set to 1 (neither a robot nor a bamboo hat type blade library), the T command is executed by the blade changing program ProgramTool;
 - +256 The table manipulator tool library has forward and backward movements;
 - +512 Table adopts a special tool disk, and the selection and rotation of the tool disk are controlled by the tool changing program ProgramTool; +1024 Table M06 and M106 functions are exchanged;
- When the +2048 table M06 is on the same line as the T command, it is executed in programming order (compatible with FANUC system),

如:M06Txx, TxxM06, TxxM06Txx;

+4096 表屏蔽加工中心的K1 键(手动主轴定位);屏蔽K1 固化功能后,在PLC 中按键K1 可以启动 M1070-M1079/M1140-M1179:

M1070-M1079/M1140-M1179 对应控制程序 ProgramUser0-ProgramUser9/ProgramUser10-ProgramUser49;例如下面 PLC 按 K1 启动 M1079(对应 ProgramUser9, 程序最后要复位 M306):



+8192 表屏蔽加工中心的 K3 键(设定当前刀套号);

+16384 表加工中心刀库开机不需要回零;

+32768 表主轴定向或 Z 轴回换刀点时刀套是否同时倒下(P16-1);

注意:

(1) 对于“+512 表采用特殊刀盘”, 可以针对斗笠式刀库, 也可针对机械手刀库。

(2) 在 ProgramTool 程序中, 当刀具号和刀套号不一致时, TAIM 和 WHEN** 和 CASET** 中均指刀套号而不是刀具号; 在 ProgramTool 中可以用 STATUSINFO 指令进行系统状态信息提示。

(3) 设为+1024 表 M06 与 M16 功能交换时, M36 功能(先选刀再换刀) 依然保持不变。

P16-1, 主轴定向或 Z 轴回换刀点时刀套是否同时倒下[1 表是, 0 表否]

P17, 斗笠式刀库换刀时 Z 轴的提升高度(mm)

P18, 斗笠式刀库换刀时提升 Z 轴的速度(mm/min)

P19, 斗笠式归刀(转盘式刀套倒下/紧刀)延时(0.1 秒)

P20, 换刀时松刀后延时时间(0.1 秒)

P20-1, 斗笠式刀库 Z 轴下降到位后紧刀前延时(0.1 秒)

P21, 刀套抬起后延时(ms)

P22, Z 轴运动与主轴抓刀/斗立刀库是否互锁[1 表是, 0 表否]

P23, 刀库前进指令 M71 是否检查 Z 轴位置[0 表是, 1 表否]

P24, 固定刀位区的最大刀具号[设置该参数后, 请初始化刀套表]

P25, 刀库数刀信号[上升沿“1000+编号”, 下降沿“2000+编号”]

P26, 刀库定位信号[1000+编号]

P27, 刀库正转输出点[1000+编号]

P28, 刀库反转输出点[1000+编号]

P29, 机械手刹车输入点[1000+编号]

For example: M06Txx, TxxM06, TxxM06Txx;

+4096 The K1 key (manual spindle positioning) of the shielded machining center; After the K1 shielding curing function is enabled, the K1 key in the PLC can start M1070-M1079/M1140-M1179:

M1070-M1079/M1140-M1179 corresponds to control programs ProgramUser0-ProgramUser9/ ProgramUser10-ProgramUser49; for example, the following PLC starts M1079 by K1 (corresponding to ProgramUser9, and the program should reset M306 at the end):



+8192 K3 key of the surface shielding machining center (set the current tool sleeve number);

+16384 machining center tool library does not need to return to zero when starting up;

+32768, whether the tool sleeve falls down at the same time when the table spindle is oriented or the Z-axis retracts the tool point (P16-1);

pay attention to :

(1) For the "+512 table with special knife disk", it can be used for both the conical knife bank and the mechanical hand knife bank.

(2) In the ProgramTool program, when the tool number and tool sleeve number are not consistent, TAIM, WHEN** and CASET** all refer to the tool sleeve number instead of the tool number; in the ProgramTool, the status information can be indicated by the STATUS INFO instruction.

(3) Set to +1024. When the function of M06 and M16 is exchanged, the function of M36 (selecting the tool first and then changing the tool) remains unchanged.

P16-1, whether the tool holder falls down at the same time when the spindle is oriented or the Z-axis retracts the tool point [1 indicates yes, 0 indicates no] P17, the lifting height of the Z-axis during the replacement of the conical tool magazine (mm)

P18, the speed of the Z-axis when changing the blade in the conical knife store (mm/min)

P19, conical return blade (rotary disc blade down/lock blade) delay (0.1 seconds)

P20, delay time after tool change (0.1 seconds)

P20-1, conical knife magazine Z-axis down position delay before cutting forward (0.1 seconds)

P21, delay after the tool sleeve is lifted (ms)

P22, whether the Z-axis movement is interlocked with the spindle gripper/drum vertical tool magazine [1 indicates yes, 0 indicates no] P23, whether the tool magazine advance command M71 checks the Z-axis position [0 indicates yes, 1 indicates no]

P24, Maximum tool number in the fixed tool position zone [After setting this parameter, initialize the tool magazine table] P25, Tool magazine count signal [Rising edge "1000+ Number", Falling edge "2000+ Number"] P26, Tool magazine positioning signal [1000+ Number]

P27, tool magazine forward output point [1000+ number]

P28, tool magazine reverse output point [1000+ number] P29, manipulator brake input point [1000+ number]

P30, 机械手转动输出点[1000+编号]

P32, 选刀方式/数刀信号[D0=0 表单向, D0=1 表双向选刀; D1-D14 详见说明书]

D0=1 表加工中心选刀方式为双向随机选刀;

D1=1 (+2) 表加工中心数刀信号检测沿反转;

D2=1 (+4) 表对加工中心刀库数刀信号进行滤波;

D3=1 (+8) 表对加工中心机械手停止信号进行滤波;

D4=1 (+16) 表加工中心刀库数刀信号和加工中心机械手停止信号采用 8ms 滤波方式[否则表采用 4ms 滤波方式];

D5=1 (+32) 表加工中心机械手刀库执行 T 指令后自动将刀套倒下, 当该功能有效时, 执行换刀指令 M6 后, 刀套不再自动抬起, 而是处在倒下的状态, 直到再执行选刀指令 T 时才自动抬起;

D6=1 (+64) 表加工中心机械手刀库执行换刀指令的最后一步抬起刀套时不检测抬起到位就执行下一段程序;

D7=1 (+128) 表执行 T 指令时不等待刀套抬起就执行下一段程序;

D8=1 (+256) 表对车床刀位检测信号进行 2ms 滤波;

D9=1 (+512) 表对车床刀位检测信号进行 4ms 滤波;

D10=1 (+1024) 表对车床刀位检测信号进行 8ms 滤波, D8-D10 可同时为 1;

D11=1 (+2048) 表对 WAT 等待输入信号进行 2ms 滤波;

D12=1 (+4096) 表对 WAT 等待输入信号进行 4ms 滤波;

D13=1 (+8192) 表对 WAT 等待输入信号进行 8ms 滤波, D11-D13 可同时为 1;

D14=1 (+16384) 表加工中心机械手刀库的机械手为滕州大地的特殊机械手。

P33, 机械手 Z 轴不输出 Y24 安全参考点(3001/3002/3003/3004 表示第 1/2/3/4 参考点)

P34, 加工中心数刀信号滤波时间(ms) [>8]

P35, 加工中心刀库转动后到检测数刀信号的间隔时间(ms) [>8]

P70, 伺服刀库回零正向旋转速度(度/min)

P71, 伺服刀库回零反向旋转速度(度/min)

P72, 伺服刀库回零后偏移距离(0.001 度)

P73, 伺服刀库选刀旋转速度(度/min)

P74, 伺服刀库使用的坐标轴(1=A; 2=B; 3=C)

P75, 伺服刀库运行加速度(度/min/s)

P76, 伺服刀库回零检测零位信号的最大距离(0.001 度)

P77, 伺服刀库回零是否检测电机零位信号(0 表不检测, 非 0 表检测, 367 表示采用绝对编码器)

P78, 伺服刀库换刀方式(-1 表示增量值电机时反向间隙补偿, -9 表示不就近换刀)

P79, 伺服刀库回零后的当前刀具号

P100, 参考点 1 的 X(mm)

P101, 参考点 1 的 Y(mm)

P102, 参考点 1 的 Z(mm)

P103, 参考点 1 的 A(mm)

P30, rotating output point of the manipulator [1000+number]

P32, tool selection mode/number of cuts signal [D0=0 for unidirectional, D0=1 for bidirectional; D1-D14 refer to the manual] D0=1 indicates that the machining center uses bidirectional random tool selection;

D1=1 (+2) indicates that the number of machining center signals is reversed;

D2=1 (+4) Table filters the number of tool sets and signals for the machining center;

D3=1 (+8) filters the stop signal of machining center manipulator;

D4=1 (+16) The signal of the number of tools in the machining center tool library and the stop signal of the machining center manipulator adopt 8ms filtering mode [otherwise, the table adopts 4ms filtering mode];

D5=1 (+32) When the machining center manipulator tool magazine executes the T command, it will automatically lower the tool sleeve. When this function is effective, after executing the tool change command M6, the tool sleeve will not be automatically lifted, but in a lowered state until the tool selection command T is executed again, then it will be automatically lifted;

D6=1 (+64) The last step of the machining center manipulator tool magazine executing the tool change instruction is to lift the tool sleeve without detecting the lifting position and execute the next program;

D7=1 (+128) means that the program is executed without waiting for the tool sleeve to lift when executing the T instruction;

D8=1 (+256) filters the lathe tool position detection signal for 2ms;

D9=1 (+512) table filters the lathe tool position detection signal for 4ms;

D10=1 (+1024) table filters the lathe tool position detection signal for 8ms, and D8-D10 can be 1 at the same time; D11=1 (+2048) table filters the WAT waiting input signal for 2ms;

D12=1 (+4096) Table filters the WAT waiting input signal for 4ms;

D13=1 (+8192) indicates that the WAT waits for 8ms filter input signal, and D11-D13 can be 1 at the same time; D14=1 (+16384) indicates that the machining center's mechanical arm is a special mechanical arm from Tengzhou Dadi.

P33, the Z-axis of the manipulator does not output the Y24 safety reference point (3001/3002/3003/3004 indicate the first/second/third/fourth reference points)

P34, machining center number of signal filter time (ms) [>]8

P35, the interval time (ms) between the rotation of the machining center tool magazine and the detection of the number of cuts [>]

P70, the forward rotation speed of the servo tool magazine to zero (°/min)

P71, reverse rotation speed of servo tool magazine back to zero (°/min)

P72, offset distance (0.001 degrees) after servo tool magazine returns to zero

P73, Servo tool library selection rotation speed (°/min)

P74, the coordinate axes used by the servo tool library (1=A; 2=B; 3=C)

P75, servo tool magazine operating acceleration (°/min/s)

P76, maximum distance of zero position signal for servo tool magazine return to zero (0.001 degrees)

P77, whether the servo tool magazine returns to zero detects the motor zero position signal (0 indicates no detection, non-0 indicates detection, 367 indicates the use of absolute encoder)

P78, servo tool magazine tool change mode (-1 indicates incremental value motor reverse clearance compensation, -9 indicates not to change the tool nearby)

P79, current tool number after servo tool magazine return to zero

P100, reference point 1 X (mm)

P101, Y (mm) of reference point 1

P102, Z (mm) of reference point 1

P103, A (mm) at reference point 1

P104, 参考点 2 的 X(mm)

P105, 参考点 2 的 Y(mm)

P106, 参考点 2 的 Z(mm)

P107, 参考点 2 的 A(mm)

P108, 参考点 3 的 X(mm)

P109, 参考点 3 的 Y(mm)

P110, 参考点 3 的 Z(mm)

P111, 参考点 3 的 A(mm)

P112, 参考点 4 的 X(mm)

P113, 参考点 4 的 Y(mm)

P114, 参考点 4 的 Z(mm)

P115, 参考点 4 的 A(mm)

(三) 速度参数

220, 显示攻丝误差脉冲数(1:表示攻丝时攻丝轴反馈脉冲数与主轴编码器脉冲数的误差, 正值表示攻丝轴超前, 负值表示主轴超前)

P230, 主轴 SP 指令 G00 定位速度(0.1rpm)。

P231, 主轴 SP 指令 G01 定位控制方式[+4 表按 F 执行,+8 表按 G90/G91 执行,+16 表由当前显示值发送脉冲]

0 表主轴回零为脉冲控制, 当主轴回零方式为脉冲控制时设置为 0;

+4 表执行 SP 指令 G01 时按当前 F 速度[度/分钟]定位(在 G00 状态下按 P230 执行), 否则按 P230 执行;

+8 表执行 SP 指令时受当前 G90/G91 的控制, 为 G90 时表绝对会归算, 为 G91 时表增量不会归算, 否则会进行归算并按就近方向运行;

+16 表按当前值开始计算终点坐标。

P232, 插补方式主轴 SP 指令定位方向[0 表正向, 1 表负向, 其余表就近方向]

该参数在 G90 或参数设成不受 G90/G91 控制时有效。

233, 主轴回零方式[1 表脉冲控制, 2 表驱动器控制(系统输出 M61 检测 M22), 16/32 表先检测/不检测 M22 再脉冲控制, 64 表先停主轴再驱动器控制]。

首先停主轴, 然后主轴准停, 根据编码器反馈计算出当前主轴位置, 然后再按参数“P230, 主轴定位 SP 指令速度(0.1rpm)”的速度定位到参数“P234, 换刀时主轴定向停止角度(0.1度)”设置的角度。如果发现定位偏差很大, 将轴类参数“P56/总线 P201, 检测主轴位置反馈”设为 16 对编码器的反馈方向反向即可。

234, 主轴定向停止角度(0.1度)

235, 主轴回零速度(0.1 rpm)

P236, 主轴定位速度(0.1 rpm)

注意:

P230-P236 要求轴类参数 P411, 攻丝控制方式[2 跟踪;3 插补]设置为插补 3。

5.3 加工中心调试

P104, X (mm) of reference point 2
P105, Y (mm) of reference point 2
P106, Z (mm) of reference point 2
P107, A (mm) at reference point 2
P108, X (mm) of reference point 3
P109, Y (mm) of reference point 3
P110, Z (mm) of reference point 3
P111, A (mm) at reference point 3
P112, X (mm) of reference point 4
P113, Y (mm) of reference point 4
P114, Z (mm) of reference point 4
P115, A (mm) at reference point 4

(3) Speed Parameters

220, shows the number of screw error pulses (1: indicates the error between the feedback pulses of the screw shaft and the pulses of the spindle encoder during screwing. A positive value indicates that the screw shaft is ahead, and a negative value indicates that the spindle is ahead)

P230, spindle SP command G00 positioning speed (0.1 rpm).

P231, spindle SP command G01 positioning control mode [press F for +4 table press G90/G91, press G90/G91 for +8 table press current display value sends pulse]

0 The table spindle return to zero is pulse control, and when the spindle return to zero mode is pulse control, it is set to 0;

+4 When executing SP command G01 on the table, it shall be positioned according to the current F speed [°/min] (P230 shall be executed under G00 state), otherwise P230 shall be executed;

+8 When the table performs SP instruction, it is controlled by the current G90/G91. When G90, the table absolute will be calculated; when G91, the table increment will not be calculated, otherwise it will be calculated and run in the nearest direction;

+16 The table starts calculating the end point coordinates from the current value.

P232, interpolation mode spindle SP instruction positioning direction [0 indicates forward, 1 indicates negative, the rest indicates near direction]. This parameter is valid when G90 or the parameter is set to be not controlled by G90/G91.

233, spindle return to zero mode [Table 1 pulse control, Table 2 driver control (system output M61 detects M22), Table 16/32 first detect/M22 then pulse control, Table 64 stop spindle first then driver control].

First, stop the spindle and then perform a spindle stop. Calculate the current spindle position based on encoder feedback, then use the "P230, spindle positioning SP command speed (0.1 rpm)" parameter to position the spindle at the angle set by the "P234, tool change spindle orientation stop angle (0.1 degrees)" parameter. If significant positioning deviation is detected, set the shaft parameter "P56/Bus P201, detect spindle position feedback" to 16 and reverse the encoder feedback direction.

234, spindle orientation stop Angle (0.1 degrees)

235, spindle return speed (0.1 rpm)

P236, spindle positioning speed (0.1 rpm) Note:

P230-P236 requires shaft parameters P411, and the tapping control mode [2 tracking; 3 interpolation] is set to interpolation 3.

5.3 Processing Center Debugging

系统状态显示的“Tab HxDx cdTef”含义,“ab”表示主轴上的刀具号,“cd”表示刀库中当前刀具的刀套位置号“ef”表示刀库中的当前刀具号。

(一) 标准斗笠式刀库加工中心

1、手动按键操作

K1 为主轴定向 (主轴定向完成时 K1 指示灯亮): 输出 M61, 检测 M22;

K2 为 Z 轴回换刀点;

K3 为设定当前刀套的刀套号;

“刀库正转”: 输出 M63, 检测 T08;

“刀库反转”: 输出 M71, 检测 T08;

2、M 指令

M71 为刀库前进, 撤消 Y24, 输出 M73, 刀具参数 P23=1;

M73 为刀库后退, 撤消 M73, 输出 Y24, 刀具参数 P23=1;

M881 同“K1 功能”;

3、换刀操作

Txx : 将主轴上的刀具归还到刀库内, 再将指令中的 Txx 换到主轴上。

M36 Txx: 以单段的方式按换刀步骤把指令中的 Txx 换到主轴上。执行完一步后处于暂停状态, 当按下运行按钮后继续执行下一步, 主要用于调试时使用。

4、主轴定向偏差设置 (主轴编码器信号必须接入系统)

1) P400=1, 换刀时检测主轴位置反馈, 设置为“0”时不检测位置反馈;

2) 设置 P401, 主轴定向时检测角度, 手动按 K1 主轴定向完成后输入 SP 值到 P401;

3) 设置 P402, 主轴定向时角度检测误差;

5、换刀流程

1) Z 轴到换刀点 (Z 轴必须先回零);

2) 主轴定向: 输出 M61, 检测回零到位 M22;

3) 刀库前进: 输出 M73, 检测前进到位 T06;

4) 主轴松刀: 输出 M10, 检测松刀到位 T02;

5) Z 轴提升;

6) 刀库正转 (或反转): 输出 M63 (或 M71), 检测 T08 刀位计数;

7) 主轴松刀: 输出 M10, 检测松刀到位 T02;

8) Z 轴下降到换刀点;

9) 主轴紧刀: 撤消 M10, 检测紧刀到位 T05;

10) 刀库后退: 输出 Y24, 检测后退到位 T07;

6、刀具参数设置

P10=1, 启动换刀程序;

P11=0, 只是 Z 轴回换刀点;

P12=2, 换刀时进给轴回第 2 参考点;

P13=1, 换刀前检测主轴定向和进给轴回换刀点, 检测 M22 信号;

P14=1, 换刀时主轴定向, 输出 M61 信号;

P15=2, 刀库类型为“斗笠式”;

The meaning of "Tab HxDx cdTef" displayed in the system status is that "ab" indicates the tool number on the spindle, "cd" indicates the tool holder position number of the current tool in the tool library, and "ef" indicates the current tool number in the tool library.

(1) Standard conical knife storage machining center

1. Manual Button Operation

K1 is spindle orientation (the K1 indicator light is on when the spindle orientation is completed): output M61 and detect M22;

K2 is the Z-axis return tool point;

K3 is the sleeve number of the current sleeve;

"刀库正转": output M63 and detect T08;

"刀库反转": output M71 and detect T08;

2、 M instruct

M71 moves forward for the tool magazine, withdraws Y24, outputs M73, and tool parameter P23=1;

M73 is for the tool magazine to retreat, cancel M73, output Y24, tool parameter P23=1;

M881 is the same as "K1 function";

3. Blade Replacement Operation

Txx: Return the tool on the spindle to the tool magazine, and then transfer the instruction Txx to the spindle.

M36 Txx: In a single-stage manner, the Txx instruction is transferred to the spindle according to the tool change steps. After completing one step, it enters a pause state and continues to execute the next step when the run button is pressed. It is mainly used for debugging.

4、 Axis Orientation Deviation Setting (the Spindle Encoder Signal Must Be Connected to the System)

1) P400=1, when changing the tool, the spindle position feedback is detected, and it is not detected when set to "0";

2) Set P401, detect the Angle when the spindle is oriented, and manually press K1 to complete the spindle orientation and input the SP value to P401;

3) Set P402, the Angle detection error when the spindle is oriented;

5. Blade Replacement Process

1) From the Z-axis to the tool change point (the Z-axis must be zeroed first);

2) Axis orientation: Output M61, detect return to zero position M22;

3) Knife library advance: Output M73, detect the forward position T06;

4) Main shaft loosening: output M10, detect the loosening position T02;

5) Z-axis lift;

6) Turn the tool forward (or reverse): output M63 (or M71), and detect T08 tool position count;

7) Main shaft loosening: output M10, detect the loosening position T02;

8) The Z-axis drops to the tool change point;

9) Master shaft tight knife: remove M10, and check the tight knife is in place T05;

10) Knife library back: Output Y24, detect back in place T07;

6. Tool Parameter Setting

P10=1, start the tool changing program;

P11=0, only the Z-axis retracts the tool point;

P12=2, feed axis returns to the second reference point when changing the tool;

P13=1, detect the spindle orientation and feed axis return to the tool change point before changing the tool, and detect the M22 signal;

P14=1, the spindle is oriented when changing the tool and outputs the signal M61;

P15=2, the tool library type is "bald hat style";

P16=0, 标准刀库; 如果+1024, 表示 M06 与 M106 指令功能交换;
P17=120, 换刀时 Z 轴的提升高度(mm);
P18=2000, 库换刀时提升 Z 轴的速度(mm/min);
P19=5, 斗笠式归刀延时(0.1 秒), 刀库前进检测 T06 到位后和主轴紧刀检测 T05 到位后延时时间;
P20=5, 换刀时松刀后延时时间(0.1 秒), 松刀检测 T02 到位后延时时间;
P20-1=5, 斗笠式刀库 Z 轴下降到位后紧刀前延时(0.1 秒), 紧刀前延时时间;
P21=5, 刀套抬起后延时(ms);
P22=1, Z 轴运动与主轴抓刀/斗笠式刀库是否互锁[1 表是, 0 表否], 设为“1”时, 表示 Z 轴运行时要检测刀库回退到位信号 T07(辅助继电器 M31)有效
P23=1, 刀库前进指令 M71 是否检查 Z 轴位置[0 表是, 1 表否], 设置为“0”时, 要求 Z 轴当前坐标必须高于 Z 轴换刀点 20mm 才允许换刀;
P24=0, 固定刀位区的最大刀具号[设置该参数后, 请初始化刀套表];
P25=1007 (X07), 刀库计数信号[上升沿“1000+编号”, 下降沿“2000+编号”];
注意:X07 计数信号 0V 有效;如果是+24V 有效则必须接 2803 反向板;
P26=1041 (X41), 刀库定位信号[1000+编号];
P27=1001 (Y01), 刀库正转输出点[1000+编号];
P28=1005 (Y05), 刀库反转输出点[1000+编号];
P100=0, 参考点 1 的 X(mm);
P101=0, 参考点 1 的 Y(mm);
P102=0, 参考点 1 的 Z(mm);
P103=0, 参考点 1 的 A(mm);
P104=0, 参考点 2 的 X(mm);
P105=0, 参考点 2 的 Y(mm);
P106=50, 参考点 2 的 Z(mm);
P107=0, 参考点 2 的 A(mm);

特别注意:

- 1、在换刀过程当由于突然断电或紧急停止等情况导致刀库乱刀时, 请在手动方式下转动刀库几个刀位后, 再用 K3 键设置当前刀套号;
- 2、在使用指令 M71 刀库前进时, 注意检查 Z 轴位置以免发生意外;
- 3、T0 号刀上不应该有刀具, 否则, 有可能发生意外;
- 4、初次装刀时, 主轴上必须有刀(如果显示 T00, 可以通常初始化刀套表后, 主轴上应为 T01 号刀)。

(二) 标准机械手刀库加工中心

1、手动按键操作

K1 为主轴定向(主轴定向完成时 K1 指示灯亮): 输出 M61, 检测 M22;
K2 为 Z 轴回换刀点;
K3 为设定当前刀套的刀套号;

“刀库正转”: 输出 M63, 检测 T08, 检测刀库定位信号 X41, 要求刀套必须抬起到位 X44;

P16=0, standard tool magazine; if +1024, it indicates the exchange of M06 and M106 command functions;
P17=120, the height of Z-axis lift (mm) when changing the tool;
P18=2000, the speed of lifting Z-axis (mm/min) when changing the tool in the library;
P19=5, the time delay of the conical return tool (0.1 seconds), and the delay time after the T06 tool magazine advance detection and the T05 spindle tight tool detection;
P20=5, the delay time after tool change (0.1 seconds), and the delay time after T02 is detected;
P20-1=5, the delay time for the tight knife advance after the conical blade magazine's Z-axis reaches its lower limit (0.1 seconds); P21=5, the delay time after lifting the blade sleeve (ms);
P22=1, Z-axis movement and whether the main shaft gripper/cap knife library are interlocked [1 is yes, 0 is no], set to "1"

At this point, when the Z-axis is running, it is necessary to detect the valid signal T07 (auxiliary relay M31) indicating the tool magazine has returned to position. When P23=1, the tool magazine forward command M71 checks the Z-axis position [0 for yes, 1 for no]. If set to "0",

The current coordinate of the Z-axis must be higher than the tool change point on the Z-axis by 20mm to allow tool change; P24=0, the maximum tool number in the fixed tool position area [after setting this parameter, please initialize the tool sleeve table];

P25=1007 (X07), tool magazine count signal [rising edge "1000+number", falling edge "2000+number"]; Note: X07 count signal 0V is valid; if +24V is valid, it must be connected to 2803 reverse board;
P26=1041 (X41), tool magazine positioning signal [1000+number];
P27=1001 (Y01), the output point of the tool library forward rotation [1000+number];
P28=1005 (Y05), tool magazine reverse output point [1000+number];
P100=0, X (mm) of reference point 1;
P101=0, Y (mm) of reference point 1;
P102=0, Z (mm) of reference point 1;
P103=0, A (mm) of reference point 1;
P104=0, X (mm) of reference point 2;
P105=0, Y (mm) of reference point 2;
P106=50, Z (mm) of reference point 2;
P107=0, A (mm) of reference point 2;

Special note:

1. During the tool changing process, when the tool magazine is disordered due to sudden power failure or emergency stop, turn several tool positions of the tool magazine in manual mode, and then set the current tool sleeve number with K3 key;
2. When using the instruction M71 and moving the tool library, pay attention to check the Z-axis position to avoid accidents;
3. There should be no tool on the T0 knife, otherwise, accidents may occur;
4. When installing the tool for the first time, there must be a tool on the spindle (if T00 is displayed, after normal initialization of the tool sleeve table, there should be a T01 tool on the spindle).

(2) Standard Manipulator Tool Library Machining Center

1. Manual Button Operation

K1 is spindle orientation (the K1 indicator light is on when the spindle orientation is completed): output M61 and detect M22;

K2 is the Z-axis return tool point;

K3 is the sleeve number of the current sleeve;

"刀库正转": output M63, detect T08, detect the tool library positioning signal X41, and require the tool sleeve to be lifted in place X44;

“刀库反转”：输出 M71，检测 T08，检测刀库定位信号 X41，要求刀套必须抬起到位 X44；

2、 M 指令

M71 为刀套倒下，撤消 Y26，输出 Y27，刀库定位检测 X41 必须到位；

M73 为刀套抬起，撤消 Y27，输出 Y26，检测刀套抬起到位 X44；

M65 为机械手无条件旋转一步，用于调试，输出 Y24；

M881 同“K1 功能”；

3、换刀操作

M06：把当前刀套的刀具换到主轴上。

Txx：将指令中的 Txx 转到当前换刀位置。

M06 Txx：把当前刀套的刀具换到主轴上，并将指令中的 Txx 转到当换刀位置为下一次换刀做准备。（先换刀再选刀）。

M106 Txx：将指令中的 Txx 转到当前换刀位置，再把当前刀套的刀具换到主轴上。（先选刀再换刀，也就是把指令中的 Txx 刀具换到主轴上）。

M36：以单段的方式按换刀步骤把当前刀套的刀具换到主轴上。执行完一步后处于暂停状态，当按下运行按钮后继续执行下一步，主要用于调试时使用。

4、主轴定向偏差设置（主轴编码器信号必须接入系统）

- 1) P400=1，换刀时检测主轴位置反馈，设置为“0”时不检测位置反馈；
- 2) 设置 P401，主轴定向时检测角度，手动按 K1 主轴定向完成后输入 SP 值到 P401；
- 3) 设置 P402，主轴定向时角度检测误差；

5、换刀流程

- 1) Z 轴到换刀点（Z 轴必须先回零）；
- 2) 主轴定向：输出 M61，检测回零到位 M22；
- 3) 倒下刀套：输出 Y27，检测倒下到位 X45，刀库定位检测 X41 必须到位；
- 4) 机械手扣刀：输出 Y24，检测扣刀到位 X42；
- 5) 主轴松刀：输出 M10，检测松刀到位 T02；
- 6) 机械手抓刀并换刀：输出 Y24，检测抓刀到位 X42；
- 7) 主轴紧刀：撤消 M10，检测紧刀到位 T05；
- 8) 机械手归位：输出 Y24，检测归位到位 X42；
- 9) 抬起刀套：输出 Y26，检测抬起到位 X44，检测不到 X44 时将报警“刀库故障”；

特别注意：

1、在换刀过程当由于突然断电或紧急停止等情况导致刀库乱刀时，请在手动方式下转动刀库几个刀位后，再用 K3 键设置当前刀套号。

2、对于该类型刀库，可以在“刀具参数”中的 NO.24 号参数中设置固定刀位区的最大刀具号，设置该参数后，最好初始化刀套表。例如设置为 8 就表示从 1 到 8 号刀套表中只能一一对应地装入相应号码的刀具，即从 1 到 8 号刀具不是随机选刀，是固定安装的。该功能可用于安装大直径刀盘的刀具。

3、T0 号刀上不应该有刀具，否则，有可能发生意外。

4、M65 指令不检查条件，只有急停才能停止，使用时请格外注意否则可能造

"Tool library reversal": output M71, detect T08, detect tool library positioning signal X41, and require the tool sleeve to be lifted in place X44;

2、 M instruct

M71 is the knife sleeve to fall, cancel Y26, output Y27, and the knife library positioning detection must be in place at X41;

M73 lifts the blade sleeve, withdraws Y27, outputs Y26, and detects that the blade sleeve is lifted in place X44;

M65 is an unconditional rotation step for the manipulator, used for debugging, and outputs Y24;

M881 is the same as "K1 function";

3. Blade Replacement Operation

M06: Change the tool of the current tool holder to the spindle.

Txx: Transfer the Txx in the instruction to the current tool change position.

M06 Txx: Change the tool of the current tool holder to the spindle, and turn the Txx in the instruction to the tool change position to prepare for the next tool change. (Change the tool first, then select the tool).

M106 Txx: Transfer the Txx tool in the instruction to the current tool change position, and then transfer the tool of the current tool holder to the spindle. (Select the tool first and then change the tool, that is, transfer the Txx tool in the instruction to the spindle).

M36: Replace the tool of the current tool holder on the spindle in a single-stage manner according to the tool changing steps. After executing one step, it is in a pause state and continues to execute the next step when the run button is pressed. It is mainly used for debugging.

4 、 Axis Orientation Deviation Setting (the Spindle Encoder Signal Must Be Connected to the System)

1) P400=1, when changing the tool, the spindle position feedback is detected, and it is not detected when set to "0";

2) Set P401, detect the Angle when the spindle is oriented, manually press K1 to complete the spindle orientation, and input the SP value to P401;

3) Set P402, the Angle detection error when the spindle is oriented;

5. Blade Replacement Process

1) From the Z-axis to the tool change point (the Z-axis must be zeroed first);

2) Axis orientation: Output M61, detect return to zero position M22;

3) Drop the knife sleeve: Output Y27, detect the drop in place X45, and the knife library positioning detection X41 must be in place;

4) Mechanical knife: Output Y24, detect the position of the knife X42;

5) Main shaft loosening: output M10, detect the loosening position T02;

6) Mechanical hand grab and replace the knife: Output Y24, detect the position of the grab X42;

7) Master shaft tight knife: remove M10, check the tight knife is in place T05;

8) Mechanical hand return: Output Y24, detect X42 to return;

9) Lift the knife sleeve: output Y26, detect the lifting position X44, if X44 is not detected, alarm "knife library fault" will be issued;

Special note:

1 、 During the process of changing the tool, when the tool magazine is disordered due to sudden power failure or emergency stop, please turn several tool positions of the tool magazine in manual mode, and then use K3 key to set the current tool sleeve number.

2、 For this type of tool magazine, the maximum tool number in the No.24 parameter under "Tool Parameters" can be set to define the fixed tool holder zone. After configuring this parameter, it's advisable to initialize the tool holder table. For example, setting it to 8 means that only tools numbered 1 through 8 can be installed in the corresponding tool holder positions. This ensures that tools from 1 to 8 are not randomly selected but fixedly installed. This feature is particularly useful for mounting tools on large-diameter tool magazines.

3、 There should be no tool on the T0 knife, otherwise, accidents may occur.

4、 The M65 instruction does not check the condition, and only an emergency stop can stop it. Please pay special attention when using it, otherwise it may cause damage

成安全事故。

6、刀具参数设置

P10=1, 启动换刀程序;

P11=0, 只是 Z 轴回换刀点;

P12=2, 换刀时进给轴回第 2 参考点;

P13=1, 换刀前检测主轴定向和进给轴回换刀点, 检测 M22 信号;

P14=1, 换刀时主轴定向, 输出 M61 信号;

P15=0, 刀库类型为“机械手”;

P16=0, 标准刀库;

P17=120, 换刀时 Z 轴的提升高度(mm);

P18=2000, 换刀时提升 Z 轴的速度(mm/min);

P19=5, 斗笠式归刀延时(0.1 秒), 刀库前进检测 T06 到位后和主轴紧刀检测 T05 到位后延时时间;

P20=5, 换刀时松刀后延时时间(0.1 秒), 松刀检测 T02 到位后延时时间;

P20-1=5, 斗笠式刀库 Z 轴下降到位后紧刀前延时(0.1 秒), 紧刀前延时时间;

P22=1, Z 轴运动与主轴抓刀/斗笠式刀库是否互锁[1 表是, 0 表否], 设为“1”时, 检测 X43(辅助继电器 M09)有效表主轴不在原点, 此时 Z 轴不能动作;

P23=1, 刀库前进指令 M71 是否检查 Z 轴位置[0 表是, 1 表否];

P24=0, 固定刀位区的最大刀具号[设置该参数后, 请初始化刀套表];

P25=1007 (X07), 刀库计数信号[上升沿“1000+编号”, 下降沿“2000+编号”];

P26=1041 (X41), 刀库定位信号[1000+编号];

P27=1001 (Y01), 刀库正转输出点[1000+编号];

P28=1005 (Y05), 刀库反转输出点[1000+编号];

P29=1042 (X42), 机械手刹车输入点[1000+编号]

P30=1024 (Y24), 机械手转动输出点[1000+编号]

P32, 选刀方式/数刀信号[0 表单向, 1 表双向选刀]

P33, 机械手 Z 轴不输出 Y24 安全参考点(3001/3002/3003/3004 表示第 1/2/3/4 参考点)

P100=0, 参考点 1 的 X(mm);

P101=0, 参考点 1 的 Y(mm);

P102=0, 参考点 1 的 Z(mm);

P103=0, 参考点 1 的 A(mm);

P104=0, 参考点 2 的 X(mm);

P105=0, 参考点 2 的 Y(mm);

P106=50, 参考点 2 的 Z(mm);

P107=0, 参考点 2 的 A(mm);

(三) 特殊斗笠式刀库加工中心

1、参数设置: 斗笠式刀具参数 P15=2、P16=16, 刀库设置为“特殊斗笠”。

2、编辑换刀程序 ProgramTool 导入系统。ProgramTool 编程从换刀过程的第 3) 步开始, 其他操作与标准斗笠式刀库相同。

It's a safety accident.

6. Tool Parameter Setting

P10=1, start the tool changing program;

P11=0, only the Z-axis retracts the tool point;

P12=2, feed axis returns to the second reference point when changing the tool;

P13=1, detect the spindle orientation and feed axis return to the tool change point before changing the tool, and detect the M22 signal;

P14=1, the spindle is oriented when changing the tool and outputs the signal M61;

P15=0, the tool library type is "mechanical hand";

P16=0, standard tool library;

P17=120, the height of Z-axis lift (mm) when changing the tool;

P18=2000, the speed of lifting Z-axis (mm/min) when changing the tool;

P19=5, the time delay of the conical return tool (0.1 seconds), and the delay time after the T06 tool magazine advance detection and the T05 spindle tight tool detection;

P20=5, the delay time after tool change (0.1 seconds), and the delay time after tool release detection T02 is in place;

P20-1=5, the delay time for advancing the clamping tool before the conical tool magazine's Z-axis reaches its lower limit (0.1 seconds); P22=1, indicating whether Z-axis movement is interlocked with spindle tool clamping/conical tool magazine [1 indicates yes, 0 indicates no]. When set to "1", it detects that auxiliary relay M09 (X43) is active, meaning the spindle is not at the origin and thus the Z-axis cannot operate; P23=1, indicating whether the tool magazine advance command M71 checks the Z-axis position [0 indicates yes, 1 indicates no];

P24=0, the maximum tool number in the fixed tool position area [After setting this parameter, please initialize the tool sleeve table];

P25=1007 (X07), tool magazine count signal [rising edge "1000+ number", falling edge "2000+ number"]; P26=1041 (X41), tool magazine positioning signal [1000+ number];

P27=1001 (Y01), the output point of the tool library forward rotation [1000+number];

P28=1005 (Y05), tool magazine reverse output point [1000+number];

P29=1042 (X42), mechanical hand brake input point [1000+number]

P30=1024 (Y24), robotic arm rotation output point [1000+number]

P32, tool selection mode/number of cuts signal [0 for unidirectional, 1 for bidirectional tool selection]

P33, the Z-axis of the manipulator does not output the Y24 safety reference point (3001/3002/3003/3004 indicate the first/second/third/fourth reference points)

P100=0, X (mm) of reference point 1;

P101=0, Y (mm) of reference point 1;

P102=0, Z (mm) of reference point 1;

P103=0, A (mm) of reference point 1;

P104=0, X (mm) of reference point 2;

P105=0, Y (mm) of reference point 2;

P106=50, Z (mm) of reference point 2;

P107=0, A (mm) of reference point 2;

(3) Special Conical Knife Storage Processing Center

1. Parameter setting: The parameters of the conical tool are P15=2 and P16=16, and the tool library is set as "special conical".

2、 ProgramTool is imported into the system. ProgramTool programming starts from step 3) of the tool change process, and other operations are the same as standard hat-type tool library.

（四）特殊机械手刀库加工中心

- 1、参数设置：机械手刀具参数 P15=0、P16=64，刀库设置为“特殊机械手”；刀具参数 P29=0、P30=0。
- 2、编辑换刀程序 ProgramM6 导入系统。ProgramM6 编程从换刀过程的第 3) 步开始，其他操作与标准机械手刀库相同。

（五）特殊排刀库加工中心

- 1、参数设置：斗笠式刀具参数 P15=1、P16=128，刀库设置为“排刀库”。
- 2、编辑换刀程序 ProgramTool 导入系统。换刀整个过程都由程序 ProgramTool 完成。

（六）特殊伺服刀库加工中心

- 1、参数设置：机械手刀具参数 P15=0、P16=32（设置为标准机械手伺服刀库）或 P16=32+64=96（设置为特殊机械手伺服刀库）。
斗笠式刀具参数 P15=2、P16=32（设置为标准斗笠式伺服刀库）或 P16=32+16=48（设置为特殊斗笠式伺服刀库）。
- 2、如果 P15=0、P16=96 时，编辑换刀程序 ProgramM6 导入系统。ProgramM6 编程从换刀过程的第 3) 步开始，其他操作与标准机械手刀库相同。
如果 P15=2、P16=48 时，编辑换刀程序 ProgramTool 导入系统。ProgramTool 编程从换刀过程的第 3) 步开始，其他操作与标准斗笠式刀库相同。
- 3、伺服刀库回零：P74=1 检测 A0 信号，P74=2 检测 B0 信号，P74=3 检测 C0 信号；如果 P77=1 时还要检测电机零位信号；如果 P77=367 时伺服刀库电机为绝对值伺服电机，此时不用回零；如果 P16 参数+16384 开机也不用回零。
- 4、伺服刀库夹紧松开输出信号 Y25，到位信号 X43，如果无夹紧松开时必须将 Y25 短接到 X43。

（七）配台湾首轮机械圆盘式机械手刀库加工中心

- 1、参数设置：刀具参数 P16=64，刀库设置为“特殊机械手”；刀具参数 P29=0、P30=0。
- 2、编辑换刀程序 ProgramM6 导入系统。ProgramM6 编程从换刀过程的第 3) 步开始，其他操作与标准机械手刀库相同。
- 3、M 指令
 - M71 为刀套倒下，撤消 Y26，输出 Y27，刀库定位检测 X41 必须到位；
 - M73 为刀套抬起，撤消 Y27，输出 Y26；
 - OUT+Y30 为机械手无条件旋转一步，用于调试，输出 Y30；
 - M881 同“K1 功能”；
- 4、刀库信号接线表（刀库信号输入经过 2803 反向）：

I/O	信号	机械手刀库加工中心	台湾首轮圆盘式刀库
X00	T01	刀库电机过载检测；	
X01	T02	主轴松刀检测	
X02	T03	润滑油位检测	
X03	T04	冷却缺液检测	

(4) Special Manipulator Tool Library Machining Center

- 1、Parameter setting: The tool parameters of the manipulator are set as P15=0 and P16=64, and the tool magazine is set as "special manipulator"; the tool parameters are set as P29=0 and P30=0.
- 2、ProgramM6, the program for changing the tool, is imported into the system. ProgramM6 starts from step 3) of the tool changing process, and other operations are the same as those of the standard manipulator tool library.

(5) Special Tooling Library Machining Center

1. Parameter setting: The parameters of the conical tool are P15=1 and P16=128, and the tool library is set as "tool row library".
- 2、The program tool ProgramTool is imported into the system. The whole process of tool change is completed by the program tool ProgramTool.

(6) Special Servo Tool Library Machining Center

1. Parameter setting: Tool parameters of the manipulator P15=0, P16=32 (set as standard manipulator servo tool library) or P16=32+64=96 (set as special manipulator servo tool library).
- Hat-type tool parameters P15=2, P16=32 (set as standard hat-type servo tool library) or P16=32+16=48 (set as special hat-type servo tool library).
- 2、When P15=0 and P16=96, the program ProgramM6 is imported into the system for the tool changing process. ProgramM6 starts from step 3 of the tool changing process, and other operations are the same as those of the standard manipulator tool library.

When P15=2 and P16=48, the program tool (ProgramTool) is imported into the system. The programming of ProgramTool starts from step 3) of the tool change process, and other operations are the same as those in the standard bucket-type tool library.

- 3 Servo tool magazine reset: Check the A0 signal when P74=1, check the B0 signal when P74=2, and check the C0 signal when P74=3. If P77=1, also check the motor zero position signal. If P77=367, the servo tool magazine motor is an absolute value servo motor, so no reset is needed. If P16 parameter +16384 is activated, no reset is required either.
- 4、The servo tool magazine clamping and releasing output signal Y25, and the arrival signal X43. If there is no clamping and releasing, Y25 must be shorted to X43.

(7) To Be Equipped With a Taiwan First-Round Mechanical Disc-Type Manipulator Tool Library Machining Center

- 1 Parameter setting: Tool parameter P16=64, tool library is set as "special manipulator"; Tool parameter P29=0, P30=0.
- 2、ProgramM6, the program for changing the tool, is imported into the system. ProgramM6 starts from step 3) of the tool changing process, and other operations are the same as those of the standard manipulator tool library.
- 3、M instruct

M71 is the knife sleeve to fall, cancel Y26, output Y27, and the knife library positioning detection X41 must be in place;

M73 lifts the blade sleeve, withdraws Y27 and outputs Y26;

OUT+Y30 is the unconditional rotation of the manipulator for debugging, and outputs Y30;

M881 is the same as "K1 function";
4. Tool library signal wiring table (the tool library signal input is reversed by 2803):

I/O	signal	Mechanical tool magazine machining center	Taiwan's first round disc knife collection
X00	T01	Overload detection of knife library motor;	
X01	T02	Shaft loosening detection	
X02	T03	Lubrication level detection	
X03	T04	Cooling and fluid loss detection	

X04	T05	主轴紧刀检测	
X05	T06		扣刀确认信号 S6（位置右）
X06	T07		原点确认信号 S7（位置左）
X07	T08	刀库计数(0V 有效)	刀库计数 S1(0V 有效)
X23	ALM1	主轴驱动器报警	
X24	ALM2	冷却电机过载检测	
X27	M22	主轴定位完成/回零到位检测	
X29	M12	机床门开关检测 要求：杂类参数 P7=1	
X30	M14	压缩空气压力检测报警	
X31	M16	主轴远程松刀按钮 要求：杂类参数 P22=1	
X40		排屑电机过载检测报警	
X41		刀库定位检测	刀库定位信号 S2
X42		刀库机械手停止检测： （机械手扣刀、机械手抓刀 换刀、机械手归位）	刹车确认信号 S5（位置中）
X43		机械手原点信号，换刀前检测 此信号为无效	
X44		刀套抬起检测 要求：换刀前必须抬起到位	回刀定位信号 S4
X45		刀套落下检测	倒刀定位信号 S3
X46		准停指令 M01 输入	
X47		主轴风机过载检测	
Y00	M61	主轴回零输出，检测回零到 位 M22	
Y01	M63	刀库正转控制输出，检测 T08 刀位计数；	换刀马达电机正转
Y02	M65	黄指示灯（停止指示灯）	
Y03	M67	红指示灯（故障指示灯）	
Y04	M69	绿指示灯（运行指示灯）	
Y05	M71	刀库反转控制输出，检测 T08 刀位计数；	换刀马达电机反转
Y06	M73		
Y07	M59	M59 吹气	
Y08	M32	润滑控制输出	
Y10	M10	松紧刀控制输出： 主轴松刀：输出 M10，检测	

X04	T05	Spindle tight knife detec- tion	
X05	T06		Check the signal confir- mation S6 (right position)
X06	T07		Origin confirmation sig- nal S7 (left position)
X07	T08	Knife count (0V valid)	Knife count S1 (0V valid)
X23	ALM1	The spindle driver alarms	
X24	ALM2	Cool motor overload de- tection	
X27	M22	Main shaft positioning is complete / zero return detection	
X29	M12	Machine door switch de- tection Request: Miscellaneous parameter P7=1	
X30	M14	Air pressure detection al- arm	
X31	M16	Main shaft remote tool release button require- ments: Miscellaneous pa- rameter P22=1	
X40		Overload detection alarm of dust removal motor	
X41		Knife library positioning detection	Knife library positioning signal S2
X42		The knife library manipu- lator stops detection: (Machine hand buckle kn- ife, machine hand grab and change knife, mach- ine hand return to posit- ion)	Brake confirmation signal S5 (in position)
X43		The mechanical tool origin signal is invalid before tool change	
X44		Raise the knife handle to detect Requirement: The blade must be lifted in place before changing	Return to the target sig- nal S4
X45		Knife sleeve drop detec- tion	Flip the target positioning signal S3
X46		Enter the stop instruction M01	
X47		Overload detection of main fan	
Y00	M61	The spindle returns to zero output and detects that the return to zero is in place M22	
Y01	M63	The tool library controls the output in forward ro- tation and detects the T08 tool position count;	Switch the motor to run forward
Y02	M65	Yellow indicator light (stop indicator light)	
Y03	M67	Red indicator light (fault indicator light)	
Y04	M69	Green indicator light (op- erating indicator light)	
Y05	M71	Turnaround control out- put of the tool library, and detect the T08 tool posit- ion count;	Reverse the swivel motor
Y06	M73		
Y07	M59	M59 blew	
Y08	M32	Lubrication control output	
Y10	M10	Looseness control outp- ut: Main shaft loosening: output M10, detection	

		松刀到位 T02; 主轴紧刀: 撤消 M10, 检测 紧刀到位 T05;	
Y11	M08	冷却控制输出	
Y12	M05	M05	
Y13	M04	主轴反转控制输出	
Y14	M03	主轴正转控制输出	
Y15	M75	主轴控制模式切换输出	
Y24			
Y25			
Y26		刀套抬起控制输出, 检测抬 起到位 X44	回刀电磁阀 S10
Y27		刀套落下控制输出, 检测倒 下到位 X45, 刀库定位检测 X41 必须到位;	倒刀电磁阀 S8
Y28			
Y29		主轴伺服驱动器使能	主轴伺服驱动器使能
Y30		刀库机械手电机控制输出: (机械手扣刀、机械手抓刀 换刀、机械手归位)	刀库机构马达电机
Y31			

(八) 例子：斗笠式加工中心示意图

如果配广数主轴伺服，注意如下：

系统 M75 接伺服驱动位置速度切换信号；

系统 M22 接伺服驱动位置速度切换完成信号；

系统 M61 接伺服驱动定向启动信号；

系统 M03/M04/M05/+10V/ALM1/+24V/0V 接于伺服驱动对应信号；

伺服驱动参数要设为位置/速度控制模式。

		The pine knife is in place T02; Master shaft tight knife: remove M10, check tight knife in place T05;	
Y11	M08	Cooling controls output	
Y12	M05	M05	
Y13	M04	Shaft reversal control output	
Y14	M03	Main shaft forward rotation control output	
Y15	M75	Switch output of spindle control mode	
Y24			
Y25			
Y26		The knife handle lifts to control the output and detects that it is in position X44	Return knife solenoid valve S10
Y27		The knife sleeve is lowered to control the output, and the detection of X45 is in place. The positioning detection of X41 must be in place;	Backward knife solenoid valve S8
Y28			
Y29		The spindle servo driver is enabled	The spindle servo driver is enabled
Y30		Motor control output of knife library manipulator: (manipulator clamping knife, manipulator grabbing and changing knife, manipulator returning)	Knife library motor
Y31			

(8) Example: Diagram of a Conical Lathe

If the number of spindle servo is allocated, pay attention to the following:

System M75 receives the position and speed switching signal of servo drive;

System M22 receives the signal of position and speed switching of servo drive;

System M61 receives servo drive directional start signal;

System M03/M04/M05/+10V/ALM1/+24V/0V is connected to the corresponding signal of the servo drive; the servo drive parameters should be set to position speed control mode.

1000M斗箭式加工中心接线示意图



1000M斗篷式加工中心接线示意图



第六篇 双通道系统

一、系统控制轴数

- 1、第一通道 X/Y/Z/A(J1/J2/J3/J4)轴对应驱动站号 1/2/3/4;
- 2、第二通道 X/Y/Z/A(J1/J2/J3/J4)轴对应驱动站号 5/6/7/8;

二、系统功能

- 1、第一、第二通道所有功能兼容单通道系统功能;
- 2、加工参数 P260/P261/P262 设置二个通道 XYZ 区域干涉安全距离,当二个通道的相应轴机床绝对坐标距离值小于此参数时出现报警,此时必须在手轮状态按复位键清楚报警,再通过手轮移动坐标轴到安全位置即可;

3、I/O 输入输出分配:

- 1) 输入 X00-X47 为两通道共同使用,使用时请合理分配;
- 2) 第一通道使用输出 Y00-Y15; Y32-Y38;
- 3) 第二通道使用输出 Y16-Y31;
- 4) 双通道系统 I/O 输入输出分配如下:

X00-X07:1T01-1T07;	X26:1TOK;
X16/X17/X18/X19/X32/X35/X21/X23:2T01-2T07;	X27:2TOK;
X09:1-L; X10:1+L; X14:2-L; X15:2+L;	
X08:1A0; X11:1Y0; X12:1X0; X13:1Z0;	
X16:2A0; X17:2Z0; X25:2Y0 X26:2X0;	
X20:1ALM; X24:1M22; X33:1ALM1; X34:1ALM2;	
X22:2ALM; X36:2ALM1; X37:2ALM2; X40:2M22;	
X41:1HALT; X42:ESTOP; X43:1KRUN;	
X44:2HALT; X45:2KRUN; X46:RUN	
Y00:1+T; Y01:1M63-T; Y02:1M65; Y03:1M67;	
Y04:1M69; Y05:1M61; Y06:1M73; Y07:1M71;	
Y08:1M32; Y09:1M79; Y10:1M10; Y11:1M08;	
Y12:1M59; Y13:1M04; Y14:1M03; Y15:1M75;	
Y16:2+T; Y17:2M63-T; Y18:2M75; Y19:2M08;	
Y20:2M61; Y21:2M65; Y22:2M67; Y23:2M69;	
Y24:2M10; Y25:2M79; Y26:2M32; Y27:2M71;	
Y28:2M03; Y29:2M04; Y30:2M73; Y31:2M61;	
ALM:驱动报警; ALM1:变频器报警; ALM2:无润滑油报警;	
1HALT/2HALT:外接暂停; ESTOP:外接急停; 1KRUN/2KRUN:外接运行;	
RUN:总运行; X0/Y0/Z0/A0: X/Y/Z/A 轴回零开关信号;	

4、M881 主轴准停:

第 1 通道:M881 输出 2M61/Y05,到位 X24,使能 2M75/Y15;

Part VI. Two-Channel Systems

I. Number of Control Axes in the System

- 1、 The first channel X/Y/Z/A (J1/J2/J3/J4) axis corresponds to the drive station number 1/2/3/4;
- 2、 The second channel X/Y/Z/A (J1/J2/J3/J4) axis corresponds to the drive station number 5/6/7/8;

2. System Functions

1. All functions of the first and second channels are compatible with those of the single channel system;
- 2、 Processing parameters P260/P261/P262 Set two channels of interference safety distance in the XYZ area. When the absolute coordinate distance value of the corresponding axis machine tool in the two channels is smaller than this parameter, an alarm will occur. At this time, the reset button must be pressed in the handwheel state to clear the alarm, and then the coordinate axis can be moved to the safe position through the handwheel;
- 3、 I/O Input/Output Allocation:
 - 1) Enter X00-X47 for two channels to be used together. Please allocate reasonably when using it;
 - 2) The first channel uses outputs Y00-Y15; Y32-Y38;
 - 3) The second channel uses output Y16-Y31;
 - 4) The dual-channel system I/O input/output is allocated as follows:

X00-X07:1T01-1T07;	X26:1TOK;
X16/X17/X18/X19/X32/X35/X21/X23:2T01-2T07;	X27:2TOK;
X09:1-L;	X10:1+L;
X08:1A0;	X11:1Y0;
X16:2A0;	X17:2Z0;
X20:1ALM;	X24:1M22;
X22:2ALM;	X36:2ALM1;
X41:1HALT;	X42:ESTOP;
X44:2HALT;	X45:2KRUN;
Y00:1+T;	Y01:1M63-T;
Y04:1M69;	Y05:1M61;
Y08:1M32;	Y09:1M79;
Y12:1M59;	Y13:1M04;
Y16:2+T;	Y17:2M63-T;
Y20:2M61;	Y21:2M65;
Y24:2M10;	Y25:2M79;
Y28:2M03;	Y29:2M04;
X14:2-L;	X15:2+L;
X12:1X0;	X13:1Z0;
X25:2Y0	X26:2X0;
X33:1ALM1;	X34:1ALM2;
X37:2ALM2;	X40:2M22;
X43:1KRUN;	
X46:RUN	
Y02:1M65;	Y03:1M67;
Y06:1M73;	Y07:1M71;
Y10:1M10;	Y11:1M08;
Y14:1M03;	Y15:1M75;
Y18:2M75;	Y19:2M08;
Y22:2M67;	Y23:2M69;
Y26:2M32;	Y27:2M71;
Y30:2M73;	Y31:2M61;

ALM: Drive alarm; ALM1: Inverter alarm; ALM2: Lubricant shortage alarm; 1HALT/2HALT: External pause; ESTOP: External emergency stop; 1KRUN/2KRUN: External operation; RUN: Total operation; X0/Y0/Z0/A0: Zero return switch signals for X/Y/Z/A axes;

4、 M881, Spindle Stop:

Channel 1: M881 outputs 2M61/Y05, X24 is in place, enabling 2M75/Y15;

第 2 通道:M881 输出 2M61/Y31, 到位 X40, 使能 2M75/Y18;

三、主轴功能

- 1) 第一通道用第 1 路模拟量+10V1, 第二通道用第 2 路模拟量+10V2;
- 2) 第二通道杂类参数 P900, 自定义加工参数对话框[1 表无效, 4 表部分, 8 表全部, +512 表两个通道 M3/M4/M5 共用一个主轴;

四、宏变量分配

两通道共用宏变量为 #900-#999, 其余宏变量为各通道独立使用; 同时宏变量 #900-#999 可以在 PLC 中调用作 CMP 比较运算, 在导入宏变量参数文件时只有第一通道才可导入宏变量 #900-#999 ;
其中, #950-953 系统指令 M890-M893 已经使用。

双通道共用辅助继电器: M346--M361, 二个通道操作同一个辅助继电器, PLC 中用该辅助继电器输出到对应通道的 Y 输出点即可; 例如: 第 2 通道执行 OUT+M346, 第 1 通道 PLC 中 M346 输出 Y32 侧第 2 通道就可以控制 Y32。

五、同步相互等待指令

- 1、M701-M779: 等待另一个通道执行同一指令时才继续向下执行程序。该 M 指令必须单独占用一行;
- 2、取消该通道等待指令: M700, 在程序执行 M700 指令后, M701-M779 将跳过不执行。该 M 指令必须单独占用一行;
- 3、M797: 启动另一个通道运行程序;
M798: 暂停另一个通道程序运行;
M799: 复位另一个通道程序运行;
M790: 启动第二通道运行第一通道的当前运行程序;
M791: 启动第二通道运行程序;
M792: 暂停第二通道程序运行;
M793: 复位第二通道程序运行。
M794: 启动第一通道运行程序
M795: 暂停第一通道程序运行;
M796: 复位第一通道程序运行;

注意: 被启动的通道如果在手动状态则自动进入自动状态并运行;

- 4、第二通道编码器反馈指令:

<1>加工参数 P35 设置:

P35, 第二通道编码器反馈 (16: 选择第一编码器, 32: 选择第二编码器)

<2>指令设置

M111 选择第一编码器, M112 选择第二编码器;

六、通道选择

- 1、按 “K1” 选择第一通道;
- 2、按 “K2” 选择第二通道;

七、面板功能

- 1、“手轮” 功能: 手轮对所有通道有效, 使用时请注意;

Channel 2: M881 outputs 2M61/Y31, X40 is in place, enabling 2M75/Y18;

3. Main Shaft Function

- 1) The first channel uses the 1st analog signal +10V1, and the second channel uses the 2nd analog signal +10V2;
- 2) The second channel miscellaneous parameters P900, custom machining parameters dialog box [Table 1 invalid, Table 4 partial, Table 8 all, +512 indicates that two channels M3/M4/M5 share one spindle;

IV. Macro Variable Allocation

The macro variables shared by the two channels are #900-#999, while the other macro variables are used independently by each channel. Meanwhile, the macro variables #900-#999 can be invoked in the PLC as CMP comparison operation. When importing the macro variable parameter file, only the first channel can import the macro variables #900-#999;

Among them, the system instructions #950-953 M890-M893 have been used.

Dual-channel shared auxiliary relay: M346--M361, two channels operate the same auxiliary relay. In the PLC, this auxiliary relay outputs to the corresponding channel's Y output point. For example: When the second channel executes OUT+M346, the PLC's M346 outputs to the second channel on the Y32 side, allowing the second channel to control Y32.

5. Synchronize Mutual Waiting Instructions

1 、 M701-M779: The program continues to execute downward only after another channel has executed the same instruction. This M instruction must occupy a separate line;

2 Cancel the channel waiting instruction: M700. After the program executes the M700 instruction, M701-M779 will skip the execution. The M instruction must occupy a separate line;

3 、 M797: Start another channel to run the program; M798: Pause another channel program to run; M799: Reset another channel program to run;

M790: Start the second channel to run the current running program of the first channel;

M791: Start the second channel running program;

M792: Pause the operation of the second channel program;

M793: Reset the second channel program to run.

M794: Start the first channel running program

M795: Pause the operation of the first channel program;

M796: Reset the first channel program to run;

Note: If the started channel is in manual state, it will automatically enter automatic state and run;

4. Feedback instruction of the second channel encoder:

<1>Processing parameter P35 setting:

P35, second channel encoder feedback (16: select first encoder, 32: select second encoder)

<2>Instruction Settings

M111 selects the first encoder, and M112 selects the second encoder;

VI. Channel Selection

1. Select the first channel according to "K1";

2. Select the second channel according to "K2";

VII. Panel Functions

1 、 Handwheel function: The handwheel is effective for all channels. Please pay attention to it when using;

2、“启动”按钮启动当前通道；

八、操作使用说明

1、两个通道有各自独立的参数包，包括关于 PLC 的文件和 I/O 配置文件均为各通道独立控制；

2、两个通道有各自独立的刀具控制和刀具长度半径补偿功能，有各自独立的螺距误差反向间隙补偿等功能；

3、在第一和第二通道状态下均可以导入导出参数和导入软件，两个通道共享用户文件存储区；

4、第一和第二通道的显示区别为抬头标题栏颜色不同，第一通道为单通道原始的蓝色，第二通道为绿色；

5、第一和第二通道的软件版本号显示有区别：第一通道显示 D1 标志；第一通道显示 D2 标志；

6、局域网传输文件和远程监控功能只能用于第一通道，第二通道不具有该网络功能。

- 2、 The "Start" button starts the current channel;

VIII. Operation Instructions

- 1 、 The two channels have their own independent parameter packages, including the files about PLC and IO configuration files, which are independently controlled by each channel;
2. The two channels have their own independent tool control and tool length radius compensation functions, as well as their own independent pitch error reverse clearance compensation functions;
3. Both the first and second channels can import and export parameters and import software, and the two channels share the user file storage area;
4. The difference in display between the first and second channels is that the color of the heading title bar is different. The first channel is the original blue of the single channel, and the second channel is green;
- 5、 The software version numbers of the first and second channels are shown to be different: the first channel shows the D1 logo; the first channel shows the D2 logo;
6. The LAN file transfer and remote monitoring functions can only be used for the first channel, and the second channel does not have this network function.